

Derivatives and Hedge Funds

Edited by
Stephen Satchell



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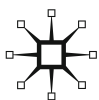
Derivatives and Hedge Funds

Edited by

Stephen Satchell

Professor of Finance, Sydney University, Australia

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Chapters originally published in *The Journal of Derivatives and Hedge Funds* by Palgrave Macmillan (various years)

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First published 2016 by
PALGRAVE MACMILLAN

Palgrave Macmillan in the UK is an imprint of Macmillan Publishers Limited, registered in England, company number 785998, of Houndmills, Basingstoke, Hampshire RG21 6XS.

Palgrave Macmillan in the US is a division of St Martin's Press LLC, 175 Fifth Avenue, New York, NY 10010.

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ISBN 978-1-349-55828-5 ISBN 978-1-137-55417-8 (eBook)
DOI 10.1057/9781137554178

This book is printed on paper suitable for recycling and made from fully managed and sustained forest sources. Logging, pulping and manufacturing processes are expected to conform to the environmental regulations of the country of origin.

A catalogue record for this book is available from the British Library.

A catalog record for this book is available from the Library of Congress.

Typeset by MPS Limited, Chennai, India.

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Preface

It is a pleasure and an honour to edit this volume, celebrating 20 years of the *Journal of Derivatives and Hedge Funds* (JDHF). The 18 chapters included in this volume represent a small sample of my favourite papers published during the life of the Journal, and a period which included my long tenure as editor. Many excellent papers have not been included; my tastes often reflect strongly idiosyncratic features and another rational decision-maker may well choose 18 completely different contributions. Broadly, the papers chosen reflect popular demand and impact as measured by such statistics as impact factor, number of citations etc., but I exercised my personal preferences a little as well. The motivation for this volume is to mark the ending of JDHF as a stand-alone entity and its merger into the *Journal of Asset Management*. I should mention that there is continuity in many respects in as much as I edited both; I shall continue to edit the merged journal and most of the stakeholders from Palgrave Macmillan, the editorial board, and hopefully contributors or readers, will continue as before.

In this time both hedge funds and derivatives have fluctuated in reputational terms; they have been blamed for the global financial crisis and been praised for the provision of liquidity in troubled times. Both topics are rather under-researched due to a combination of data and secrecy issues. JDHF has been one of the few outlets for industry-orientated research in these areas; hopefully this book will reach the audience who will appreciate the journal's special contributions.

I have divided the volume into two sections; the first focuses on Hedge Funds and the second on Markets, Prices and Products. This way of dividing the chapters helps overcome the problems associated with discrete taxonomy. Thus the chapter by Raphael Lam and Marco Rossi on Sovereign Wealth Funds is included in the second section, while Bernd Scherer's chapter on commodity trading advisors (CTAs) is included in the first section.

Bernd provides two computational alternatives to find the optimal number of CTAs in a real-world setting where frictional costs of diversification, the amount of assets under management, risk aversion and the state dependence on hedge fund payoffs matter to investors. A final marginal case may be Damir Tokic, whose chapter investigates the traders' futures-only positions, options-only positions and futures-and-options-combined positions in crude oil to draw some implications with respect to the behaviour of traders during the 2008 oil bubble. He finds clearly different investment behaviours for different manager types. He considers, inter alia, Money Managers, Swap Dealers and Producers/Merchants/Processors.

The other contributors to the hedge fund section fall into two groups, what I would describe as the highly non-linear versus the rest. We have two good chapters on the use of higher moments in a hedge fund context, one by Greg Bergh and Paul van Rensburg and the other by Ryan J. Davies, Harry M. Kat and Sa Lu. Both chapters use higher moments to construct hedge fund portfolios, topics with applications; a little-explored application being its potential use in high net-worth investment, where preferences over higher moments can be used as a way of characterising investors.

Our final group of essays on hedge funds includes work by Roland Füss, Dieter G. Kaiser and Zeno Adams who examine the conditional volatility characteristics of daily management-style returns and compare different out-of-sample forecasts of different Value at Risk (VaR) approaches. According to the criteria adopted, the generalised auto-regressive conditional heteroscedasticity (GARCH-type) VaR outperforms the other VaRs for most of the hedge fund-style indices.

Meredith Jones investigates whether smaller, younger hedge funds offer stronger performance than larger, older hedge funds. Indices created with six subsets of hedge fund data (small, medium, large, young, mid-age and older funds, as defined herein) and Monte Carlo simulations are used to examine the performance, volatility and risk profiles of each fund group. This approach to investigate performance in terms of duration is highly innovative and has broad application. Samuel Manser and Markus M. Schmid, in a related chapter to Meredith Jones, investigate the persistence of raw and risk-adjusted returns for equity long/short hedge funds. Only limited evidence of persistence is found for raw returns. Funds with the highest raw returns last year continue to outperform over the subsequent year, although not significantly, while there is no persistence in returns beyond one year. In contrast, performance persistence based on risk-adjusted return measures such as the Sharpe Ratio and in particular an alpha from a multifactor model, is more easily detected.

Lastly, among the hedge fund chapters, I mention François-Éric Racicot and Raymond Théoret who ask the question: is there a negative correlation between beta and the market risk premium in the hedge fund sector? Previous researchers have found such a phenomenon in the mutual fund industry. The study reveals that there is generally little evidence of this in the hedge fund industry. There is an apparent beta puzzle for the distressed securities and short seller strategies, but it seems rational for those strategies to increase their beta when the market risk premium is low: business opportunities are then much more important for them. This chapter uses a variety of innovative econometric tools.

In Part II, the title – Markets, Pricing and Products – reflects a certain heterogeneity which is evident in the contents. One of the literature's early contribution to empirical market-micro structure is a paper published in JDHF in 2005 by Roy A. Batchelor, Amir H. Alizadeh and Ilias D. Visvikis. This looks at the relation between bid-ask spreads (BASs) and price volatility

in forward markets. In particular it looks at the forward freight agreement (FFA) market developed in the 1990s and which grew very fast as the main derivatives market offering agents in the shipping and transportation industry a risk management instrument. This chapter examines the relationship between expected volatility and BASs in this derivatives market where the underlying asset is a service rather than physical or financial assets. The study employs a two-step modelling specification. In the first step, the GARCH specification is used to model the volatility of the FFA prices; in the second step, the relationship between expected conditional volatility (led by one day) and the current BAS using the generalised method of moments (GMM) approach is investigated. The results indicate that there is a positive relationship between BASs, and expected price volatility in three out of four FFA contracts, after other factors are controlled.

By contrast the second chapter, by Geng Deng, Joshua Mallett and Craig McCann, models autocallable structured products. Since it was first introduced in 2003, the number of autocallable structured products in the United States has increased exponentially. The autocall feature causes the product to be redeemed if the reference asset's value rises above a pre-specified call price. Because an autocallable structured product matures immediately if it is called, the autocall feature reduces the product's duration and expected maturity. This chapter presents a flexible Partial Differential Equation framework to model autocallable structured products. The framework allows products with either discrete or continuous call dates. The analysis is highly sophisticated involving computational procedures and closed form solutions.

The third chapter by Christian L. Dunis, Jason Laws and Ben Evans continues the pattern of technical sophistication using non-linear cointegration techniques to investigate the gasoline crack spread. Alarmed readers will be relieved to discover that cracking here is a chemical process and the spread to cracking (relative to crude prices) can be viewed as the profit margin gained by cracking crude oil, and it is argued in the chapter that any non-linearity can be interpreted in the context of the effect on market participants. Further, a number of non-linear neural networks are used to forecast the gasoline crack spread. The architectures used are multilayer perceptron, recurrent neural networks and higher-order neural networks; these are benchmarked against a fair value non-linear cointegration model. The results show, first, that the spread does indeed exhibit asymmetric adjustment, with movements away from fair value being nearly three times larger on the downside than on the upside. The time-series asymmetric adjustment is a topic of current research especially post GFC.

Christos Floros and Dimitrios V. Vougas are the authors of the next chapter which focuses on various techniques to investigate the relationship between information and the volatility of the FTSE/ASE-20 and FTSE/ASE Mid 40 indices in Greece. The results for the FTSE/ASE-20 index suggest that

futures trading has led to decreased stock market volatility (negative effect), but the results for the FTSE/ASE Mid 40 index indicate that the introduction of stock index futures has led to increased volatility (positive effect), while the estimations of the unconditional variances indicate lower market volatility after the introduction of stock index futures. Furthermore, the results show that good news has a more rapid impact on FTSE/ASE-20 stock return volatility. For the FTSE/ASE Mid 40 index, the results suggest that news is reflected in prices more slowly, while old news has a less persistent effect on prices. The impact of the creation of futures markets on existing stock market volatility is a perennial problem in finance and this chapter dissects this question very thoroughly.

We note, at this point, a later chapter by Mohammad G. Robbani et al., closely related to these two.

A chapter that directly addresses some of the critical issues underlying the GFC is the essay by Andreas A. Jobst. It addresses popular concerns about complex structured finance techniques, such as customised single tranche and hybrid collateralised debt obligations with overlay structures, against the background of tightening credit spreads and greater dislocation in the correlation market. Subsequent warnings about the resilience of credit risk transfer to systemic crisis, however, hardly extended beyond indistinct assessments of how derivatives might propagate asset shocks across different capital market segments. This chapter defines structured finance in order to inform a more specific debate about the regulatory challenges posed by the assembly of asset exposures and credit risk transfer in complex structured finance transactions that marry considerations of profitability and diversification alike. It is titled 'A Primer on Structured Finance'; I found it very valuable in understanding an area that has only recently appeared in textbooks.

The next chapter, by Michael L. McIntyre and David Jackson, could be called a core options chapter as it addresses the practice of covered call writing which involves selling call options while owning the underlying asset. This strategy, in theory, is just reshuffling of deck-chairs on the titanic of put-call parity. The authors examine the empirical performance of an investment strategy that uses covered call writing to enhance the income from long positions in 27 stocks that are included in the FT-SE 100 index. Using data for the period January 1994–December 1999 they show that, contrary to theory, in most instances covered call positions generate returns that exceed returns generated by buy-and-hold strategies.

In the next chapter, Lei Meng and Owain ap Gwilym address credit default swap trading before the GFC as it was published in 2007. They study single name reference entities and find a prevalence of five-year maturities, US\$5 million and US\$10 million notional amounts, senior-ranked underlying debt and modified restructuring clauses. They find that the majority of reference entities had credit ratings from at least one international rating agency, and the average credit quality of the universe fluctuated during the sample

period. Trading activity demonstrated an inverse U shape across the week. In some sense, this thoughtful chapter is a victim of history as the general perception and popularity of CDS products have changed dramatically.

Mohammad G. Robbani and Rafiqul Bhuyan also contribute a chapter on the impact of the introduction of derivative markets on asset markets. The effect of the introduction of futures and options on the Dow Jones Industrial Average index on the volatility and trading volume of its underlying stocks is examined. Traditional measures and GARCH specification show that the levels of volatility and trading volume significantly increased after the introduction of futures and options on the index. The study provides new evidence in support of the argument that futures trading attracts uninformed or irrational traders along with the rational or informed traders, which causes an increase in stock return volatility. This may indicate that, even though the market may become more liquid, the destabilising effect brought by irrational traders in both the cash and futures markets outweighs the beneficial liquidity effect. This chapter suggests potentially detrimental effects of derivative introduction and certainly is evidence that JDHF is not just a trade journal publishing only positive news for our stakeholders.

Finally, we end with a technically sophisticated chapter by Sascha Wilkens, which addresses issues of option pricing with mixture distributions. The use of flexible distributional assumptions in an option pricing framework has been analysed extensively in recent years. Based on a large set of transaction data from 1999 and 2000, this chapter readdresses the suitability of flexible distributions for forecasting and hedging purposes of DAX and Euro-Bund-Future options traded in the Eurex. Implied risk-neutral distributions (RND) are derived via the original Black and Scholes model, the series expansion approach of Corrado and Su and models based on mixtures of two and three log-normal distributions. While, especially for DAX options, the more complex RND proves to be superior to the log-normality assumption within the Black-Scholes model in terms of forecasting prices, he finds evidence that delta-neutral hedge portfolios are best derived from simple Black-Scholes hedge ratios.

Overall I hope this book encourages further research in practitioner-orientated research in the above areas. In my continuing role as Editor of the *Journal of Asset Management*, I plan to continue publishing similar material.

Stephen Satchell

Part I

Hedge Funds

1

Frictional Costs of Diversification: How Many CTAs Make a Diversified Portfolio?

Bernd Scherer

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How many commodity trading advisors (CTAs) are needed to arrive at a diversified portfolio? We provide two computational alternatives to find the optimal number of CTAs in a real-world setting where frictional costs of diversification, the amount of assets under management, risk aversion and the state dependence on hedge fund payoffs matter to investors.

Introduction

With the establishment of modern portfolio theory, researchers have started to test how well its normative diversification advice is reflected in observed portfolios. Early studies focused on equity markets and tried to answer the question: ‘How many stocks make a diversified portfolio?’ Elton and Gruber (1977), Statman (1987), Newbould and Poon (1993), O’Neal (1997) and Statman (2004) all come to different conclusions about the optimal number of stocks in a naively diversified (randomly selected stocks with equal weighting in the absence of conditioning information) portfolio. The recommended holdings range between 10 and 300 stocks. However, even these

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numbers are high relative to the accounts of individual investors, which often contain only a handful of stocks as well as large holdings in their own company stocks. On the back of these results, Statman (2004) coined the term ‘behavioural portfolio theory’, that is, the attempt to ‘rationalize’ the apparent under-diversification of individual investors. In his view, individual investors divide their total wealth into mental buckets according to their investment goals. Equities fall into the top portfolio layer that reflects the investors’ demand for lottery tickets. Recent support for this has been provided by Frazzini and Pedersen (2010), who find that leverage aversion will cause investors to arrive at under-diversified portfolios that concentrate on the more volatile stocks.

The popularity of hedge funds, as both an investment vehicle and an object of academic interest, has created interest in the question: ‘How many hedge funds (commodity trading advisors (CTAs), managed futures, etc.) make a diversified portfolio?’. Despite well-documented differences in hedge fund return distributions (most notably non-normality and non-linearity with respect to underlying risk factors) and hedge fund investment costs, virtually all studies heavily borrowed the methodologies designed for individual stock portfolios and applied them to hedge funds. In short, this amounts to a two-step procedure:

1. Simulate random portfolios of size $n=1, \dots, N$ and record the evolution of volatility, SHARPE-ratio or correlation with an already diversified index to trace out a diversification curve, that is, a functional relationship between portfolio standard deviation and portfolio size.
2. Decide when the marginal improvement in the above-mentioned statistic becomes ‘small’. What ‘small’ means is usually left to eyeballing the diversification curve, that is, it relies on the researcher’s subjective statement.

Henker and Martin (1998), Amin and Kat (2002) and Lhabitant and Learned (2002) are examples of this approach. The number of hedge funds they deem optimal ranges between 5 and 25. Despite the arbitrariness of the above approach, Brown *et al* (2011) claim that fund of funds exhibit excess diversification.

We see several shortcomings in the above papers. First, no attempt is made specifying the frictional costs of adding another fund into a portfolio. In the absence of these costs, it is always optimal to naively diversify across all possible investments. Samuelson (1967) made this point early on by stating that investors should diversify as much as possible, aware of the tradeoff between diversification and its costs. Frictional costs arise from fixed monitoring costs per additional funds, as well as the loss of bargaining power for fee rebates when diversifying among too many funds. Second, assets under management do not enter the decision-making problem, even

though fixed costs can be spread more easily across a large pool of assets. Clearly it makes a very practical difference whether a decision-maker with 10 million USD or 100 million USD asks for the optimal number of assets to invest in. Third, the reduction in volatility is most valuable for investors with high risk aversion, while investors with low risk aversion will be less willing to incur frictional diversification costs for a reduction in volatility they value only very little. Finally, but most importantly, volatility for an investment will not differ if we reshuffle returns across different states of the world. However, investors have a preference for investments that pay well in bad states (where wealth is down) of the world. Such an investment might be more valuable or offer more protection than an asset that offers a higher SHARPE-ratio or lower volatility. Diversifications studies on hedge funds remain silent on this topic. This is most relevant for CTAs that, due to their trend-following trading style and money management techniques, offer portfolio insurance properties. For a risk-averse investor, it will now matter most how well his portfolio of CTAs performs in those months where he values insurance most highly. Consequently, the normative advice of these papers is limited at best.

Another more recent motivation of our work is the flood of papers motivated by Demiguel *et al* (2009). The authors show that equal weighting ($1/n$) is preferable to mean variance optimization if the SHARPE ratio differences between assets are small (adjusted for sample size). This situation is likely to be given for CTAs that are notoriously known for both large return dispersion and little to no persistence as documented in Bhardwaj *et al* (2008). None of the $1/n$ papers discusses frictional diversification costs and consequently the optimal number of assets is imposed rather than derived.

The next section first reviews the existing methodology used in diversification studies. We then extend the traditional mean variance framework to account for frictional costs of diversification, differences in assets under management and risk aversion to arrive at a closed form solution for the optimal number of assets. This method works well under the assumptions of normality and for investors who show no interest in the conditional nature of hedge fund returns. However, for investors who care whether losses are realized in good or bad times, we evaluate portfolios of CTAs for different state price deflators (assuming investors exhibit power utility). This allows us to evaluate whether a portfolio of CTAs is properly diversified, where diversification means the extraction of 'CTA-beta', that is, the ability to protect from a fall in risky assets. The latter sections apply both methods on CTAs. The final section concludes.

Optimal diversification revisited

Diversification curves

Diversification curves trace out the relation between the expected risk (or, more generally, performance measure) of a portfolio of hedge funds (or more

general assets) that consists of n randomly selected and equally weighted constituents. The brute force approach is to repeatedly sample n funds (out of a total of N funds). For the j th sampling we get a set $S_j(n)$ that contains n index numbers (out of N) to calculate the portfolio return

$$R(n)_{m,j} = \sum_{i \in S_j(n)} \frac{1}{n} R_{i,m} \quad (1)$$

for all $m=1, \dots, M$ scenarios, that is, observations on a databank. This exercise is repeated many times, that is, we compute the variance of (1), that is,

$$\sigma(n)_j^2 = \sum_{m=1}^M \left(R(n)_{m,j} - \frac{1}{M} \sum_{m=1}^M R(n)_{m,j} \right)^2 \quad (2)$$

and average it over all J samplings. However, it was shown early on by Elton and Gruber (1977) that a simulation is not needed and we can instead replace time-intensive simulations with the formula for the expected variance of a naively diversified portfolio.

$$\sigma^2(n) = \frac{\bar{\sigma}^2}{n} + \left(1 - \frac{1}{n}\right) \bar{\sigma}^2 \bar{\rho} \quad (3)$$

where $\bar{\sigma}^2$ and $\bar{\rho}$ represent average variance and correlation across all N assets, that is, for the whole universe rather than the subset of n stocks. The rationale for this surprising result is that (3) is equivalent to the expectation of all possible n out of N permutations of index numbers. Mathematically, (3) is a monotonically decreasing function of n that converges to $\bar{\sigma}^2 \bar{\rho}$; hence, optimal diversification is only achieved by holding the full universe. A decision to hold less than n assets is typically imposed by *ad hoc* assumptions on the desired degree of reduction in diversifiable risk and is therefore not satisfactory.

Mean variance-based performance measure

Rather than the previous *ad hoc* approach, we first suggest starting to model optimal diversification as a decision-making problem for a standard mean variance investor and ask: ‘How many hedge funds make a diversified portfolio?’ In contrast to previous work, our decision maker will trade off the marginal benefits from diversification against their marginal costs. For an investor using naive diversification (that is, in the absence of conditioning information) the only choice parameter is the number of equally weighted funds she intends to include, that is, we assume that the investor tries to optimize

$$\mu(n) - \lambda \sigma^2(n) - n \frac{f}{aum} \quad (4)$$

where $\mu(n)$ and $\sigma^2(n)$ are the expected return and risk for an equally weighted portfolio of size n , λ denotes the investor's risk aversion and f/aum represents the additional fixed costs as a fraction of assets under management per additional fund.

While adding funds to a portfolio, we can expect portfolio risk to fall, while average portfolio return is expected to remain constant, that is $(d\mu(n)/dn)=0$.¹ We can therefore focus on the impact of increasing portfolio size on portfolio risk and diversification costs. For the optimal number of funds in a portfolio, marginal benefits and costs need to be balanced. Hence we can write

$$\frac{f}{aum} = -\lambda \frac{d\sigma^2(n)}{dn} \quad (5)$$

As seen in the previous section, the expected variance for an equally weighted portfolio can be equally written as $\sigma^2(n) = \bar{\sigma}^2 / n + (1 - (1/n))\bar{\sigma}^2\bar{\rho}$, so we can find an explicit solution for the marginal change in risk

$$\frac{d\sigma^2(n)}{dn} = \frac{1}{n^2} \bar{\sigma}^2 (\bar{\rho} - 1) \quad (6)$$

Substituting (6) into (5) we arrive at

$$\frac{f}{aum} = -\lambda \frac{1}{n^2} \bar{\sigma}^2 (\bar{\rho} - 1) \quad (7)$$

which can be solved for the optimal n .

$$n^* = \sqrt{\lambda \bar{\sigma}^2 (1 - \bar{\rho}) \left(\frac{f}{aum} \right)^{-1}} \quad (8)$$

We see that the optimal number of funds increases with rising risk aversion (λ), rising average volatility ($\bar{\sigma}^2$), falling average correlation ($\bar{\rho}$), falling frictional costs (f) and rising assets under management (aum). The objective of (8) is to provide insight into the determinants of naïve diversification with frictional diversification costs. Linearizing (8) via logs will also yield a testable model for observed fund of fund data. The model suffers from two shortcomings. First, how should an investor solve the above problem when instead he has some information (return forecasts) on individual funds? In this case we need to solve a quadratic program with mixed integer constraints using dedicated software like NUOPT for SPLUS/R. A formulation is given in Appendix C. Second, it is clear that (8) will provide little guidance when hedge fund data are non-normally distributed. This is dealt with in the next section.

State price deflators

The previous section established the optimal number of hedge funds in closed form for a mean variance decision-maker. Even if the returns for our assets would all be individually normal (which we know they are not) the approach in the previous section would fail. The main drawback of the solution in the previous sub-section arises from hedge fund returns that display a non-linear relationship to risky assets, as documented by Fung and Hsieh (2001). This is particularly true for CTAs that create part of their attraction from their ability to perform well if risky assets are down for a prolonged period of time. This feature of CTA returns is important to real-world investors but gets lost in SHARPE-ratio or volatility measures. Imagine we simply reshuffle hedge fund returns, that is, we take the returns of hedge fund strategies, and rearrange them randomly (without replacement) along the timeline. This will affect neither their SHARPE-ratio nor their skewness or volatility, but it will alter their attractiveness to real-world investors as suddenly their returns conditioned on the returns of equity and bond markets will change.

We follow Chen and Knez's (1996) seminal work and use state price deflators for performance measurement to address this problem. A state price deflator is a stochastic discount factor that applies a separate discount rate to each state of the world. States where economy-wide wealth is down and therefore marginal utility is up should carry a larger importance for risk-averse investors. Hence we can generically express the state price deflator Λ_m for state m , a negative function of wealth in state m (W_m), and a positive function of risk aversion (γ)

$$\Lambda_m = F\left(\gamma, W_m\right)_{\substack{(+), \\ (-)}} \quad (9)$$

The exact functional form for (9) and its calibration (choice of γ) to real data is described in Appendices A and B. Once we know the state price deflator, we can price any asset (or derivative claim) by multiplying the state price deflator with the corresponding payoff to the asset under consideration and build the expectation under the real-world probability measure P . The value of an n fund portfolio with return $R(n)_{m,j}$ in state m after frictional costs $\phi(n)$ in simulation run j is given by)

$$V(n)_j = E^P\left(\Lambda_m \cdot (1 + R(n)_{m,j})\right) - \phi(n) \quad (10)$$

Averaging across many (J) we get

$$EV(n) = \bar{V}(n) = \frac{1}{J} \sum_{j=1}^J V(n)_j \quad (11)$$

Consequently, funds that pay off well when wealth is low will get a higher valuation even when their average returns are identical or even

lower than those hedge funds that synchronize their losses with losses in the economy.

In order to model the frictional costs of diversification, we introduce two cost functions. The first simply models the increase in fixed costs that arise when an additional fund is added to a portfolio. These costs can be thought of as monitoring costs (hire qualified analysts, consultants, buy software, data, or incur other due diligence costs like costs of flights, hotel). All this is consumed in annual fixed costs f per additional fund; hence, costs per months as a fraction of assets under management (aum) are given by

$$\phi(n) = -n \frac{f \cdot aum^{-1}}{12} \quad (12)$$

In addition, we introduce rebates (θ) to reflect buying power if an investor focuses on investing larger sums in a smaller number of funds.

$$\phi(n) = -n \frac{f \cdot aum^{-1}}{12} + b \cdot \log\left(\frac{aum}{n}\right) \cdot \frac{\theta}{12} \quad (13)$$

The size of the rebate depends on $\log(aum/n)$, that is, it rises with the average fund size. We apply a scaling factor b to control the range of rebates. Typically one would set b equal to the inverses of the log of the maximum mandate size in a given study, such that the maximum mandate gets a full rebate $\theta/12$ while all other mandates get a lower rebate. We use this valuation measure to calculate where $(dEV(n)/dn)=0$ to arrive at the optimal number of CTAs in a portfolio.

Case study #1: Optimal Number of CTAs in a Mean Variance Framework

Our first case study looks at potential differences in optimal diversification when diversifying across large or small CTAs. The data for this case study are summarized in Table 1.1. We use monthly data from the Barclays Managed Futures database stretching from January 2007 to January 2012, where size is defined as average assets under management (AUM) between January 2011 and January 2012. Small funds are defined as funds with AUM smaller than 100 million. No attempt has been made to filter out micro CTAs (CTAs with less than 5 million under management) so the number of funds is large and investing in this quintile might not be entirely realistic. Medium-size CTAs are defined by AUM ranging between 100 and 750 million, while large CTAs are defined by AUM above 750 million.²

Comparing CTAs across the size dimension reveals some interesting differences. First, large CTAs display the lowest average volatility (13 per cent) and the highest average correlation (0.8). The high correlation might be due

Table 1.1 Managed futures – size versus key characteristics

Definition	Small	Medium	Large
	AUM smaller than 100 million	AUM between 100 and 750 million	AUM larger than 750 million
# of funds	332	137	60
Average volatility	19.98	14.55	13.0
Average internal correlation	0.33	0.46	0.80
Average Sharpe	0.50	0.62	0.56
Median Sharpe	0.55	0.56	0.53
5% Percentile Sharpe	–0.21	0.08	0.07
95% Percentile Sharpe	1.22	1.40	1.25
1/ <i>n</i> Sharpe	0.96	0.91	0.78

The table presents average volatility (aggressiveness), average correlation (diversification benefit), average SHARPE (skill) as well as the SHARPE of an equally weighted portfolio for managed futures bucketed into three size categories. Our data stretch from January 2007 to January 2012. Size is defined as average AUM between January 2011 and January 2012.

Source: Barclays Managed Futures Database.

to the fact that large CTAs originate from the same set of ideas, as some of them are just early offspring of other large CTAs. Smaller CTAs try instead to bring in new ideas as they can't afford to be perceived as 'me too' products. This is reflected in their low internal correlation (0.33). Small CTAs want to grow with performance and their large volatility (19.98 per cent) reflects the call-options-like feature of hedge fund business models. Large funds, on the other hand, display lower volatility. This reflects the fact that they have more to lose from drawdowns (large redemptions), as well as the fact that their size makes it difficult to operate with the same level of aggressiveness as their smaller peers.

All this should clearly affect the optimal number of CTAs in an investor's portfolio. The basic information in Table 1.1 contains all inputs needed to apply (8). We start with an example. Imagine a small investor with 10 million assets under management willing to invest in large CTAs. Her risk aversion is 2 and she faces fixed costs of 15 000 USD per additional fund.³ How many CTAs will this investor find optimal? The optimal solution becomes

$$\sqrt{2 \cdot 0.13^2 \cdot (1 - 0.8) \left(\frac{0.015}{10} \right)^{-1}} = 2.12 \quad (14)$$

We can broaden our calculations to varying levels of assets under management and risk aversion, as displayed in Figure 1.1 where various 'iso-diversification'-curves are shown. These curves describe the location of aum and risk aversion combinations that would yield the same optimal number

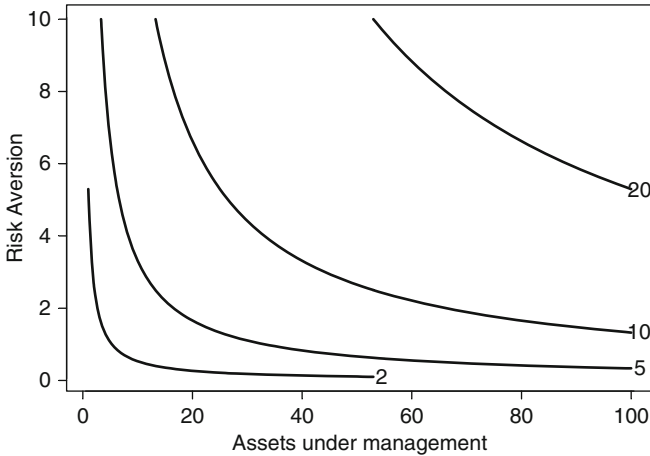


Figure 1.1 Optimal number of CTAs for a mean variance investor. The contour plot displays the optimal number of CTAs using (8) as a function of both risk aversion and assets under management for the universe of large CTAs, that is, $\bar{p}=0.80$, $\bar{\sigma}^2=0.13$ and $f=0.015$

of funds. Each curve describes the tradeoff between risk aversion and assets under management. A small number of assets in a portfolio are constant with either a very large investor and very low risk aversion or a very small investor and very high risk aversion.

Investing in large funds – that are often lookalikes – has limited pay-offs and so the number of funds should be kept small unless risk aversion becomes large.

Case study #2: optimal number of CTAs in contingent claims framework

In this section we apply the more general state price deflator approach to a set of CTA returns in order to find the optimal number of CTAs. Our starting point is using the implied risk aversion of a power utility investor who invests 100 per cent of his wealth in the US equity market to define a representative investor. We use 10 years of monthly data from January 2002 to December 2011 where the one-month US risk-free rate and US equity market excess returns come from the Fama and French research database. We calibrate the state price deflator as described in Appendix B and plot it across time. The result of this is shown in Figure 1.2.

The importance of performing well in down markets in order to score a high valuation measure is evident from the figure. Hedge funds that perform poorly in times of down markets (in general short volatility strategies) will be penalized for this behaviour with respect to investor valuation. This effect

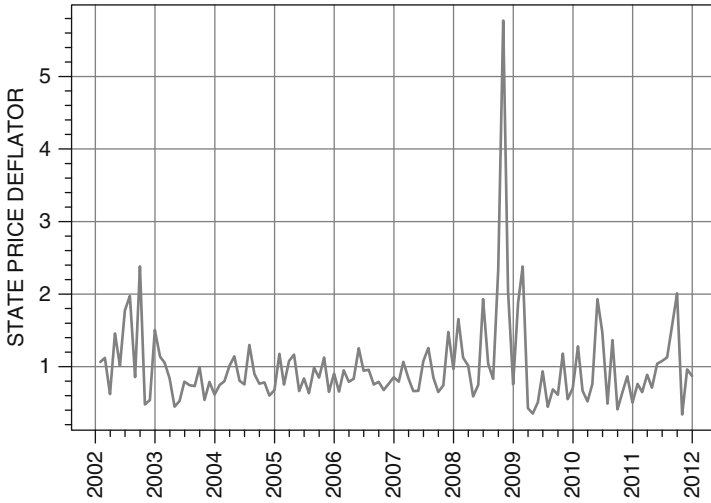


Figure 1.2 State price deflator

We plot the implied state price deflator for an economy where the representative display power utility investor finds it optimal to hold 100 per cent in US equities. Our data span 10 years of monthly data from January 2002 to December 2011. The one-month US risk-free rate and US equity market excess returns come from the FAMA/FRENCH research database.

will become even more pronounced for higher risk aversions, where our valuation measure becomes more and more centric in a few down markets.

Suppose we now want to use the above state price deflators to find the optimal number of CTAs for the period January 2002 to December 2011. First, we select all funds from the Barclays Managed Futures databank that show a complete 10-year series of monthly returns and are described in the databank as technical/ systematic/diversified funds. Second, we also require those funds to be denominated in USD. After applying these criteria we are left with 74 funds. Imagine we use cost function (12), that is, a cost function that only incorporates the stepwise increase in fixed costs once a new fund is added to the portfolio. We set $\gamma=2.3$, $f=0.01$, $aum=50$ and $J=50\,000$, that is, we perform 50000 simulations for a power utility investor with risk aversion of 2.3, 10000 USD stepwise fixed costs and 50 million assets under management. Figure 1.3 plots the distribution (as a boxplot) of our valuation metrics given as a function of n , adding two new funds at a time.

The highest portfolio valuation is reached for four funds, that is, our investor finds it optimal to allocate 12.5 million per fund. We can interpret the valuation measure as monthly alpha of around 70 bps versus a 100 per cent investment in US equities with alpha of 0 per cent. In order to generalize this result, we repeat the above analysis for investors with 10, 50, 100 and 500 million assets under management, and risk aversions of 2.3, 10 and 50 for both cost functions.⁴ The results are shown in Table 1.2.

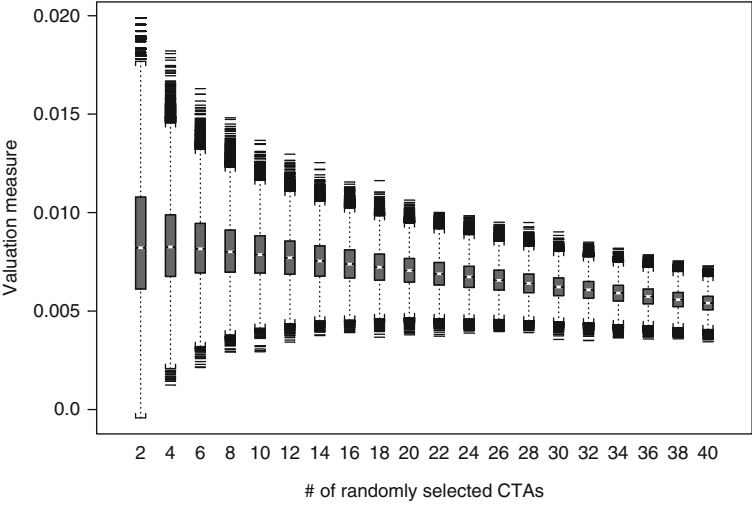


Figure 1.3 State price deflator
Optimal diversification curve according to our valuation metric in (10) for $\gamma=2.3$, $f=0.01$, $aum=50$ and $J=50000$.

Table 1.2 How many CTAs make a diversified portfolio?

	Risk aversion		
	2.3	10	50
Panel A			
AUM			
10	2	4	14
50	4	10	36
100	8	12	>40
500	22	>40	>40
Panel B			
AUM			
10	2	4	18
50	2	8	24
100	2	10	22
500	4	12	>40

We display the optimal number of CTAs for investors with varying risk aversion and assets under management. Frictional costs of diversification are assumed to follow either (12) with 10000 USD per additional fund (Panel A) or (13) with a rebate θ of 50bps and maximum account size for rebates of 250 million (Panel B).

Panel A provides several interesting features. As in the mean variance case, we see that higher AUM allow a larger number of CTAs. Equally, as the fixed costs of investing into an additional CTA simply become lower (as a percentage of AUM), the number of funds increases with assets under management. Higher risk aversion also means that an investor is willing to give up more return (pay for additional CTAs) in order to reduce portfolio volatility. The optimal number of CTAs rises with risk. For large risk aversions this seems excessive. An investor with 10 million AUM and risk aversion of 50 would still be willing to invest in 14 funds, that is, to incur costs of 1.4 per cent of portfolio size without expectations of increased returns. Where does this come from? Note that higher risk aversion also means that the state price deflator becomes very large in a few extreme down markets. Why is diversification so important in those markets, that is, why do tail risk-sensitive investors need many CTAs? The answer to this can be found in Figure 1.4, which plots the cross-sectional volatility of CTA returns over time.

We see that the dispersion of returns is highest (that is, the correlation between CTAs is lowest) in extreme market states. This is worrying for CTA investors as they can't be sure that diversifying into only a few CTAs will provide protection for asset owners. In extreme months, some funds will have the optimal trades on, while some are still on the other side as their long-term models are caught out by trend reversals. Markets with high volatility,

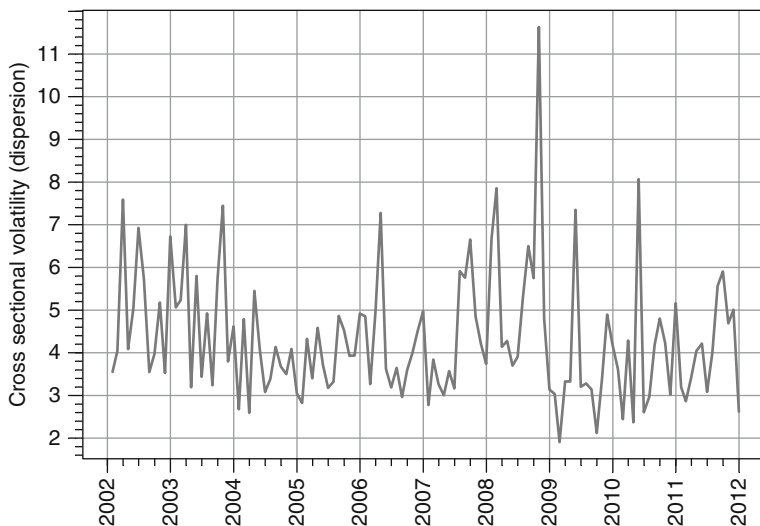


Figure 1.4 Cross-sectional volatility (dispersion) of CTA returns

Cross-sectional volatility (dispersion) of CTA returns from January 2002 to December 2011. All funds are selected from the Barclays Managed Futures databank and show a complete 10-year series of monthly returns.

that is, large unanticipated swings in asset prices, will amplify even small differences in positions and the CTA universe will display inhomogeneous performance in times of crisis.

Investors with high risk aversion should allocate into CTAs as a group (as CTAs provide tail insurance) and invest in many CTAs of this group to ensure that a CTA portfolio actually provides tail insurance in down markets.

Conclusion

This article shows that in order to find the optimal number of CTAs in a portfolio (or any group of assets) it does not suffice to simply calculate/plot portfolio volatility as a function of portfolio size and choose the optimal number where additional diversification becomes 'small'. Surely we must be able to do better than that. Instead, the frictional costs of diversification, the amount of assets under management, the degree of risk aversion and the state dependence on hedge fund payoffs matter to investors. This article presented two practical and easy to implement methods that allow us to calculate the optimal number of equally weighted CTAs for investors who can't distinguish between CTAs. We found that portfolios containing more than 40 CTAs do not necessarily display over-diversification. They can simply arise out of a combination of large assets under management, high risk aversion and low frictional diversification costs. We also find that valuation measures based on realistic preferences lead investors to portfolios containing more rather fewer funds. This contrasts with earlier claims that adding more funds increases tail risk. We find that that CTA dispersion increases in extreme down markets as small differences in positions become amplified. Highly risk-averse investors should therefore diversify more.

Notes

1. Assume a universe of hedge funds with average return μ . Sampling all funds without replacement generates in every sampling the average universe return such that the average trivially equals μ again. On the other hand, sampling one fund many times creates a return that is different for every sampling, but the average converges against μ , where convergence depends on the number of samplings.
2. Clearly our choice of data induces survivorship bias, but the case studies' aim is to add an example to our concept rather than deal with the many implementation details real-world investors have to face.
3. What does a risk aversion of 2 practically mean? We know from standard portfolio theory that a mean variance investor optimally allocates a fraction $w = \mu \lambda^{-1} \sigma^{-2}$ of his wealth into the risky asset, where μ and σ^2 denote expected return and risk, while λ describes risk aversion. For $\mu=0.08$, $\sigma=0.2$, a risk aversion $\lambda=2$ translates into an optimal allocation of 100 per cent into the risky asset. Low risk aversions model investors that are willing to take substantial risks, while high risk aversions lead to less risky portfolios.

4. Note that for higher risk aversions our valuation can no longer be interpreted as alpha. Instead we can interpret our deflators as a subjective valuation rather than a no arbitrage valuation measure.

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Appendix A

State price deflators for performance measurement

Let Λ_m denote the state price deflator for $m=1, \dots, M$ equally likely states of the world, defined by the marginal utilities for a representative investor

$$\Lambda_m = \frac{u'(W_m)}{\frac{1}{M} \sum_{m=1}^M u'(W_m)} (1+c)^{-1} \quad (\text{A.1})$$

where $u()$ denotes the preferences of a representative investor and W_m denotes the wealth of the representative investor in state m . The riskless rate is given by c . We use equally likely states of the world as this facilitates our empirical work using equally time spaced return from a data bank. The valuation of a *single* CTA with return R_m in state m in (A.1) is found by applying the state price deflator across all states of the world

$$V = \frac{1}{M} \sum_{m=1}^M \Lambda_m (1 + R_m) - 1 \quad (\text{A.2})$$

This valuation measure will put more weight on returns that occur in states where Λ_m is high, that is, in those states of the world where marginal utility is high or equivalently where wealth in the economy is low. Consequently, funds that pay off well when wealth is low will get a higher valuation – even when their average returns are identical – than those hedge funds that synchronize their losses with losses in the economy. We need to find $EV(n)$, that is, our expected valuation measure for a n fund portfolio. Suppose in the j th draw we sample n funds, that is, we sample a set $S_j(n)$ that contains n index numbers, where each index number identifies a given CTA. For a single draw j we can now value a CTA portfolio with n randomly drawn funds according to

$$V(n) = \frac{1}{M} \sum_{m=1}^M \Lambda_m \left(1 + \sum_{i \in S_j(n)} \frac{1}{n} R_{i,m} - \phi(n) \right) - 1 \quad (\text{A.3})$$

where $\phi(n)$ is defined as in (12) or (13). All draws are sampled independently as CTA returns do neither exhibit statistically significant serial correlation (in which case we would have employed sampling from blocks of random lengths) nor temporal dependency in their second moment. This process is repeated across $j=1, \dots, J$ samplings and the expected value of an n fund portfolio is given by

$$EV(n) = \frac{1}{J} \sum_{j=1}^J \left(\frac{1}{M} \sum_{m=1}^M \Lambda_m \times \left(1 + \sum_{i \in S_j(n)} \frac{1}{n} R_{i,m} - \phi(n) \right) - 1 \right) \quad (\text{A.4})$$

We use this valuation measure to calculate where $(dEV(n)/dn)=0$ to arrive at the optimal number of CTAs in a portfolio. Note that the only randomness in our simulations arises from $S_j(n)$.

Appendix B

Calibration of a state price deflator

To apply state price deflators to real-world data we need to make assumptions on the investor's utility function and on the asset our representative investor holds. Without apology, we assume power utility, that is, $u(1+R_{US,m})=(1/1-\gamma)(1+R_{US,m})^{1-\gamma}$ where marginal utility is given by

$$u' = (1+R_{US,m})^{-\gamma} \quad (\text{B.1})$$

Our representative investor is further assumed to hold 100 per cent of his wealth in US equities. Calibration involves choosing γ to satisfy the first-order condition of a utility-maximizing investor as given in (B.2)

$$\frac{1}{m} \sum_{m=1}^M (1 + R_{US,m})^{-\gamma} (R_{US,m} - c) = 0 \quad (\text{B.2})$$

which is a simple numerical exercise. We arrive at $\gamma=2.3$ for the above-described data set. Our implied risk aversion is low, as the past 10 years of US equity returns provide a SHARPE-ratio of just 0.22. Any investor willing to accept that position must display low aversion to risk, that is, be willing to accept a small return per unit of risk. Substituting the expression for marginal utility given by (B.1) into (A.1) and noting that $W_m = 1 + R_{US,m}$ we get (B.3)

$$\Lambda_m = \frac{(1 + R_{US,m})^{-2.3}}{\frac{1}{m} \sum_{m=1}^M (1 + R_{US,m})^{-2.3}} (1 + c)^{-1} \quad (\text{B.3})$$

This exercise can be applied to other one parameter utility functions. Researchers can also use bootstrapping methods to test for the statistical significance of estimates.

Appendix C

Frictional diversification costs and optimal number of assets with return information and mean variance preferences

Let μ_i and w_i denote the expected return and weight for asset i and σ_{ij} the covariance between asset i and j . We now need to maximize

$$\max_{w, \delta} \sum_{i=1}^n w_i \mu_i - \left(\sum_{i=1}^n \delta_i \right) \frac{f}{aum} \quad (\text{C.1})$$

under the following constraints

$$\sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} \leq \bar{\sigma} \quad (\text{C.2})$$

$$\sum_{i=1}^n w_i = 1 \quad (\text{C.3})$$

$$w_i \leq M \delta_i \quad (\text{C.4})$$

$$0 \leq w_i \leq 1 \quad (\text{C.5})$$

$$\delta_i \in \{0, 1\} \quad (\text{C.6})$$

Equation (C.1) models the expected portfolio return given return forecasts as well as the number of funds ($\sum_{i=1}^n \delta_i$) multiplied by the monitoring/selection costs, f/aum , per fund (as a percentage of AUM). Here δ_i is defined as an integer variable (taking on either a value of 0 or 1) as in (C.6). As an asset is either in or out we pin down the value of this integer value by (C.4), where M is a 'large' number (for example, 10). If an asset is included (even at tiny size), the inequality is only satisfied for $\delta_i=1$. As soon as it leaves the optimal solution δ_i must assume a value of zero. Equation (C.2) and (C.5) are the usual budget and individual position constraints. The system (C.1)–(C.6) can be solved with a quadratic solver allowing integer constraints as in NUOPT™ for SPLUS™.

2

Crude Oil Futures Markets: Another Look into Traders' Positions

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We investigate the traders' futures-only positions, options-only positions and futures-and-options-combined positions in crude oil to draw some implications with respect to the behavior of traders during the 2008 oil bubble. We find that: (iv) the Money Manager group primarily speculated in crude oil markets via options on crude oil futures; (v) the Swap Dealer group generally behaved as a typical institutional long-only investor in crude oil—they were long crude oil futures and hedged with the long protective put options on crude oil futures; (vi) the Producer/Merchant/Processor/User group, in addition to being a short hedger, possibly acted as a counterparty in crude oil options markets; (vii) Other Reportable traders possibly engaged in the protective call risk management strategy; and (viii) the Non-Reportable traders group actually showed the most sophisticated, almost neutral, hedged position in crude oil futures and options on crude oil futures.

Introduction

Many studies have analyzed the positions of traders in commodity futures markets (Chatrath *et al*, 1997; Roon *et al*, 2000; Wang, 2002; Weiner, 2002; Wang, 2003; Klitgaard and Weir, 2004; Sanders *et al*, 2004; Moulton, 2005;

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Till, 2009; Yung and Liu, 2009; Irwin and Sanders, 2010; Sanders and Irwin, 2010; Sanders and Irwin, 2011; Stoll and Whaley, 2011; Tokic, 2011). These studies usually test for the relationships between the traders' positions and the price/volatility of the underlying commodity futures contracts.

The data on weekly positions of traders, published by the US Commodity Futures Trading Commission (CFTC) in the Commitment of Traders (COT), the Commodity Index Traders and the Disaggregated Commitment of Traders (DCOT) reports, is the standard database in similar studies. The DCOT Report, which is the final version of several revisions to the legacy COT Report, was created with the mission to increase the transparency in crude oil futures markets. Thus, most studies investigating the 2008 oil bubble use the DCOT Report as the source of data on traders' positions. Masters (2008) heavily influenced the creation of the DCOT Report when he urged the US Congress to investigate the manipulation of crude oil futures prices during the 2008 oil bubble by the Swap Dealer category.

The DCOT Report publishes weekly positions of traders (each Friday the report reports the positions for Tuesday of the same week) by their respective categories and separates the report into the futures-only positions and futures-and-options combined positions. Most authors analyze the futures-and-options combined data, although some authors analyze the futures-only data. For example, Sanders *et al* (2004) use the futures-only COT data citing the high correlation between the futures-and-options-combined data and futures-only data during their sample interval 1995–1999. To the best of our knowledge, no study has estimated the options-only positions and analyzed them.

Furthermore, we also noticed that existing studies do not even discuss (or analyze) the relationship between the futures-only and futures-and-options combined data, perhaps assuming that the correlation cited by Sanders *et al* (2004) still holds during the period covered by the DCOT data, June 1996–December 2010. Thus, in this study (to fill the gap in the literature) we: (i) estimate the options-only positions in crude oil and (ii) take another look into traders' futures-only positions, options-only positions and futures-and-options-combined positions to investigate any significant differences between these positions and draw some implications with respect to the behavior of traders during the 2008 oil bubble.

Our analysis produces several important findings that, to the best of our knowledge, were overlooked by the existing literature: (i) the options-only positions are highly and negatively correlated with the futures-only positions; (ii) the options-only positions are highly and positively correlated with the combined positions; and (iii) the futures-only positions are barely correlated or negatively correlated with the combined positions, which shows that Sanders *et al's* (2004) findings do not apply during the period covered by the DCOT Report data. We also find that: (iv) the Money Manager

group primarily speculates in crude oil markets via options on crude oil futures; (v) the Swap Dealer group generally behaved as a typical institutional long-only investor in crude oil—they were long crude oil futures and hedged with the long protective put options on crude oil futures; (vi) the Producer/Merchant/Processor/User (P/M/P/ U) group, in addition to being a short hedger, possibly acts as a counterparty in crude oil options markets; (vii) Other Reportable traders possibly engage in the protective call risk management strategy; and (viii) the Non-Reportable traders group actually shows the most sophisticated, almost neutral, hedged position crude oil futures and options on crude oil futures.

We briefly discuss the DCOT Report in the next section, and subsequently define some basic futures/options strategies that can clarify some of our implications. The subsequent section presents the analysis of each trader group's positions, and at the end we summarize with the key implications.

The Dcot Report: Brief Background

The Disaggregated Commitments of Traders Report, or the DCOT Report, is a weekly report published by the US CFTC. The CFTC is an independent agency, created by the US Congress in 1974, with the mandate to regulate commodity futures and option markets in the United States. The CFTC separates traders into the following categories based on their responses on the CFTC Form 40 (see ctfc.com for definitions):

- A 'P/M/P/U' is an entity that predominantly engages in the production, processing, packing or handling of a physical commodity, and uses the futures markets to manage or hedge risks associated with those activities.
- A 'Swap Dealer' is an entity that deals primarily in swaps for a commodity and uses the futures markets to manage or hedge the risk associated with those swaps transactions. The swap/dealer's counterparties may be speculative traders, like hedge funds, or traditional commercial clients that are managing risk arising from their dealings in the physical commodity.
- A 'Money Manager', for the purpose of the DCOT Report, is a registered commodity trading advisor, a registered commodity pool operator or an unregistered fund identified by CFTC. These traders are engaged in managing and conducting organized futures trading on behalf of clients.
- Every Other Reportable trader that is not placed into one of the other three categories is placed into the 'Other Reportables' category.
- Non-Reportable traders' positions are derived by subtracting the total long 'Reportable Positions' and the total short 'Reportable Positions' from the total open interest.

The CTFC publishes the weekly positions of these traders in futures-only and futures-and-options-combined formats. Open interest is the total of all

futures and/or options contracts entered into and not yet offset by a transaction, by delivery, by exercise and so on. The aggregate of all long open interest is equal to the aggregate of all short open interest.

A trader's position is defined as the open interest held by that trader. For the futures-and-options-combined report, option open interest and traders' option positions are computed on a futures-equivalent basis using delta factors supplied by the exchanges. The term 'delta' is defined as the rate of change of the option price with respect to the price of the underlying asset. Long call and short put open interest are converted to long futures-equivalent open interest. Likewise, short-call and long-put open interest are converted to short futures-equivalent open interest. A trader's long and short futures-equivalent positions are added to the trader's long and short futures positions to give 'combined-long' and 'combined-short' positions. We further discuss the different speculation/hedging strategies using futures and options in the next section.

Basic Futures and Options Strategies

Futures and options on futures can be used as the financial tools for speculation and hedging in commodities. In this section, we briefly discuss the key strategies that crude oil traders can use to speculate/hedge. First, we discuss the uncovered directional strategies used by traders that speculate on directional changes in crude oil prices. Second, we discuss the risk management strategies that can be used to hedge the speculative directional positions. Lastly, we discuss the basic hedging strategies by commercial market participants.

Directional uncovered strategies

- Long futures contract strategy
Traders speculating that the price of crude oil would rise within the specific time frame (intraday to longer term) can buy crude oil futures—go long crude oil. They would have to offset their open long position by going short, or by selling the contract for profit or loss before the contract expires. The futures account broker requires the trader to post the initial margin to buy crude oil futures, which is usually between 5 per cent and 10 per cent of the notional value of crude oil; thus, heavy leverage is involved. These contracts are market-to-market daily and the broker can require that a customer deposits more funds into the account if the account value drops below the maintenance margin level—the margin call.
- Short futures contract strategy
Traders speculating that the price of crude oil would fall within the specific time frame (intraday to longer term) can sell crude oil futures—go short crude oil. They would have to offset their open short position by going long, or by buying back the contract for profit or loss before the contract expires. The short position is equivalent to the long positions with

respect to the initial margin (heavily leveraged), the maintenance margin and the margin call procedure.

- **Long call option strategy**
Traders speculating that the price of crude oil would rise within the specific time frame can also buy a call option on crude oil futures—a long call position. A call option gives the right to buy crude oil at the specific strike price, within the specific time period. Call option buyers have to pay the premium or price for the option. If the option expires worthless, the call option buyer can only lose the premium paid; thus, the losses are limited and there are no margin calls. On the other hand, the profit potential with call options is theoretically unlimited.
- **Long put option strategy**
Traders speculating that the price of crude oil would fall within the specific time frame can also buy a put option on crude oil futures—long put. A put option gives the right to sell crude oil at the specific strike price, within the specific time period. The long put position is equivalent to the long call position with respect to the premium paid, the limited loss and very large gain potential (although not unlimited since the price of crude oil cannot drop below 0) and no margin calls.
- **Short call option strategy**
Traders speculating that the price of crude oil would not rise as much (above the strike price) within the specific time frame can sell a call option—a short call position. Their profit is limited to the premiums received, whereas their loss potential is theoretically unlimited.
- **Short put option strategy**
Traders speculating that the price of crude oil would not fall as much (below the strike price) within the specific time frame can sell the put option—a short put position. Their profit is limited to the premiums received, whereas their loss potential is very large.
- **Directional option spreads strategies**
Options traders have many different strategies to speculate on direction on crude oil prices by combining long and short positions across different strike prices, different expirations (bull spreads, bear spreads, strangles, straddles, calendar spreads and so on).

Hedged speculative positions (risk management)

- **Protective put (portfolio insurance) risk management strategy**
Long put options can also be used in combination with long futures positions, as insurance against unexpected drop in crude oil prices. Speculators that bet on rising oil prices by going long crude oil futures can hedge their long position by buying put options—the protective put strategy.
- **Protective call risk management strategy**
Long call options can also be used in combination with short futures positions, as insurance against unexpected rise in crude oil prices. Speculators

that bet on falling oil prices by going short crude oil futures can hedge their short position by buying call options—the protective call strategy.

- **Covered call strategy**
Speculators that hold long futures positions can increase their cash inflow by selling call options against their long futures position. Thus, their profits will be limited if crude oil prices rise above the strike price of the call option, and their losses will be lessened by the amount of premium collected from short call positions if crude oil prices fall.
- **Delta hedging strategy**
More sophisticated speculators can engage in a dynamic delta-hedging strategy, for example by selling call options and buying optimal quantities of crude oil futures as oil prices rise and selling crude oil futures as crude oil prices fall, to create a 0-delta position.

Basic commercial hedging strategies

- **Short hedge**
Commercial producers/merchants of crude oil can hedge their inventories (either in ground or in storage) by selling futures contracts against the inventory, thus fixing the profit margin. These participants are long the physical commodity such as crude oil, and short the commodity futures. The motive for short hedging can range from budgetary issues (for oil-producing countries) to profit issues (for commercial businesses) (see Swidler *et al*, 1999; Overdahl, 1987).
- **Long hedge**
Commercial consumers of crude oil can hedge their anticipated future need for crude oil inventories by buying futures contracts, thus fixing the profit margin. These participants are long futures contract and short cash. The motive for short hedging can also be due to budgetary issues (for oil-consuming countries) or profit issues (for commercial businesses).

Analysis of Traders' Positions

We first plot the weekly futures-and-options-combined positions for each trader and make some initial observations about their trading demand/behavior. Next, we plot the futures-only positions and options-only positions and expand on our initial observation about the trading behavior of each trader. We compute options-only positions by deducting the futures-only positions from the combined positions Equation (1).

$$Positions_{combined} - Positions_{futures} = Positions_{options} \quad (1)$$

Next, we confirm our observations about the positions of traders with the descriptive statistics and the cross-correlations between the

futures-and-options-combined positions, futures-only positions and options-only positions. Finally, we test for the lead—lag relationships between the futures-and-options-combined positions, futures-only positions and options-only positions using the Granger causality test. We are interested to see, for example, whether the lagged futures-only positions Granger cause the current options-only positions, and vice versa (Granger, 1969). In essence, we run bivariate regressions in Equation (2).

$$\begin{aligned} y_t &= \alpha_0 + \alpha_1 y_{t-1} + \cdots + \alpha_i y_{t-i} + \beta_1 x_{t-1} + \cdots + \beta_i x_{t-i} + \varepsilon_t \\ x_t &= \alpha_0 + \alpha_1 x_{t-1} + \cdots + \alpha_i x_{t-i} + \beta_1 y_{t-1} + \cdots + \beta_i y_{t-i} + u_t \end{aligned} \quad (2)$$

for all possible pairs of (x, y) series in the group. The reported F -statistics are the Wald statistics for the joint Equation (3).

$$\beta_1 = \beta_2 = \cdots = \beta_i = 0 \quad (3)$$

for each equation. The null hypothesis is that x does not Granger-cause y in the first regression and that y does not Granger-cause x in the second regression. The pairwise Granger causality test requires all variables to be stationary, or time independent and mean reverting. We run the Augmented Dickey-Fuller test Equation (4) to determine whether all of our variables are stationary.

$$\Delta y_t = a_0 + a_2 t + \gamma y_{t-1} + \sum_{i=2}^p \beta_i \Delta y_{t-1+i} + \varepsilon_t \quad (4)$$

The Augmented Dickey–Fuller test is repeated for variables non-stationary in levels, transformed into first differences (see Table 2.A1 in Appendix for the results of the augmented Dickey–Fuller test). We test for the optimal lag level using the AIC criteria, and the most optimal level appears to be at 1 week lag. We verified our results using different lags, but in most cases we find no pairwise causality at any lag level.

Analysis of the Money Manager group's positions

We plot the futures-and-options-combined positions of the Money Manager group in Figure 2.1 and observe the following:

1. the Money Manager group was net long crude oil during the entire period, which suggests that Money Managers acted as long-only investors in crude oil;
2. Money Managers had a relatively modest net long position in crude oil in 2006; in the second half of 2007, Money Managers sharply increased their net long positions and held those positions right into the top of the oil bubble in July of 2008;

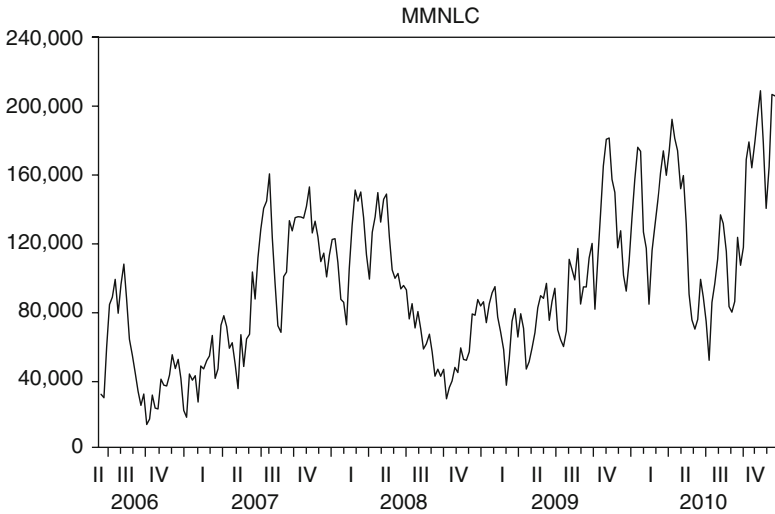


Figure 2.1 Futures-and-options-combined positions for the Money Manager group MMNLC-Money Manager net long combined.

3. after the bubble burst, in the second half of 2008, Money Managers sharply reduced their net long positions, suggesting that they almost perfectly played the 2008 oil bubble—got in early, got out in time;
4. Money Managers started to increase their net long positions as the price of crude oil bottomed during the period from late 2008 to early 2009;
5. these new net long positions eventually exceeded the net long positions in 2007, suggesting that Money Managers aggressively invested in crude oil in 2010.

We plot the futures-only and options-only positions of the Money Manager group in Figure 2.2 and observe the following:

1. the Money Manager group had a very large net long position in crude oil futures in 2006 and the second half of 2007;
2. the Money Manager group apparently hedged their large net long position in futures with a large net short position in crude oil options, which suggests that Money Managers were possibly engaged in a protective put risk management strategy—long futures and long put options;
3. Money Managers were reducing their net long positions from the second half of 2007 to the second half of 2008 after the oil bubble bust;
4. Money Managers were first offsetting their protective puts as they were reducing their net long futures positions, and then in the second half of 2007 they actually possibly engaged in a speculative long call strategy

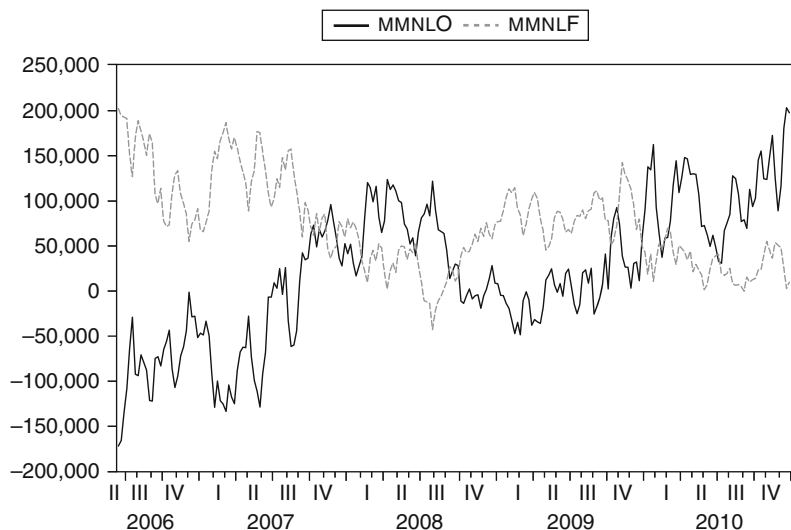


Figure 2.2 Futures-only and options-only positions for the Money Manager group
MMNLO—Money Manager net long options, MMNLF—Money Manager net long futures.

(all the way to the top of the 2008 oil bubble) in a clear trend of rising net long interest, which possibly indicates the speculative positive feedback trading exclusively in options markets;

5. after the 2008 oil bubble bust, Money Managers kept their futures net long positions relatively modest and stable, while they continued to significantly increase their net long positions in options.

On the basis of our analysis of Figures 2.1 and 2.2, we conclude that Money Managers chased oil prices primarily with call options, which makes sense knowing that by buying call options losses are limited, whereas gains are theoretically unlimited. Thus, a long call is a perfect strategy for chasing a bubble (which could get much larger, but also bust at any time). In addition, a very large combined net long position at the end of 2010 was also primarily due to a very large net long position in options, suggesting that oil prices were in the bubble again, or Money Managers anticipated a spike in crude oil prices due to the situation in the Middle East (the Arab Spring). We would also like to point to a change in behavior for the Money Manager group from a relatively conservative long-futures-long-puts portfolio insurance strategy to a full-blown crude oil speculator via long-futures and particularly long-options strategies (this observation was discussed in United Nations, 2011 report.).

The descriptive statistics in Table 2.1 confirm that Money Managers were on average net long crude oil futures and crude oil options. While their futures-and-options-combined position has never dropped to negative

Table 2.1 Descriptive statistics for the Money Manager group

	MMNLC	MMNLF	MMNLO
Mean	94 571.57	73 368.54	21 203.03
Median	87 579.50	68 283.50	25 029.00
Maximum	208 915.0	202 221.0	203 377.0
Minimum	12 710.00	-42 735.00	-171 863.0
Standard deviation	44 628.63	49 931.16	76 447.59
Skewness	0.438 621	0.503 556	-0.166 260
Kurtosis	2.508 664	2.733 424	2.481 650
Jarque-Bera	10.02 540	10.76 292	3.760 967
Probability	0.006 653	0.004 601	0.152 516
Sum	22 508 034	17 461 713	5 046 321
Sum of squared deviations	4.72E+11	5.91E+11	1.39E+12
Observations	238	238	238

MMNLO – Money Manager net long options, MMNLF – Money Manager net long futures, MMNLC – Money Manager net long combined.

Table 2.2 Correlation matrix for the Money Manager group

	MMNLC	MMNLF	MMNLO
MMNLC	1.000 000	-0.305 022	0.783 004
MMNLF	—	1.000 000	-0.831 209
MMNLO	—	—	1.000 000

MMNLO – Money Manager net long options,
MMNLF – Money Manager net long futures,
MMNLC – Money Manager net long combined.

territory, both futures-only positions and options-only positions were net short at some point.

The correlation matrix in Table 2.2 confirms the argument that Money Managers primarily speculate in crude oil via options since their futures-and-options-combined position is highly and positively (0.78) correlated with their options-only position. The options-only position is highly and negatively (-0.83) correlated with the futures-only position, partially due to protective put hedging and also due to the fact that Money Managers speculated primarily with options. The futures-and-options-combined position is negatively correlated (-0.30) with the futures-only position for crude oil.

Next, we question whether there is any pairwise Granger causality between the positions. Table 2.3 shows that the futures-only position Granger causes the options-only position with 1 week lag. In addition, we see bidirectional Granger causality between the combined position and the options-only position. This confirms our argument that Money Managers used options to strategically speculate on the crude oil while reducing their futures positions.

Table 2.3 Pairwise Granger causality tests for the Money Manager group

Lags: 1		
Null hypothesis	F-statistics	Probability
MMNLF does not Granger cause MMNLC	1.31 425	0.2528
MMNLC does not Granger cause MMNLF	0.60 120	0.4389
RMMNLO does not Granger cause MMNLC	3.64 321*	0.0575
MMNLC does not Granger cause RMMNLO	5.19 524**	0.0236
RMMNLO does not Granger cause MMNLF	0.23 615	0.6275
MMNLF does not Granger cause RMMNLO	4.63 156**	0.0324

*significant at 1% level; **significant at 5% level

MMNLO – Money Manager net long options,

MMNLF – Money Manager net long futures,

MMNLC – Money Manager net long combined.

Analysis of the Swap Dealer group's positions

We plot the futures-and-options-combined positions of the Swap Dealer group in Figure 2.3 and observe the following:

1. the Swap Dealer's combined net long position was generally in a range from close to 0 to around 80 000 contracts, except in 2010 when the Swap Dealer's combined position turned net short by close to 80 000 contracts;
2. the Swap Dealer was reducing its net long position from the beginning of 2007 to the second half of 2008; thus, there is no evidence of any speculative behavior during the oil bubble of 2008. On the contrary, the Swap Dealer group missed the opportunity to profit from the crude oil bubble.

We plot the futures-only and options-only positions of the Swap Dealer group in Figure 2.4 and observe the following:

1. the Swap Dealer was net long crude oil futures, except in 2006;
2. the Swap Dealer was net short crude oil options, except in 2006;
3. the plot of the net long futures-only positions and the net long options-only positions looks like a 'mirror image', suggesting that Swap Dealer hedged its net long position in futures with put options—a protective put portfolio insurance risk management strategy;
4. the Swap Dealer group was actually increasing its net long futures-only position leading to the top of the 2008 oil bubble, which suggests some speculative positive feedback trading (Delong *et al*, 1990).

On the basis of our analysis of Figures 2.3 and 2.4, we conclude that the Swap Dealer group was primarily a long-only investor in crude oil futures, perfectly hedged with a protective put as the portfolio insurance risk management

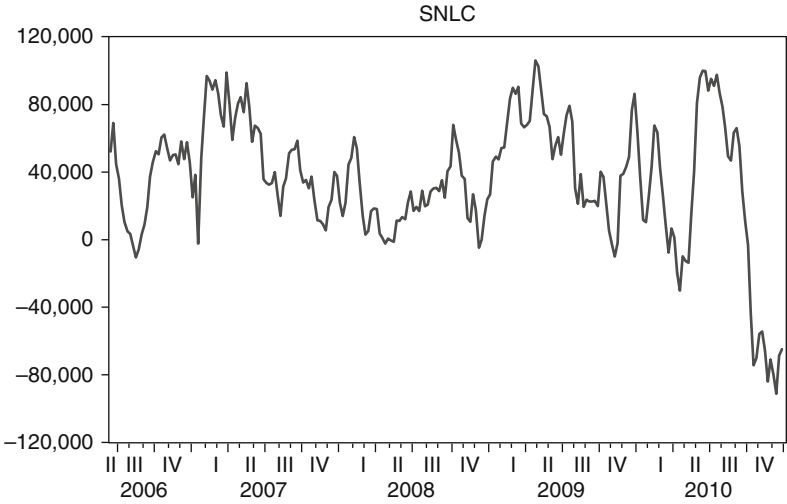


Figure 2.3 Futures-and-options-combined positions for the Swap Dealer group
SNLC-Swap Dealer net long combined.

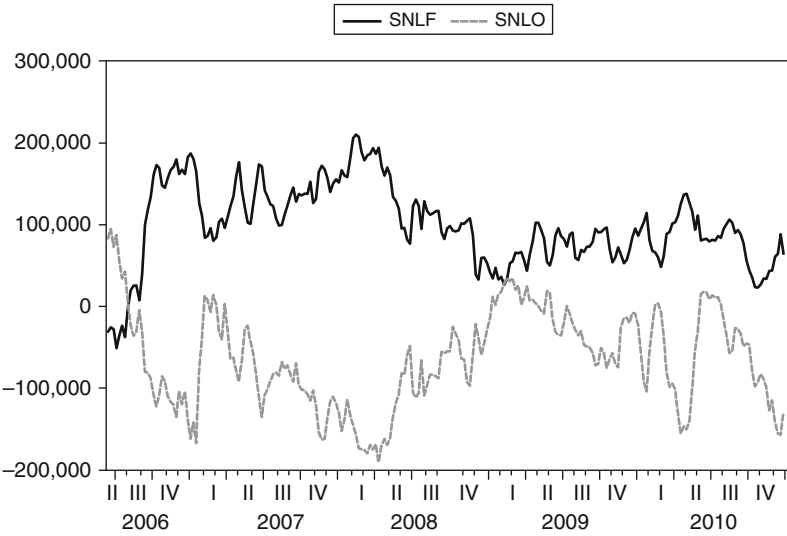


Figure 2.4 Futures-only and options-only positions for the Swap Dealer group
SNLO-Swap Dealer net long options, SNLF-Swap Dealer net long futures SNLC-Swap Dealer net
long combined.

strategy. But we also point to some positive feedback trading in crude oil futures leading to the top of the 2008 oil bubble, which is unobservable in the combined net long positions due to heavy hedging activity in options markets. Table 2.4 confirms that on average the Swap Dealer group was net long crude oil futures and net short crude oil options. In further support of this, Table 2.5 shows a high and negative correlation between the net long futures-only position and net long options-only position (-0.77). The correlation between the combined position and the options-only position is (0.54) and between the combined position and the futures-only position the correlation is (0.11). Further analysis of the Swap Dealer's positions shows no pairwise Granger-type causality between positions (Table 2.6), further supporting the notion that Swap Dealers were primarily hedging their futures positions with options, and not engaged in any strategic speculative futures-options strategy.

Analysis of the P/M/P/U group's positions

It is important to note that the P/M/P/U group should be engaged in (as a commercial participant) either a long hedge or short hedge. We plot the futures-and-options-combined positions of the P/M/P/U group in Figure 2.5 and observe the following:

1. the P/M/P/U group was net short crude oil during the entire period, which suggests that members of the P/M/P/U group were mostly Producers/

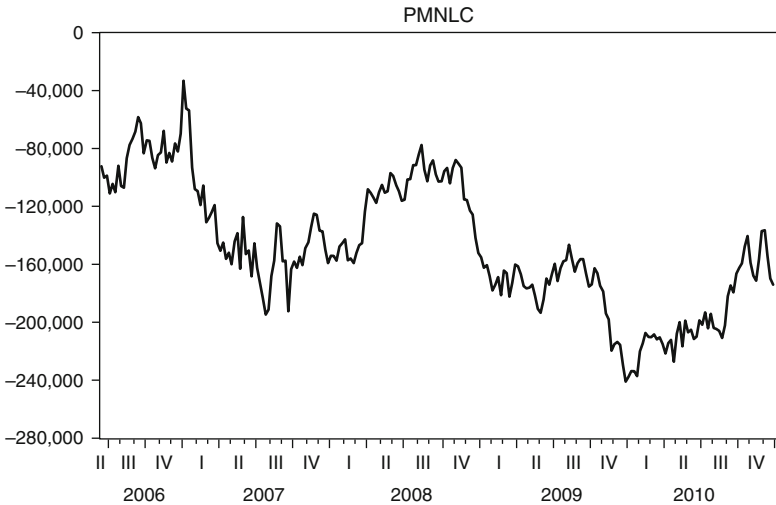


Figure 2.5 Futures-and-options-combined positions for the P/M/P/U group
PMNLC–P/M/P/U net long combined.

Table 2.4 Descriptive statistics for the Swap Dealer group

	SNLC	SNLF	SNLO
Mean	35 342.84	99 596.79	-64 253.95
Median	37 244.50	95 607.50	-64 099.50
Maximum	106 176.0	210 415.0	94 702.00
Minimum	-91 292.00	-51 150.00	-190 789.0
Standard deviation	37 739.14	50 189.48	59 451.01
Skewness	-0.769 272	-0.213 625	0.069 821
Kurtosis	4.138 043	3.102 346	2.439 937
Jarque-Bera	36.31 745	1.914 085	3.303 937
Probability	0.000 000	0.384 027	0.191 672
Sum	8 411 597	23 704 037	-15 292 440
Sum of squared deviations	3.38E+11	5.97E+11	8.38E+11
Observations	238	238	238

SNLO – Swap Dealer net long options, SNLF – Swap Dealer net long futures, SNLC – Swap Dealer net long combined.

Table 2.5 Correlation matrix for the Swap Dealer group

	SNLC	SNLF	SNLO
SNLC	1.000 000	0.107 915	0.543 691
SNLF	—	1.000 000	-0.775 712
SNLO	—	—	1.000 000

SNLO – Swap Dealer net long options, SNLF – Swap Dealer net long futures, SNLC – Swap Dealer net long combined.

Table 2.6 Pairwise Granger causality tests for the Swap Dealer group

Lags: 1			
Null hypothesis	Observations	F-statistics	Probability
SNLF does not Granger cause SNLC	237	0.53 840	0.4638
SNLC does not Granger cause SNLF		0.08 999	0.7645
SNLO does not Granger cause SNLC	237	0.53 840	0.4638
SNLC does not Granger cause SNLO		0.03 159	0.8591
SNLO does not Granger cause SNLF	237	0.08 999	0.7645
SNLF does not Granger cause SNLO		0.03 159	0.8591

SNLO – Swap Dealer net long options, SNLF – Swap Dealer net long futures, SNLC – Swap Dealer net long combined.

Merchants with the long position in the physical crude oil, hedged with the short position in crude oil futures/options;

2. the net short position is in a downtrend, possibly suggesting an increase in crude oil inventories in 2010;
3. the P/M/P/U group was actually reducing their net short positions from 2007 leading into the 2008 oil bubble peak, which suggests some unwilling positive feedback trading (see Tokic, 2011).

We plot the futures-only and options-only positions of the P/M/P/U group in Figure 2.6 and observe the following:

1. the P/M/P/U group was net short crude oil futures during the entire period;
2. we see a significant reduction in futures-only net short positions from the 2007 into the 2008 oil bubble top and after the 2008 oil bubble bust, suggesting that the P/M/P/U group engaged in positive feedback trading in crude oil futures and contributed to the rising oil prices in 2007 and 2008, and also provided some support to the crashing crude oil prices post the bubble bust;
3. the P/M/P/U group hedged their significant net short futures-only position in 2006 and 2007 with call options—a protective call risk management strategy—or possibly sold put options to hedging Swap Dealers (see Figure 2.2);

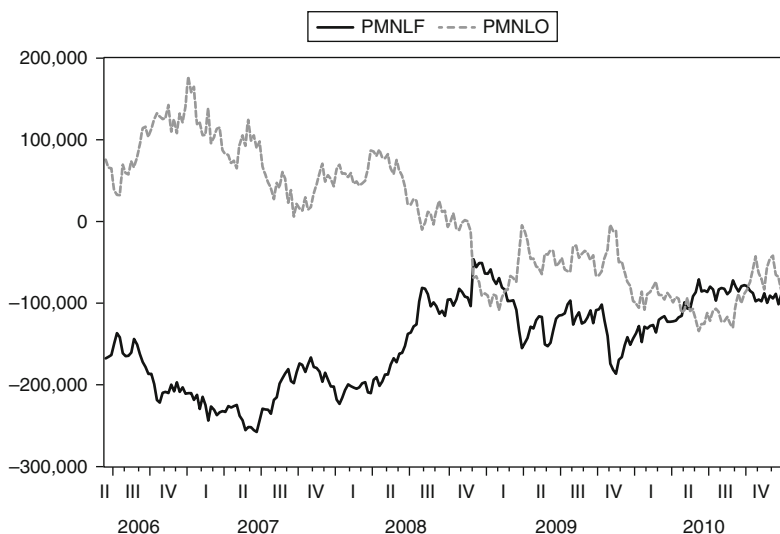


Figure 2.6 Futures-only and options-only positions for the P/M/P/U group
PMNLO—P/M/P/U net long options, PMNLF—P/M/P/U net long futures.

4. post the 2008 oil bubble bust, the P/M/P/U group kept their net short futures-only position relatively modest and stable;
5. however, the P/M/P/U group became net short crude oil options, possibly suggesting that the P/M/P/U group was selling call options and collecting premiums from the Money Manager group (see Figure 2.4);
6. thus, there is a noticeable change in behavior of the P/M/P/U group from the option hedged net short futures-only position to a counterparty in speculative crude oil options markets.

Table 2.7 shows that the P/M/P/U group was on average net short crude oil futures, and only negligibly on average net long crude oil options. However, the descriptive statistics can be misleading in this case. We showed that the P/M/P/U group was possibly a counterparty in a speculative options market by selling call options to the speculative Money Manager group and possibly selling put options to the Swap Dealer. In support, Table 2.8 shows a very high and positive correlation between the options-only positions and the combined positions (0.75) and a very high and negative correlation between the options-only and futures-only positions (−0.84). The futures-only position is actually negatively correlated with the combined position (−0.28). Thus, it is very important to acknowledge the importance of activity of the P/M/P/U group in crude oil options markets. Table 2.9 shows no Granger-type causality between the options-only and futures-only positions, suggesting that the P/M/P/U group did not engage in a strategic (speculative or hedging) futures-options strategy. In support of this, the options-only positions and the futures-only positions both Granger-cause the combined positions.

Table 2.7 Descriptive statistics for the P/M/P/U group

	PMNLC	PMNLF	PMNLO
Mean	−147916.7	−148928.5	1011.790
Median	−155765.5	−142412.0	891.5000
Maximum	−33011.00	−46420.00	177327.0
Minimum	−240744.0	−257895.0	−134121.0
Standard deviation	44599.27	53832.43	79004.27
Skewness	0.110408	−0.171176	0.101281
Kurtosis	2.243195	1.825294	1.794999
Jarque–Bera	6.163339	14.84663	14.80616
Probability	0.045883	0.000597	0.000609
Sum	−35204172	−35444978	240806.0
Sum of squared deviations	4.71E+11	6.87E+11	1.48E+12
Observations	238	238	238

RPMNLO – P/M/P/U net long options (first difference), RPMNLF – P/M/P/U net long futures (first difference), RPMNLC – P/M/P/U net long combined (first difference).

Table 2.8 Correlation matrix for the P/M/P/U group

	PMNLC	PMNLF	PMNLO
PMNLC	1.000 000	-0.282 114	0.756 746
PMNLF	—	1.000 000	-0.840 645
PMNLO	—	—	1.000 000

RPMNLO – P/M/P/U net long options (first difference),
RPMNLF – P/M/P/U net long futures (first difference),
RPMNLC – P/M/P/U net long combined (first difference).

Table 2.9 Pairwise Granger causality tests for the P/M/P/U group

Lags: 1

Null hypothesis	Observations	F-statistics	Probability
RPMNLF does not Granger Cause RPMNLC	236	3.18 388	0.0757
RPMNLC does not Granger Cause RPMNLF		0.17 486	0.6762
RPMNLO does not Granger Cause RPMNLC	236	3.18 388	0.0757
RPMNLC does not Granger Cause RPMNLO		0.00 301	0.9563
RPMNLO does not Granger Cause RPMNLF	236	0.17 486	0.6762
RPMNLF does not Granger Cause RPMNLO		0.00 301	0.9563

RPMNLO – P/M/P/U net long options (first difference), RPMNLF – P/M/P/U net long futures (first difference), RPMNLC – P/M/P/U net long combined (first difference).

Analysis of the Other Reportable group's positions

We plot the futures-and-options-combined positions of the Other Reportable group in Figure 2.7 and observe the following:

1. Other Reportable crude oil traders were switching between modest net long and net short combined positions during the period of rising oil prices from 2006 to the first half of 2008;
2. the Other Reportable group significantly increased their net long positions after the oil bubble burst and remained net long crude oil for the rest of the period, although with increased volatility.

We plot the futures-only and options-only positions of the Other Reportable group in Figure 2.8 and observe the following:

1. the Other Reportable group was actually net short crude oil futures in 2006, 2007 and the beginning of 2008, and turned net long crude oil futures right into the peak of the 2008 oil bubble top, which suggests short covering by Other Reportable traders (or unwilling positive feedback trading);

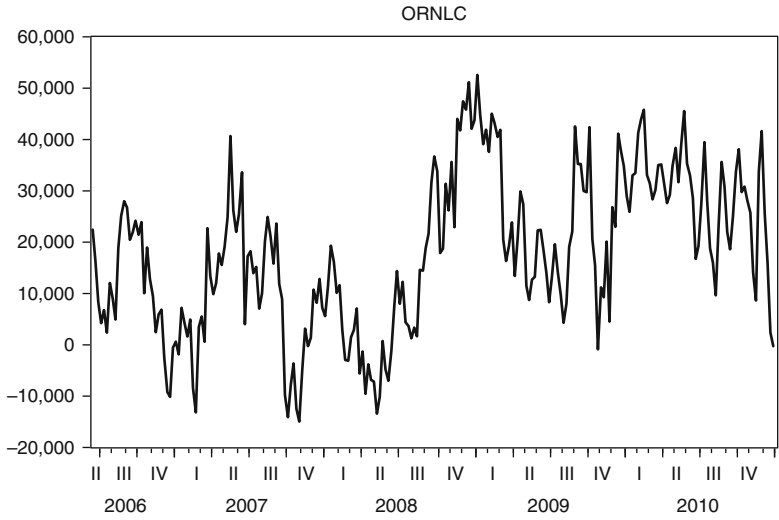


Figure 2.7 Futures-and-options-combined positions for the Other Reportable group
ORNLC–Other Reportable net long combined.

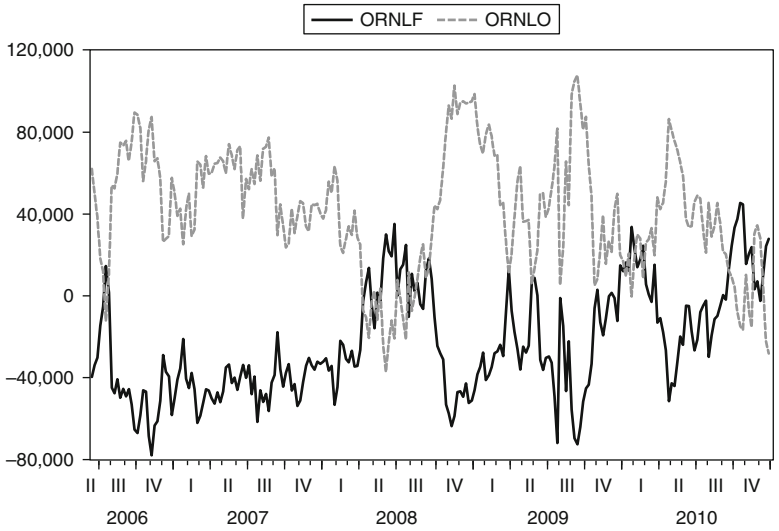


Figure 2.8 Futures-only and options-only positions for the Other Reportable group
ORNLO–Other Reportable net long options, ORNLF–Other Reportable net long futures.

2. post the 2008 oil bubble bust, the Other Reportable group remained mostly net short crude oil futures;
3. during the entire period, Other Reportable traders hedged their net short positions in crude oil futures with long positions in crude oil options—the protective call risk management strategy. The plot of the options-only and the futures-only positions resembles a mirror image.

The descriptive statistics in Table 2.10 confirm that the Other Reportable group was on average net long crude oil options and net short crude oil futures. Furthermore, the correlation matrix in Table 2.11 shows a very high and negative correlation between the options-only position and the futures-only position (-0.86), further supporting the ‘mirror image’ observation of the protective call risk management. The options-only position is also positively correlated with the combined position (0.47) whereas the combined position is not correlated with the futures-only position (0.03), which further shows the importance of options activity in crude oil markets for Other Reportable traders. We do not find Granger-type causality among pairs, which rules out

Table 2.10 Descriptive statistics for the Other Reportable group

	ORNLC	ORNLF	ORNLO
Mean	17 898.26	-23 698.83	41 597.09
Median	18 284.00	-30 280.00	42 494.00
Maximum	52 579.00	45 447.00	107 841.0
Minimum	-14 999.00	-77 986.00	-37 127.00
Standard deviation	15 276.13	26 743.07	30 281.97
Skewness	0.023 769	0.452 636	-0.172 448
Kurtosis	2.262 715	2.351 395	2.544 319
Jarque–Bera	5.413 006	12.29 871	3.238 765
Probability	0.066 770	0.002 135	0.198 021
Sum	425 9785	-5 640 322	9 900 107
Sum of squared deviations	5.53E+10	1.70E+11	2.17E+11
Observations	238	238	238

ORNLO – Other Reportable net long options, ORNLF – Other Reportable net long futures, ORNLC – Other Reportable net long combined.

Table 2.11 Correlation matrix for the Other Reportable group

	ORNLC	ORNLF	ORNLO
ORNLC	1.000 000	0.038 619	0.470 357
ORNLF	—	1.000 000	-0.863 653
ORNLO	—	—	1.000 000

ORNLO – Other Reportable net long options, ORNLF – Other Reportable net long futures, ORNLC – Other Reportable net long combined.

any speculative lead-lag futures-options strategy and supports the protective call hedging by the Other Reportable group (see Table 2.12).

Analysis of the Non-Reportable group's positions

We plot the futures-and-options-combined positions of the Non-Reportable group in Figure 2.9 and observe the following:

1. the Non-Reportable group was mostly net short crude oil before the 2008 oil bubble peak, and mostly net long crude oil post the 2008 oil bubble peak.

Table 2.12 Pairwise Granger causality tests for the Other Reportable group

Lags: 1

Null hypothesis	Observations	F-statistics	Probability
ORNLF does not Granger cause ORNLC	237	0.40826	0.5235
ORNLC does not Granger cause ORNLF		0.04414	0.8338
ORNLO does not Granger cause ORNLC	237	0.40826	0.5235
ORNLC does not Granger cause ORNLO		0.00016	0.9901
ORNLO does not Granger cause ORNLF	237	0.04414	0.8338
ORNLF does not Granger cause ORNLO		0.00016	0.9901

ORNLO—Other Reportable net long options, ORNLF—Other Reportable net long futures, ORNLC—Other Reportable net long combined.

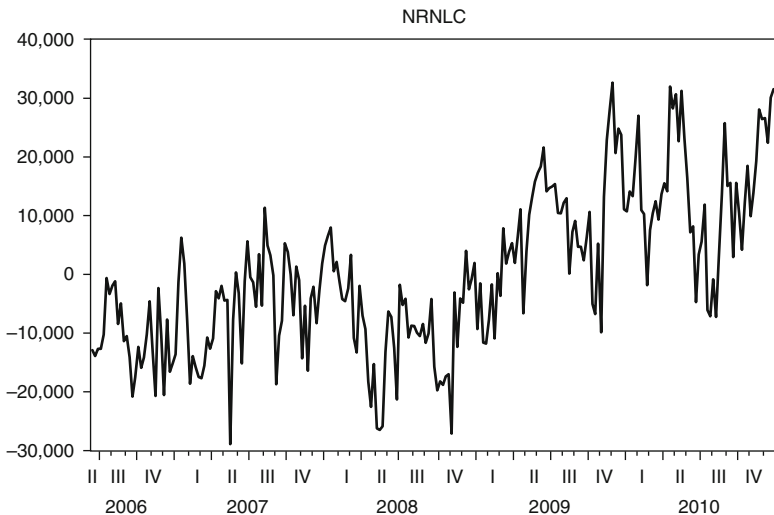


Figure 2.9 Futures-and-options-combined positions for the Non-Reportable group
NRNLC—Non-Reportable net long combined

We plot the futures-only and options-only positions of the Non-Reportable group in Figure 2.10 and observe the following:

1. the Non-Reportable group was actually mostly net long crude oil futures before the 2008 oil bubble peak and mostly net short crude oil futures post the 2008 oil bubble peak;
2. the Non-Reportable group hedged their net long futures position with the net short option positions—a protective put risk management strategy (portfolio insurance). We see a rising trend of options-only position and a falling trend of futures-only position, intersecting right at the top of the 2008 oil bubble.

The descriptive statistics in Table 2.13 show that Non-Reportable traders had on average a very small net long combined position, and on average they were net short futures and net long options, which indicates an almost perfectly hedged (close-to-zero delta) position in crude oil futures and options. The correlation matrix in Table 2.14 confirms a very high and negative correlation between the options-only and the futures-only positions (-0.81), and a very high and positive correlation between the options-only and combined positions (0.80), while the combined position is negatively correlated with the futures-only position (-0.30), which further shows the importance of options positions even for the Non-Reportable group. We

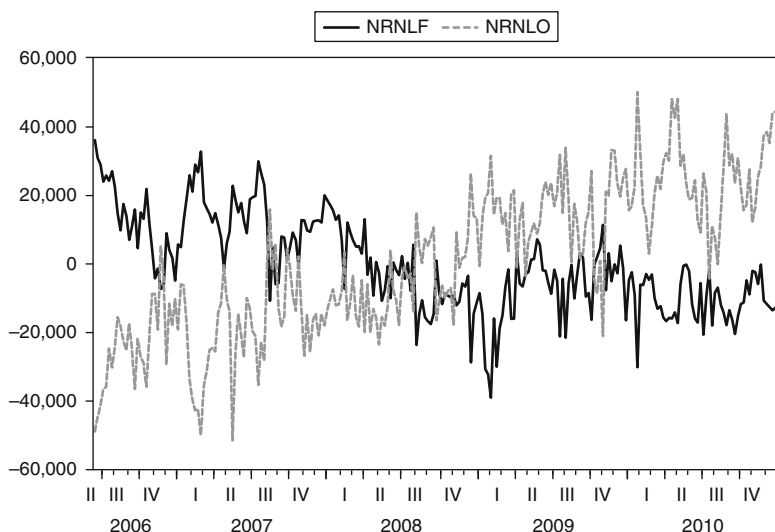


Figure 2.10 Futures-only and options-only positions for the Non-Reportable group
NRNLO—Non-Reportable net long options, NRNLF—Non-Reportable net long futures

Table 2.13 Descriptive statistics for the Non-Reportable group

	NRNLC	NRNLF	NRNLO
Mean	103.8193	-338.0252	441.8445
Median	-1605.500	-2031.000	-583.0000
Maximum	32 653.00	36 061.00	50 065.00
Minimum	-28 931.00	-39 109.00	-51 608.00
Standard deviation	13 395.45	13 652.17	21 820.03
Skewness	0.370 708	0.216 811	-0.017 121
Kurtosis	2.605 791	2.796 245	2.312 402
Jarque-Bera	6.992 231	2.276 320	4.700 139
Probability	0.030 315	0.320 408	0.095 363
Sum	24 709.00	-80 450.00	105 159.0
Sum of squared deviations	4.25E+10	4.42E+10	1.13E+11
Observations	238	238	238

NRNLO – Non-Reportable net long options, NRNLF – Non-Reportable net long futures,
NRNLC – Non-Reportable net long combined.

Table 2.14 Correlation matrix for the Other Reportable group

	NRNLC	NRNLF	NRNLO
NRNLC	1.000 000	-0.301 552	0.802 579
NRNLF	—	1.000 000	-0.810 796
NRNLO	—	—	1.000 000

NRNLO – Non-Reportable net long options, NRNLF – Non-Reportable net long futures,
NRNLC – Non-Reportable net long combined.

find no Granger-type causality among positions, which supports the notion that Non-Reportable traders use options only to hedge their futures positions (see Table 2.15).

Summary and key implications

Behavior of traders

Our analysis shows that:

1. the Money Manager group primarily speculates in crude oil markets via options on crude oil futures. They possibly engaged in positive feedback trading leading to the top of the 2008 oil bubble by increasing their net long options-only positions, an activity unobservable in futures-only positions and the combined positions;
2. the Swap Dealer group generally behaves as a typical institutional long-only investor in crude oil – with the long crude oil futures position hedged with the protective put options on crude oil futures;

Table 2.15 Pairwise Granger causality tests for the Non-Reportable group

Lags: 1			
Null hypothesis	Observations	F-statistics	Probability
NRNLF does not Granger cause NRNLC	237	2.07 674	0.1509
NRNLC does not Granger cause NRNLF		1.61 747	0.2047
NRNLO does not Granger cause NRNLC	237	2.07 674	0.1509
NRNLC does not Granger cause NRNLO		0.09 274	0.7610
NRNLO does not Granger cause NRNLF	237	1.61 747	0.2047
NRNLF does not Granger cause NRNLO		0.09 274	0.7610

NRNLO – Non-Reportable net long options, NRNLF – Non-Reportable net long futures,
NRNLC – Non-Reportable net long combined.

- the P/M/P/U group is a short hedger; however, we also raise a possibility that this group acts as a counterparty in crude oil options markets, by selling put options to hedgers (Swap Dealers, for example) and call options to speculators (Money Managers, for example);
- Other Reportable traders are generally net short crude oil futures, and they hedge their short futures position by being net long crude oil futures options – possible protective call risk management strategy;
- the Non-Reportable traders group, which is supposed to represent small noise traders, actually shows the most sophisticated hedged position – almost neutral, perfectly hedged position in crude oil futures and options.

Importance of the options-only position

The cross-correlation analysis between the combined positions, the futures-only position and the options-only positions shows that:

- the options-only positions are highly and negatively correlated with the futures-only positions for all groups of traders (the range for the coefficient of correlation is from (-0.77) to (-0.86));
- the options-only positions are highly and positively correlated with the combined positions for all groups of traders (the range for the coefficient of correlation is from (0.47) to (0.80));
- the futures-only positions show low and even negative correlations with the combined positions (the range for the coefficient of correlation is from (-0.30) to (0.10)).

Key implications

This article shows that it is necessary to fully analyze the combined-futures-and-options data, the futures-only data and the options-only data on

positions of traders in crude oil futures markets to get a better understanding on how crude oil traders really behave. The implications can be very important for regulators brainstorming the possible regulation of crude oil futures markets, as some patterns that emerge in options-only positions are generally unobservable in combined or futures-only positions.

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Appendix

Table 2.A1 Augmented Dickey–Fuller unit root test

Variable	Level	First difference
ORNLC	−4.21***	—
ORNLF	−3.74***	—
ORNLO	−3.91***	—
NRNLC	−3.18**	—
NRNLF	−3.81***	—
NRNLO	−2.63*	—
PMNLC	−2.13	−17.02***
PMNLF	−1.35	−15.40***
PMNLO	−1.42	−17.55***
MMNLC	−2.69*	—
MMNLF	−3.16**	—
MMNLO	−2.56	−14.16***
SNLC	−3.28**	—
SNLF	−3.52***	—
SNLO	−3.92***	—

Notes: *significant at 1% level; **significant at 5% level; ***significant at 10% level.

3

Fund of Hedge Funds Portfolio Selection: A Multiple-Objective Approach

Ryan J. Davies, Harry M. Kat and Sa Lu

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Harry M. Kat is Professor of Risk Management and Director of the Alternative Investment Research Centre at the Sir John Cass Business School at City University in London. Before returning to academia, he was Head of Equity Derivatives Europe at Bank of America in London, Head of Derivatives Structuring and Marketing at Banc One in Tokyo and Head of Derivatives Research at MeesPierson in Amsterdam. He is a member of the editorial board of *The Journal of Derivatives*, *The Journal of Alternative Investments* and *The Journal of Wealth Management*.

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Practical applications Hedge funds exhibit complex, non-normal return distributions. In this context, it is difficult for investors to determine how much capital to allocate across different hedge fund strategies. Standard mean-variance portfolio theory and performance measures based on it (for example, the Sharpe ratio) may be inappropriate. The paper proposes an alternative portfolio allocation technique based on polynomial goal

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programming (PGP) that is simple to implement and computationally robust.

This chapter develops a technique for fund of hedge funds to allocate capital across different hedge fund strategies and traditional asset classes. Our adaptation of the polynomial goal programming optimisation method incorporates investor preferences for higher return moments, such as skewness and kurtosis, and provides computational advantages over rival methods. We show how optimal allocations depend on the interaction between strategies, as measured by covariance, co-skewness and co-kurtosis. We also demonstrate the importance of constructing 'like for like' representative portfolios that reflect the investment opportunities available to different-sized funds. Our empirical results reveal the importance of equity market neutral funds as volatility and kurtosis reducers and of global macro funds as portfolio skewness enhancers.

Introduction

As hedge funds continue to become more and more popular with investors, the amount of assets under their management has steadily grown, from around \$40 billion in 1990 to an estimated \$1839 billion by September 2007. Most investors do not invest in individual hedge funds directly, but invest in so-called funds of hedge funds (FoHF) instead. In return for a typically not-insignificant fee, FoHF (claim to) take care of the many unavoidable, time-consuming and complex issues that come with investing in a highly opaque asset class such as hedge funds. Although FoHF have been around for quite some time, it is still unclear how FoHF should optimally allocate capital across various hedge fund strategies.¹ In this paper, we show how a simple allocation technique based on polynomial goal programming (PGP) is particularly well-suited to dealing with the complex return distributions of hedge funds and their practical institutional constraints.

Amin and Kat,² Anson³ and others show that hedge fund returns are substantially more complex than common stock and bond returns. Not only do hedge fund return distributions tend to exhibit significant skewness and kurtosis, they also tend to display significant co-skewness with the returns on other hedge funds as well as equity. As a result, standard mean-variance portfolio theory (as well as performance measures based on it, such as the Sharpe ratio) is inadequate when dealing with portfolios of (or including) hedge funds – a more extensive model is required.⁴

Here, we construct a PGP optimisation model that is able to balance multiple conflicting and competing hedge fund allocation objectives: maximising expected return while simultaneously minimising return variance, maximising skewness and minimising kurtosis. We show how changes in investor preferences lead to different asset allocations across hedge fund strategies and across asset classes (hedge funds, stocks and bonds). The PGP

model provides guidance on how much capital, if any, should be allocated to each hedge fund strategy. In this way, a fund of funds can incorporate the investment goals of its target investors to help determine whether it should hold a wide cross-section of strategies (as the majority of FoHF do), or instead should focus its expertise on a few strategies or a single strategy.

We proceed as follows. The next section formulates optimal hedge fund portfolio selection within a four-moment framework as a multiple-objective problem. The following section describes the data. The penultimate section provides illustrative empirical results. The final section concludes. The Appendix outlines the procedure used for unsmoothing the raw hedge fund return data.

Portfolio Selection in a Four-Moment Framework

The PGP approach was first used in finance by Tayi and Leonard⁵ to facilitate bank balance sheet management. It has subsequently been used by Lai,⁶ Chunhachinda *et al.*,⁷ Sun and Yan⁸ and Prakash *et al.*⁹ to solve portfolio selection problems involving a significant degree of skewness. Here, we adapt the basic PGP approach to the context of FoHF portfolio selection. In order to incorporate more information about the non-normality of returns, we augment the dimensionality of the PGP portfolio selection problem from mean-variance-skewness to mean-variance-skewness-kurtosis. Later, we describe how we construct representative portfolios for each hedge fund strategy.

Consider an environment with $n+1$ assets. Each of the assets 1, 2, ..., n is a portfolio of hedge funds selected in a manner described below to represent a typical portfolio of funds drawn from each of the n hedge fund strategy classifications. Each strategy portfolio has a random return $\tilde{R}_i$. We impose a no short-sale requirement: negative positions in the portfolios of hedge funds are not allowed. Asset $n+1$ is the risk-free asset, with rate of return r for both borrowing and lending.

Let x_i denote the percentage of wealth invested in the i th asset and let $\mathbf{X}=(x_1, x_2, \dots, x_n)^\top$. Corresponding to $\tilde{\mathbf{R}}=(\tilde{R}_1, \tilde{R}_2, \dots, \tilde{R}_n)^\top$ is a positive definite $n \times n$ variance-covariance matrix $\mathbf{V}$. The percentage invested in the risk-free asset is determined by $x_{n+1}=1-\mathbf{I}^\top \mathbf{X}$, where $\mathbf{I}$ is an $n \times 1$ identity vector. As the portfolio decision depends on the *relative* percentage invested in each asset, the portfolio choice $\mathbf{X}$ can be rescaled and restricted on the unit variance space (that is, $\{\mathbf{X}|\mathbf{X}^\top \mathbf{V} \mathbf{X}=1\}$). Then, the portfolio selection problem may be stated as the following multiple-objective programming problem:

$$\text{Maximise } Z_1=E[\mathbf{X}^\top \tilde{\mathbf{R}}]+x_{n+1}r, \quad (1)$$

$$\text{maximise } Z_3=E[\mathbf{X}^\top (\tilde{\mathbf{R}}-E[\tilde{\mathbf{R}}])]^3, \quad (2)$$

$$\text{minimise } Z_4=E[\mathbf{X}^\top (\tilde{\mathbf{R}}-E[\tilde{\mathbf{R}}])]^4, \quad (3)$$

subject to $\mathbf{X}^\top \mathbf{V} \mathbf{X} = 1; \mathbf{X} \geq 0;$

$$x_{n+1} = 1 - \mathbf{I}^\top \mathbf{X}. \quad (4)$$

where portfolio expected return is Z_1 , skewness is Z_3 and kurtosis is Z_4 .

Given an investor's preferences among objectives, a PGP can be expressed instead as

$$\text{Minimise } Z = (1 + d_1)^\alpha + (1 + d_3)^\beta + (1 + d_4)^\gamma, \quad (5)$$

$$\text{subject to } E[\mathbf{X}^\top \tilde{\mathbf{R}}] + x_{n+1}r + d_1 = Z_1^* \quad (6)$$

$$E[\mathbf{X}^\top (\tilde{\mathbf{R}} - E[\tilde{\mathbf{R}}])]^3 + d_3 = Z_3^*, \quad (7)$$

$$-E[\mathbf{X}^\top (\tilde{\mathbf{R}} - E[\tilde{\mathbf{R}}])]^4 + d_4 = -Z_4^*, \quad (8)$$

$$d_1, d_3, d_4 \geq 0, \quad (9)$$

$$\mathbf{X}^\top \mathbf{V} \mathbf{X} = 1; \mathbf{X} \geq 0;$$

$$x_{n+1} = 1 - \mathbf{I}^\top \mathbf{X}; \quad (10)$$

where $Z_1^* = \text{Max}\{Z_1 | \mathbf{X}^\top \mathbf{V} \mathbf{X} = 1\}$ is the mean return for the optimal mean-variance portfolio with unit variance, $Z_3^* = \text{Max}\{Z_3 | \mathbf{X}^\top \mathbf{V} \mathbf{X} = 1\}$ is the skewness value of the optimal skewness-variance portfolio with unit variance and $Z_4^* = \text{Min}\{Z_4 | \mathbf{X}^\top \mathbf{V} \mathbf{X} = 1\}$ is the kurtosis value of the optimal kurtosis-variance portfolio with unit variance; and where α , β and γ are the non-negative investor-specific parameters representing the investor's subjective degree of preferences on the mean, skewness and kurtosis of the portfolio return. The specification of our objective function in (5) ensures that it is monotonically increasing in d_1 , d_3 and d_4 for all possible values.

In summary, solving the multiple-objective PGP problem involves a two-step procedure. First, the optimal values for Z_1^* , Z_3^* and Z_4^* , expected return, skewness and kurtosis, respectively, are each obtained within a unit variance two-space framework. Subsequently, these values are substituted into the conditions (6)–(8), and the minimum value of (5) is found for a given set of investor preferences $\{\alpha, \beta, \gamma\}$ within the four-moment framework.

All the optimal portfolios obtained above are composed of risky assets (hedge fund strategy portfolios) and the risk-free asset, in order to ensure the uniqueness of each optimal portfolio. To capture an investor that is fully invested in hedge funds, we rescale the portfolio $\mathbf{X}$ such that the total investment is 1 (that is, so that $x_{n+1} = 0$). Let $\gamma_i = x_i / (x_1 + x_2 + \dots + x_n)$ be the

percentage invested in the i th asset in the optimal portfolio Y . In the context of FoHF portfolios, γ_i is the capital weight allocated to each hedge fund strategy in the optimal hedge fund portfolio.

When we investigate the asset allocation strategies for portfolios of stocks, bonds and hedge funds, there are $n+3$ assets in the world: n representative portfolios, one for each hedge fund strategy; the S&P500 index, representing stocks; the Salomon Brothers 7-Year Government Bond Index (SALGVT7), representing bonds; and the risk-free asset. A no short-sale restriction is imposed for hedge fund portfolios only. Negative positions in stocks and/or bonds are allowed for.

Our PGP framework can be thought about in economic terms. Investors' utility will be augmented by a positive first moment (expected return), positive third moment (skewness) and negative fourth moment (kurtosis). The investor preference parameters α , β , and γ are directly associated with the marginal rate of substitution, which measures the desirability of forgoing one objective in order to gain another (conflicting) objective. For example, the marginal rate of substitution between expected return and skewness is given by $((\partial Z/\partial d_1)/(\partial Z/\partial d_3)) = (\alpha(1+d_1)^{\alpha-1})/(\beta(1+d_3)^{\beta-1})$ and the marginal rate of substitution between expected return and kurtosis is given by $((\partial Z/\partial d_1)/(\partial Z/\partial d_4)) = (\alpha(1+d_1)^{\alpha-1})/(\gamma(1+d_4)^{\gamma-1})$.

Thus, our approach allows users a simple, transparent method to specify their heterogeneous preferences for higher moments. This contrasts with standard portfolio optimisation based on a specific utility function.¹⁰ Recall that even the standard mean-variance utility function of Markowitz¹¹ and Sharpe¹² may be viewed as an approximation to the more basic von Neumann–Morgenstern utility function and more particularly to the isoelastic family of utility functions.¹³

Unfortunately, these functions do not provide an exact preference ordering for risky portfolios using the first three (or higher) moments of portfolio returns. To isolate the impact of each moment, these non-polynomials are typically expanded using a Taylor series approximation. From an academic perspective, this is easily accommodated.¹⁴ In practice, however, it is difficult for investors of hedge funds to describe their 'utility function'. Cremers *et al*¹⁵ use a full-scale optimisation approach to show that different specifications of investor preferences (power utility, bilinear utility and S-shaped value functions) imply considerable differences in the effect of higher return moments on optimal hedge fund allocations. Our simple approach largely mitigates these difficulties.

It is also important to highlight the advantages of using our approach over using a one-stage linear objective function, such as

$$\max E[R_p] - \lambda_1 \text{ Variance } (R_p) + \lambda_2 \text{ Skewness } (R_p) - \lambda_3 \text{ Kurtosis } (R_p) \quad (11)$$

This linear objective function requires portfolio weights to be optimised over four dimensions simultaneously – potentially a very difficult computational

problem. Worse yet, because skewness and kurtosis are the third and fourth return moments scaled by variance, each of the terms in the objective function interact with each other in complex ways, making trade-offs difficult to interpret. Thus, as the possible asset space increases in size, it becomes increasingly likely that a numerical solver will solve for a *local* maximum (or minimum) rather than the *global* maximum of (11).

In contrast, our PGP approach improves computational tractability by conducting the optimisation in two stages. In the first stage, optimisation is conducted in two-dimensional space, thereby ensuring that a numerical solver always locates the global maximum (minimum) value of each moment. Then, in the second stage, optimisation is conducted relative to these known targets (an easier computational problem). Investors can specify their preferences relative to these targets, allowing them more control and greater insights about the potential trade-offs. Throughout this process we set variance equal to 1 (which can later be rescaled) – in effect, this means our optimisation uses unscaled return moments, rather than skewness and kurtosis. This further improves computational tractability and also leads to better investment choices (see Brulhart and Klein¹⁶ for compelling evidence of this).

A natural question to ask is how our PGP approach performs out of sample relative to other approaches. Some evidence is provided by Anson *et al.*,¹⁷ who use our PGP optimisation approach to the CalPERS' hedge fund portfolio. They run the optimisation process quarterly, and impose a 3-month lag between the optimisation and the implementation (because of quarterly liquidity constraints). They find that out-of-sample performance of the optimised portfolios is better than that of mean-variance portfolios and that increasing investor preference parameters for the third and fourth return moment improves these moments out of sample. Although this evidence is certainly encouraging, we caution readers that higher return moments can be driven by rare outliers, and therefore a more detailed out-of-sample analysis would require a longer history of hedge fund data than that currently available.

Finally, before proceeding to our empirical results, we mention that another advantage of the PGP approach is that it can be adapted to embed other investor goals. For instance, based on an earlier version of this paper, at least one hedge fund has already adapted the PGP optimisation function to incorporate a value-at-risk (VaR) measure. Thus, the method could incorporate features of other hedge fund allocation techniques, such as the mean-modified VaR optimisation procedure proposed by Favre and Galeano.¹⁸

Other important approaches in the hedge fund allocation literature include Lamm's Cornish–Fisher expansion,¹⁹ Terhaar *et al.*'s factor model,²⁰ Alexander and Dimitriu's statistical factor model approach,²¹ Cvitanic *et al.*'s manager ability uncertainty framework,²² Amenc and Martellini's out-of-sample model²³ and Popova *et al.*'s benchmark over/under performance construction.²⁴

Strategy Classification and Data

Hedge fund investment strategies tend to be quite different from the strategies followed by traditional money managers. In principle every fund follows its own proprietary strategy, which means that hedge funds are a very heterogeneous group. It is, however, customary to ask hedge fund managers to classify themselves into one of a number of different strategy groups depending on the main type of strategy followed. We concentrate on seven main classes of funds. The numbers in square brackets indicate the estimated market share of each strategy group in terms of assets under management based on the June 2002 TASS asset flows report:

Long/Short Equity [43 per cent]: Funds that simultaneously invest on both the long and the short side of the equity market. Unlike equity market neutral funds, the portfolio may not always have zero market risk. Most funds have a long bias.

Equity Market Neutral [7 per cent]: Funds that simultaneously take long and short positions of the same size within the same market, that is, portfolios are designed to have zero market risk. Leverage is often applied to enhance returns.

Convertible Arbitrage [9 per cent]: Funds that buy undervalued convertible securities while hedging (most of) the intrinsic risks.

Distressed Securities [11 per cent]: Funds that trade the securities of companies in reorganisation and/or bankruptcy, ranging from senior secured debt to common stock.

Merger Arbitrage [8 per cent]: Funds that trade the stocks of companies involved in a merger or acquisition, buying the stocks of the company being acquired while shorting the stocks of its acquirer.

Global Macro [9 per cent]: Funds that aim to profit from major economic trends and events in the global economy, typically large currency and interest rate shifts. These funds make extensive use of leverage and derivatives.

Emerging Markets [3 per cent]: Funds that focus on emerging and less mature markets. These funds tend to be long only because in many emerging markets short selling is not permitted and futures and options are not available.

The database used in this study covers the period June 1994–May 2001 and was obtained from Tremont TASS, which is one of the best known and largest hedge fund databases available. Our database includes the Asian, Russian and LTCM crises as well as the end of the IT bubble and the first part of the bear market that followed. As of May 2001, the database contains monthly net-of-fee returns on a total of 2183 hedge funds and FoHF. Reflecting the tremendous growth of the industry as well as a notoriously

high attrition rate, only 264 of these hedge funds had 7 or more years of data available.

As shown in Amin and Kat,²⁵ concentrating on surviving funds only will not only overestimate the mean return on individual hedge funds by around 2 per cent but will also introduce significant biases in estimates of standard deviation, skewness and kurtosis. To avoid this problem, we decide not to work with the raw return series of the 264 survivor funds but instead to create 348 7-year monthly return series by, starting off with the 348 funds that were alive in June 1994, replacing every fund that closed down during the sample period by a fund randomly selected from the set of funds alive at the time of closure following strategy of the same type and of similar size and age. We do not include FoHF or dedicated short bias funds in our sample.

This replacement procedure implicitly assumes that in case of fund closure, investors are able to roll from one fund into the other at the reported end-of-month net asset value and at zero additional costs. This somewhat underestimates the true costs of fund closure to investors for two reasons. First, when a fund closes shop, its investors have to look for a replacement investment. This search takes time and is not without costs. Second, investors may get out of the old and into the new fund at values that are less favourable than the end-of-month net asset values contained in the database. Unfortunately, it is impossible to correct for this without additional information.

As hedge funds frequently invest in, to various degrees and combinations, illiquid exchange-traded and difficult-to-price over-the-counter securities, hedge fund administrators can have great difficulty in marking a portfolio to market at the end of the month to arrive at the fund's net asset value. Having difficulty obtaining an accurate value for illiquid assets, most will rely on 'old' prices or observed transaction prices for similar but more liquid assets. Such partial adjustment or 'smoothing' produces systematic valuation errors, which tend not to be diversified away, resulting in serial correlation in monthly returns and underestimation of their true standard deviations. In this paper we follow the approach of Brooks and Kat,²⁶ outlined in the Appendix, to unsmooth hedge fund returns and thereby reconcile stale price problems. Table 3.1 provides a statistical summary of reported and unsmoothed individual hedge fund returns. Looking at the 1-month autocorrelations, the smoothing problem is especially acute among convertible arbitrage and distressed securities funds. This is plausible as the securities held by these funds tend to be highly illiquid. As well, the unsmoothing produces standard deviations that are substantially higher than those calculated from reported returns, especially in convertible arbitrage and distressed securities where we observe a rise of around 30 per cent. In what follows, we concentrate on the unsmoothed returns.

Table 3.1 offers some other insights as well. Funds in different strategy groups tend to generate quite different returns, which confirms that the

Table 3.1 Statistical summary of reported and 'unsmoothed' hedge fund returns and stock/ bond returns

	N	Mean	SD	Skewness	Kurtosis	AC(1)	AC(2)	AC(3)	AC(4)	J-B (%)
<i>Based on reported returns</i>										
Convertible arbitrage	24	0.96	3.01	-1.14	5.93	0.30	0.15	0.09	0.02	91.7
Distressed securities	29	0.89	2.37	-0.78	6.36	0.25	0.08	-0.04	0.02	86.2
Equity market neutral	12	0.54	2.70	-0.41	2.82	0.20	0.03	0.05	0.05	50.0
Global macro	46	0.77	5.23	1.06	7.63	0.11	0.01	-0.00	-0.03	84.8
Long/short equity	172	1.34	5.83	0.00	3.35	0.09	-0.00	0.01	-0.03	69.2
Merger arbitrage	18	1.17	1.75	-0.50	4.96	0.10	-0.00	0.00	-0.03	77.8
Emerging markets	47	0.22	7.85	-0.86	5.79	0.10	-0.01	-0.00	-0.02	68.1
S&P500	—	1.36	4.39	-0.83	1.11	-0.11	-0.05	0.03	-0.06	—
Bond index	—	0.59	0.84	0.24	1.39	0.22	0.12	0.06	0.02	—
<i>Based on 'unsmoothed' returns</i>										
Convertible arbitrage	24	0.96	3.99	-0.91	5.46	0.00	-0.03	-0.01	-0.02	79.2
Distressed securities	29	0.91	3.06	-0.67	6.60	0.01	-0.03	-0.01	-0.02	86.2
Equity market neutral	12	0.55	3.06	-0.39	2.94	0.01	-0.03	-0.01	-0.02	50.0
Global macro	46	0.76	5.35	1.03	7.16	0.01	-0.02	-0.01	-0.03	82.6
Long/short equity	172	1.37	6.35	0.01	3.19	0.00	-0.03	-0.00	-0.03	68.6
Merger arbitrage	18	1.17	2.06	-0.46	4.65	0.00	-0.03	-0.01	-0.02	77.8
Emerging markets	47	0.23	9.63	-0.91	5.91	0.01	-0.03	-0.01	-0.02	66.0

Reported values are calculated from monthly net-of-all-fee returns and averaged across funds. 'Unsmoothed' returns are reported returns adjusted for possible state price effects using the technique described in the Appendix. Kurtosis measures excess kurtosis. First-order to fourth-order autocorrelation is given by AC(1)–AC(4). The final column reports the percentage of funds within each strategy that reject the null hypothesis of a normal return distribution under a Jarque-Bera (J-B) test with a 5 per cent significance level.

(self-)classification used has significant discriminatory power. From the table it is also clear that the risk profile of the average hedge fund cannot be accurately described by standard deviation alone. The table reports that the majority of funds in each strategy group reject the null hypothesis of a normal return distribution under a Jarque–Bera test at the 5 per cent significance level. All strategy groups exhibit non-zero skewness and excess kurtosis, with global macro being the only strategy producing positive skewness.

Most existing research on optimal hedge fund allocation, including Lamm,¹⁹ Morton *et al.*,²⁷ Amenc and Martellini²³ and Cvitanic *et al.*,²² uses well-known hedge fund indices (obtained from, for example, HFR or CSFB-TASS) to represent the different hedge fund strategy classes. Given the different number of funds in each index, however, different indices will achieve different levels of diversification. As shown in Amin and Kat²⁸ and Davies *et al.*,²⁹ hedge fund portfolio return properties vary substantially with the number of hedge funds included in the portfolio. For instance, an index constructed from only 10 funds will typically have significantly higher variance than a similar index constructed from 100 funds. An index composed of more funds is therefore likely to be allocated more capital. This higher allocation, however, results because this *index* has less specific risk than other indices based on a smaller number of funds, rather than because this *strategy* has lower risk.

To compare ‘like for like’, we construct representative portfolios containing the same number of funds for each hedge fund strategy. Specifically, we consider representative portfolios of 5 and 15 funds to capture the feasible investment possibilities of small- and large-sized FoHF. In practice, FoHF have to deal with minimum investment requirements, typically ranging from \$100 000 to \$500 000 per fund. For smaller FoHF, this forms a significant barrier to diversification. In contrast, large FoHF typically spread their investments over a relatively large number of managers to prevent the fund from becoming the dominant investor in any one particular fund.³⁰

We construct the representative portfolios for a given strategy as follows. First, we randomly sample 5000 portfolios of a given size (5 or 15 funds). We calculate each portfolio’s mean, standard deviation, skewness and kurtosis and take the average of each moment over the 5000 portfolios. The representative portfolio is then selected from the 5000 random portfolios in order to minimise the sum of the ranked differences across each of the four average moments. Table 3.2 provides the return characteristics of the representative portfolios thus obtained. In unreported results, we also considered representative portfolios of 10 and 20 funds. Our results show that as the number of funds in portfolio increases, standard deviations fall substantially. This indicates relatively low correlation between funds within the same strategy group, that is, a high level of fund-specific risk. Also, when the number of funds increases, portfolio return distributions become more skewed, indicating a high degree of co-skewness between funds within the same strategy group. Diversification is no longer a free lunch: Investors pay for a lower

Table 3.2 Statistical summary of returns for representative portfolios

	Mean	SD	Skewness	Kurtosis
<i>Small investors (each strategy portfolio has 5 funds)</i>				
Convertible arbitrage	0.96	2.84	-0.83	5.11
Distressed securities	0.89	2.23	-1.89	11.00
Equity market neutral	0.50	1.75	-0.40	1.53
Global macro	0.75	3.37	0.64	2.41
Long/short equity	1.38	4.03	-0.21	2.10
Merger arbitrage	1.16	1.54	-1.08	6.62
Emerging markets	0.25	7.54	-1.15	5.80
<i>Large investors (each strategy portfolio has 15 funds)</i>				
Convertible arbitrage	0.95	2.35	-1.00	4.65
Distressed securities	0.90	2.03	-2.55	14.31
Equity market neutral	0.52	1.31	-0.65	1.26
Global macro	0.75	2.79	0.76	1.64
Long/short equity	1.37	3.55	-0.23	1.83
Merger arbitrage	1.17	1.37	-1.71	8.98
Emerging markets	0.22	7.19	-1.27	5.93

Representative portfolios are obtained by first randomly sampling 5000 portfolios of a given size (5 or 15 funds). The mean, standard deviation, skewness and kurtosis of each randomly sampled portfolio's return series are calculated. Then, the average of each moment over the 5000 portfolios is taken. The representative portfolio is then selected from the 5000 random portfolios to minimise the sum of the ranked differences across each of the four average moments. Reported values are calculated from monthly net-of-all-fee returns and averaged across funds. Kurtosis measures excess kurtosis.

standard deviation by accepting a lower level of skewness. The only exception is global macro, where lower standard deviations go hand in hand with higher levels of skewness.

In practice, managers of FoHF do not select hedge funds by random sampling. This said, the fact that many spend a lot of time and effort to select the funds they invest in does not necessarily mean that in many cases a randomly sampled portfolio is not a good proxy for the portfolio that is ultimately selected. There is no evidence that some FoHF are able to consistently select future outperformers, nor is there any evidence of specific patterns or anomalies in hedge fund returns. When properly corrected for all possible biases, there is no significant persistence in hedge fund returns, nor is there any significant difference in performance between older and younger funds, large and small funds, and so on. In addition, older funds may be more or less closed to new investment, implying that expanding fund of funds are often forced to invest in funds with little or no track record. The fund prospectus and manager interviews may provide some information, but in most cases this information will be sketchy at best and may add more noise than actual value.

Finally, our analysis uses the sample average of the 90-day US T-bill rate, $r=0.423317$ per cent on a monthly basis, as the risk-free rate. Although for the most part we focus on the portfolio selected by a FoHF that neither borrows nor lends, the risk-free rate is necessary to determine the optimal portfolio as it reflects the leverage possibilities available to a FoHF investor.

Empirical Results

We now use PGP optimisation to obtain optimal portfolios for 10 different sets of investor preferences, for both FoHF portfolios and portfolios of stocks, bonds and hedge funds. These preference sets are chosen to illustrate the extent to which investors must trade off different moments, to determine which hedge fund strategies are crucial in determining overall portfolio performance and to see how allocations change if we impose capital constraints on each hedge fund strategy.

Trade-off between multiple objectives

The more importance investors attach to a certain moment, that is, the greater the preference parameter for this moment, the more favourable the value of this moment statistic will tend to be in the optimal portfolio. Investor preferences (α, β, γ) determine the relative importance of the difference between the values for expected return, skewness and kurtosis obtained in mean-variance-skewness-kurtosis space and their corresponding optimal values obtained in mean-variance (Z_1^*), skewness-variance (Z_3^*) and kurtosis-variance space (Z_4^*). Figure 3.1 shows that the difference, $d_1 = Z_4^* - E[\mathbf{X}^T \hat{\mathbf{R}}] + \chi_{n+1} r$, decreases monotonically as investors' preference for expected return (α) increases, holding $\beta=1$ and $\gamma=0.5$ fixed. It also provides the analogous results for skewness (holding $\alpha=1$ and $\gamma=0.5$ fixed) and kurtosis (holding $\alpha=1$ and $\beta=1$ fixed). Based on these results, we choose realistic values for the preference parameters: $\alpha, \beta \in \{0(\text{none}), 1(\text{low}), 2(\text{medium}), 3(\text{high})\}$ and $\gamma \in \{0(\text{none}), 0.25(\text{low}), 0.5(\text{medium}), 0.75(\text{high})\}$, to capture investors with no, low, medium and high preference, respectively, for the applicable return moment.

Table 3.3 provides the return characteristics of PGP optimal portfolios for small and large investors for different sets of investor preferences over expected return (α), skewness (β) and kurtosis (γ). Portfolio A with $(\alpha, \beta, \gamma) = (1, 0, 0)$ corresponds with the mean-variance efficient portfolio. Expected return is relatively high and standard deviation low, which in mean-variance terms makes for a highly attractive portfolio. Looking beyond mean and variance, however, we see that the skewness and kurtosis properties of this portfolio are extremely unattractive. This confirms the point raised by Amin and Kat² that mean-variance optimisers may be nothing more than skewness minimisers.

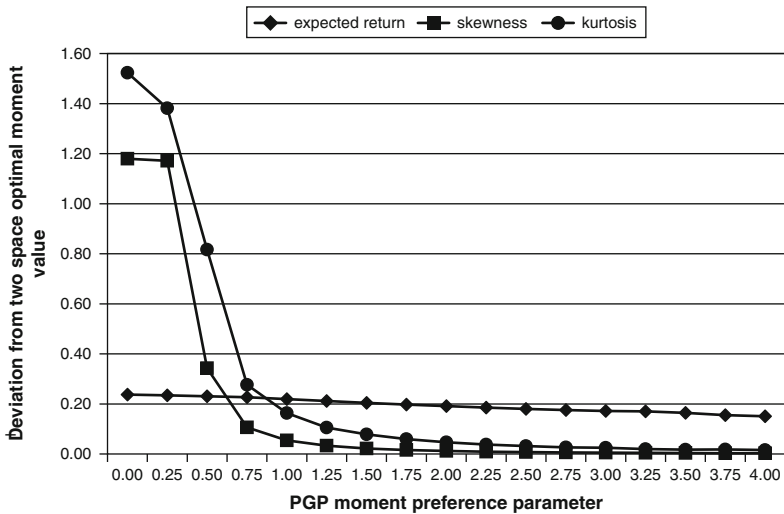


Figure 3.1 This figure illustrates how the deviation from the two-space optimal expected return varies as we vary the large investor's preference parameter over the given moment, holding the investor's preference over the other moments constant. For expected return, we hold $\beta=1$ and $\gamma=0.5$ constant and plot the deviation d_1 versus the preference parameter over expected return (α). The deviation is measured relative to 0.97, which is the highest obtainable expected return when variance is held constant at 1. For skewness, we hold $\alpha=1$ and $\gamma=0.5$ constant and plot the deviation d_3 versus the preference parameter over skewness (β). The deviation is measured relative to 0.85, which is the highest obtainable skewness value when variance is held constant at 1. For kurtosis, we hold $\alpha=1$ and $\beta=1$ constant and plot the deviation d_4 versus the preference parameter over kurtosis (γ). The deviation is measured relative to -0.10 , which is the lowest obtainable kurtosis value when variance is held constant at 1

Portfolios B–J show that variations in investor preferences will change the risk-return characteristics of the optimal portfolio to quite an extent. These results reinforce the trade-offs illustrated in Figure 3.1 and show that as one moment statistic improves, at least one of the other three moment statistics will tend to deteriorate. Compare, for example, portfolio E, $(\alpha, \beta, \gamma)=(1, 3, 0.25)$, with portfolio H, $(\alpha, \beta, \gamma)=(2, 3, 0.25)$. These two portfolios have the same level of preference over skewness and kurtosis. Despite this, the higher preference over expected return in portfolio H leads to a higher expected return at the cost of a higher standard deviation. The same phenomenon can be observed by comparing portfolio G, $(\alpha, \beta, \gamma)=(2, 1, 0.75)$, with portfolio H. Higher preference for skewness and lower preference for kurtosis cause the high kurtosis and high standard deviation of portfolio H to be traded in for a higher expected return and substantially higher skewness.

The above observation that hedge fund moment statistics tend to trade off against each other is quite an interesting one. Despite the fact that hedge

Table 3.3 Moment statistics and asset allocation across strategy classes for optimal FoHF portfolios

	Portfolio									
	A	B	C	D	E	F	G	H	I	J
α	1.00	1.00	1.00	3.00	1.00	1.00	2.00	2.00	3.00	3.00
β	0.00	1.00	1.00	1.00	3.00	1.00	1.00	3.00	2.00	1.00
γ	0.00	0.00	0.25	0.25	0.25	0.75	0.75	0.25	0.25	0.50
<i>Small investors (representative strategy portfolios have 5 funds)</i>										
Moments										
Expected return	1.07	0.85	1.38	0.83	0.79	0.70	0.66	0.84	0.79	0.77
SD	1.30	2.78	4.03	1.42	2.40	1.59	1.46	2.84	1.70	1.38
Skewness	-1.25	0.66	0.21	0.25	0.63	0.27	0.23	0.66	0.51	0.15
Kurtosis	6.07	1.96	2.10	1.01	1.76	-0.26	-0.27	1.72	1.35	0.05
Allocation										
Convertible arbitrage	0.05	0.04	0.00	0.10	0.13	0.11	0.13	0.03	0.09	0.11
Distressed securities	0.00	0.00	0.00	0.00	0.04	0.00	0.10	0.00	0.00	0.00
Equity market neutral	0.10	0.01	0.00	0.27	0.12	0.52	0.51	0.04	0.28	0.41
Global macro	0.05	0.78	0.00	0.32	0.66	0.28	0.25	0.78	0.43	0.25
Long/short equity	0.00	0.10	1.00	0.00	0.05	0.09	0.00	0.15	0.01	0.04
Merger arbitrage	0.80	0.07	0.00	0.30	0.00	0.00	0.00	0.00	0.19	0.19
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>Large investors (representative strategy portfolios have 15 funds)</i>										
Moments										
Expected return	0.97	0.75	0.76	0.85	0.73	0.66	0.77	0.75	0.78	0.79
SD	0.92	1.51	1.40	1.13	1.69	1.35	0.92	1.50	1.26	0.93
Skewness	-1.29	0.83	0.80	0.31	0.85	0.42	0.19	0.83	0.72	0.21
Kurtosis	6.19	1.42	1.29	1.13	1.42	0.19	0.29	1.40	1.20	0.45
Allocation										
Convertible arbitrage	0.02	0.01	0.04	0.09	0.00	0.16	0.09	0.02	0.03	0.08
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equity market neutral	0.30	0.32	0.34	0.27	0.32	0.45	0.47	0.32	0.34	0.44
Global macro	0.00	0.49	0.44	0.30	0.56	0.35	0.19	0.49	0.38	0.20
Long/short equity	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00
Merger arbitrage	0.68	0.18	0.18	0.34	0.09	0.00	0.25	0.17	0.25	0.29
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00

Reported values are calculated from monthly net-of-all-fee returns. Investors assign capital to representative portfolios of convertible arbitrage, distressed securities, equity market neutral, global macro, long/short equity, merger arbitrage and emerging market hedge funds. Optimal allocations are based on investors' preferences over expected return (α), skewness (β) and kurtosis (γ) using the PGP technique.

funds often follow highly active, complex strategies, hedge fund returns seem to exhibit the same type of trade-offs typically observed in the underlying securities markets, where prices are (thought to be) explicitly set to generate this type of phenomenon. Hedge funds therefore appear unable to dodge the rules of the game and seem to pick up a lot more from the markets that they trade than their well-cultivated market neutral image may suggest.

Figure 3.2 shows the feasible set of portfolios and the resulting mean-variance efficient frontier as well as the mean-variance coordinates of some

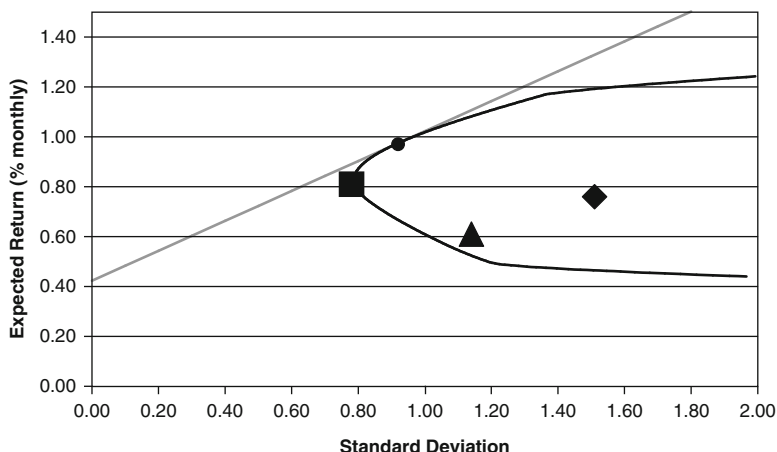


Figure 3.2 This figure illustrates the feasible set of portfolios and the efficient frontier in a mean-variance framework for large investors. The square point indicates the optimal portfolio for $(a, \beta, \gamma) = (1, 0, 1)$. The triangle point indicates the optimal portfolio for $(a, \beta, \gamma) = (0, 0, 1)$. The diamond point indicates the optimal portfolio for $(a, \beta, \gamma) = (1, 1, 0)$

specific optimal portfolios, including the mean-variance-skewness efficient portfolio (portfolio B with $(a, \beta, \gamma) = (1, 1, 0)$), and the mean-variance-kurtosis efficient portfolio (with $(a, \beta, \gamma) = (1, 0, 1)$). Doing so demonstrates a key point of our analysis: if investor preferences over skewness and kurtosis are incorporated into the portfolio decision, then in mean-variance space the optimal portfolio may well lie below the mean-variance efficient frontier. The reason is, of course, that expected return, skewness and kurtosis are conflicting objectives. Portfolios with relatively high skewness and low kurtosis will tend to come with a relatively low expected rate of return and vice versa.

Even though investor preferences over variance (or standard deviation) are not explicitly specified in our objective function, variance still plays a key role in the trade-off interaction. In the first stage of our PGP optimisation, the optimal values of Z_1^* , Z_3^* and Z_4^* are each obtained by seeking the best trade-off between variance and return, variance and skewness, and variance and kurtosis, respectively. In the second stage of the PGP optimisation, we obtain the optimal portfolio that has the best possible expected return, skewness and kurtosis with the relative trade-offs between them determined by how close their values are to Z_1^* , Z_3^* and Z_4^* with the difference ‘penalty’ determined by investor preferences. Standard deviation is therefore essentially the trade-off counterpart to every moment in the optimisation process.

Optimal allocation across hedge fund strategies

Table 3.3 also reports the optimal allocation weights across the different hedge fund strategies for different sets of investor preferences over expected

return, skewness and kurtosis. When the analysis is limited to mean-variance space (portfolio A), merger arbitrage is allocated a dominant 80 per cent. This, however, fully reflects merger arbitrage's comparatively low volatility and high return during our data period. The attractive mean-variance characteristics of merger arbitrage come at the cost of unfavourable skewness and kurtosis properties. When preference for skewness and kurtosis is introduced, the allocations change dramatically. The allocation to merger arbitrage drops to a much lower level, whereas global macro and equity market neutral take over as the dominant strategies, irrespective of investor size or preferences. Convertible arbitrage and long/ short equity tend to receive relatively small allocations here and there. No money is allocated to distressed securities and emerging markets.

The allocations in Table 3.3 are not at all in line with strategies' means and variances as reported in Table 3.2. Purely based on mean and variance, one would expect a much higher allocation to merger arbitrage, convertible arbitrage, long/ short equity and distressed securities. Part of the explanation lies in the skewness and kurtosis values reported in Table 3.2. Global macro combines positive skewness with low kurtosis. Merger arbitrage, convertible arbitrage, and especially distressed securities, however, exhibit exactly the opposite characteristics.

It is well known that in diversified portfolios, the marginal return characteristics of the assets involved only play a relatively minor role in determining the return characteristics of the portfolio. To explain the above allocations we therefore must look not only at the various strategies' marginal return properties (as given by Table 3.2) but also, and especially, at the way they are related to each other. Doing so may explain why for example in the PGP optimal portfolios, equity market neutral, which offers a low expected return and significant negative skewness, receives a higher allocation than long/ short equity, which offers a high expected return and less skewness.

As shown in Davies *et al.*,³¹ long/short equity tends to exhibit negative co-skewness and high co-kurtosis with other strategies. This means that in a portfolio context, the (negative) impact of long/short equity on portfolio skewness and kurtosis will be stronger than that evident from its marginal statistics. Equity market neutral on the other hand tends to exhibit low covariance and low co-kurtosis with other strategies. This makes this strategy attractive as a volatility and kurtosis reducer, which is reflected in the allocations, especially when preference for kurtosis is high as in portfolio F and portfolio G. Global macro tends to exhibit positive co-skewness with other strategies and thereby acts as portfolio skewness enhancer, which explains why this particular strategy picks up by far the highest allocations, particularly when there is a strong preference for skewness such as in portfolio E and portfolio H. Contrary to global macro, distressed securities display strong negative co-skewness with other strategies, which explains the complete lack of allocations to the latter strategy.

From an economic perspective, none of the above comes as a complete surprise. Although many hedge funds do not invest directly in equities, a significant drop in stock prices is often accompanied by a widening of credit spreads, a significant drop in market liquidity, higher volatility, and so on. As hedge fund returns are highly sensitive to these factors, most of them will perform poorly when there is a fall in stock prices, which will technically show up as negative co-skewness.³² The recent bear market provides a good example. Over the 3 years that stock prices dropped, overall hedge fund performance (as measured by the main indices) was virtually flat. The main exceptions to the above are equity market neutral and global macro funds. For equity market neutral funds, maintaining market neutrality is one of their prime goals, which makes them less sensitive to market moves than other funds. Global macro funds tend to take views on macro-economic events and are generally thought to perform best when markets drop and/or become more volatile, which is confirmed by their positive co-skewness properties. Convertible arbitrage funds, which are long convertibles, will suffer when stock markets come down. On the other hand, they will benefit from the simultaneous increase in volatility. Overall, this provides convertible arbitrage with relatively moderate risk characteristics, which in turn explains the allocations to this particular strategy.³³ At this stage, it is important to emphasise that the PGP technique is general enough to accommodate an investor that wishes to replace historic return distributions with her own beliefs. For instance, an investor could first use the method proposed by Black and Litterman³⁴ to combine market equilibrium-implied returns with her own subjective views and then use the resulting mixed estimate of expected returns to re-centre the distribution of hedge fund returns while maintaining its overall shape (in terms of variance, skewness and kurtosis).³⁵ Clearly, this approach would lead to different allocations than those reported here. For example, it would likely increase expected returns for emerging market funds and thereby cause this strategy to receive a higher allocation under some preference parameters.

Next, we consider the sensitivity of the allocations to constraints on standard deviation, skewness and kurtosis.³⁶ Part A of Tables 3.4 and 3.5 illustrates optimal allocations for portfolios that have the same preferences over objectives as those in Table 3.3, but under the constraint that the portfolio standard deviation is 10 per cent less than its unconstrained value. Part B illustrates the case in which the value of each portfolio's skewness is constrained to be 10 per cent higher than its counterpart in Table 3.3, and part C illustrates the case in which each portfolio's kurtosis is constrained to be 10 per cent lower than its counterpart in Table 3.3. In the tables, equity market neutral's role in reducing portfolio standard deviation and kurtosis and global macro's role as a skewness enhancer are evident. Allocations to global macro increase when portfolio skewness is constrained, whereas allocations to equity market neutral increase when standard deviation or kurtosis is constrained.

Table 3.4 Asset allocation for optimal hedge fund portfolios with constrained portfolio standard deviation, skewness or kurtosis for small investors

	Portfolio									
	A	B	C	D	E	F	G	H	I	J
α	1.00	1.00	1.00	3.00	1.00	1.00	2.00	2.00	3.00	3.00
β	0.00	1.00	1.00	1.00	3.00	1.00	1.00	3.00	2.00	1.00
γ	0.00	0.00	0.25	0.25	0.25	0.75	0.75	0.25	0.25	0.50
(A) SD constrained (10% improvement)										
Convertible arbitrage	0.06	0.05	0.07	0.09	0.00	0.12	0.11	0.05	0.13	0.06
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.07
Equity market neutral	0.21	0.07	0.33	0.35	0.68	0.53	0.47	0.13	0.30	0.63
Global macro	0.05	0.70	0.47	0.25	0.07	0.24	0.22	0.69	0.35	0.05
Long/short equity	0.00	0.07	0.13	0.00	0.01	0.00	0.00	0.13	0.01	0.00
Merger arbitrage	0.68	0.10	0.00	0.31	0.24	0.00	0.20	0.00	0.21	0.18
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(B) Skewness constrained (10% improvement)										
Convertible arbitrage	0.09	0.01	0.07	0.10	0.01	0.11	0.12	0.01	0.08	0.11
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equity market neutral	0.13	0.00	0.33	0.26	0.00	0.52	0.50	0.00	0.16	0.40
Global macro	0.13	0.89	0.48	0.34	0.89	0.29	0.28	0.90	0.53	0.26
Long/short equity	0.00	0.12	0.13	0.00	0.11	0.09	0.11	0.12	0.06	0.02
Merger arbitrage	0.65	0.00	0.00	0.30	0.00	0.00	0.00	0.00	0.17	0.21
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
(C) Kurtosis constrained (10% improvement)										
Convertible arbitrage	0.08	0.04	0.07	0.12	0.02	0.11	0.14	0.04	0.08	0.10
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00
Equity market neutral	0.13	0.03	0.33	0.29	0.12	0.53	0.54	0.12	0.24	0.42
Global macro	0.04	0.78	0.48	0.38	0.71	0.27	0.25	0.69	0.48	0.25
Long/short equity	0.00	0.14	0.13	0.16	0.15	0.09	0.03	0.15	0.09	0.02
Merger arbitrage	0.75	0.01	0.00	0.04	0.00	0.00	0.00	0.00	0.11	0.21
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Panel A illustrates optimal allocations for portfolios that have the same preferences over objectives as those in Table 3.3, but under the constraint that the portfolio standard deviation is 10 per cent less than its unconstrained value. Panel B illustrates the case in which the value of each portfolio's skewness is constrained to be 10 per cent higher than its counterpart in Table 3.3, and panel C illustrates the case in which each portfolio's kurtosis is constrained to be 10 per cent lower than its counterpart in Table 3.3. Each optimal portfolio is constructed under investors' preferences over expected return (α), skewness (β) and excess kurtosis (γ) using the PGP technique, selected from representative portfolios of convertible arbitrage, distressed securities, equity market neutral, global macro, long/short equity, merger arbitrage and emerging market funds. Each representative strategy portfolio has five funds. Bold numbers indicate that the strategy capital loading has increased relative to the optimal unconstrained portfolio.

We have shown that global macro and equity market neutral funds have important roles to play in FoHF portfolios. More generally, our analysis suggests that it may be optimal for FoHF to concentrate on just a few specific strategies rather than diversify across a large variety of them. In practice, strategy-focused FoHF, however, are a lot less common than well-diversified FoHF.

Table 3.5 Asset allocation for optimal hedge fund portfolios with constrained portfolio standard deviation, skewness or kurtosis for large investors

	Portfolio									
	A	B	C	D	E	F	G	H	I	J
α	1.00	1.00	1.00	3.00	1.00	1.00	2.00	2.00	3.00	3.00
β	0.00	1.00	1.00	1.00	3.00	1.00	1.00	3.00	2.00	1.00
γ	0.00	0.00	0.25	0.25	0.25	0.75	0.75	0.25	0.25	0.50
(A) SD constrained (10% improvement)										
Convertible arbitrage	0.03	0.04	0.04	0.00	0.02	0.14	0.07	0.02	0.03	0.03
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Equity market neutral	0.39	0.49	0.38	0.46	0.34	0.47	0.57	0.36	0.38	0.62
Global macro	0.01	0.13	0.38	0.23	0.50	0.30	0.00	0.42	0.32	0.00
Long/short equity	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
Merger arbitrage	0.57	0.34	0.20	0.32	0.15	0.06	0.36	0.19	0.26	0.32
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00
(B) Skewness constrained (10% improvement)										
Convertible arbitrage	0.03	0.01	0.00	0.05	0.00	0.18	0.09	0.00	0.02	0.07
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equity market neutral	0.33	0.32	0.30	0.38	0.30	0.26	0.47	0.31	0.33	0.43
Global macro	0.00	0.49	0.62	0.26	0.61	0.48	0.19	0.61	0.44	0.20
Long/short equity	0.00	0.00	0.03	0.00	0.02	0.05	0.00	0.02	0.00	0.00
Merger arbitrage	0.65	0.18	0.06	0.31	0.08	0.00	0.25	0.06	0.21	0.29
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00
(C) Kurtosis constrained (10% improvement)										
Convertible arbitrage	0.03	0.16	0.03	0.13	0.00	0.10	0.07	0.02	0.05	0.08
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equity market neutral	0.31	0.46	0.34	0.83	0.32	0.50	0.57	0.34	0.36	0.44
Global macro	0.00	0.34	0.47	0.03	0.58	0.17	0.00	0.48	0.36	0.19
Long/short equity	0.00	0.00	0.05	0.00	0.07	0.00	0.00	0.03	0.00	0.00
Merger arbitrage	0.66	0.00	0.12	0.01	0.03	0.22	0.36	0.14	0.24	0.29
Emerging markets	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Panel A illustrates optimal allocations for portfolios that have the same preferences over objectives as those in Table 3.3, but under the constraint that the portfolio standard deviation is 10 per cent less than its unconstrained value. Panel B illustrates the case in which the value of each portfolio's skewness is constrained to be 10 per cent higher than its counterpart in Table 3.3, and panel C illustrates the case in which each portfolio's kurtosis is constrained to be 10 per cent lower than its counterpart in Table 3.3. Each optimal portfolio is constructed under investors' preferences over expected return (α), skewness (β) and excess kurtosis (γ) using the PGP technique, selected from representative portfolios of convertible arbitrage, distressed securities, equity market neutral, global macro, long/short equity, merger arbitrage and emerging market funds. Each representative strategy portfolio has 15 funds. Bold numbers indicate that the strategy capital loading has increased relative to the optimal unconstrained portfolio.

Constraints on capital allocations

Our framework easily accommodates restrictions on the allocations to a hedge fund strategy. To illustrate, we consider the case where the allocation to each hedge fund strategy is constrained to be no more than 30 per cent of total capital ($x_i \leq 0.30 \forall i$). Table 3.6 displays the resulting optimal allocations.

Table 3.6 Moment statistics and asset allocation across strategy classes for optimal fund of hedge fund portfolios with global constraints on capital investment

	Portfolio									
	A	B	C	D	E	F	G	H	I	J
α	1.00	1.00	1.00	3.00	1.00	1.00	2.00	2.00	3.00	3.00
β	0.00	1.00	1.00	1.00	3.00	1.00	1.00	3.00	2.00	1.00
γ	0.00	0.00	0.25	0.25	0.25	0.75	0.75	0.25	0.25	0.50
<i>Small investors (representative strategy portfolios have 5 funds)</i>										
Moments										
Expected return	0.90	0.82	0.81	0.82	0.81	0.84	0.76	0.81	0.82	0.82
SD	1.28	1.39	1.40	1.38	1.40	1.76	1.61	1.39	1.39	1.42
Skewness	-1.26	0.22	0.22	0.21	0.22	0.08	0.04	0.22	0.22	0.19
Kurtosis	3.47	0.74	0.64	0.80	0.67	0.12	0.55	0.69	0.73	0.59
Allocation										
Convertible arbitrage	0.19	0.12	0.15	0.10	0.14	0.19	0.30	0.13	0.12	0.14
Distressed securities	0.15	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00
Equity market neutral	0.24	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Global macro	0.07	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Long/short equity	0.04	0.00	0.00	0.00	0.00	0.19	0.01	0.00	0.00	0.02
Merger arbitrage	0.30	0.28	0.25	0.30	0.26	0.02	0.02	0.27	0.28	0.24
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
<i>Large investors (representative strategy portfolios have 15 funds)</i>										
Moments										
Expected return	0.88	0.83	0.83	0.83	0.83	0.81	0.82	0.83	0.83	0.83
SD	1.00	1.12	1.12	1.12	1.12	1.15	1.14	1.12	1.12	1.12
Skewness	-1.32	0.39	0.39	0.39	0.39	0.32	0.34	0.39	0.39	0.39
Kurtosis	3.63	0.91	0.86	0.91	0.91	0.59	0.63	0.91	0.91	0.85
Allocation										
Convertible arbitrage	0.18	0.10	0.11	0.10	0.10	0.17	0.16	0.10	0.10	0.11
Distressed securities	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equity market neutral	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Global macro	0.06	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Long/short equity	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Merger arbitrage	0.30	0.30	0.29	0.30	0.30	0.23	0.24	0.30	0.30	0.29
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Reported values are calculated from monthly net-of-all-fee returns. Investors assign capital to representative portfolios of convertible arbitrage, distressed securities, equity market neutral, global macro, long/short equity, merger arbitrage and emerging market hedge funds. Optimal allocations are based on investors' preferences over expected return (α), skewness (β) and kurtosis (γ) using the PGP technique. The capital weight for each hedge fund strategy is constrained to be between 0 and 30 per cent.

When allocations are constrained, the degree of variation in return parameters achievable is quite a lot less than in the unconstrained case. This underlines that significant improvements in portfolio skewness and kurtosis can only be achieved by restricting the number of hedge fund strategies. In comparison with Table 3.3, we notice that the previously dominant capital weights on equity market neutral funds and global macro funds are now

forced down to 30 per cent, with merger arbitrage and convertible arbitrage picking up the difference. As in the case without constraints, the model continues to avoid distressed securities, long/short equity and emerging markets.

Portfolios of stocks, bonds and hedge funds

Until now, we have studied FoHF portfolios in isolation, that is, implicitly assuming that investors will not invest in anything else than the FoHF. In practice, however, investors will mix FoHF in with their existing portfolio. The means that preferably the optimal fund of fund portfolio should be derived from a wider framework, including stocks and bonds, to take account of the relation between hedge funds, stocks and bonds. This is what we do in this section.

Table 3.7 reports the moment statistics of optimal portfolios constructed with stocks, bonds and hedge fund strategies for small and large investors. Although the addition of stocks and bonds results in optimal portfolios with less kurtosis and higher skewness than the corresponding optimal portfolios of hedge fund only, we observe similar behaviour. Again, the mean-variance efficient portfolio has relatively unattractive skewness and kurtosis properties, which improve when explicit preference for higher moments is introduced. The improvement comes at a cost, however, as moment statistics tend to trade off against each other. An improvement in one statistic can only be obtained by accepting deterioration in one or more others.

Table 3.7 reveals that the bond index is the primary recipient of capital, receiving at least 40 per cent weight in most instances. In stark contrast, the stock index is sold short, irrespective of the investor preference parameters. This result is consistent with the observation of Amin and Kat² and Davies *et al*³¹ that hedge funds mix far better with bonds than with stocks. Whereas the co-skewness between stocks and most hedge fund strategies is negative, the co-skewness between bonds and hedge funds is generally higher and the co-kurtosis lower. A long bonds position and a short position in stocks combined with positive holdings in hedge funds will increase portfolio skewness and reduce kurtosis. As a result, optimal portfolios have to rely less on equity market neutral and global macro to perform these tasks, which allows them to diversify into strategies such as long/short equity and merger arbitrage. Even in this new context, however, no allocation is given to distressed securities and emerging markets strategies.

Table 3.8 shows the asset allocation across stocks, bonds and hedge funds under the constraint that capital weights are between 0 and 30 per cent for any hedge fund strategy and that the capital weights are within 730 per cent for both the stock index and the bond index. In general, these constraints seem to have quite an impact on the optimal portfolio moment statistics, reflecting a substantial change in the underlying allocations. With the bond allocation restricted to 30 per cent, the optimal portfolios turn increasingly

Table 3.7 Moment statistics and optimal asset allocation across stocks, bonds and hedge fund strategies

	Portfolio									
	A	B	C	D	E	F	G	H	I	J
α	1.00	1.00	1.00	3.00	1.00	1.00	2.00	2.00	3.00	3.00
β	0.00	1.00	1.00	1.00	3.00	1.00	1.00	3.00	2.00	1.00
γ	0.00	0.00	0.25	0.25	0.25	0.75	0.75	0.25	0.25	0.50
<i>Small investors (representative strategy portfolios have 5 funds)</i>										
Moments										
Expected return	0.83	0.87	0.72	0.76	0.72	0.60	0.74	0.56	0.75	0.75
SD	0.71	3.20	0.87	0.63	0.88	0.75	0.61	0.77	0.73	0.62
Skewness	-0.14	0.88	0.70	0.13	0.71	0.16	0.11	0.48	0.48	0.10
Kurtosis	0.94	3.48	1.17	-0.21	1.22	-0.81	-0.40	0.82	0.61	-0.33
Allocation										
Convertible arbitrage	0.06	0.19	0.06	0.07	0.06	0.10	0.08	0.14	0.06	0.08
Distressed securities	0.00	0.03	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.00
Equity market neutral	0.06	0.00	0.16	0.09	0.16	0.17	0.13	0.15	0.12	0.11
Global macro	0.02	0.00	0.19	0.06	0.19	0.04	0.05	0.00	0.13	0.05
Long/short equity	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Merger arbitrage	0.45	0.00	0.19	0.32	0.19	0.00	0.27	0.00	0.27	0.29
Emerging markets	0.00	0.10	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.00
S&P500	-0.04	-0.72	-0.02	-0.05	-0.02	-0.03	-0.04	-0.10	-0.04	-0.05
Bond index	0.46	0.40	0.42	0.51	0.42	0.62	0.52	0.80	0.46	0.52
<i>Large investors (representative strategy portfolios have 15 funds)</i>										
Moments										
Expected return	0.86	0.87	0.83	0.83	0.77	0.63	0.83	0.85	0.89	0.83
SD	0.65	1.59	1.37	0.72	1.38	0.83	0.67	1.55	1.23	0.67
Skewness	-0.48	1.12	1.06	0.25	0.99	0.12	0.02	1.12	0.88	0.01
Kurtosis	0.83	2.53	1.45	0.04	1.38	-0.71	-0.46	2.11	2.90	-0.44
Allocation										
Convertible arbitrage	0.05	0.00	0.05	0.12	0.04	0.12	0.13	0.00	0.04	0.12
Distressed securities	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equity market neutral	0.16	0.00	0.00	0.00	0.00	0.16	0.00	0.00	0.00	0.00
Global macro	0.00	0.31	0.35	0.12	0.42	0.07	0.00	0.34	0.13	0.00
Long/short equity	0.00	0.39	0.25	0.00	0.15	0.00	0.00	0.35	0.30	0.00
Merger arbitrage	0.54	0.27	0.22	0.48	0.22	0.00	0.48	0.24	0.39	0.47
Emerging markets	0.00	0.00	0.00	0.00	0.01	0.05	0.00	0.00	0.00	0.00
S&P500	-0.07	-0.30	-0.21	-0.13	-0.19	0.02	-0.11	-0.27	-0.25	-0.10
Bond index	0.32	0.33	0.35	0.41	0.34	0.58	0.51	0.33	0.40	0.51

Reported values are calculated from monthly net-of-all-fee returns. Investors assign capital to representative portfolios of convertible arbitrage, distressed securities, equity market neutral, global macro, long/short equity, merger arbitrage, emerging market hedge funds, the S&P500 index and the Salomon Brothers 7-Year Government Bond US Index. Optimal allocations are based on investors' preferences over expected return (α), skewness (β) and kurtosis (γ) using the PGP technique.

Table 3.8 Moment statistics and optimal asset allocation across stocks, bonds and hedge fund strategy classes with global constraints on capital investment

	Portfolio									
	A	B	C	D	E	F	G	H	I	J
α	1.00	1.00	1.00	3.00	1.00	1.00	2.00	2.00	3.00	3.00
β	0.00	1.00	1.00	1.00	3.00	1.00	1.00	3.00	2.00	1.00
γ	0.00	0.00	0.25	0.25	0.25	0.75	0.75	0.25	0.25	0.50
<i>Small investors (representative strategy portfolios have 5 funds)</i>										
Moments										
Expected return	0.81	0.75	0.78	0.77	0.73	0.71	0.77	0.76	0.77	0.76
SD	0.81	1.18	1.47	0.86	1.09	1.16	0.95	0.99	0.92	0.81
Skewness	-0.72	0.75	0.65	0.36	0.71	0.38	0.21	0.63	0.51	0.22
Kurtosis	1.11	1.84	0.61	0.45	1.21	-0.29	-0.31	1.18	0.98	0.05
Allocation										
Convertible arbitrage	0.12	0.06	0.08	0.09	0.07	0.09	0.09	0.07	0.08	0.09
Distressed securities	0.05	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.00
Equity market neutral	0.19	0.15	0.14	0.19	0.21	0.30	0.24	0.18	0.17	0.23
Global macro	0.04	0.30	0.30	0.16	0.25	0.15	0.13	0.23	0.20	0.13
Long/short equity	0.00	0.00	0.24	0.00	0.00	0.21	0.13	0.00	0.00	0.00
Merger arbitrage	0.30	0.21	0.09	0.30	0.17	0.08	0.25	0.25	0.30	0.30
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
S&P500	-0.01	-0.02	-0.15	-0.04	-0.01	-0.18	-0.13	-0.03	-0.05	-0.05
Bond index	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
<i>Large investors (representative strategy portfolios have 15 funds)</i>										
Moments										
Expected return	0.79	0.87	0.82	0.76	0.87	0.70	0.77	0.87	0.84	0.75
SD	0.63	1.41	1.21	0.67	1.41	0.88	0.71	1.41	1.18	0.60
Skewness	-0.32	1.05	0.97	0.44	1.05	0.47	0.16	1.05	0.95	0.04
Kurtosis	0.11	1.95	1.18	0.51	1.94	-0.38	-0.46	1.94	1.25	-0.35
Allocation										
Convertible arbitrage	0.10	0.04	0.06	0.08	0.05	0.07	0.08	0.05	0.06	0.09
Distressed securities	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
Equity market neutral	0.26	0.01	0.06	0.23	0.01	0.30	0.30	0.01	0.04	0.30
Global macro	0.01	0.30	0.30	0.14	0.30	0.15	0.09	0.30	0.29	0.04
Long/short equity	0.02	0.30	0.21	0.00	0.30	0.14	0.09	0.30	0.19	0.00
Merger arbitrage	0.30	0.30	0.26	0.30	0.30	0.14	0.30	0.30	0.30	0.30
Emerging markets	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00
S&P500	-0.03	-0.25	-0.19	-0.06	-0.26	-0.13	-0.11	-0.26	-0.19	-0.05
Bond index	0.30	0.30	0.30	0.30	0.30	0.30	0.25	0.30	0.30	0.30

Reported values are calculated from monthly net-of-all-fee returns. Investors assign capital to representative portfolios of convertible arbitrage, distressed securities, equity market neutral, global macro, long/short equity, merger arbitrage, emerging market hedge funds, the S&P500 index and the Salomon Brothers 7-Year Government Bond US Index. Optimal allocations are based on investors' preferences over expected return (α), skewness (β) and kurtosis (γ) using the PGP technique. The capital weight for each hedge fund strategy is constrained to be between 0 and 30 per cent, and the capital weights for the stock and bond indices are both constrained to be within ± 30 per cent.

towards equity market neutral and global equity to control variance, skewness and kurtosis. Investment in equity market neutral and global macro funds seems to make a similar contribution to the overall portfolio return distribution as an investment in the bond index.

Conclusion

This paper has incorporated investor preferences for higher moments into a PGP optimisation function. This allows us to solve for multiple competing (and often conflicting) hedge fund allocation objectives within a four-moment framework. Our empirical analysis has yielded a number of conclusions, the most important being that

- Hedge fund return moment statistics tend to trade off against each other in much the same way as in the underlying securities markets. Despite following often complex strategies, hedge funds therefore appear unable to dodge the rules of the game. This is in line with the results of Amin and Kat³⁷ who conclude that when taking the entire return distribution into account, there is nothing superior about hedge fund returns.
- Introducing preferences for skewness and kurtosis in the portfolio decision-making process may yield portfolios far different from the mean-variance optimal portfolio, with much less attractive mean-variance characteristics. This again emphasises the various trade-offs involved.
- Equity market neutral and global macro funds have important roles to play in optimal hedge fund portfolios, thanks to their attractive covariance, co-skewness and co-kurtosis properties. Equity market neutral funds act as volatility and kurtosis reducers, whereas global macro funds act as skewness enhancers.
- Especially in terms of skewness, hedge funds and stocks do seem not to combine very well. This suggests that investors may be better off using hedge funds to replace stocks instead of bonds, as appears to be current practice.

Acknowledgements

We are grateful for discussions with Gaurav Amin, Anca Dimitriu, Rob Grauer, Peter Klein, Mark Kritzman, Alex Russo, Steven Thorley and Eric Weigel. We thank seminar participants at the Northern Finance Association meetings, the Financial Management Association annual meetings, the Financial Management Association European meetings, the Gutmann Center Symposium on Hedge Funds and the Alternative Investment Conference in Montebello. We thank Hans de Ruiter and ABP Investments for generous support and Tremont TASS (Europe) Limited for supplying the hedge fund data.

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Appendix

Procedure to 'unsmooth' data

The observed (or smoothed) value V_t^* of a hedge fund at time t can be expressed as a weighted average of the underlying (true) value at time t , V_t , and the smoothed value at time $t-1$, V_{t-1}^* : $V_t^* = \alpha V_t + (1-\alpha) V_{t-1}^*$. Let B be the backshift operator defined by $B^L x_t = x_{t-L}$.

Define the following lag function, $L_t(a)$, which is a polynomial B , with different coefficients for each of the $t=1, \dots, 12$ appraisal cohorts:

$$L_t(\alpha) = \frac{t}{12} + \sum_{l=1}^{\infty} \left[(1-\alpha)^{L-1} \left(\frac{12-t}{12} \right) + (1-\alpha)^L \left(\frac{t}{12} \right) \right] B^L.$$

Let r_t and r_t^* denote the true underlying (unobservable) return and the observed return at time t , respectively. The monthly smoothed return is given by $r_t^* = \alpha L_m(\alpha) r_t$. We then can derive:

$$\begin{aligned} r_t^* &= \alpha r_t + (1-\alpha) r_{t-1}^* = \alpha r_t + \alpha(1-\alpha) r_{t-1} \\ &\quad + \alpha(1-\alpha)^2 r_{t-2} \dots \end{aligned} \tag{A1}$$

Here we implicitly assume that hedge fund managers use a single exponential smoothing approach. This yields an unsmoothed series with zero first-order autocorrelation: $r_t = \alpha^{-1} (r_t^* - (1-\alpha) r_{t-1}^*)$. As the stock market indices have around zero autocorrelation coefficients, it seems plausible in the context of the results above to set $1-\alpha$ equal to the first-order autocorrelation coefficient. The newly constructed return series, r_t , has the same mean as r_t^* , and zero first-order autocorrelation (aside from rounding errors), but with higher standard deviation.

4

A Primer on Structured Finance

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Received (in revised form): 7th June, 2007

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Practical applications

The article helps uninitiated market participants, policy makers and regulators understand the technical aspects and economic characteristics of

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structured finance in a bid to promote the development of market practices and prudential standards that guarantee market stability without encumbering the welfare-enhancing effect of risk sharing and efficient market pricing.

On the heels of last year's downgrades of General Motors Corp. and Ford Motor Co. to noninvestment grade status, the haircut unwinding of exposed collateralised debt obligations reverberated in mounting regulatory unease about current risk measurement standards of derivatives and their impact on financial stability. Probable knock-on effects of the recent bankruptcies of US auto parts supplier Delphi and two major airlines have fuelled groundswell concern about complex structured finance techniques, such as customised single-tranche and hybrid collateralised debt obligations with overlay structures, against the background of tightening credit spreads and greater dislocation in the correlation market. Subsequent warnings about the resilience of credit risk transfer to systemic crisis, however, hardly extended beyond indistinct assessments of how derivatives might propagate asset shocks across different capital market segments. This brief paper defines structured finance in order to inform a more specific debate about the regulatory challenges posed by the assembly of asset exposures and credit risk transfer in complex structured finance transactions that marry considerations of profitability and diversification alike.

Introduction

The traditional view of financial stability concentrates analytical efforts on the identification of increasing vulnerabilities prior to stress from individual failure in institutions, markets and infrastructure, assuming that the financial system is in equilibrium and adjusts when it experiences a shock. As opposed to this conventional approach, the presence of systemic vulnerabilities warrant the monitoring of ongoing developments in evolving areas of financial markets where the impact of unsustainable imbalances on financial stability is deemed most severe and widespread but difficult to measure, leave alone forecast. Given the increasing sophistication of financial products, the diversity and global reach of financial institutions, as well as the growing interdependence of financial markets and services, such areas exposed to extreme scenarios are likely to be found in financial innovation, where market forces and participants are left to their own devices and when complex incentive structures encourage greater risk taking in a benign economic environment but entail more adverse economic consequences when stress occurs.

One case in point is the development of the structured finance market, which has benefited from the attractive proposition of greater risk diversification. On the heels of prominent corporate downgrades in the US in April 2005, corporate defaults, such as auto parts supplier Delphi (see

Box 4.1), and the recent jitters in the US sub-prime mortgage market during the first quarter of 2007, the haircut unwinding of exposed *collateralised debt obligations* (CDOs) and other securitisation transactions reverberated in mounting regulatory unease about current risk measurement standards of derivatives and the possible knock-on effects of the structured finance market on other investment classes. After the fallout of the high-profile bankruptcies of automotive supplier Delphi as well as Northwest and Delta airlines in November 2005, investors and regulators began to worry about the systemic resilience of complex structured finance techniques (such as customised single-tranche hybrid CDOs with overlay structures) – especially against the background of tightening credit spreads and greater dislocation in the correlation market. The subsequent drumbeat of warnings about the impact of leveraged structured claims on financial stability in times of stress, however, hardly extended beyond indistinct assessments of the mechanics of structured finance markets and the ability of different structured finance products to propagate asset shocks across different capital market segments.

This brief paper defines structured finance in order to inform a more specific debate about the regulatory challenges posed by the assembly of asset exposures and credit risk transfer in complex structured finance transactions that marry considerations of profitability and diversification alike.

Definition of Structured Finance

Structured finance encompasses all advanced private and public financial arrangements that serve to efficiently refinance and hedge any profitable economic activity beyond the scope of conventional forms of on-balance sheet securities (debt, bonds, equity) at lower capital cost and agency costs from market impediments and liquidity constraints. In particular, most structured investments (i) combine traditional asset classes with contingent claims, such as risk transfer derivatives and/or derivative claims on commodities, currencies or receivables from other reference assets, or (ii) replicate traditional asset classes through synthetication or new financial instruments. Structured finance is invoked by financial and nonfinancial institutions in both banking systems and capital markets if either (i) established forms of external finance are unavailable (or depleted) for a particular financing need, or (ii) traditional sources of funds are too expensive for issuers to mobilise sufficient funds for what would otherwise be an unattractive investment based on the issuer's desired cost of capital. Structured finance offers issuers enormous flexibility to create securities with distinct risk-return profiles in terms of maturity structure, security design, and asset type, providing enhanced return at a customised degree of diversification commensurate to an individual investor's appetite for risk. Hence, structured finance contributes

to a more complete capital market by offering any mean–variance trade-off along the efficient frontier of optimal diversification at lower transaction cost. The increasing complexity of the structured finance market, and the ever-growing range of products being made available to investors, however, invariably create challenges in terms of efficient management and dissemination of information.

The premier form of structured finance is predicated on capital market-based risk transfer (save for loan sales, asset swaps and natural hedges through bond trading (see Figure 4.1)), whose two major asset classes include *asset securitisation* (which is mostly used for funding purposes) and *credit derivative* transactions (as hedging instruments) permit issuers to devise almost an infinite number of ways to combine various asset classes in order to both transfer asset risk between banks, insurance companies, other money managers and nonfinancial investors in order to achieve greater transformation and diversification of risk.

Asset securitisation describes the process and the result of converting (or monetising) a pool of designated financial assets into tradable liability and equity obligations as contingent claims backed by identifiable cash flows from the credit and payment performance of these asset exposures.¹ From an issuer perspective, securitisation registers as an alternative, market-based source of refinancing profitable economic activity in lieu of intermediated debt finance. Securitisation seeks to substitute capital market-based finance for credit finance by sponsoring financial relationships without the lending and deposit-taking capabilities of banks (*disintermediation*).

Issuers raise funds via securitisation in order to improve their capital management and liquidity position without increasing the capital base. The implicit risk transfer of securitisation allows issuers to benefit from more cost-efficient terms of high-credit quality finance without increasing their on-balance sheet liabilities or compromising the profit-generating capacity of assets. Investors in securitisation have a wider choice of high-quality investments at their disposal, whose market valuation engenders greater overall efficiency and liquidity of capital markets. Securitisation, however, involves a complex structured finance technology, which commands significant initial investment of managerial and financial resources.

Aside from being a flexible and efficient source of funding, the off-balance sheet treatment of securitisation also serves (i) to reduce both economic cost of capital and regulatory minimum capital requirements as a balance sheet restructuring tool (*regulatory and economic motive*) and (ii) to diversify asset exposures (especially interest rate risk and currency risk). The generation of securitised cash flows from a diversified asset portfolio represents an effective method of redistributing asset risks to investors and broader capital markets (*transformation and fragmentation of asset exposures*). As opposed to ordinary debt, a securitised contingent claim on a promised portfolio

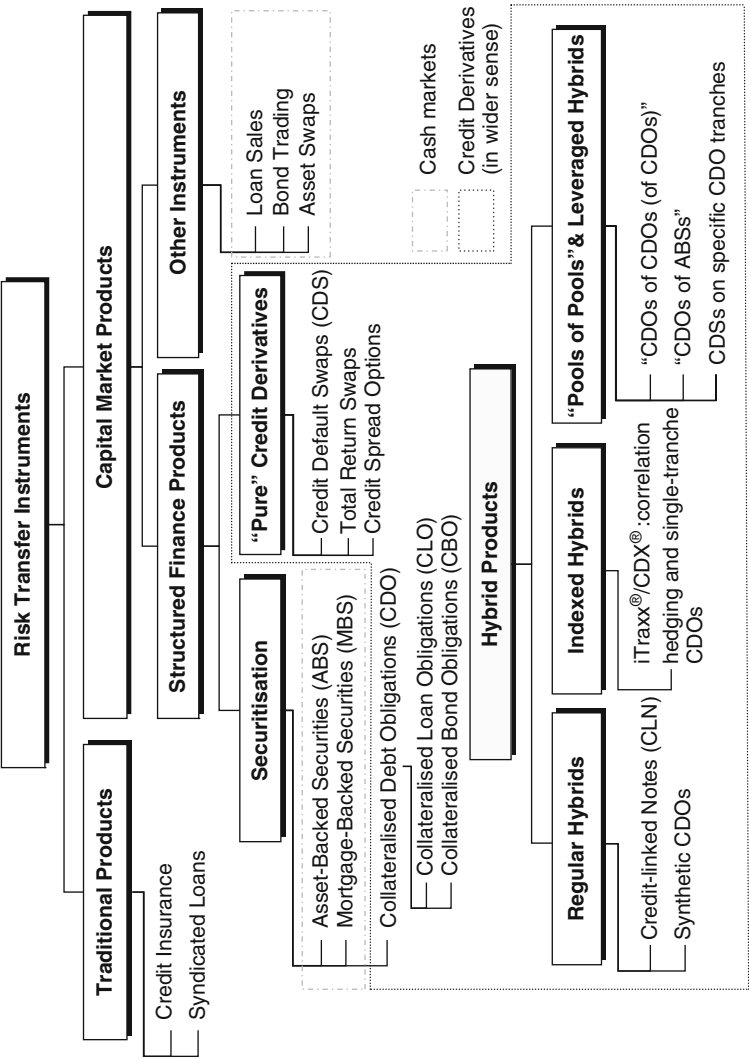


Figure 4.1 Overview of risk transfer instruments

performance affords investors at low transaction costs to quickly adjust their investment holdings due to changes in personal risk sensitivity, market sentiment and/or consumption preferences.

Derivatives in general are financial contracts on a pre-determined payoff structure of securities, indices, commodities or any other assets of varied maturities. Derivatives assume economic gains from both risk shifting and efficient price discovery² by providing hedging and low-cost arbitrage opportunities. Risk diversification improves the pricing of risk, increases stability at all levels of the financial system, and enhances general welfare. In addition to their capacity of eliminating or attenuating risk, derivatives also supplement cash markets as alternatives to trading underlying assets.

Credit derivatives are financial instruments that isolate and transfer credit risk. As a common working principle, they involve the sale of contingent credit protection for pre-defined credit events and/or asset performance. In their basic concept, the sale of credit derivatives severs the link between the loan origination and associated credit risk, but leave the original borrower–creditor relationship intact. The protection buyer of a credit derivative hedges specific credit risk at the expense of periodic premium payments to the protection seller, who assumes the credit exposure of the underlying transaction.³ The significance of credit derivatives lies less in their market share next to other derivative instruments (eg interest rate and foreign exchange derivatives) but in their ability to supplement traditional ways of hedging credit risk through the transfer of credit-related exposures to a third party. Other, noncredit derivative-based forms of credit risk transfer include credit insurance, syndicated loans, loan sales, bond trading and asset swaps (Box 4.1).

Box 4.1 The lessons of Delphi case and recovery rate products

The economic fallout caused by the bankruptcy of Delphi did not result from inappropriate risk management and speculation, but reflected inefficiencies in the microstructure of derivative markets at the time of settlement.⁴ When doubts about Delphi's creditworthiness emerged, the prospect of a shortage of deliverable debt increased the price (and reduced the attendant average recovery rate) of CDS-based credit protection beyond the level that might have otherwise been justified by the expected repayment from debt resolution (implied by rating agencies' estimates of Delphi's ultimate recovery rate or settlement prices of comparable firms). Higher transaction costs incurred by scarcity of collateral deterred trading of physical delivery CDS contracts (for lack of demand), because protection buyers would have had to settle contracts on

overpriced collateral, thereby discounting their recovery value implied by par value compensation through the CDS contract. Cash settled CDS contracts would not have implied such recovery risk.

Sellers of credit protection via CDS contracts are constantly exposed to recovery risk, while protection buyers are faced indirectly with recovery rate risk if the underlying asset (or a surrogate *cheapest-to-deliver* (CTD) asset) trades above the fair market price (suggested by the projected recovery rate) due to limited asset diversity when market liquidity is poor.

Several products permit investors to hedge recovery risk separately from default risk in derivative contracts. As opposed to a plain vanilla CDS contract, where the protection seller is exposed to recovery rate risk upon default of the underlying reference asset, a *fixed recovery rate CDS*, for instance, eliminates the uncertainty on the recovery rate by fixing a specific recovery value over the maturity of the CDS contract. After a credit event, the protection buyer is entitled to a cash settlement equal to 100 minus the pre-specified, fixed recovery rate. Recovery CDS with a fixed recovery rate set to zero are referred to as zero recovery CDS. Such products could serve as pricing benchmarks in the wake of a credit event, fostering a more efficient settlement process.

A *recovery lock* is a cash-neutral forward contract that fixes the recovery rate irrespective of the settlement price of the underlying reference asset. In practice, a recovery lock is structured by means of two opposite trades on the same reference entity. Protection sellers hedge themselves against recovery rate risk of their long position in a plain vanilla CDS by purchasing protection through a fixed recovery CDS. If the implicit recovery rate of the conventional CDS contract concurs with the fixed recovery rate, the premium payments on the transactions wash out and net to zero. In this case, if the reference entity defaults, the protection buyer of the fixed recovery CDS delivers the defaulted debt to the recovery seller and receives compensation equal to 100 minus the pre-specified, fixed recovery rate, which, in turn, pays off the compensation claim under the issued plain vanilla CDS contract. If the premium payments of the two trades differ, for example the actual recovery of the underlying asset drops below the fixed recovery rate at the time of default, the protection buyer reinvests higher premium income from the short position of a plain vanilla contract into fixed recovery rate protection on a larger notional value.

We distinguish between credit derivatives in the *narrower* and in a *wider sense*.^{5,6} The broader classification of derivatives includes (i) *pure* credit derivatives, such as *credit default swaps* (CDSs), *total return swaps* and *credit spread options*, which fall within the narrow definition of credit derivatives,

as well as (ii) *hybrid* and *securitisation* products with constituent credit derivative elements, such as *traditional* or *true sale* CDOs of bonds and loans and *partially funded* or *unfunded* structured finance transactions, for example *credit-linked notes* (CLNs), *synthetic* CDOs and other credit derivative-based *hybrid products* (see Figure 4.1). Hybrid structures employ securitisation⁷ for refinancing (through cash flow restructuring) and tranche-specific credit risk transfer⁸ (though the sale of credit protection or the issuance of *leveraged super-senior* (LSS)^{9,10} tranches). These *hybrid* and *securitisation* products are considered *credit derivatives in a wider sense*, which usually condition the repayment of securitised debt on a defined credit event in a bilateral hedge (in the case of CLNs), the premium income generated from credit protection sold on reference assets (in the case of synthetic CDOs), or the returns from investing and/or writing credit protection (long position on credit risk) on diversified pools of securitisation transactions of CDOs and/or *asset-backed securities* (ABSs) ('pools of pools') or newly formed CDSs and collateralised debt indices (eg the Dow Jones *iTraxx*® and the *iBoxx*® index),^{11–13} and the composite *ABX*® indices of CDS on ABS (ABCDs).¹⁴

CDOs have been the fastest growing area of structured finance.¹⁵ The CDO market has rapidly evolved since 1996, when the sector first gained significant prominence after its inception in the late 1980s, when some US banks started using CDOs as expedient risk-transfer mechanism. CDOs have now become a globally accepted structured finance technique, spanning the US, Europe and large parts of Asia. CDOs are investment vehicles that allow issuers to refinance the purchase of debt instruments by repackaging them into different slices of risk and maturity. While CDOs use the same structuring technology as ABS to convert a large, diversified pool of exposures into tradable commercial papers (*tranches*), their underlying collateral pool typically includes a wider and more diverse range of heterogeneous reference assets, such as senior secured bank loans, high yield bonds and CDSs, as opposed to more homogenous titles, such as home equity loans and credit card receivables.¹⁶

A CDO transaction is arranged and administered like a 'managed fund' with debt and equity holders, where a designated reference portfolio represents a diversified exposure to one or more asset classes from different issuers and/or industry sectors. Conversely, CDO investors sell credit protection to issuers against default on a portion of underlying reference assets. Generally, managers of CDO collateral portfolios choose a certain degree of diversification for a pre-specified risk-return profile subject to limits and guidelines that are determined by the issuers, rating agencies and investors at the commencement of the transaction.

In a CDO structure asset managers can increase assets under management while locking in committed funds and achieving some protection from market value volatility. While *cash CDOs* are backed by a collateral of actual bonds and loans as reference assets, whose legal title is transferred to

the purchaser, issuers of *synthetic CDOs* enlist wads of credit derivatives and various third-party guarantees to create unfunded or partially funded, highly leveraged investment from synthetic claims on the performance of designated credit exposures.¹⁷ CDOs involve either *cash flow* or *arbitrage* mechanisms to either fund expected principal and interest payments or expected trading and sales activity. CDOs enable issuers to achieve a broad range of financial goals, which include the off-balance sheet treatment of securitised exposures, reduced minimum regulatory capital requirements and access to alternative sources for asset funding and liquidity support. The conventional security design of CDOs assumes a typical three-tier securitisation structure of junior, mezzanine and senior tranches, which concentrates expected losses in a small first loss position as equity claim, which bears the majority of the credit exposure and is frequently covered by a junior CDS, shifting most unexpected risk to larger, more senior tranches, which display distinctly different risk profiles.^{18,19} This risk sharing arrangement induces a leverage effect on constituent tranches, whose distinct risk-return profiles can be tailored to specific investment preferences.²⁰

Boundaries between structured finance and conventional finance

The flexible nature of structured finance straddles the indistinct boundary between traditional fixed income products, debentures and equity on one hand and derivative transactions on the other hand. Notwithstanding the ostensible difficulties of defining structured finance, a *functional* and *substantive* differentiation seems to be most instructive for guiding an informed demarcation between the most salient properties of structured and conventional forms of external finance. The following definition reflects such a proposition if we compare two financial arrangements:

- (a) Investment instruments are motivated by the *same or similar financial objective* from both the issuer's and the investor's point of view, but a *dissimilar legal and functional implementation* requires a *different valuation*.
- (b) Investment instruments are motivated by the *same or similar financial objective* and are *substantively equivalent* (ie they share a close equilibrium price relation and the same investor pay-off profile), but differences in *transaction structure and/or security design* necessitate a *different valuation*.

In the first case, *pure credit derivatives* are clear examples of structured products, which allow very specific and capital-market priced credit risk transfer (see Figure 4.1). Credit insurance and syndicated loans share the same financial objective; however, they do *not* constitute an arrangement to create a new risk-return profile from existing or future reference assets. In the same vain, *mortgage-backed securities* (MBSs) and *Pfandbrief*-style covered mortgage

bonds represent different functional and legal methods of securitisation with the same financial objective. Although both refinancing techniques convert homogenous pools of mortgage claims into negotiable securities, they represent two distinct forms of debt securities issued on the same type of underlying reference asset either off-balance sheet (*asset-backed securitisation*), on-balance sheet ('*Pfandbrief-style*' securitisation) or even through *synthetic securitisation*.

The *Pfandbrief* is the most prominent deal structure for securitised mortgage loans in Europe, which matches the importance of MBS in the US by issuance volume, trading activity and historical track record (see Box 4.2). In contrast to the US, where the market for MBS has had a longstanding

Box 4.2 The definition of *Pfandbrief* transactions

Although many European countries have already put in place legal frameworks for *Pfandbrief*-style products, the German *Pfandbrief* (literally 'letter of pledge') is the eponym of this type of covered mortgage bond.²¹ Although the creation of the first *Pfandbrief* instrument was attributed to an executive order of Frederick II of Prussia in 1769,²² it was only when the *Mortgage Bank Law* was passed in 1899 that the *Pfandbrief* took its present form. The first legal guidance for the issuance of *Pfandbrief*-style products was actually adopted in France in 1852 with the *Loi sur l'obligation foncière et communale*. The oldest mortgage credit market can be traced to Denmark, when the Great Fire of 1789 created vast demand for housing finance in its wake. In Sweden, a mortgage market has existed at least since 1860 under the legal provisions of the more general Law on Credit Companies, but no specific mortgage bank law has been issued so far.

While the *Pfandbrief* is a classical on-balance sheet refinancing tool of mortgages and public loans with both origination and issuance completed by one and the same entity, MBS transactions are off-balance sheet transactions and involve at least one more party (besides the mortgage originator). *Pfandbrief* serve primarily as funding instruments, whereas MBS issues are also employed for credit risk transfer and balance sheet restructuring, with the aim of efficient management of economic and regulatory capital. Originators of MBS sell contingent claims on asset cash flows in order to remove and legally segregated ('bankruptcy remote') reference portfolio of securitised assets from the balance sheet. In contrast, in a typical German *Pfandbrief* transaction, reference assets are 'ring-fenced' on the balance sheet of government-licensed issuers and repayments to investors are independent from the repayment of securitised assets. Issuers of *Pfandbrief* deals are fully liable with their registered capital if reference assets fail to generate sufficient cash flows for the repayment of investors. Hence, this arrangement implies a double

protection of investors against credit risk and the solvency of the issuer. Given the value of this institutional guarantee depending on the issuer's financial strength, *Pfandbrief* transactions generally receive high ratings. However, in comparison, MBS transactions are devoid of any institutional guarantee and solely return cash flows generated from the pool performance of the designated reference portfolio. Issuers of MBS transactions compensate investors for the higher asset exposure due to the lack of institutional protection by including various kinds of internal and external liquidity and credit support, such as bridge-over facilities, surety bonds, third-party guarantees, excess spreads, over-collateralisation and reserve accounts. Finally, *Pfandbrief* issues are typically subject to stringent federal laws (requiring a weighted average loan-to-market (or appraised) value (LTV) of at least 60 per cent as a statutory benchmark), while *private-label* MBS issues are free from these legal requirements, except in so-called *agency-MBS* in the US, where the quasi-government agencies *Fannie Mae* (FNMA), *Freddie Mac* (FHLMC) and *Ginnie Mae* (GNMA) provide institutional guarantees in return for certain restrictions imposed on mortgages eligible for purchase in MBS structures.

In general, *Pfandbrief* transactions represent a very secure and liquid asset class of fixed income instruments with an established track record and cyclical resilience. MBS issues are equally liquid (at least in the US market) and feature an unchallenged degree of structural flexibility allowing for customised features and investor arrangements, such as variations to amortising repayment (in contrast to bullet repayment structures of *Pfandbrief* issues) and security design.

tradition since the first half of the 1980s,²³ off-balance sheet securitisation via MBS is a relatively recent development in Europe and has gained traction only over the last years, with issuance amounts being still relatively low compared to established on-balance sheet securitisation via covered mortgage bonds or *Pfandbrief*. ABS issues have caught up with *Pfandbrief* transactions. The *Pfandbrief* market is still the biggest segment of the euro-denominated private debt market and rivals in size the individual government bond markets in Europe.^{24–25} Whereas originators of *Pfandbrief* issues retain securitised assets on their balance sheet, issuers of MBSs sell assets to a separate legal entity (such as trust, fund and corporation), commonly referred to as a *special purpose vehicle* (SPV), which refinances the acquisition of the assets by issuing debt (eg bonds or commercial paper) or equity claims on the back on these reference assets. The designated assets are considered securitised insofar as their cash performance serves to secure any repayment obligation to investors. Alternatively, *synthetic securitisation* represents a compound form of structured finance, which amalgamates properties of both asset-backed securitisation and credit derivatives in one coherent

structure. Synthetic securitisation does not involve the transfer of assets, but serves to hedge the credit risk to which the originator is exposed. The originator merely transfers the credit risk through the use of *funded* (eg credit-linked notes (CLNs)) or *unfunded* (eg CDSs) credit derivatives or guarantees, in which the counterparty agrees upon specific contractual covenants to cover a predetermined amount of losses.²⁶

In the second case, for instance, *Islamic finance* falls squarely within the domain of structured finance instrument whenever religious constraints require the replication of conventional interest-bearing assets through structural arrangements of two or more contingent claims (see Figure 4.2). Islamic finance is limited to financial relationships involving entrepreneurial investment, subject to the prohibition of (i) interest earnings (*riba*) and money lending, (ii) sinful activity (*haram*),²⁷ such as direct or indirect association with lines of business involving alcohol, tobacco, pork products, firearms and gambling, and (iii) the speculative trade or exchange of money for debt *without* an underlying asset transfer (*gharar*).²⁸ As opposed to conventional finance, where interest represents the contractible cost for funds over a pre-specified lending period, in Islamic finance, both financiers and borrowers to share the business risk (and returns) from investment in religiously acceptable services, trade or products (in adherence to lawful activities) (*halal*), where profits are not guaranteed *ex ante*, but only accrue if the investment itself yields income. Islamic financial transaction assigns to investors clearly identifiable rights and obligations for which they are entitled to receive commensurate return.²⁹

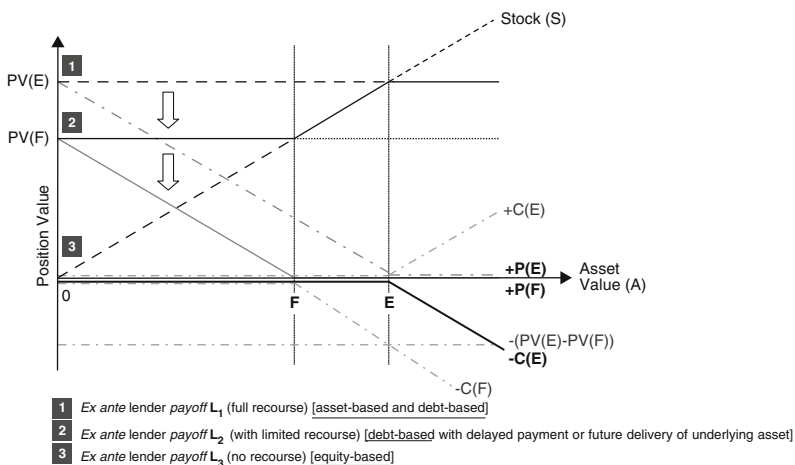


Figure 4.2 The pay-off profile under put-call parity of the three basic Islamic finance transactions

In light of these moral impediments to both 'passive' investment and interest as form of compensation, *shariah*-compliant lending in Islamic finance requires the replication of interest-bearing, conventional finance via structural arrangements of contingent claims. Although Islamic and conventional finance are *equivalent in terms of substance* and yield the same lender and investor pay-offs (ie equilibrium price equivalence) at the inception of the transaction, they differ in legal form and might require a different valuation due to dissimilar transaction structures and/or security design.

As opposed to conventional lending, Islamic finance substitutes a temporary transfer of an asset to the lender for a permanent transfer of funds to the borrower as a source of indebtedness. We distinguish between three basic forms of Islamic financing methods for both investment (eg plant, equipment, machines) and trade finance:^{30–32} (i) *synthetic (mortgage) loans* (debt-based)³³ through a sale-repurchase agreement or back-to-back sale of borrower- or third party-held assets, (ii) *operating or finance leases* (asset-based)³⁴ through a lease- buyback agreement or purchase order, and (iii) *profit-sharing contracts* (equity-based)³⁵ of future assets (see Figure 4.2). As opposed to equity-based contracts, both debt- and asset-based contracts (in their generic form) are initiated by a temporary transfer of existing assets from the borrower to the lender or the acquisition of third-party assets by the lender on behalf of the borrower.³⁶ These different forms of Islamic finance combine two or more contingent claims to replicate the risk-return trade-off of conventional lending contracts or equity investment without any contractual guarantee of investment return or payments by reference to an interest rate as time-dependent cost of funds.³⁷

This unsecured arrangement constitutes entrepreneurial investment on part of the financier, who receives returns from direct participation in uncertain asset performance in the form of state-contingent payments according to an agreed schedule and amount.³⁸ The specific lending arrangement underlying each type of Islamic finance represents a different form of a put-call parity³⁹-based replication of interest income, which re-characterises the rate of return of conventional investments in a religiously acceptable manner.

The three main types of Islamic finance are only distinct as to the attribution of economic benefits from the use of an existing or future asset owned by the lender. In *asset-based* Islamic finance for investment or trade, the borrower leases from the lender one or more assets A at current stock value S , which the lender has previously acquired either from the borrower or a third party. The lender writes a call option C with strike price E to the borrower to acquire the asset after time T , subject to the promise (put option P) of full payment E of the current asset price plus an agreed premium in the form of total rental payments over the investment period, which amounts to a present value of $PV(E)$ of risky debt at maturity. If the lender has full recourse (ie he retains ownership until maturity T , when the borrower can

exercise the right to acquire the asset), also the put option has a strike price E , which ensures that the borrower's failure to fully repay entitles the lender to sell the asset to compensate for the financial shortfall. This arrangement amounts a collateralised loan with maturity T and a fully collateralised principal amount, which is equivalent to the current purchase price of the desired asset. According to put-call parity, the lender's position at maturity is $S - C(E) + P(E) = PV(E)$,⁴⁰ which equals the present value of the principal amount and interest of a conventional loan. In a more realistic depiction, the combination of a held put and a written call option on the same strike price does not represent a simple forward contract on the underlying asset, but a series of forward contracts over multiple rental payment dates, each of which obliging the holder to renew the periodic call option of purchasing the asset at maturity as a buyback or allowing the lender to resell the asset at final maturity.⁴¹

In *debt-based* and *equity-based* Islamic finance, the payoff profiles are similar. In *debt-based* Islamic finance, borrower indebtedness from a sale-repurchase agreement of an asset with current value $PV(E)$ implies a premium payment to the lender for the use of funds over the investment period T . As opposed to an *asset-based* arrangement, the lender relinquishes same asset ownership right after inception, if there is a mismatch of funding and repayment reducing the option value of possible recourse (written put) to $P(F)$, so that *ex ante* lender payoff L_2 is now $S - C(E) + P(F) = PV(F) - (PV(E) - PV(F)) - C(F) + C(E)$. In *equity-based* Islamic finance, the lender (ie investor) is fully repaid only if the investment project generates high enough returns to repay the initial investment amount and the premium payment in return for investment risk until maturity T . This arrangement precludes any recourse by the lender. If the investor owns 100 per cent equity of investment S , *ex ante* lender payoff L_3 is $S - C(E) = PV(E) + P(E)$.

Conclusion

Against the background of rising regulatory unease about the evolution of derivative markets, we argue that a clear-cut definition of structured finance helps substantiate more viable debate about the resilience of credit risk transfer to financial shocks. Structured finance encompasses all advanced private and public financial arrangements that serve to efficiently refinance and hedge any profitable economic activity beyond the scope of conventional forms of on-balance sheet securities (debt, bonds and equity) in the effort to lower cost of capital and to mitigate agency costs of market impediments on liquidity. Especially, the distinction of the various methods of credit risk transfer through credit derivatives in a wider and narrower sense as well as securitisation transactions illustrates the need for more comprehensive and creative regulatory considerations that take on board the heterogeneity of institutions, markets, and infrastructure.

Disclaimer

The views expressed in this paper are those of the author and should not be attributed to the International Monetary Fund, its Executive Board, or its management. Any errors and omissions are the sole responsibility of the author. An earlier version of this paper is available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=832184.

References and Notes

1. Jobst, A. (2006a) 'Sovereign Securitization in emerging markets', *Journal of Structured Finance*, Vol. 12, No. 3, pp. 2–13.
2. Derivatives help "discover" the fair market price (spot and future) of certain assets or risks in instances of high transaction costs, poor liquidity due to the dispersion of markets, limited asset supply or the conglomeration of many risks into one whole asset.
3. In a *cash-settled* CDS, the protection seller is required to make a settlement payment in the amount of the difference between the notional principal and the market price of the underlying bond or the reduced recovery value of the defaulted bank credit. Alternatively, in what has increasingly become the market norm, *physical settlement* CDSs oblige the protection seller to accept the reference asset (or any other eligible collateral asset, such as *cheapest-to-deliver* (CTD) bonds) against payment of their par value. Unlike credit insurance contracts, credit derivatives are negotiable and attract large secondary trading.
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5. Jobst, A. (2006b) 'Correlation, price discovery and co-movement of ABS and equity', *Derivatives Use, Trading & Regulation*, Vol. 12, No. 1–2, pp. 60–101.
6. Effenberger, D. (2003) 'Frankfurt Voice: Credit Derivatives—Implications for Credit Markets', Deutsche Bank Research.
7. This feature does not apply to plain vanilla *asset-backed securities* (ABS) and *mortgage-backed securities* (MBS).
8. Although the transformation and fragmentation of credit risk through securitisation brings greater diversification within the financial system, the structural complexity arising from multi-layered security designs, diverse amortisation schedules and the state-contingent funding of synthetic credit risk transfer might obfuscate actual riskiness of these investments and inhibit provident investment. The tradability of credit risk facilitates the synthetic assembly and dynamic adjustment of credit portfolios via secondary markets, but numerous counterparty links established in the commoditisation of securitised asset risk and wads of derivative claims also create systemic dependence susceptible to contagion on the risk of heightened aggregate moral hazard. This prospect of leveraged investment in synthetic structures seems to be particularly troubling when investors take on more risks for yield during times of compressed spreads and rising default rates when credit cycles approach their turning-point. Moreover, the contingent liability of credit derivatives as credit protection of securitised assets requires the protection seller to put up liquidity only if a credit event occurs.
9. 'Market developments testify to "structural substitution" in complex *hybrid* CDOs. After CDO- squareds ("CDOs of CDOs") led to the narrowing of mezzanine spreads in the CDO market, *leveraged super-senior tranches* (SS) with *marked-to-market* (MTM) loss- and spread-based triggers emerged in a significant number

of synthetic CDOs. Most recently, investment banks with significant mezzanine ABS inventory also began to employ *leveraged super-senior (LSS) tranches* in synthetic CDOs as an alternative method of hedging specific—and not diversified—mezzanine tranche exposure by offloading senior risk instead of selling credit protection or delta hedging. With the leveraged super-senior concept becoming exhausted, CDO managers are now introducing overlay structures that bring in other sources of risk, such as foreign exchange rates, inflation and commodity price linkages in order to juice up investor yields.¹⁷ *Super-senior tranches* themselves are usually secured by a CDS as a means of improving the marketability of issued claims.

10. Jobst, A. (2005) 'Risk Management of CDOs During Times of Stress', *Derivatives Week*, Euromoney, London, 28 November.
11. Both the Dow Jones *iTraxx*® Europe index of *credit default swap* (CDS) contracts and the *iBoxx*® index of collateralised (fixed-rate) debt obligations have inaugurated the first round of emerging standardisation. A recent survey paper by Cousseran and Rahmouni¹² states that the development of liquid pricing benchmarks has greatly contributed to the maturity of the market for *collateralised debt obligations* (CDOs) as regards 'an improvement in the transparency of the market, as tranche prices are continuously quoted; a significant increase in market liquidity [y], allowing protection to be bought and sold at a lower cost [y], an improvement in the management of market participants' risk, as they now have access to daily valuations from which they can obtain [y] levels of implied correlation [y], [and] a broadening of the investor base to new market participants such as hedge funds, which use these instruments for their sophisticated trading strategies (correlation trading).' Large parts of the ABS market in Europe have, however, shed little of their frequently deplored opacity. Market observers do not find a dramatic change in market transparency due to standardisation as much as they acknowledge changing hedging patterns of ABS issues in its wake.¹³ For instance, CDOs are generally structured to meet specific investor needs. In the past, issuers would hedge unbalanced positions of customised CDOs through complex subordinated, multi-tranche structures ('transaction-based'), whose complexity inhibited transparent asset pricing. When the Dow Jones *iTraxx*® Europe index was created in June 2004 from the merger between two existing CDS indices, large issuers began to offer standard (single) CDO tranches on the *iTraxx*® index, which replicate the behaviour of synthetic CDO claims on constituent names of the Dow Jones *iTraxx*® index. These standardised (synthetic) CDO claims on liquid indices now offer a base correlation measure ('CDO delta') with the actual equity prices of (underlying) reference assets and constitute a dynamic 'market-based' hedge for issuers of bespoke and mostly privately transacted single-tranche transactions (arranged for single investors). Most recently, issuers also began to offer multi-tranche transactions with mezzanine tranches indexed to equity prices and tranche-specific CDS contracts on any retained CDO interest.
12. Cousseran, O. and Rahmouni, I. (2005) 'The CDO Market—Functioning and Implications in Terms of Financial Stability', *Banque de France Financial Stability Review*, No. 6, pp. 43–62.
13. Tsui, E. (2005) 'Asia Poised for Take-Off in CDOs', *Financial Times*, Capital Markets, 15 July.
14. The introduction of CDS indices contributed to hedging patterns thanks to greater standardisation. In the past, issuers would hedge unbalanced positions of customised CDOs through complex subordinated, multi-tranche structures ('transaction-based'), whose complexity inhibited transparent asset pricing. When

the Dow Jones iTraxx® Europe index was created in June 2004 from the merger between two existing CDS indices, large issuers began to offer standard (single) CDO tranches on the iTraxx® index, which replicate the behaviour of synthetic CDO claims on constituent names of the Dow Jones iTraxx® index. These standardised (synthetic) CDO claims on liquid indices now offer a base correlation measure ('CDO delta') with the actual equity prices of (underlying) reference assets and constitute a dynamic 'market-based' hedge for issuers of bespoke and mostly privately transacted single-tranche transactions (arranged for single investors). Most recently, issuers also began to offer multi-tranche transactions with mezzanine tranches indexed to equity prices and tranche-specific CDS contracts on any retained CDO interest.

15. The annual issuance volume worldwide has grown more than fourfold from US\$ 48.1 billion in 1997 to US\$ 193.5 billion by the end of 2004. In 2004, one out of four new CDO deals was synthetic (US\$ 143.7 billion cash CDOs vs US\$ 49.8 billion funded synthetic CDOs), up from one out of 20 in 1997 (US\$ 45.5 billion cash CDOs vs US\$ 2.7 billion funded synthetic CDOs). With new asset classes being securitised in CDO transactions, the market size of outstanding CDO tranches has reached an estimated US\$ 800 billion by the end of 2004, which does not include privately placed transactions and unreported unsold tranches of bespoke, single-tranche synthetic CDOs.
16. Jobst, A. (2005) 'Tranche pricing in subordinated loan securitization', *Journal of Structured Finance*, Vol. 11, No. 2, pp. 64–96.
17. Shepherd, B. (2005) 'The Synthetic CDO Shell Game', *Investment Dealer's Digest*, 16 May.
18. Jobst, A. (2005) 'Need for Vigilance by CDO Investors', *Financial Times*, Comments & Letters, 4 November.
19. Jobst, A. (2005) 'Investors Must Heed Those CDO Risks', *Financial Times*, Comments & Letters, 19 April.
20. Although investors should expect the same returns for CDOs as for similar credit risk exposure in plain vanilla debt, their risk profile of CDO tranches varies dramatically in response to changes in the valuation of the underlying (reference) asset.
21. Also Spain, Denmark and Sweden have established a long track record in the issuance of *Pfandbrief*-style investment products.
22. Skarabot, J. (2002) 'Securitization and Special Purpose Vehicle Structures', Working Paper, Haas School of Business, University of California at Berkeley. April.
23. The first ABS issue in its modern form was completed by *Sperry Corporation*, which issued computer lease-backed notes in 1985 (Kendall and Fishman, 1996).
24. Mastroeni, O. (2005) 'Pfandbrief-Style Products in Europe', Occasional Papers Series, European Central Bank, Frankfurt/Main.
25. Jobst, A. (2006c) 'European securitization: A GARCH model of secondary market spreads', *Journal of Structured Finance*, Vol. 12, No. 1, pp. 55–80.
26. Thus, synthetic arrangements effectively sidestep possible legal constraints associated with different loan characteristics and jurisdictions, mainly because most or all of the securitised assets are never sold to capital market investors.
27. Other, less relevant sinful activity under Islamic law in this context include hoarding, miserliness and extravagance.
28. These distinctive features derive from two religious sources, which aim at an equitable system of distributive justice: (i) the *sharia'ah* (or *shariah*), which comprises the *qur'an* (literally, 'the way') and the sayings and actions of the prophet

- Mohammed recorded in a collection of books known as the *sahih hadith*, and (ii) the *fiqh*, which represents Islamic jurisprudence based on a body of laws deduced from the *shariah* by Islamic scholars.
29. While the elimination of interest is fundamental to Islamic finance, *shariah*-compliant investment behaviour also aims to eliminate exploitation pursuant to Islamic law. Note also that Islamic law does not object to payment for the use of an asset. In fact, the earning of profits or returns from assets is encouraged.
 30. See Archer and Karim²⁷ as well as Iqbal and Llewellyn²⁸ for an in-depth analysis.
 31. Archer, S. and Karim, R.A., (eds.) (2002) 'Islamic Finance: Growth and Innovation', Euromoney Books, London.
 32. Iqbal, M. and Llewellyn, D., (eds.) (2002) 'Islamic Banking and Finance: New Perspective on Profit-Sharing and Risk', Edward Elgar Publishing, Ltd, Cheltenham/UK.
 33. Islamic debt instruments come in the form of *murabaha* (or *murabahah*) (cost-plus sale), *salam* (deferred delivery sale), *bai bithaman ajil* (BBA) (deferred payment sale), *istina* (or *istisna*) (pre-delivery, project finance) and *quard al-hasan* (benevolent loan) contracts, which create borrower indebtedness from the purchase and resale contract of an asset in lieu of interest payments. Interest payments are implicit in an installment sale with instantaneous or future title transfer for promised payment of agreed sales price in the future.
 34. In Islamic asset or quasi-debt instruments (*al-ijarah* leasing notes) the lender leases an asset to the borrower for a specified rent and term. The lessor (ie financier) acquires the asset either from the borrower or a third party (at the request of the borrower) and leases it to the borrower for an agreed sum of rental payable in installments according to an agreed schedule. The legal title of the asset remains with the financier throughout the tenure of the transaction, who bears all the risk associated with the ownership of the asset. The asset is returned to the borrower for the original sale price or the negotiated market price unless otherwise agreed (as opposed to debt-based contracts, which require a higher re-purchase price inclusive of quasi-interest payments). If the lessee does not exercise the option to buy the assets at a pre-determined price at maturity, the lender will dispose of it in order to realise the salvage value.
 35. In Islamic profit-sharing contracts (*mudharaba* and *musharaka*), lenders and borrowers agree to share any gains of profitable projects based on the degree of funding or ownership of the asset by each party.
 36. In a debt-based *synthetic loan*, the borrower repurchase the assets from the lender at a higher price than the original sales price, whereas borrowers under a *lease-back agreement* repurchase the assets at the same price at the end of the transaction and pay quasi-interest in the form of leasing fees for the duration of the loan.
 37. El Qorchi, M. (2005) 'Islamic Finance Gears Up', *Finance and Development* (December), *International Monetary Fund (IMF)*, Vol. 42, No. 4, pp. 46–49.
 38. The underlying asset transfer of Islamic lending arrangements provides collateralisation until the lender relinquishes ownership at the maturity date. In equity-based Islamic investments lenders do not have any recourse unless pre-mature termination enables the lender to recover some investment funds from the salvage value of project assets.
 39. The relationship between the put and call values of a European option on a nondividend paying stock of a traded firm can be expressed as $PV(E)+C=S+P$. $PV(E)$ denotes the present value of a risky debt with a face value equal to exercise price E , which is continuously discounted by $\exp(-rT)$ at an interest rate r over T

number of years. In our case of a lending transaction, the share price S represents the asset value of the funded investment available for the repayment of the debt obligation at future value E .

40. The lease payments by the borrower are received wash out in this representation.
41. The call option is extendible in that the borrower has the right to renew the option to eventually acquire the asset by making the required rental payments or retiring any upcoming obligation according to the investment contract. The borrower pays a periodic premium for the call option to compensate the lender for the short position on the underlying asset until final repayment at maturity. The put option $+P(E)$ represents a series of cash-neutral, risk-free hedges of the lender's credit risk exposure. In corporate finance, borrowers (ie managers) would pay debt investors (ie lenders) a spread over the risk-free return (implied in the coupon yield) as option premium of their put on default risk if the asset value is insufficient to existing debt E (strike price). As opposed to holders of risky corporate debt with payoff $PV(E) - P(E)$, financiers of such lending transactions own the underlying asset and hold a long put position on the firm value, which reflects the lender's full recourse for each installment repayment during the term of the transaction if the asset value S falls below the promised repayment level E .
42. Kendall, L.T. and Fishman, M.J. (1996). '*A primer on securitization*', MIT Press, Cambridge, MA.

5

Value at Risk, GARCH Modelling and the Forecasting of Hedge Fund Return Volatility

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Practical applications

The Value at Risk (VaR) approach is widely used within the asset and risk management of traditional and alternative investments. From the investor's point of view, an adequate VaR model should indicate how the positions of the hedge fund portfolio should be sized for the best protection against downside risk. Due to the skewness and excess kurtosis of daily financial return distributions, the normal VaR has its drawbacks particularly when it is applied to hedge funds. In addition to the Cornish–Fischer VaR, which explicitly accounts for non-normally distributed returns, we examine the conditional volatility characteristics of daily hedge fund management style returns. The inclusion of time-varying conditional volatility into the VaR measurement enables us to trace the actual return process more effectively. We show that such a GARCH-type VaR is a superior measure of downside risk, especially for trading strategies that exhibit negative skewness and excess kurtosis and offer daily pricing.

This paper examines the conditional volatility characteristics of daily management style returns and compares the out-of-sample forecasts of different Value at Risk (VaR) approaches, namely, the normal, Cornish–Fisher (CF), and the so-called GARCH-type VaR. The examination of the conditional volatility of hedge fund styles and composite returns shows important differences concerning persistence, mean reversion and asymmetry in the period under consideration. Hedge fund returns exhibit significant negative skewness and excess kurtosis, which cannot be captured in the normal VaR whereas the CF-VaR results in a systematic downward shift of the conventional VaR. The GARCH-type VaR, however, includes the time-varying conditional volatility and is able to trace the actual return process more effectively. Since the forecast performance cannot detect which of the three VaR types can match the time-varying risk adequately, an adjusted hit ratio takes the size of the hits as well as the average VaR into account. According to this, the GARCH-type VaR outperforms the other VaRs for most of the hedge fund style indices.

Introduction

Since the breakdown of Long Term Capital Management (LTCM), risk management and the transparency of hedge funds have become dramatically important and evolved into outstanding fields for practitioners and academic researchers. Value at Risk (VaR) is one of the most important concepts widely used for risk management by banks and financial institutions. Since VaR can be easily computed by capturing risk in only one figure, it has gained increasing popularity in the past. Although there are several forms of financial risk, we focus on market risk in this paper, that is, the unexpected changes in stock returns.

The literature on VaR has become quite expansive (eg Hendricks¹, Beder², Marshall and Siegel³). However, the conventional VaR assumes that returns follow a normal or conditional normal distribution. Particularly, in the case of skewed and fat-tailed returns, the assumption of normality leads to substantial bias in the VaR estimation and results in an underestimation of volatility.

In contrast to mutual funds, different trading instruments, such as arbitrage, leverage and short selling, characterise hedge funds. These trading instruments are highly dynamic and often exhibit low systematic risk (Fung and Hsieh⁴). Since hedge funds use options or option-like trading strategies or strategies that lose money during down-market phases, they may generate non-normal payoffs. In addition, Liang⁵ emphasises the higher Sharpe ratios and lower market risk as well as the higher abnormal returns of hedge funds investments. Moreover, it has been well documented that monthly return distributions of most hedge fund indices show extremely high negative skewness, positive excess kurtosis and, significantly, positive first-order serial correlation. In the context of the frequently used mean-variance approach, these return properties inevitably result in an underestimation of the true volatility.

Favre and Galeano⁶ suggest a modified method of VaR by implementing a Cornish-Fisher (CF) expansion, which is used to control for skewness and kurtosis.⁷ Agarwal and Naik⁸ introduce a mean-conditional VaR (CVaR) framework that explicitly accounts for negative tail risk.⁹ As the conventional VaR refers only to the frequency of extreme events, the CVaR focuses on both frequency and size of losses in the case of extreme events. Kellezi and Gilli¹⁰ introduce a risk capital measure based on the Extreme Value Theory (EVT). The EVT focuses only on extreme values, that is, the tail of the distribution, rather than the whole distribution. However, Danielsson and de Vries¹¹ show that the EVT can be accurately used only for very extreme events and often does not provide good results at more conventional 5 per cent VaR levels. Furthermore, EVT assumes an identically and independently distributed (iid) framework that is not consistent with most financial data.

In this paper, we use a GARCH-type VaR by modelling and forecasting conditional volatility, using GARCH and EGARCH, and then implementing the time-varying volatility in the VaR. In doing so, we also control for skewness and kurtosis. Volatility forecasting is important not only in risk management and market timing for single hedge funds, but also in the context of portfolio diversification including hedge funds. The knowledge of future volatilities allows portfolio managers to control the risk temporally, for example, sell an asset or portfolio before a dramatic increase in volatility takes place (Engle and Patton¹²). Furthermore, by means of information on the volatility process in general, and the development of volatility in particular, the risk pricing of the market can be determined.

To our knowledge, there are no empirical studies that introduce GARCH-type forecasts into the conventional VaR framework to simultaneously account for time-varying volatility, serial correlation, skewness and kurtosis in hedge fund returns.

This paper is organised as follows: Firstly, the next section describes different hedge funds strategies from the data provider, *Standard & Poor's*, according to the different management styles. The concepts of conventional VaR and CF expansion are then briefly introduced. Following this, the stylised facts of volatility and the two conditional volatility models, $GARCH(p,q)$ and $EGARCH(p,q)$, that should capture these features are discussed. Subsequently the conditional variances for the hedge funds styles under consideration are estimated using alternative model specifications, and their volatility characteristics are analysed. The GARCH-type models are then applied to estimate the daily VaR of the different hedge fund styles. The accuracy of one-step-ahead VaR forecasts is evaluated by different ratios that measure the distance between the observed and forecasted VaR values. Some concluding remarks are offered in the final section.

Hedge Funds Styles and their Strategies

Investment strategies used by hedge funds tend to be quite different from those of traditional mutual funds. Basically, every hedge fund embarks on its own preferred strategy, which leads to a very heterogeneous asset class 'hedge funds'. However, hedge funds may be classified into a number of different strategy groups depending on the main type of strategy followed (Kat and Lu¹³). In referring to market exposure as classification criteria, Purcell and Crowley¹⁴ distinguish between three different styles of hedge funds. These broader categories encompass the relative value, the event-driven and the opportunistic strategy. Based on market exposure to traditional asset classes, Agarwal and Naik¹⁵ also classify hedge fund strategies into categories: directional and non-directional. Ineichen¹⁶ conducts a subdivision on the hedge funds styles: arbitrage, event-driven and directional (see Figure 5.1). In this study, we adapt the classification proposed by the latter to match the system of the hedge funds index provider S&P, as we use their indices later in the empirical section of this paper.

Arbitrage management style

The 'arbitrage' strategies, also known as 'relative value' or 'non-directional' strategies, try to take advantage of temporarily wrong valuations between different financial instruments and attempt to offer their investors very low market exposure. Independent of the market situation, the strategy aims at achieving absolute returns by avoiding or reducing market (beta) risk, industry risk, interest rate risk (duration), etc. Thus, this so-called 'equity

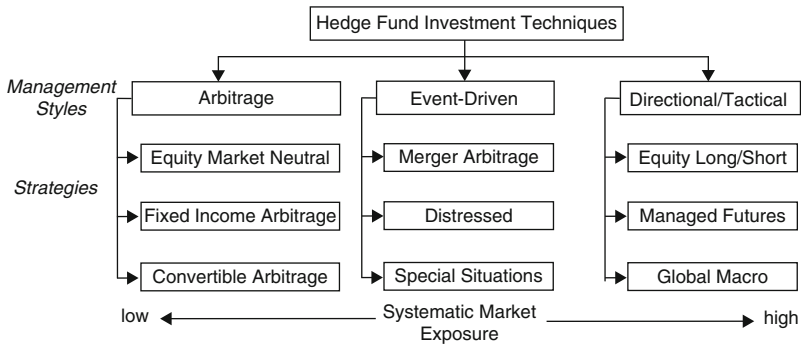


Figure 5.1 Hedge fund styles and strategies

Source: Adapted from Ineichen.¹⁶

market neutral' strategy takes advantage of short-term pricing inefficiencies between stocks or groups of stocks, which usually behave identically. In applying this mean-reverting strategy, the directional market risk is eliminated or neutralised as far as possible so that the beta of the overall situation is ideally zero. Capocci's¹⁷ analysis comes to the conclusion that most of the market neutral funds are not significantly exposed to the equity market, but tend to be more exposed during bear market than during bull market without being negatively impacted.

The strategy 'convertible arbitrage' exploits the differences in the value between a convertible bond and the underlying stock. It does this by buying (selling) the underrated (overrated) convertible bond and, at the same time, also selling (buying) the stock. Agarwal *et al.*¹⁸ show that the key risk factors in convertible arbitrage strategies are: equity (and volatility) risk, credit risk and interest rate risk. They also demonstrate that the risk-adjusted returns of convertible arbitrage hedge funds are affected by mismatches between supply of and demand for convertible bonds. They also show that convertible arbitrageurs escape some of the losses experienced by long-only convertible bond mutual funds by changing their risk exposures in response to the Long-Term Capital Management (LTCM) crisis.

'Fixed income arbitrage' funds search for false valuations or anomalies of bonds in order to achieve arbitrage profits in securities of different maturities, credit ratings and volatilities (with high leverage factors). The overall situation should have a duration of zero, that is, the interest rate risk should be neutralised (immunisation). The advantage of this strategy is the attainment of liquidity and credit risk premiums. Leverage is also applied to increase absolute returns.¹⁹ Fung and Hsieh,²⁰ as well as Jaeger and Wagner²¹ show that the fixed income arbitrage strategy bears a risk profile similar to

a short option, with a risk of significant losses but otherwise steady returns. They also demonstrated that the heaviest losses of fixed income arbitrage occur in 'flight to quality' scenarios, when credit spreads suddenly widen, liquidity evaporates and emerging markets fall sharply. In their analysis of the risk and return characteristics of fixed income arbitrage strategies, Duarte *et al.*²² find that these tend to produce significant alphas after controlling for traditional bond and equity market risk factors. Furthermore, even after taking into account the typical hedge fund fees, these alphas remain significant, and some fixed income arbitrage strategies actually produce positively skewed returns.

Event-driven management style

The goal of the 'event-driven' strategy is to take advantage of price anomalies triggered by pending or upcoming firm transactions such as mergers, restructurings, liquidations and insolvencies. The success of this strategy comes from a false judgement of the situation and the uncertainty of other investors in the case of take-overs, reorganisations or management buyouts. 'Merger arbitrage' invests simultaneously in long and short positions by purchasing the stocks of the company being taken over and at the same time, selling those of the take-over company. Generally, stocks that are the object of a take-over gain in value while stocks of the take-over company fall in value. According to Mitchell and Pulvino,²³ merger arbitrage strategies display rather high correlations to the equity markets when the latter declines and, in turn, display low correlations when stocks trade up or sideways. In other words, the payout profile of merger arbitrage strategies corresponds directly to a short position in a put option on announced merger deals.

The sub-strategy 'distressed securities' invests in assets of companies that have financial or operational problems, are bankrupt or are expecting such an economic situation. Such a situation means reorganisation, insolvencies, liquidations and/or other restructurings of companies. A typical strategy is to buy the stocks with a realised backwardation that results from the tight situation. The stocks are then kept until the process of restructuring is completed and the value of the company has appreciated. Depending on the style of the fund manager, investments are made in bank or corporate loans, pecuniary claims, equity shares, preference shares and warrants. The risk factors of distressed securities strategies come with a simple set of exposures to credit, equity (particularly small cap equity) and liquidity risks. Jaeger and Wagner²¹ show that with an alpha between 3 and 4 per cent per annum, distressed securities funds and its peers in the event-driven discipline offer the highest level of alpha in the hedge fund industry.

The success of the 'special situations' strategy results from the ability to correctly judge and take advantage of a broad range of corporate events such as spin-offs, index reshufflings, insolvency reconstruction and share buy-backs. This strategy also includes closed-end fund arbitrage, which involves

the purchase and hedging of closed-end funds that may be trading at significantly different levels of their net asset values (NAVs).

Directional/tactical management style

The directional management styles, which are characterised by significant market exposure (long and short), contain the strategy indices equity hedge, commodity trading advisors (CTAs)/managed futures and global macro. 'Equity hedge' or 'long/short' equity funds are basically comprised of long positions in stocks that are hedged at any time through tactical short selling and/or stock index options. Most equity hedge funds have a long bias (Kat and Lu¹³). Besides stocks, equity hedge funds can also be invested in other assets on a limited scale. According to Fung and Hsieh²⁴ as well as Jaeger and Wagner,²¹ most of the equity hedge managers have exposure to both the broad equity market and, in particular, to small cap stocks. They argue that it may be easier for equity hedge managers to find opportunities in a rising market, and to go short in large cap stocks and buy small ones instead.

'Managed futures' funds take long and short positions in liquid and listed futures contracts, for example, foreign exchange, interest rates, stock market indices and commodities. Most of the CTAs pursue a trend following strategy in which a trend is replicated to where one can earn from increasing as well as decreasing prices. The results of Schneeweis and Spurgin²⁵ indicate that the technical trading rule and market momentum variables explain a significant part of the sources of managed futures funds returns. According to Jaeger and Wagner,²¹ 'managed futures' hedge fund strategy is the only one that displays a negative alpha. Fung and Hsieh²⁶ show that the returns of trend following funds during extreme market shifts can be explained by a combination of primitive trend following strategies on currencies, commodities, three-month interest rates and US bonds, but not by strategies on stock indices. Kat²⁷ shows that when investors account for 'managed futures' in conventional portfolios, this allows them to achieve a very substantial degree of overall risk reduction at limited costs. The author concludes that adding managed futures to a portfolio of stocks and bonds reduces a portfolio's standard deviation more efficiently than other hedge funds strategies do, without the undesirable side-effects of skewness and kurtosis. In relevant literature, for example, Schneeweis and Spurgin,²⁸ Schneeweis,²⁹ Ineichen,¹⁶ Kat,²⁷ Lamm,³⁰ considered managed futures or CTAs as a separate alternative investment class rather than as a hedge fund strategy. However, as some index providers, for example, Barclay, CS/Tremont, EurekaHedge or S&P, also publish a managed futures strategy index, this strategy is also considered in our analysis as well.

The strategy of 'global macro' is based on macroeconomic top-down analysis, where fund managers try to profit from major economic trends and events in the global economy. Typically, large shifts in interest rates, currency, bonds and equity prices are quickly utilised by making extensive use

of leverage and derivatives (Kat and Lu¹³). Jaeger and Wagner²¹ show that global macro managers do better in strong bond markets, have exposure to the risk characteristic of managed futures and also have some non- linear exposure to the broad equity market.

Conventional VaR and the CF Expansion

The VaR is a downside risk measurement used widely by financial institutions for internal and external purposes, with the attractive characteristics of expressing the risk in only one figure. It is the estimated loss of an asset that within a given period, normally 1 or 10 days, will only be exceeded by a certain small probability θ (mainly 1 or 5 per cent). Thus, with a probability of 95 per cent, the one day 5 per cent VaR shows the negative return that will not be exceeded within this day:

$$\text{prob}[\text{return}_t < -\text{VaR}_t | \Omega_t] = \theta \quad (1)$$

where Ω_t denotes the information set available at time t .

Statistically speaking, the 5 per cent quantile of the probability density function of the asset returns is considered. Assuming that the returns are normally distributed under a foregoing observation period of one year, the VaR can be calculated as the deviation of the value from the distribution function η times the standard deviation σ from its mean μ :

$$\text{VaR} = -(\eta \cdot \sigma - \mu) \quad (2)$$

In general, a higher probability value and a longer period increase the VaR. Hence, in this paper, a period of one day and a probability of 95 per cent are assumed.

The VaR is based on the assumption of normally distributed returns. However, financial asset and particularly hedge fund returns often feature excess kurtosis and negative skewness, which make the probability of extreme returns more likely than under a normal distribution. Thus, applying the VaR under the assumption of normality can lead to a systematic underestimation of the actual risk. Favre and Galeano⁶ apply a CF expansion by adjusting the critical value according to the distribution function η :

$$\eta_{CF} = \eta + \frac{1}{6}(\eta^2 - 1)S + \frac{1}{24}(\eta^3 - 3\eta)K - \frac{1}{36}(2\eta^3 - 5\eta)S^2 \quad (3)$$

with S and K as the skewness and the excess kurtosis of the empirical distribution, respectively. If the return distribution is normal, that is, S and K are equal to zero, then $\eta_{CF} = \eta$ according to equation (2). In contrast, for

non-normally distributed hedge fund returns, it can be expected that the VaR threshold obtained from CF expansion will be more accurate in comparison with the estimated threshold from conventional VaR.

Models of conditional price volatility

Generally speaking, a volatility model has to be able to forecast the variance of a time series of returns preferably well. Thus, to forecast the quantiles for VaR applications, the predictability of volatility is required. Engle and Patton¹² mention common stylised facts of asset price volatility process that should also hold for hedge fund prices. Many of these stylised facts can be viewed as properties of the volatility forecasts, and can be taken as a starting point when examining its consistency with the time series data.

If current returns or current volatility shocks have an influence on the expected variance several periods in the future, then the volatility process of returns is called *persistent*. This property is based on the observation of volatility clustering in the price process whereupon clusters of high and low price changes occur over time. Mandelbrot³¹ and Fama³² empirically conclude that large stock price changes are often followed by large price variation, and there are periods in which one can observe consecutive small changes.

Volatility is referred to as mean reverting (or stationary) if, after a period of high or low volatility, a reversion to a normal level of risk eventually occurs over time. Mean reversion of volatility is thus regarded as the normal level of volatility that is always reached after short-run deviations. In this way, the conditional variance fluctuates around the unconditional volatility depending on positive and negative differences.

Many volatility models assume that the variance of asset prices is influenced asymmetrically by positive and negative innovations. Hence, it is very unlikely that positive and negative shocks have the same effect on the conditional volatility of assets or hedge fund returns. The influence of the sign of the price innovation on volatility, and the negative correlation between asset returns and changes in volatility, is called leverage effect or risk premium effect. The leverage effect describes the circumstance when decreasing stock prices increase the debt to equity ratio which, in turn, results in a higher volatility of returns for shareholders. The risk premium effect suggests that increased volatility caused by news reduces the demand for the stock because of risk-averse market participants (Engle and Patton¹²). Besides purely endogenous effects, which are normally considered in univariate volatility models by only using information from the history of a time series, other exogenous factors (like interest rates, exchange rates, etc.) and/or deterministic events (like announcements by companies, economic news, time-of-day effects, etc.) can be relevant for the volatility of a time series (Bollerslev and Melvin,³³ Engle *et al.*,³⁴ Engle and Mezrich³⁵).

GARCH(p, q) model

The conditional mean μ_t and the conditional variance h_t of a time series of returns R_t are defined as the mean and the variance of a random variable with their expectation influenced by the knowledge of another random variable:

$$\begin{aligned}\mu_t &= E[r_t | \Omega_{t-1}] \text{ and} \\ h_t &= E[(r_t - \mu_t)^2 | \Omega_{t-1}]\end{aligned}\quad (4)$$

with μ_t being the mean of r_t given the information from the past period $t-1$, Ω_{t-1} .

This means that R_t is generated by the following process:

$$R_t = \mu_t + \sqrt{h_t} \varepsilon_t,$$

with:

$$E[\varepsilon_t | \Omega_{t-1}] = 0$$

and

$$\text{Var}[\varepsilon_t | \Omega_{t-1}] = 1$$

A GARCH(p, q) model assumes that the returns R_t in the mean equation is composed of the conditional mean μ and the price innovation ε_t .³⁶

$$R_t = \mu + \varepsilon_t,$$

with

$$\varepsilon_t | \Omega_{t-1} \sim N(0, h_t) \quad (6)$$

However, to account for the serial dependence structure in the hedge fund returns, which are often caused by infrequent trading, the mean equation is modelled by an ARMA process:

$$R_t = \mu_t + \sum_{k=1}^m \phi_k r_{t-k} + \sum_{l=1}^n \theta_l \varepsilon_{t-l} + \varepsilon_t \quad (7)$$

The conditional variance h_t in a GARCH(p, q) model is defined as a function of the past squared error terms ε_{t-j}^2 and the conditional variance of past periods h_{t-i} .³⁷

$$h_t = \omega + \sum_{j=1}^q \alpha_j \varepsilon_{t-j}^2 + \sum_{i=1}^p \beta_i h_{t-i}, \quad (8)$$

with $\varepsilon_t = R_t - \mu$

The parameters in equation (8) have to meet the following restrictions: $\omega > 0$, α_j , and $\beta_i > 0$ for $i = 1, \dots, p$ and $j = 1, \dots, q$. The introduction of this non-negativity condition is necessary in order to exclude a negative variance. In addition, to provide stationarity the following condition must hold:³⁸

$$\sum_{j=1}^q \alpha_j + \sum_{i=1}^p \beta_i < 1. \quad (9)$$

If the sum of α_j and β_i have values close to one, the volatility is highly persistent, that is, volatility has a very long 'memory'. The condition of stationarity also includes the mean reverting property if the sum of the ARCH and GARCH term is significantly less than one. Thus, it appears that although the return volatility has quite a long memory, the volatility process nevertheless returns to its mean. One can test for mean reversion by comparing the average (stationary) unconditional mean of the GARCH (p, q) process with the sample estimate of the unconditional variance. The unconditional mean of the GARCH(p, q) process is calculated as the ratio between ω and $(1 - \alpha_j - \beta_i)$.

Previous tests have shown that a lag structure of $p = 1$ and $q = 1$ is already sufficient to attain adequate estimation results. Thus, the choice of parameters in the empirical analyses in the next section are restricted to a GARCH(1,1) model.³⁹ For estimating the parameter vector $\theta = (\omega, \alpha_j, \beta_i)$, the maximum likelihood technique is applied under the assumption that the residuals ε_t follow a conditional Gaussian (normal) distribution.⁴⁰

Asymmetric conditional volatility models

As already mentioned, not only the magnitude but also the sign of an innovation can influence volatility. Hence, a relationship between the volatility of returns and the returns themselves is assumed, to have a negative sign, that is, decreasing asset returns lead to an increasing volatility and vice versa (Engle and Ng⁴¹).

In relevant literature, several approaches are used to model the leverage effect. There are the QGARCH(p, q), Quadratic GARCH and the NGARCH(p, q) model and the Non-linear Asymmetric GARCH, among other GARCH modifications (Sentana,⁴² Engle and Ng⁴¹). The EGARCH(p, q) model, developed by Nelson,⁴³ assumes normally distributed residuals and can be expressed as:

$$\begin{aligned} \log(h_t) = & \omega + \sum_{i=1}^p \beta_i \log(h_{t-i}) \\ & + \sum_{j=1}^q \alpha_j \left(\left| \frac{\varepsilon_{t-j}}{\sqrt{h_{t-j}}} - \sqrt{\frac{2}{\pi}} \right| + \gamma_j \frac{\varepsilon_{t-j}}{\sqrt{h_{t-j}}} \right) \end{aligned} \quad (10)$$

Unlike the linear GARCH(p, q) model in equation (8), the conditional variance in the EGARCH model is formulated in logarithms. In doing so, all restrictions become irrelevant, particularly the non-negativity condition for the parameters of the conditional variance.⁴⁴ On the one hand, the unconditional standard deviation is considered by the parameter a_j , that includes the effects of price innovations j periods in the past, and on the other by the term $\gamma_j a_j$ that considers the effect of the sign. In a model comparison where a number of different ARCH, GARCH, EGARCH and other more complex semi-parametric and non-parametric models were compared, Pagan and Schwert⁴⁵ could provide empirical evidence that a simple EGARCH(1,1) model is already a sufficient parameterisation to adequately model the dynamics of price innovations. As a result of the modelling of volatility clustering, positive parameter values should result for a_1 and, in order to take the leverage effect into account, negative values result for γ_1 .

GARCH-type VaR

Although the CF expansion includes the deviations from a normal distribution, it ignores the volatility clustering of the returns. However, as ARCH effects establish the fact that one bad day with highly negative returns makes a consecutive bad day more likely than without ARCH effects, the unconditional standard deviation s is replaced by the estimated conditional standard deviation of the GARCH process:

$$VaR_t = -(\eta \cdot \sqrt{h_t} - \mu_t) \quad (11)$$

Empirical analysis of stochastic volatility

Data

For the empirical analysis of the stochastic volatilities, daily data from the S&P hedge fund index series (SPHG) are utilised, as this is the hedge fund database with the longest daily track record, starting in September 2002. However, the use of daily indices has several drawbacks. Firstly, all available daily hedge fund indices are based on managed account platforms. In their empirical analysis on hedge funds based on managed accounts, Haberfelner *et al.*⁴⁶ show that the managed account composite with 0.37 exhibits a significant lower Sharpe ratio than the Eurekahedge composite (1.08). Moreover, investors obtain a premium of 8.19 per cent per annum for not going the managed account way. Thus, one can conclude that the SPHG benchmark for the hedge fund universe suffers from selection bias as managers of account platform have strict requirements in terms of transparency, liquidity and investability. Secondly, from the investor's perspective, 'traditional' hedge fund indices also suffer from survivorship, selection and self-selection biases. However, it is important to mention that these are non-investable hedge fund indices. Fung and Hsieh⁴⁷ argue that fund of hedge

fund data are less prone to these data biases than non-investable hedge fund indices. Drawing on conclusions from Fung and Hsieh⁴⁸, we are of the opinion that the investable hedge fund indices used in this study are good proxies for the advancement of diversified hedge fund portfolios.

Descriptive statistics

The observation period ranges from 30th September, 2002 until 31st May 2006. During this period, 926 return observations resulted, whereby the last month serves as an out-of-sample forecast period. By using daily data, a sufficiently large sample is available that makes the existence of ARCH effects, and the occurrence of volatility clustering, in particular, a requirement for estimating GARCH models.

Denoting P_t as the price or the NAV of a particular hedge fund style at the time t , the continuous return for the period $t-1$ until t is calculated as follows:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \cdot 100 \quad (12)$$

Table 5.1 gives an overview of the return and risk characteristics of the various style indices denominated in US dollar. Thereby, the mean as well as the standard deviation are provided as annualised values. As can be seen, the performance of the particular strategies in the observed period is very heterogeneous and altogether not favourably compared to the other financial assets shown in the lower part of the table. For instance, the style event-driven has the highest annual return with 9.20 per cent, followed by the directional style with 7.01 per cent, while the style arbitrage (2.79 per cent) and the strategy managed futures (4.73 per cent) exhibit the lowest total returns. None of the hedge fund indices exceed the returns of the other financial assets with the exception of bonds.⁴⁹

In reference to the second moment, the hedge fund composite and the three style indices exhibit the lowest volatilities, whereas managed futures simultaneously display extraordinary variability and low annualised returns. However, using only volatility as a risk measurement can only be reasonable if it can be assumed that the observed returns are normally distributed. According to the test statistics of the Jarque–Bera normality test, the null hypothesis of normally distributed returns can only be accepted for the index of small capitalised companies. In contrast, the evidence of leptokurtosis or positive excess kurtosis of hedge fund returns supports the existence of ARCH effects in these time series. Most of the other financial assets are also leptokurtically distributed, which is a typical characteristic of daily equity returns. Furthermore, the composite, directional and managed futures indices are negatively skewed.

Table 5.1 Descriptive statistics, 09/30/2002–04/28/2006

Asset	Indices	Mean $\bar{r}_{ann.}$ (in %)	Volatility $\sigma_{ann.}$ (in %)	Skewness	Kurtosis	Jarque–Bera test
Hedge fund strategies	SPHG composite	6.396	2.076	-0.262	3.965	45.456**
	SPHG arbitrage	2.791	3.272	0.109	3.836	28.101***
	SPHG event driven	9.204	2.090	0.086	5.402	218.39***
	SPHG directional	7.013	4.767	-0.242	4.097	54.146***
	SPHG managed futures	4.733	16.193	-0.380	4.258	75.538***
Financial assets	S&P 500	14.906	14.207	0.332	5.284	213.10***
	NASDAQ composite	16.904	20.789	0.239	4.800	130.66***
	Wilshire growth	12.799	14.062	0.269	4.299	74.456***
	Wilshire value	19.092	14.946	0.236	5.785	300.59***
	Wilshire small cap	23.077	16.688	-0.104	3.198	3.115*
	Wilshire large cap	15.405	13.941	0.312	5.178	193.43***
	J.P. Morgan bond index	2.030	4.810	-0.276	4.550	105.384***
	GS commodity index	18.084	23.661	0.081	3.403	7.087**

Notes: Based on daily continuously compounded returns for 904 observations. ***, **, * for significance at 99, 95 and 90% confidence levels (rejection of the normal distribution); managed futures contains 839 observations covering the period from 01/02/2003 to 04/28/2006.

As can be seen in Table 5.2, the returns of the various styles of the SPHG hedge fund indices do not show high autocorrelation coefficients, but some of them are still highly significant. As mentioned, the marking-to-market problem emerges in cases of positive autocorrelation, where the true standard deviation is underestimated.

In real estate literature, several desmoothing techniques exist in order to adjust data series from positive autocorrelation caused by infrequent trading.⁵⁰ In this paper, autocorrelation is utilised to model the return- generating process in the mean equation of the GARCH models. In addition, with first-order negative correlation coefficients as they appear for arbitrage and managed futures, the standard deviations of the unsmoothed time series would decrease. Table 5.3 shows the estimated ARMA processes for the composite, arbitrage and directional index. As event-driven and managed futures do not exhibit serial correlation, the mean equation simply consists of a constant term.

Finally, autocorrelation diagnostic tests for the residuals were computed to check the adequacy of the estimated ARMA models. In addition, the Ljung–Box test of squared residuals and the LM test control for ARCH effects exist only for the arbitrage and event-driven styles. In this context, notice that tests of volatility clustering are of low power, that is, they allow for type-II error. However, as mentioned earlier, the existence of fat tails permits the estimation of ARCH/ GARCH models. Another test of ARCH effects to examine volatility clustering consists of the Ljung–Box test for squared returns, which is shown in Table 5.4. With the exception of composite and managed futures, the strategies thus have dependencies in the squared return- time

Table 5.2 Autocorrelations

Strategy indices	Lag(1)	Lag(2)	Lag(3)	Lag(4)	Lag(5)	Lag(10)	Lag(15)	Lag(20)
<i>SPHG composite</i>								
ACF	0.071**	0.049	0.041	0.034	0.051	0.018	0.024	-0.030
PACF	0.071**	0.044	0.035	0.027	0.044	0.015	0.002	-0.050
<i>SPHG arbitrage</i>								
ACF	-0.189**	-0.107**	0.010	-0.001	0.046	0.026	0.047	0.020
PACF	-0.189**	-0.148**	-0.044	-0.025	0.039	0.026	0.065	0.020
<i>SPHG event driven</i>								
ACF	0.056	0.016	0.058	0.027	0.043	0.098	0.098**	0.039
PACF	0.056	0.013	0.057	0.021	0.039	0.042	0.089**	0.022
<i>SPHG directional</i>								
ACF	0.084**	-0.021	-0.011	0.000	-0.006	-0.013	-0.043	-0.028
PACF	0.084**	-0.028	-0.007	0.001	-0.007	-0.006	-0.061**	-0.025
<i>SPHG managed fut.</i>								
ACF	-0.023	-0.036	-0.003	0.064	-0.057	0.005	-0.024	-0.026
PACF	-0.023	-0.037	-0.005	0.063	-0.055	-0.003	-0.024	-0.033

Table 5.3 ARMA-modelling

Indices	SPHG composite	SPHG arbitrage	SPHG event driven	SPHG directional	SPHG managed fut
Model	ARMA (1,1)	ARMA (1,2)	—	ARMA (1,2)	—
$\hat{\mu}$	0.028*** (0.007)	0.012** (0.005)	0.037*** (0.004)	0.030*** (0.010)	0.019 (0.035)
$\hat{\phi}_1$	0.947*** (0.034)	-0.222*** (0.033)	—	0.879*** (0.153)	—
$\hat{\theta}_1$	-0.913*** (0.044)	—	—	-0.794*** (0.156)	—
$\hat{\theta}_2$	—	-0.155*** (0.034)	—	-0.082** (0.035)	—
AIC	-1.233	-0.378	—	0.438	—
SIC	-1.217	-0.359	—	0.460	—
Q(5)	0.598	3.269	8.480	0.270	7.828
Q ² (5)	4.803	46.940***	14.278**	4.080	1.241
LM-test	1.068	9.461***	3.393*	2.161	0.033

Notes: Based on daily continuously compounded returns for 904 observations; managed futures 839 observations 01/02/2003 to 04/28/2006; standard errors are presented in parenthesis; ***, **, * for significance at 99%, 95% and 90% confidence levels; Q(5) and Q²(5) are the Ljung-Box statistics for residuals and squared residuals at lag five.

Table 5.4 Test for ARCH effects—squared returns

Indices	$\hat{\rho}_1$	$\hat{\rho}_2$	$\hat{\rho}_3$	$\hat{\rho}_4$	$\hat{\rho}_5$	$\hat{\rho}_{10}$	$\hat{\rho}_{15}$	LB(5)	LB(10)	LB(15)
SPHG	0.007	0.039	0.007	-0.010	0.030	0.038	0.042	2.371	7.692	19.255
composite										
SPHG	0.174	0.077	0.072	0.067	0.157	0.113	0.100	64.22***	93.61***	108.9***
arbitrage										
SPHG	0.045	0.080	0.009	0.050	0.007	0.019	0.045	10.09*	20.80**	23.62**
event-driven										
SPHG	0.046	-0.017	0.049	0.010	0.001	0.034	0.049	4.509	16.55*	31.74***
directional										
SPHG	0.004	0.003	0.029	0.026	0.007	0.030	0.006	1.331	3.845	11.72
managed fut.										

Notes: ***, **, * for significance of Ljung–Box statistics at 99, 95 and 90% confidence level of squared daily returns; LB denotes the Ljung–Box test statistics at lag 5, 10 and 15, respectively.

series. This suggests the existence of a time- dependent conditional variance. To analyse the stochastic properties of those return-time series, a symmetric as well as an asymmetric volatility model is applied in the next section.

Parameter estimation

Since volatility clustering is a specific characteristic of speculative prices, the conditional second moment is modelled using a variation of the GARCH process originally suggested by Bollerslev³⁷. By estimating these conditional volatility models for the particular hedge fund style indices, it is possible, in the second step, to check the consistency of certain properties of volatility that are again important for the forecasting of volatility.

The ARMA–GARCH results are given in Table 5.5. Approximate maximum likelihood estimates of the parameters in the model were obtained by using the Berndt *et al.*⁵¹ algorithm. Standard errors are computed using the robust method of Bollerslev–Wooldridge. The estimation results show $\hat{\alpha}$ that the coefficients are significant at the 5 per cent level with the exception of the directional style, that is, ARCH effects in the return-time series of the hedge funds exist. The parameter restrictions are fulfilled for all hedge fund indices; even though the volatility of returns has a long memory, the volatility processes are still mean-reverting ones. For the directional strategy and the managed futures index, the persistence of the volatility, measured as the sum of $\hat{\alpha}$ and $\hat{\beta}$, is much higher compared to the other hedge fund indices. An intuitive approach for measuring the persistence in volatility processes is the computation of the half-life period (HLP) of a shock on the process, that is, the length until half of the volatility, generated by a price innovation, is decomposed. The HLP, which is computed as $HLP = \log(0.5) / [\log(\hat{\alpha} + \hat{\beta})]$, takes 17 days for the composite index and 25 days for the event driven style, whereby the HLP of 47 and 51 days for managed futures and directional style index are much higher.

Table 5.5 ARMA-GARCH (p,q) model for hedge fund indices

Parameter	SPHG composite	SPHG arbitrage	SPHG event-driven	SPHG directional	SPHG ₂ managed fut ₃ .
<i>Mean equation</i>					
$\hat{c}$	0.030*** (0.008)	0.009** (0.004)	0.038*** (0.004)	0.032*** (0.011)	0.023** (0.009)
$\hat{\phi}_1$	0.951*** (0.026)	-0.180*** (0.035)	—	0.901*** (0.076)	—
$\hat{\theta}_1$	-0.911*** (0.035)	—	—	-0.835*** (0.086)	—
$\hat{\theta}_2$	—	-0.168*** (0.037)	—	-0.060* (0.035)	—
<i>Variance equation</i>					
$\hat{\omega}$	0.001 (0.001)	0.001 (0.001)	0.0001 (0.000)	0.001*** (0.001)	0.001* (0.001)
$\hat{\alpha}$	0.040** (0.016)	0.057*** (0.018)	0.043** (0.019)	0.019* (0.010)	0.023** (0.011)
$\hat{\beta}$	0.920*** (0.039)	0.918*** (0.029)	0.929*** (0.035)	0.968*** (0.012)	0.963*** (0.016)
AIC	-1.247	-0.433	-1.237	0.409	0.418
SIC	-1.215	-0.401	-1.215	0.446	0.439
LogL	568.94	201.41	562.88	-177.50	-184.89
Q(5)	0.518	4.174	11.276**	0.427	4.844
Q ² (5)	1.486	1.146	2.974	2.182	2.507
ARCH-LM test	0.000	0.001	0.023	0.273	0.210
J.B.	39.691***	12.808***	188.41***	23.065***	27.333***
$\hat{\alpha} + \hat{\beta}$	0.960	0.975	0.973	0.987	0.985
HLP	17.151	27.618	25.163	51.368	47.370
$\bar{\sigma}_{ann.}$ (in %)	2.08	3.13	2.12	4.92	4.91

Notes: Based on daily continuously compounded returns for the time period 09/30/2002 to 04/28/2006 (904 observations); ***, ** and * for the significance of the GARCH coefficients at the 99, 95 and 90% confidence level; Bollerslev-Wooldridge robust standard errors are presented in parenthesis.

However, since the sum of the GARCH terms are still smaller than one, all strategies show a mean-reverting behaviour. The property of mean reversion exists if the unconditional variance from the GARCH(1,1) process is close to the estimated unconditional sample variance. The unconditional variance from the GARCH models is computed as the ratio of the constant $\hat{\omega}$ and the expression $(1 - \hat{\alpha} - \hat{\beta})$. For instance, for the arbitrage style an unconditional variance of 0.0391 results which corresponds to an annualised volatility of $\bar{\sigma}_{ann.} = 3.128$ per cent that is thus approximately the same as the estimated sample volatility (3.272) from Table 5.1. Taken together, all unconditional volatilities acquired from the GARCH processes are close to their sample estimators except for the managed futures index. The Ljung-Box test for

Table 5.6 ARMA-EGARCH(p,q) model for hedge fund indices

Parameter	SPHG composite	SPHG arbitrage	SPHG event-driven	SPHG directional	SPHG managed fut. ₂
<i>Mean equation</i>					
$\hat{c}$	0.031** (0.013)	0.008* (0.004)	0.038*** (0.004)	0.034** (0.013)	0.032*** (0.010)
$\hat{\phi}_1$	0.976*** (0.015)	-0.187*** (0.035)	—	0.947*** (0.044)	—
$\hat{\theta}_1$	-0.934*** (0.021)	—	—	-0.873*** (0.056)	—
$\hat{\theta}_2$	—	-0.156*** (0.037)	—	-0.062* (0.035)	—
<i>Variance equation</i>					
$\hat{\omega}$	-0.797*** (0.271)	-0.204*** (0.077)	-0.631** (0.294)	-1.366*** (0.415)	-1.647*** (0.538)
$\hat{a}$	0.142*** (0.052)	0.134*** (0.040)	0.171*** (0.060)	0.140* (0.074)	0.131* (0.078)
$\hat{\gamma}$	-0.110*** (0.038)	-0.009 (0.022)	-0.058 (0.040)	-0.172*** (0.055)	-0.175*** (0.056)
$\hat{\beta}$	0.833*** (0.061)	0.970*** (0.017)	0.876*** (0.065)	0.483*** (0.165)	0.362* (0.217)
AIC	-1.255	-0.434	-1.241	0.423	0.426
SIC	-1.218	-0.397	-1.214	0.465	0.452
LogL	573.71	203.15	565.84	-182.80	-187.46
Q(5)	0.502	4.314	8.832	1.471	5.313
Q ² (5)	1.112	0.833	3.363	2.761	3.393
ARCH-LM test	0.082	0.009	0.158	0.001	0.008
J.B.	20.441***	11.439***	179.70***	24.586***	24.775***

Notes: Based on daily continuously compounded returns for the time period 09/30/2002 to 04/28/2006 (904 observations); ***, ** and * for the significance of the EGARCH coefficients at the 99, 95 and 90% confidence levels; Bollerslev–Wooldridge robust standard errors are presented in parenthesis.

testing on dependencies of the residuals $Q(5)$ and squared residuals $Q^2(5)$ as well as the ARCH–LM(1) test, confirm that no linear or non-linear dependencies exist, and that the models take the heteroscedasticity and the changing unconditional and conditional variance in the return-time series into account. Finally, the Jarque–Bera statistic suggests that skewness and kurtosis in the standardised residuals are reduced but not completely eliminated.

Through assessment of an ARMA–EGARCH(1,1) model, an examination of asymmetric effects on the conditional variance is conducted. The coefficient $\hat{\gamma}$ from Table 5.6 shows that for the arbitrage and event-driven management styles the sign of a price innovation has no influence on the conditional variance, whereas for the composite, directional and managed futures index the coefficients for the negative standardised residuals in t increases the volatility at time $t+1$. Comparing these results with the

descriptive statistic from Table 5.1, one can easily see that asymmetric effects on conditional volatility are absent mainly in those style indices that do not have (significant) negative skewness coefficients.

A comparison between the estimates of conventional and CF-VaR for the five hedge fund indices is shown in Figure 5.2. The underestimation can be seen most clearly for directional and managed futures where negative skewness and excess kurtosis reach relatively high values. In addition, the estimations of GARCH-type VaR can trace the return process much better than the conventional VaR. Furthermore, it suggests that the actual risk compared to the VaR, under the normality assumption, is much higher in down-market phases, as can be seen from the event-driven style at the end of 2005. In contrast, during tranquil periods the normal VaR overestimates the actual risk position as can be seen from the arbitrage style for the year 2004. Thus, the conventional and CF-VaR are quite static, whereas the GARCH-type VaR is more sensitive to changes in the return process.

Conditional variance forecasts

The parameters of the estimated GARCH models, as well as the observations of the squared residuals and the conditional variance, can be used to compute variance forecasts of the hedge fund style returns. Using 28th April, 2006 as the last sample observation, a recursive daily out-of-sample forecast for the next six days is computed. For a GARCH(1,1) model, the forecast for the period $T+1$ is $h_{T+1} = \hat{\omega} + \hat{\alpha}\varepsilon_T^2 + \hat{\beta}h_T$ with ε_T^2 and h_T being the last observation of the squared returns and the conditional variance at the end of the sample period. For the forecasts from $T+2$ to $T+6$, the expected values of the squared residuals and the conditional variance have to be computed according to the following equation:

$$h_{T+2} = \hat{\omega} + \hat{\alpha}E_T[\varepsilon_{T+1}^2] + \hat{\beta}E_T[h_{T+1}] \quad (13)$$

which is denoted for the general case $T+x$ as $h_{T+x} = \hat{\omega} + (\hat{\alpha} + \hat{\beta})h_{T+x-1}$. Thus, only the estimated parameters and the conditional variance of the last forecast period are used for the forecast of the next period. The forecast of the conditional variance of the EGARCH models for the period $T+1$ is calculated as:

$$h_{T+1} = \exp(\hat{\omega}) \cdot \hat{h}_T^{\hat{\beta}} \cdot \exp\left(\hat{\alpha} \left| \frac{\varepsilon_T}{\sqrt{h_T}} + \hat{\gamma} \frac{\varepsilon_T}{\sqrt{h_T}} \right| \right) \quad (14)$$

While the general form for $T+x$ with $x > 1$ is described by:

$$h_{T+x} = \exp(\hat{\omega}) \cdot \hat{h}_{T+x-1}^{\hat{\beta}} \times \left[\exp\left(\frac{(\hat{\gamma} + \hat{\alpha})^2}{2}\right) \Phi(\hat{\gamma} + \hat{\alpha}) + \exp\left(\frac{(\hat{\gamma} - \hat{\alpha})^2}{2}\right) \Phi(\hat{\alpha} - \hat{\gamma}) \right] \quad (15)$$

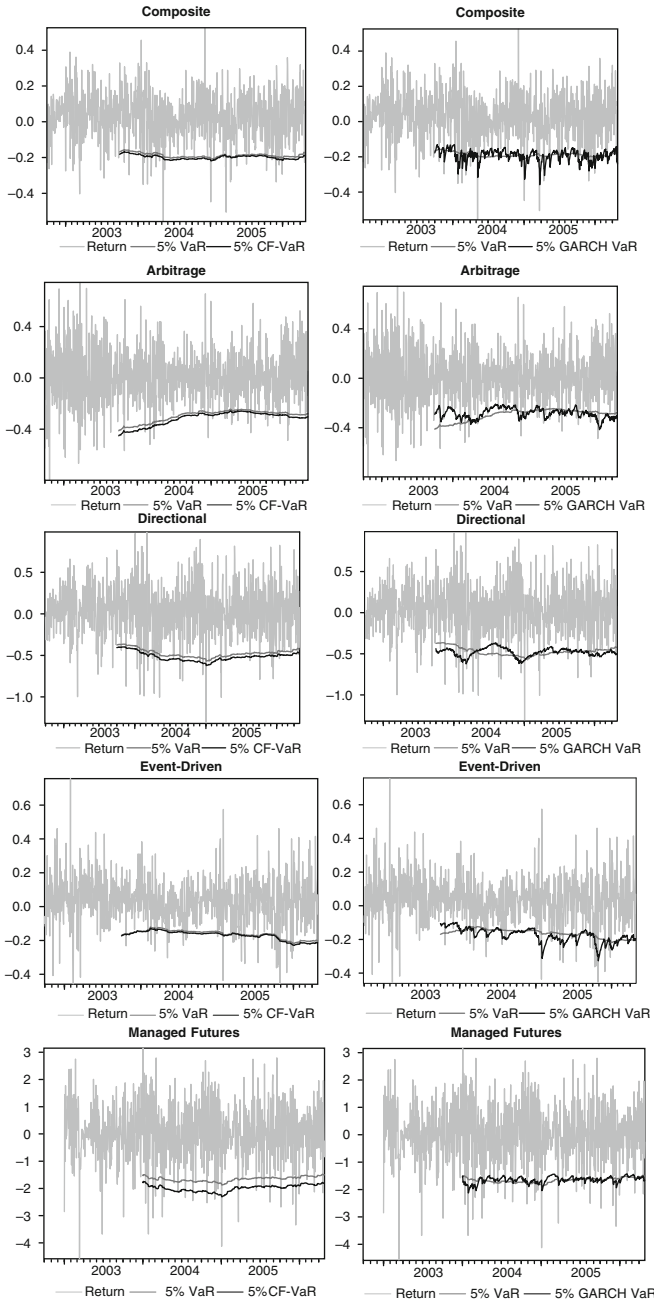


Figure 5.2 Comparison of the different VaR estimations

Notes: The continuously compounded returns start on 10/02/2002, whereas the VaR needs a prior observation period of one year and starts on 10/01/2003. Accordingly, for managed futures, the return series starts on 01/02/2003 and the VaR on 02/01/2004.

with $\Phi(\hat{\gamma} + \hat{a})$ and $\Phi(\hat{a} - \hat{\gamma})$ as the values of the distribution function of the standard normal distribution. The forecasts for arbitrage, event-driven and managed futures using a GARCH(1,1) model and for composite and directional using an EGARCH(1,1) model are shown in Table 5.7. In order to judge the quality of the forecasts, the Theil inequality coefficient (TIC) is also shown. Depending on the value of the conditional variance at the end of the observation date the conditional variance increases or decreases slowly to its unconditional value because of the high $(\alpha + \beta)$ values which are close to one, and because shocks are unexpected and therefore not included in the forecast equation. In this context, only the arbitrage style overestimates the volatility of the next six days, whereas for the other indices, conditional volatility is underestimated in comparison to the unconditional annualised standard deviation. For instance, the forecast volatility of the composite index for the next day is 2.399 per cent, which is almost lower than the estimated sample volatility of 2.076 per cent in Table 5.1.

Furthermore, the values of the TIC suggest that the forecasts for the event-driven and the composite indices can be considered clearly superior to the random walk, whereas the forecasts for arbitrage, managed futures and directional indices are quite close to what a random walk would predict.

Comparison of VaR forecasts

The forecasted conditional variance of the GARCH(1,1) and EGARCH(1,1) models can be implemented in the VaR equation in order to obtain forecasts for the GARCH-type VaR. This requires that the standard deviation σ_t from equation (2) be replaced by the square root of the forecasted conditional GARCH-Variance h_t from equations (7) and (9): $VaR_{T+x} = -\eta \cdot \sqrt{\hat{h}_{T+x}} - \hat{\mu}_{T+x}$. For the conditional variance h_t , a one month out-of-sample forecast from 28th April, 2006 to 31st May, 2006 containing 23 observations is computed. Furthermore, the return process for the five indices are modelled and forecasted using an ARIMA process. The one-step ahead forecast can be calculated as:

$$\hat{r}_{T+1} = \hat{\phi}_0 + \sum_{k=1}^m \hat{\phi}_k r_{T+1-k} + \sum_{l=1}^n \hat{\theta}_l \varepsilon_{T+1-l} \quad (16)$$

Table 5.7 Conditional volatility forecasts

	T+1 (%)	T+2 (%)	T+3 (%)	T+4 (%)	T+5 (%)	T+6 (%)	TIC
SPHG composite	2.399	2.350	2.309	2.277	2.249	2.227	0.8315
SPHG arbitrage	3.025	3.027	3.030	3.032	3.035	3.037	0.9274
SPHG event-driven	2.181	2.179	2.178	2.176	2.175	2.174	0.6924
SPHG directional	4.800	4.841	4.819	4.808	4.803	4.800	0.8979
SPHG managed futures	16.46	16.43	16.41	16.38	16.36	16.34	0.9844

Notes: For a comparison to the unconditional volatility of Table 5.1, the forecast values are represented as annualised conditional volatility; TIC denotes the Theil's inequality coefficient.

with $\hat{\phi}_0$ as a constant and $\hat{\phi}_k$ and $\hat{\theta}_l$ as the estimated AR and MA parameters of order m and n , respectively. From the forecasted returns a mean $\hat{\mu}_t$ and, in the case of the normal VaR, a standard deviation $\hat{\sigma}_t$, as well as the skewness and the excess kurtosis in the case of the CF-VaR, can be obtained. Thus, the observation period of one year can move ahead one month to now include the forecasted month. In Table 5.8, the three different sample VaRs for the five indices are compared with their forecasts. To evaluate the performance of the forecasts, the Theil inequality coefficient (TIC), as well as the heteroscedasticity adjusted mean absolute error (HMAE) and heteroscedasticity adjusted root mean squared error (HRMSE), are applied. A relatively volatile period is chosen for the forecast month in order to observe how the

Table 5.8 One-month value at risk forecasts

Indices	TIC			HMAE			HRMSE		
	VaR	CF-VaR	GARCH-type VaR	VaR	CF-VaR	GARCH-type VaR	VaR	CF-VaR	GARCH-type VaR
SPHG composite	0.0152	0.0382	0.1532	0.0797	0.0657	0.2859	0.0085	0.0065	0.1342
SPHG arbitrage	0.0206	0.0196	0.0478	0.0403	0.0380	0.0728	0.0424	0.0404	0.0975
SPHG event-driven	0.0236	0.0236	0.1093	0.0360	0.0387	0.1236	0.0496	0.0498	0.2350
SPHG directional	0.0429	0.0433	0.1228	0.0699	0.0719	0.2722	0.0899	0.0910	0.2809
SPHG managed futures	0.0262	0.0296	0.0485	0.0393	0.0459	0.0782	0.0539	0.0611	0.0101

Notes: The Theil's inequality coefficient (TIC) is normalised between 0 and 1, where lower values indicate better forecast performances. The TIC is defined as:

$$TIC = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (sample VaR_i - forecasted VaR_i)^2}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (sample VaR_i)^2} + \sqrt{\frac{1}{n} \sum_{i=1}^n (forecasted VaR_i)^2}}$$

The heteroscedasticity adjusted MAE and RMSE are computed as:

$$HMAE = \frac{1}{n} \sum_{i=1}^n \left| 1 - \frac{sample VaR_i}{forecasted VaR_i} \right|$$

and

$$HRMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(1 - \frac{sample VaR_i}{forecasted VaR_i} \right)^2},$$

respectively.

forecasts match the actual changes in the VaR. Nevertheless, the VaR and the CF-VaR change only marginally over this period, which might explain the lower TIC values, particularly for composite, arbitrage and managed futures index. Forecasting a time series, which is more or less stable over time, results inevitably in adequate forecast performance when evaluated by TIC, HMAE and HRMSE. For the GARCH-type VaR, the performance ratios are much higher, especially for the very volatile composite index, and the directional and event-driven styles. This can be traced back to the time-varying property of the GARCH-type VaR models that display the return process much better in comparison with the other alternative VaR approaches.

However, Table 5.8 only exhibits the forecast performances and cannot show which of the three VaR types can actually match the time-varying risk adequately. To answer this question, the number of hits, that is, the number of times the returns fall below the actual VaR, would have to be compared with the forecasted 5 per cent VaR. However, the size of the forecast period (23 observations) is clearly too short if a hit is to be expected 5 per cent of the time. In addition, taking into account that ARIMA and GARCH models can only be used for short-term forecasts, no reliable forecast period with the necessary number of observations can be made. However, one could ask which of the three VaR types generally model the time-varying risk adequately over the whole in-sample period (1st October 2003 to 28th April 2006). From the investor's point of view, the adequate VaR model should indicate how the positions of the hedge fund portfolio should be sized for the best protection against downside risk. This means that investors who include hedge funds in their asset allocation weigh portfolio positions with a low VaR, relative to the VaR of the whole portfolio, more heavily than those with a higher VaR. For this reason the first term of equation (17) does include not only the numbers of hits, $\tilde{n}$, but also their value that is measured by the magnitude of the hit. The more the VaR is exceeded by the hit value, the stronger are the deviations weighted in the adjusted hit ratio. In addition, these squared deviations are weighted by the average value of the VaR position described by the second term:

$$\begin{aligned}
 HIT_{adj.} = & \frac{1}{n} \sum_{i=1}^{\tilde{n}} (HIT\ value_i - VaR_i)^2 \\
 & \times \left| \frac{1}{n} \sum_{i=1}^n VaR_i \right| \quad \forall VaR_i < 0
 \end{aligned} \tag{17}$$

with n as the number of total observations. A lower adjusted hit ratio indicates a better VaR.

As can be seen in Table 5.9, the GARCH-type VaR outperforms the other two VaRs in the case of composite, event-driven and directional strategies. For event driven strategy, one can see the GARCH-type VaR's ability to

Table 5.9 Adjusted Hit ratios of the alternative value at risk approaches

$\times 10^{-3}$	VaR	CF-VaR	GARCH-type VaR
SPHG Composite	0.1447	0.1337	0.1134
SPHG Arbitrage	0.1269	0.0958	0.1491
SPHG Event-driven	0.1216	0.1142	0.0945
SPHG Directional	2.5586	2.2395	2.1815
SPHG Managed futures	85.992	58.960	86.824

follow the return process much better than the other two VaRs, which then leaves the investor better prepared in case of a hit. Furthermore, an investor using the GARCH-type VaR is often warned prior to very negative returns due to a sharp increase in the VaR, whereas the other two VaRs do not show strong reactions. At the same time, the GARCH-type VaR decreases quickly after high volatility periods as can be seen for event-driven and directional strategies. Indeed, as in the case of arbitrage and managed futures strategies, where the GARCH-type VaR is dominated by the other two VaRs, the GARCH-type VaR sometimes tends to be too 'optimistic', decreasing too soon and so leading to hits that do not occur for the other two VaRs. Although the forecast performance for the GARCH-type VaR is not as accurate as that of the other VaRs, it dominates the other two VaRs in terms of the adjusted hit ratio.

Conclusions

In this paper, alternative volatility models such as GARCH(p,q) and EGARCH(p,q) for particular hedge fund strategies were estimated and compared with each other. The examination of the conditional volatility of hedge fund strategy returns shows important differences concerning persistence, mean reversion and asymmetric effects among the strategies considered. The knowledge of the conditional volatility of hedge fund returns can be used to enhance the VaR estimation and forecast. While the normal VaR is generally based on the assumption of normally distributed returns, the CF expansion tends to result in a systematic downward shift in the VaR estimates. Furthermore, both VaR measurements are very inertial, meaning that they do not react immediately to small or large price changes. In contrast, the GARCH-type VaR is able to trace the return process. However, skewness and kurtosis are not completely eliminated by the GARCH modelling and, thus, some bias remains in the VaR estimations and forecasts. Performance ratios indicate an inferior forecast ability compared to the normal and CF-VaR. This is due to the fact that the GARCH-type VaR is more sensitive to changes in the return process. However, by introducing an adjusted hit ratio,

one can demonstrate that the GARCH-type VaR adjusts much better to the time-varying risk. Hence, by allowing such a temporal risk control, GARCH-type VaR offers an enhanced protection against downside risk in a portfolio including hedge funds.

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6

Index Futures Trading, Information and Stock Market Volatility: The Case of Greece

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Practical applications

The introduction of a futures market and, in particular, the impact of futures on stock market volatility is a long debate. Previous studies show that the futures market leads to an increase in market depth and a decrease in volatility. This is due to the more rapid rate at which information is reflected in prices and speculation. Other studies suggest that a decrease in cash market volatility is due to an increase in market liquidity. Empirical studies for UK and US financial markets do not conclude clearly whether the introduction

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of futures stabilises or destabilises the underlying spot market. It is therefore important for practitioners to look at the link between information (news) and volatility. To the authors' knowledge, this is the first study that examines this effect of the Greek futures market on stock market volatility.

This paper examines the effect of futures trading on the volatility of the underlying spot market. It focuses on various techniques to investigate the relationship between information and the volatility of the FTSE/ASE-20 and FTSE/ASE Mid 40 indices in Greece. The results for the FTSE/ASE-20 index suggest that futures trading has led to decreased stock market volatility (negative effect), but the results for the FTSE/ASE Mid 40 index indicate that the introduction of stock index futures has led to increased volatility (positive effect), while the estimations of the unconditional variances indicate lower market volatility after the introduction of stock index futures. Furthermore, the results show that good news has a more rapidly impact on FTSE/ASE-20 stock return volatility. For the FTSE/ASE Mid 40 index, the results suggest that news is reflected in prices more slowly, while old news has a less persistent effect on prices. These findings are helpful to financial managers dealing with Greek stock index futures.

Introduction

The debate on the impact of futures on stock market volatility is still controversial. In other words, the issue of whether the futures markets affect underlying spot markets is not widely accepted. Some past critics of index futures agree that the introduction of stock index futures increases stock market volatility. Others report no significant volatility effect associated with the introduction of stock index futures. In general, the link between futures trading and stock market volatility has become very complex. Several studies have empirically examined the long-term relationship across the decades. Most of them compare the volatility of the spot market before and after the introduction of futures trading, using econometric models. According to Bologna and Cavallo,¹ there are two theories in the literature about the relationship between futures markets and underlying spot markets. The first theory supports the argument that futures trading destabilises the underlying spot market by increasing its volatility. In practice, volatility increases through speculation or arbitrage strategies. Then, an increase in interest rates and the cost of capital, leading to a reduction in the value of investments, is quite possible. From a financial viewpoint, the price volatility usually depends on the arrival of new information in the market. If the market is efficient, the price reflects this new information, but the literature presents arguments that futures markets increase market depth² and reduce spot market volatility. One possible explanation why futures markets reduce the spot market volatility is financial risk.

The majority of papers in the literature consider the effect of futures trading on stock market volatility using either an unconditional measure of

volatility or GARCH models. GARCH models spot and futures market volatility. This paper uses both techniques. In addition, most of papers in the literature look at data from highly developed markets such as the US and UK.

This paper examines two newly introduced Greek stock index futures on the FTSE/ASE-20 and FTSE/ASE Mid 40. It addresses any stabilisation/destabilisation effects and the impact of futures trading on the associated stock market indices. In this respect, the findings are very important in the absence of any previous work examining the effect of futures trading on the Greek spot index markets.

The plan of the paper is as follows. In the second section, the theoretical debate and literature review are presented. The third section outlines the methodology, while the fourth section describes the data. The fifth section presents empirical results from various econometric models, and the final section concludes the paper and summarises the findings.

Theoretical Debate and Literature Review

Futures contracts are attractive to investors and speculators because they have low margin requirements and low transaction costs. The above does not imply, however, that futures are destabilising. Apparently, several theoretical arguments have been used over the years to explain why derivatives markets in general might affect the volatility of the spot market (the underlying asset market). Theoretically, when futures are introduced, one can buy or sell futures, options, commodities or other products. This may influence spot prices, if the producers respond to new information. Various information-based models analyse and explain whether the opening of a futures market may stabilise or destabilise stock prices. This literature includes papers by Stein,³ Subrahmanyam⁴ and Newbery.⁵

In addition, according to Chari *et al.*⁶ futures markets may stabilise or destabilise prices. Since futures and spot prices are usually closely related, futures are attractive to speculators. Antoniou and Holmes⁷ argue that the arrival of futures trading depends on speculators' information. In other words, when speculators have perfect information, the introduction of futures markets stabilises prices. But when speculators have a noisy signal, there is a destabilising effect. In some cases, however, the introduction of a futures contract leads to a more stable spot price.

Since futures markets have a higher degree of leverage than cash markets do, they are more likely to attract uninformed speculative investors and thus destabilise cash markets by increasing volatility. But futures markets increase the overall market depth and informativeness, are important for price discovery, allow the transfer of risk and may actually reduce spot volatility.

Index futures contracts have several advantages over the underlying stock index. These enable holders to trade on accessible credit and reduce costs using several strategies. Index futures have some disadvantages, however,

such as a non-beneficial increase in price variability; see Hodgson and Nicholls.⁸

Several studies examine the relationship and behaviour between stock market volatility and index futures volatility. For instance, Edwards⁹ examines stock market volatility before and after the introduction of futures. He finds that the introduction of the S&P500 futures contract has reduced cash market volatility. Hodgson and Nicholls,⁸ however, examine the effect of index futures on Australian sharemarket volatility, and argue that the introduction of trading in index futures has not affected the long-term volatility of the underlying spot share market. Further, Chang *et al.*¹⁰ point out that futures trading increases spot portfolio volatility (with no volatility spillovers). In addition, Aggarwal¹¹ and Harris¹² find that the post-futures period is more volatile. Harris¹² reports that the volatility of S&P500 stocks increased, relative to the volatility of stocks in a control system. Also, Lockwood and Lim¹³ find that cash stock market volatility increased after stock index futures trading. Also, Maberly *et al.*¹⁴ find that volatility increased in the S&P500 index.

Further, for the New York Stock Exchange (NYSE), Chang *et al.*¹⁵ examine the effects on futures volatility and price changes. Empirical results show that there is an increase in volatility only at the close of the futures market. By contrast, when the NYSE closes, volatility falls significantly. Another interesting point is that higher volatility in the spot market increases the demand for further hedging in stock index futures.¹⁶

Since analysis of the introduction of index futures usually depends on the choice of pre- and post-futures periods,¹⁴ the effect of future trading on volatility can be modelled using GARCH-family models. Specifically, one can examine whether the existence of futures trading has any effect on volatility, using a conditional variance model with a dummy variable.

First, Chan *et al.*¹⁷ examine the intra-day volatility spillovers between S&P500 index stock and futures markets using GARCH models and show a strong cross-market dependence in the volatility process. Then, Antoniou and Foster¹⁸ analyse the effect of futures trading on Brent crude oil spot price volatility (for 1988). From GARCH analysis, they find an increase in informational spot market efficiency and a decrease in the old-news term.

Another approach is by Antoniou and Holmes.⁷ They use GARCH models to examine the relationship between trading, information and volatility in the FTSE 100 stock index futures (using data from November 1980 to October 1991). They argue that it is better to model the volatility following a GARCH (1,1) process as the conditional variance of the error term is a linear function. Hence, they use a GARCH (1,1) process with a dummy variable to investigate the impact of futures trading on spot market volatility. Their results show that the volatility of the spot market with futures is greater than that without futures. For the whole period, the analysis with GARCH (1,1) suggests that there is an increase in spot price volatility on

futures. For the sub-periods, the GARCH parameters are significantly different from zero and, also, for both pre-and post-futures trading there is an increase in the unconditional variance, and therefore an increase in volatility and a definite impact of trading in futures on spot price volatility has been found. Furthermore, Brorsen¹⁹ and Gulen and Mayhew²⁰ suggest that volatility increases in many markets. Gulen and Mayhew²⁰ examine stock market volatility before and after the introduction of equity index futures trading in 25 countries using the GJR-GARCH, the Non-linear GARCH and the EGARCH models. In addition, Holmes²¹ examines the impact of FTSE Eurotrack futures trading on spot volatility using daily closing price indices for the period June 1990–April 1994. His results show that futures contracts have a beneficial impact on price discovery in the spot market, even when traded market is thin. By contrast, Robinson²² provides evidence of a stabilising effect of futures trading activity on the volatility of the FTSE 100 index. Antoniou *et al.*,²³ however, examine further the impact of futures trading on FTSE 100 stock index volatility. Now using the GJR-GARCH (1,1) model with a dummy variable, they conclude that futures trading has a negative impact on the volatility of the stock market. Further, Butterworth²⁴ examines the link between volatility and news for the FTSE Mid 250 index using symmetric and asymmetric GARCH methods. He argues that the new information after the onset of futures is reflected in prices less rapidly and, therefore, an increase in the persistence of volatility is quite possible.

In addition, Rahman²⁵ examines the impact of index futures trading on the volatility of component stocks for the Dow Jones Industrial Average (DJIA). The data cover the period April–June 1997 (pre-futures) and April–June 1998 (post-futures). Rahman²⁵ uses the simple GARCH (1,1) model to estimate the conditional volatility of intra-day returns. The empirical results confirm that there is no change in conditional volatility from pre- and post-futures periods.

Another good representation for volatility modelling is the EGARCH model. Koutmos and Tucker²⁶ use the EGARCH model and show that, for both stock and futures markets, bad news increases volatility more than good news. Further, Pericli and Koutmos²⁷ use an EGARCH model and find that the volatility of the S&P500 index has decreased after the introduction of futures trading (for S&P500). Darrat *et al.*²⁸ examine the volatility and test the relationships between volatilities in the spot and futures markets. The sample spans the period after the stock market crash of October 1987 (ie November 1987–November 1997). They use an EGARCH model to measure spot returns volatility and futures returns volatility. According to their empirical results, ‘futures trading is not responsible for cash market volatility’. Therefore, a possible explanation is that an unstable spot market could affect volatility in the index futures market.

Further, Lien and Tse²⁹ examine the effect and the relationship between cash settlement and cash-futures prices using a bivariate GARCH model with

a dummy variable (for measuring those effects). Their empirical results show that the futures market leads the cash market and that the cash settlement has never had an impact on cash and futures returns (ie the coefficient of the dummy variable is found to be statistically insignificant). In addition, Kyriacou and Sarno³⁰ examine the relationship between UK spot market volatility and futures trading using the GARCH process. The time series spans from 1 October 1992 to 29 December 1995. The empirical results suggest that spot market volatility could be modelled by a GARCH (1,1), and volatility is characterised as statistically reliable. Kyriacou and Sarno³⁰ argue that futures trading affects spot market volatility in 'opposite directions'.

Gulen and Mayhew²⁰ examine the time series behaviour of stock indices for 25 countries, using several GARCH models. As they point out, in the US and Japan, futures trading leads to an increase in conditional volatility. Bologna and Cavallo¹ examine the effect of the introduction of stock index futures, that is whether it reduces stock market volatility (for Italian market). Their empirical results show that the introduction of stock index futures affects the volatility of the spot market. Also, the results from GARCH (1,1) models—for pre-futures and post-futures sub-periods—suggest that the index futures market reduces volatility. In addition, the 'old news' parameter from the presence of stock index futures trading has had a different (less) impact for stock market volatility estimation.

Pilar and Rafael³¹ analyse the effect of the introduction of futures in the Spanish stock market using a GJR model. The results indicate a decrease in volatility after the introduction of futures. Another empirical approach on the effect of futures trading on volatility is that from Figuerola-Ferretti and Gilbert.³² They use Aluminium transaction prices to examine whether futures trading affects volatility. To do so, they use error-correction models and the GARCH (1,1) regression model. In addition, they report the results of a VAR model and display impulse response analysis to track the effects of a shock to each of the volatilities. Their results show that volatility decreases in the post-period.

Chiang and Wang³³ examine the impact of futures trading on Taiwan spot index volatility. This study also discusses the macroeconomic effects on spot volatility and the asymmetric effects of futures trading on spot price volatility behaviour. The data cover the period from 5 January 1995 to 10 March 2000. The asymmetric time-varying GJR volatility model is employed. Their empirical results show that futures trading of Taiwan Index Futures has major impacts on spot price volatility, while the trading of MSCI Taiwan futures has no effects except the asymmetric response behaviour. Bae *et al.*³⁴ examine the effect of the introduction of index futures trading in the Korean markets on spot price volatility. The results show that there is an increase in spot price volatility after the introduction of futures trading. These findings are in line with Ryoo and Smith³⁵ for the Korean market. They find that futures trading increases the speed at which information is reflected

in spot market prices, reduces the persistence of information and increases spot market volatility. Recently, Antoniou *et al.*³⁶ test the hypothesis that the introduction of index futures has increased positive feedback trading in the spot markets of six industrialised nations. Their findings support the view that futures markets help stabilise the underlying spot markets by reducing the impact of feedback traders and attracting more rational investors.

Overall, it is evident that the majority of studies discussed conclude that futures may have either a positive or a negative impact on stock market volatility. Also, a decrease or increase in volatility after the introduction of futures is found. Hence, the debate on the impact of futures trading on stock market volatility does not give clear conclusions.

To analyse the impact of futures trading on Greek stock market volatility, several GARCH models are estimated for both sub-periods, thereby allowing a comparison of volatility before and after the introduction of futures trading. Several GARCH models are used to find which model fits the data best. Also, one needs to know how these GARCH specifications model volatility before and after the introduction of futures trading. Notice that GARCH models link information (news) to volatility.

Methodology

Past empirical papers show that the effect of future trading on volatility can be modelled using conditional variance models. In other words, one can examine whether the existence of futures trading has any effect on volatility by using a (GARCH) volatility model with a dummy variable (taking the value zero pre-futures and one post-futures). If the dummy is statistically significant, the existence of futures trading has an impact on spot market volatility (see Antoniou *et al.*²³).

Many studies analyse the 'futures effect' using the standard GARCH(1,1) model^{1, 24} or the GJR model, which tests for the^{23,24,31} presence of asymmetries.

To analyse the effect of future trading on stock market volatility, GARCH models are also employed. The GARCH(p, q) model captures better the tendency of returns to exhibit volatility clustering, incorporating heteroscedasticity into the estimation procedure. In this model, positive and negative past values have a symmetric effect on the conditional variance. The standard GARCH(1,1) model with dummy variable can be expressed as follows

$$\begin{aligned} R_t &= \mu + \varepsilon_t \\ \sigma_t^2 &= \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + cD_t + u \end{aligned} \quad (1)$$

The dummy variable D_t takes zero value for the pre-futures period and one for the post-futures period. This dummy allows one to determine whether futures prices are related to any change in the spot market volatility. When the coefficient of the dummy variable is positive (negative), there is a positive (negative) effect of futures trading on volatility. In addition, assuming that

markets are efficient, a_1 (the ARCH parameter) can be viewed as a 'news' coefficient, while β_1 (the GARCH parameter) can be viewed as 'old news' and the persistence coefficient (see Antoniou and Holmes⁷). Furthermore, according to Butterworth²⁴, an increase (decrease) in a_1 suggests that news is reflected in prices more rapidly (slow). A reduction in β_1 suggests that old news has a less persistent effect on prices changes. In addition, an increase in β_1 suggests greater persistence. Also, when the sum $a_1 + \beta_1$ approaches unity, volatility shocks are persistent.

Other specifications of the GARCH(p,q) include the exponential GARCH (EGARCH) and threshold GARCH (TGARCH). Both models capture volatility asymmetry.

- *EGARCH model.* The variance equation of Exponential GARCH(1,1) model with a dummy variable is the following

$$\log(\sigma_t^2) = \omega + a_1 \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + a_2 \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + a_3 \log(\sigma_{t-1}^2) + cD_i \quad (2)$$

- *TGARCH(1,1) model.* The variance equation with a dummy variable is given by

$$\sigma_t^2 = \omega + a_1 \varepsilon_{t-1}^2 + a_2 \varepsilon_{t-1}^2 d_{t-1} + a_3 \sigma_{t-1}^2 + cD_i \quad (3)$$

Good news ($\varepsilon_t < 0$) and bad news ($\varepsilon_t > 0$) have an impact on a_1 and $a_1 + a_2$, respectively. In other words, a negative innovation (shock) has a greater impact than a positive innovation on volatility. Also when $a_2 > 0$, the leverage effect exists. When $a_2 \neq 0$ and significant, the news impact is asymmetric.

The methodology is based on the following two parts: First, we test for the impact of futures trading on stock market volatility using the above models with a dummy variable. The next step consists of examining and comparing the volatility parameters across two sub-periods: the pre-futures period and the post-futures period. GARCH models are estimated with the Marquardt algorithm, using the Heteroscedasticity Consistent Covariance option (see also Floros and Vougas³⁷). The conditional mean of the data is also estimated using ARMA(p,q) models (when necessary). Both orders are determined by the AIC. Finally, the unconditional variance is also estimated.

Data Description

Daily closing prices for the FTSE/ASE-20 index are used over the period September 1997–July 2001. For the FTSE/ASE Mid 40 index, the daily closing prices are used over the period December 1999–July 2001. Closing prices for stock indices were obtained from *Datastream*.

Next, the main statistics (mean and standard deviations) of the returns for the sub-periods before and after the introduction of futures trading are reported. Table 6.1 contains information for the FTSE/ASE-20 index, and Table 6.2 for the FTSE/ASE Mid 40 index.

It is clear that daily standard deviations (SD) have changed little. For both periods before and after the introduction of futures trading the SD falls slightly. This means that futures trading may not destabilise the underlying spot market. A more detailed empirical investigation needs to be carried out, however, using GARCH-volatility models.

Empirical Results

FTSE/ASE-20 index

In Table 6.3, all GARCH models with a dummy variable are reported. There is a negative effect of futures trading on stock market volatility. The negative effect is statistically significant and, therefore, there is a decrease in volatility associated with the introduction of futures.

The results presented in Table 6.3 show that the introduction of FTSE/ASE-20 stock index futures has an effect on the volatility of the underlying

Table 6.1 Statistics for FTSE/ASE-20 log

Sample period	N	Mean	SD
Pre-futures	502	0.002084	0.024643
Post-futures	500	-0.001346	0.019884

Table 6.2 Statistics for FTSE/ASE Mid 40 log returns

Sample period	N	Mean	SD
Pre-futures	36	-0.003228	0.037806
Post-futures	393	-0.002826	0.023835

Table 6.3 Results of GARCH models for FTSE/ASE-20 index

Model	Coeff. on dummy	t-ratio
MA(1)-TGARCH	-5.48E-05	-2.0743*
MA(1)-GARCH (1,1)	-5.34E-05	-2.1660*
MA(1)-EGARCH	-0.1070	-1.9461*
GARCH (1,1)	-5.71E-05	-2.2703*
EGARCH	-0.1134	-2.1201*
TGARCH	-6.06E-05	-2.2261*

*Significant at the 5 per cent level.

Table 6.4 FTSE/ASE-20 index

A. Pre-futures period: MA(1)-TGARCH			B. Post-futures period: MA(1)-TGARCH		
Mean equation	Coefficient	t-statistic	Mean equation	Coefficient	t-statistic
μ	0.0017	1.5196	μ	-0.0018	-2.0068*
MA(1)	0.1893	4.1803*	MA(1)	0.1616	3.3140*
Variance equation			Variance equation		
ω	0.0001	2.8118*	ω	7.16E-05	2.9744*
a_1	0.0228	0.4726	a_1	0.1364	1.7603*
a_2	0.3387	2.9968*	a_2	0.1486	0.9729
a_3	0.4990	3.7119*	a_3	0.6114	6.4064*

*Significant at the 5 per cent level.

spot market. This result is in line with the finding of Bologna and Cavallo¹ for the Italian Stock Exchange and Antoniou *et al.*²³ for several stock indices from different countries.

The next step is to examine and compare the values of volatility parameters for the pre-futures and the post-futures periods.

The results from the MA(1)-TGARCH model are presented in Table 6.4. It is clear that some GARCH parameters are statistically significant at the 5 per cent level (with the exception of the ARCH parameter). In the pre-futures period, the news parameter is not statistically significant, while the leverage effect exists. Therefore, negative shocks have a greater impact on volatility. In the post-futures period, however, the leverage effect term is not significant, indicating that there is no asymmetric effect. The leverage effect changes from 0.3387 (pre-futures) to 0.1479 (post-futures). According to Table 6.4, there has been an increase in both ARCH and GARCH parameters. An increase in the ARCH parameter suggests that good news has a greater impact. Also, an increase in the GARCH parameter indicates that old news has a greater persistent effect on price changes. Notice that similar findings arise from the TGARCH model.

The results from other GARCH models are presented in Table 6.5 for the pre-futures and post-futures period (*t*-statistics in the parentheses).

Both GARCH(1,1) and MA(1)-GARCH (1,1) reveal similar results. Specifically, all parameters are non-negative (and statistically significant), indicating that GARCH(1,1) models are well specified³⁸. Thus, there have been significant changes in the volatility structure of the FTSE/ASE-20 spot market after the introduction of futures trading. In addition, the evidence indicates an increase in the ARCH parameter, which suggests that news is reflected in prices more rapidly. Also, a decrease in the GARCH parameter suggests that old news has a less persistent effect on price changes. Therefore, old news will have a lower impact on today's price changes. The sum of the coefficients a_1 and β_1 changes from 0.9015 (pre-futures) to 0.7979 (post-futures) for the

Table 6.5 Estimation results of GARCH models for FTSE/ASE-20 index

A. Pre-futures period Models	ω	a_1	a_2	a_3	β_1
MA(1)-GARCH(1,1)	6.00E-05 (2.4661)*	0.1021 (2.5213)*			0.7994 (12.899)*
MA(1)-EGARCH	-2.7736 (-2.6371)*	0.2530 (2.7142)*	-0.1999 (-3.2260)*	0.6562 (4.7544)*	
GARCH(1,1)	4.63E-05 (2.4088)*	0.0885 (2.4346)*			0.8377 (15.703)*
EGARCH	-3.0397 (-2.7150)*	0.2834 (2.6735)*	-0.1824 (-2.7899)*	0.6226 (4.2267)*	
TGARCH	0.0001 (2.7282)*	0.0477 (0.8196)	0.2816 (2.4640)*	0.5075 (3.7449)*	
B. Post-futures period					
MA(1)-GARCH(1,1)	7.82E-05 (2.6819)*	0.2108 (2.7970)*			0.5871 (5.2739)*
MA(1)-EGARCH	-1.8263 (-2.6472)*	0.3521 (3.1675)*	-0.0769 (-0.9191)	0.8028 (9.7076)*	
GARCH(1,1)	8.34E-05 (2.9061)*	0.2327 (2.9225)*			0.5586 (5.1236)*
EGARCH	-1.8680 (-2.8931)*	0.3813 (3.4953)*	-0.0779 (-1.0286)	0.7999 (10.291)*	
TGARCH	7.62E-05 (3.0339)*	0.1530 (1.9538)*	0.1566 (1.0782)	0.5853 (6.0002)*	

* Significant at the 5 per cent level.

MA(1)-GARCH(1,1), and from 0.9262 (pre-futures) to 0.7913 (post-futures) for the simple GARCH(1,1) model. Hence, the persistence of shocks from the pre-futures period to the post-futures period is reduced, indicating increased market efficiency. This is also confirmed by the reduction of the GARCH parameter (β_1). Notice that, the difference in the sum of the GARCH coefficients is unlikely to be either statistically or economically significant. It is quite unlikely that 0.9015 is statistically significant from 0.9262. In the same line, the difference of the two coefficients is not economically significant. Nevertheless, the sign of MA(1) coefficient may be useful in capturing non-synchronous, infrequent trade in the FTSE-20 index.

From EGARCH(1,1) models, there is an increase in a_1 and a_3 parameters. Also, the leverage effect term in both MA(1)-EGARCH(1,1) and EGARCH(1,1) models is negative. In the pre-futures period, the leverage effect term is statistically different from zero, indicating the existence of leverage in stock returns during the sample period. In the post-futures period, the leverage effect term is not significant.

FTSE/ASE Mid 40 index

Next, estimates of the effect of futures trading on stock market volatility for the FTSE/ASE Mid 40 index are reported. Table 6.6 shows that there is a positive effect of futures trading on stock market volatility based on the AR(3)-TGARCH(1,1) and AR(3)-EGARCH(1,1) models. This effect is statistically significant, and there is an increase in volatility associated with futures introduction. The other GARCH models indicate a decrease in volatility (ie negative effect). This effect is not statistically significant in all cases.

The results from the AR(3)-TGARCH(1,1) model are presented in Table 6.7. For this model, the leverage effect term is positive and statistically significant. So leverage exists, and there is an asymmetric effect to the news. In addition, the a_1 parameter changes from -0.1797 (pre-futures) to 0.0747 (post-futures). Nevertheless, both t ratios indicate that these parameters are insignificant. The GARCH parameter (a_3), however, decreases (from 1.0449 to 0.7761), indicating that old news has a less persistent effect on prices and, therefore, old news will have less impact on today's price changes.

Table 6.6 Results of GARCH models for FTSE/ASE MID 40 index

Model	Coeff. on dummy	t ratio
AR(3)-TGARCH	4.11E-05	1.7904*
AR(3)-GARCH (1,1)	-4.69E-07	-0.0111
AR(3)-EGARCH	0.0461	2.3694*
GARCH (1,1)	-3.83E-05	-0.6211
EGARCH	-0.0133	-0.2739
TGARCH	-1.60E-05	-0.3209

*Significant at the 5 per cent level.

Table 6.7 FTSE/ASE MID 40 index AR(3)-TGARCH: dependent variable R

Mean equation	Coefficient	t-statistic	Variance equation		
A. Pre-futures period					
μ	-0.0110	-4.6507*	ω	-1.53E-05	-0.9185
AR(1)	0.1501	1.1929	a_1	-0.1797	-1.0374
AR(2)	-0.3403	-2.7025*	a_2	0.5088	2.2920*
AR(3)	-0.0972	-0.9300	a_3	1.0449	6.2609*
B. Post-futures period					
μ	-0.0039	-3.4578*	ω	4.49E-05	1.9556*
AR(1)	0.1804	3.4773*	a_1	0.0747	1.3070
AR(2)	-0.1454	-2.4790*	a_2	0.1655	1.6189
AR(3)	0.0709	1.1719	a_3	0.7761	10.572*

*Significant at the 5 per cent level.

For the TGARCH model, leverage is positive but not statistically significant. Also, both ARCH and GARCH parameter coefficients fall. This means good news has a relatively slow impact with low persistence (there is a decrease in a_3). Table 6.8 reports the estimation results of various GARCH models before and after the introduction of stock index futures.

For GARCH models, all parameters are non-negative, indicating the superiority of the GARCH(1,1) models. In both AR(3)-GARCH(1,1) and GARCH(1,1), there is a decrease in the ARCH parameter, which suggests that news is being reflected in prices slowly. The ARCH coefficient parameters are statistically significant, but estimates of the GARCH parameters differ. For AR(3)-GARCH(1,1), there is increase in the GARCH parameter (from 0.7050 to 0.7089), which suggests that old news has a greater persistent effect on price changes. For the simple GARCH(1,1) model, the reduction of the GARCH parameter suggests that old news has a less persistent effect, and that old news has a lower impact on today's price changes. Also, as for the FTSE/ASE-20 index, the sum $\alpha_1 + \beta_1$ reduces from 1.0116 (pre-futures) to 0.8756 (post-futures) for the AR(3)-GARCH(1,1) and from 1.0232 (pre-futures) to 0.8766 (post-futures) for the simple GARCH(1,1) model. Therefore, there is an increase in market efficiency. In all cases, the coefficient parameters are statistically significant.

Finally, EGARCH models give interesting information about the futures trading effect. In particular, for AR(3)-EGARCH, the leverage effect term changes from 0.4484 to -0.1033. For the simple EGARCH model, the leverage effect term changes from -0.3723 to -0.0948. Notice that in the pre-futures period, $a_2 < 0$ and significant, indicating that a leverage effect exists in stock returns during the sample period. Hence, a negative shock increases the conditional variance.

Table 6.8 Estimation results of GARCH models for FTSE/ASE MID 40 index

Models	ω	a_1	a_2	a_3	β_1
A. Pre-futures period					
AR(3)-GARCH(1,1)	-1.57E-05 (-0.3699)	0.3065 (1.8786)*			0.7050 (6.1171)*
AR(3)-EGARCH	-9.2383 (-2.8020)*	-0.5784 (-1.2277)	0.4484 (1.8194)*	-0.4212 (-0.8926)	
GARCH(1,1)	-4.14E-05 (-0.7668)	0.2827 (2.0505)*			0.7404 (6.1231)*
EGARCH	0.5160 (1.5135)	-0.2515 (-0.5529)	-0.3723 (-3.421512)*	1.0355 (121.33)*	
TGARCH	-6.55E-05 (-1.0970)	0.1439 (1.3037)	0.2010 (0.7889)	0.8024 (5.9325)*	
B. Post-futures period					
AR(3)-GARCH(1,1)	7.03E-05 (2.0132)*	0.1667 (2.7657)*			0.7089 (6.7465)*
AR(3)-EGARCH	-0.4974 (-2.4167)*	0.2026 (3.2929)*	-0.1033 (-1.8453)*	0.9542 (39.541)*	
GARCH(1,1)	7.41E-05 (2.1626)*	0.1884 (2.9855)*			0.6882 (6.8682)*
EGARCH	-0.9229 (-2.5212)*	0.2936 (3.4892)*	-0.0948 (-1.5247)	0.9066 (20.884)*	
TGARCH	5.37E-05 (2.0751)*	0.1009 (1.7309)*	0.1609 (1.5108)	0.7401 (9.3872)*	

* Significant at the 5 percent level.

Unconditional variance

In most GARCH (1,1) models, the ARCH and GARCH parameters are non-negative. Also since the sum $a_1 + 1$ for the GARCH (1,1) model is less than one, the models have finite unconditional variances (see also Bologna and Cavallo¹). The unconditional variance (σ^2) has the form

$$\sigma^2 = \frac{\omega}{1 - a_1 - \beta_1} \quad (4)$$

Comparing parameters across the two sub-periods, it is found that for the FTSE/ASE-20 contract there is an increase in the ARCH parameter, while the GARCH parameter decreases. Now, for the MA(1)-GARCH(1,1) model, the unconditional variance is equal to 0.000609651 for the pre-futures period and 0.000387044 for the post-futures period. In addition, for the simple GARCH(1,1) model, the unconditional variance is equal to 0.000628171 for the pre-futures period and 0.000399753 for the post-futures period. In other words, the unconditional variance in the post-futures period is lower than that in the pre-futures period. This indicates lower market volatility after the introduction of stock index futures in the Greek stock market.

For the FTSE/ASE Mid 40 index, the unconditional variance is still lower in the post-futures period. More specifically, for the AR(3)-GARCH(1,1) model, the unconditional variance is equal to 0.001347408 for the pre-futures period and 0.000565422 for the post-futures period. Also, for the simple GARCH(1,1) model, the unconditional variance changes from 0.001898816 to 0.000600954. Thus, the unconditional variance in the post-futures period is lower than that of the pre-futures period. In other words, the volatility of the Greek stock market diminished after the introduction of stock index futures trading.

This finding is in line with what Bologna and Cavallo¹ found for the Italian stock market. Finally, the impact of futures trading on the rate at which information is incorporated into spot prices is illustrated, and the effect of the shock to both the pre- and post-futures periods is plotted. Figures 6.1 and 6.2 plot the effect of the shock for the FTSE/ASE-20, and Figures 6.3 and 6.4 plot the effect of the shock for the FTSE/ASE Mid 40. From these figures it is clear that the impact of the shock is more persistent in the post-futures period than in the pre-futures period for both indices.

Summary and conclusions

The introduction of a futures market and, in particular, the impact of futures on stock market volatility has generated a long debate. Previous studies have shown that the futures market leads to an increase in market depth and a decrease in volatility. This is due to the more rapid rate at which information is reflected in prices and speculation. Other studies suggest that a

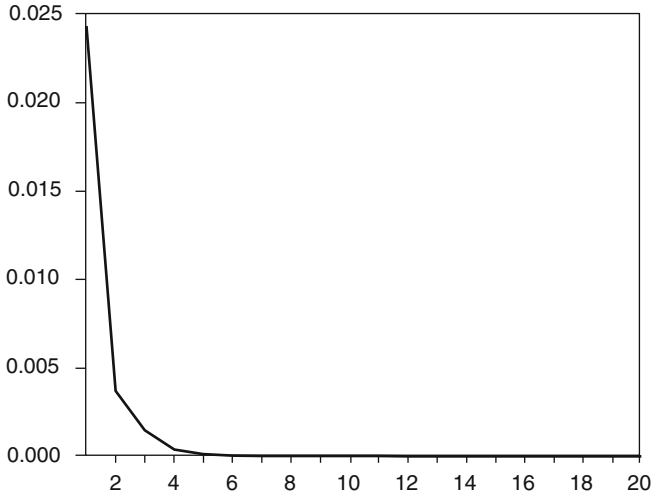


Figure 6.1 Effect of a 1 SD shock on spot price volatility before the onset of futures trading (FTSE/ASE-20)

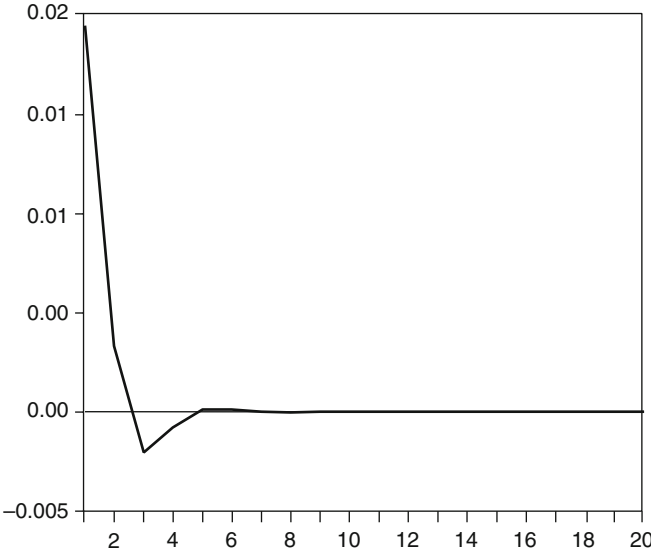


Figure 6.2 Effect of a 1 SD shock on spot price volatility after the onset of futures trading (FTSE/ASE-20)

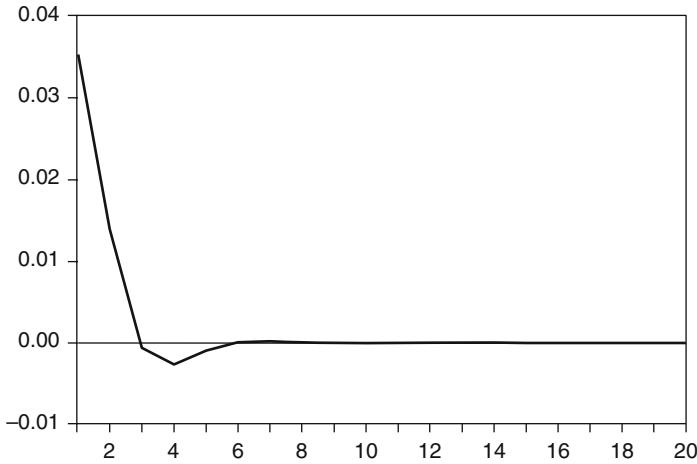


Figure 6.3 Effect of a 1 SD shock on spot price volatility before the onset of futures trading (FTSE/ASE Mid 40)

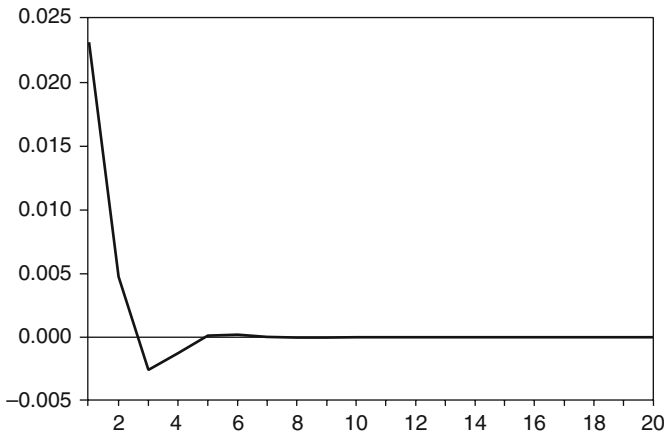


Figure 6.4 Effect of a 1 SD shock on spot price volatility after the onset of futures trading (FTSE/ASE Mid 40)

decrease in cash market volatility is due to an increase in market liquidity. Empirical studies for UK and US financial markets do not conclude clearly whether the introduction of futures stabilises or destabilises the underlying spot market.

This study has analysed the effect of the impact of the introduction of futures on other stock prices on the Greek Stock Exchange. To the authors'

knowledge, this is the first study that has examined the effect of the Greek futures market on stock market volatility.

A significant indicator of this effect is spot market volatility. To analyse the relationship between stock index futures and stock market volatility, several GARCH models were used to model the FTSE/ASE-20, FTSE/ASE Mid 40 and General ASE indexes.

For the FTSE/ASE-20, the results of the effect of futures trading suggest that there has been a negative effect on spot price volatility during the period. For the sub periods, we find that good news has a more rapid impact on stock return volatility, and that the persistence of shocks is reduced, indicating increased market (pricing) efficiency. This is not surprising, as the FTSE/ASE-20 futures market is highly liquid. In addition, the results suggest that old news has either a greater or a less-persistent effect on price changes. The fact that noise traders³⁹ are the principal users of the FTSE/ASE-20 contract indicates that volatility becomes more persistent, so an increase in persistence is not surprising. Further, GARCH(1,1) models show that futures trading improves the speed of information flow to the spot market. This is in line with arguments from Antoniou and Holmes⁷ for the FTSE 100 index.

For the FTSE/ASE Mid 40 index, the empirical results are mixed. Asymmetric AR(3)-EGARCH and AR(3)-TGARCH models show a positive effect on price volatility during the period examined. GARCH-type models (where the conditional mean equation just includes a constant term) show a negative (but not significant) effect, however. Furthermore, there has been a decrease in both the ARCH and GARCH parameters, indicating that news is being reflected in prices more slowly, and that old news has a less persistent effect on prices. The decrease in the coefficient on past variance shows that, after the onset of futures trading, spot market volatility is not important to spot market participants (see Antoniou and Foster¹⁸). The main reason is that the onset price risk can be hedged in the futures market. Further, the TGARCH and EGARCH specifications show that the leverage effect exists and, thus, there is an asymmetric effect to the news. Also, the EGARCH model shows that a negative shock increases the conditional variance.

In summary, the evidence suggests a negative effect of futures trading on Greek stock market volatility (however, this is not very strong for FTSE/ASE Mid 40 index). This is confirmed by the estimation of three different types of GARCH specifications and unconditional variances. In particular, the unconditional variance in the post-futures period is found to be lower than that of the pre-futures period. This indicates lower market volatility after the introduction of stock index futures. This is consistent with Bologna and Cavallo¹ for the Italian Stock Exchange.

Finally, future research should test (1) whether the introduction of stock index futures affects the volume-volatility relationship in the spot market, using GARCH and other stochastic volatility models, and (2) for the presence of causal relationships, using VAR representation and Granger causality tests.

References and Notes

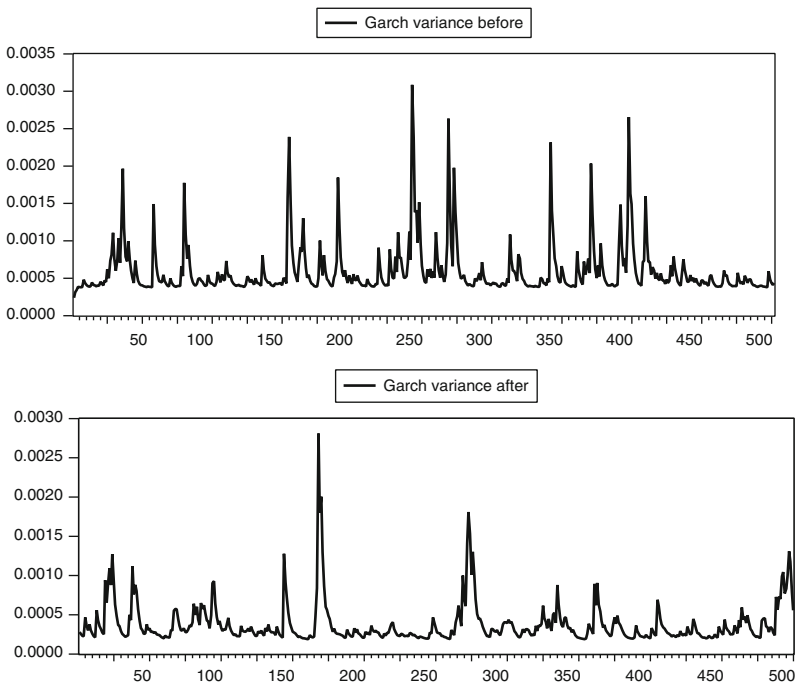
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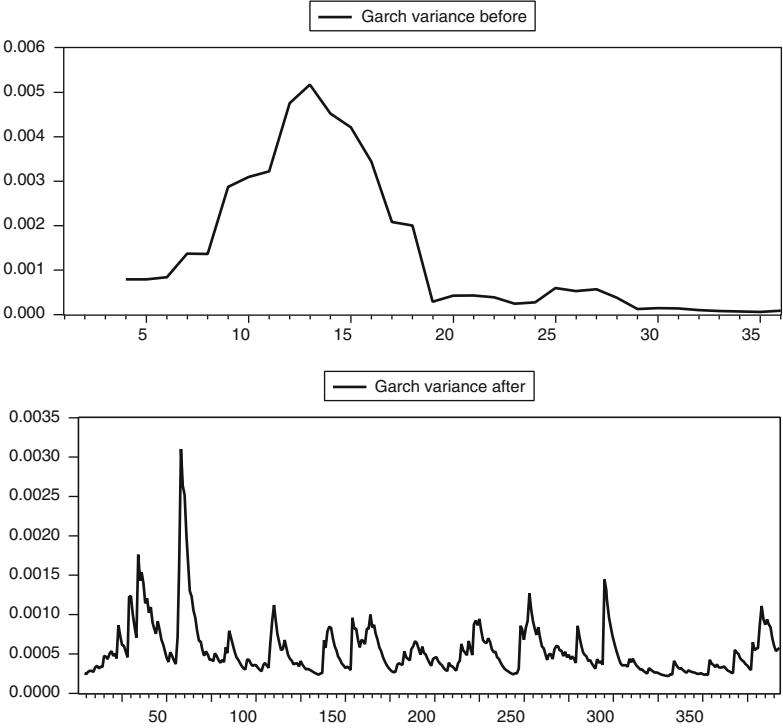
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Appendix 1

A GARCH variance series before and after the introduction of the FTSE/ASE-20 stock index futures



B GARCH variance series before and after the introduction of the FTSE/ASE Mid 40 stock index futures



7

Modelling and Trading the Gasoline Crack Spread: A Non-Linear Story

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Practical applications

The research below has three main practical applications.

Firstly, the ability to predict WTI-GAS spread displays obvious advantages to oil refiners, whose profit maximisation depends on knowing when to buy and sell stocks of either WTI crude or unleaded gasoline. Refining when the WTI-GAS spread is large will lead to higher operating profits than refining when the WTI-GAS spread is small.

Secondly, the tests of non-linear cointegration show whether upside and downside moves are significantly different. In this case this was interpreted as the ability of major oil companies to control the price of unleaded gasoline. This method could be used for other spreads indicative of profit margins such as the soybean crush spread (the difference between soybeans and soybean products), spark spread (the difference between natural gas and electricity) or the frac spread (the difference between natural gas and propane).

Finally, from the perspective of a futures trader, the ability to predict the direction of the WTI-GAS spread using a simple fair value model as described

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here could be easily implemented as a comparatively low risk trading strategy. Further it is also demonstrated that the use of filters can enhance the risk/return profile of this strategy.

This paper investigates the gasoline crack spread time series, using the non-linear cointegration method developed by Enders and Granger (1998, 'Unit-root Tests and Asymmetric Adjustment with an Example Using The Term Structure of Interest Rates', Journal of Business and Economic Statistics, Vol. 19, pp. 166–176). The spread can be viewed as the profit margin gained by cracking crude oil, and therefore any non-linearity can be interpreted in the context of the effect on market participants. Further, a number of non-linear neural networks are used to forecast the gasoline crack spread. The architectures used are multilayer perceptron, recurrent neural networks and higher order neural networks, these are benchmarked against a fair value non-linear cointegration model. The final models are judged in terms of out-of-sample annualised return and drawdown, with and without a number of trading filters. The results show, first, that the spread does indeed exhibit asymmetric adjustment, with movements away from fair value being nearly three times larger on the downside than on the upside. Secondly, the best trading model of the spread is the higher order neural network with the threshold filter, owing to a superior out-of-sample risk/return profile.

Introduction

The motivation for this paper emanates from events in the oil markets c. 2005. Price rises in the level of crude oil cause increasing prices of crude oil products, such as unleaded gasoline. This slows down growth and hurts the economy.^{1,2} In contrast many oil-refining companies have listed some of the largest profits in history.^{3,4}

Further motivation for this research is the non-linear analysis of the gasoline crack spread. The ability to identify non-linearities in a time series can clearly influence the decision of which models should be chosen to forecast the time series accurately, and also the type of trading filter that should be employed.

This case seems to be ideal for the investigation of any non-linearity in the spread time series and, furthermore, whether such asymmetries are biased in favour or against oil refining companies. The question of whether the consumer of unleaded gasoline has been getting a 'fair deal' seems valid. Discovering (or otherwise) any non-linearity in the spread can answer this question.

The gasoline crack spread can be interpreted as the profit margin gained by processing crude oil into unleaded gasoline. It is simply the monetary difference between West Texas Intermediate crude oil and Unleaded Gasoline, both of which are traded on the New York Mercantile Exchange (NYMEX). The spread is calculated as shown in equation (1)

$$S_t = GAS_t - WTI_t \quad (1)$$

where S_t is the price of the spread at time t (in \$ per barrel), GAS_t is the price of unleaded gasoline at time t (in \$ per barrel), and WTI_t is the price of West Texas Intermediate crude oil at time t (in \$ per barrel).

A few large oil companies govern the pricing relationship between crude oil and unleaded gasoline. These companies are some of the largest in the world and, of these companies, BP, Shell and Exxon Mobil dominate the market. It is possible, therefore, for these companies to act in their own interests and not necessarily in the interests of the end user.

The spread time series for the in-sample period (1st January, 1995–25th April, 2003) is shown in Figure 7.1.

It is evident from Figure 7.1 that the spread shows mean reversion around approximately \$5 per barrel, but it is also evident that the spread shows asymmetry, with seemingly larger moves occurring on the upside of the long-term ‘fair value’ than on the downside.⁵ This could be a reason to expect the presence of non-linear cointegration, which is further explained in the fourth section of this paper.

This mean reversion is due to the fact that the spread is representative of a profit margin (the margin for refining crude oil into unleaded gasoline). The most common refining ratio for crude oil is 5:3:2, that is 5 parts crude oil to 3 parts unleaded gasoline and 2 parts heating oil. This refining ratio means that crude oil products other than unleaded gasoline, particularly heating oil, can play significant roles in defining the pricing relationship between crude oil and unleaded gasoline. In addition, the range of products of crude oil is large and varied (gasoline, liquefied petroleum gas, naphtha, kerosene, gasoil and fuel oil, to name just the fuels). Fluctuating demand for

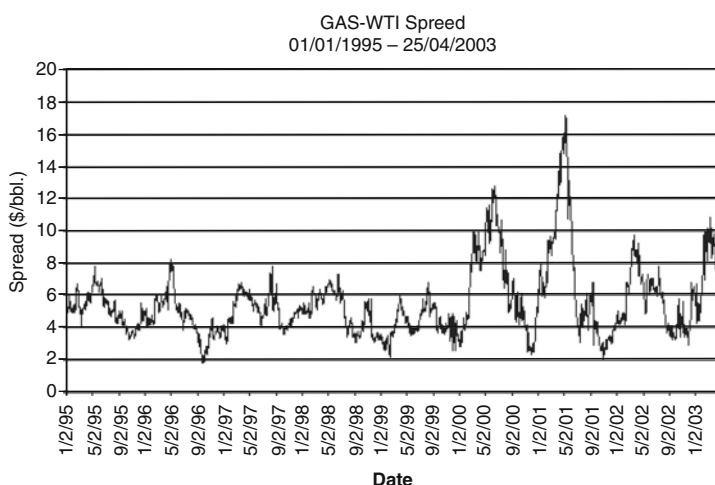


Figure 7.1 Gasoline crack spread price 1st January, 1995–25th April, 2003

these products can cause the gasoline crack spread to move away from the long-term fair value.

With such an unusual relationship, both economically and physically, the possibility exists that the movements of the spread are asymmetric. With this in mind, the hypothesis that the relationship between crude oil and unleaded gasoline exhibits non-linear adjustment is tested. It is a widely held view that, because of their market domination, adjustments in favour of oil refiners should be larger than adjustments that cause decreases in the refining margin. This is explained more fully below.

A fair value model is developed along the same lines as that of Evans *et al.*⁶ This is used as a benchmark for other non-linear models such as multi-layer perceptron (MLP), recurrent neural networks (RNN) and higher order neural networks (HONN). The models are used to forecast ΔS_{ν} , the daily change in the spread.

Finally, the correlation filter of Evans *et al.*⁶ is investigated and benchmarked against a more traditional threshold filter and, if the cointegration exhibits asymmetry, an asymmetric threshold filter. The exact specifications of these filters are included in the fifth section of this paper.

This paper is set out as follows: the next section details some of the relevant literature; the third section explains the data and methodology; the fourth section defines the trading models used; the fifth section defines the filters that have been employed; and the sixth and final sections give the results and conclusions, respectively.

Literature Review

Cointegration was first introduced by Engle and Granger.⁷ The technique is to test the null hypothesis that any combination of two series contains a unit root. If the null hypothesis is refuted and the conclusion is that a unit root does not exist, the combination of the two series is cointegrated. This is explained further below. As explained in the previous section, the spread may exhibit larger moves in one direction than in the other, this is known as asymmetry. Since the traditional unit root test has only one parameter for the autoregressive estimate, it assumes upside and downside moves to be identical or symmetric.

Non-linear cointegration was first introduced by Enders and Granger,⁸ who extended the unit root test by considering upside and downside moves separately, thus allowing for the possibility of asymmetric adjustment. The exact specification of this model is shown below.

This technique has been employed by, among others, Enders and Dibooglu,⁹ who test purchasing power parity (PPP) and find 'cointegration with threshold adjustment holds for a number of European countries on a bilateral basis'. Further, they conclude that 'central banks attempt to influence certain types of exchange rate movements and not others would seem to be a *prima facie* case against any type of symmetric adjustment'.⁹

Boucher¹⁰ investigates rational bubbles in the US and French stock markets, concluding that 'conventional cointegration tests fail to detect a long-run relationship between stock prices, dividends and proxies of the risk premium for US sample period 1953:1–2003:2, while the asymmetric cointegration tests uncover them'.

The trading of cointegrated time series is investigated by Evans *et al.*,⁶ who develop a simple fair value model, as described below. They use this model to trade the WTI–Brent spread, finding that the fair value model competes very well against some more sophisticated models owing to the low level of trading activity and, therefore, transaction costs.

With the increasing processing power of computers, rule-induced trading has become far easier to implement and test. Kaastra and Boyd¹¹ investigate the use of neural networks (NN) for forecasting financial and economic time series. They conclude that the large amount of data needed to develop working forecasting models involved too much trial and error. In contrast, Chen *et al.*¹² study the 30-year US Treasury bond using a NN approach. The results prove to be good, with an average buy prediction accuracy of 67 per cent and an average annualised return on investment of 17.3 per cent.

Krishnaswamy *et al.*¹³ attempt to show the development of NN as modelling tools for finance. In turn, they cite valuable contributions from Kryzanowski *et al.*¹⁴, Refenes *et al.*¹⁵, Bansal and Viswanathan¹⁶ and Zrilli¹⁷ in the field of stock market and individual stock prediction, proving that not only do NN outperform linear regression models, but that NN are 'superior in dealing with structurally unstable relationships, notably stock market returns'.¹³ This research kick-started the search for increasingly more advanced NN architectures.

Recurrent networks or Elman networks were first developed by Elman¹⁸ and possess a form of error feedback, which is further explained in the subsection 'Recurrent neural network'. These networks are generally better than MLP networks, but they do suffer from long computational times.¹⁹ According to Saad *et al.*,²⁰ however, compared with other architectures this should not matter much: 'RNN has the capability to dynamically incorporate past experience due to internal recurrence, and it is the most powerful network of the three in this respect ... but its minor disadvantage is the implementation complexity'.²⁰

Higher order NN were first introduced by Giles and Maxwell²¹ and were called 'tensor networks', although the extent of their use in finance is limited. Knowles *et al.*²² show that, despite shorter computational times and limited input variables on the EUR/USD time series, 'the best HONN models show a profit increase over the MLP of around 8 per cent'. A significant advantage of HONN is detailed in Zhang *et al.*:²³ 'HONN models are able to provide some rationale for the simulations they produce and thus can be regarded as 'open box' rather than 'black box'. Moreover, HONN

are able to simulate higher frequency, higher order non-linear data, and consequently provide superior simulations compared to those produced by ANN (Artificial NN)'.

In recent years, there has been an expansion in the use of computer trading techniques, which has once again called into doubt the efficiency of even very liquid financial markets. Lindemann *et al.*²⁴ suggest that it is possible to achieve abnormal returns on the Morgan Stanley High Technology 35 index, using a Gaussian mixture NN model. Lindemann *et al.*²⁵ justify the use of the same model to trade the EUR/USD exchange rate successfully, an exchange rate noted for its liquidity.

This paper investigates the use of the fair value model as a trading tool in the gasoline crack spread market, but also as a benchmark against more state of the art models such as MLPNN, recurrent NN (RNN) and HONN, which are described more fully below.

Data and Methodology

Data

The dataset used is the daily closing price data of the NYMEX West Texas Intermediate (WTI) for the crude oil and NYMEX Unleaded Gasoline (GAS). With both markets trading on the same exchange and closing at identical times, the problem of non-simultaneous pricing is avoided. Figure 7.1 shows the gasoline crack spread pricing series. The spread between the two pricing series is calculated as shown in equation (1).

The returns of this series are then calculated as follows:

$$\Delta S_t = \left[\frac{(GAS_t - GAS_{t-1})}{(GAS_{t-1})} - \frac{(WTI_t - WTI_{t-1})}{(WTI_{t-1})} \right] \quad (2)$$

where ΔS_t is the percentage return of spread at time t .

Forming the returns series in this way means that it is possible to present results with more conventional percentage return/risk profiles. This methodology was used by Butterworth and Holmes²⁶ and Evans *et al.*⁶ to calculate percentage spread returns.

The dataset has been split into two sets: in sample and out of sample. They are shown in Table 7.1.

Table 7.1 In-sample and out-of-sample dates

Data set	Dates	No. of observations
In-sample	1st January, 1995–25th April, 2003	2,170
Out-of-sample	28th April, 2003–1st January, 2005	440

Table 7.2 Training and test period dates

Period of in-sample	Dates	No. of observations
Training	1st January, 1995–17th August, 2001	1,730
Test	20th August, 2001–25th April, 2003	440

In the case of the NN models, the in-sample dataset was further divided into two periods. They are shown in Table 7.2.

The reason for the further segmentation of the in-sample dataset is to avoid overfitting. As described later in this paper, the networks are trained to fit the training dataset and stopped when returns on the test dataset are maximised.

Rollovers

Using non-continuous time series brings a unique problem, since any long-term study will require a continuous series. If a trader takes a position on a futures contract, which subsequently expires, he can take the same position on the next available contract. This is called rolling forward. The problem with rolling forward is that two contracts of different expiry but the same underlying may not (and invariably do not) have the same price. When the roll-forward technique is applied to a futures time series, it will cause the time series to exhibit periodic blips in the price of the contract. While the cost of carry (which actually causes the pricing differential) can be mathematically taken out of each contract, this does not leave us with a precisely tradable futures series.

As this study is dealing with futures spreads, both contracts have been rolled forward on the same day of each month (irrespective of the exact expiry dates). The cost of carry, which is the cause of the price difference between the cash and futures price is determined by the cost of buying the underlying in the cash market now and holding until the futures expiry. Since the cost of storage of both underlying is similar, they will approximately offset each other. We are left with a tradable time series with no cost of carry effect.

Transactions costs

In order to assess the returns of each model realistically, they are assessed in the presence of transactions costs. The transactions costs are calculated from an average of five bid–ask spreads on WTI and GAS (ten in total), taken from different times of the trading day. These are 0.094 per cent for WTI and 0.1004 per cent for GAS. Therefore, on the spread, there is a total round trip transaction cost of 0.1944 per cent. Since commission fees are generally small and varied, they have not been considered here.

Trading models

The following section contains descriptions of the trading rules employed. First, the non-linear fair value model is used to predict the value of $\Delta\mu_t$, the daily change in the cointegration equation residuals: This is used as a proxy for $\Delta\tilde{S}_t$, as the actual returns of the spread are very similar to the actual returns of the cointegration residuals,²⁷ or $\Delta S_t \approx \Delta\mu_t$. This is then used as a benchmark to test the ability of three NN models to predict ΔS_t . The use of this benchmark is justified by the obvious asymmetry of the time series, as is apparent from Figure 7.1.

Non-linear cointegration

Enders and Granger⁸ extend the Dickey-Fuller test²⁸ to allow for the unit root hypothesis to be tested against an alternative of asymmetric adjustment. Here, this is developed from its simplest form; consider the standard Dickey-Fuller test

$$\Delta\mu_t = \rho\mu_{t-1} + \varepsilon_t \quad (3)$$

where ε_t is a white noise process. The null hypothesis of $\rho=0$ is tested against the alternative of $\rho \neq 0$. $\rho=0$ indicates that there is no unit root, and therefore μ_t is a stationary series. If the series μ_t are the residuals of a long-run cointegration relationship as indicated by Johansen,²⁹ this simply results in a test of the validity of the cointegrating vector (the residuals of the cointegration equation should form a stationary series).

The extension provided by Enders and Granger⁸ is to consider the upside and downside moves separately, thus allowing for the possibility of asymmetric adjustment. Following this approach leads to equation (4)

$$\Delta\mu_t = I_t\rho_1\mu_{t-1} + (1 - I_t)\rho_2\mu_{t-1} + \varepsilon_t \quad (4)$$

where I_t is the zero-one 'heaviside' indicator function. This paper uses the following specification

$$I_t = \begin{cases} 1, & \text{if } \mu_{t-1} \geq 0 \\ 0, & \text{if } \mu_{t-1} < 0 \end{cases} \quad (5)$$

Enders and Granger⁸ refer to the model defined above as threshold autoregressive (TAR).

The null hypothesis of symmetric adjustment is $(H_0: \rho_1 = \rho_2)$, which can be tested using the standard F -test (in this case the Wald test), with an additional requirement that both ρ_1 and ρ_2 do not equal zero. If $\rho_1 \neq \rho_2$,

Table 7.3 Results of non-linear cointegration

	ρ_1	ρ_2	$\rho_1 = \rho_2$
Coefficient	-0.012957	-0.038508	-
F-stat	-	-	5.1936
p-value	-0.0021	-0.0002	0.0228

cointegration between the underlying assets is non-linear. In this case a trading model as described in Evans *et al.*⁶ will be misspecified and equation (4) should be used to predict $\Delta\mu_t$.

A fair value trading model has been developed from this procedure. In this model, equation (4) is used to estimate values of $\Delta\mu_t$ from values of μ_{t-1} . The values of ρ_1 and ρ_2 are estimated for the in-sample period and fixed for the out-of-sample period, the actual values of ρ_1 and ρ_2 are shown in Table 7.3.

Multi-layer perceptron

The reference NN model used in this paper is the MLP. The MLP network has three layers; they are the input layer (explanatory variables), the output layer (the model estimation of the time series) and the hidden layer. The number of nodes in the hidden layer defines the amount of complexity that the model can fit. The input and hidden layers also include a bias node (similar to the intercept for standard regression), which has a fixed value of 1 (see Lindemann *et al.*²⁵ and Krishnaswamy *et al.*).¹³

The network processes information as shown below:

1. The input nodes contain the values of the explanatory variables (in this case lagged values of the change in the spread).
2. These values are transmitted to the hidden layer as the weighted sum of its inputs.
3. The hidden layer passes the information through a non-linear activation function and onto the output layer.

The connections between neurons for a single neuron in the net are shown in Figure 7.2, where $x_t^{[n]}$ ($n=1,2,?,k+1$) are the model inputs (including the input bias node) at time t (in this case these are lags of the spread); $h_t^{[m]}$ ($m=1,2, \dots, m+1$) are the hidden nodes outputs (including the hidden bias node); $\Delta\tilde{S}_t$ is the MLP model output (predicted the percentage change in the spread at time t); u_{jk} and w_j are the network weights; $\mathcal{O}$ is the transfer sigmoid function $S(x)=1/1+e^{-x}$; $\mathcal{O}$ is a linear function $F(x)=\Sigma_i x_i$.

The error function to be minimised is

$$E(u_{jk}, w_j) = \frac{1}{T} \sum [\Delta S_t - \Delta S_t(u_{jk}, w_j)]^2$$

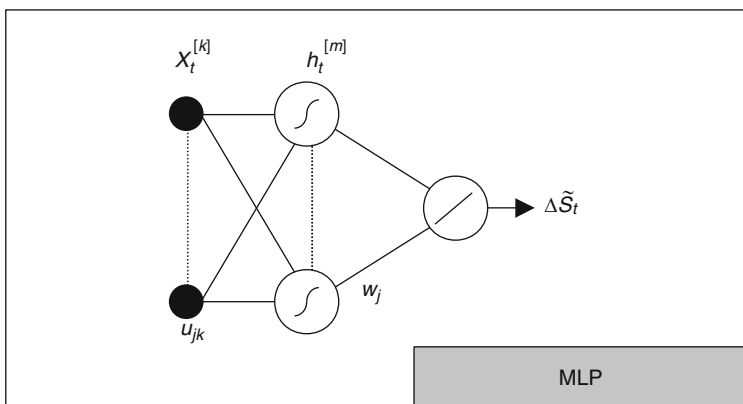


Figure 7.2 A single output, fully connected MLP model

where ΔS_t is the target value (the actual percentage change in the spread at time t).

Recurrent Neural Network

While a complete explanation of the recurrent network is beyond the scope of this paper, a brief explanation of the significant differences between RNN and MLP architectures is presented below. For an exact specification of the recurrent network, see Elman.¹⁸

A simple recurrent network has activation feedback, which embodies short-term memory (see, for example, Elman¹⁸). The advantages of using recurrent networks over feedforward networks, for modelling non-linear time series, has been well documented in the past (see, for example, Adam *et al.*³⁰). As described in Tenti,¹⁹ however, ‘the main disadvantage of RNN is that they require substantially more connections, and more memory in simulation, than standard backpropagation networks’, thus resulting in a substantial increase in computational time. Recurrent neural networks, however, can yield better results in comparison with simple MLPs owing to the additional memory input.

Connections of a simple recurrent network are shown in Figure 7.3.

The state/hidden layer is updated with external inputs, as in the simple MLP, but also with activation from previous forward propagation, shown as ‘Previous State’ in Figure 7.3. In short, the RNN architecture can provide more accurate outputs because the inputs are (potentially) taken from all previous values.

The Elman network in this study uses the transfer sigmoid function, error function and linear function, as described for the MLP architecture above. This has been done in order to be able to draw direct comparisons between the architectures of both models.

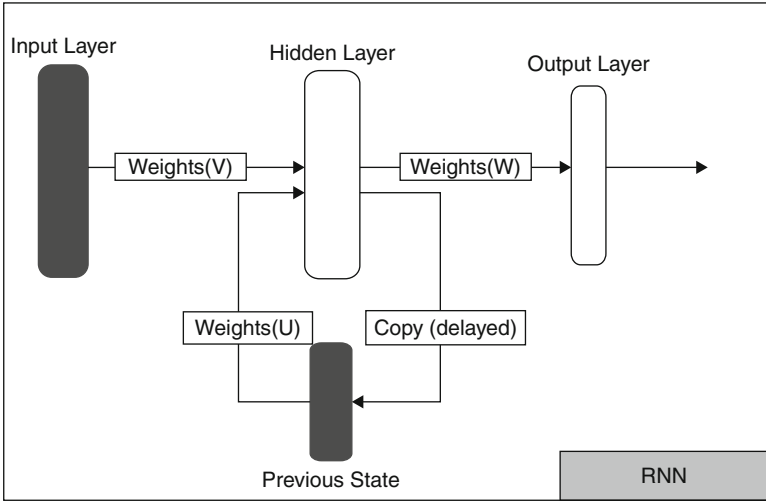


Figure 7.3 Architecture of Elman or RNN

Higher Order Neural Network

Higher order NN were first introduced by Giles and Maxwell,²¹ who referred to them as ‘tensor networks’. While they have already experienced some success in the field of pattern recognition and associative recall, they have not been used extensively in financial applications. The architecture of a three input second-order HONN is shown in Figure 7.4:

Higher order NN use joint activation functions; this technique reduces the need to establish the relationships between inputs when training. Furthermore, this reduces the number of free weights and means that HONN can be faster to train than even MLPs. Because the number of inputs can be very large for higher order architectures, however, orders of 4 and over are rarely used.

Another advantage of the reduction of free weights means that the problems of overfitting and local optima affecting the results can be largely avoided.²² For a complete description of HONN, see Giles and Maxwell.²¹

The HONN in this study uses the transfer sigmoid function, error function and linear function as described for the MLP architecture above. This has been done in order to be able to draw direct comparisons between the architectures of the models.

Neural network training procedure

The training of the network is of utmost importance, as it is possible for the network to learn the training subset exactly (commonly referred to as

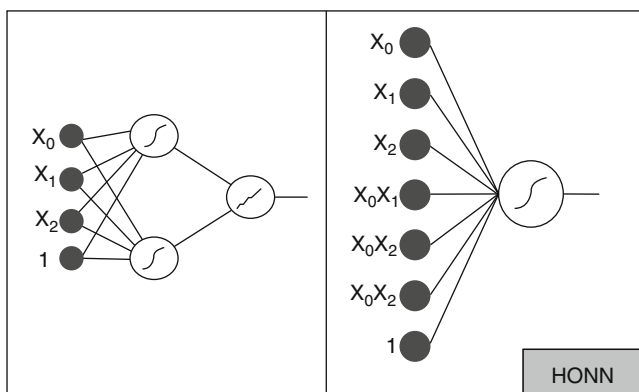


Figure 7.4 Left, MLP with three inputs and two hidden nodes; right, second-order HONN with three inputs

overfitting). For this reason, the network training must be stopped early. This is achieved by dividing the dataset into three different components (as shown in Table 7.2). First, a training subset is used to optimise the model, and the ‘back propagation of errors’ algorithm is used to establish optimal weights from the initial random weights. Secondly, a test subset is used to stop the training subset from being overfitted. Optimisation of the training subset is stopped when the test subset is at maximum positive return. These two subsets are the equivalent of the in-sample subset for the fair value model. This technique will prevent the model from overfitting the data, while also ensuring that any structure inherent in the spread is captured.

Finally, the out-of-sample subset is used to simulate future values of the time series, which for comparison is the same as the out-of-sample subset of the fair value model.

Since the starting point for each network is a set of random weights, a committee of ten networks has been used to arrive at a trading decision (the average estimate decides on the trading position taken). This helps to overcome the problem of local minima affecting the training procedure. The trading model predicts the change in the spread from one closing price to the next; therefore, the average result of all ten NN models was used as the forecast of the change in the spread or ΔS_t .

This training procedure is identical for all the NN used in this study.

Trading filters

A number of filters have been employed to refine the trading rules, they are detailed in the following section.

Threshold filter

With all the models in this study, predicting the percentage change in the spread ($\Delta\tilde{S}_t$), the threshold filter X is as follows

- If $\Delta\tilde{S}_t > X$ then go, or stay, long the spread,³¹
- If $\Delta\tilde{S}_t < -X$ then go, or stay, short the spread,³¹
- If $-X < \Delta\tilde{S}_t < X$, then stay out of the spread

where $\Delta\tilde{S}_t$ is the model's predicted spread return, and X is the level of the filter (optimised in-sample).

With accurate predictions of the spread, it should be possible to filter out trades that are smaller than the level of the filter, thus improving the risk/return profile of the model.

Asymmetric threshold filter

As defined earlier, the relationship between WTI and GAS may not be the same, depending on whether the spread is above or below the fair value. With this in mind, an alternative filter is proposed: the asymmetric filter. The formalism is shown below:

- If $\Delta\tilde{S}_t > |\rho_1| * X$ then go, or stay, long the spread,³¹
- If $\Delta\tilde{S}_t < -|\rho_2| * X$ then go, or stay, short the spread,³¹
- If $-|\rho_2| * X < \Delta\tilde{S}_t < |\rho_1| * X$, then stay out of the spread,

where $\Delta\tilde{S}_t$ and X are described above, and, the values of ρ_1 and ρ_2 are estimated in equation (4) over the in-sample period and fixed for the out-of-sample period.

Correlation filter

As well as the application of threshold and asymmetric threshold filters, the spreads were filtered in terms of correlation. The idea is to enable the trader to filter out periods of static spread movement (when the correlation between the underlying legs is increasing) and retain periods of dynamic spread movement (when the correlation of the underlying legs of the spread is decreasing). This was done in the following way.

A rolling Z -day correlation of the daily price changes of the two futures contracts is produced for the two legs of the spread. The Y -day change of this series is then calculated. From this a binary output of either 0, if the change in the correlation is above X , or 1, if the change in the correlation is below X . X is the filter level. This is then multiplied by the returns series of the trading model.

Using this filter, it should also be possible to filter out sudden moves away from fair value which are generally harder to predict than moves back to fair value. Figure 7.5 shows the entry and exit points of the filter with $X=0$.

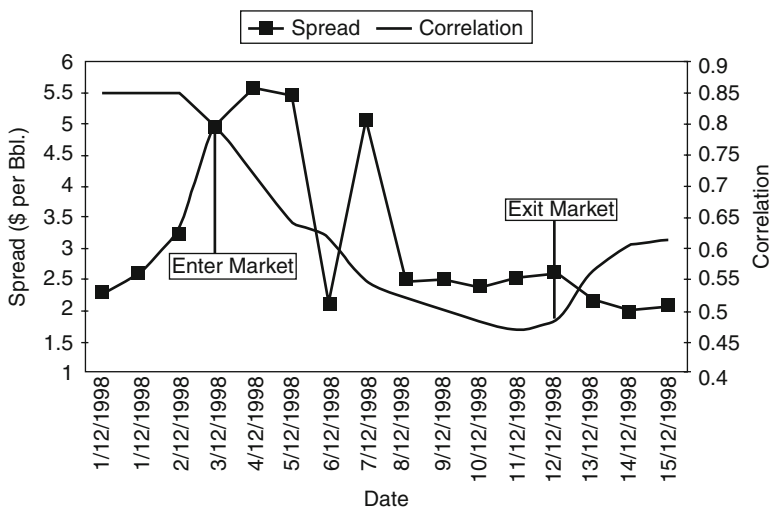


Figure 7.5 Operation of the correlation filter

Figure 7.5 shows that we enter the market the day after the change in correlation, ΔC , is below zero (ie $\Delta C < 0$), and exit the market the day after the change in correlation is above zero (ie $\Delta C > 0$). Doing this, one can filter out not only periods when the spread is stagnant, but also the initial move away from fair value, which for a mean-reverting asset is less predictable than the move back to fair value.

There are several optimising parameters which can be used for this type of filter, namely the length of correlation lag, period of correlation change and amount of correlation change. For this study the correlation lag (Z) is set to 30 days, and the period of correlation change (Y) to one day.³² The only optimising parameter used was the amount of correlation change. Formally, the correlation filter X_c can be written as

If $\Delta C < X_c$, then take the decision of the trading rule,
 If $\Delta C > X_c$, then stay out of the market.

where ΔC is the change in correlation, and X_c is the size of the correlation filter.

Results

The following section shows the results of the empirical investigation. The filters have been optimised in sample in order to maximise the Calmar ratio, defined by Jones and Baehr³³ as

$$\text{Calmar Ratio} = \frac{\text{Return}}{|\text{Max DD}|} \quad (6)$$

where *Return* is the annualised return of trading model, and *MaxDD* is the maximum drawdown of the trading model defined as

$$\text{Maximum drawdown} = \text{Min} \left[\Delta S_t - \text{Max} \left(\sum_{t=1}^n \Delta S_t \right) \right] \quad (7)$$

Equation (7) is given a high priority, as futures are naturally leveraged instruments. This statistic gives a good measure of the amount of return that can be expected for the amount of investment capital needed to finance a strategy. Furthermore, unlike the Sharpe ratio, which assumes that large losses and large gains are equally undesirable, the Calmar ratio defines risk as the maximum likely loss and is therefore a more realistic measure of risk-adjusted return.

Fair value non-linear cointegration

The cointegration vector of the in-sample period is shown below

$$\mu_t = (0.826234 * GAS) - WTI + (0.000116 * trend) \quad (8)$$

where *trend* is a linear trend included to account for any inflationary effects on the spread. Equation (8) was estimated using the Johansen method as detailed in Johansen.²⁹

The results of the regression of equation (4) are shown in Table 7.3, columns 2 and 3, the result of the Wald test for $\rho_1 = \rho_2$ is shown in Table 7.3, column 4.

The result shows that the two coefficients are significant in the estimation of equation (4). They are also significantly different at the 5 per cent level (shown by the Wald test results). This confirms the presence of non-linearity in the cointegration relationship between WTI and GAS. The authors can confirm that, for the in-sample data period, the magnitude of movements of the spread above the cointegration fair value is nearly three times smaller than that of the spread below fair value.

The in-sample and out-of-sample trading results of the non-linear cointegration fair value model are shown in Table 7.4.

Table 7.4 shows that all the filters improve the in-sample leverage factor over and above that of the unfiltered model and therefore should be considered as selected over the unfiltered model.

Table 7.5 shows that only one of the selected filters improves the out-of-sample leverage factor over and above that achieved by the unfiltered model. It should be noted that both the threshold and correlation filters show poorer performance out of sample than the unfiltered model, therefore proving to be erroneous selections.

Table 7.4 Non-linear fair value in-sample trading results^a

Filter	Ann.Ret (%)	Ann.StDev (%)	MaxDD (%)	Calmar	# Trades
BTC	19.12	22.34	-28.81	0.6638	3.8163
UnFiltered	17.66	22.35	-29.00	0.609	3.8163
Threshold	24.01	19.31	-25.57	0.9389	12.176
Correl	17.82	22.10	-28.83	0.6182	4.9673
Asymm	19.77	21.77	-28.83	0.6857	8.6019

^aBTC: results before transactions costs are taken into account, without any filters being employed. All subsequent rows take transactions costs into account.

UnFiltered: results of the unfiltered model.

Threshold: results with the threshold filter applied to the model.

Correl: results with the correlation filter applied to the model.

Asymm: results with the asymmetric filter applied to the model.

Ann.Ret: annualised return of the model.

Ann.StDev.: annualised standard deviation of the model.

MaxDD: maximum drawdown of the model, the maximum loss of the model during the sample period.

Calmar: Calmar ratio. Equation (6), gives a ratio for the amount of return for probable capital input.

#Trades: average number of trades per year.

Table 7.5 Non-linear fair value out-of-sample trading results

Filter	Ann.Ret (%)	Ann.StDev (%)	MaxDD (%)	Calmar	# Trades
BTC	19.12	22.34	-28.81	0.6638	3.8163
BTC	43.66	23.70	-17.94	2.4331	8.5175
UnFiltered	40.35	23.68	-17.94	2.2486	8.5175
Threshold	38.38	22.08	-16.12	2.3814	14.979
Correl	32.96	23.73	-17.94	1.8369	10.8671
Asymm	37.24	23.45	-17.93	2.0768	12.042

Multi-layer perceptron network

The trading results of the MLP model are shown in Table 7.6: all three filtered models outperform the unfiltered model, and therefore any of the three could potentially have been selected.

Table 7.7 shows that, of the filters selected, the threshold and asymmetric filters improve the out-of-sample Calmar ratio of the model.

Recurrent neural networks

The trading results of the RNN model are shown in Table 7.8: all filters could potentially have been chosen, because all filters improve the in-sample performance of the RNN model over and above that achieved by the unfiltered model.

Table 7.6 MLP in-sample results

Filter	Return (%)	Stdev (%)	MaxDD	Calmar	# Trades
BTC	31.62	22.61	-34.86	0.907	92.51
UnFiltered	13.65	22.67	-47.42	0.2878	92.51
Threshold	9.91	12.27	-20.18	0.4911	53.63
Correl	13.96	22.64	-47.38	0.2946	94.14
Asymm	6.79	7.16	-8.14	0.8344	18.8

Table 7.7 MLP out-of-sample results

Filter	Return (%)	Stdev (%)	MaxDD (%)	Calmar	# Trades
BTC	19.91	23.66	-33.70	0.5907	96.79
UnFiltered	1.31	23.84	-45.75	0.0287	96.79
Threshold	6.32	11.68	-10.98	0.5761	48.68
Correl	0.97	23.84	-46.34	0.021	97.94
Asymm	2.26	7.26	-6.33	0.3574	24.05

Table 7.8 RNN in-sample trading results

Filter	Return (%)	Stdev (%)	MaxDD (%)	Calmar	# Trades
BTC	25.88	22.64	-35.54	0.7282	87.4
UnFiltered	8.90	22.72	-56.70	0.1569	87.4
Threshold	6.76	9.95	-14.80	0.4569	39
Correl	9.22	22.71	-54.04	0.1706	88.1
Asymm	5.60	6.71	-7.90	0.7095	21.47

Table 7.9 shows that the threshold and asymmetric filters would have been bad selections. Although the correlation filter improves the in-sample Calmar ratio, this is not the case out of sample, but it cannot be considered a bad selection.

Higher order neural networks

The trading results of the HONN model are shown in Table 7.10: all three filters could have been chosen for the in-sample period, because they all improve on the performance of the unfiltered model.

Table 7.11 shows that, of the filters potentially selected from the in-sample performance, only the correlation filter proves to be a bad selection.

Overall, the best-performing model before transactions costs is the fair value model, which shows higher out-of-sample returns and therefore a better ability to predict the direction of the spread.

Table 7.9 RNN out-of-sample trading results

Filter	Return (%)	Stdev (%)	MaxDD (%)	Calmar	# Trades
BTC	40.73	23.56	-18.41	2.2118	100.23
UnFiltered	21.36	23.62	-19.19	1.1128	100.23
Threshold	15.92	15.16	-16.20	0.9829	97.36
Correl	21.36	23.62	-19.19	1.1128	100.23
Asymm	4.93	11.97	-17.82	0.2765	60.14

Table 7.10 HONN in-sample trading results

Filter	Return (%)	Stdev (%)	MaxDD (%)	Calmar	# Trades
BTC	21.43	22.66	-46.22	0.4637	107.83
UnFiltered	0.48	22.76	-60.08	0.008	107.83
Threshold	5.64	7.27	-8.49	0.6638	9.87
Correl	11.48	13.85	-28.11	0.4082	108.88
Asymm	1.97	3.11	-1.67	1.1812	2.79

Table 7.11 HONN out-of-sample trading results

Filter	Return (%)	Stdev (%)	MaxDD (%)	Calmar	# Trades
BTC	37.41	23.58	-21.09	1.7738	105.38
UnFiltered	16.93	23.70	-30.70	0.5513	105.38
Threshold	7.39	5.64	-0.97	7.6097	10.88
Correl	9.22	15.42	-22.50	0.4095	105.38
Asymm	2.84	4.77	-1.82	1.5596	4.58

Further, the results show that, before transactions cost are taken into account, the performance of the NN models in predicting the direction of the spread lie largely in order of computational time. The one exception is the HONN, which outperforms the MLP both in and out of sample, despite having far shorter computational times.

The results of the filters show improvements in the out-of-sample trading statistics over and above that achieved by the unfiltered model in 6 of the 12 times a filter is selected. Contrasting this to the threshold filter for which the same ratio is 3 out of 4, shows that the threshold filter is the most consistent method of filtering trading models on this spread. The threshold filter improves the out-of-sample Calmar ratio by an average of about 8 times. This is compared to about 3 for the asymmetric filter and about -0.2 for the correlation filter.

Conclusions

The results clearly show that the relationship between WTI and GAS is non-linear. This is evidenced by the results of the Wald test, showing $\rho_1 \neq \rho_2$.

Further, since ρ_1 is smaller in absolute terms than ρ_2 , movements below fair value tend, on average, to be larger than movements above fair value. The widely believed theory that refiners are holding onto a large refining margin, as presented by Blum³ and Macalister *et al.*⁴ is proved to be false for the period concerned here. It can be concluded that any asymmetry in the time series is not in favour of oil refiners.

If the aim is to model ΔS_t , or the change in the spread, the best model is the fair value model. This is evidence by the largest out-of-sample profits without a filter and before transactions costs are taken into account, indicating a superior ability to predict the sign of ΔS_t . It is also worth noting that the HONN outperformed the MLP out of sample, despite shorter computational times and limited variables. The authors feel that HONN should be investigated further in reference to financial markets.

The effect of transactions costs is extremely large on very active models such as the three NN investigated here, resulting in the fair value model producing the best out-of-sample performance after transactions costs. In fact, the level of transactions costs for the fair value model is 3.31 per cent p.a., in comparison with 18.59 per cent for the MLP, 19.37 per cent for the RNN and 20.49 per cent for the HONN (indicating an average of 17.03, 92.45, 99.65 and 105.38 trades per year, respectively).

Of the filters investigated, the best model is the threshold filter, which easily outperforms the asymmetric and correlation filters, proving to be the best out-of-sample performer on three of the four models. Furthermore, the threshold filter improves the out-of-sample Calmar ratio by an average factor of about 8 in contrast to about 3 for the threshold and about -0.2 for the correlation filters. The result that this filter is the best filter is surprising, given the clear non-linearity of the time series.

Finally, the best model in terms of out-of-sample Calmar ratio is the HONN with a threshold filter, with a Calmar ratio of 7.6097. Making the choice of a trading model on the basis of the in-sample Calmar ratio, however, would have led to the HONN model with the asymmetric filter: out of sample, this model performs satisfactorily, with a Calmar ratio of just over 1.5.

This paper proves that, contrary to the hypothesis raised by Blum³ and Macalister *et al.*,⁴ consumers of unleaded gasoline have been getting a 'fair deal' over the sample period concerned. Further work could include the testing of asymmetric adjustment on other spreads that are reflective of profit margins, such as the soybean crush spread or spark spread.

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8

The Relation between Bid–Ask Spreads and Price Volatility in Forward Markets

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Practical applications

Forward freight agreement (FFA) contracts have become the main risk management instrument in the sea transportation industry. In recent years, there has been a huge increase in the numbers of shipping companies, commodity trading houses and financial institutions trading FFAs. This study examines the relationship between bid–ask spread and expected volatility in the freight market. The results of this study provide a better understanding of the movements of FFA prices and their consequent effect on transactions

Reprinted from *Derivatives Use, Trading & Regulation*, 11(2): 105–125, 2005, ‘The Relation between Bid–Ask Spreads and Price Volatility in Forward Markets’, by Roy A. Batchelor, Amir H. Alizadeh and Ilias D. Visvikis. With kind permission from Henry Stewart Publications. All rights reserved.

costs. Market agents using information on the behaviour of bid–ask spreads can have a better insight about the timing of their FFA transactions and the future direction of the FFA market, as a widening bid–ask spread corresponds to an anticipation of increased future volatility.

The forward freight agreement (FFA) market developed in the 1990s and is growing very fast as the main derivatives market offering agents in the shipping and transportation industry a risk management instrument. This paper examines the relationship between expected volatility and bid–ask spreads (BASs) in this derivatives market where the underlying asset is a service rather than physical or financial assets. The study employs a two-step modelling specification. In the first step, the GARCH specification is used to model the volatility of the FFA prices; in the second step, the relationship between expected conditional volatility (led by one day) and the current BAS using the generalised method of moments (GMM) approach is investigated. The results indicate that there is a positive relationship between BASs, and expected price volatility in three out of four FFA contracts, after other factors are controlled.

Introduction

The forward freight agreement (FFA) market was developed in the 1990s and is growing very fast as the main derivatives market offering agents in the shipping and transportation industry a risk management instrument. FFAs agreements are derivatives contracts traded in an over-the-counter (OTC) market where two parties must agree to do business with each other while accepting credit risk from the other party.¹ The primary advantage of an OTC market is that the terms and conditions are tailored to the specific needs of the two parties. Since this market is a private market in which the general public does not know that the transaction was done, it does not normally require initial, maintenance and variation margins, which are common in the futures organised exchanges.²

The aim of the formalisation of the FFA market during the 1990s was to provide a mechanism for hedging freight rate risk in the dry-bulk and wet-bulk sectors of the shipping industry. FFAs agreements are principal-to-principal contracts between a seller and a buyer to settle a freight or hire rate for a specified quantity of cargo or type of vessel for usually one, or a combination of, the major trade routes. Currently, FFA contracts have as the underlying asset spot freight rates in routes of the Baltic Panamax Index (BPI), the Baltic Handymax Index (BHMI), the Baltic Capesize Index (BCI) and the Baltic International Tanker Routes Index (BITR). One counterparty takes the view that the price of an agreed freight route, at an agreed time, will be higher in the future. He/she buys FFA contracts (charterer) in order to sell them in the future at the higher price and thus controls for the possibility of paying higher spot rates in the future. The other party takes the opposite

position and sells FFA contracts (shipowner). Settlement is made on the difference between the contracted price (forward price) and the average price for the route selected in the index over the last seven working days.

The growth of transactions in the FFA market is also evidenced in Figure 8.1, which presents the estimated notional amount of freight contracts traded in the FFA market. The graph also illustrates the decline in annual volume of previously traded freight futures contracts in the London International Financial Futures Exchange (LIFFE) known as the 'Baltic International Freight Future Exchange' (BIFFEX). It can be seen that the trading volume of BIFFEX contracts has been dropping steadily, particularly since 1995, when trading in FFA contracts really started to grow owing to the ineffectiveness of BIFFEX contracts in risk management.

In the OTC FFA market, there are no official organised exchanges, but there is a network of shipbrokers who act as FFA brokers; transactions occur only when buy and sell ask orders are matched. The shipbrokers supply the market by quoting daily bid and ask prices simultaneously against which market orders can be executed. In the trading process, interest in buying or selling forward contracts is relayed by telephone or a computerised order-entry system by the shipbrokers to all potential traders. On receiving the replies, the shipbrokers try to match the bid and ask prices by continuously negotiating with the two parties. If an agreement is reached, the contract is fixed. During this process, the daily bid and ask prices are directly observable and, therefore, there is no need to estimate them as in other derivatives markets. Several procedures have been proposed for the estimation of the BAS

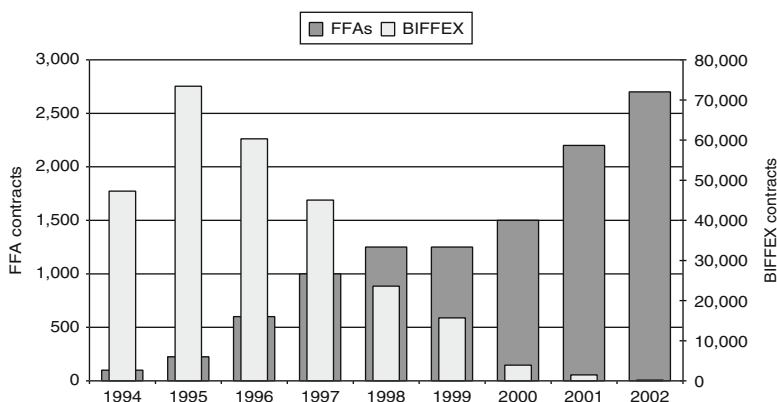


Figure 8.1 Volume of transactions in the FFA and BIFFEX markets. Reported figures are market estimates. It is estimated that, on average, one FFA contract has the same monetary value as approximately 75 BIFFEX contracts

FFA, Forward Freight Agreement; BIFFEX, Baltic International Freight Future Exchange.

Source: Simpson, Spence and Young (SSY) Futures.

(and its components) when it is not directly observable (see, for example, Bhattacharya,³ Roll,⁴ Choi *et al.*,⁵ Thompson and Waller,⁶ George *et al.*,⁷ Laux and Senchack⁸ and Chu *et al.*⁹). For a formal discussion of the alternative BAS estimators see Ding.¹⁰

Transaction costs are usually ignored in asset pricing theories but they are an important consideration in investors' investment decisions. One significant cost is the BAS. Brokers match buy and sell contracts, and the price charged for this service is known as the BAS—the difference between the buying (bid) and selling (asked) price per contract. This is normally regarded as compensation to brokers for providing liquidity services in a continuously traded market. The mark-up charged by brokers in the financial markets, as in any other market, is a function of the operational efficiency of the brokers and the nature of the product. Tinic and West¹¹ argue that there is a positive relationship between spreads and price volatility on the grounds that the greater the variability in price, the greater the risk associated with the performance of the function of the brokers. Intuitively, unambiguous good or bad news regarding the fundamentals of the price of the asset should have no systematic effect on the spread. Both the bid and the ask prices should adjust in the same direction in response to the traders receiving buy or sell orders that reflect the particular news event. Greater uncertainty regarding the future price of the asset, however—as associated with greater volatility of the price of the asset—is likely to result in a widening of the spread (see Bollerslev and Melvin¹²).

The nature and the behaviour of the BASs have been examined thoroughly in the financial markets (see McNish and Wood¹³ for equity; Bessembinder¹⁴ and Bollerslev and Melvin¹² for foreign exchange; and Kalimipalli and Wargha¹⁵ for bond markets). The relationship between BAS and volatility in derivatives markets has also been examined in a number of studies (for example, Laux and Senchack,⁸ Ma *et al.*,¹⁶ Wang *et al.*,¹⁷ Ding¹⁰ and Wang and Yau,¹⁸ among others). BAS, which is a component of the transaction costs related to derivatives trading, is an important issue because the low cost of trading is often cited as one rationale for the existence of derivatives markets. High transaction costs will normally affect market participants' abilities to trade quickly and cheaply. Therefore regulators (Forward Freight Agreement Brokers Association—FFABA) will need to consider how their policy decisions may affect the volatility of the market and, consequently, the BASs.

There have been a number of studies on the behaviour of the FFA prices levels and volatilities,^{19–21} yet there is no evidence on the relationship between volatility and the transaction cost in this derivatives market. Therefore, the purpose of this study is to investigate what impact an anticipated increase in FFA price volatility will have on transaction costs in terms of BAS. Extant literature that provides some possible answers to the previous question includes those studies on the relationship between BASs and price

volatility (see, for example, Benston and Hagerman,²² Stoll,²³ Copeland and Galai²⁴ and McNish and Wood,¹³ among others).

This study contributes to the existing literature in a number of dimensions. First, it examines the relationship between BAS and price volatility in a derivatives market where the underlying asset is a service rather than physical or financial assets. The fact that the underlying asset is a service implies that the usual cost-of-carry relationship between spot and forward prices is not valid, and the relationship must be only through agents' expectations. This also implies that there is no arbitrage link between the spot and futures prices and no inventories are held by brokers and/or market makers. Secondly, a two-step modelling specification is employed in order to ensure robust inferences on the relationships between variables. In the first step, the GARCH specification is used to model the volatility of FFA prices. This specification is consistent with a return distribution which is leptokurtic (speculative prices), it also allows for a long-term memory (persistence) in the variance of the conditional return distributions. The GARCH model is known to be capable of mimicking observed statistical characteristics of many time-series of return on financial assets.²⁵ The second step investigates whether the expected conditional volatility (led by one day) has a significant positive relationship with the current BAS using the generalised method of moments (GMM) approach.²⁶

Thirdly, volatility in the several markets of the shipping industry are subject to sudden movements which are, at best, only partially predictable. A better understanding of the movements in FFA prices, and the consequent effect on transaction costs may provide important information and insights for market agents about the timing of trades, the sentiment and the future direction of the FFA market. For example, a widening of the BAS may discourage market agents from participating and trading, as it may indicate a period of high volatility. More specifically, traders, speculators, hedgers and arbitrageurs alike are interested in extracting information from these variables to discover how their reaction to new information can be used in predicting future prices. From a policy perspective, the issue is important because of its implications for the analysis of market liquidity and its relationship with risk. For example, Demsetz²⁷ argues that BAS can be considered a proxy for market liquidity, since a market is regarded as liquid when large transactions can be executed with a small impact on prices. In addition, agents and regulators are interested in knowing what impact changes in these variables have on market activity. Furthermore, there are special features in these contracts which do not appear in other derivatives markets, making this study more interesting. (For example, the investigation of the issue in an OTC forward market rather than a futures market, which has not been done before, primarily owing to the lack of available data.)

The remainder of this study is organised as follows. The second section presents the literature review. The third section discusses the research

methodology. A description of the data and some preliminary statistics are then presented in the fourth section. The empirical results are presented in the fifth section and the final section summarises this paper.

Literature review

Demsetz²⁷ characterizes the BAS as the cost of obtaining immediacy; the right to transact without significant delay. Microstructure theory implies that BASs must cover three costs incurred by providers of immediacy: inventory carrying costs;²⁴ asymmetric information costs;²⁸ and order processing costs.^{27,29} The inventory component should be the cost to the market-maker of maintaining open positions or demanding liquidity from other market participants and is positively related to risk. According to this view, volatility increases price risk and thereby pushes up spreads.¹² The asymmetric information costs component may be positively correlated with price volatility and competition would affect the total size of the spread inversely.¹⁴ O'Hara²⁹ provides a comprehensive literature and guide to the most influential theoretical work in market microstructure.

Although there are differences in the theoretical arguments, all the above empirical studies conclude that BASs are positively related to price volatility (when price change was measured over short intervals, eg daily, while the relationship became insignificant for price changes measured over longer intervals, eg monthly). Of the three different types of costs, the asymmetric information cost is the most relevant in the FFA market. Order processing costs are relatively low, and shipbrokers do not sustain any inventory carrying costs as they do not hold inventories of forward contracts. The BAS must be wider to protect brokers from the costs of providing liquidity to informed traders, which can affect the brokers unfavorably.²⁴ In this respect, the BAS may vary with both the timing of information arrival and the uncertainty of the information flow. Moreover, the BAS might be positively related to the amount of information coming to the market.^{30,31}

If information arrives sequentially, the more informed participants will trade first and the less informed participants will trade later. Because informed traders who acquire positive (negative) information are willing to bid (ask) a higher (lower) price to buy (sell), the spread may change according to the trading behaviour of the parties who possess private information. The trader's perceived exposure to private information determines how he/she will respond to large versus small orders and to the arrival of market-generated and other publicly available information. With regard to the uncertainty of information flows, it has been argued that less informed traders seek protection from the generation and ownership of private information in the market by requiring a higher risk premium.³⁰ This adverse selection hypothesis suggests that the level of BASs should be related to the uncertainty of the information flow in the market. As the broker attributes a

positive probability to the order being generated from informed traders, the BAS widens and, therefore, may signal the arrival of new information. Saar³² investigates the role of demand uncertainty from a different perspective, ie uncertainty about preferences and endowments of the investors' population in introducing information content to the order flow. He shows that demand uncertainty increases both the BAS and price volatility.

In the equity market, McNish and Wood¹³ report that New York Stock Exchange (NYSE) equity BASs widen (decrease) with underlying volatility (trading volume and trade size) over time. Wang *et al.*,¹⁷ using direct estimates of the BAS, examine the intra-day relationship of BASs and price volatility in the S&P500 Index futures market and control for information effects. They find that BASs and price volatility are jointly determined and positively related. In the bond market, Kalimipalli and Wargha¹⁵—using an autoregressive conditional duration (ACD) model, which provides input for an ordered probit model for observed BAS—find a significant positive (negative) relationship between latent volatility (trading volume proxy) and observed BAS. When repeating the exercise using a GARCH specification instead of the ACD model, they report that their findings are robust to alternative specifications.

In the foreign exchange market, Fieleke³³ reports a positive relationship between the rate of change in the exchange rate and the cost of transacting. Overturf³⁴ finds a positive relation between BAS and price volatility measured by its standard deviations. The latter study further suggests that the uncertainty regarding the rate of change in exchange rates tends to widen the BAS. Boothe³⁵ finds that various measures of risk and transactions volume have an impact on BASs and, in particular, he provides evidence for a positive relationship between the level of uncertainty regarding futures prices and BASs. Bollerslev and Melvin¹² report a positive relationship between latent volatility and observed BAS on the Deutschmark/dollar exchange market. Using a similar framework, Gwilym *et al.*³⁶ find a positive relationship between BAS for stock index options traded on the LIFFE market and the volatility of the underlying stock market index. Ding¹⁰ investigates intra-day and daily determinants of BASs in the foreign exchange futures market and argues that the number of transactions is negatively related to the BAS, whereas volatility in general is positively related to it.

Another strand of research on the relationship between BAS and price volatility concentrates on the trading hours of derivatives markets. For example, Amihud and Mendelson³⁷ demonstrate the existence of a positive relationship at the closing hour of the NYSE, while Brock and Kleidon³⁸ show that periodic market closure causes greater transaction demand at the open and close of trading. This greater transaction demand at open and close increases asks and lowers bids (widens the spread)—so that the spreads follow a U-shaped pattern throughout the day—and also increases trading activity (volume). Subrahmanyam³⁹ and Foster and Viswanathan⁴⁰ also predict

higher BASs at open and close because the presence of informed traders increases the adverse information component of the spread.

Research Methodology

Most of the previous empirical studies have concentrated on explaining the determinants of BASs utilising two classes of factors. The first class includes:

1. activity variables such as volume and order size;
2. various measures of risk;
3. competition in market-making, such as the numbers of brokers trading in the asset;
4. the number of transactions; and
5. institutional ownership.

The second class of factors is related to the features of exchanges and the financial characteristics of assets. Extensive literature reviews are provided by Benston and Hagerman,²² Hasbrouck,³¹ Stoll⁴¹ and McNish and Wood,¹³ among others.

The measurement of price volatility is a difficult task, and many different measurement procedures have been employed in the literature. These can be subdivided into those which have used historical volatility and those which have used a forecast of the volatility. The latter are those which use the implied volatilities derived through option prices. The definition of historical price volatility employed in any particular study depends on the frequency of the available observations (ie transactions data, closing prices) and the length of period for which the volatilities are to be computed (ie days or months). It is often taken as the variance of the logarithm of the daily price relatives. This has the advantage that, as the level of the prices alters over time, the variance of the logarithm of the price relatives is more likely to be stationary than is the variance of alternative volatility measures.⁴²

A shortcoming of the earlier studies is the way price volatility is computed. Board and Sutcliffe⁴² have shown that studies based on the historical estimate of volatility are sensitive to the measures of volatility used. Recent studies, however, indicate that most of the financial price series exhibit non-linear price dependencies. For example, it is possible for FFA prices to be linearly unrelated and yet be non-linearly dependent. The general evidence suggests that dependencies work through the conditional variance (and other even-ordered moments), rather than being a result of certain mis-specified first order dynamics.⁴³ Engle's ARCH model⁴⁴ and Bollerslev's GARCH model⁴⁵ can capture such time variation in return distributions. There is a great deal of evidence in various financial markets that the conditional variance from the ARCH class of models provides a superior estimate

of price variability (see Bollerslev *et al.*⁴⁶ for a review). ARCH processes allow the examination of the structure and the characteristics of volatility, explicitly address the issue of time dependence in the variance and, therefore, overcome problems associated with heteroscedasticity in the data. In order to derive an estimate of the FFA volatility, the following AR-GARCH(1,1) model is employed:

$$\Delta F_t = \varphi_0 + \sum_{i=1}^{p-1} \varphi_i \Delta F_{t-i} + \varepsilon_t; \varepsilon_t \sim \text{iid}(0, h_t) \quad (1a)$$

$$h_t = a_0 + a_1 h_{t-1} + \beta_1 \varepsilon_{t-1}^2 \quad (1b)$$

where F_t is the natural logarithm of the daily FFA price change (average mid-point of the bid-ask quotes); Δ is the first-difference operator; and ε_t are the residuals that follow a normal distribution with mean zero and time-varying variance h_t . Bollerslev²⁵ shows that GARCH(1,1) adequately fits many economic time-series. Models are also estimated using the percentage BAS—defined as $(\text{Ask}-\text{Bid})/[(\text{Ask}+\text{Bid})/2]$. The results are qualitatively unaffected, however, and thus in the ensuing analysis the models using the differenced BAS are reported. After ensuring that the model is well specified, following Bessembinder¹⁴ and Galati,⁴⁷ one-step ahead conditional volatility estimates (h_{t+1}) can be constructed. Following a common practice in the literature, the GARCH model is fitted on the entire time-series, thus yielding in-sample forecasts. Ideally, volatility implied in FFA option prices could be used, since there is evidence in other markets that it outperforms GARCH models in providing forecasts of future volatility.⁴⁸ FFA option contracts, however, are currently not very liquid.

To analyse the relationship between expected volatility and current BAS, the BASs are regressed against variables that represent risk, information, a dummy variable that serves to measure non-trading intervals, and a lagged BAS. To evaluate the importance of the approach of non-trading intervals in determining BASs, following Bessembinder, a non-trading indicator variable is included, which is set equal to one on Fridays and on the last trading day before a bank holiday in the UK.¹⁴ The results, however, yield insignificant coefficients of the dummy variable in all routes and are therefore excluded from the ensuing analysis:

$$\text{BAS}_t = \beta_0 + \beta_1 h_{t+1} + \beta_2 \text{BAS}_{t-1} + \beta_3 \Delta F_t + u_t; u_t \sim \text{iid}(0, h_t) \quad (2)$$

where risk is defined as the one-step ahead conditional volatility (h_{t+1}) from a GARCH(1,1) model, information effects are evaluated by the first-difference FFA price series (ΔF_t) and BAS_t is the difference of the natural logarithm of the ask quote minus the natural logarithm of the bid quote ($\ln(\text{Ask}_t) - \ln$

(Bid_t)). Ding¹⁰ proposes an alternative method for evaluating information effects using a price dummy variable, calculated as follows. First, the median transaction price is identified from the entire time-series. The FFA price of each day is then compared with the overall median price. If the FFA price is greater than the median price, the dummy variable is assigned a value of one. Otherwise, a value of zero is assigned. Ding,¹⁰ however, points out that the first-difference price series, rather than the price dummy, may generally provide more information. Thus, only the results containing the first-difference price series are reported here.

Two main problems occur when examining the relationship between volatility and BAS. First, it is readily seen that the use of BAS will result in simultaneity bias leading to inconsistent ordinary least squares (OLS) estimates. In order to overcome the simultaneity problem, Harvey⁴⁹ points out that lagged values of the endogenous variables should be used, because they are classified—together with exogenous variables—as predetermined. The second difficulty concerns the presence of heteroscedasticity implying inefficient standard errors. Thus, the model should be estimated using the GMM approach proposed by Hansen.²⁶ The GMM approach allows an instrument to be used for BAS,⁵⁰ thereby avoiding any simultaneity bias when lagged BASs are used as instruments. The GMM also has the additional advantage of yielding heteroscedasticity and autocorrelation consistent estimates (as proposed by Newey and West⁵¹) in the process.

The use of the first-difference FFA price series in the model assists in examining the relationship between informational uncertainty and BASs. If high price levels result from informed trading, the relationship between price levels and BASs should be positive, as it is reported in most empirical studies.^{23,24} In general, large broker spreads are attributed to the risk of adverse selection or uniform trading, while a negative relationship is argued to be the result of the presence of scale economies in the market. The latter is because, when prices are high, the dollar volume of transactions is large, which means brokers require lower BAS to cover their costs.¹³

Description and Properties of Data

From the creation of the FFA market on 1st February, 1992, to 1st November, 1999, the 11 Panamax and Capesize voyage and time-charter routes of the Baltic Freight Index (BFI) served as the underlying assets of the FFA trades in the dry-bulk sector of the shipping industry. After the latter date, with the exclusion of the Capesize routes and with the index renamed as BPI, the underlying assets of the FFA contracts are Panamax routes. The composition of the BPI, as it stood in January 2001, is presented in Table 8.1. The data sets that are used consist of daily FFA and BAS prices in Panamax Atlantic routes 1 and 1A from 16th January, 1997 to 31st July, 2000 and daily FFA and BAS prices in Panamax Pacific routes 2 and 2A from 16th January, 1997 to

Table 8.1 Baltic Panamax Index (BPI)—route definitions

Routes	Route description	Cargo	Vessel size (dwt)	Weightings in bpi
1	1–2 safe berths/anchorages US Gulf (Mississippi River not above Baton Rouge) to ARA (Antwerp, Rotterdam, Amsterdam).	Light grain	55,000	10%
1A	Transatlantic (including ECSA) round of 45–60 days on the basis of delivery and redelivery Skaw Passero range.	T/C	70,000	20%
2	1–2 safe berths/anchorages US Gulf (Mississippi River not above Baton Rouge)/1 no combo port to South Japan.	HSS	54,000	12.5%
2A	Basis delivery Skaw Passero range, for a trip via Gulf to the Far East, redelivery Taiwan-Japan range, duration 50–60 days.	T/C	70,000	12.5%
3	1 port US North Pacific/1 no combo port to South Japan.	HSS	54,000	10%
3A	Transpacific round of 35–50 days either via Australia or Pacific (but not including short rounds such as Vostochny/Japan), delivery and redelivery Japan/South Korea range.	T/C	70,000	20%
4	Delivery Japan/South Korea range for a trip via US West Coast—British Columbia range, redelivery Skaw range, duration 50–60 days.	T/C	70,000	15%

Each shipping route is given an individual weighting to reflect its importance in the worldwide freight market.

Routes 1A, 2A, and 3A refer to time-charter (T/C) contracts, while 1, 2, 3, and 4 refer to voyage contracts. The vessel size is measured by its carrying capacity (dwt—deadweight tonnes) and includes the effective cargo, bunkers, lubricants, water, food rations, crew and any passengers. HSS, heavy grain, soya and sorghum.

Source: Baltic Exchange.

10th August, 2001. All price data are from Clarkson Securities Limited. The FFA price series are transformed into natural logarithms. The FFA prices are always those of the nearby contract. To avoid thin markets and expiration effects, however, there is a rollover to the next nearest contract one week before the nearby contract expires, as there is sufficient liquidity in the nearby contract up to a few days before its maturity date.

Summary statistics for the daily logarithmic first-difference FFA prices and of the BAS prices for the four Panamax routes are presented in Table 8.2.

Table 8.2 Descriptive statistics of logarithmic first difference FFA and BAS prices ($\ln(\text{Ask}_t) - \ln(\text{Bid}_t)$)

	N	SD	Skew	KURT	Q(24)	Q ² (24)	J-B	ADF (lags)	PP(6)
Panel A: Route 1 FFA and BAS price series (16th January, 1997 to 31st July, 2000)									
FFA	896	0.0239	-0.151	5.429	44.466	34.183	1,096.7	-31.722 (0)	-32.070
BAS	897	0.0441	1.103	4.327	3,236.0	2,698.3	247.548	-8.773 (0)	-8.517
Panel B: Route 1A FFA and BAS price series (16th January, 1997 to 31st July, 2000)									
FFA	896	0.0301	-0.037	4.708	35.083	50.891	822.28	-29.547 (0)	-29.936
BAS	897	0.0606	0.828	3.813	5,689.8	5,506.7	127.294	-5.516 (2)	-6.792
Panel C: Route 2 FFA and BAS price series (16th January, 1997 to 10th August, 2001)									
FFA	1,150	0.0178	0.285	12.711	45.426	56.827	4,534.59	-31.632 (0)	-31.727
BAS	1,151	0.0105	1.369	6.208	2,452.3	1,420.4	852.907	-12.979 (1)	-16.837
Panel D: Route 2A FFA and BAS price series (16th January, 1997 to 10th August, 2001)									
FFA	1,150	0.0278	0.984	15.266	48.906	50.905	7,394.89	-31.084 (0)	-31.176
BAS	1,151	0.0381	1.534	6.499	8,170.1	7,666.8	1,038.29	-7.113 (2)	-9.628

All series are measured in logarithmic first differences.

N is the number of observations.

SD is the standard deviation of the series.

Skew and Kurt are the estimated centralised third and fourth moments of the data; their asymptotic distributions under the null are $\sqrt{T}\hat{a}_3 \sim N(0,6)$ and $\sqrt{T}(\hat{a}_4 - 3) \sim N(0,24)$, respectively.

Q(24) and Q²(24) are the Ljung-Box⁵³ Q statistics on the first 24 lags of the sample autocorrelation function of the raw series and of the squared series, respectively; these tests are distributed as $\chi^2(24)$.

J-B is the Jarque-Bera⁵² test for normality, distributed as $\chi^2(2)$.

ADF is the Augmented Dickey-Fuller⁵⁴ test. The ADF regressions include an intercept term; the lag-length of the ADF test (in parentheses) is determined by minimising the Schwartz Bayesian Information Criterion (SBIC).⁵⁶

PP is the Phillips and Perron⁵⁵ test; the truncation lag for the test is in parentheses.

The 5% critical value for the ADF and PP test is -2.89.

Jarque-Bera⁵² tests indicate departures from normality for FFA and BAS prices in all routes. The Ljung-Box⁵³ Q(24) and Q²(24) statistics on the first 24 lags of the sample autocorrelation function of the raw series and of the squared series indicate significant serial correlation and the existence of heteroscedasticity, respectively. After the Augmented Dickey-Fuller⁵⁴ (ADF) and Phillips-Perron⁵⁵ (PP) unit root tests are applied on the daily log first-difference FFA price series, the results indicate that in all routes the log first-difference FFA price series are stationary. The results of the unit root tests on the levels of the BAS series indicate that all BAS price series are stationary.

The BASs for routes 1, 1A, 2 and 2A, respectively are presented in Figures 8.2-8.5, providing a visual representation of the transactions costs induced by the shipbrokers. The figures show that the maximum BAS for is \$0.25 per tonne for route 1, \$0.35 per day for route 1A, \$0.09 per tonne for route 2, and \$0.29 per day for route 2A. Moreover, after about September 1999, the BASs for routes 2 and 2A start to narrow significantly. This, and the small BAS figures in route 2, can be explained by the fact that routes 2 and 2A

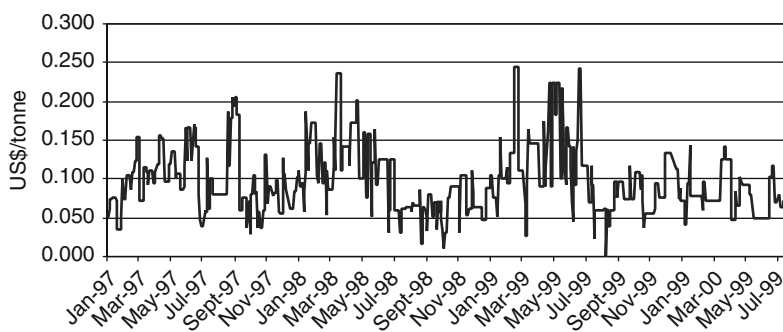


Figure 8.2 Route 1 BAS series; sample period 16th January, 1997 to 31st July, 2000

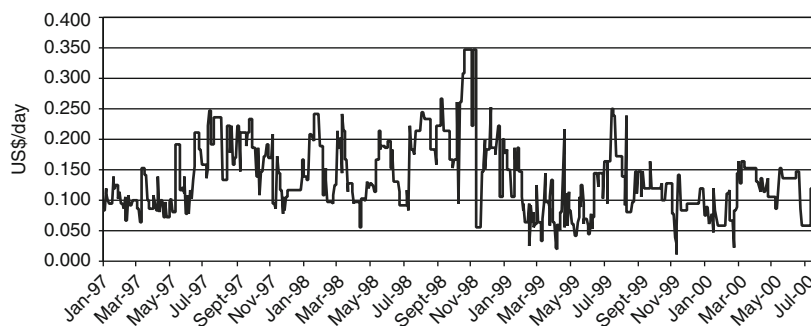


Figure 8.3 Route 1A BAS series; sample period 16th January, 1997 to 31st July, 2000

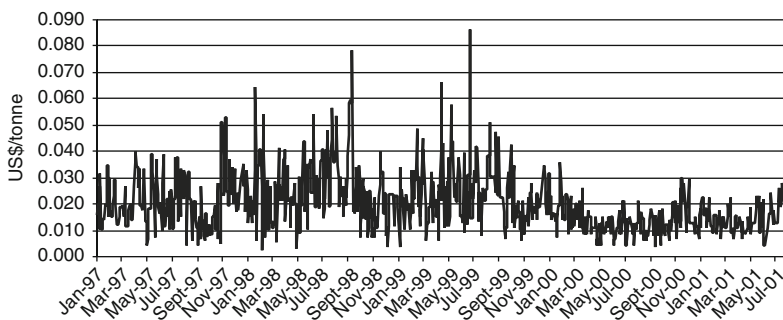


Figure 8.4 Route 2 BAS series; sample period 16th January, 1997 to 10th August, 2001

concentrate most of the FFA trading interest in the Panamax sector. Thus, shipbrokers can report narrow BASs as shipowners and charterers agree to fix FFA contracts after a few negotiations only. Figures 8.6–8.9 show the historical volatility (standard deviation) of daily percentage FFA price changes, computed over moving windows of 20 days, and the BASs for routes 1, 1A, 2 and 2A, respectively. The figures show a positive relationship between volatility and BAS in most cases, which is clearer and more consistent in routes 2 and 2A. Formal empirical analysis is needed, however, for the significance of the above inference.

Empirical Results

In order to model the volatility of the FFA prices, AR-GARCH(1,1) models are estimated. The most parsimonious specification for each model is estimated by excluding insignificant variables. The quasi-maximum likelihood estimates of the GARCH models of FFA rates for each route are presented in Table 8.3. The diagnostic tests on the standardised residuals and squared

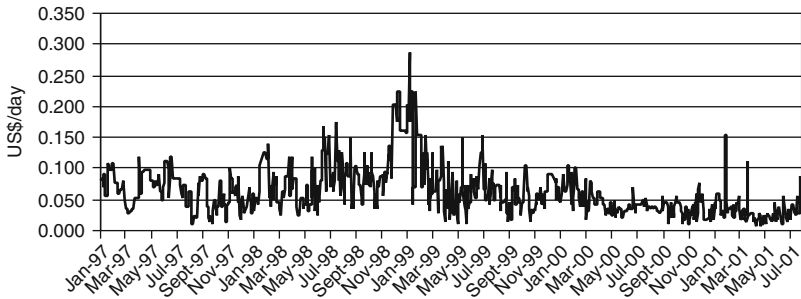


Figure 8.5 Route 2A BAS series; sample period 16th January, 1997 to 10th August, 2001

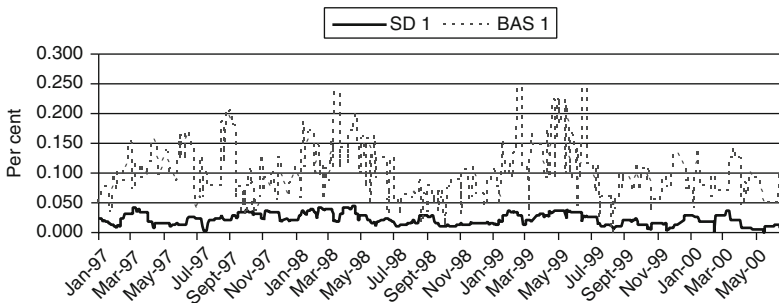


Figure 8.6 Route 1 BAS and historical volatility; sample period 16th January, 1997 to 4th July, 2000

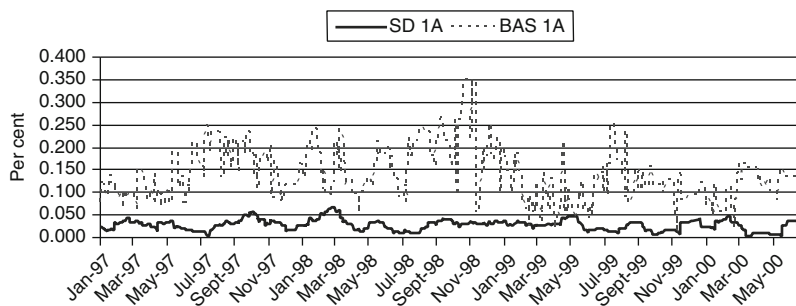


Figure 8.7 Route 1A BAS and historical volatility; sample period 16th January, 1997 to 4th July, 2000

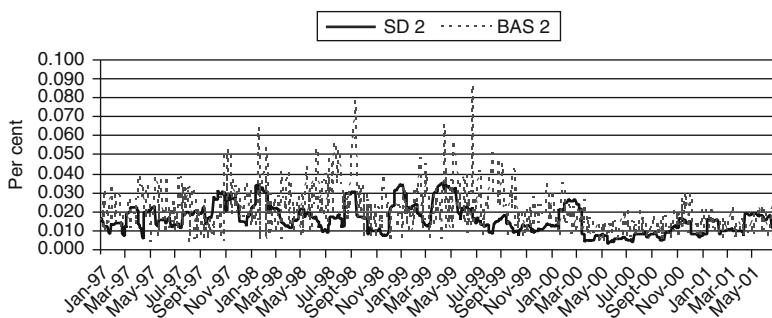


Figure 8.8 Route 2 BAS and historical volatility; sample period 16th January, 1997 to 16th July, 2001

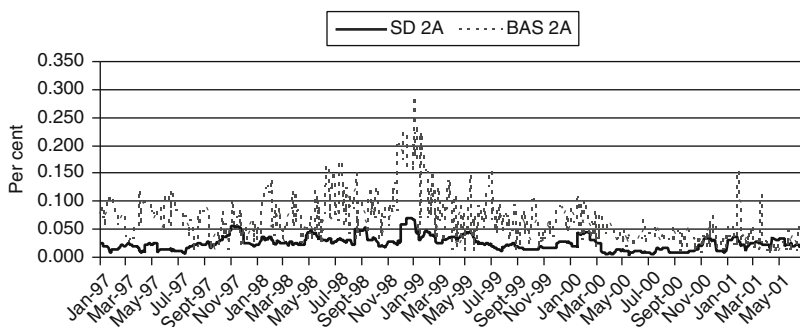


Figure 8.9 Route 2A BAS and historical volatility; sample period 16th January, 1997 to 16th July, 2001

Table 8.3 GARCH model estimates of the FFA conditional volatility

$$\Delta F_t = \varphi_0 + \sum_{i=1}^{p-1} \varphi_i \Delta F_{t-i} + \varepsilon_t; \varepsilon_t \sim \text{iid}(0, h_t) \quad (1a)$$

$$h_t = a_0 + a_1 h_{t-1} + \beta_1 \varepsilon_{t-1}^2 \quad (1b)$$

	Route 1A (16th January, 1997–31st July, 2000)	Route 2 (16th January, 1997–10th August, 2001)	Route 2A (16th January, 1997–10th August, 2001)
Panel A: Coefficient estimates			
Mean equation			
φ_0	0.012 (0.972)	–0.0002 (–0.425)	–0.0004 (0.512)
φ_1	–0.079* (–2.541)	0.065* (2.339)	0.099* (3.638)
Variance equation			
a_0	1.1E–05 (0.637)	1.7E–06 (0.964)	7.1E–06 (1.142)
a_1	0.969* (28.715)	0.981* (93.471)	0.970* (67.859)
β_1	0.011 (1.362)	0.013** (1.997)	0.021* (2.459)
Panel B: Residual diagnostic			
LL	2,079.5	3,032.7	2,532.7
Skewness	0.004	0.422	1.207
Kurtosis	13.260	13.704	16.567
J-B	3,925.8	5,519.8 [0.000]	9,090.9
Q(12)	15.550 [0.159]	15.288 [0.170]	19.313 [0.056]
Q ² (12)	5.046 [0.929]	4.165 [0.965]	5.131 [0.925]
ARCH(5)	0.259 [0.935]	0.242 [0.944]	0.712 [0.615]
ARCH(12)	0.386 [0.969]	0.339 [0.982]	0.417 [0.957]

Persistence	0.980	0.943	0.994	0.991
UV	0.000425	0.000772	0.000283	0.000211
Sign bias	-0.545 [0.586]	-1.115 [0.265]	-0.924 [0.356]	-0.635 [0.525]
Negative size bias	0.627 [0.531]	0.562 [0.575]	0.352 [0.725]	0.298 [0.766]
Positive size bias	0.270 [0.787]	-0.086 [0.932]	-0.565 [0.572]	-0.272 [0.786]
Joint test for three effects	0.161 [0.922]	0.496 [0.685]	0.623 [0.601]	0.215 [0.886]

Figures in parentheses () and in squared brackets [.] indicate *t*-statistics and exact significance levels, respectively. * and ** indicate significance at the 5% and 10% levels, respectively.

The GARCH process is estimated with the QMLE. The BHHH algorithm maximised the QMLE.

LL is the Log-Likelihood. J-B is the Jarque-Bera⁵² normality test. Q(12) and Q²(12) are the Ljung-Box⁵³ tests for 12th order serial correlation and heteroscedasticity in the standardised residuals and in the standardised squared residuals, respectively ARCH(.) is the Engle's⁵⁴ *F*-test for autoregressive conditional heteroscedasticity.

Persistence is defined as the degree of convergence of the conditional volatility to the unconditional volatility after a shock and is calculated as $\alpha_1 + \beta_1$.

UV is the unconditional volatility estimate of the GARCH models, measured as $(a_0)/(1 - \alpha_1 - \beta_1)$.

The test statistics for the Engle and Ng⁵⁷ tests are the *t*-ratio of *b* in the regressions: $\hat{e}o^2_t = a_0 + \alpha_1 Y_{t-1} + \omega_t$ (sign bias test); $\hat{e}o^2_t = a_0 + \alpha_1 Y_{t-1} \varepsilon_{t-1} + \omega_t$ (negative size bias test); $\hat{e}o^2_t = a_0 + \alpha_1 Y_{t-1} \varepsilon_{t-1} + \omega_t$ (positive size bias test), where $\hat{e}o^2_t$ are the squared standardised residuals $\hat{e}^2_t/\sigma_t$, Y_{t-1} is a dummy variable taking the value of one when ε_{t-1} is negative and zero otherwise, and $Y_{t-1} = 1 - Y_{t-1}$. The joint test is based on the regression $\hat{e}o^2_t = a_0 + \alpha_1 Y_{t-1} + a_2 Y_{t-1} \varepsilon_{t-1} + a_3 Y_{t-1} \varepsilon_{t-1} + \omega_t$.

The joint test $H_0: \alpha_1 = \alpha_2 = \alpha_3 = 0$, is an *F* test with 95% critical value of 2.60.

FFA, Forward Freight Agreement.

standardised residuals indicate that models are well specified with no asymmetries, and there are no linear and non-linear dependencies, respectively. The estimated implied kurtosis indicates the presence of excess kurtosis in the standardised residuals in all investigated routes. As a result, the Jarque-Bera⁵² test rejects normality in all routes. In routes 1 and 1A, the coefficients of the lagged variance (α_1) are significant, suggesting that there is a persistence effect in price volatility, while the coefficients of the lagged error-terms (β_1) are insignificant. In routes 2 and 2A, the coefficients of the lagged variance and the lagged error terms are significant at conventional significance levels. The persistence estimates of the conditional volatility reveal the presence of a near-integrated GARCH (IGARCH) process in all trading routes, with persistence estimates close to, but slightly less than, unity.²⁵

After estimating the GARCH(1,1) models and ensuring that they are well-specified, one step ahead conditional volatility estimates (h_{t+1}) are extracted for each trading route. The results of ADF and PP unit root tests on the daily one step ahead conditional volatility estimates indicate that the conditional volatility series are stationary for all routes (not reported). The BASs are then regressed against one step ahead conditional volatilities, current first-difference FFA returns and lagged BAS, to investigate the relationship between BAS and expected volatility. The results from the GMM regressions are presented in Table 8.4, panel A. The diagnostic tests indicate the existence of serial correlation and heteroscedasticity in most cases, and thus justify the use of the GMM approach. The adjusted *R*-squares of 0.711 for route 1, 0.782 for route 1A, 0.407 for route 2 and 0.683 for route 2A show that 71.1 per cent, 78.2 per cent, 40.7 per cent and 68.3 per cent, respectively, of the variation in daily BASs are explained by the independent variables.

Consistent with the findings in the literature, the coefficient on the GARCH variance forecast (β_1) is positive and statistically significant in routes 1, 2 and 2A, suggesting that expected volatility has predictive power in determining BASs through its effect on asymmetric information costs. This result was expected, as anticipated large price changes may be correlated with the presence of information traders, and FFA brokers might increase the BAS to compensate for expected losses when trading with informed traders. In terms of magnitude, the elasticity of BASs with respect to price volatility is higher in route 1 (20.409) than those in routes 2 (13.517) and 2A (5.839). These results are in accordance with Figures 8.1–8.4, which indicate that in routes 2 and 2A the BASs are significantly narrower than in route 1, as routes 2 and 2A concentrate most of the FFA trading interest in the Panamax sector. The finding of the β_1 coefficient being negative and insignificant in route 1A is in stark contrast to the findings in the literature, and is possibly explained by the infrequent FFA trading activity.

The coefficients of lagged BASs (β_2) are positive and significant at the 1 per cent level. This suggests that the dynamic adjustment of the BAS is not usually completed in a one day period for the selected forward contracts. The

Table 8.4 GMM estimates of the relationship between BAS and price volatility

$$\text{BAS}_t = \beta_0 + \beta_1 h_{t-1} + \beta_2 \text{BAS}_{t-1} + \beta_3 \Delta F_t + u_t; u_t \sim \text{iid}(0, h_t)$$

Explanatory variables	Route 1 (16th January, 1997–30th July, 2000)	Route 1A (16th January, 1997–30th July, 2000)	Route 2 (16th January, 1997–9th August, 2001)	Route 2A (16th January, 1997–9th August, 2001)
Panel A: Volatility measured as the conditional variance of GARCH models				
β_0	0.005 (1.095)	0.019* (3.295)	0.004* (6.313)	0.008* (4.452)
β_1	20.409* (2.176)	–3.524 (0.657)	13.517* (4.948)	5.839* (3.175)
β_2	0.827* (31.421)	0.884* (37.958)	0.573* (12.002)	0.801* (27.456)
β_3	–0.098 (–1.191)	0.0256 (0.333)	–0.052* (–2.081)	0.068 (1.317)
Diagnostics				
R^2	0.711	0.782	0.407	0.683
Q(12)	19.048 [0.087]	30.793 [0.002]	45.401 [0.000]	54.154 [0.000]
Q ² (12)	64.458 [0.000]	28.599 [0.005]	185.82 [0.000]	140.69 [0.000]
Panel B: Volatility measured as the rolling variances				
β_0	0.014* (6.611)	0.017* (5.235)	0.007* (8.929)	0.010* (7.639)
β_1	5.242* (2.640)	–0.120 (–0.121)	2.978* (2.773)	4.530* (5.039)
β_2	0.826* (29.962)	0.883* (36.913)	0.599* (15.206)	0.783* (29.699)
β_3	–0.099 (–1.220)	0.036 (0.472)	0.040* (1.993)	0.065 (1.252)
Diagnostics				
R^2	0.710	0.779	0.392	0.687
Q(12)	18.348 [0.106]	31.310 [0.002]	59.030 [0.000]	54.781 [0.000]
Q ² (12)	57.074 [0.000]	27.520 [0.006]	150.01 [0.000]	134.39 [0.000]

Figures in parentheses (.) and in squared brackets [.] indicate *t*-statistics and exact significance levels, respectively.

* and ** denote significance at the 5% and 10% levels, respectively.

Volatility, in panel A, is defined as the one-step ahead conditional variance of the FFA prices, computed from a well-specified GARCH(1,1) model.

Volatility, in panel B, is defined as the one-step ahead variance of percentage FFA price changes, computed over moving-windows of 20 days.

Q(12) and Q²(12) are the Ljung–Box⁵³ tests for 12th order serial correlation and heteroscedasticity in the residuals and in the squared residuals, respectively.

R^2 is the adjusted *R*-squared of the regression. The GMM method uses a weighting matrix ($A = \hat{\Omega}^{-1}$) that is robust to heteroscedasticity and autocorrelation of unknown form. The covariance matrix ($\hat{\Omega}$) is defined as: $\hat{\Omega} = \hat{\Gamma}(0) + \left(\sum_{j=1}^{T-1} k(j, q) [\hat{\Gamma}(j) - \hat{\Gamma}'(j)] \right)$, where, the kernel (*k*) is set to Bartlett functional form, and the truncation lag window (*q*) is set to Newey–West fixed bandwidth selection criterion. BAS, bid–ask spread; FFA, Forward Freight Agreement; GMM, general method of moment.

coefficient of the first-difference FFA price series (β_3) is found to be negatively significant in route 2 only. In the other three routes investigated, the β_3 coefficient is insignificant. This finding, in route 2, dominates the presence of any asymmetric information trading. It is, therefore, consistent with the presence of trading economies in the FFA market of route 2. It also supports the results of McNish and Wood¹³ for the stock market and of Ding¹⁰ for the

currency futures market. Copeland and Galai²⁴ argue that higher price levels in the stock market are associated with larger spreads because of a higher informational uncertainty due to bidding up of prices by informed traders. By contrast, the findings of lower spread levels when prices increase support the notion of the presence of economies of scale (when prices are high, the dollar volume of transactions rises, leading to a lowering of brokers' required BAS to cover their costs) in trading FFA contracts in route 2.

In order to verify the previous inferences, the relationship between BASs and volatility is further estimated where, as a measure of historical volatility, the one step ahead variances of daily percentage FFA price changes are used, computed over moving windows of 20 days (approximately one trading month). The results, presented in Table 8.4, panel B, are in accordance with previous results, as the coefficients of the statistically constructed measure of volatility (β_1) are positive and statistically significant in routes 2 and 2A. In route 1A, as expected, the β_1 coefficient is negative and insignificant. The coefficients of lagged BASs (β_2) are positively significant at the 1 per cent level and the coefficients of the first-difference FFA price series (β_3) are found to be negatively significant in route 2 only.

Conclusion

The microstructure of the freight forward market differs in several ways from that of the often examined derivatives markets, providing an interesting alternate market for developing and testing microstructure theories. This paper utilises a two-step model that attempts to explain some of the empirical regularities cited in the microstructure literature. The study contributes to the general literature by examining an OTC forward market, extending the concepts associated with forward prices to non-storable commodities (eg services), with no explicit storage relationship linking spot and forward prices. In addition, a feature of this market is higher transaction costs in spot compared with the FFA market. Some new evidence is provided on interactions between expected volatility and BASs from finding that FFA spreads vary with proxies for asymmetric information costs, including alternative risk forecasts. More specifically, results indicate that there is a positive relationship between BASs and expected price volatility in routes 1, 2 and 2A, after other factors are controlled. By contrast, in route 1A no significant relationship between BASs and expected volatility is observed. This finding may be explained by the thin trading of the FFA contracts in the latter route.

The results of this study can provide a better understanding of the movements of FFA prices and the consequent effect in transactions costs. Market agents using the information on the behaviour of the BASs can gain better insight into the timing of their FFA transactions and the future direction of the FFA market, as a widening BAS corresponds to an anticipation of increased future volatility. As a policy implication, FFABA should consider how its future

policy decisions may affect the volatility of the market and, consequently, the BASs. Although this study investigated and identified some key determinants of BASs in the freight forward market, it recognises the possibility that others may exist (ie trading volume). In general, however, risk is thought to be a stable determinant and is found to support those of previous studies.

References and notes

1. The credit risk associated with forward contracts can take the form of risk that occurs when one party is not performing, on the expiration date, the obligations relative to a change in the value of the forward contract from zero. If, during the life of the contract, the forward price continually mirrors the spot price, there is negligible credit risk associated with the forward contract and the contract can be sold at the market price.
2. In futures markets, the trader is required to place an initial margin with the clearing-house, which is an amount of money on a per contract basis and is set at a size to cover the clearing house against any losses which the trader's new position might incur during the day. Moreover, futures contracts are mark-to-market at the end of each trading day. That is, the resulting profit or loss is settled on that day. Traders are required to post a variation margin in order to cover the extent to which their trading positions show losses.
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Part II

Markets, Pricing and Products

9

Introduction of Futures and Options on a Stock Index and Their Impact on the Trading Volume and Volatility: Empirical Evidence from the DJIA Components

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Reprinted from *Derivatives Use, Trading & Regulation*, 11(3): 246–260, 2005, 'Introduction of Futures and Options on a Stock Index and Their Impact on the Trading Volume and Volatility: Empirical Evidence from the DJIA Components', by Mohammad G. Robbani and Rafiqul Bhuyan. With kind permission from Palgrave Macmillan Ltd. All rights reserved.

Dr Bhuyan's research interests are options and futures, information economics and empirical finance.

Practical applications

The results of this study show that the volatility of the market has significantly increased after the start of futures and options trading on the Dow Jones Industrial Average (DJIA) index. This finding will be of great interest to market participants, especially to those investors who trade stocks that are part of the DJIA index. The higher volatility, as evidenced in the study, should mean higher required rates of return on the underlying stocks. Since investors always look for a stock with minimum risk for a given level of return, the findings of this study will help them to determine the price of the target stock. The findings of this study can also be applied for any future stock index that will be traded in the options or futures markets. The findings on the increased daily trading volume after the introduction of futures and options can also be used by investors to make their investment decisions. A higher trading volume of an asset indicates a high level of liquidity which, in turn, ensures fair pricing. Therefore, it can be a great comfort for the investors that the stock will be priced fairly whenever they want to buy or sell.

The effect of the introduction of futures and options on the Dow Jones Industrial Average index on the volatility and trading volume of its underlying stocks is examined. Traditional measures and generalised autoregressive conditional heteroscedastic (GARCH) specification show that the levels of volatility and trading volume significantly increased after the introduction of futures and options on the index. The study provides new evidence in support of the argument that futures trading attracts uninformed or irrational traders along with the rational or informed traders, which causes an increase in stock return volatility. This may indicate that, even though the market may become more liquid, the destabilising effect brought by irrational traders in both the cash and futures markets outweigh the beneficial liquidity effect.

Introduction

Futures and options on the Dow Jones Industrials Average (DJIA) index have been trading on the Chicago Board of Trade since 6th October, 1997. As the most widely quoted and followed benchmark in the US stock market, the DJIA contains 30 blue chip stocks that have a total market value of more than \$2 trillion and represent roughly one-fifth of the total market value of all US stocks. (These estimates are made by the Chicago Board of Trade based on the market values of the Dow 30 stocks at the time when the index futures and options were introduced.) One of the major concerns for both practitioners and academics alike is how the introduction of the index

futures and options would affect the volatility of the underlying stock. The effect of stock index futures and options on the stock market has long been a subject of debate, and DJIA futures trading may help spark a new round of debate on this old issue. The different views on this issue can be no better illustrated than by the article published in the 'Abreast of the Market' column of the *Wall Street Journal* on the first trading day of the index futures and options. The front page article reported that Michael Schwartz, Chief Options Strategist at Oppenheimer & Co., believed that the volatility would increase tremendously. Patrick Catania, Executive Vice President at Chicago Board of Trade, argues otherwise, saying that volatility would not increase and the opposite was likely to be true. (For details, see the article on page A1 of the *Wall Street Journal*, 7th October, 1997.)

The purpose of this paper, therefore, is to address whether the introduction of futures and options on DJIA has any impact on the volatility and volume of the DJIA index components. Using both parametric and non-parametric methods, the paper empirically examines whether the introduction of DJIA futures and options trading correlates with any significant change in the volatility and trading volume of the underlying stocks.

A large number of theoretical and empirical studies have examined the effect of stock index futures and options on the volatility of the underlying spot market. Although the academic findings of these studies thus far have not produced any conclusive evidence on this issue, the popular public perception seems to be that index futures and options trading will increase the volatility in the stock market. The negative perception of the stock index futures and options has especially been strengthened after the highly publicised financial debacles of Orange County and Barings Inc.

Theoretically, the impact of stock index futures and options on the stock market volatility is still not clear. The linkage between these derivatives markets and the stock market is generally established through arbitraging activities. The results, however, depend, to a large extent, on what assumptions are made about the arbitrageurs. The important assumption is whether index futures and options trading brings in more informed or uninformed investors to the stock market. One school of thought argues that arbitraging or speculating activities in the futures markets add more informed traders to the stock market, thereby increasing the liquidity and decreasing the volatility of the market. The other school of thought asserts that index futures and options introduce more uninformed or irrational traders in both the derivatives and stock markets in search of short-term gains, therefore increasing the stock market volatility. Stein¹ develops a model and shows that futures trading by poorly informed investors or speculators in fact destabilises the stock market and therefore increases its volatility. By contrast, Danthine² presents a model which implies that futures markets help improve market depth and reduce volatility, since the cost responding to mispricing is reduced for informed traders. In another study, Weller and Yano³ use a general

equilibrium model to study the effect of stock index futures trading on the volatility of the stock market and conclude that stock market volatility may decrease when index futures are introduced.

The empirical findings on this issue have also been inconclusive. Edwards⁴ finds that the market volatility in the S&P500 was greater before the advent of index options and futures trading than that after. Bessembinder and Seguin⁵ indicate that active futures markets are associated with decreased rather than increased stock market volatility. Schwert⁶ shows that the introduction of stock index futures and options has not increased the stock market volatility. More recent studies, such as that of Pericli and Koutmos,⁷ have provided further evidence that index futures and options do not lead to increases in stock market volatility. Some other studies, by contrast, have documented evidence supporting the opposite argument. Maberly *et al.*⁸ argue that the introduction of stock index futures increases the volatility of the stock market. Brorsen⁹ also shows that stock market volatility increases after the advent of index futures trading, which causes a reduction in autocorrelations in the spot and derivatives markets. Investigating this issue for international markets, researchers have examined some major foreign stock index futures and options and found results similar to those obtained by Maberly *et al.*⁸ In short, these studies show that the underlying stock market volatility tends to increase after the introduction of futures and options trading on the market indices. Rahman¹⁰ examines the impact of trading in DJIA index futures and options on the conditional volatility of component stocks. His result suggests that there is no structural change in the conditional volatility due to the introduction of futures and options. When option listing is studied on individual stocks, there are significant number of studies, including Conrad,¹¹ Detemple and Jorion,¹² Kumar *et al.*,¹³ Sorescu,¹⁴ and Pilar and Rafael,¹⁵ which argue that option listing has a significant effect on price and volatility.

Similar to other stock market indices, the effect of DJIA index futures and options trading on the underlying stocks should be an empirical one. The empirical results should be of great significance to general investors. If the market structure after the index futures and options trading remains the same, higher volatility in the market should mean higher required rates of return on the underlying stocks. It is possible that the change in volatility, if any, may be the result of some market structural changes rather than the introduction of index futures and options. This study does not intend to address the causality of this issue. Rather, this research focuses on investigating whether, empirically, the volatility and trading volume of the underlying stocks have increased significantly since the introduction of futures and options trading on the DJIA index. Because of the inconclusive empirical results from the studies on other stock indices, the results of this study on the DJIA index should provide additional insights into the existing literature.

Data and Methodology

The information on the 30 DJIA stocks is obtained from the online Dow Jones News Retrieval System. The data items for these stocks include daily high, low and closing prices and daily trading volumes. The data period covers 7th October, 1996 to 6th October, 1998, in order to have exactly one year of data available for pre-futures and post-futures trading with 253 and 254 trading days, respectively.

The daily returns on the 30 individual stocks are computed based on the daily closing prices. Both the traditional method of volatility estimation and the generalised autoregressive conditional heteroscedastic (GARCH) technique are used. Using the traditional method, both parametric and non-parametric tests are applied to identify whether there is any shift in the volatility after futures and options trading was introduced on the DJIA index. The volatility is measured as the standard deviation of the daily returns for individual stocks. The average trading volume of pre-futures and post-futures periods for all individual stocks are computed and tested for any significant shift by using both parametric and non-parametric techniques. Both prices and trading volume are adjusted for stock splits that took place during the period indicated above. In addition, a volatility measure which considers the daily high and low prices is also employed. This technique, known as Parkinson's efficient variance estimator, takes into consideration the daily high and low prices and is measured as:

$$\frac{[\ln(H_t) - \ln(L_t)]^2}{4 \ln 2} \quad (1)$$

where H_t and L_t are the daily high and low prices, respectively. This measure provides additional insights into the understanding of stock return volatility through intra-day price movements, rather than the daily closing price changes. Table 9.1 provides some basic returns and volatility measures for the underlying stocks.

The statistical methods used to test the impact of futures and options trading on the underlying stocks include the two-sample F -test for the two measures of volatility (standard deviation and high-low variance), and the t -test for the return and trading volume. These techniques, which test whether there is any significant change in the average daily return, average volatility and average daily trading volume, are applied for each of the individual stocks. Then, to test whether there is a significant change in return, volatility and trading volume during the post-futures period compared with the pre-futures period for the 30 stocks as a group, the paired t -test and the Wilcoxon signed-rank test were applied. In order to perform these tests, average return, standard deviation of stock returns, daily high-low variance and the mean trading volume before and after the DJIA index

Table 9.1 Basic statistics for the 30 Dow Jones Industrial Stocks^a

		Whole period	Pre-futures period	Post-futures period
Returns	Mean	0.0713	0.1349	0.0008
	SD	1.9377	1.6867	2.1481
	Minimum	-0.0530	-0.0402	-0.1450
	Maximum	0.1819	0.2599	0.2146
Volume	Mean	2,318,495	2,044,339	2,591,568
	SD	1,371,120	1,288,596	1,528,781
	Minimum	504,136	447,204	560,845
	Maximum	5,598,343	5,177,039	6,507,393

^aThe whole time period covers 17th March, 1997–11th March, 1998. The pre-futures period is from 17th March, 1997 to 5th October, 1997, and the post-futures period is from 7th October, 1997 to 11th March, 1998. The measures for returns are percentages and the measures for volumes are in hundreds. Both these measures are on a daily basis.

futures trading date are computed for each stock. The difference between these two tests is that the Wilcoxon test does not require the normality assumption for the testing variable, whereas the *t*-test does. Both tests are applied to check the robustness of the analysis irrespective of the underlying assumptions.

As some of the studies indicate that volatility is created because of the activity of the speculators in the market, it is important to investigate how flow of information and volatility are related. Most of the previous studies, however, do not explicitly investigate this relationship. Since with the change of information volatility of stock price also changes, we attempt to model this relationship as a conditional variance using the GARCH model developed by Engle¹⁶ and Bollerslev.¹⁷ Unlike ordinary least squares regression, GARCH specifically incorporates the conditional variance as a linear function of the lagged squared residuals and lagged conditional variances. In other words, this specification shows how current volatility is affected by past volatility. A GARCH model with orders *p* and *q* can be written as

$$R_t = \lambda_0 + \lambda_1 R_{mt} + \varepsilon_t, \varepsilon_t | \Psi_{t-1} \sim N(0, h_t) \quad (2)$$

$$h_t = \alpha_0 + \sum_{i=1}^p a_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j h_{t-j} \quad (3)$$

where Equation (2) is the conditional mean equation, and Equation (3) is the conditional variance equation. R_t is the individual stock return and R_{mt} is the market portfolio return. The daily percentage change in S&P500 index is used to proxy the market portfolio. (Since we are testing the overall market reaction due to the introduction of futures and options trading on the DJIA index, the S&P500 index was chosen as the appropriate proxy for the market portfolio.) Ψ_{t-1} is the past information set, and h_t represents the variance. The residual

term ε_t is conditional on the information set and is assumed to be normally distributed with mean zero and variance h_t . Both Equations (2) and (3) are jointly estimated by maximising the log-likelihood function. GARCH(1,1) is applied to the process to estimate the measures of volatility. (In the empirical model, GARCH(1,1) was used, since the log-likelihood ratio with $p=1$ and $q=1$ is the maximum. Additionally, using GARCH(p,q) with $p+q \geq 3$ produces unstable estimates and the marginal benefit is very small.) Under this specification, the values of a_0 , a_1 and β_1 show whether the immediate past information is influencing the present volatility of stock returns.

These parameters are estimated for the sub-samples of both pre-futures and post-futures periods. If the values of a_0 , a_1 and β_1 for the post-futures period are significantly higher than those for the pre-futures period, this may indicate that there has been an increase in the way current variance is affected by the past. This kind of shift in the return generating process is generally due to the presence of some important event, which can alter the way the information is processed in the market.

Empirical Test Results

Traditional results

Tables 9.2–9.6 show the results of the tests which examine whether there is any change in return, volatility and trading volume during the post-futures period compared with the pre-futures period for the DJIA stocks. Table 9.2 shows the results of the F -test, comparing the volatility of the 30 individual stocks before and after index futures trading. Two types of volatility measures are specified: variance of daily individual stock returns and the Parkinson's daily high–low variance estimator. The F -test is used to examine whether the post-futures variance has increased significantly by comparison with the pre-futures variance. The p -values for these tests are provided to show the significance of the tests. The results show that volatility—as measured by daily stock return variance—increased significantly after introducing the index futures and options on the DJIA. Of the 30 underlying individual stocks, 28 stocks showed an increase in stock return. Of these 28 stocks, the increase in volatility of 25 stocks is significant at the five per cent level and of the two stocks whose volatilities decreased, only one stock is significant at the five per cent level. To measure the economic significance, the figures in Table 9.2 are converted to standard deviations, which is the commonly used measure of stock return volatility. The average increase in volatility for all 30 stocks from the pre-futures to post-futures period is about 62 per cent. The average increase in volatility for the 28 stocks is about 70 per cent, while the average decrease for the two stocks is about 18.5 per cent. These results indicate that the volatility of the underlying stocks increased significantly after the introduction of DJIA index futures and options.

To examine whether the increased volatility is accompanied by any change in returns, the two-sample t -test is used to compare the mean returns

Table 9.2 Individual DJIA stocks' volatility before and after DJIA index futures trading^a

Company	Variance (close-to-close)			Variance (high-low)		
	After	Before	p-value	After	Before	p-value
Alcoa	5.69	2.13	0.0000	2.42	1.82	0.0017
Allied Signal	4.16	2.26	0.0000	4.77	1.84	0.0000
American Express	6.37	3.59	0.0000	4.15	2.64	0.0021
ATT	5.13	3.02	0.0000	3.31	2.58	0.0339
Boeing	6.11	2.95	0.0000	3.86	1.96	0.0002
Caterpillar	5.95	3.07	0.0000	4.42	1.94	0.0000
Chevron	3.26	2.09	0.0002	2.53	1.76	0.0000
Coca-Cola	3.89	2.73	0.0027	3.20	2.04	0.0038
Disney	5.23	1.89	0.0000	3.62	1.66	0.0002
DuPont	5.40	3.00	0.0000	4.13	2.50	0.0000
Exxon	2.69	2.29	0.0976	2.57	2.17	0.0781
General Electric	3.44	2.39	0.0020	3.02	2.02	0.0014
General Motors	3.72	2.28	0.0000	2.74	1.84	0.0000
Goodyear	3.63	1.50	0.0000	2.56	1.46	0.0000
HP	6.46	4.96	0.0186	4.58	3.88	0.0354
IBM	4.85	3.82	0.0306	3.23	2.86	0.1469
International Paper	4.89	3.27	0.0009	3.94	2.66	0.0008
Johnson & Johnson	2.83	3.12	0.2196*	2.44	2.48	0.4368*
Kodak	3.88	3.68	0.3340	2.85	2.08	0.0026
McDonald's	3.97	1.93	0.0000	2.70	1.67	0.0005
Merck	3.99	3.30	0.0670	2.94	2.36	0.0444
MMM	3.21	2.05	0.0002	2.33	1.63	0.0002
Morgan, J.P.	6.34	2.31	0.0000	4.00	1.66	0.0000
Philip Morris	3.85	5.08	0.0142*	2.87	3.95	0.0392*
Proctor & Gamble	4.40	2.70	0.0000	3.36	2.24	0.0005
Sears	6.42	3.21	0.0000	4.51	2.34	0.0000
Traveler	9.03	5.18	0.0000	5.80	3.74	0.0005
Union Carbide CP	4.06	2.25	0.0000	3.21	1.65	0.0000
United Technology	3.62	2.02	0.0000	2.41	1.79	0.0012
Wal-Mart	4.89	3.28	0.0008	4.42	2.56	0.0002
Average	4.61	2.84	0.0000	3.42	2.26	0.0000

^aThe post-futures and pre-futures variances are computed over the periods after and before the introduction of index futures trading using daily stock returns. The bold *p*-values indicate the significance of the tests at the 5 per cent level. The *p*-values with an asterisk imply that the volatility decreased after index futures trading.

before and after the index futures trading. Contrary to the evidence found in the case of volatility change, most of the stocks in this research show a significant decline in the return during the post-futures period compared with the pre-futures period. Since the main objective is to investigate the effect on the volatility and trading volume, the test results on the change in the

Table 9.3 Individual DJIA stocks' trading volumes before and after DJIA index futures trading^a

Company	Post-futures volume	Pre-futures volume	p-value
Alcoa	948,890	706,878	0.0000
Allied Signal	1,582,440	639,646	0.0000
American Express	1,558,040	1,403,384	0.0135
ATT	5,361,527	5,177,039	0.2083
Boeing	4,472,504	2,262,191	0.0000
Caterpillar	1,536,617	813,816	0.0000
Chevron	1,399,375	1,178,249	0.0000
Coca-Cola	3,559,766	3,397,542	0.1239
Disney	3,041,902	1,389,308	0.0000
DuPont	2,968,578	1,663,325	0.0000
Exxon	3,294,694	2,436,233	0.0000
General Electric	4,828,320	3,564,899	0.0000
General Motors	2,577,379	2,479,568	0.1239
Goodyear	560,844	447,204	0.0000
Hewlett Packard	3,409,062	3,154,183	0.0577
IBM	4,258,971	3,823,329	0.0038
International Paper	1,337,568	1,243,394	0.0380
Johnson & Johnson	2,323,359	2,494,191	0.0073*
Kodak	1,528,392	1,521,012	0.4773
McDonald's	2,275,683	2,326,919	0.3017*
Merck	3,035,837	2,795,800	0.0154
MMM	1,050,375	920,649	0.0004
Morgan, J.P.	1,035,618	705,091	0.0000
Philip Morris	6,507,393	4,685,686	0.0000
Proctor & Gamble	2,193,536	1,271,451	0.0000
Sears	1,862,932	1,416,620	0.0000
Traveler	4,584,587	2,495,974	0.0000
Union Carbide CP	685,114	566,608	0.0003
United Technology	830,756	651,660	0.0000
Wal-Mart	3,136,967	3,698,013	0.0000
Average	2,591,568	2,044,339	0.0000

^aThe average daily trading volumes in post-futures and pre-futures periods are computed over 220 trading days around the introduction of the DJIA index futures. The bold *p*-values indicate the significance of the tests at the 5 per cent level. The *p*-values with an asterisk imply that the volatility decreased after index futures trading.

return of individual stocks are not presented here. The results from the pair-wise *t*-test and the Wilcoxon test for the sample are provided in Table 9.4.

Table 9.2 also provides the results from the Parkinson's efficient high-low variance estimator as a measure of stock return volatility. The results are similar to those found using the standard deviation technique. The results from Parkinson's method are consistent with those from standard deviation in the sense that the same 28 stocks showed an increase and two stocks showed a decrease in volatility. The only difference is that the increase in

Table 9.4 Comparison of return, risk and volume before and after futures trading^a

	Post-futures	Pre-futures	Wilcoxon statistic	<i>p</i> -value of paired <i>t</i> -test
Returns	0.08	13.49	-4.27	0.00
SD	2.15	1.69	4.62	0.00
Volatility-hi/lo	3.43	2.26	4.45	0.00
Trading volume	2,591,568	2,044,339	4.12	0.00

^aTwo volatility measures have been used both for pre- and post-futures periods. The first is the standard deviation of the daily stock returns based on daily closing prices and the second is the Parkinson's efficient high-low variance estimator. The bold numbers indicate that the difference is significant at least at the 5 per cent level.

Table 9.5 Estimation from GARCH(p,q) model^a

Company	Post-futures			Pre-futures		
	α_0	α_1	β_1	α_1	α_1	β_1
Alcoa	0.00006	0.1866	0.7209	0.00036	-0.0246	-0.7079
Allied Signal	0.00008	0.2268	0.5895	0.00038	-0.0448	-0.7014
American Express	0.00005	0.1837	0.7355	0.00056	0.0926	-0.6757
ATT	0.00035	0.3783	-0.0253	0.00052	0.0659	-0.8218
Boeing	0.00020	0.1467	0.5327	0.00048	-0.0228	-0.6161
Caterpillar	0.00017	0.2250	0.5029	0.00022	0.1469	0.1427
Chevron	0.00008	0.0930	0.6726	0.00018	-0.0427	0.1609
Coca-Cola	0.00010	0.4273	0.3476	0.00023	0.1825	-0.0170
Disney	0.00029	0.2431	0.1779	0.00024	0.0633	-0.3380
DuPont	0.00038	0.0673	0.2246	0.00009	0.1786	0.5055
Exxon	0.00008	0.2078	0.5131	0.00003	0.1355	0.2152
General Electric	0.00002	0.2040	0.7417	0.00042	0.0537	-0.8254
General Motors	0.00007	0.1389	0.6831	0.00041	0.05379	-0.8984
Goodyear	0.00021	0.2650	0.1685	0.00007	0.02480	0.4894
HP	0.00098	-0.0269	-0.4976	0.00046	0.0798	0.0000
IBM	0.00011	0.3686	0.4484	0.00019	0.03037	0.4774
International Paper	0.00034	0.3648	-0.0270	0.00001	0.04333	0.9342
Johnson & Johnson	0.00008	0.0960	0.6365	0.00024	0.2189	-0.0083
Kodak	0.00053	0.0792	-0.4725	0.00051	0.0716	-0.4736
McDonald's	0.00032	0.1069	-0.1154	0.00230	0.0799	-0.2704
Merck	0.00023	0.1389	0.2748	0.00011	-0.0494	0.7266
MMM	0.00041	-0.0236	-0.2618	0.00043	0.0454	-0.9960
Morgan, J.P.	0.00002	0.0623	0.9073	0.00001	0.04021	0.9318
Philip Morris	0.00012	0.1828	0.4989	0.00034	0.1273	0.2021
Proctor & Gamble	0.00024	0.0985	0.6786	0.00027	-0.0575	0.0544
Sears	0.00069	0.0712	-0.1610	0.00003	0.0546	0.8378
Traveler	0.00034	0.0184	0.4621	0.000,017	0.03548	0.9329
Union Carbide	0.00006	0.0792	0.7852	0.00016	0.0817	0.2000
United Technology	0.00002	0.1503	0.8099	0.00018	0.1023	0.0088
Wal-Mart	0.00032	0.5088	-0.1212	0.00035	0.1830	-0.2364
Average	0.00023	0.1756	0.3477	0.00033	0.0650	-0.0256

^aThis table presents the estimated parameters of the GARCH model. GARCH(1,1) is estimated for each of the stocks in the DJIA index. The daily percentage change in S&P500 index value is used as the explanatory variable. The bold parameters are significant at least at the 5 per cent level.

Table 9.6 Comparison of GARCH(1,1) parameters before and after futures trading^a

	Post-futures	Pre-futures	Wilcoxon statistic	<i>p</i> -value of paired <i>t</i> -test
a_0	0.00023	0.00033	-1.3367	0.1245
a_1	0.1756	0.0650	3.5686	0.0000
β_1	0.3477	-0.0256	2.6019	0.0036

^aThe parameters of the GARCH(1,1) model are tested using both parametric and non-parametric tests. The tests used are the *p*-value of the paired *t*-test and the Wilcoxon statistic. The bold numbers indicate that the difference between post-futures and pre-futures parameters is significant at least at the 5 per cent level.

the volatility of IBM is significant using the standard deviation method, while it is insignificant using Parkinson's method, while Merck has an insignificant increase using standard deviation method, but increases significantly using Parkinson's method. The average increase in volatility for all 30 stocks is about 51.80 per cent. The average change in volatility for the 28 stocks with increased volatility is 59.05 per cent, while the average change for two stocks with decreased volatility is -17.42 per cent.

Table 9.3 presents the results of the *t*-test, examining the effect of introducing index futures and options on the trading volume of the underlying individual stocks. It is interesting to see whether index futures and options trading have led to increases in the trading volumes of the underlying stocks. The test may also help ascertain whether the increased volatility can be partially attributed to an increase in trading volume. The trading volumes for the stocks are adjusted for stock split factors over the time period. That is, if a stock has a 2-for-1 stock split, the trading volume for that stock is multiplied by a factor of 2 for trading volumes before the occurrence of the split. Then, for each stock, a two-sample *t*-test is applied to examine whether the average trading volume after the introduction of the index futures has increased significantly. The *p*-values of these tests are shown in the table with significant values at the five per cent level in bold. As in Table 9.2, an asterisk indicates that the post-futures trading volume has decreased for that stock. Table 9.3 shows that average trading volume increased for 28 stocks and decreased for only two stocks after index futures trading started. Of the 28 stocks, the increases for 23 stocks are significant at the five per cent level; of the two stocks that decreased, the decrease for one stock is significant at the five per cent level. The average increase in the daily trading volume from the pre-futures to post-futures period for all 30 stocks is about 26.77 per cent. The average change for the 28 stocks with increased daily volume is 29.44 per cent, while the average decrease in daily trading volume for the remaining two stocks is -4.61 per cent. These results strongly indicate that the daily trading volume of the underlying individual stocks have increased significantly after the introduction of the DJIA index futures and options.

The pairwise *t*-test and the Wilcoxon test were also used to test the significance of the difference in post-futures and pre-futures values of return,

standard deviation, high-low variance and trading volume for the 30 stocks. These results are presented in Table 9.4. Jointly analysing Tables 9.2, 9.3 and 9.4, one finds that there is an evident positive association between the change in trading volume and the change in volatility. Of the 28 stocks showing an increase in volatility in return, 27 stocks also show an increase in daily trading volume. There is only one stock (McDonald's) that shows an increase in volatility but a decrease in trading volume and one stock (Philip Morris) that shows a decrease in volatility with an increase in trading volume. These results provide strong support for the argument that volatility and trading volume have a positive association and are consistent with the findings from previous studies that greater volatility may lead to greater trading volume (see, for example, Bessembinder and Seguin⁵).

Econometric results (GARCH model)

Given the evidence that, using traditional methods, that there has been an increase in the volatility of underlying stocks after futures trading on the DJIA index, further testing of how past information and volatility may have influenced the current volatility of those stocks will shed additional light. For this purpose, the GARCH(1,1) model was used. The parameters of the variance equation (Equation 3) provide explanations on how past information and volatility have influenced current volatility. (The parameters of the return equation [Equation 2] are not presented here, since the objective in this section is to test how past information affects current volatility.) The estimated parameters for a_0 , a_1 and β_1 of both pre-futures and post-futures periods are presented in Table 9.5. Any change in a_0 during the post-futures period compared with the pre-futures period shows whether there is a change in unconditional variance. The average change in the value of a_0 is -0.0001 , which is not statistically significant, implying that the unconditional variance has remained the same since the start of futures trading on the DJIA index. During the post-futures period, however, 17 stocks showed significant values of a_0 compared with 13 stocks during the pre-futures period.

Similarly, a change in the value of a_1 relates to the change in the impact of past news on the current stock price changes. For example, an increase in a_1 indicates a greater impact of yesterday's news on the current price changes. The average change in the value of a_1 is 0.1106 , with the p -value of the paired t -test equal to zero and the Wilcoxon statistic equal to 3.57 . The number of stocks with significant values of a_1 during the post-futures period was 20, whereas only six stocks had significant values of a_1 during the pre-futures period. These results imply that the impact of past news on the current price change has increased significantly during the post-futures period compared with the pre-futures period. This shift, again, can take place only when any major event occurs in the market. One such event is the introduction of futures trading on the DJIA. a_1 is the coefficient that relates to the lagged

error term ($p=1$); therefore, the increase in the value of α_1 means that the effect of yesterday's difference in expected and actual return has increased today's volatility of stock returns. In other words, the change in the value of α_1 shows that today's stock return volatility is caused by the yesterday's information changes.

Conversely, β_1 is the coefficient of lagged variance ($q=1$) which reflects the information prior to yesterday's price changes and thus shows the effect of the news that arrived before yesterday. The average increase in the value of β_1 is 0.3733, which is significant using both the t -test (p -value 0.0036) and the Wilcoxon test (2.60). The number of stocks with a significant value of β_1 is 19 during the post-futures period compared with ten during the pre-futures period. These results imply that past information has a significant impact on the change in volatility of underlying stocks after the introduction of futures trading on the DJIA. Therefore, combining the results of α_1 and β_1 , the implication is that there has been a significant change in the process of volatility formation of the underlying stocks due to futures trading on the DJIA.

As a whole, the GARCH test shows a significant increase in the volatility during the post-futures period compared with the pre-futures period. The results of the pairwise t -test and Wilcoxon test for α_0 , α_1 and β_1 for the pre-futures and post-futures periods are presented in Table 9.6. As shown in the table, the average difference in α_0 between the post- and pre-futures periods is not statistically significant, whereas the average differences in α_1 and β_1 are highly significant. As mentioned earlier, α_0 is the measure of unconditional volatility and α_1 and β_1 are measures of conditional volatility. Even though the unconditional variance did not change after the introduction of futures trading, the conditional variances have increased significantly. This result reinforces the contention that the increase in volatility found using the traditional volatility estimation is due to the increase in the conditional volatility, ie the volatility that is induced by some important event such as futures trading on the DJIA.

Conclusion

A large number of prior studies have investigated the impact of derivatives on the financial markets, especially on the underlying assets. Results of research of both a theoretical and an empirical nature are inconclusive. The introduction of DJIA futures and options in October, 1997, provided an opportunity to investigate this issue empirically in order to bring more insights into this debate.

The empirical results of this study show that the volatility of the market significantly increased after the start of futures and options trading on the DJIA index. On an individual basis, a majority of the 30 DJIA stocks showed an increase in volatility after the introduction of futures and options on

the index, with significant increases for a large number of them. On a sample basis, both parametric and non-parametric tests displayed significant increases in volatility measures after trading of futures and options on the index. Efforts were made to study whether stock returns and trading volume also increased over this period. A significant increase in the daily trading volume was found for 23 stocks in the index after the introduction of futures and options trading. There is evidence, however, that the average daily rate of return on these 30 underlying stocks decreased significantly.

The increase in the volatility found using a traditional measure is also tested using the GARCH(1,1) model to identify the process of volatility formation. The results from the GARCH model are similar to results from the traditional measure. This shows that the impact of past news (a_1) and past variance (β_1) on the current price change is significantly higher during the post-futures period than during the pre-futures period. The difference in the values of a_0 that measures the unconditional variance between the two sub-periods, however, is not statistically different. Therefore, the results from the GARCH model indicate that the volatility increase during the post-futures period is due to the increase in conditional volatility, not due to the change in unconditional volatility.

The empirical analysis of this study provides new evidence in support of the argument that futures trading attracts uninformed or irrational traders along with the rational or informed traders, which causes an increase in stock return volatility. This may indicate that, even though the market may become more liquid, the destabilising effect brought by irrational traders in both cash and futures markets outweigh the beneficial liquidity effect. It is possible that the increase in volatility may be due to some structural changes in the financial market as a whole rather than to the introduction of futures and options trading, which is worth further study. Nonetheless, this study provides new findings about the volatility change of the 30 Dow Jones Industrial stocks after futures and options trading on the index. It also provides further insight into the effects of derivatives on the underlying stocks.

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10

The Characteristics and Evolution of Credit Default Swap Trading

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Practical applications

This paper analyses the characteristics of credit default swap (CDS) trading. The findings inform market participants of the existence of clustering in the terms of CDS contracts, and imply possible means for improved order execution by avoiding rarely used terms. The findings of seasonality in CDS trading activity are informative in relation to the timing of trades.

This is the first systematic empirical study of the characteristics and the evolution of credit default swap (CDS) trading. We study single name reference entities and find a prevalence of five-year maturities, US\$5 million and US\$10 million notional amounts, senior-ranked underlying debt and modified restructuring clauses. We find increased trading up to 2003 and an increased ratio of trades to quotes in recent years. A decrease in daily volume since 2004 is likely to be caused by the recent increased use of CDS index-related products. The

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majority of reference entities have credit ratings from at least one international rating agency, and the average credit quality of the universe fluctuates during the sample period. Trading activity demonstrates an inverse U shape across the week.

Introduction

Credit derivatives allow users to manage credit risk by isolating credit risk from underlying financial assets. Although small relative to other derivative and security markets, the credit derivative market has become one of the fastest-growing derivative markets since the late 1990s. The British Bankers Association's (BBA) survey suggests that the global credit derivatives market increased in size (measured by notional amount outstanding) from around US\$151 billion in 1997 to US\$514 billion in 1999.¹ Credit default swaps (CDS) account for a large share of the market. According to the BBA survey,¹ around 50 per cent of the market was in CDS, while Patel² finds that CDS account for 72 per cent of the market. The International Swaps and Derivatives Association's (ISDA) market survey indicates that the outstanding notional amount of CDS was US\$12.43 trillion by the end of June 2005.³ Because of the market's rapid growth, CDS have attracted increasing attention in academic research. In recent years, there has also been an increase in CDS index trading, which has provided further depth to the market.

There have been a number of recent empirical studies on CDS that focus on recovery forms, pricing determinants, performance of pricing models⁴ and interaction between the CDS market and other markets.⁵ Commonly, only CDS premiums are used in these empirical studies, although some features of CDS such as maturity and notional amount may be considered in data filtering. Market information other than the actual prices of CDS transactions is largely ignored. However, information such as the rank of reference assets under CDS and the time when a quote is placed or a trade is closed, as well as time-varying features such as changes in volume can be valuable to the understanding of the characteristics of the CDS market and how it has been evolving. This paper aims to explore these previously under-researched aspects.

There are sporadic descriptions of CDS data in the literature. To the best of our knowledge, there is only one previous (nonacademic) empirical study by FitchRatings that systematically analyses the properties of CDS data.⁶ FitchRatings uses data obtained from GFI and Creditex, two leading brokers in the CDS market, to analyse (i) growth in the number of reference entities quoted (as an alternative measure of market growth⁷); (ii) trends in bid-ask spreads (as a proxy for liquidity) and the degree to which liquidity varies over time; and (iii) overall trends in trading efficiency (measured by the ratio

of the number of bid prices to the number of offer prices). The time period of their sample is from January 2002 to September 2004.

Aunon-Nerin *et al.*⁸ use CDS data from an undisclosed broker. They describe (i) the industry distribution of reference entities; (ii) the top ten corporate and top ten sovereign reference entities between January 1998 and February 2000. Houweling and Vorst⁹ describe the distribution of currency, notional amount and maturity of CDS between May 1999 and January 2001. Their data are obtained from Creditex and CreditTrade (a broker), and from a number of commercial and investment banks. Ericsson *et al.*¹⁰ graphically depict the daily frequency of bid and offer quotes and the distribution of maturities of a CreditTrade CDS data set from January 1999 to December 2002. Hull *et al.*¹¹ describe the maturity profile and report the annual number of bid and offer quotes, annual number of reference entities and the level of CDS premiums. They use data from GFI from January 1998 to May 2002. In summary, all the above literature reports the characteristics of CDS data before 2003.

This paper is the first systematic empirical study dedicated to analysis of the characteristics of CDS data. By analysing statistical features of the key factors that define a CDS transaction, we provide a distinctive insight into the practice of CDS trading. Further, the analysis of the time-varying features of CDS trading and evidence on the microstructure features will lead to a clearer understanding of the evolution of the CDS market. The findings may also provide supportive evidence to the increasing importance of CDS index trading.

The CDS data used in this paper are a sample of American market quoted prices (hereafter referred to as 'quotes') and traded prices (hereafter referred to as 'trades') from the Market Prices database of CreditTrade.¹² CreditTrade divides Market Prices into America, Europe, Asia and Japan market prices. American market prices are defined as prices on reference entities from North America and Latin America. Therefore, the prices are not necessarily obtained from the US. The whole data sample used in this paper covers the period from 10th June, 1997 to 3rd March, 2005. The total number of reference entities in the data set is 945, including 72 banks, 846 corporates and 27 sovereigns. Risk Magazine reports that GFI, Creditex and CreditTrade were the most well-known CDS data providers at that time.¹³ The CDS data service of CreditTrade covered 1,500 reference entities, and the coverage of Creditex and GFI's data services was 1,400 reference entities and 1,700 reference entities, respectively. This indicates that CreditTrade is among the leading participants in this CDS data services sector. Also, our data sample represents a substantial proportion of CreditTrade activity.

The remainder of this paper is organised as follows: The next section investigates the profile of the CDS market by examining the embedded features of CDS; the subsequent section analyses the availability of credit ratings for the reference entities quoted or traded in the market; the penultimate

section explores trading activity and CDS indices and the final section concludes the paper.

CDS market profile

Currency, rank and restructuring

Table 10.1 presents results discussed here. The US dollar is the prevailing currency for American reference entities (North America and Latin America), accounting for more than 99 per cent of all the CDS in our database. There are also a small proportion of CDS contracts denominated in euro (about 0.1 per cent). CDS denominated in other currencies, that is Japanese yen, Singapore dollar, Thai Baht and Caribbean dollar, are negligible. Aunon-Nerin *et al.*⁸ and Houweling and Vorst⁹ also report US\$ as the most frequently quoted currency.

In the whole sample, 98.3 per cent of reference debt is senior-ranked, while only 1.5 per cent is subordinated debt¹⁴ (see Table 10.1). This phenomenon may be due to several factors. First, subordination involves analysis of sensitivity of the default to the value of the debt, which makes the pricing of CDS more complex.¹⁵ Secondly, FitchRatings suggests that structured finance securities usually involve a list of 'soft' credit events,¹⁶ and ISDA is still developing standardised structured finance documentation.

Table 10.1 also shows the use of restructuring clauses in CDS quotes. The prevailing use of modified restructuring (MR) in the American entities' market is consistent with FitchRatings' finding. MR is used in the majority of cases, followed by Old Restructuring (OR), which accounts for 6.8 per cent in the whole sample.¹⁷

Table 10.1 also identifies the distribution of credit type. More than 80 per cent of the trades and quotes are on corporate reference entities. Sovereign entities and bank entities account for approximately 2/3 and 1/3 of the rest, respectively. US reference entities are dominant, with 87.2 per cent in the whole sample. Four Latin American sovereigns, that is Brazil, Mexico, Columbia and Venezuela are heavily represented. They account for 85.2 per cent (38,266 of 44,929) of sovereign quotes and trades, and 10.2 per cent (38,266 of 376,260) of the whole sample.

Notional amount

Figure 10.1 shows the distribution of the most frequently used 12 notional amounts, which account for more than 99 per cent of all the quotes and trades.¹⁸ Notably, the most popular five notional amounts, that is, \$5 million, \$10 million, \$2 million, \$3 million and \$20 million, account for more than 95 per cent of all the quotes. Figure 10.2, however, shows that the \$5 million notional amount started to dominate the market only since 2002. Before 2002, the use of \$10 million as notional amount prevailed. This is consistent

Table 10.1 Currency, rank, restructuring and credit type

	Frequency	Per cent
<i>Currency</i>		
USD	375,760	99.8671
EUR	478	0.1270
JPY	17	0.0045
SGD ^a	2	0.0005
THB ^a	2	0.0005
XCD ^a	1	0.0003
<i>Rank</i>		
Senior	369,927	98.32
Subordinate	5,556	1.48
N/A	777	0.21
<i>Restructuring</i>		
MR ^b	339,980	90.36
OR ^b	25,739	6.84
NR ^b	10,204	2.71
MMR ^b	335	0.09
<i>Credit type</i>		
Corporate	305,198	81.11
Sovereign	44,929	11.94
Bank	26,133	6.95

^aSGD=Singapore dollar, THB=Thai Baht, XCD=East Caribbean Dollar.

^bMR=modified restructuring; OR=old restructuring; NR=no restructuring; MMR=modified modified restructuring.

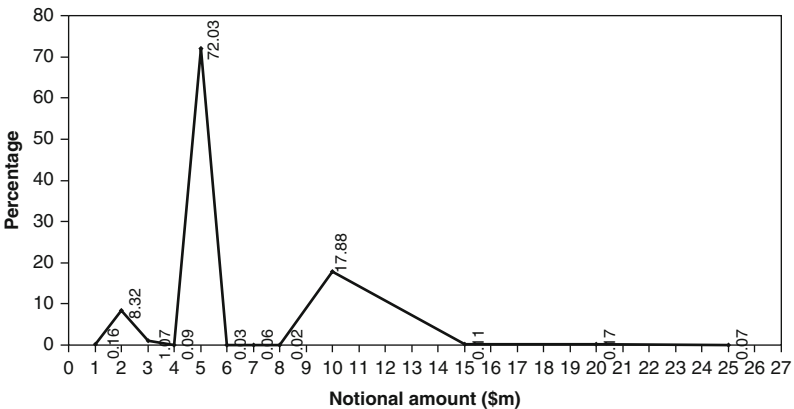


Figure 10.1 Distribution of notional amount

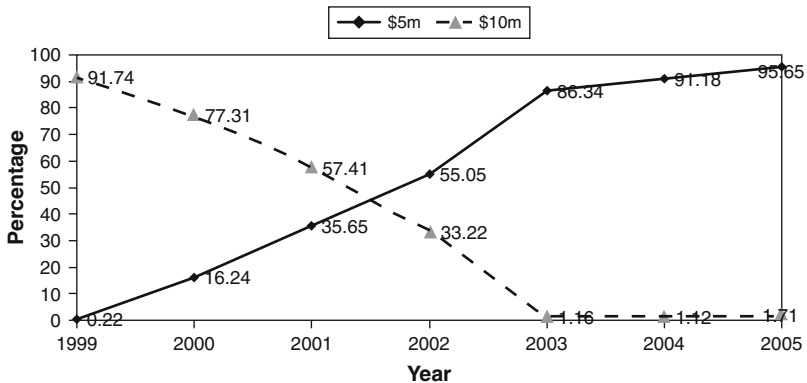


Figure 10.2 Percentage of the use of \$10 million and \$5 million as notional amount
 *Before 2003, there are CDS observations in the data set without notional amount information, and all the notional amounts in 1997 and 1998 are not available.

with Houweling and Vorst's⁹ finding that \$10 million is the most frequently quoted notional amount, given that their sample period is from May 1999 to January 2001.

The increased clustering at \$5 million amounts may be due to some popular trading strategies in the CDS index market. As investors initially establish a large position in a CDS index and hedge out only a few undesirable reference entities, the notional amounts of single-name CDS they quote may be corresponding to their partial hedging needs and is only a fraction of the initial position in the CDS index. Another potential explanation for this phenomenon is that due to the popularity of \$5 million as the notional amount, traders may prefer to quote several \$5 million CDS instead of one quote of the desired notional amount in order to access greater liquidity.

Maturity

Figure 10.3 shows the distribution of the most frequently quoted 12 maturities for CDS, which account for around 95 per cent of the whole sample. The frequency is more dispersed than in the case of notional amount. Maturities around five years are the most frequently quoted five maturities, and these account for 83 per cent of the whole sample. 3-year, 2-year, 10-year, 1-year and 7-year maturities are also among the ten most frequently quoted maturities (in descending order). This is similar to Houweling and Vorst's⁹ finding, except that they find that 10-year maturity is the third most common. Hull *et al.*¹¹ and Ericsson *et al.*¹⁰ also report that 5-year maturity CDS represent the majority of the data used in their empirical tests. Figure 10.4 shows that the use of 5-year maturity and maturities around five years (60–64 months) was increasing until 2002 and levelled off subsequently.

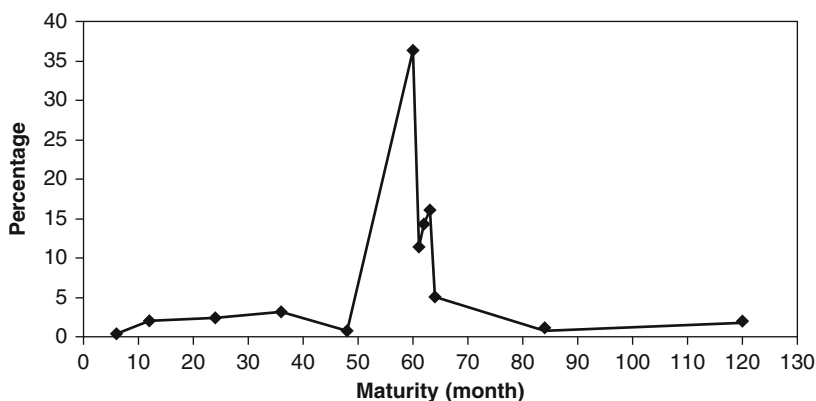


Figure 10.3 Distribution of maturity

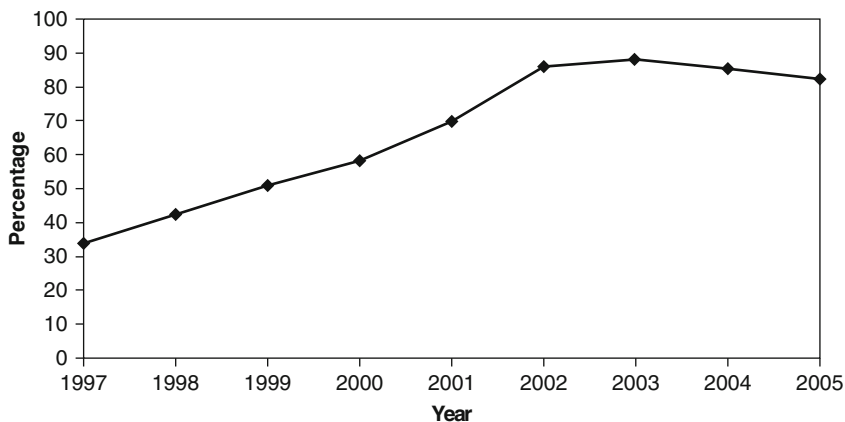


Figure 10.4 Percentage of the use of 5-year* as maturity

*60–64 month maturities are the most popular maturities in the whole sample and are all included as '5 years' in this figure.

Table 10.2 shows the time-varying composition of maturities of CDS. In general, the proportion of below 5-year maturities has been declining throughout the whole period, while the proportion of the above 10-year maturities has been very low and stable. There has been a modest increase in the proportion of maturities that lie between five and ten years in the last three years.

The phenomenon that maturities of CDS are less clustered than notional amounts of CDS may reflect the fact that breakdown of a target notional amount into \$5 million units is horizontal.¹⁹ This can even diversify

Table 10.2 Time-varying composition of maturity

Maturity (months)	Percentage								
	1997	1998	1999	2000	2001	2002	2003	2004	2005
<12	32.346	19.539	8.383	6.783	5.763	3.294	1.731	1.590	1.175
13–59	27.654	33.020	33.614	28.031	21.010	8.891	7.624	8.724	7.494
60–64	33.827	42.491	50.954	58.111	69.880	86.030	88.261	85.325	82.379
65–120	6.173	4.949	6.709	6.940	3.264	1.713	2.227	4.225	8.577
121–240	0.000	0.000	0.340	0.135	0.084	0.072	0.145	0.132	0.375
>240	0.000	0.000	0.000	0.000	0.000	0.000	0.009	0.001	0.000

counterparty risk in CDS transactions if the protection is offered by a number of protection sellers. On the other hand, roll-over of 5-year CDSs to reach a target maturity is vertical and involves ‘re-investment risk’. If using a 5-year roll-over strategy, the premium that a protection buyer/seller wants to pay/receive may be different at the end of five years. An increase/ decrease in premium will be disadvantageous for the protection buyer/seller.

Availability of credit ratings for reference entities and credit quality of the universe

Unlike previous empirical studies on the determinants of CDS premiums (eg Aunon-Nerin *et al.*⁸) that use credit ratings as an independent variable, analysis of the availability of ratings for reference entities here is to investigate to what extent it is an eligibility criterion for a reference entity to be quoted in the CDS market. The credit rating data for the reference entities are from Financial Times Credit Ratings International.

Among the 945 reference entities, 462 (48.8 per cent) have ratings from FitchRatings, 848 (89.6 per cent) have ratings from Moody’s Investors Service and 858 (90.7 per cent) have ratings from Standard & Poor’s Corporation (S&P). There are also 293 (30.0 per cent) reference entities that have credit ratings from at least one of six other international rating agencies. These rating agencies are, in descending order of frequency, Dominion Bond Rating Service, Duff & Phelps, Thomson BankWatch, Canadian Bond Rating Service,²⁰ Japan Rating & Investment Information and Japan Credit Rating Agency (see Panel A of Table 10.3).

There are 885 (93.6 per cent) reference entities that have ratings from at least one of the three leading rating agencies, and exactly the same number of reference entities have at least one rating. A total of 823 (87.0 per cent) reference entities have ratings from both Moody’s and S&P, 451 (47.7 per cent) have ratings from both FitchRatings and Moody’s, 451 (47.7 per cent) have ratings from both FitchRatings and S&P and 442 (46.7 per cent) have ratings from all the three leading rating agencies (see Panel B of

Table 10.3 Availability of credit ratings

Panel A—Availability of ratings from different agencies

Fitch	Moody's	S&P	Others	1	2	3	4	5	6
462	848	858	293	59	172	122	14	10	61
48.8%	89.6%	90.7%	30.0%	6.2%	18.2%	12.9%	1.5%	1.1%	6.5%

Panel B—Additional features

At least 1 from main 3^a	At least 1^b	Moody's S&P	Fitch Moody's	Fitch S&P	Three leading	Short	Watch
885	885	823	451	451	442	549	624
93.6%	93.6%	87.0%	47.7%	47.7%	46.7%	58.0%	66.0%

1=Canadian Bond Rating Service; 2=Dominion Bond Rating Service; 3=Duff & Phelps; 4=Japan Rating & Investment Information; 5=Japan Credit Rating Agency; 6=Thomson BankWatch.

^aRatings from at least one of the three leading rating agencies.

^bRatings from at least one rating agency.

Table 10.3). There are 549 (58.0 per cent) reference entities that have short-term ratings.

A total of 624 (66.0 per cent) reference entities are on agencies' rating watch lists at some time during the sample period (see Panel B of Table 10.3). To compare with the rated universe in general, we identified from Financial Times Credit Ratings International that 39.3 per cent of issuers (excluding the CDS reference entities in our sample) were on a watch list some time during November 1999 to March 2005. The fact that a relatively high proportion of CDS reference entities are on watch lists reflects the ongoing debate surrounding the accuracy of credit ratings versus market-based indicators of credit risk. Changes in CDS premiums, which reflect point-in-time changes in a firm's financial condition, are increasingly used as a market-based indicator for credit risk,²¹ that is when there is a market-perceived change in a firm's credit quality, trading of CDS on this firm's outstanding debt becomes more active and CDS premiums also change accordingly. Watch lists, which reflect point-in-time estimates of a firm's credit quality, are issued by rating agencies as a prior warning, to reconcile the dilemma of achieving both accuracy and stability of ratings. Hamilton and Cantor²² find that when rating outlooks and watches are taken into consideration, the correlation between credit ratings and CDS premiums is higher. Therefore, it is perhaps not surprising that a majority of CDS reference entities are on agency watch lists.

In our CDS data set, 140 reference entities are quoted in CDS with short maturities, ie less than 12 months. One hundred and five (75 per cent) of these reference entities have short-term ratings during the period. The majority (85.7 per cent) of the 140 reference entities are also on rating watches during the period they are quoted or traded.

Table 10.4 Time-varying credit quality of the reference entities in our sample^a

Panel A—Percentage of reference entities within different rating bands

Rating bands	Percentage								
	1997	1998	1999	2000	2001	2002	2003	2004	2005
Aaa-Aa3	8.03	21.21	14.38	9.73	10.09	10.92	4.54	4.55	6.90
A1-A3	18.07	33.96	46.48	33.14	36.43	35.04	23.11	18.55	9.97
Baa1-Baa3	11.65	19.70	21.42	34.56	38.83	42.60	61.22	46.39	37.80
Ba1-Ba3	15.26	14.27	5.25	2.12	4.07	7.59	6.16	15.17	20.70
B1-B3	46.99	10.86	12.47	20.45	10.17	3.83	4.54	15.29	24.54
Caa1-C	0.00	0.00	0.00	0.00	0.41	0.02	0.43	0.06	0.08

Panel B—Average rating of all reference entities

Average	10.20	13.15	13.25	12.38	12.95	13.31	12.39	11.44	10.58
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^a In Panel A, the percentage is calculated based on reference entities that have Moody's ratings available. In Panel B, the calculation of average rating is based on the following recoding: each of Moody's rating categories is assigned a number, with Aaa being 21 and C being 1.

Table 10.4 presents the annual distribution of reference entities that belong to different credit rating bands. The credit rating data used here are from Moody's. The distribution changes substantially across the sample period (except for the lowest band 'Caa1-C'). Reference entities with A1 or above ratings account for a declining proportion since 2000. The Baa rating band has been the most heavily represented since 2000. There is a large proportion of Ba- and B-rated entities at the beginning and the end of the sample period. Each of the rating categories is assigned a number (Aaa as 21 through to C as 1) to calculate the average rating of all the reference entities. Notably, the average rating has been declining since 2002. The annual average ratings, however, generally float within the 'Baa1-Baa3' rating band.

Trading activity and CDS indices

Quotes and trades, reference entities and trading activity

Hull *et al.*¹¹ found that the annual number of CDS quotes and trades in their data set increased from 4,759 in 1998 to 125,000 in 2002. Ericsson *et al.*¹⁰ demonstrate the trend of the changes in the daily number of CDS quotes and trades graphically. FitchRatings find that the number of unique reference entities increased between January 2002 and September 2004.⁶ On average, the weekly number of quoted reference entities increased from 325 in the first quarter of 2002 to 581 in the third quarter of 2004. The number of reference entities quoted in the market measures, to some extent, the growth of the CDS market. As the use of CDS becomes increasingly popular, additional participants engage in trading and the scope of demand and supply of protection is wider.

Table 10.5 presents the summary statistics of the number of reference entities in our sample from 10th June, 1997 to 3rd March, 2005. First, the average number of reference entities quoted daily increased steadily from 1997 to 2002, and then stabilised. The maximum daily number of reference entities quoted has been rising throughout the period.

Investigating the ratio of the number of CDS bid prices to that of offer prices reveals a balance at approximately 1:1 since late 1999 (see Figure 10.5). These results are, however, inconsistent with FitchRatings' finding that there is a structural imbalance in the CDS market,⁶ that is the number of bids to buy protection consistently exceeds the number of offers to sell protection. The probable explanation for this is the difference in the data samples. The data used in this paper are obtained from CreditTrade covering all the US reference entities quoted from June 1997 to March 2005, while FitchRatings' data are obtained from GFI group and Creditex covering all the reference entities quoted from January 2002 to September 2004.⁶

Figure 10.6 demonstrates that quotes substantially outnumber trades. Linking to the previous finding that the demand and supply for protection

Table 10.5 Daily number of reference entities and trading activity

	1997	1998	1999	2000	2001	2002	2003	2004	2005
Mean	2.83	4.52	14.50	20.69	32.96	53.59	50.73	50.19	56.42
Standard deviation	1.76	3.10	7.56	9.14	16.52	16.18	17.97	19.57	22.67
Minimum	1	1	1	1	1	1	1	1	4
Maximum	10	14	38	57	88	90	102	101	114
Sum ^a	331	963	3625	5277	8340	13345	12784	12297	2426

^aSum is the sum of daily number of reference entities in a year.

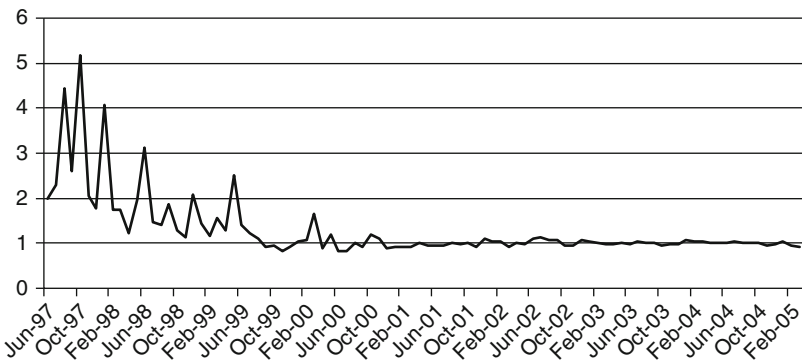


Figure 10.5 Ratio of number of bids to number of offers (monthly)

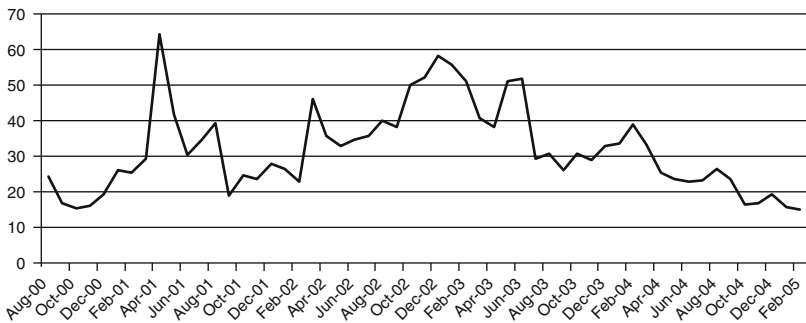


Figure 10.6 Ratio of monthly quotes to monthly trades

were quite balanced since late 1999, it can be implied that the large number of untraded quotes is not attributable to a bid-offer imbalance. Figure 10.6 also shows that the quotes/trades ratio has fallen towards the end of the sample period. One explanation for the falling trend may be that the launch of CDS benchmark indices such as Trac-X series, iBoxx series and iTraxx series helped to improve accuracy in CDS pricing.

Growth of CDS index products and the potential implications for single-name CDS trading

According to FitchRatings' credit derivatives market surveys,^{23,24} the growth rate of CDS index and index-related products has skyrocketed from 49 per cent in 2003 to 900 per cent in 2005. Meanwhile, the growth rate of single-name CDS has been declining year by year. The BBA predicted that the share of index-related products in the credit derivatives market would rise from 11 to 17 per cent between 2004 and 2006, while the share of single-name CDS would fall from 51 to 42 per cent.²⁵

Although there is no hard evidence of a direct causal relationship between the rapid growth of the CDS index market and the slowing of growth of the single-name CDS market, it is well known that CDS indices have some advantages over single-name CDS. Following the merger of Trac-X and iBoxx to iTraxx, the number of market makers has been rising and liquidity has been improving. The introduction of these CDS indices is also a direct product of market participants' demand for pricing benchmarks as the CDS indices consist of highly liquid reference entities. This, according to some practitioners, has created a virtuous circle: investors with surplus cash on a short-term basis are attracted to iTraxx, as the bid-ask spreads are narrow. The participation of these investors brings even more liquidity to the market.

A passive investment strategy of tracking a CDS index, which allows investors to maintain positions in a number of reference entities, is reported

to be the most popular strategy with CDS index products.²⁵ A parallel exists with stock indices that provide opportunities to small investors who lack 'stock-picking' capability. CDS indices attract smaller investors who may not have internal resources to research all the individual reference entities.

The inter-day pattern of volume

In our database, there are in total 58,828 trades and quotes on Mondays, 84,749 on Tuesdays, 87,365 on Wednesdays, 82,614 on Thursdays and 62,680 on Fridays. The total number of quotes and trades thus demonstrates an inverse U shape throughout weekdays. Previous studies explain such an inter-day pattern of volume in securities markets. On the assumption that private information is received throughout the week while public information is received on weekdays only, Foster and Viswanathan²⁶ demonstrate in their model that (i) due to the information advantage (in particular received during weekends), informed traders trade on Mondays even if market makers tend to widen bid-ask spreads to mitigate the information asymmetry problem; (ii) uninformed traders increase their trades after Monday as more public information becomes available; and (iii) liquidity traders avoid trading on Mondays when the bid-ask spreads are the widest.

Although there is a lack of previous empirical studies, the information asymmetry problem in the CDS market is widely discussed.²⁷⁻²⁹ Banks who have private information about a reference entity's creditworthiness will exploit this advantage by buying protection before the information goes public. As a result, banks are warned by financial regulators to set up firewalls between their trading and lending desks.³⁰ If Foster and Viswanathan's²⁶ assumption is valid in the CDS market, their finding can explain why the volume is the lowest on Monday and increases towards Wednesday. The decline in the volume after Wednesday may be due to the fact that liquidity and uninformed traders are not willing to maintain large positions during the weekend due to the unavailability of public information, and therefore they reduce their exposures.

Conclusion

First, the analysis addresses stylised facts of the CDS market. More than 99 per cent of the CDS in our sample are quoted in US dollars. The vast majority of the debt underlying CDS transactions is senior-ranked, a phenomenon that may be attributed to the fact that the pricing for CDS with structured underlying assets is much more complicated. There are few CDS transactions without restructuring clauses and the prevailing restructuring clause with US reference entities is MR. Sovereign CDS cluster on four Latin American countries, namely Brazil, Mexico, Colombia and Venezuela. In a departure from previous findings, we show that the most frequently quoted notional amount is US\$5 million, which may be attributable to the popular partial

hedging practice in CDS index trading. The most popular maturity is around five years. The clustering of maturities, however, is much less than the clustering in notional amounts.

Secondly, we investigate the availability of credit ratings and the credit quality of the reference entities in our sample. The vast majority of the reference entities quoted or traded in the market have ratings from at least one of the three leading rating agencies: Moody's, S&P and Fitch. The availability of ratings from six other international rating agencies is limited and reference entities are unlikely to have ratings from these rating agencies unless they also have ratings from the three main rating agencies. Each year, there are significant changes in the rating status of reference entities in the sample. The average rating has been declining since 2002.

Finally, we consider the time-varying characteristics of the data sample. The number of reference entities has grown in the sample period. There is a 'structural imbalance' in this sample of the CDS market up to 2001, whereby bids for protection consistently exceed offers of protection. While the number of quotes substantially exceeds the number of trades, there is a clear change since 2003, which could be attributed to the launch of a number of CDS benchmark indices. The growth of CDS index trading provides some supportive evidence for the patterns observed in trading activity. In addition, the pattern of volume demonstrates an inverse U shape across the week, with volume increasing from Monday, reaching its peak on Wednesday and then declining afterwards.

Acknowledgments

The authors acknowledge valuable comments from Haibin Zhu of the Bank for International Settlements, Antonio Di Cesare of Banca d'Italia, and from seminar participants at the Centre for Empirical Finance, University of Wales, Aberystwyth.

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11

The Performance Persistence of Equity Long/Short Hedge Funds

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Practical applications

Investors often allocate capital to specific hedge funds on the basis of the funds' track records, which implies that they expect some persistence in the funds' performance. In this context, large-scale studies analysing the performance persistence of hedge funds provide important insights on the expected success probability of such investment behaviour. From a practical point of view, the methodological approach chosen in this paper, which consists of repeatedly forming a portfolio based on observable information and then tracking it for the next period, represents a trading strategy that is conceptually easily implementable. However, lockup and redemption periods impede

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the implementation of such trading strategies. Consequently, an annual frequency of reforming the portfolios makes the strategies more feasible with respect to transaction costs and lockup periods, especially for funds of funds which often face more favourable conditions than individual investors.

This paper examines the persistence of raw and risk-adjusted returns for equity long/short hedge funds using the portfolio approach of Hendricks et al. Only limited evidence of persistence is found for raw returns. Funds with the highest raw returns last year continue to outperform over the subsequent year, although not significantly, while there is no persistence in returns beyond 1 year. In contrast, we find performance persistence based on risk-adjusted return measures such as the Sharpe Ratio and in particular an alpha from a multifactor model. Funds with the highest risk-adjusted performance continue to significantly outperform in the following year. The persistence does not last longer than 1 year except for the worst performers. Funds with significant risk-adjusted returns show less exposure to the market and have high raw returns and low volatility. These results are robust to adjustments for stale prices and sub-period analysis.

Introduction

Studies examining the persistence of hedge fund performance vary greatly in their conclusions owing to different methodologies, databases, investigation periods and performance measures.¹ This paper does not consider various approaches to clarify the picture, but, instead, focuses on a particularly flexible one. Every period, hedge funds are sorted into portfolios according to characteristics in the last period, and the portfolios are then tracked for the next period. After the tracking period, the sorting is repeated. This approach has been used in the mutual fund literature by Hendricks *et al*² and Carhart,³ among others, and has several advantages. First, portfolio betas may be more stable than betas of individual funds because time-varying betas can offset each other on the portfolio level. This is particularly relevant for hedge funds: as they have fewer restrictions on borrowing, shorting, the use of derivatives and so on, they typically follow highly opportunistic strategies that lead to time-varying risk exposures. Secondly, beta measurement is more precise owing to diversification of idiosyncratic risk and long time series for the portfolio returns. Finally, suppose there is only a very small autocorrelation in fund returns. Given the high return variance, it is difficult to detect this correlation by looking at individual funds. As in the case of momentum strategies, we have to buy a portfolio of last period's winners, not just one fund, to find persistence.

This paper extends the current literature in several directions. First, by restricting the sample to equity long/short hedge funds, the number of (risk) factors can be greatly reduced while the explanatory power of the factor models is maximised. In fact, the widely used factor models generally exhibit a

very high explanatory power for the equity long/short strategy – in absolute terms as well as relative to other strategies.^{4–6} Given only monthly observations of hedge fund returns, a low number of parameters is desirable, as it renders inference more robust owing to conservation of degrees of freedom and mitigation of multicollinearity problems. Secondly, and related, prior studies generally pool different hedge fund strategies and analyse them jointly.^{7,8} This approach can be susceptible to the problem of model misspecification: alpha portfolios may contain funds of the same strategy because neglected risk factors for this strategy show up as alpha. Hence, results focusing on one particular strategy may be more robust. Thirdly, possible nonlinear market exposure is accounted for by including option returns as risk factors. Agarwal and Naik^{5,9} find that the systematic risk exposure of hedge funds can include option-based strategies. No such exposure has been found, however, for equity long/short hedge funds on the aggregate level.^{5,6} Individual funds, on the other hand, may well show nonlinear market exposure and it may be important to account for nonlinearities when evaluating performance. Finally, we consider sorts based on regression coefficients. In order to examine the nature of return persistence, it is important to know whether the momentum loading of funds persists and whether these funds outperform.

The analysis in this paper is interesting for several reasons. First, repeatedly forming a portfolio based on observable information and tracking it for the next period represents a trading strategy that is conceptually easily implementable. Practically, however, the typical investor faces lockup periods prohibiting him from exploiting these strategies. Still, they may be interesting for funds of funds that are able to waive lockup periods.⁷ Secondly, performance persistence may be more important for hedge funds than for mutual funds owing to their higher attrition rates.¹⁰ By looking at funds' transition probabilities of moving from one portfolio to another and by tracking a portfolio over not only 1 year, but over several years, one gets a clearer picture of the persistence.

After a short overview of the literature, the factor model is introduced and applied on the index level. Next, we present the data set and sample selection procedures. The discussion of the results is the main part and focuses on the persistence of raw returns as well as risk-adjusted performance measures. In order to mitigate spurious results, various robustness tests are performed before a conclusion is reached.

Literature overview

Although the performance of mutual funds has been extensively studied in the academic literature, there is a growing body of literature on hedge fund performance persistence. Even more than in the mutual fund literature, there is a large variance in the conclusions drawn by the studies examining the persistence of hedge fund performance. A recent overview of the literature on persistence of hedge fund performance is provided by Eling.¹

He shows that these studies differ widely in methodology, database, investigation period, performance measures and conclusions. To obtain a clearer picture, at a minimum, persistence of raw returns and risk-adjusted returns has to be distinguished.

Harri and Brorsen¹¹ report short-term persistence of 3–4 months, with the biggest effect in the first month based on simple regressions of returns on lagged returns. Over the quarterly horizon (portfolios are reformed quarterly), Boyson and Cooper⁷ obtain a monthly return spread of more than 1.14 per cent using a pool of all hedge fund strategies. For the annual horizon, Baquero *et al*¹² find an insignificant spread of an annual 4.9 per cent and an annual spread of 11.5 per cent at the quarterly horizon. Clearly, the smaller the horizon, the stronger the persistence in performance as measured by raw returns. Overall, the literature is in favour of short-term persistence over horizons of up to 6 months, but is mixed with respect to annual persistence.¹

The evidence on the persistence of risk-adjusted performance is particularly mixed as there are various methods and performance measures. Amenc *et al*¹³ show that different models strongly disagree on the risk-adjusted performance of hedge funds because there is a large dispersion of alphas across models. Still, they generally tend to rank the funds in a similar way. Common factor choices for hedge fund (risk) factors are the three Fama–French factors; Carhart's³ momentum factor; commodity, bond and volatility factors; and factors representing returns to technical trading strategies such as the Fung and Hsieh¹⁴ primitive trend-following factors¹⁵ or the Agarwal and Naik⁵ option factors.¹⁶ Persistence can be examined using simple autoregressions, Spearman's rank correlation, and contingency tables as the most prominent methods. The evidence is mixed, however. For example, while Agarwal and Naik⁹ find no persistence beyond the quarterly horizon, Kosowski *et al*¹⁷ and Edwards and Caglayan¹⁸ report annual persistence.

One prominent method introduced by Hendricks *et al*² and used in the mutual fund literature^{3,19} consists of repeatedly forming portfolios of funds based on lagged characteristics and tracking them for the next period. Boyson and Cooper⁷ sort on characteristics other than returns or alpha for hedge funds. Although they find no persistence when sorting on past performance alone, considering manager tenure as well leads to the finding of quarterly persistence. The persistence is mainly concentrated in the poor performers. Considering all hedge fund strategies jointly, however, they use up to 20 factors to end up with an R^2 of around 0.75. Capocci⁸ sorts portfolios based on additional properties of the return distribution. He finds that portfolios of funds with the highest lagged Sharpe Ratio deliver positive alpha. The same holds for funds with the lowest volatility and, to a lesser extent, funds with low market beta. Sorting portfolios based on higher moments of the return distribution or lagged alpha, however, does not detect a significant alpha spread between top and bottom portfolios.

Model

Portfolio formation

Our approach to form portfolios is based on Hendricks *et al.*² On 1 January of each year,

portfolios of hedge funds are formed based on specific characteristics during the formation period and are tracked for the subsequent year. The portfolios are then reformed. The portfolios are numbered from 1 to 10, where portfolio 1 contains the 10 per cent of funds with, for example, the highest lagged returns, and portfolio 10 contains the 10 per cent of funds with the lowest lagged returns. The weights of the funds in the portfolios are equal and readjusted whenever a fund disappears during the tracking year. To provide additional information, the deciles 1 and 10 are further subdivided into terciles, indicated by capital letters A, B and C. Also, a portfolio that is long in portfolio 1 and short in portfolio 10 (long 1A and short 10C) is analysed.

In this paper, the tracking horizon is 1 year for two reasons. First, quarterly or even monthly reforming of a portfolio would be difficult to implement as a trading strategy owing to lock-in periods. Secondly, because of illiquidity or managed prices, hedge fund managers have leeway in marking their positions for month-end reporting. This flexibility can be used to artificially reduce return volatility or market beta. The resulting returns spuriously show short-term autocorrelation known as 'stale prices',^{20,21} which shows up as short-term persistence in returns. Frequent portfolio reforming may therefore just take up this autocorrelation. Focusing on the 1-year horizon alleviates this problem. In the robustness section, we also explicitly control for 'stale prices'.

Factor benchmark and alpha

The investment flexibility of hedge funds makes performance assessment more difficult. Style-drift and exposure to a wide range of asset classes, possibly even nonlinear, must be accounted for. Given the limited amount of data, this task can only be dealt with on a rudimentary level. Possible nonlinearities are accounted for by including option-based strategies, using a rolling-window approach, and performing an analysis of sub-periods, which may mitigate the effects of style-drift.

As hedge funds typically employ dynamic strategies, the static CAPM is an inappropriate benchmark. For equity long/short funds, Fung and Hsieh⁶ show that the overall market and the spread between large and small cap stocks account for over 80 per cent of return variation on the index level. Combined with a momentum factor, they find it unlikely to have omitted an important risk factor. Hence, equity long/short returns can be well captured using the Fama–French factors and a momentum factor. Agarwal and Naik⁵ and Fung and Hsieh¹⁴ show that hedge funds can have exposure to option returns on standard asset classes. Specifically, Agarwal and Naik⁵ use

returns of rolling-over put and call options on the S&P 500 index. Hence, we extend our factor model by including Agarwal and Naik's out-of-the-money call and out-of-the-money put option factors on the S&P 500 index. This can be represented with the sar's³ four-factor model.

$$R_{i,t} - R_{f,t} = \alpha + \beta_{i, RMRF} RMRF_t + \beta_{i, SMB} SMB_t + \beta_{i, HML} HML_t + \beta_{i, MOM} MOM_t + \beta_{i, OTMC} OTMC_t + \beta_{i, OTMP} OTMP_t$$

The left-hand side is the excess return of portfolio i in month t . $RMRF$ is the excess return on the market portfolio, proxied by the value-weighted return on all NYSE, AMEX and NASDAQ stocks minus the 1-month Treasury Bill rate. SMB is the excess return of the factor-mimicking portfolio for size, HML is the excess return of the factor-mimicking portfolio for book-to-market equity and MOM is the excess return of the factor-mimicking portfolio for 1-year momentum.²² Finally, $OTMC$ and $OTMP$ are the out-of-the-money call and out-of-the-money put option factors on the S&P 500 index as used in Agarwal and Naik.^{5,23}

The option factors turned out to be hardly significant in all of our regression specifications, did not increase the model's explanatory power and showed no specific pattern. In addition, their inclusion only had minor effects on the other coefficients for all the regressions reported. As the option factors are currently available only until January 2005 and would therefore shorten our sample period by 11 months, we excluded them from our factor models, as reported in the tables (with the exception of Table 11.1). All regressions on the portfolio as well as the single fund level, however, turned out to be robust to the inclusion of these option factors.

Table 11.1 Long/short index of CISDM database

	Alpha	RMRF	HML	SMB	MOM	OTMC ($\times 10^{-3}$)	OTMP ($\times 10^{-3}$)	R ² _{adj.}
<i>Panel A: Including option-based factors</i>								
Long/short index	0.33***	0.41***	0.06*	0.20***	0.07***	0.57	-0.94	0.82
<i>Panel B: Excluding option-based factors</i>								
Long/short index	0.34***	0.45***	0.06*	0.20***	0.07***	—	—	0.82

The excess returns of the equity long/short index from CISDM are regressed on the following factors over the period 1994–2005: $RMRF$, HML and SMB are the Fama–French²⁷ factors for the market proxy and factor-mimicking portfolios for book-to-market and size. MOM is a factor-mimicking portfolio for 1-year momentum. The first regression (Panel A) additionally includes two option-based factors, $OTMC$ and $OTMP$, which are the out-of-the-money call and out-of-the-money put option factors on the S&P 500 index as used in Agarwal and Naik.⁵ Newey–West²⁸ corrected t -values are represented by ***, * for 99%, 90% significance, respectively.

In unreported tests, we also attempt to control for market timing abilities by including a squared term of the market factor²⁴ or option-like factors as in Henriksson and Merton.²⁵ These additional factors are, however, hardly ever statistically significant and do not affect the other coefficients in our factor model. Therefore, we do not report results based on the alternative risk models in the tables.

Performance at the index level

The CISDM equity long/short index shows the median performance of equity long/short hedge funds reporting to the CISDM database. The hedge fund index is not investable, however, and may suffer from various biases, which are discussed below. Figure 11.1(a) compares its performance to the S&P 500. Obviously, the average equity long/short hedge fund shows strong market exposure until the year 2000, where it was smart to reduce the exposure. After the market downturn, the hedge funds again took advantage of rising stock prices. The performance of the long/short strategy matches very closely the performance of the average hedge fund across all strategies. This reflects the dominance of equity long/short hedge funds in the database, as other strategies performed much differently. Overall, the strategy clearly outperformed the market benchmark and had a long exposure to the market, as Figure 11.1(b) shows.

As argued above, the market is an inappropriate benchmark. Applying the factor model above shows that the index outperforms the more demanding multifactor benchmark. Panel A of Table 11.1 displays a monthly outperformance of 0.33 per cent. The factor loadings show significant exposure to the market, small stocks and last year's winner stocks. The exposure to value stocks is very small, as the marginally significant *HML* loading shows. These coefficients are consistent with Agarwal and Naik⁵ and Fung and Hsieh.⁶ The option factors are not able to capture any nonlinear market exposure, however. For comparison reasons, we re-estimate the same regression equation without the option factors. In fact, the results in Panel B of Table 11.1 show that the alpha and all factor loadings remain basically unchanged.²⁶

Data

Data on hedge fund returns were obtained from CISDM. The CISDM database contains returns for both surviving and defunct funds. Two thousand two hundred and ten (901 for equity long/short) non-active funds are included out of a total of 4390 (1693 for equity long/short), excluding funds of funds. It is well known that fund databases suffer from various biases. This paper tries to mitigate a survivorship bias by including defunct funds in the analysis and by excluding data before 1994, when CISDM did not keep track of defunct funds.²⁹ For the resulting sample period from January 1994 to December 2005, a total of 1649 funds are classified as equity long/

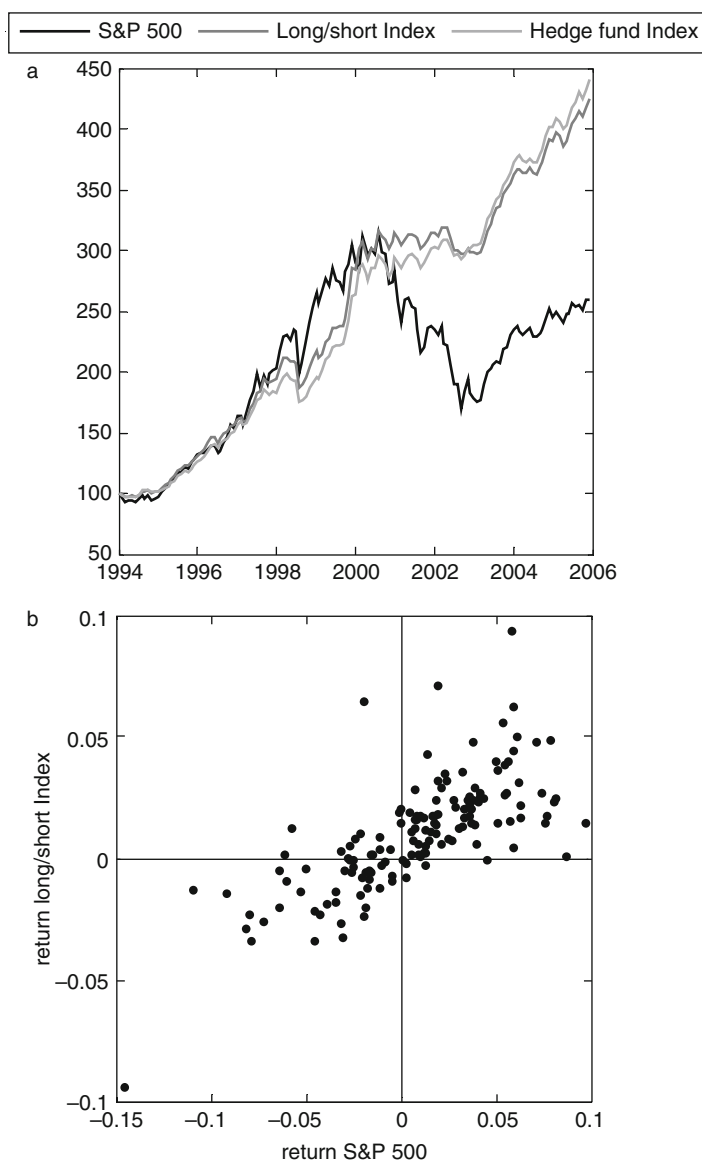


Figure 11.1(a) Performance of equity long/short strategy

The figure shows the S&P 500 and the CISDM long/short index and overall hedge fund index. All indices are normalised to 100 in the beginning of 1994.

Figure 11.1(b) Equity long/short versus market returns.

The scatterplot shows the S&P 500 monthly return versus the CISDM long/short index return from 1994 until the end of 2005.

short funds. To mitigate the backfill bias, we use the standard procedure and delete the first 12 observations for each fund in the database.

CISDM includes monthly net-of-fee returns, assets under management, inception date, self-declared strategy, and more fund characteristics. Many characteristics are only rarely provided by the funds. The following list shows the selection criteria for the funds that are included in our final sample.

- Funds that appear multiple times were deleted. This was considered to be the case when funds had the same name and company ID. In addition, only the funds in USD were kept in the sample, resulting in the deletion of 85 funds.
- The period considered ranges from January 1994 until December 2005, resulting in 144 monthly observations. Only data from 1994 onwards are used, as CISDM had not started to keep track of either life or defunct funds before this. This results in the exclusion of 35 funds that have no observations during this period.
- To correct for the backfill bias, the first 12 observations of every fund are excluded. In addition, at least 12 observations are needed to sort the funds in the formation period. Hence, 423 funds with less than 2 years of data for the time period considered are deleted.

This results in a sample of 1150 funds. The time span is long enough to cover more than a business cycle and contains a variety of different market environments: the bull market in the 1990s, the bear market from 2000 until 2003, and events such as the Asian crisis in 1997 and the collapse of LTCM in the wake of the Russian financial crisis in 1998.

Empirical Results

Persistence of raw returns

An important question for hedge fund investors is that of whether past performance is indicative of future performance. So far, private and institutional investors clearly allocate more funds to past good performers. For mutual funds, Sirri and Tufano³⁰ find large inflows into last year's winners and large withdrawals from last year's worst performers. For hedge funds, Fung *et al*³¹ document that alpha funds attract more capital inflows than beta-only funds and Agarwal *et al*³² find that '... funds with persistently good (bad) performance attract larger (smaller) inflows compared to those that show no persistence' (p. 2). Hence, in the presence of performance persistence, an investor may be able to realise superior performance.

Table 11.2 reports the performance persistence in raw returns based on 10 portfolios of funds that are sorted according to their lagged 1-year raw returns. The table reveals significant variation in mean returns. The mean returns tend to decrease in portfolio rank, resulting in an annualised spread

Table 11.2 Portfolios of hedge funds formed on lagged 1-year raw returns

Portfolio	Excess return (s.d.)	Alpha	RMRF	HML	SMB	MOM	R ² _{adj.}
1A	0.95 (0.65)	0.10	0.73***	-0.45***	0.48***	0.54***	0.64
1B	0.69 (0.50)	-0.08	0.73***	-0.09	0.34***	0.25**	0.49
1C	1.09 (0.38)	0.58**	0.52***	-0.11	0.47***	0.23***	0.66
1 (high)	0.92 (0.46)	0.22	0.68***	-0.20*	0.44***	0.35***	0.69
2	0.93 (0.34)	0.29*	0.63***	-0.06	0.32***	0.20***	0.82
3	0.99 (0.24)	0.43***	0.50***	0.08	0.23***	0.14***	0.75
4	0.93 (0.23)	0.41***	0.42***	0.02	0.24***	0.13***	0.83
5	0.76 (0.24)	0.20	0.50***	0.14**	0.21***	0.07**	0.72
6	0.78 (0.21)	0.33***	0.47***	0.09***	0.18***	0.05**	0.74
7	0.59 (0.23)	0.10	0.45***	0.16***	0.25***	-0.02	0.81
8	0.50 (0.22)	0.17	0.37***	0.06	0.18***	-0.07***	0.76
9	0.60 (0.23)	0.31**	0.42***	0.14**	0.18***	-0.14***	0.68
10 (low)	0.53 (0.36)	0.26	0.61***	0.10	0.21***	-0.26***	0.63
10A	0.62 (0.35)	0.31	0.62***	0.17	0.18**	-0.23***	0.55
10B	0.56 (0.30)	0.30	0.51***	0.15	0.18***	-0.14**	0.38
10C	0.38 (0.54)	0.16	0.70***	-0.01	0.25***	-0.42***	0.56
1-10	0.39 (0.50)	-0.04	0.07	-0.30	0.23*	0.60***	0.48
1A-10C	0.56 (0.72)	-0.06	0.04	-0.43	0.23	0.96***	0.51

Each January from 1995 until 2005, decile portfolios based on lagged 1-year returns are formed. Portfolio 1 contains funds with the highest 1-year returns, portfolio 10 funds with the lowest 1-year returns. Portfolios 1 and 10 are further divided into terciles. The portfolios are equally weighted and the weights are readjusted whenever a fund is delisted. *RMRF*, *HML* and *SMB* are the Fama—French²⁷ factors for market proxy and factor-mimicking portfolios for book-to-market and size. *MOM* is a factor-mimicking portfolio for 1-year momentum. Newey—West²⁸ corrected *t*-values are represented by ***, **, * for 99%, 95%, 90% significance, respectively.

of 4.78 per cent between portfolios 1 and 10. This spread is not significant, however, with a monthly mean of 0.39 per cent and standard deviation of 0.50 per cent. Moreover, the mean returns do not decrease monotonically. Because of the tendency for winners to remain winners, there seems to be weak evidence for 'hot hands'² in equity long/ short hedge funds for the 1-year horizon. This value is smaller than Carhart's³ annual 8 per cent spread for mutual funds and Capocci *et al*'s¹⁰ 7.6 per cent obtained by pooling all hedge fund strategies.

Our results indicate that, assuming normality for simplicity, the probability that last year's winners exhibit a positive return over the next month is $P(X > 0) = 1 - \phi((0 - 0.92)/0.46\sqrt{T}) = 0.57$, where T is equal to 132. Moreover, the extreme portfolios have a higher return variance than portfolios in the inner deciles, possibly because of higher leverage or generally riskier strategies. This is consistent with Herzberg and Mozes,³³ who find that the funds with the highest past returns have the highest volatility.

When we alternatively adjust the raw returns for risk by using an alpha from a Carhart³ four-factor model, the spread in performance disappears. In

fact, portfolio 10 has a (insignificantly) higher alpha than portfolio 1. It is interesting to see that the significant alphas are located in the inner deciles (portfolios 3, 4, 6 and 9), whereas the portfolios of last year's winners do not outperform against the benchmark model. Instead, they show more return variation, which eventually places them in more extreme deciles. The alpha portfolios, however, show relatively little return variance. This indicates that, on average, alpha funds are characterised not only by high returns, but also by low volatility.³⁴ In fact, portfolios 3, 4 and 6 have the three highest Sharpe Ratios, and portfolio 9 the sixth highest Sharpe Ratio (not reported).

Looking at the factor loadings, the coefficient on *RMRF* is around 0.5 and shows no specific pattern. With respect to *HML*, there seems to be a tendency of last year's winners to be more growth-oriented, whereas other deciles have more of a value focus, though coefficients are generally small and insignificant. More revealing is the pattern of the *SMB* coefficients. All portfolios have positive exposure, which confirms earlier findings that long/short funds are small cap-oriented.^{5,6} Last year's winners are strongly exposed to small cap stocks. This exposure decreases in the decile numbering but remains positive. The spread between portfolios 1 and 10 is marginally significant and explains some of the spread in raw returns: some funds continue to earn higher returns because they are more exposed to small stocks and capture their premium. The largest spread, however, is in the momentum loadings. There is a monotonic decrease from 0.35 for portfolio 1 to -0.26 for portfolio 10, with the extreme portfolios 1A and 10C showing the strongest exposure. Given the monthly return spread of 0.39 per cent and the mean of the momentum factor of 0.84 per cent, the spread in the loading of 0.6 more than accounts for the difference in returns. The momentum factor explains the return spread by identifying last year's winners as the holders of last year's winning stocks. The identified patterns for *SMB* and *MOM* are very similar to mutual funds, where Carhart³ reports a spread of 0.30 for *SMB* and 0.38 for *MOM* between portfolios 1 and 10.

Overall, the model does a good job at explaining the returns of portfolios based on lagged fund returns. The R^2 ranges from 0.63 to 0.83 for the decile portfolios. This is lower than Carhart's four-factor model for mutual funds, where the R^2 s are above 0.9. This is not surprising, however, given the diversity of individual hedge fund strategies for this particular style.

When a fund has been sorted into a portfolio and tracked for a year, it can either stay in that portfolio, move to another portfolio or it may have stopped reporting during the year. To visualise the probability of a fund to move to portfolio j (or stop reporting), given that it is in portfolio i , we construct a contingency table, which is presented in Figure 11.2(a). The figure shows that in general winners tend to remain winners and losers tend to remain losers. Last year's winners are, however, after portfolio 10, the most probable group to move to the loser portfolio. The same is true for last year's

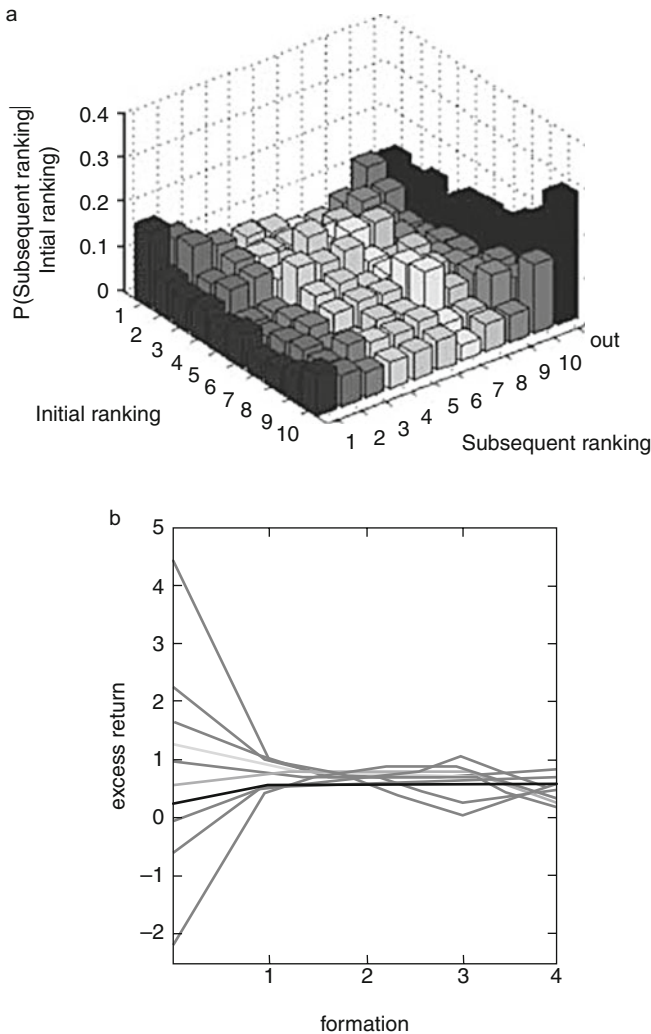


Figure 11.2(a) Contingency table of initial and subsequent ranking (lagged returns)
 Each year, funds are sorted into decile portfolios based on their past 1-year return. The bars indicate the transition probability of a fund moving from decile i into decile j or stop reporting.
Figure 11.2(b) Postformation returns of portfolios sorted on lagged 1-year returns
 Each year, funds are sorted in decile portfolios based on lagged 1-year returns. The figure shows the monthly excess returns of these portfolios in the formation year and subsequent years.

losers: They are, after portfolio 1, the most probable group to end up as this year's winners. This is in accordance with the high return variance of the extreme portfolios. There is also the tendency of the inner decile funds to remain in the inner deciles, which is consistent with their low return variance. The probability of stopping to report shows an increasing pattern with respect to last year's ranking and is highest for last year's losers. This is in line with Liang,³⁵ Brown *et al.*,³⁶ and Baquero *et al.*¹² who find that hedge funds with low past performance are likely candidates for liquidation. Overall, however, hedge funds move a great deal between portfolio deciles and the detected patterns are not very strong.

If funds move frequently lot between portfolios, one expects the persistence to be short-term in nature. Figure 11.2(b) shows the average return of the portfolios in the years following the formation period. The return spread shrinks drastically in the first year after ranking. This is the weak 'hot hands' effect as identified in the table. Following the portfolios beyond 1 year shows no persistence at all, not even for the bad performers.

Detection and persistence of alpha

Forming portfolios based on lagged returns does not seem to be a useful way to detect alpha portfolios. Instead, one would like a characteristic that provides a monotonic change of alpha across the deciles. Hence, we alternatively sort funds into portfolios based on lagged alphas. To provide reasonable estimates, alpha is calculated using the past 24 months of data. Table 11.3 provides the regression results for the decile portfolios, with portfolio 1 containing the funds with the highest lagged alpha and portfolio 10 containing the funds with the lowest lagged alpha.

This sorting is able to identify a sizeable and significant alpha spread of a monthly 0.79 per cent. Although the pattern is not perfectly monotonic, significant alphas are clearly located in the portfolios containing the funds with high lagged alpha. Persistence in alpha is especially pronounced for the extreme portfolios 1A and 10C. Portfolio 1A, however, seems to be poorly explained by the factors.

The table also shows that the portfolios with significant alphas have the highest returns. In fact, sorting based on lagged alpha produces a larger return spread than sorting based on lagged returns. The spread is a significant monthly 0.52 per cent compared to an insignificant 0.39 per cent for the portfolios based on lagged returns. Hence, lagged alpha seems to provide more information about both future alpha and future raw returns. With respect to factor exposure, alpha portfolios are generally less exposed to the market. This is also shown by the 1-minus-10 portfolio at the bottom of Table 11.3, where the difference in the *RMRF* coefficient is reported to be negative and significant at the 1 per cent level.

Alternatively, we sort the funds based on lagged 3-year alphas. The alpha spread between portfolios 1 and 10, however, decreases to an insignificant

Table 11.3 Portfolios of hedge funds formed on lagged 2-year alpha

Portfolio	Excess return (s.d.)	Alpha	RMRF	HML	SMB	MOM	R ² _{adj.}
1A	1.38 (0.41)	1.37***	0.32**	-0.23*	0.02	-0.08	0.24
1B	0.80 (0.40)	0.44	0.35***	-0.14	0.34***	0.15**	0.43
1C	1.23 (0.36)	0.75***	0.66***	0.16**	0.28***	-0.05	0.67
1 (high)	1.02 (0.32)	0.71***	0.48***	-0.03	0.24***	-0.01	0.63
2	0.82 (0.27)	0.40***	0.44***	0.04	0.29***	0.08**	0.71
3	0.85 (0.25)	0.47***	0.49***	0.10**	0.23***	-0.00	0.78
4	0.71 (0.23)	0.31***	0.42***	0.10**	0.24***	0.06**	0.74
5	0.51 (0.26)	0.10	0.47***	0.06	0.27***	0.05**	0.79
6	0.41 (0.26)	0.09	0.45***	0.03	0.24***	-0.03	0.78
7	0.74 (0.25)	0.23*	0.54***	0.18***	0.23***	0.07***	0.80
8	0.79 (0.26)	0.42***	0.51***	0.08*	0.16***	0.00	0.75
9	0.42 (0.29)	-0.11	0.60***	0.14***	0.25***	0.06*	0.81
10 (low)	0.51 (0.42)	-0.08	0.80***	0.03	0.25***	0.06	0.77
10A	0.68 (0.36)	0.07	0.64***	0.18*	0.23***	0.12*	0.55
10B	0.78 (0.40)	0.33	0.68***	0.09	0.26***	-0.06	0.59
10C	-0.19 (0.69)	-0.95***	1.18***	-0.24*	0.24***	0.17*	0.69
1-10	0.52 (0.24)	0.79***	-0.32***	-0.06	-0.01	-0.07**	0.23
1A-10C	1.57 (0.60)	2.32***	-0.85***	0.01	-0.22*	-0.25***	0.41

Each January from 1996 to 2005, decile portfolios based on lagged 2-year alphas are formed. Portfolio 1 contains funds with the highest 2-year alpha, portfolio 10 funds with the lowest 2-year alpha. Portfolios 1 and 10 are further divided into terciles. The portfolios are equally weighted and the weights are readjusted whenever a fund is delisted. *RMRF*, *HML* and *SMB* are the Fama-French²⁷ factors for market proxy and factor-mimicking portfolios for book-to-market and size. *MOM* is a factor-mimicking portfolio for 1-year momentum. Newey-West²⁸ corrected *t*-values are represented by ***, **, * for 99%, 95%, 90% significance, respectively.

0.44 per cent. There are at least two potential explanations for this decrease. First, by requiring funds to have at least 3 years of data to estimate 3-year alphas, the portfolios contain fewer funds and the estimates become less precise. In fact, the R^2 for portfolio 1 becomes as low as 0.44. Secondly, a fund's alpha 3 years ago may provide little information about today's alpha. Sorting on lagged 1-year alpha, on the other hand, gives a significant 0.67 per cent spread in alpha between portfolios 1 and 10. The funds' alphas are estimated very imprecisely, however, and the portfolio alphas show no monotonic pattern. This suggests that alpha persistence is only a short-run phenomenon. The following passage takes a closer look at this. Figure 11.3(a) shows the transition probabilities of funds between portfolios sorted on lagged alphas. There is a strong trend for funds to remain in the same portfolio or to move to adjacent portfolios. This is particularly true for the extreme deciles. Portfolio 1, containing the funds with the highest lagged alphas, keeps over 30 per cent of its funds in the following year. This confirms the results of Jagannathan *et al.*,³⁷ who find that approximately 20 per cent of abnormal performance relative to the style benchmark over a 3-year period

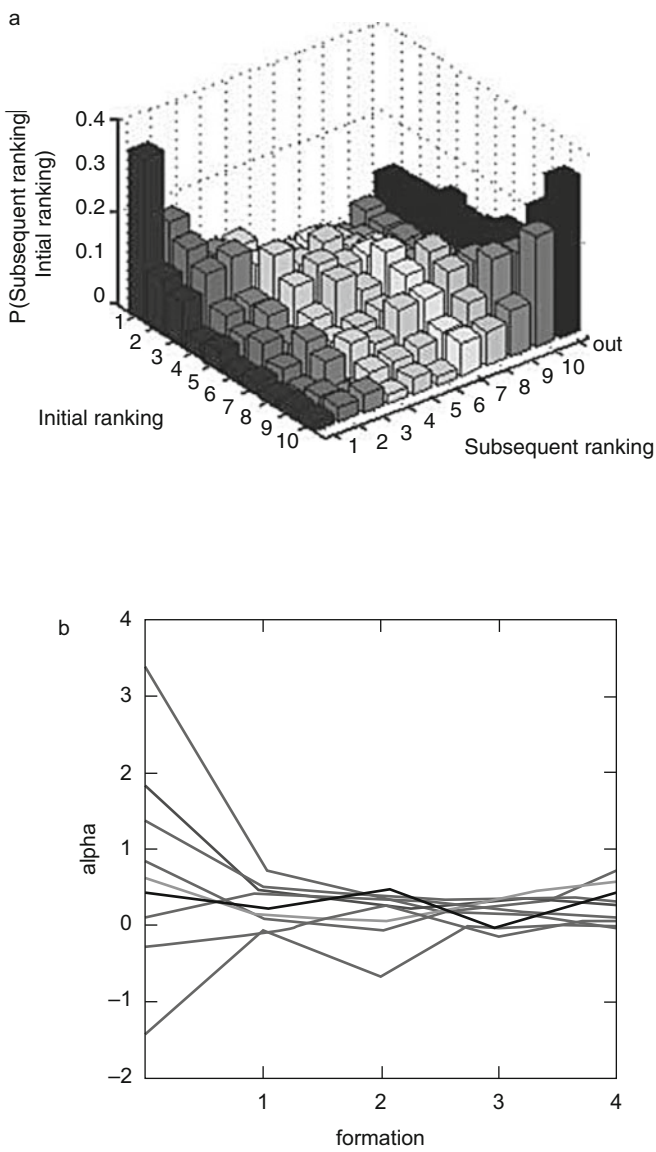


Figure 11.3(a) Contingency table of initial and subsequent ranking (lagged alpha)
Each year, funds are sorted into decile portfolios based on their past 2-year alphas. The bars indicate the transition probability of a fund moving from decile i into decile j or stop reporting.

Figure 11.3(b) Postformation alpha of portfolios sorted on lagged 2-year alpha
Each year, funds are sorted in decile portfolios based on lagged 2-year alphas. The figure shows the monthly excess returns of these portfolios in the formation year and subsequent years.

spills over to the next 3-year period. This persistence is stronger among top funds than among bottom funds. Moreover, the figure shows that it is very unlikely for a high-alpha fund to become a low-alpha fund in the subsequent year or vice versa. This stands in sharp contrast to high-return funds, as shown above. The probability that a fund stops reporting decreases in lagged alpha.

This confirms the findings of Fung *et al.*,³¹ who show that alpha funds exhibit substantially lower liquidation rates than beta-only funds.

Figure 11.3(b) shows the alphas of the portfolios over several years. Similar to sorting portfolios based on lagged returns, the spread decreases after the formation period, though not as strongly. There is clearly more persistence in alpha than in raw returns for the first year. As in Kosowski *et al.*,¹⁷ abnormal performance is significant and persists over 1 year. While much of the persistence in alpha is gone after 1 year, the worst performers continue to underperform for 2 years. The alpha spread even widens to 0.97 per cent. But both the winner and loser portfolios have alphas insignificantly different from zero after the tracking year and the alpha spread is only significant at the 5 per cent level. Hence, portfolios based on lagged alphas should be reformed annually, as there is no outperformance 2, 3 and 4 years after picking the high-alpha funds.

The results are similar when we alternatively sort the funds based on the *t*-value of lagged alphas, which is the same as sorting based on the appraisal ratio. This sorting procedure identifies a highly significant alpha spread of 0.84 per cent and a return spread of 0.58 per cent. Carhart,³ however, argues that using the same model for sorting and performance evaluation can pick up a model bias. For example, if the factor exposures are estimated too low or too high for a fund, this shows up as persistent over- or underperformance relative to the factor model. The problem is similar to an omitted factor. Hence, it is important to keep in mind this potential shortcoming when interpreting the results reported in this section.

Persistence of the Sharpe Ratio

Finally, we sort the funds into portfolios based on lagged Sharpe Ratios. The results in Table 11.4 show that this does a better job at detecting alpha portfolios than sorting based on raw returns or return variance. The results based on alpha sorting as reported in Table 11.3 are, however, substantially stronger. Table 11.4 shows that for portfolios 1–6, the alpha is positive and significant, whereas it is insignificant for portfolios 7, 8 and 10. In addition, portfolio 1 generates the highest alpha, whereas portfolios 7, 8 and 10 generate the lowest alphas. For space reasons, we do not report the transition probabilities of funds between portfolios sorted on lagged Sharpe Ratios or the Sharpe Ratios of the portfolios over several years (equivalent to Figure 11.3(a) and (b) for the alpha persistence).³⁸

Table 11.4 Portfolios of hedge funds formed on lagged 1-year Sharpe Ratio

Portfolio	Excess return (s.d.)	<i>Alpha</i>	<i>RMRF</i>	<i>HML</i>	<i>SMB</i>	<i>MOM</i>	R^2_{adj}
1A	0.69 (0.18)	0.49***	0.15***	-0.02	0.10**	0.09***	0.23
1B	0.72 (0.19)	0.35**	0.25***	0.04	0.28***	0.12***	0.62
1C	1.01 (0.24)	0.63***	0.36***	-0.03	0.18***	0.16***	0.51
1 (high)	0.79 (0.17)	0.45***	0.25***	0.02	0.19***	0.12***	0.60
2	0.97 (0.23)	0.44***	0.41***	0.02	0.23***	0.18***	0.76
3	0.81 (0.24)	0.26**	0.49***	0.05	0.22***	0.14***	0.78
4	0.90 (0.26)	0.43***	0.50***	-0.00	0.26***	0.09***	0.77
5	0.88 (0.28)	0.36***	0.50***	-0.01	0.31***	0.13***	0.82
6	0.92 (0.27)	0.31**	0.56***	0.13***	0.28***	0.08**	0.76
7	0.64 (0.31)	0.07	0.70***	0.05	0.29***	0.02	0.86
8	0.54 (0.31)	-0.03	0.60***	0.09	0.27***	-0.02	0.81
9	0.71 (0.29)	0.35**	0.56***	0.10	0.19***	-0.12**	0.67
10 (low)	0.48 (0.29)	0.18	0.47***	0.13	0.20***	-0.18***	0.63
10A	0.86 (0.36)	0.59**	0.62***	0.07	0.07	-0.12*	0.51
10B	0.23 (0.37)	0.11	0.43***	0.09	0.17**	-0.25**	0.39
10C	0.32 (0.31)	-0.02	0.44***	0.29***	0.22***	-0.18***	0.38
1-10	0.31 (0.28)	0.26	-0.26***	-0.12	-0.01	0.30***	0.43
1A-10C	0.37 (0.35)	0.50	-0.30***	-0.31***	-0.12	0.27***	0.26

Each January from 1995 until 2005, decile portfolios based on lagged 1-year Sharpe Ratios are formed. Portfolio 1 contains funds with the highest 1-year Sharpe Ratio, portfolio 10 funds with the lowest 1-year Sharpe Ratio. Portfolios 1 and 10 are further divided into terciles. The portfolios are equally weighted and the weights are readjusted whenever a fund is delisted. *RMRF*, *HML* and *SMB* are the Fama-French²⁷ factors for market proxy and factor-mimicking portfolios for book-to-market and size. *MOM* is a factor-mimicking portfolio for 1-year momentum. Newey-West²⁸ corrected *t*-values are represented by ***, **, * for 99%, 95%, 90% significance, respectively.

Robustness Checks

Sub-period analysis

In order to assess the robustness of these results, we conduct the analyses for sub-periods. Prior research examines the relation between hedge fund performance and market conditions. Using a conditional benchmark, Kat and Miffre³⁹ conclude that abnormal performance is countercyclical. Capocci *et al.*,¹⁰ however, find that hedge funds show stronger outperformance in bullish times. Here, March 2000 is set as the cutoff point that divides the sample into a bull market period from January 1994 to March 2000 and a period with a bear market from April 2000 to December 2005. This also allows the factor loadings to vary between these two samples. In fact, Kosowski *et al.*¹⁷ find a breakpoint in the year 2000 for most hedge fund return indices.

The results for lagged 2-year alpha sorts are shown in Table 11.5 and confirm the previous findings for the sub-periods. Alpha portfolios have generally higher returns and lower factor exposures, especially for the market. Again,

Table 11.5 Sub-period analysis of portfolios of hedge funds formed on lagged 2-year alpha

Portfolio	Excess return (s.d.)	Alpha	RMRF	HML	SMB	MOM	R ² _{adj.}
<i>1996:1–2000:3</i>							
1 (high)	1.87 (0.47)	1.15***	0.45**	−0.14	0.19**	0.07	0.48
10 (low)	1.62 (0.64)	0.32	1.00***	0.06	0.23**	0.09	0.69
1–10	0.25 (0.42)	0.83**	−0.55***	−0.21	−0.04	−0.02	0.27
<i>2000:4–2005:12</i>							
1 (high)	0.40 (0.42)	0.26	0.50***	0.11***	0.25***	−0.04	0.76
10 (low)	−0.32 (0.53)	−0.43*	0.76***	0.08	0.28***	0.04	0.81
1–10	0.72 (0.27)	0.69***	−0.26***	0.04	−0.04	−0.09	0.17

Each January from 1996 to 2005, decile portfolios based on lagged 2-year alphas are formed. Portfolio 1 contains funds with the highest 2-year alpha, portfolio 10 funds with the lowest 2-year alpha. The portfolios are equally weighted and the weights are readjusted whenever a fund is delisted. *RMRF*, *HML* and *SMB* are the Fama–French²⁷ factors for market proxy and factor-mimicking portfolios for book-to-market and size. *MOM* is a factor-mimicking portfolio for 1-year momentum. Newey–West²⁸ corrected *t*-values are represented by ***, **, * for 99%, 95%, 90% significance, respectively.

alpha persistence is much stronger than return persistence (not reported) over the 1-year horizon. Buying funds with the highest lagged alpha and reforming annually delivers high returns, relatively low market exposure and positive alpha. For example, while the mean monthly market return over the second sub-period was −0.16 per cent, portfolio 1 with low market exposure earned a monthly return of 0.4 per cent and positive (but insignificant) alpha. In addition, Table 11.5 shows that, looking at the excess returns, hedge funds have clearly performed much better in the bull-market period. Given their net long market exposure, this is not surprising.⁴⁰

Correction for ‘stale prices’

The presence of ‘stale prices’ because of hard-to-price assets or managed prices can artificially reduce fund statistics such as volatility or beta. If, for example, the market falls near the end of the month and an illiquid asset does not trade, the drop in price will not show up until the following month. In addition, the lack of market prices may leave hedge funds with ‘flexibility’ in how they mark such positions for (month-end) reporting. Consequently, both illiquidity and managed prices can lead to asynchronous price reactions. This can spuriously show up as alpha because the contemporaneous factors will lose explanatory power. As equity long/short hedge funds invest in listed stock, this problem is expected to be of minor importance for the results in our study. Nevertheless, we check the robustness of our results with respect to stale prices by aid of the procedure proposed by Getmansky *et al.*,²¹ which allows adjustment of the raw return series for potential smoothing. We estimate their MA(2) model for all equity long/short hedge funds in the

Table 11.6 Portfolios of hedge funds formed on lagged 2-year alpha

Portfolio	Excess return (s.d.)	Alpha	RMRF	HML	SMB	MOM	R ² _{adj.}
1A	1.34 (0.43)	1.33***	0.34**	-0.26*	-0.01	-0.06	0.23
1B	0.74 (0.36)	0.36	0.39***	0.07	0.35***	0.03	0.36
1C	1.22 (0.41)	0.75***	0.67***	-0.01	0.31***	0.01	0.68
1 (high)	0.98 (0.33)	0.66***	0.50***	-0.06	0.24***	0.00	0.65
2	0.87 (0.27)	0.43***	0.46***	0.09*	0.30***	0.06	0.70
3	0.83 (0.24)	0.43***	0.49***	0.13***	0.23***	0.00	0.81
4	0.70 (0.25)	0.31***	0.44***	0.05	0.25***	0.06**	0.75
5	0.68 (0.26)	0.28***	0.46***	0.02	0.25***	0.07**	0.81
6	0.26 (0.30)	-0.12	0.53***	0.06	0.27***	-0.03	0.73
7	0.70 (0.26)	0.22**	0.53***	0.11***	0.23***	0.08***	0.83
8	0.79 (0.27)	0.37**	0.55***	0.09*	0.17***	0.01	0.75
9	0.47 (0.29)	-0.08	0.63***	0.15***	0.22***	0.08**	0.82
10 (low)	0.45 (0.44)	-0.13	0.83***	-0.03	0.26***	0.07	0.79
10A	0.28 (0.38)	-0.26	0.66***	0.07	0.27***	0.08	0.60
10B	1.15 (0.41)	0.69**	0.65***	0.06	0.22***	0.00	0.50
10C	-0.26 (0.71)	-0.99***	1.20***	-0.30**	0.21**	0.18*	0.71
1-10	0.53 (0.25)	0.80***	-0.33***	-0.03	-0.02	-0.07**	0.25
1A-10C	1.59 (0.62)	2.32***	-0.86***	0.04	-0.22*	-0.24***	0.40

The fund returns are modified according to the Getmansky *et al*²¹ correction that accounts for 'stale prices'. Each January from 1996 to 2005, decile portfolios based on lagged 2-year alphas are formed. Portfolio 1 contains funds with the highest 2-year alpha, portfolio 10 funds with the lowest 2-year alpha. The portfolios are equally weighted and the weights are readjusted whenever a fund is delisted. *RMRF*, *HML* and *SMB* are the Fama-French²⁷ factors for market proxy and factor-mimicking portfolios for book-to-market and size. *MOM* is a factor-mimicking portfolio for 1-year momentum. The fund returns are modified according to the Getmansky *et al*²¹ correction that accounts for 'stale prices'. Newey-West²⁸ corrected *t*-values are represented by ***, **, * for 99%, 95%, 90% significance, respectively.

CISDM database to obtain the smoothing parameters (or teta coefficients) based on which we construct 'unsmoothed' return series. The results are reported in Table 11.6 for the alpha sorts, and show that the correction leaves the results qualitatively unchanged.⁴¹

Conclusion

This paper investigates the performance persistence of equity long/short hedge funds. We find returns to show very little persistence at the annual horizon, irrespective of the length of the formation period. In addition, the observed persistence can be fully explained by factor exposures. In fact, it is mainly driven by holdings of last year's winner stocks. Hence, funds with the highest return last period are generally no-alpha funds. Moreover, best and worst performing funds tend to switch their places often, as they are very volatile.

There are, however, more promising criteria to select outperforming funds. Sharpe Ratio, market beta, and, in particular, alpha are more useful characteristics to sort funds because they are more persistent than raw returns. Although not reported in the paper, we find that sorting on lagged Sharpe Ratio identifies portfolios with the highest Sharpe Ratio, which are also more likely to have alpha. Sorting on lagged market beta or alpha identifies the largest and most significant alpha spread over the subsequent period. Past alpha also provides the most information about future returns by discovering the most significant spread in raw returns.

These selection strategies are associated with less risk than selection based on raw returns. Funds with high alpha are very unlikely to have a low alpha in the next period: instead, they will likely stay in the same or adjacent portfolio. Also, there is more persistence in alpha at the annual horizon. No persistence in alpha has, however, been detected for longer horizons except that funds with the lowest alpha continue to have the lowest alpha for 2 years.

There are at least three reasons that render such trading strategies impractical though. First, the option of going short a portfolio of hedge funds to get the alpha spread is not available. Secondly, the transaction and administrative costs of implementing such a strategy may outweigh its benefits. Thirdly, lockup or redemption periods imposed by funds can make frequent disinvestment impossible. Selecting funds with the highest lagged alpha, however, already produces a significant monthly alpha of 0.71 per cent, whereas the spread between portfolios 1 and 10 is 0.78 per cent. Going short portfolio 10 is therefore not necessary to get most of the benefit.⁴² Secondly, the annual frequency of reforming the portfolios is relatively low. This makes the strategies more feasible with respect to transaction costs and lockup periods, especially for funds of funds facing more favourable conditions than individual investors.

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 42. Using the more recent subsample (2000–2005), however, the largest part of the alpha spread comes from the negative alpha of portfolio 10.

12

Examination of Fund Age and Size and Its Impact on Hedge Fund Performance

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Practical applications

Hedge funds are an increasingly popular investment option for both high net worth and institutional investors. With an increasing number of funds from which to choose, it is imperative that investors find ways to narrow down their investment options and evaluate managers.

This paper attempts to discover whether smaller, younger hedge funds offer stronger performance than larger, older hedge funds. Using indices created with six subsets of hedge fund data (small, medium, large, young, mid-age and older funds, as defined herein) and Monte Carlo simulations, we examine the performance, volatility and risk profiles of each fund group. Derivatives Use, Trading & Regulation (2007) 12, 342–350. doi:10.1057/palgrave.dutr.1850052

Introduction

There is an old adage that states ‘Age and treachery will always overcome youth and skill’. However, it is questionable whether or not this maxim is indeed true when it comes to hedge fund investments. The issue of hedge fund age and size, and its potential impact on performance has been a topic of frequent debate over the past five years. Certainly, hedge funds—large and small, or new and old—have experienced their fair share of both positive and negative publicity. For example, Julian Robertson’s Tiger Funds drew attention to the potential pitfalls surrounding large, established funds, when after decades of successful performance, he shut down his flagship

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fund due in large part to the difficulties he encountered in managing his massive fund. On the other hand, smaller, younger funds like Integral Capital Management have made headlines for both operational and performance woes. This paper will attempt to examine the performance of hedge funds, classified into six size and age subcategories, to determine if any one group has a performance advantage over the others.

Supersize Me?

In many areas, bigger is perceived as better. However, in hedge funds, the opposite seems to be the case. We created three size-based hedge fund indices by first combining the hedge fund performance records from the Hedge Fund Research, HedgeFund.net, Altvest from InvestorForce and Barclays Global HedgeSource databases into a single 'master' database. Duplicate hedge fund records as well as records for funds of funds were removed. Reports were then run to find the monthly return and monthly fund size for each fund from January 1996 to July 2006. Based on its then-current fund size, all funds were recategorised each month and divided into three classes: funds with less than or equal to \$100 m under management; funds with over \$100 m up to \$500 m under management; and funds with over \$500 m. A simple mean of all monthly returns in each of the three categories was calculated for each month. If a fund did not have a reported fund size in a given month, it was not included in any of the size-based indices for that month. Funds denominated in a non-US currency, with a very different value from the US dollar (USD) (eg JPY, CHF, NOK and SEK), were also excluded from the size-based indices, as it was not clear whether the monthly fund sizes reported for such funds were denominated in USD or in the native currency, making accurate size categorisation problematic. The sample of funds included in each of the three indices varied from month to month. On average, the small-sized index contained, 1,790.5 funds per month, whereas the medium-sized and large-sized indices contained 479.9 and 136.5 funds per month, respectively. In all three cases, the earlier monthly samples contained less funds than later samples. The three size-based indices that were created using this information are shown below in Figure 12.1.

As the summary risk-reward tables below show, small funds clearly provide the best investment option, given that 'best' is simply defined as providing maximised returns. For example, the small funds index in our study provided an annualised return of 15.46 per cent while maintaining an annualised standard deviation of 6.31 per cent and annualised downside deviation of 2.58 per cent. In comparison, medium-sized funds produced an annualised return of 12.50 per cent with slightly lower annualised standard and downside deviations of 5.89 and 2.10 per cent, respectively. The large fund index in the study produced the lowest annualised return of 11.93 per cent and also the lowest annualised standard deviation at 5.72 per cent. Its

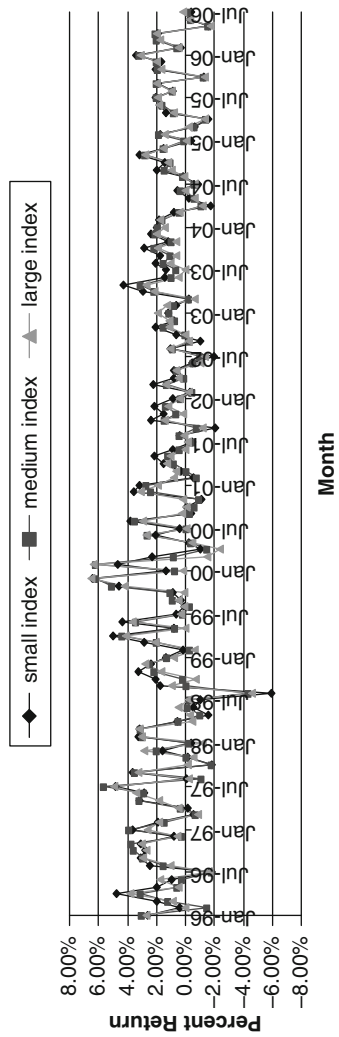


Figure 12.1 Performance of small, medium and large hedge funds

annualised downside deviation of 2.16 per cent falls in the middle of the three groups, although it is very similar to that of the medium-sized fund index in the study (Tables 12.1–12.3).

The pattern of smaller funds outperforming larger funds is repeated when examining Monte Carlo simulations performed on the created indices. The simulations were run five years forward, based on the full historical data of each index (January 1996 to July 2006), using the S&P 500 Index (Total

Table 12.1 Small HF index (up to \$100m)

Risk table	Month	Quarter	Annualised
Compound return	1.20%	3.66%	15.46%
Arithmetic mean	1.22%	3.76%	N/A
Standard deviation	1.82%	3.71%	6.31%
Semi deviation	1.87%	3.92%	6.49%
Gain deviation	1.30%	2.90%	4.50%
Loss deviation	1.08%	1.70%	3.72%
Down deviation (5.00%)	0.90%	1.41%	3.13%
Down deviation (2.00%)	0.81%	1.15%	2.79%
Down deviation (0%)	0.75%	0.98%	2.58%
Sharpe (2.00%)	0.58	0.88	2.01
Sortino (5.00%)	0.88	1.73	3.06
Sortino (2.00%)	1.29	2.76	4.47
Sortino (0%)	1.62	3.72	5.6
Skewness	-0.2	-0.08	
Kurtosis	1.19	-0.09	

Table 12.2 Medium HF index (over \$100m up to \$500m)

Risk table	Month	Quarter	Annualised
Compound return	0.99%	2.99%	12.50%
Arithmetic mean	1.00%	3.07%	N/A
Standard deviation	1.70%	3.26%	5.89%
Semi deviation	1.50%	2.93%	5.20%
Gain deviation	1.37%	2.84%	4.76%
Loss deviation	0.78%	1.71%	2.70%
Down deviation (5.00%)	0.78%	1.18%	2.72%
Down deviation (2.00%)	0.67%	0.93%	2.33%
Down deviation (0%)	0.61%	0.79%	2.10%
Sharpe (2.00%)	0.49	0.79	1.7
Sortino (5.00%)	0.74	1.49	2.56
Sortino (2.00%)	1.22	2.69	4.22
Sortino (0%)	1.63	3.8	5.64
Skewness	0.49	0.48	
Kurtosis	1	0.9	

Table 12.3 Large HF index (over \$500m)

Risk table	Month	Quarter	Annualised
Compound return	0.94%	2.86%	11.93%
Arithmetic mean	0.96%	2.92%	N/A
Standard deviation	1.65%	2.88%	5.72%
Semi deviation	1.48%	2.74%	5.14%
Gain deviation	1.35%	2.37%	4.66%
Loss deviation	0.88%	1.35%	3.04%
Down deviation (5.00%)	0.79%	1.12%	2.75%
Down deviation (2.00%)	0.69%	0.85%	2.38%
Down deviation (0%)	0.62%	0.69%	2.16%
Sharpe (2.00%)	0.48	0.84	1.66
Sortino (5.00%)	0.68	1.46	2.34
Sortino (2.00%)	1.13	2.79	3.92
Sortino (0%)	1.51	4.12	5.23
Skewness	0.43	0.29	
Kurtosis	1.48	0.65	

Return) as the market benchmark, and a 2 per cent annualised risk-free rate of return and minimum acceptable return (MAR). Using the Bootstrap method, we ran 1,000 simulations with quarterly rebalancing, which provided a minimum and maximum predicted return for each index. As shown below, the maximum simulated return for the large hedge fund index was 21.57 per cent, while the medium hedge fund index showed a maximum simulated return of 23.57 per cent, and the small hedge fund index produced a maximum simulated return of 27.49 per cent. The mean simulated annualised returns followed the same pattern: best for the small index, worst for the large index (Tables 12.4–12.6).

However, while the small fund index in the study produced, and had the potential to produce in the future, a higher annualised return, it also had a higher volatility profile than did the larger- and medium-sized indices. In Tables 12.1, 12.2 and 12.3 above, we have already noted that the smaller funds had the highest standard and downside deviation of the three fund groups. Using the same Monte Carlo simulation methods as above, we can extrapolate potential maximum drawdowns for the three indices. The maximum simulated drawdown of the small fund index is –14.23 per cent, while the medium-sized fund index came in significantly lower at –11.02 per cent, and the large fund index fell in the middle with a maximum simulated drawdown of –12.30 per cent (Tables 12.7–12.9).

The question therefore arises, why is performance higher for smaller funds? A number of factors may be responsible for the higher performance. First, smaller funds, because they have less capital to invest, can select only from their best investment ideas. Their positions are generally smaller, so they can move more nimbly and attract less attention to their strategic

Table 12.4 Small fund index Monte Carlo simulation—annualised return

All portfolio statistics	Annualised return
Number of simulations	1,000
Mean	15.44%
Median	15.43%
Standard deviation	3.14%
Maximum	27.49%
Minimum	5.11%

Table 12.5 Medium fund index Monte Carlo simulation—annualised return

All portfolio statistics	Annualised return
Number of simulations	1,000
Mean	12.51%
Median	12.44%
Standard deviation	2.85%
Maximum	23.57%
Minimum	4.16%

Table 12.6 Large fund index Monte Carlo simulation—annualised return

All Portfolio statistics	Annualised return
Number of simulations	1,000
Mean	11.94%
Median	11.91%
Standard deviation	2.74%
Maximum	21.57%
Minimum	4.74%

Table 12.7 Small fund index Monte Carlo simulation—drawdown

All portfolio statistics	Maximum drawdown
Number of simulations	1,000
Mean	-4.49%
Median	-3.80%
Standard deviation	2.26%
Maximum	-14.23%
Minimum	-0.59%

Table 12.8 Medium fund index Monte Carlo simulation—drawdown

All portfolio statistics	Maximum drawdown
Number of simulations	1,000
Mean	−3.77%
Median	−3.57%
Standard deviation	1.68%
Maximum	−11.02%
Minimum	−0.92%

Table 12.9 Large fund index Monte Carlo simulation—drawdown

All portfolio statistics	Maximum drawdown
Number of simulations	1,000
Mean	−3.84%
Median	−3.54%
Standard deviation	1.79%
Maximum	−12.30%
Minimum	−0.66%

moves than can large funds. Smaller funds may also be able to exploit small market inefficiencies and opportunities that larger funds may have to ignore. In the mean time, as assets increase, managers may be forced to change the strategy that made them initially successful by resorting to one of the following tactics to keep capital in play:

- Look outside the manager’s area of expertise for additional investment opportunities.
- Concentrate more money into top tier investment ideas.
- Move from top investment ideas to second, third or fourth tier opportunities.
- Keep a large amount of the funds in cash, thereby lowering the returns.
- Farm out portions of the investment management process to other sub-advisors who may not have the manager’s level of experience, credentials or back office infrastructure and
- Spend more time managing the organisation and less time managing the funds, leaving the trading in the hands of less-seasoned personnel.

Additionally, a considerable portion of large funds’ assets may be from institutional investors, such as pension funds, prompting large fund managers to invest more conservatively than small fund managers, who mainly handle the assets of less risk-averse, high net worth individuals.

Of course, there are disadvantages to investing with small funds as well. Our small fund index exhibited the highest volatility of the three. Fund

infrastructure may be weak, and operational problems can plague or even destroy a fund. The fund assets may be concentrated with a few limited partners, which can cause a cascading effect if just one or two investors redeem, and the manager has to liquidate positions prematurely. Finally, smaller fund managers may take more risks owing to a desire to establish a strong track record and attract additional assets, or because of a lack of established risk-control measures.¹ Meanwhile, larger funds will often have a more well-developed infrastructure and back office, well-defined risk-control measures, steady performance that caters to their institutional limited partners and a diversified investor-base.^{2,3} To choose the best fund, it is therefore critical that small, medium and large funds be fully evaluated before an investment is made.

Youth over experience?

Another popular subject of debate is whether hedge funds really do produce their best returns early in their history, regardless of assets under management. To try to shed light on this issue, three age-based indices from the same master hedge fund database referred to above, were created from the Hedge Fund Research, HedgeFund.net, Altvest from InvestorForce and Barclays Global HedgeSource databases. Reports were then run to find the monthly return for each fund from January 1996 to July 2006. Based on its then-current fund age all funds were recategorised each month and divided into three classes: funds with less than a two-year track record, funds with two to four years of performance and funds with more than four years of performance. A simple mean of all monthly returns in each of the three categories was calculated for each month. The indices that were created using this information are shown below in Figure 12.2.

Like the smallest funds, the youngest funds in this study were the strongest performers. The young fund index in the study produced an annualised return of 17.50 per cent, while maintaining an annualised standard deviation of 5.97 per cent and an annualised downside deviation of 2.24 per cent. In comparison, the mid-age fund index produced an annualised return of 14.10 per cent, with annualised standard and downside deviations of 6.39 and 2.81 per cent, respectively. The index composed of oldest funds in the study produced the lowest annualised return of 11.84 per cent and also the lowest annualised standard deviation, of 6.32 per cent. Its downside deviation of 2.85 per cent falls in the middle of the three groups, although, again, it is so close to that of the mid-age index that the difference may not be significant (Tables 12.10–12.12).

The pattern of younger funds outperforming older funds is repeated when examining Monte Carlo simulations performed on the created indices. As with the size-based indices, the simulations were run five years forward, based on the full history of the indices from January 1996 to July 2006. The S&P 500 Index (Total Return) was again used as the market benchmark, with a 2 per cent annualised risk-free rate of return and MAR. Using the Bootstrap

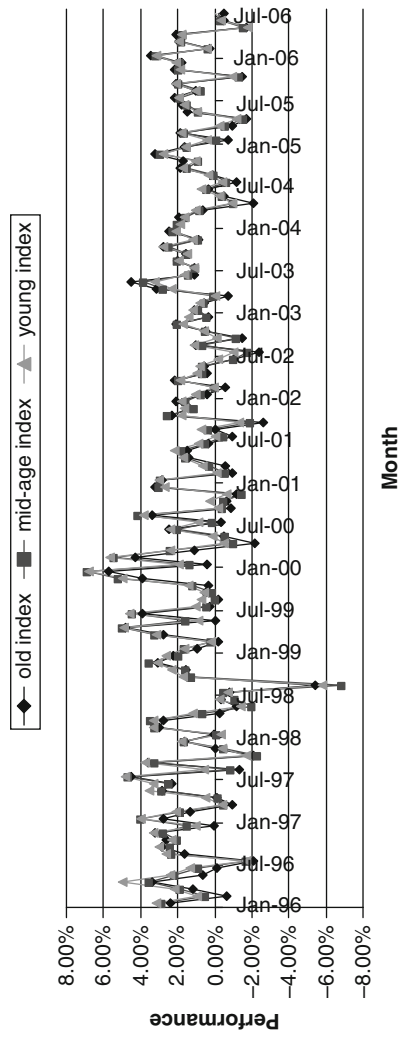


Figure 12.2 Monthly performance of various age hedge funds

Table 12.10 Young hedge fund index (up to 2 Yr)

Risk table	Month	Quarter	Annualised
Compound return	1.35%	4.11%	17.50%
Arithmetic mean	1.37%	4.21%	N/A
Standard deviation	1.72%	3.63%	5.97%
Semi deviation	1.73%	3.70%	5.99%
Gain deviation	1.31%	2.92%	4.55%
Loss deviation	1.13%	1.70%	3.92%
Down deviation (5.00%)	0.78%	1.13%	2.70%
Down deviation (2.00%)	0.70%	0.90%	2.41%
Down deviation (0%)	0.65%	0.78%	2.24%
Sharpe (2.00%)	0.7	1.02	2.42
Sortino (5.00%)	1.21	2.55	4.21
Sortino (2.00%)	1.71	4	5.91
Sortino (0%)	2.09	5.31	7.24
Skewness	-0.11	0.15	
Kurtosis	2.18	0.45	

Table 12.11 Mid-age hedge fund index (2–4 Years)

Risk table	Month	Quarter	Annualised
Compound return	1.11%	3.35%	14.10%
Arithmetic mean	1.12%	3.46%	N/A
Standard deviation	1.84%	3.89%	6.39%
Semi deviation	1.87%	4.01%	6.48%
Gain deviation	1.34%	3.11%	4.64%
Loss deviation	1.20%	1.93%	4.15%
Down deviation (5.00%)	0.97%	1.59%	3.35%
Down deviation (2.00%)	0.87%	1.32%	3.02%
Down deviation (0%)	0.81%	1.15%	2.81%
Sharpe (2.00%)	0.52	0.76	1.8
Sortino (5.00%)	0.72	1.33	2.5
Sortino (2.00%)	1.08	2.17	3.74
Sortino (0%)	1.36	2.9	4.71
Skewness	-0.21	0.13	
Kurtosis	2.36	0.61	

method, we ran 1,000 simulations with quarterly rebalancing, which provided a minimum and maximum simulated return for each index. As shown below, the maximum simulated return for the old hedge fund index was 23.35 per cent, while the mid-age hedge fund index showed a maximum simulated return of 25.70 per cent, and the young hedge fund index produced a maximum simulated return of 27.62 per cent. Mean simulated returns followed the same pattern (Tables 12.13–12.15).

Table 12.12 Old hedge fund index

Risk table	Month	Quarter	Annualised
Compound return	0.94%	2.84%	11.84%
Arithmetic mean	0.95%	2.93%	N/A
Standard deviation	1.83%	3.50%	6.32%
Semi deviation	1.88%	3.58%	6.50%
Gain deviation	1.24%	2.63%	4.29%
Loss deviation	0.98%	1.72%	3.40%
Down deviation (5.00%)	1.01%	1.60%	3.49%
Down deviation (2.00%)	0.89%	1.30%	3.10%
Down deviation (0%)	0.82%	1.12%	2.85%
Sharpe (2.00%)	0.43	0.7	1.5
Sortino (5.00%)	0.53	1	1.82
Sortino (2.00%)	0.86	1.8	2.99
Sortino (0%)	1.14	2.53	3.95
Skewness	-0.14	-0.09	
Kurtosis	0.32	-0.37	

Table 12.13 Young fund index Monte Carlo simulation—annualised return

All portfolio statistics	Annualised return
Number of simulations	1,000
Mean	17.50%
Median	17.48%
Standard deviation	3.01%
Maximum	27.62%
Minimum	7.80%

Table 12.14 Mid-age fund index Monte Carlo simulation—annualised return

All portfolio statistics	Annualised return
Number of simulations	1,000
Mean	14.10%
Median	14.11%
Standard deviation	3.16%
Maximum	25.70%
Minimum	4.23%

Table 12.15 Old fund index Monte Carlo simulation—annualised return

All portfolio statistics	Annualised return
Number of simulations	1,000
Mean	11.82%
Median	11.81%
Standard deviation	3.08%
Maximum	23.35%
Minimum	2.18%

Table 12.16 Young fund index Monte Carlo simulation—drawdown

All portfolio statistics	Maximum drawdown
Number of simulations	1,000
Mean	-3.74%
Median	-2.75%
Standard deviation	2.28%
Maximum	-13.04%
Minimum	-0.53%

Table 12.17 Mid-age fund index Monte Carlo simulation—drawdown

All portfolio statistics	Maximum drawdown
Number of simulations	1,000
Mean	-4.95%
Median	-3.92%
Standard deviation	2.72%
Maximum	-17.46%
Minimum	-1.00%

Table 12.18 Old fund index Monte Carlo simulation—drawdown

All portfolio statistics	Maximum drawdown
Number of simulations	1,000
Mean	-5.28%
Median	-5.15%
Standard deviation	2.19%
Maximum	-14.84%
Minimum	-1.45%

However, unlike the smallest funds in the study, which exhibited the highest historical and simulated future returns, as well as the highest historical and simulated future volatility and drawdowns among the three size-based indices, the young fund index maintained the highest returns, without a corresponding impact on simulated future drawdowns. In fact, it is the mid-age hedge fund index that had the highest simulated maximum drawdown of the group, at least suggesting that those funds exhibit somewhat of a 'sophomore slump', before moving towards the more institutional profile of their older peers. Using the same Monte Carlo simulation methods as above, we can extrapolate potential maximum drawdowns for each group of funds. The maximum simulated maximum drawdown of the young fund index is -13.04 per cent, while the old fund index came in at -14.84 per cent, and the mid-age fund group posted a maximum simulated maximum drawdown of -17.46 per cent (Tables 12.16–12.18).

Conclusion

While performance is certainly not the only metric by which to select a hedge fund investment, an analysis of hedge fund performance for small, medium and large hedge funds, as well as of young, mid-age and older funds, suggests that investors who wish to maximise return should start their search by looking for younger, smaller funds. Investors who wish to maximise capital preservation should begin their hedge fund screening with larger, older funds. However, it is always important to consider the qualitative aspects of any fund, in addition to historical and simulated future returns, before making an investment.

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13

Great in Practice, Not in Theory: An Empirical Examination of Covered Call Writing

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Practical applications

Covered call writing is a common investment approach among individual investors and money managers. Although these investors often argue for the practice on the grounds that income from covered call writing can enhance overall returns during periods when stock prices are languishing, it had not been previously demonstrated that the practice is empirically supported. The practical application of this paper is that it provides an empirical grounding that investors can use when deciding on the merits of a covered call writing investment strategy.

We examine the empirical performance of an investment strategy that uses covered call writing to enhance the income from long positions in 27 stocks that are included in the FT-SE 100 Index. Using data for the period January 1994–December 1999 we show that, contrary to theory, in most instances covered call positions generate returns that exceed returns generated by buy-and-hold strategies.

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Introduction

There are various mutual funds in existence¹ that claim to enhance returns to unit holders through writing call options on investments held in the funds. This strategy is called ‘covered call writing’ when the number of options written on a stock does not exceed the number of shares of that stock held in the portfolio. Proponents of the covered call strategy assert that the portfolio benefits from a price increase² and the call premium under conditions when the stock is called out, and benefits from the call premium if the stock is not called out. In the latter instance, income from writing the call mitigates the drag on returns due to stagnancy or decline in the price of a portfolio member.

While these arguments are superficially appealing, closer inspection suggests that covered call writing may militate against enhancing portfolio returns. To see this, think of a covered call position as a portfolio comprised of a long position in the underlying financed in part by a short position in an option on the same underlying. Using a and b to denote positive portfolio weights where a is greater than 1, and $a-b=1$, the initial condition is the following:

$$\underbrace{aS_t - bC_t}_{\text{Covered Call}} = \underbrace{S_t}_{\text{Buy and Hold}} \quad (1)$$

where S_t is the price of the underlying asset and C_t is the price of the call. In a Black and Scholes^{3,4} world, the option can be replicated using a long position in δ_t units of the underlying asset, where $0 < \delta_t < 1$, and a short position in a riskless bond so that:

$$C_t = \delta_t S_t - B_t \quad (2)$$

Substituting the expression in (2) into (1) gives:

$$aS_t - b(\delta_t S_t - B_t) = (a - \delta_t b)S_t + bB_t \quad (3)$$

The expression in (3) indicates that entering a covered call position amounts to increasing one’s position in a riskless bond in substitution for the underlying. In most cases one would think of the expected return on the underlying exceeding the return on the riskless bond, so writing covered calls typically would reduce expected returns.

Monte Carlo Simulation

To illustrate the case in which the expected return on the underlying exceeds the return on the riskless bond ($E[r_u] > r_f$), we constructed a 50,000 path Monte Carlo simulation using five economic specifications. The economies

are the Black–Scholes³ economy (BS), the Cox⁵ constant elasticity of variance economy (CEV), the Geske⁶ compound option-pricing model (LEV), the Merton⁷ jump-diffusion economy (JD), and the Heston⁸ stochastic volatility economy (HES). Details of each of these economies are presented in the Appendix. In each of these economies, we adopted covered call strategies using options with, respectively, 30, 60, and 90 days to maturity.

We chose model parameters, shown in Table 13.1, so that in each economy the price of a 30-day, at-the-money European style call option approximated the Black–Scholes price with volatility set at 15 per cent. In Table 13.1, the percentages associated with strike prices K_1 and K_2 denote the degree of moneyness⁹ at which the near-the-money and out-of-the-money options were written in the simulations. Note that the comparison of covered call strategies to the buy-and-hold strategies in this paper are all ‘within model’ and not ‘across models’ so it is not necessary to establish a correspondence between models that goes beyond the qualitative similarity implied by the above approach.

As set out in Figure 13.1, the pattern of covered call writing we adopted in the simulations matches the pattern that is feasible using transactions data. Although Monte Carlo simulations permit one to assume the existence of options with any desired maturities, we elected to render our simulations consistent with the coverage pattern that can be achieved with transactions data. For the stocks included in this study (see Table 13.4 below), option prices

Table 13.1 Parameter values

	Economies				
	Black–Scholes	Leveraged	CEV	Heston	Jump/Diffusion
S_0	100	100	100	100	100
V	—	150	—	—	—
$M(t, T)$	—	50	—	—	—
$T-t$	—	10 years	—	—	—
K_1	102%	102%	102%	102%	102%
K_2	105%	105%	105%	105%	105%
μ	0.12	0.095	0.12	0.12	0.12
R	0.05	0.05	0.05	0.05	0.05
Q	0	0	0	0	0
a	—	—	4/3	—	0.13
$\sqrt{V}$	0.15	0.10	$\sqrt{0.5}$	0.15	$\sqrt{0.16}$
q	—	—	—	0.15	—
k	—	—	—	20.0	0
s	—	—	—	0.10	—
r	—	—	—	-0.5	—
l	—	—	—	—	2 jumps/year
b	—	—	—	—	$\sqrt{0.006}$

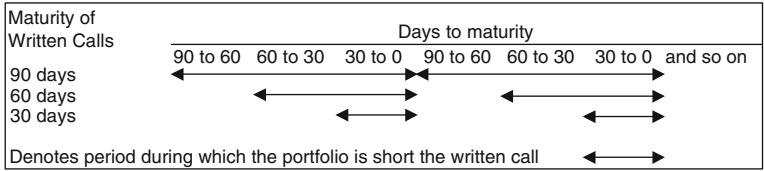


Figure 13.1 Coverage pattern of written calls

were quoted for options with 90-day increments in time to maturity. This permitted the coverage pattern for written calls that is shown in Figure 13.1. For example, in the 30-day written call protocol, we left the position uncovered (ie no call was written) for 60 days until the shortest maturity call that was quoted on the market had 30 days left to maturity. At this point, we wrote a 30-day call at the quoted market price. In the next 90-day cycle, we followed the same protocol. Similarly, the 60-day options left the covered call position uncovered one-third of the time.

The simulation results are set out in Table 13.2. In addition to including strategies involving 30, 60, and 90-day options, respectively, we tracked covered call positions for two levels of strike price. In the first case, we assumed that the written option possessed a strike price, K_1 , 2 per cent out of the money. In the second case we assumed a strike price, K_2 , 5 per cent out of the money. The results indicate that in all cases, the mean returns from the covered call positions were less than for the buy-and-hold positions. Also, in all cases, the volatilities of returns from the covered call positions were less than for the buy-and-hold positions.

Table 13.3 presents data on the frequency with which returns from the covered call positions exceeded returns from the buy-and-hold positions. Notwithstanding the results for the mean shown in Table 13.2, Table 13.3 shows that in qualitative terms, the 'losing' covered call strategy outperforms the buy-and-hold strategy relatively frequently. Arguably, participants in a market which exhibits this characteristic could develop a favourable perception of the merits of covered call writing even though it is a losing strategy overall.

Data

We obtained the data used in this paper from the LIFFE Euro-Out Products tick data disc. We randomly selected 30 companies from among the constituents of the FT-SE 100 Index as on 31st December, 1999. The LIFFE Euro-Out Products tick data disc includes option price¹⁰ data for a wide variety of traded options along with contemporaneous prices for the underlying. We deleted one company from the study because it was unclear whether a stock split reported by the company had been properly accounted for in the underlying price time series on the data disc.¹¹ We deleted two companies

Table 13.2 Summary of simulation results

	30 day		60 day		90 day	
	K ₁ (%)	K ₂ (%)	K ₁ (%)	K ₂ (%)	K ₁ (%)	K ₂ (%)
<i>Panel A: Covered call returns as a percent of buy-and-hold returns</i>						
BS	93.7	98.7	84.8	93.9	75.5	87.3
CEV	93.7	98.9	84.4	94.5	74.5	87.9
LEV	94.1	98.9	85.5	94.6	76.6	88.5
JD	94.2	99.3	85.6	94.8	76.4	88.2
HES	93.5	98.8	84.0	93.9	74.0	86.8
<i>Panel B: Covered call volatility as a percent of buy-and-hold volatility</i>						
BS	90.2	97.5	75.8	88.7	55.6	75.6
CEV	90.7	98.4	75.9	90.7	55.4	78.3
LEV	91.0	98.0	76.8	90.4	56.9	78.2
JD	90.7	96.3	76.7	87.6	58.1	75.1
HES	90.6	97.7	76.9	90.0	57.6	77.8

Table 13.3 Percentage of paths with covered call returns in excess of buy-and-hold returns

	90 day	
	K ₁ (%)	K ₂ (%)
BS	21	38
CEV	17	39
LEV	22	41
JD	27	41
HES	21	39

because of insufficient numbers of quoted option prices. Table 13.4 sets out the names and stock symbols of the companies included in the study.

Table 13.5 reports the returns, volatility, and the coefficient of correlation between returns on the individual stocks and the FT-SE 100 Index for the sample data. The reported returns are the annualised log-differences of share price at the beginning and end of the timeframe the study covers. The reported standard deviations of returns are the annualised standard deviations of daily returns over the timeframe the study covers. The reported correlations are also based on daily returns observations.

The coefficients of correlation for all companies were positive. Although a more sophisticated estimation of time varying coefficients of correlation, and a broader definition of the market portfolio would be desirable, the historical coefficients are suggestive of the possibility that each of the stocks is a positive beta stock in a capital asset pricing model sense. This being the case, the return expected on each stock exceeds the return on a riskless

Table 13.4 Companies included in the sample and related stock symbols

Stock symbol	Company name	Stock symbol	Company name
ALD	Allied Domecq PLC	MS	Marks and Spencer PLC
BBL	Barclays PLC	PO	Peninsular and Oriental Steam Navigation Company PLC
BOT	Boots Company PLC	RTZ	Rio Tinto PLC
BP	British Petroleum Company PLC	RUT	Reuters Holdings PLC
BSS	Bass PLC	RYL	Royal and Sun Alliance Insurance Group PLC
CAD	Cadbury Schweppes PLC	SAN	Sainsbury (J) PLC
CIR	Blue Circle Industries PLC	STL	Corus Group PLC (formerly British Steel PLC)
CUA	Commercial General Union PLC	TAR	Tarmac PLC
CW	Cable and Wireless PLC	TCO	Tesco PLC
DIX	Dixon's Group PLC	TW	Thames Water PLC
GXO	Glaxo Wellcome PLC	ULV	Unilever PLC
HSB	HSBC Holdings PLC	VOD	Vodafone Air Touch PLC
ICI	Imperial Chemical Industries PLC	ZCA	Zeneca Group PLC
LMO	LASMO PLC		

Table 13.5 Realised returns, volatilities, and correlation with the FTSE 100 Index

Company	Annual returns (%)	Volatility (%)	Correl. coeff.	Company	Annual returns (%)	Volatility (%)	Correl. coeff.
ALD	-3.43	24.09	0.39	MS	0.35	34.69	0.41
BBL	17.11	29.35	0.65	PO	-7.07	26.51	0.43
BOT	0.35	23.19	0.38	RTZ	7.89	25.02	0.44
BP	20.92	23.65	0.29	RUT	9.51	25.64	0.44
BSS	-1.20	44.85	0.37	RYL	14.63	33.41	0.58
CAD	5.82	26.07	0.37	SAN	5.42	32.95	0.37
CIR	5.60	23.60	0.33	STL	-4.18	26.06	0.28
CUA	0.86	31.45	0.58	TAR	3.94	33.55	0.27
CW	7.19	29.80	0.54	TCO	-5.12	32.56	0.39
DIX	11.30	33.00	0.28	TW	15.92	26.57	0.30
GXO	27.26	32.48	0.60	ULV	4.88	22.15	0.53
HSB	14.86	28.03	0.69	VOD	6.94	24.31	0.52
ICI	16.13	31.86	0.36	ZCA	33.43	33.11	0.50
LMO	-3.03	29.43	0.24				

bond. Consequently, as shown in (3), our *ex ante* expectation is that mean returns for the buy-and-hold strategies should exceed mean returns for the covered call strategies.¹²

For each stock in the study, we tracked a buy-and-hold position concurrent with a covered call position. In each case, we did so for three different

maturities of written option with the coverage patterns following those described in Figure 13.1. For each maturity, we tracked a covered call strategy for two levels of moneyness. For the first level, we used options with a strike price out of the money, but nearest to the at-the-money level. For the second level, we used the next available out-of-the-money strike price. We wrote options in a quantity equal to the holdings in the underlying at the beginning of each of the coverage periods shown in Figure 13.1 and used the option premiums from the written calls to increase the number of units of the underlying in the covered call position. If the stock was called out, we deleted units from our position equal in amount to the number of written calls in the call-out, and used the proceeds from the call-out (strike price times the number of written calls in the call-out) to acquire as many units of the underlying as possible at the price of the underlying at the time of the call-out. At the end of the study period, we calculated the annualised returns for the covered call position and the buy-and-hold position, respectively. We did not take account of transaction costs involved in either investment strategy because the goal of this paper is to examine the comparative properties of the underlying and options on the underlying rather than to determine relative economic merits.

Results and discussion

Table 13.6 summarises our results. In all but one of the scenarios, covered call writing strategies produced better returns more frequently than the buy-and-hold strategy. We consider the statistical significance of these outcomes using the ‘sign test for matched pairs’ and the ‘Wilcoxon signed rank test’. These statistical methods and the expression of the null hypothesis in the context of them are discussed below.^{13,14}

The sign test for matched pairs applies in situations in which there is either a failure or a success arising from repeated independent trials and there is a constant probability, p , of success. The test permits inference about the hypothesised probability, p . In the current context, the trials are the attempts to earn superior returns through the buy-and-hold strategy versus the covered call strategy. One would hypothesise that, if each strategy is equally effective, there would be an equal chance that one strategy would outperform the other at each comparison point. Thus, in the context of the sign test for matched pairs, the null hypothesis is the following:

H_0 : The probability that buy and hold outperforms covered call writing for each attempt is $\frac{1}{2}$.

The assumptions underlying the test are the following:

1. the outcomes can be classified as a success or a failure
2. the probability of a success remains constant from trial to trial and
3. the trials are independent.

Table 13.6 Annualised returns from covered call writing compared to the buy-and-hold strategy

	Covered call						Buy and Hold
	30 Day		60 Day		90 Day		
	A (%)	B (%)	A (%)	B (%)	A (%)	B (%)	
<i>Panel A</i>							
ALD	-4.20	-3.12	0.18	-1.34	1.13	-0.47	-3.43
BBL	13.99	15.84	16.55	18.94	7.78	12.34	17.11
BOT	2.45	2.06	4.74	3.00	2.54	1.71	0.35
BP	23.14	23.31	21.90	22.71	20.95	22.63	20.92
BSS	-10.51	-10.88	-1.85	-3.19	-4.96	-3.61	-1.20
CAD	3.25	6.47	8.96	8.30	7.94	8.12	5.82
CIR	7.47	7.79	8.72	9.27	8.70	9.72	5.60
CUA	7.21	3.01	6.29	4.93	7.84	5.34	0.86
CW	5.11	8.36	8.36	9.25	8.60	8.87	7.19
DIX	6.37	8.20	11.36	13.12	7.63	8.95	11.30
GXO	25.46	27.17	22.30	25.03	11.78	16.03	27.26
HSB	13.12	15.88	13.65	15.88	6.91	11.46	14.86
ICI	8.29	11.35	12.76	15.34	6.87	8.66	16.13
LMO	-4.81	-1.17	-0.74	-0.41	-0.39	-1.51	-3.03
MS	4.79	4.30	8.34	6.17	8.24	5.85	0.35
PO	-7.12	-6.99	-5.37	-6.21	-4.61	-6.51	-7.07
RTZ	8.53	10.81	11.79	12.01	8.03	10.40	7.89
RUT	6.89	8.67	6.57	7.72	6.01	8.33	9.51
RYL	18.58	16.53	21.91	20.78	18.06	18.32	14.63
SAN	3.67	7.25	7.81	8.92	4.16	5.97	5.42
STL	-3.49	-3.83	-1.05	-0.88	-0.11	-2.25	-4.18
TAR	1.06	2.50	5.41	6.81	8.97	8.14	3.94
TCO	2.48	-1.28	0.34	-0.28	0.45	-0.79	-5.12
TW	19.72	18.31	18.38	18.90	19.38	20.40	15.92
ULV	2.51	4.28	3.83	3.70	4.04	4.55	4.88
VOD	10.75	8.72	7.20	8.77	6.53	7.71	6.94
ZCA	30.39	33.91	23.06	28.18	22.99	25.10	33.43
<i>Panel B</i>							
VWP	10.93	11.90	11.82	12.78	9.26	10.42	11.50
EWP	10.35	11.45	10.83	11.90	8.75	9.81	10.95
<i>Panel C</i>							
Winners	11	19	19	21	16	18	

A denotes the nearest the money strategy, and B denotes the next-nearest the money strategy.

There were no ties in the tests. The null hypothesis is that the probability of a success or failure is equal for each comparison of investment protocols. The hypothesised probability is, therefore, constant.

For larger samples ($N > 20$), the test statistic z for the sign test for matched pairs can be determined in relation to either P_M , where P_M is the number of occurrences of the more frequent outcome divided by the number of trials N , or in relation to P_L , where P_L is the number of occurrences of the less

frequent outcome divided by the number of trials N . There are therefore two possible z -scores, Z_M determined using P_M , and Z_L determined using P_L . It is easy to show that for hypothesised $p = \frac{1}{2}$, $Z_M = -Z_L$ so, from an inference point of view, and consistent with the two-sided hypothesis stated above, it makes no difference whether one bases the statistical test on P_M or P_L . The sole effect of basing the z -score calculation on P_M versus P_L is that in the first instance the correction for continuity in the z -score calculation is $-\frac{1}{2}N$, whereas in the second instance it is $+\frac{1}{2}N$. Expressions for both possible z -scores are set out below:

$$\begin{aligned} Z_M &= \frac{(P_M - p) - 1/2N}{\sqrt{pq/N}} \\ Z_L &= \frac{(P_L - p) + 1/2N}{\sqrt{pq/N}} \end{aligned} \quad (4)$$

In equations (4), Z_M and Z_L are the usual z -score associated with the unit normal distribution. p is the hypothesised probability, and q is $1-p$. The probability associated with the test statistic $z_M(z_L)$ is interpreted as the probability that P_M or greater (P_L or less) was obtained by chance given a true probability p equal to its hypothesised value.

The Wilcoxon signed rank test is also a test for population differences in matched pairs. It takes account not only of the number of occurrences of a particular outcome but also takes account of the rank of the absolute size of each difference. This test, therefore, uses more information from a particular test than the sign test for matched pairs. As in the preceding case, comparison of individual returns performance is between the buy-and-hold strategy and the covered call strategy.

The assumptions underlying the Wilcoxon signed rank test are the following:

1. Define Z_i such that $Z_i \equiv r_{CC} - r_{BSH}$ where r_{BSH} denotes returns from the buy-and-hold strategy and r_{CC} denotes returns from the covered call strategy, both in respect of stock i . The assumed underlying model is:

$$Z_i = \theta + e_i \quad (5)$$

In (5), θ is the hypothesised differential in investment strategies and the e_i are unobservable random variables.

2. The unobservable random variables e_i are mutually independent.

3. Each e_i comes from a continuous population that is symmetric about zero. The populations do not need to be identical.

The null hypothesis in the context of the Wilcoxon signed rank test is:

$$H_0: \theta=0$$

To calculate the test statistic T , rank¹⁵ the absolute values of Z_i so that there is a rank value R_i corresponding to each Z_i . Define the indicator variable ψ_i to equal one if Z_i is positive and zero otherwise. If $\sum \psi_i < \sum (1 - \psi_i)$, the test statistic T is $\sum R_i \psi_i$. Otherwise, it is $\sum R_i (1 - \psi_i)$

For large samples, T is normally distributed with:

$$E[T] = \frac{N(N+1)}{4} \quad (6)$$

and

$$\text{VAR}[T] = \frac{N(N+1)(2N+1)}{24} \quad (7)$$

The test score is:

$$z = \frac{T - E[T]}{\sqrt{\text{VAR}[T]}} \quad (8)$$

In (7), T is the test statistic calculated as described above. In (8), z is the usual z -score associated with the unit normal distribution. If z is less than (greater than) zero, the probability associated with the test statistic z is interpreted as the probability of obtaining the test statistic or less (or more) given that the populations of returns differentials are symmetrical with identical median.

As with the sign test for matched pairs, inferences are identical whether the test statistic z is determined from T calculated using the less frequently occurring outcome, as above, or if z is determined from T calculated using the more frequently occurring outcome. As a consequence, and consistent with the two-sided nature of the hypothesis stated above, one need not make an *ex ante* selection of a 'preferred' or 'winning' investment protocol to calculate the test statistic. Define T_L equal to T calculated using the less frequently occurring outcome, T_M equal to T calculated using the more frequently occurring outcome, and corresponding z -scores z_L and z_M . It can

be shown that $z_L = -z_M$. This being the case, there can be no difference in inference in relation to the two-sided hypothesis stated above resulting from calculating z_L rather than z_M , or vice versa.

Given all of the assumptions set out above, the Wilcoxon signed rank test constitutes a test for location. If the error terms cannot be assumed to be symmetrically distributed, the Wilcoxon signed rank test constitutes a joint test for the median and for symmetry (cf. Gibbons and Chakraborti,¹⁶ pp. 168–169). In this circumstance, the null hypothesis in the context of the Wilcoxon signed rank test is:

H_0 : The population of differences in investment protocols is symmetric with median $\theta=0$.

If the joint null hypothesis stated above is rejected, it is not possible to determine whether the failure is attributable to one or both elements of the null hypothesis.

To facilitate discussion, we arbitrarily use the term winner to identify cases where returns from covered call writing exceed returns from the buy-and-hold strategy. The column headings A and B in Table 13.7 denote, respectively, the nearest the money option and the next-nearest the money option. Panel A in Table 13.7 summarises the results of the sign test for matched pairs. The results for 30-day options next-nearest the money, and 60-day options both nearest and next-nearest the money, and 90-day options next-nearest the money indicate that the covered call writing strategy produced an unusual number of winners. The results for 60-day options next-nearest the money indicate that the probability that the realised number of winners occurred by chance conditional on an hypothesised probability of 50 per cent was

Table 13.7 Results of statistical tests

	30 Day		60 Day		90 Day	
	A	B	A	B	A	B
<i>Panel A: Sign test for matched pairs</i>						
Winners	11	19	19	21	16	18
Losers	16	8	8	6	11	9
Max	16	19	19	21	16	18
z	0.7698	1.9245	1.9245	2.6943	0.7698	1.5396
p -value	0.221	0.027	0.027	0.004	0.221	0.062
<i>Panel B: Wilcoxon signed rank test</i>						
T	168	115	102	58	182	148
z	-0.5045	-1.7778	-2.0902	-3.1473	-0.1682	-0.9850
p -value	0.307	0.038	0.018	0.001	0.433	0.162

just 0.4 per cent. These results suggest that the hypothesis under the sign test for matched pairs is not supported by the data for at least some of the covered call writing protocols.

Panel B of Table 13.7 summarises the results for the Wilcoxon signed rank test. In each case the test statistic z is negative. The probability associated with the test statistic z is interpreted as the probability of obtaining the test statistic or less given that the populations of returns differentials are symmetrical with identical median. For the most part, the results in Panel B support those obtained in Panel A with the possible exception of the result for 90-day next-nearest the money. The test result under the latter test is less extreme than for the former test. The difficulty with the Wilcoxon signed rank test is that its interpretation is unclear if the error terms in the model cannot be assumed symmetrically distributed. The stylised facts of returns distributions suggest that there is the possibility of asymmetrically distributed stock returns. This therefore calls into question whether the hypothesis fails due to asymmetry or due to a median $\theta \neq 0$.

Conclusions and future research

In this paper, we present results of a comparison of two investment protocols. In the first instance, the investment protocol involves a buy-and-hold strategy. In the second instance, it involves writing covered calls on a long position in an underlying stock. We conduct the comparison in synthesised economies to demonstrate the likely outcome if the real economy were consistent with the theoretical models in the synthesised economies. We also conduct the comparison using transactions data. In the latter case, we find a surprising number of cases in which the covered call strategy produces greater returns than the buy-and-hold strategy, with some evidence that this outcome is statistically significant. This result is surprising considering the clear evidence in the synthesised economies that this result does not occur when tested over many repeated trials. The evidence from the synthesised economies indicates, however, that even though the buy-and-hold strategy produces greater returns when an average is taken over many repeated trials, a qualitative assessment of the covered call strategy suggests it performs well relatively frequently.

References and Notes

1. cf. the PRO-AMS, PRO-AMS US, and Triax CARTS funds, for example.
2. In practice, covered call writing involves writing options that are somewhat out-of-the-money.
3. Black, F. and Scholes, M. (1973) 'The Pricing of Options and Corporate Liabilities', *Journal of Political Economy*, Vol. 81, pp. 637–654.
4. We recognise that there are various limitations of the Black–Scholes pricing approach, but we submit that the approach is qualitatively appropriate in this instance.

5. Cox, J.C. (1975) 'Notes on Option Pricing 1: Constant Elasticity of Variance Diffusions', unpublished manuscript, Stanford University.
6. Geske, R. (1979) 'The Valuation of Compound Options', *Journal of Financial Economics*, Vol. 7(March), pp. 63–81.
7. Merton, R.C. (1976) 'Option Pricing When Underlying Stock Returns are Discontinuous', *Journal of Financial Economics*, Vol. 3, pp. 125–144.
8. Heston, S.L. (1993) 'A Closed-Form Solution for Options With Stochastic Volatility With Application to Bond And Currency Options', *Review of Financial Studies*, Vol. 6, pp. 327–343.
9. By moneyness, we mean the extent to which the option is out of the money.
10. Prices are in respect of American style options. Although the options used in the study possessed the early exercise feature, there is no evidence to suggest that early exercise occurred during the life of these options.
11. We also calculated results including this company and our best estimate of appropriate treatment for the apparent stock split. There was no material difference in the conclusions of the study.
12. Since both strategies include a long position in the underlying, we did not explicitly take account of dividends in the cash flows arising from each of the respective positions.
13. Hollander, M. and Wolfe, D.A. (1973) 'Nonparametric Statistical Methods', John Wiley & Sons, New York.
14. Hollander and Wolfe¹³ present and explain both of the statistical methods used in this paper. The comments on the statistical methods in this section follow the comments and analysis in their Chapters 2 and 3.
15. The ranking is in ascending order so that the smallest absolute value receives a rank of one.
16. Gibbons, J.D. and Chakraborti, S. (1992) 'Nonparametric Statistical Inference, 3rd edn. Marcel Dekker, Inc, New York.

Appendix 1

Black-scholes

In this model, the price of the underlying asset S is assumed to follow the stochastic process:

$$dS = \mu S dt + \sigma S dZ \quad (A1)$$

The discrete time version of (A1) is:

$$S(t_{i+1}) = S(t_i) \exp[(\mu - \nu/2)\Delta t + (\sqrt{\nu}x\sqrt{\Delta t})] \quad (A2)$$

where x is a random draw from the unit normal distribution.

Leveraged model

The Geske⁶ compound option pricing model is used to model option pricing in the context of a firm with financial leverage. Geske⁶ assumes that firm value, V , follows the stochastic process:

$$dV = \mu V dt + \sqrt{\nu} V dZ \quad (A3)$$

He also assumes that the firm has issued a pure discount bond that gives the holder the right to amount M at maturity T . Denote its value at the present time t by $M(t, T)$ and at maturity by $M(T, T)$. The firm is not permitted to issue *pari passu* or senior ranking claims, nor is it permitted to make distributions or share repurchases prior to T . Under this specification, the bondholders can be viewed as 'owning' the firm with the common shareholders having an option to acquire ownership at T upon payment to the bondholders of the amount $M(T, T)$. A call option on a common share is therefore an option on an option. Geske⁶ shows that the value of a common share, S , is given by the Black–Scholes call-option pricing formula and develops a pricing expression for the value of a call option on S .

The following is the discrete time version of (A3):

$$V(t_{i+1}) = V(t_i) \exp[(\mu - \nu / 2)\Delta t + (\sqrt{\nu x} \sqrt{\Delta t})] \quad (\text{A4})$$

where x is a random draw from the unit normal distribution.

Constant elasticity of variance

The CEV model⁵ is distinguished by its assumption that the underlying follows the stochastic process:

$$dS = \mu S dt + \sqrt{S} \sigma^{1/2} dZ \quad (\text{A5})$$

The discrete time version of (A5) is:

$$S(t_{i+1}) = S(t_i) + \mu S(t_i) \Delta t + \sqrt{S(t_i)} \sigma^{1/2} \sqrt{\Delta t} x \quad (\text{A6})$$

where x is a random draw from the unit normal distribution.

Stochastic volatility

Heston⁸ (see also Knoch, 1992) specifies geometric Brownian motion for the underlying and a square root process with a mean reversion term for the volatility parameter as follows:

$$dS = \mu S dt + \sqrt{S} dZ_1 \quad (\text{A7})$$

$$dv = \kappa[\theta - v] dt + \sigma \sqrt{v} dZ_2 \quad (\text{A8})$$

Here, μ , κ , θ , and σ are constants and dZ_1 and dZ_2 are standard Wiener processes whose increments have instantaneous correlation ρ . The discrete time versions of (A7) and (A8) are:

$$S(t_{i+1}) = S(t_i) \exp[(\mu - \nu(t_i) / 2)\Delta t + (\sqrt{\nu(t_i)} x_1 \sqrt{\Delta t})] \quad (\text{A9})$$

and,

$$\nu(t_{i+1}) = \nu(t_i) + \kappa(\theta - \nu(t_i))\Delta t + (\sigma \sqrt{\nu(t_i)} x_2 \sqrt{\Delta t}) \quad (\text{A10})$$

where x_1 and x_2 are determined by a random draw from the bivariate normal distribution with mean zero, unit variance, and correlation coefficient ρ .

Jump-diffusion

Merton's⁷ model is based on geometric Brownian motion augmented by a Poisson process. He thinks of the Poisson distributed event as the arrival of important information about the stock that affects its price. He assumes arrivals are independently and identically distributed. The probability of an event occurring during an interval of time h , where h can be arbitrarily small, can be described as follows:

$$\begin{aligned}\Pr\{\text{event does not occur in the time interval } (t, t+h)\} &= 1 - \lambda h + O(h), \\ \Pr\{\text{event occurs once in the time interval } (t, t+h)\} &= \lambda h + O(h), \\ \Pr\{\text{event occurs more than once in the time interval } (t, t+h)\} &= O(h)\end{aligned}\quad (\text{A11})$$

where $O(h)$ is the asymptotic order symbol defined by $\psi(h) = O(h)$ if $\lim_{h \rightarrow 0} [\psi(h)/h]$ is equal to zero, and λ is the mean number of arrivals per unit of time. If a Poisson event occurs, the effect on the price of the stock is SY where Y is a drawing of a random variable Y from a distribution where all Y are greater than or equal to zero, and successive draws of the random variable Y are independently and identically distributed. The stochastic differential equation for the price of the underlying asset that reflects this combined jump and diffusion process is:

$$dS = (\alpha - \lambda k)S \, dt + \sqrt{v}S \, dZ + S \, dq \quad (\text{A12})$$

where, following Merton's⁷ notation, α is the instantaneous expected return on the stock, v is the instantaneous variance of the return conditional on no jumps in stock price occurring, dZ is, as before, a standard Wiener process, $q(t)$ is the independent Poisson process described in (A11), dq and dZ are independent, $\kappa \equiv E[Y-1]$ where $Y-1$ is the random variable percentage change in the stock price if the Poisson event occurs, and E is the expectation operator over the random variable Y . Merton⁷ points out that a closed form solution exists if the random variable Y has a log-normal distribution.

Let the variance of the logarithm of Y equal β^2 and define $\gamma \equiv \log(1+\kappa)$. Define the random variable:

$$X_n \equiv \prod_n Y_n \quad (\text{A13})$$

with the condition that:

$$X_0 \equiv 1 \quad (\text{A14})$$

The variable n is the number of Poisson jumps that occur during the life of the option. X_n is distributed lognormally with $E[X_n] = \exp[n\gamma]$ and the variance of the logarithm of X_n equal to $\beta^2 n$. Merton⁷ defines:

$$f_n(S, T-t) = C(S, T-t, K, \sigma_n^2, r_n) \quad (\text{A15})$$

where

$$\sigma_n^2 \equiv v + \frac{n\beta^2}{(T-t)} \quad (\text{A16})$$

and

$$r_n \equiv r - \lambda k + \frac{n\gamma}{(T-t)} \quad (\text{A17})$$

$C(\cdot)$ is the Black-Scholes pricing formula with the five input variables identified in (A15). The option pricing expression that follows from the stochastic specification in (A12), unpriced jump risk, and lognormally distributed random variables Y_n is:

$$F(S, T-t) = \sum_{n=0}^{\infty} \frac{e^{-\lambda'(T-t)} (\lambda'(T-t))^n}{n!} f_n(S, T-t) \quad (\text{A18})$$

where

$$\lambda' \equiv \lambda(1+\kappa) \quad (\text{A19})$$

The discrete time version of (A12) is:

$$S(t_{i+1}) = S(t_i) X_n(\Delta t_i) \exp[(\mu - v/2 - \lambda\kappa)\Delta t + (\sqrt{v}x_1\sqrt{\Delta t})] \quad (\text{A20})$$

In (A20) $X_n(\Delta t_i)$ is the product of all of the n jumps Y_n that occur within the i th time interval Δt_i . $X_n(\Delta t_i)$ is determined by first determining the number of jumps, n , in the time interval Δt_i based on a random draw from the Poisson distribution with jump intensity parameter $\lambda\Delta t$. The size of each of the n jumps is determined by making n independent random draws from the distribution of Y . If there are no jumps in Δt_i , then $X_n(\Delta t_i) = X_0(\Delta t_i) = 1$, as in (A4). If one jump occurs in Δt_i , $X_n(\Delta t_i) = X_1(\Delta t_i) = Y_1$. If more than one jump occurs in Δt_i , that is $n > 1$, $X_n(\Delta t_i)$ is computed according to (A13). The random variable Y in the version of the jump-diffusion model which leads to the pricing expression in (A18) is distributed lognormally with $E[Y] = 1+k$, and $\text{VAR}[\ln(Y)] = \beta^2$. This is the model that is implemented in the simulations discussed in the body of the paper. In the simulations, draws are made from the distribution of Y by first generating a random variable x_2 where $x_2 \sim N(0,1)$ and defining:

$$x'_2 \equiv \beta x_2 + \ln(1+k) - \frac{\beta^2}{2} \quad (\text{A21})$$

Y is defined in the following way:

$$Y \equiv e^{x'_2} \quad (\text{A22})$$

The transformations in (A21) and (A22) ensure that $x'_2 \sim N(\ln(1+k) - (\beta^2/2), \beta^2)$ and Y is distributed lognormally with mean $1+k$.

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Hedge Funds and Higher Moment Portfolio Selection

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Practical applications

The paper aids sophisticated investors and analysts in understanding the implications of the higher moment characteristics of hedge fund return distributions. This is useful in the application of portfolio construction of fund of hedge funds as well as in determining the appropriate allocation to hedge funds within a traditional portfolio of equities and bonds. The paper also highlights less apparent risks and advantages in particular hedge fund strategies.

Notwithstanding the central limit theorem, the returns of several hedge fund indices are found to exhibit distributional characteristics inconsistent with normality. Using world hedge fund index and asset class data from 1994 to 2004, this study empirically compares the results of the Markowitz mean-variance optimisation technique with a higher moment methodology recently proposed by Davies et al. This comparison is conducted both when constructing fund-of-hedge-fund portfolios and when determining an appropriate weighting to apply when adding hedge funds to the traditional asset classes of equities, bonds and cash. The descriptive statistics show that, in particular, the hedge fund strategies of Fixed Income Arbitrage and Event-driven Opportunities, despite displaying low volatility, exhibit latent higher moment risk in the form of negative skewness and high kurtosis. These two higher moments collectively suggest an increase in the probability of

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extreme adverse returns to the investor that is not revealed in traditional mean–variance analysis. Confirming the findings of Amin and Kat and Lo, Jarque-Bera tests find that only two out of the 14 hedge fund indices used in this study are normal at the 5 per cent level. Applying Markowitz mean–variance portfolio selection to an array of published hedge fund indices produces fund-of-fund portfolios with higher ex post returns but naïve exposure to undesirable higher moment risks. When the higher moments of hedge fund index return distribution are accounted for in the portfolio optimisation algorithm, the resultant portfolios have improved diversification and higher moment statistics. This study confirms the findings of Davies et al. and Feldman et al. that Global Macro and Equity Market Neutral strategies are crucial constituents in a fund-of-hedge-funds portfolio. When constructing multi-asset class portfolios that include an allocation to hedge funds, the results show that mean–variance optimisation significantly over-allocates to the hedge fund class in comparison to when skewness and kurtosis are also taken into account. The higher moment-optimised portfolios all outperform the mean–variance comparatives when evaluated on an Omega function basis.

Introduction

Two key tasks faced by industry practitioners are that of asset allocation and that of strategy allocation within the hedge fund universe. It is well documented (see Cvitanić *et al.*,¹ Agarwal and Naik,² Amenc and Martellini³ and Amin and Kat⁴) that hedge funds are marked by their heterogeneity and unusual statistical properties. This makes the use of conventional methods of portfolio construction subject to question and necessitates the investigation of a more sophisticated approach to inform the construction of appropriate and efficient portfolios. This paper compares and evaluates the results of two related optimisation procedures. First, the classic mean–variance portfolio optimisation of Markowitz⁵ and secondly, a recent approach introduced by Davies *et al.*⁶ utilising Polynomial Goal Programming (PGP) to optimise portfolios return distributions for higher moments to include mean, variance, skewness and kurtosis for a given set of investor preferences. This comparison will be presented in the context of both a fund-of-hedge-fund strategy allocation as well as the asset allocation problem of what proportion to allocate to hedge funds in a balanced portfolio.

The remainder of the paper is organised as follows: The second section presents a brief theoretical overview of the relevance of skewness and kurtosis, and performance appraisal measures that take these features into account. The third section provides a review of the prior literature relating to portfolio construction involving hedge funds, both in terms of an asset allocation decision and when constructing a fund-of-hedge-funds portfolio. Thereafter, the data, descriptive statistics and methodology are presented. The analysis is applied to the two problems of (i) fund-of-hedge-fund strategy allocation and (ii) the asset allocation decision with hedge funds. Finally, the eighth section concludes.

The Higher Moments of a Probability Distribution

A normal distribution is fully described by its mean and variance and it is widely used, due to its mathematical tractability. The normal distribution has its theoretical motivation in the Central Limit Theorem, which states that the average distribution of an increasing number of independent variables approaches a normality if certain conditions are fulfilled. These conditions can be summarised as follows:

1. The mean and standard deviations or the processes generating the returns should be stationary over time.
2. The processes generating the returns should be independent of each other rather than a function of general systematic factors.

Johnson *et al.*⁷ comment: *'It is fairly obvious that neither of these conditions is strictly true for hedge funds and it is in part for this reason that the "fat-tails" appear in the distributions of hedge fund strategy returns. For example, systematic trend-followers depend on the existence of trends in various financial markets so that the returns of managers operating this strategy will tend to exhibit a high degree of interdependence and notable time structure.'*

In terms of quantifying the nature of non-normality, skewness is a commonly used measure of symmetry (or more precisely, a lack of symmetry):

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_j - \bar{x}}{\sigma} \right)^3$$

where $\bar{x}$ is the mean, σ is the standard deviation and n is the number of data points in the sample. The skewness of a normal distribution is zero. From a portfolio perspective, investors prefer portfolios with higher (right-skewed) skewed distributions.

Kurtosis is a measure of how the relative peakedness or flatness of a distribution compares with the normal distribution, with higher values implying fatter tails. The standard formula for calculating excess portfolio kurtosis is:

$$\text{Kurtosis} = \left\{ \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{x_j - \bar{x}}{\sigma} \right)^4 \right\} - \frac{3(n-1)^2}{(n-2)(n-3)} \quad (2)$$

where $\bar{x}$ is the mean, σ is the standard deviation and n is the number of data points in the sample. The kurtosis value of a normal distribution is three.

The relevance of non-normality in returns from an investors' perspective is that his utility function is influenced by the values of these higher moments. As stated by Athayde and Flores:⁸ *'In general, investors will prefer high values for odd moments and low ones for even moments. The former can be*

seen as a way to decrease extreme values on the side of losses and increase them on the gains. The latter can be justified by the fact that even moments measure dispersion, and therefore volatility; something undesirable because it increases the uncertainty of returns'. In other words, investors would like to maximise the first and third moments (mean and skewness) and minimise the second and fourth (variance and kurtosis).

In the ensuing analysis, two risk measures that take account of the non-normality of returns are applied. The Sortino Ratio was introduced by Sortino and Price⁹ and is essentially a derivative of the Sharpe Ratio. The Sortino Ratio substitutes standard deviation for the downside deviation (or semi-standard deviation) statistic. This means that the measure does not penalise upside volatility. The downside deviation can be measured from any given point but is usually measured either below the risk-free rate or zero. In this study, the Sortino Ratio is defined as the excess portfolio return beyond the risk-free rate per unit of downside volatility as measured by the semi-standard deviation.

$$\text{Sortino Ratio} = \frac{E(r_p) - r_f}{\sigma_{dp}} \quad (3)$$

where $E(r_p)$ denotes the expected return of the portfolio, r_f the risk-free rate and σ_{dp} the semi-standard deviation of the portfolio return series. And the semi-standard deviation is $\sigma_{dp} = (1/n-1)\sum(E_{dr}-r_f)^2$, where E_{dr} denotes returns below the risk-free rate r_f .

The Omega function was developed by Shadwick and Keating¹⁰ and incorporates all the higher moments into a performance evaluation. The function also takes into consideration a 'threshold level' above which an investor would be satisfied with the absolute return and vice versa. The objective of the authors was to find a 'universal performance measure'. Unlike other performance measures such as Sharpe or Sortino (which only consider the volatility and downside volatility of returns, respectively), the Omega function was designed to take the entire return distribution into account.

The Omega function is defined as follows:

$$\Omega(r) = \frac{\int_r^b [1 - F(x)]dx}{\int_a^r F(x)dx} \quad (4)$$

where x is the random one-period return on an investment, r is a threshold selected by the investor and a and b denote the upper and lower bounds of the return distribution, respectively. The Omega ratio is effectively the area of the distribution above the threshold level divided by the area below the threshold level. This is an important measurement tool for portfolios

that include hedge funds. From a risk-adjusted perspective, it is critical that performance is assessed in the context of the potential increased probability of large extreme losses in hedge funds. Kazemi *et al.*¹¹ show that for ease of calculation, the Omega function can also be expressed as the ratio of the price of a long European call option on the investment divided by the price of a long European put option, where the strike price is the investor threshold level.

Prior Research

The allocation to hedge funds in a balanced portfolio

In a study of the risk and return benefits of traditional portfolios with a hedge fund allocation, Schneeweis *et al.*¹² construct portfolios including hedge funds using the Markowitz mean–variance model. The authors use returns series data from hedge fund data provider EACM, the S&P500 large-cap equity index and the Salomon Brothers Government/ Corporate Bond Index over the period 1990–1998. They find that under historical market conditions, a portfolio of hedge funds offers improved risk and return characteristics when pooled with traditional stock as well as balanced (multi-asset) portfolios. The authors specifically state that ‘...*the low correlation between stock, bond markets, and a wide variety of alternative investments makes the results (improved risk and return opportunities) for the inclusion of various hedge fund strategies ... consistent across a wide variety of stock and bond portfolios*’. Their findings are supportive of the hypothesis that an inclusion of hedge funds in the investment opportunity set enhances the efficient frontier and resultant investor utility. Using this methodology with no allocation constraints often leads to large allocations to hedge funds (ie in excess of 90 per cent).

In a related work, Schneeweis and Georgiev¹³ replace the Salomon Brothers Government/ Corporate Bond Index with the Lehman Brothers Bond Index, and the data cover a longer period, from 1990–2001. They conclude that hedge funds offer the opportunity to reduce portfolio variance and enhance portfolio returns in economic environments in which traditional stock and bond investments offer limited opportunities. They also note that the allocation to hedge funds under this mean–variance framework, however, may be ‘... *affected by the historical high returns achieved by hedge funds in the first half of the 1990’s*’. Amenc and Martellini³ caution that portfolio optimisation procedures are very sensitive to differences in expected returns. They caution that portfolio optimisers typically allocate the largest proportion of capital to the asset class for which the estimation error in the expected returns is the greatest.

The conventional mean–variance approach above is also criticised by numerous other investigations, including Cvitanic *et al.*,¹ Agarwal and Naik,² Amenc and Martellini³ and Amin and Kat.⁴ These studies observe that mean–variance portfolio optimisation makes the key assumption of normal

asset return distributions. Lo¹⁴ states that 'hedge-fund returns are highly non-normal, ie, they are asymmetrically distributed, highly skewed, often multi-modal, and with fat tails that imply many more tail events than the normal distribution would predict'. Research conducted by Amin and Kat⁴ finds that 'The return distribution of a number of hedge fund indices appears to be highly skewed'. Amin and Kat⁴ also find that only 14.1 per cent of the individual hedge fund returns are normal, utilising a Jarque-Bera test for normality at the 5 per cent significance level.

Amin and Kat⁴ conduct a study with the objective of examining the effects of diversification by adding hedge funds to a traditional stock and bond portfolio. More specifically, they study the change in the portfolio return distribution with the hedge fund augmentation. They find similar results to Schneeweis and Georgiev,¹³ and Schneeweis *et al.*¹²: that the inclusion of hedge funds significantly improves the portfolios mean-variance characteristics. They also, however, find that portfolios constructed of equities and hedge funds do not combine well into truly low risk portfolios as this lowers the skewness and increases the kurtosis of the portfolio. The authors note that '*... in terms of skewness hedge funds and equity do not mix very well. In economic terms, the data suggest that when things go wrong in the stock market, they also tend to go wrong for hedge funds. In a way, this makes sense. A significant drop in stock prices will often be accompanied by a widening of a multitude of spreads, a drop in market liquidity, etc.*'

Constructing fund-of-hedge-fund portfolios

Several recent studies investigate the construction of the optimal fund-of-hedge-funds portfolio. Fund-of-hedge-funds are often seen by investors as an efficient manner to access hedge fund manager capability. Industry data provider Hedge Fund Research (HFR) statistics show that fund-of-hedge-funds currently hold 30 per cent of the estimated \$650 billion invested in hedge funds globally, as of December 2003.

Amin and Kat⁴ analyse the performance of baskets of hedge funds ranging in size from 1 to 20 funds. Using 1,721 hedge funds (drawn from the Tremont TASS database) from June 1994 to May 2001, they show that increasing the number of funds can be expected to lead not only to a lower volatility, but also, and less appealingly, to lower skewness and increased correlation with the S&P500. Most of this change occurs for relatively small portfolios holding less than 15 hedge funds and, thereafter, holding additional funds seems to have little effect on the portfolio's return distribution.

Lhabitant and Learned¹⁵ investigate the same question using a naïvely diversified (equal-weighted) Monte Carlo simulation on a database of 6,985 hedge funds. They find that increasing the number of hedge funds (from 1 to 50 funds) in a portfolio reduces the return distribution symmetry and increases kurtosis. The authors find that most of the diversification benefits are delivered with a small number of hedge funds (5–10 funds).

Feldman *et al.*¹⁶ develop a simulation-based optimisation method for the construction of optimal fund-of-hedge-fund portfolios that is based on the skewness and kurtosis of returns. Vector autoregression (VAR) methods are used to model the relations among asset returns. Investor preferences are represented by a group of utility functions that integrate both risk and loss aversion. Results suggest that the returns to Market-Neutral and Global Macro funds have distributional characteristics that make them attractive investment vehicles for risk and loss-averse investors.

Davies *et al.*¹⁷ explore the interaction of the higher order co-moments and their impact on portfolio construction. They specifically focus on the higher co-moments between various hedge fund strategies, particularly co-skewness and co-kurtosis, and observe that *'...diversification deteriorates skew and improves kurtosis in most strategies. Skewness in all strategies, kurtosis in all but distressed securities and merger arbitrage funds are reduced when moving from the individual fund level to the portfolio level. This implies a tradeoff between variance-skewness-kurtosis in hedge fund portfolios. Thus, mean-variance optimal criteria can lead to sub-optimal portfolios in the presence of skewness and kurtosis'*. Davies *et al.*¹⁷ draw a similar conclusion as Feldman *et al.*¹⁶ that Market-Neutral funds and Global Macro funds have a key role in optimal hedge fund portfolios. In addition, the authors conclude that Market-Neutral funds are kurtosis reducers while Global Macro funds are skewness enhancing. They find that as more funds are included, portfolio volatility (standard deviation) and skewness fall. Davies *et al.*¹⁷ note: *'Risk and skewness reduction both occur at a decreasing rate, with the reduction in portfolio skewness occurring at a much slower speed. Since positive skewness is generally a desirable trait, there is a clear trade-off between skewness and risk'*.

This finding concurs with that of Lhabitant and Learned¹⁵ and is the rationale behind their conclusion to limit the number of funds within a fund-of-hedge-fund portfolio to 5–10 funds. Davies *et al.*¹⁷ find that as the number of funds contained in the fund-of-hedge-funds increases, *'... portfolio expected skewness depends only on the coskewness between three different funds'* and that following the same rationale *'portfolio expected standard deviation depends only on covariance and portfolio expected kurtosis depends only on the cokurtosis between four different funds. The influence from individual fourth central moment, cokurtosis between two different funds and three different funds on expected portfolio kurtosis tends to zero'*. Both studies agree that as a number of Event-Driven type strategies are included, the kurtosis of the portfolio will increase (the fund-of-hedge-funds becomes more likely to be affected by a systematic shock, eg LTCM, the failure of a mega-merger, etc).

In a related work, Davies *et al.*⁶ utilise a PGP technique to construct fund-of-hedge-fund portfolios adjusting for investor preferences with respect to competing objectives in terms of mean, variance, skewness and kurtosis. The findings from this study confirm their earlier work above and provide a useful framework for optimising hedge fund portfolios. It is this methodology

that is used in this study to investigate optimal allocations to hedge funds within a traditional portfolio, as well as allocation among hedge fund strategies within a fund-of-hedge-funds.

Data

The data utilised in this study consist of monthly world hedge fund index data and long-only market index data from January 1994 until the end of June 2004.¹⁸ Hedge fund index data are provided by CSFB Tremont and the market index data by Morgan Stanley Capital International (MSCI) and Lehman Brothers. The hedge fund data below are organised according to the UBS Warburg classifications (Ineichen)¹⁹ with corresponding weightings in the Composite index. The Event-Driven category has three sub-indices for which no weightings are available.

CSFB tremont world hedge fund composite 100 per cent

Relative value indices:

CSFB Tremont Hedge fund convertible arbitrage (%)	4.7
CSFB Tremont Hedge fund equity market neutral (%)	4.4
CSFB Tremont Hedge fund fixed income arbitrage (%)	7.8

Event-driven indices:

CSFB Tremont Hedge fund event driven (%)	21.6
CSFB Tremont Hedge fund distressed securities	
CSFB Tremont Hedge fund event driven multi-strategy	
CSFB Tremont Hedge fund risk arbitrage	

Opportunistic indices:

CSFB Tremont Hedge fund managed futures (%)	5.8
CSFB Tremont Hedge fund global macro (%)	12.7
CSFB Tremont Hedge fund long/short equity (%)	26.3
CSFB Tremont Hedge fund dedicated short bias (%)	0.5
CSFB Tremont Hedge fund emerging markets (%)	3.7

Other:

CSFB Tremont Hedge fund multi-strategy (%)	12.6
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CSFB Tremont is a major provider of hedge fund data and compiled the first asset-weighted hedge fund indices. These hedge fund indices use the TASS database as the source of the individual hedge fund data. TASS is one of the leading providers of individual hedge fund data. The CSFB Tremont indices also have minimum criteria for inclusion into the index: a minimum of US \$10 million assets under management, a minimum one-year track record and current audited financial statements. The index is calculated and rebalanced monthly. Funds are reselected on a quarterly basis as necessary. To minimise survivorship bias, funds are not removed from the index until they are fully liquidated or fail to meet the financial reporting requirements.

Liang²⁰ finds survivorship bias in hedge fund return data from January 1992 through to December 1996. The author, however, concludes that, on a risk-adjusted basis, the average hedge fund outperformed the average mutual fund and that the outperformance cannot be explained by survivorship bias. Amin and Kat⁴ find that concentrating on surviving funds only will overestimate the mean return on individual hedge funds by approximately 2 per cent and will introduce significant biases in estimates of the standard deviation, skewness and kurtosis. Specifically, they point to: '*... significant underestimation of the standard deviation and kurtosis as well as overestimation of the skewness of individual hedge fund returns*'. Studies relating to survivorship bias with respect to hedge fund returns have not been extended to that of hedge fund indices as used in this study. Ineichen¹⁹ addresses this issue as follows: '*For hedge funds, it is unclear if survivorship bias inflates returns of hedge fund indices. Poor, as well as stellar performing hedge funds, exit the database. Poor hedge funds exit because of poor performance. Stellar hedge funds can close to new partners and, as a result of good performance, stop reporting returns to the data vendor. Hedge funds report their performance on a voluntary basis. This self-selection bias may partially offset the survivorship bias caused by the disappearance of poorly performing funds*'.

Survivorship bias in hedge fund index data is beyond the scope of this paper. And as there is a lack of any conclusive research on the matter, as well as efforts by CSFB Tremont to minimise the impact of survivorship bias in their index data, all empirical research will use the published data in its original format.

The MSCI World Index is a free float-adjusted market capitalisation index that measures global developed market equity performance. As of December 2003, the MSCI World Index consisted of the following 23 developed market country indices: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Hong Kong, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, the United Kingdom and the United States.

The MSCI World Sovereign Bond Index is a market capitalisation index that is designed to measure global developed market fixed interest performance. Over 50 per cent of the index consists of debt originating from Japan (28.7 per cent), the United States (19.5 per cent) and Germany (9.8 per cent). Cash returns are proxied using the Lehman Brothers Cash Composite. Long-only index data are sourced from the Bloomberg database. All indices in this study are denominated in US dollars.

Descriptive Statistics

As this paper focuses on hedge fund portfolio optimisation incorporating four moments, all the data series are analysed according to moments and Jarque-Bera test statistic and can be found in Table 14.1.

It can be seen from the statistics in Table 14.1 that most hedge fund strategies (with the exception of the Dedicated Short-Bias strategy) have

Table 14.1 Descriptive statistics including the first four moments and the Jarque-Bera test statistic

	Mean (%)	Standard Deviation (%)	Skewness	Kurtosis	Jarque-Bera	Normal at 5 per cent	Normal at 1 per cent
MSCI world equity	0.54	4.16	-0.57	3.47	8.12	No	Yes
MSCI world sovereign bond	0.52	1.91	0.37	3.49	4.15	Yes	Yes
Lehman Brothers cash composite	0.34	0.15	-0.69	2.06	14.73	No	No
CSFB Tremont Hedge fund composite	0.89	2.40	0.10	4.77	16.76	No	No
CSFB Tremont Hedge fund convertible arbitrage	0.81	1.37	-1.47	6.60	113.52	No	No
CSFB Tremont Hedge fund dedicated short bias	-0.13	5.10	0.92	5.17	42.37	No	No
CSFB Tremont Hedge fund distressed securities	1.07	1.97	-2.75	19.13	1525.22	No	No
CSFB Tremont Hedge fund event driven	0.92	1.71	-3.46	25.83	2988.13	No	No
CSFB Tremont Hedge fund event driven multi-strategy	0.83	1.80	-2.68	19.83	1637.60	No	No
CSFB Tremont Hedge fund emerging markets	0.68	5.03	-0.57	6.71	79.04	No	No
CSFB Tremont Hedge fund fixed income arbitrage	0.56	1.12	-3.24	19.35	1624.54	No	No
CSFB Tremont Hedge fund managed futures	0.58	3.54	0.04	3.41	0.94	Yes	Yes
CSFB Tremont Hedge fund global macro	1.17	3.42	-0.02	5.07	22.48	No	No
CSFB Tremont Hedge fund long/short equity	0.98	3.11	0.24	6.38	61.10	No	No
CSFB Tremont Hedge fund multi-strategy	0.73	1.27	-1.27	6.45	96.04	No	No
CSFB Tremont Hedge fund equity market-neutral	0.83	0.88	0.25	3.21	1.56	Yes	Yes
CSFB Tremont Hedge fund risk arbitrage	0.67	1.26	-1.31	9.19	236.98	No	No

The Jarque-Bera statistic is distributed as a χ^2 distribution with two degrees of freedom. Its critical values at the 5 per cent and 1 per cent confidence levels are 5.991 and 9.210, respectively. Therefore, the null hypothesis of normality is rejected when the Jarque-Bera statistic has a higher value than the corresponding critical value at the respective confidence level.

a higher monthly mean return than either global equities or bonds over the sample period. The top three performing hedge fund strategies in absolute terms over the period are Global Macro, Distressed Securities and Long/Short Equity. The worst performing hedge fund strategy is that of Dedicated Short-Bias. All of these strategies can be categorised as opportunistic under the UBS Warburg classification discussed earlier in the fourth section.

From a volatility perspective, the least volatile hedge fund strategy is that of Equity Market Neutral and the only series less volatile is that of US cash, as displayed in Table 14.1. This is quite intuitive given that these portfolios are largely devoid of market risk, having roughly equal long and short positions in related securities. The most volatile strategy is Dedicated Short-Bias. It is notable and logical that the 'opportunistic' strategies are the most volatile: Dedicated Short-Bias, Emerging Markets, MSCI World Equity, Managed Futures, Global Macro and Long/Short Equity are all directional strategies. This group is followed by mostly 'Event-Driven' strategies and is then followed by the 'relative value' strategies.

The descriptive statistics become more remarkable when looking at the third and fourth moments (ie skewness and kurtosis). Most of the hedge fund return series are negatively skewed and all strategies exhibit some level of excess kurtosis (ie greater than three). The most positively skewed strategy is that of Dedicated Short-Bias. This is most likely due to the funding mechanics of these portfolios.²¹

The most negatively skewed is that of the Event-Driven category. This has been noted in prior research by Agarwal and Naik² who state: *'We find ... the Event Arbitrage index showing significant factor loading on risk factor corresponding to writing a OTM put option on S&P 500 index ... this result is intuitive as Event Arbitrage strategy involves the risk of deal failure. A larger fraction of deals fail when markets are down and the Event Arbitrage strategy incurs losses. In contrast, when markets are up a larger proportion of deals go through and the strategy makes profits. But the profits are unrelated to the extent by which the market goes up. Thus, the payoff to Event Arbitrage strategy resembles that obtained by writing a naked put option on the market'*.

From a kurtosis point of view, the strategy with the highest kurtosis is that of Event-Driven, followed by its substrategy Event Driven Multi-Strategy and then Fixed Income Arbitrage.

What is interesting to note about these two strategies (Event Driven and Fixed Income Arbitrage) is that they are also marked by the most extreme cases of negative skewness and positive kurtosis. As observed by Davies *et al.*:¹⁷ *'Compounded by a high kurtosis (leptokurtosis), a negative skewed return distribution produces much higher possibilities for extreme events. ...that in most strategies, negative expected skewness goes with leptokurtosis on both individual fund and portfolio levels. Thus, it is preferable to analyse these two moments in tandem'*. Another point of interest is that, under a mean-variance

framework, volatility (variance or standard deviation) is used as a proxy for risk in circumstances where the return series can be characterised by a normal distribution. Where this is not the case, risk is better accounted for by the standard deviation in conjunction with the higher moments of the probability distribution. Where series have low standard deviations, they are often combined with low/negative skewness and high kurtosis.

Finally, all return series are subjected to the Jarque-Bera test statistic:

$$\text{Jarque} - \text{Bera} = N \left(\frac{S^2}{6} + \frac{(K - 3)^2}{24} \right) \quad (5)$$

where N is the number of observations, S is the skewness and K is the kurtosis of the series.

As evident in Table 14.1, most hedge fund index data are found to be not normally distributed, with the exception of Managed Futures and Equity Market Neutral, motivating the use of the higher moment optimisation technology described in the sixth section.

Methodology

Two portfolio construction approaches are applied in this paper. First, a conventional Markowitz mean-variance optimisation is employed, and secondly, a mean-variance-skewness-kurtosis (MVSK) optimisation. The second portfolio construction approach utilised in this study closely follows the PGP methodology of Davies *et al.*⁶ This facilitates the incorporation of both investor preferences beyond the mean-variance space (to higher moments) as well as a more complete representation of the probability distribution to effect efficient portfolio construction with hedge funds. This study is distinguished by using hedge fund index data as opposed to the single manager data used in Davies *et al.*⁶ In addition, this study augments this work by also addressing what proportion of a balanced portfolio (ie a portfolio consisting of equities, bonds and cash) should be invested in the hedge fund class.

PGP is useful in solving problems where multiple and competing objectives are present. As previously discussed, investors would like to maximise the first and third moments (mean and skewness) and minimise the second and fourth (variance and kurtosis). Davies *et al.*⁶ formulate the portfolio construction question as a multiple objective programming problem:

$$\text{Maximise } Z_1 = X^T \tilde{R} \quad (6)$$

$$\text{maximise } Z_3 = \frac{T}{(T-1)(T-2)} \times \sum \left[\frac{(X^T (\tilde{R} - \hat{R}))^3}{\sqrt{X^T V X}} \right] \quad (7)$$

$$\text{minimise } Z_4 = \left\{ \frac{T(T+1)}{(T-1)(T-2)(T-3)} \times \sum \left[\frac{(X^T(\tilde{R} - \hat{R}))}{\sqrt{X^T V X}} \right]^4 \right\} - \frac{3(T-1)^2}{(T-2)(T-3)} \quad (8)$$

$$\text{subject to } X^T V X = A \quad (9)$$

where, $X^T = (x_1, x_2, \dots, x_n)$ and x_i is the capital weight percentage of the portfolio invested in the i th asset. The asset can be a risky asset or risk-free. The T superscript denotes the transpose of the array in a matrix formula. T is the number of observations in the time series (in this paper all series have 126 observations). Z_1 is the formula for portfolio mean return, $X^T V X$ is portfolio variance, Z_3 is portfolio skewness and Z_4 is excess kurtosis.²² A denotes the level of variance pre-specified in the optimisation.

Combining the objectives in 6, 7, 8 and 9 into a single objective statement, a PGP can be expressed as:

$$\text{Minimise } Z = (1 + d_1)^\alpha + (1 + d_3)^\beta + (1 - d_4)^\gamma \quad (10)$$

$$\text{subject to } X^T \tilde{R} + d_1 = Z_1^* \quad (11)$$

$$\frac{T}{(T-1)(T-2)} \sum \left[\frac{(X^T(\tilde{R} - \hat{R}))}{\sqrt{X^T V X}} \right]^3 + d_3 = Z_3^* \quad (12)$$

$$\left\{ \frac{T(T+1)}{(T-1)(T-2)(T-3)} \sum \left[\frac{(X^T(\tilde{R} - \hat{R}))}{\sqrt{X^T V X}} \right]^4 \right\} - \frac{3(T-1)^2}{(T-2)(T-3)} + d_4 = Z_4^* \quad (13)$$

$$d_1, d_3 \geq 0 \quad (14)$$

$$d_4 \leq 0 \quad (15)$$

$$X^T V X = A \quad (16)$$

where α , β and γ are the non-negative investor preferences for the mean, skewness and kurtosis of the portfolio return series. Z_1^* is the mean return for the optimal mean–variance portfolio with a specified variance; Z_3^* is the skewness value of the optimal skewness–variance portfolio with specified variance and Z_4^* is the kurtosis value of the optimal kurtosis–variance portfolio with specified variance.

By construction, the mean return for an optimal MVSK portfolio will be lower than the mean return for an optimal mean–variance portfolio. Similarly, skewness for an optimal MVSK portfolio will be lower than that

of an optimal skewness–variance portfolio. Therefore, d_1 and d_3 represent positive deviations from Z_1^* and Z_3^* . Similarly for kurtosis, d_4 represents the negative deviation from Z_4^* .

Solving the PGP is a two-step process. First, the optimal values for Z_1^* (expected return), Z_3^* (skewness) and Z_4^* (kurtosis), respectively are solved for a pre-specified level of variance. Secondly, these optimal values are substituted into restrictions 11, 12 and 13 and a minimum value is found for the objective formula 10. Davies *et al.*⁶ use their model to solve for optimal fund-of-hedge-fund portfolios under the further constraint of optimising for a variance of one. This study extends their work by comparing the outcome of the MVSK optimisation with the mean–variance methodology for varying levels of volatility.

Results

The empirical results of this chapter are presented in two sections. The first section reports the fund-of-hedge-fund optimisation results. The second section presents the results with respect to an optimal asset allocation, including a traditional assets and a hedge fund portfolio. The sections are also divided into results obtained under a mean–variance framework and those obtained under the PGP MVSK methodology. The section concludes with a comparative performance evaluation.

Fund-of-hedge-funds optimisation

This section presents results using data from the CSFB Tremont hedge fund indices. All CSFB Tremont indices are included in this analysis except the Composite index, as the objective of this section is to derive an optimal composite.

Mean-variance optimisation

As stated in the previous section, a variance minimisation technique was used in this procedure. Minimum variance portfolios are found for 21 reference points of return in this hedge fund set. The points are derived by creating 20 equidistant points between the minimum average monthly return²³ and the maximum average monthly return²⁴ of all the indices in the set. The results are presented in Table 14.2.

Table 14.2 splits the output into two panels: Panel A shows descriptive statistics of the output while Panel B presents the allocation in portfolios 1 through 21. From Table 14.2 and Figure 14.1 it can be seen that only portfolios 12 through 21 are part of the efficient frontier. For portfolios 1 through 11 there exists a portfolio on the minimum-variance frontier for which there is a point of higher return for the same quantum of volatility (standard deviation). This means that portfolios 1–11 are not an element of the efficient frontier set.

Table 14.2 Optimal fund of hedge fund portfolios under a mean–variance framework

Panel A: Portfolio expected return and risk statistics					
Portfolio	Expected return (%)	Variance (%)	Standard Deviation (%)	Skewness	Kurtosis
1	-0.13	0.26	5.08	0.93	2.30
2	-0.07	0.20	4.46	0.83	1.80
3	0.00	0.15	3.87	0.67	1.18
4	0.06	0.11	3.33	0.46	0.48
5	0.13	0.08	2.86	0.18	-0.14
6	0.19	0.06	2.45	0.10	-0.10
7	0.26	0.04	2.06	-0.02	0.10
8	0.32	0.03	1.68	-0.15	0.38
9	0.39	0.02	1.32	-0.36	1.07
10	0.45	0.01	1.01	-0.67	2.24
11	0.52	0.01	0.78	-0.97	2.86
12	0.58	0.00	0.64	-0.77	1.58
13	0.65	0.00	0.57	-0.39	-0.27
14	0.71	0.00	0.59	-0.16	-0.28
15	0.78	0.00	0.66	-0.09	0.53
16	0.85	0.01	0.78	-0.18	2.10
17	0.91	0.01	0.98	-0.57	4.45
18	0.98	0.02	1.26	-1.18	7.59
19	1.04	0.03	1.60	-1.57	9.04
20	1.11	0.04	2.01	-1.18	6.15
21	1.17	0.12	3.40	-0.02	2.20

(continued)

Table 14.2 Continued

Panel B: Percentage allocation to hedge fund strategy in fund-of-hedge-fund portfolio												
Convertible arbitrage (%)	Dedicated short-bias (%)	Distressed securities (%)	Emerging markets (%)	Equity market- neutral (%)	Event driven (%)	Event multi- strategy (%)	Fixed income arbitrage (%)	Global macro (%)	Multi- strategy (%)	Risk arbitrage (%)	Long/ short equity (%)	Managed futures (%)
0	100	0	0	0	0	0	0	0	0	0	0	0
0	92	0	8	0	0	0	0	0	0	0	0	0
0	84	0	16	0	0	0	0	0	0	0	0	0
0	76	0	24	0	0	0	0	0	0	0	0	0
0	68	0	32	0	0	0	0	0	0	0	0	0
0	58	0	27	0	0	0	15	0	0	0	0	0
0	48	0	22	0	0	0	27	0	0	3	0	0
0	39	0	17	0	0	0	31	0	0	13	0	0
0	31	0	12	0	0	0	34	0	0	23	0	0
0	22	0	7	0	0	0	37	0	0	33	0	1
0	14	0	3	0	0	0	37	0	3	39	0	4
0	10	0	1	17	0	0	29	0	6	32	0	3
0	8	0	0	36	0	0	21	0	8	24	1	1
0	7	8	0	46	0	0	11	0	10	14	1	0
0	7	17	0	55	0	0	1	1	12	5	1	0
0	4	22	0	60	0	0	0	4	9	0	0	0
0	0	26	0	62	0	0	0	8	4	0	0	0
0	0	41	0	45	0	0	0	14	0	0	0	0
0	0	59	0	21	0	0	0	21	0	0	0	0
0	0	64	0	0	0	0	0	36	0	0	0	0
0	0	0	0	0	0	0	0	100	0	0	0	0

Panel A presents the portfolio mean return, standard deviation and the higher moments while Panel B shows the allocation to the individual hedge fund strategies.

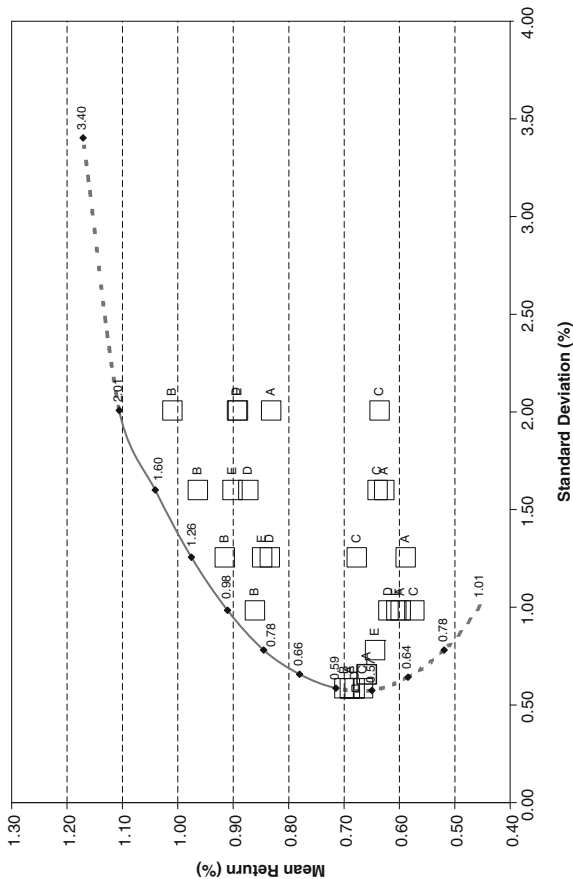


Figure 14.1 Optimal fund-of-hedge-fund portfolios minimum-variance frontier with comparative MVSX portfolios. The solid line indicates the segment of the frontier for which MVSX portfolios are modeled, while the broken line denotes the remainder of the minimum-variance frontier. The squares plot the expected mean return and standard deviation for the MVSX portfolios. It is clear that optimising for higher moments while holding variance (or standard deviation) constant results in a deterioration of mean return. Investor profiles A-E are presented in Table 14.3.

In the inefficient portfolios 1 through 5, the mean–variance optimisation initially allocates capital to Dedicated Short-Bias and Emerging Markets. The mean–variance model uses Dedicated Short-Bias as a means of initially reducing portfolio return. These two strategies have a correlation coefficient value of -0.63 and thus the Emerging Markets exposure reduces portfolio volatility. From portfolios 6 to 11, these two allocations are reduced in favour of Fixed Income Arbitrage and Risk Arbitrage (with small allocations to Multi-Strategy and Managed Futures).

An examination of the efficient portfolios 12–21 shows that almost all these portfolios contain Equity Market-Neutral allocations as well as Distressed Securities. Only at the extreme levels of expected return do Global Macro funds play a role. By construction, the efficient frontier is increasing in volatility from the minimum variance point to the point of maximum return. The mean–variance model, however, does not evaluate the impact of higher moments on portfolio design. It must be noted from Table 14.2 that from the minimum-variance point skewness initially increases (from portfolio 13 to 16) and then decreases. Portfolio kurtosis initially falls from the minimum-variance point (portfolio 13) and then increases (portfolio 15). Under a mean–variance regime, portfolio 13 is the minimum-variance portfolio. This can also be expressed as the lowest risk portfolio under this framework. Taking higher moments into account may yield a slightly different result, as portfolios 13 and 14 have more favourable third and fourth moments (higher skewness and still low kurtosis). Thus it can be argued that depending on particular investor preferences, either portfolio 13 or 14 could, in fact, be the minimum risk portfolio.

Examining the more volatile portfolios (16–19), it can also be argued that the mean–variance framework does not provide the full risk picture. In these portfolios, it can be seen that skewness decreases and kurtosis increases. As noted in the fifth section, this is an unfavourable combination as this increases the likelihood of more severe negative returns.

PGP optimisation for MVSK

Upon calculating the mean–variance efficient frontier, 20 equidistant standard deviation points along the frontier are used as ‘anchors’ to enable comparison with the PGP regime. Furthermore, these particular anchor points are along the section of the efficient frontier beyond the minimum variance portfolio and before the maximum expected return portfolio. The standard deviation anchors are 0.59 per cent, 0.66 per cent, 0.78 per cent, 0.98 per cent, 1.26 per cent, 1.60 per cent and 2.01 per cent.

Utilising these anchor points, PGP-optimised MVSK portfolios are modelled for five different profiles of investor preferences with respect to expected return, skewness and kurtosis. α denotes investor preference over expected return, while β and γ denote preference for skewness and kurtosis, respectively. The five modelled profiles (labelled A–E) for each anchor point are displayed in Table 14.3.

Table 14.3 Preference scenarios in PGP MVSK optimisation

	A	B	C	D	E
α	1	3	1	3	2
β	3	1	0	2	3
γ	0	0	3	1	1

Under this methodology, α , β and γ denote investor preferences for mean return, skewness and kurtosis, respectively. These preferences form part of the objective function Z . Three denotes that a relatively high level of investor utility is derived from this moment, 2 a medium level and 1 a low level. Zero indicates no preference.

Figure 14.1 presents a section of the original mean–variance frontier calculated in Mean–variance optimisation section. The frontier is augmented by the addition of the square indicators, which show the expected return/standard deviation point for an MVSK-optimised portfolio. The square indicators are marked by their labels (A–E), which denote the respective profiles to which they refer. It can be noted from Figure 14.1 that the MVSK-optimised portfolios differ substantially in return from the mean–variance efficient frontier portfolios. It is also clear that scenarios B and E generally map closer to the efficient frontier, while those of A and C map further away. These results are intrinsic to the MVSK model, for under scenarios B and E greater preference is placed on portfolio return (α (mean or expected return preference) values of 3 and 2 respectively). On the other hand, scenarios A and C have maximum preference for one of the higher moments (β (skewness preference) value of 3 and γ (kurtosis preference) value of 3, respectively).

A number of notable observations can be made from Figure 14.1. First, all MVSK portfolios appear below the mean–variance efficient frontier. This shows that the optimisation of a fund-of-hedge-fund portfolio in the MVSK space is one of the competing objectives. Therefore, there is a consistent trade-off between the four moments. This finding is the same as that of Davies *et al.*⁶ In other words, holding variance constant at a pre-specified level and optimising for the other three moments must lead to deterioration in the expected portfolio return. If this were not the case, optimising for skewness and kurtosis would be at no cost to the investor and would effectively constitute a ‘free lunch’.

Secondly, as the standard deviation increases along the efficient frontier, the divergence between the mean–variance-optimised portfolio and the MVSK-optimised portfolios increases. The reason for this is two-fold. As the volatility of the portfolio increases, an offsetting large reduction in expected return must be sacrificed in order to improve the skewness and kurtosis of the portfolio. Furthermore, under the mean–variance regime, the portfolios optimised beyond the minimum-variance portfolio initially have improving higher moment risk statistics. From portfolio 15 onwards, the optimal mean–variance portfolios have deteriorating skewness and kurtosis values.

Thus, in order to improve these attributes, an ever larger return forfeit is required.

The portfolios also differ substantially in composition. Using the standard deviation anchor point of 1.26 per cent as an example, the output from the model is displayed in Table 14.4. First, it must be noted that the MVSK portfolios A–E all have higher levels of skewness and lower levels of kurtosis than the mean–variance portfolio. Secondly, the MVSK are substantially different in their composition.

With the exception of the 1 per cent allocation in portfolio C, all the MVSK award no weighting to the Distressed Securities category. This is in stark contrast with the mean–variance portfolio, which has a 41 per cent holding. Furthermore, with the exception of some inconsequential allocations, almost none of the optimal portfolios include strategies that exhibit the hazardous combination of negative skewness and high kurtosis, that is, Distressed Securities, Event Driven, Event Driven Multi-Strategy, Fixed Income Arbitrage

Table 14.4 Comparison of optimal fund-of-hedge-fund portfolios under a mean–variance regime and those under an MVSK framework

Scenario	A	B	C	D	E	Mean–variance
α	1	3	1	3	2	3
β	3	1	0	2	3	0
γ	0	0	3	1	1	0
Mean	0.59	0.91	0.68	0.83	0.85	0.98
Variance	1.58	1.58	1.58	1.58	1.58	1.58
Skewness	0.75	0.72	–0.01	0.53	0.61	–1.18
Kurtosis	1.88	1.47	–0.65	0.21	0.49	7.59
Standard deviation	1.26	1.26	1.26	1.26	1.26	1.26
Convertible arbitrage (%)	0	0	10	10	2	0
Dedicated short-bias (%)	21	0	2	0	0	0
Distressed securities (%)	0	0	1	0	0	41
Emerging markets (%)	0	0	12	0	0	0
Equity market-neutral (%)	49	69	7	28	36	45
Event driven (%)	0	0	3	0	0	0
Event driven multi-strategy (%)	0	0	7	0	0	0
Fixed income arbitrage	0	0	17	0	0	0
Global macro	1	27	0	9	12	14
Multi-strategy (%)	16	0	7	14	15	0
Risk arbitrage (%)	0	0	3	0	0	0
Long/short equity (%)	0	0	0	20	19	0
Managed futures (%)	14	4	28	19	16	0

The portfolios are constructed under the MVSK PGP model depending on the investor preferences specified. A simple mean–variance portfolio can be run as a special case with the preferences of maximising return with no preference for either skewness or kurtosis. For the purpose of this comparison, a fixed portfolio standard deviation of 1.26 per cent per month is used.

and Risk Arbitrage. For example, Distressed Securities by their very nature imbue a high probability of bankruptcy, while Fixed Income Arbitrage bears credit risk—so well borne out by the LTCM²⁵ and more recent sub-prime disasters.

All portfolios include an allocation to the Equity Market Neutral category. As shown in the fifth section, the Equity Market Neutral index exhibits relatively low levels of volatility and kurtosis, as well as close to zero skewness. By pairing off similar long and short positions, systematic risks are reduced, yielding a truly low-risk strategy.

Performance evaluation

If MVSK portfolios are in fact more efficient than mean–variance portfolios, then performance appraisal measures should reflect this. For the optimised portfolios under the anchor point 1.26 per cent, three performance functions (Sharpe and Sortino ratio as well as the Omega function) are calculated and presented in Table 14.5.

By construction, the mean–variance-optimised portfolio has a superior Sharpe ratio to the MVSK portfolios. Examining the performance under the Sortino ratio, all MVSK portfolios are superior, with the exception of portfolio A. Given that the ‘risk denominator’ in the Sortino ratio is downside deviation, additionally optimising for the higher moments of skewness and kurtosis should provide some benefit. Portfolio A optimises heavily on the skewness preference and this appears to have lowered the overall return substantially below that of the other portfolios, resulting in a lower Sortino score. As mentioned, the Omega function observes the mass of a probability density function above a pre-determined threshold level—in this case, the risk-free rate for which the Lehman Brothers US Cash Composite is a proxy. According to this measure, the MVSK portfolios all outperform the mean–variance portfolio.

Asset allocation with hedge funds optimisation

This section attempts to determine what proportion of their assets investors should allocate in a traditional portfolio that includes an allocation to hedge funds. The model utilises the MSCI World Equity and World Sovereign Bond indices to proxy for diversified global equity and bond portfolios. Furthermore, the data series from the CSFB Tremont Composite Hedge Fund Index is employed as a proxy for a well-diversified fund of hedge funds.

Mean–variance optimization

The same technique as applied in the previous section is utilised with the above data set. Once more, minimum-variance portfolios are found for 20 equidistant points of return among the four assets.²⁶ The results of this mean–variance model are presented in Table 14.6 and graphically in Figure 14.2. Examining these portfolios shows that unlike the previous minimum–variance frontier, this frontier has no inefficient segment.

Table 14.5 Performance measures for optimal fund-of-hedge-fund portfolios

Investor profile	A	B	C	D	E	Mean—variance
Sharpe ratio	0.20	0.45	0.27	0.39	0.40	0.50
Sortino ratio	0.41	1.15	0.59	1.10	1.11	0.50
Omega	1.010996695	1.010996695	1.010996694	1.010996695	1.010996689	1.010909474
Omega rank	1	3	4	2	5	6

The performance measures are presented below for different investor preference scenarios for the anchor point of 1.26 per cent standard deviation. The Sharpe Ratio shows excess return (above the risk-free rate) per unit of volatility. The Sortino Ratio shows excess return (above the risk-free rate) per unit of downside volatility. The Omega function is a ratio of the area above to the area below a threshold level, of a probability distribution. The threshold level used is that of the risk-free rate to ensure comparability with the other ratios. The Omega rank refers to the ranking of the Omega function in descending order with 1 indicating the most preferred portfolio.

Table 14.6 Optimal balanced portfolios under a mean–variance framework

Panel A: Portfolio expected return and risk statistics

Portfolio number	Expected return (%)	Variance (%)	Standard deviation (%)	Skewness	Kurtosis
1	0.34	0.00	0.15	−0.70	−0.93
2	0.37	0.00	0.18	−0.15	−0.51
3	0.39	0.00	0.25	0.11	−0.05
4	0.42	0.00	0.35	0.12	0.25
5	0.45	0.00	0.44	0.09	0.41
6	0.48	0.00	0.54	0.05	0.51
7	0.50	0.00	0.65	0.02	0.57
8	0.53	0.01	0.75	0.00	0.61
9	0.56	0.01	0.85	−0.02	0.64
10	0.59	0.01	0.96	−0.03	0.66
11	0.62	0.01	1.06	−0.04	0.68
12	0.64	0.01	1.16	−0.05	0.69
13	0.67	0.02	1.27	−0.06	0.70
14	0.70	0.02	1.37	−0.07	0.71
15	0.73	0.02	1.48	−0.08	0.72
16	0.75	0.03	1.58	−0.08	0.76
17	0.78	0.03	1.71	−0.04	1.01
18	0.81	0.03	1.85	0.00	1.29
19	0.84	0.04	2.02	0.04	1.54
20	0.86	0.05	2.20	0.08	1.74
21	0.89	0.06	2.39	0.10	1.90

Panel B: Percentage allocation to investment in diversified portfolio

Hedge fund composite (%)	World equity (%)	World sovereign bond (%)	US cash composite (%)
0	0	0	100
4	0	3	93
8	0	6	86
12	0	9	79
16	0	11	73
20	0	14	66
25	0	16	59
29	0	19	52
33	0	21	46
37	0	24	39
41	0	27	32
45	0	29	26
49	0	32	19
54	0	34	12
58	0	37	6
62	0	38	0
70	0	30	0
77	0	23	0
85	0	15	0
92	0	8	0
100	0	0	0

Panel A reflects portfolio expected return and the higher moment risk statistics while Panel B shows the detailed allocation to individual assets.

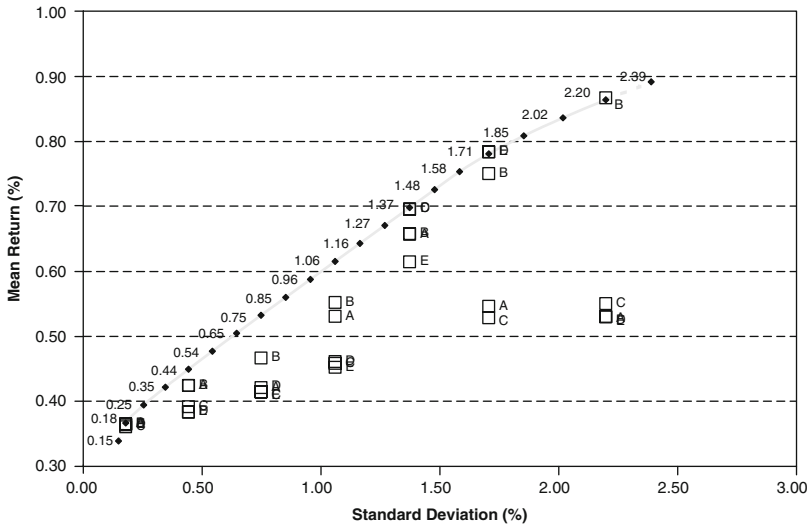


Figure 14.2 Optimal balanced portfolios under a mean–variance framework with comparative MVSK portfolios

The solid line indicates the segment of the frontier for which MVSK portfolios are modelled, while the broken line denotes the remainder of the minimum-variance frontier. The squares plot the expected mean return and standard deviation for the MVSK portfolios. It is clear that optimising for higher moments while holding variance (or standard deviation) constant results in a deterioration of mean return. Investor profiles A–E are presented in Table 14.3.

The optimal mean–variance portfolios make no allocation to the World Equity index at all. This may be due to the World Equity and World Sovereign Bond indices having very similar average monthly returns over the sample period but World Sovereign Bond having a much lower level of volatility than World Equities.²⁷ Under a framework that rewards expected returns per unit of volatility, this is an intuitive explanation.

The optimal portfolios initially allocate capital to cash and gradually increase exposure to the World Sovereign Bond index and the Hedge Fund Composite as volatility increases. The Sovereign Bond allocation increases in proportion until portfolio 16 and then declines to zero. This indicates that beyond portfolio 16, the correlation or covariance benefit from a volatility reduction perspective is surpassed by the return offered by the hedge fund composite. The US Cash Composite allocation decreases consistently until portfolio 16 where it reaches zero.

Examining the risk moments of variance, skewness and kurtosis, it can be seen that volatility rises across the frontier while the skewness and kurtosis values are more dynamic. Skewness initially increases across portfolios 1–4, then declines until portfolio 16 and then increases to the maximum return

portfolio. Kurtosis generally increases with variance across this optimisation but increases more rapidly from portfolio 16 onwards.

In terms of portfolio composition, the MVSK and mean–variance portfolios are substantially different. Using the standard deviation anchor point of 1.06 per cent as an example, the output is presented in Table 14.7.

Again, all the MVSK portfolios have a lower level of return at each volatility level than the mean–variance portfolio. For the MVSK portfolios that have no preference for either higher moment (viz. scenarios A, B and C), that particular moment is worse than the mean–variance portfolio. It is interesting to note that some of the MVSK portfolios do allocate to the World Equity asset, unlike the mean–variance portfolio, which holds no World Equity. This shows that its inclusion in a diversified portfolio has a benefit of either improving skewness or reducing kurtosis when blended with these other investments.

A key observation is the fact that the MVSK portfolios all allocate substantially lower weightings to hedge funds than the mean–variance portfolio. In fact, scenario E makes no allocation to the Hedge Fund Composite at all. This is in stark contrast to the findings above as well as to those by prior research covered in the third section, where mean–variance portfolios allocate large amounts of capital to hedge funds in balanced diversified portfolios. Furthermore, comparing the mean–variance allocation in Table 14.7 to profile B (where merely a unit of preference for positive skewness is added) reveals that the hedge fund allocation almost halves from 41 per cent to

Table 14.7 Comparison of optimal balanced portfolios under a mean–variance regime and under an MVSK framework

Scenario	A	B	C	D	E	Mean–variance
α	1	3	1	3	2	3
β	3	1	0	2	3	0
γ	0	0	3	1	1	0
Mean	0.53	0.55	0.46	0.46	0.45	0.62
Variance	1.12	1.12	1.12	1.12	1.12	1.12
Skewness	0.21	0.15	–0.27	–0.01	0.21	–0.06
Kurtosis	1.08	1.08	–0.08	0.27	0.68	0.72
Standard deviation	1.06	1.06	1.06	1.06	1.06	1.06
Hedge fund composite (%)	16	21	5	2	0	41
World equity (%)	0	0	20	15	6	0
World sovereign bond (%)	54	52	26	41	53	27
US cash composite (%)	30	28	48	41	41	33

The portfolios are constructed under the MVSK PGP model depending on the investor preferences specified. A simple mean–variance portfolio can be run as a special case with the preferences of maximising return with no preference for either skewness or kurtosis. For the purpose of this comparison, a fixed portfolio standard deviation of 1.06 per cent per month is used.

21 per cent as a result of a relatively minor change in utility function! It is clear that considering higher moments leads to a more tempered allocation to hedge funds.

Performance evaluation

Using the same format as before, Table 14.8 presents the three performance criteria for the anchor point 1.06 per cent.

The performance criteria are not as clear in expressing support for the MVSK portfolios. As in the previous section, the mean–variance portfolio should always have the best Sharpe ratio by construction.

The Omega function was again calculated using the risk-free cash return as a threshold level in order to make it comparable with the Sharpe and Sortino ratios. The Omega function ranks all the MVSK portfolios above that of the mean–variance portfolio. This is the case for all anchor points across the mean–variance frontier. The mean–variance portfolio, however, also shows a superior Sortino ratio when compared with the MVSK portfolios. It is remarkable that scenario B displays the most attractive Sharpe and Sortino ratios within the MVSK portfolios but also ranks lowest when evaluated on the Omega function criterion. Scenario C, which has the highest Omega score, in fact has the most negative skewness, but compensates by also having the lowest kurtosis.

Conclusion

This paper finds that almost all hedge fund strategies have non-normal distributions. In contrast, the Jarque-Bera test for the normality of the equity and bond indices was unable to reject the null hypothesis of normality at the 5 per cent level. Many hedge fund strategies (especially the Event-Driven category and Fixed Income Arbitrage) have negatively skewed returns as well as leptokurtosis ('fat-tails'). Through analysis of these two higher moments together, it can be concluded that these strategies have a substantially higher probability of extreme losses than suggested by a normal distribution. As an example, research by Agarwal and Naik² has shown that the payoff to the broad category of Event-Driven hedge fund strategies is akin to short put option positions on the market. This means that this strategy will yield low volatility consistent returns until the occurrence of an extreme adverse event, whereupon the strategy will make substantial losses. Based on the features of the probability distribution of this strategy, more specifically significant negative skewness and a high level of kurtosis, these extreme adverse events occur more often than suggested by a normal distribution. A similar rationale can be made for other hedge fund strategies, and Fixed Income Arbitrage in particular.

It is found that the results of an optimisation under a conventional mean–variance regime and those of the MVSK framework are substantially

Table 14.8 Performance measures for optimal balanced portfolios

Investor profile	A	B	C	D	E	Mean—variance
Sharpe ratio	0.18	0.20	0.11	0.11	0.10	0.26
Sortino ratio	0.31	0.33	0.18	0.22	0.18	0.41
Omega	1.015485507	1.015485497	1.015485535	1.015485499	1.015485526	1.015383107
Omega rank	3	5	1	4	2	6

The performance measures are presented below for different investor preference scenarios for the anchor point of 1.06 per cent standard deviation. The Sharpe Ratio shows excess return (above the risk-free rate) per unit of volatility. The Sortino Ratio shows excess return (above the risk-free rate) per unit of downside volatility. The Omega function is a ratio of the area above to the area below a threshold level, of a probability distribution. The threshold level used is that of the risk-free rate to ensure comparability with the other ratios. The Omega rank refers to the ranking of the Omega function in descending order with 1 indicating the most preferred portfolio.

different. Comparing portfolios with the same level of volatility shows that the MVSK portfolios deliver lower returns than mean–variance optimised portfolios on a mean–variance plane. This illustrates that further optimising for higher moments has an (*ex post*) cost and that measuring the returns of portfolios that include hedge funds on a purely mean–variance level is deceptive. Similarly, it has been shown that optimising on a naïve mean–variance basis can introduce the hazard of reducing skewness and increasing kurtosis in a portfolio further raising the probability of extreme adverse events. The compositions of the mean–variance portfolios and MVSK portfolios are also substantially different. The MVSK portfolios tend to be more diversified than the mean–variance portfolios. Further, the MVSK portfolios generally reduce allocations to the Event-Driven category of hedge funds. It has been shown that when assembling optimal fund-of-hedge-funds portfolios, a substantial reduction in risk from changes in the higher moments can be achieved with relatively little decrease in mean return. This paper concurs with previous studies of Feldman *et al.*¹⁶ and Davies *et al.*⁶ that Global Macro and Equity Market-Neutral strategies are key building blocks in fund-of-hedge-fund construction. From an asset allocation perspective, the MVSK optimisation indicates that hedge fund allocations should be lower than those suggested by the mean–variance framework.

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21. In a Dedicated Short-Bias portfolio, all short positions generate cash that in turn earns interest—this creates a positive drift to the performance of the portfolio.
22. It must be noted that the skewness (Z_3) and kurtosis (Z_4) formulae above correct minor errors in the original formulae applied by Davies *et al.*⁶ The authors contacted Sa Lu, a co-author of the Davies *et al.*⁶ paper, and she confirmed the appropriateness of the formulae applied above. A further difference is that Davies *et al.*⁶ solve for optimal fund-of-hedge-fund portfolios subject to a unity variance constraint.
23. Minimum average monthly return of all the series is –0.13 per cent from the Dedicated Short-Bias index.
24. Maximum average monthly return of all the series is 1.17 per cent from the Global Macro index.
25. LTCM was a hedge fund managed by ex-Salomon Brothers bond traders and Nobel laureates where the fund was highly leveraged and followed a strategy of being long high-yield debt and short sovereign treasuries. During the events of the Russian debt default and the Asian crisis of 1998—the fund collapsed and was rescued by a group of Wall Street investment banks.
26. Minimum average monthly return of all the series is 0.34 per cent from the Lehman Brothers US Cash Composite and the maximum average monthly return of all the series is 0.89 per cent from the Hedge Fund Composite index.
27. The MSCI World Equity index has a monthly standard deviation of 4.16 per cent and monthly mean return of 0.54 per cent, while the MSCI World Sovereign Bond index has a monthly standard deviation of 1.91 per cent and monthly mean return of 0.52 per cent.

15

Sovereign Wealth Funds– Investment Strategies and Financial Distress

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This article discusses the recent developments with regard to sovereign wealth funds (SWFs) and presents a simple model that illustrates their potential impact on global financial stability during periods of market distress. The model incorporates key institutional features of SWFs such as investment strategies, size and disclosure standards that could have a bearing on how shocks are transmitted across financial markets. The model suggests an asymmetric impact of SWFs during financial turmoil and booms, and provides an analytical framework to inform policy discussion with regard to the governance of SWFs.

Introduction

Sovereign wealth funds (SWFs) are generally defined as investment funds owned and managed by national governments.¹ SWFs are not a new

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phenomenon, but, as many emerging markets and commodities-exporting countries have accumulated large stocks of foreign assets, effective wealth management has increasingly become an important public sector responsibility. Accordingly, the number of SWFs has been rising and assets under management have increased by more than half in the last decade, thereby exceeding assets managed by hedge funds and accounting for between one-fourth and one-third of all foreign assets held by sovereigns. SWFs have become increasingly important not only within their own countries, but also in the global financial system.

Against this backdrop and that of the recent financial turmoil, officials from recipient countries have noted that SWFs may pose additional risks to the stability of financial markets, as the role of governments in the global financial system has expanded while the investment strategy of SWFs may respond to political objectives. On the other hand, SWFs have been perceived as offering economic benefits to the global financial system by allowing greater diversification and risk-sharing than the reserve management practices of central banks. It is not surprising, therefore, that public and private leaders in both SWF sponsor countries and recipient countries recognized that the international system would benefit from enhanced clarity regarding the principles and practices followed by both sides. In this regard, SWFs have recently formed an International Working Group (IWG) under the auspices of the International Monetary Fund (IMF).

The issue of whether SWFs are a source of additional risks or a stabilizing force, particularly at times of financial distress, has gained momentum, but little analytical work is available. This article develops a framework that incorporates the investment strategies of SWFs and other market participants and their reactions to shocks, allowing for a discussion of how the interplay among market participants potentially affects the stability of the international financial system. It also provides insights into policy discussion on the governance of SWFs, and highlights areas for future empirical work.

This article is organized as follows. It starts by reviewing recent developments, and pointing out the rapid growth in SWFs' activities, the incentive for risk-sharing and the limited disclosure standards of SWFs. The section after that presents the model framework, and the penultimate section draws some policy implications and highlights areas for future empirical work. The final section concludes.

Background

The number and size of SWFs have risen remarkably in the past decade, and are expected to grow further (Figure 15.1). The number of SWFs has doubled during the last decade, and, of the top 10 largest SWFs, nearly half were established after 2000. Assets under management, which are currently about US\$3 trillion above those managed by hedge funds (\$1.9 trillion) and

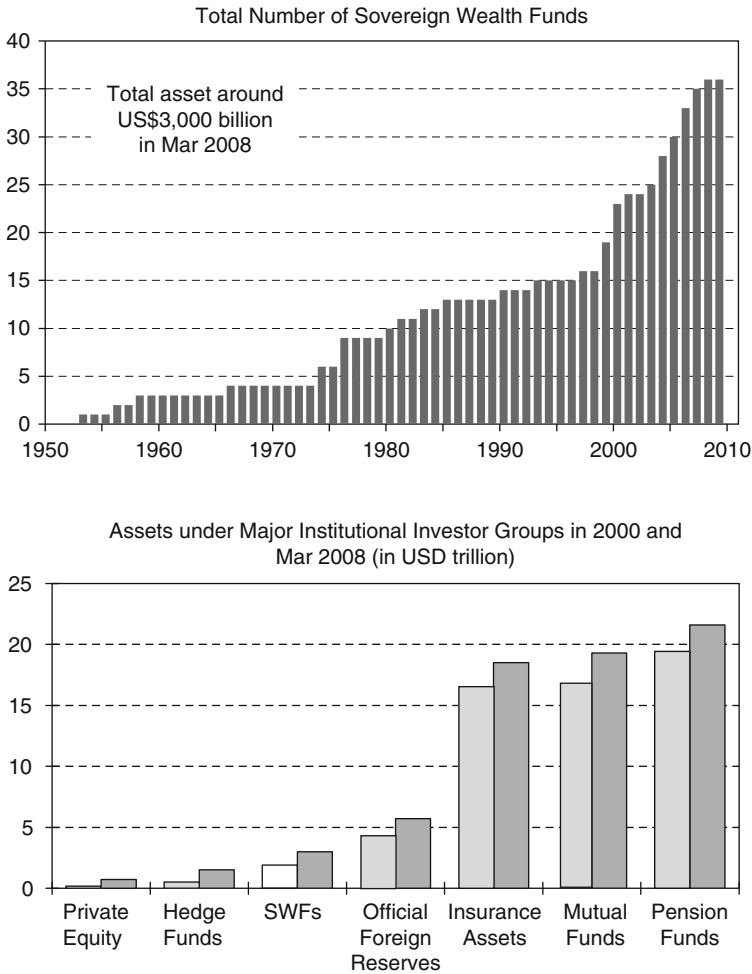


Figure 15.1 Recent developments of sovereign wealth funds

Source: IMF, McKinsey Global Institute, Morgan Stanley, various websites, and authors' estimates.

private equity funds, are estimated to reach \$5 trillion by 2010, and surge to \$10 trillion by 2015.²⁻⁴ This estimated growth is much larger than for most other institutional investors such as pension funds or mutual funds.

SWFs have mostly been established in emerging markets and developing countries in recent years (Table 15.1), responding to a period of buoyant oil and commodities prices and strong economic growth and capital inflows.⁵⁻⁷ These countries have accumulated a large amount of foreign exchange reserves and, against a backdrop of low money markets returns, have found

Table 15.1 List of sovereign wealth funds across countries as of June 2008

Country ^a	Names of SWFs	Type	Year established	US (billion)	Geographical coverage	Portfolio allocation
Algeria	Revenue Regulation Fund	Oil	2000	47	—	—
Angola	Reserve Fund for Oil	Oil	2007	0	—	—
Australia	Australian Future Fund	Non-commodity	2004	59	Mainly local	Only 18% in international equities
Azerbaijan	State Oil Fund of the Republic of Azerbaijan	Oil	1999	5	Global asset allocation	Equities and fixed-income
Bahrain	Bahrain Mumtalakat Holding Company	Oil	2006	10	Regional	Equities and corporate
Botswana	Pula Fund	Diamond	1966	7	—	Equities and fixed-income
Brazil ^b	TBA	TBA	2008	8–10*	Mainly local	Equities and corporates
Brunei	Brunei Investment Authority General Reserve Fund	Oil	1983	30	Global asset allocation	Major asset class
Canada	Alberta Heritage Savings Trust Fund	Oil	1976	17	Global asset allocation	Major asset class
Chile	Economic and Social Stabilization Fund; Pension Reserve Fund	Copper	1985	16	Stabilization	Fixed-income
China	State Foreign Exchange Investment Corporation/China Investment Corporation	Non-commodity	2007	200–311	Global asset allocation	Unknown; major asset class

(continued)

Table 15.1 Continued

Country ^a	Names of SWFs	Type	Year established	US (billion)	Geographical coverage	Portfolio allocation
East Timor	Timor-Leste Petroleum Fund	Oil	2005	3	Global asset allocation	Major asset class
India ^b	TBA	TBA	TBA	5*	—	—
Iran	Oil Stabilisation Fund	Oil	1999	13	Global asset allocation	Equities and fixed-income
Ireland	National Pensions Reserve Fund	Non-commodity	2001	31	Global asset allocation	Equities and fixed-income
Kazakhstan	Kazakhstan National Fund	Oil	2000	22	—	—
Kiribati	Revenue Equalization Reserve Fund	Phosphates	1956	0	—	—
Korea	Korea Investment Corporation	Exports	2005	30	Global asset allocation	Major asset class
Kuwait	Kuwait Investment Authority; General Reserve Fund; Future Generations Fund	Oil	1953	250	Global asset allocation	Major asset class
Libya	Libyan Arab Foreign Investment Company	Oil	1981	50	Regional	—
Malaysia	Khazanah Nasional	Non-commodity	1993	25.7	Local	Equities and corporates
Mauritania	National Fund for Hydrocarbon Reserves	Oil and gas	2006	0	—	—
New Zealand	New Zealand Superannuation Fund	Non-commodity	2003	14	Global asset allocation	Major asset class
Nigeria	Excess Crude Account	Oil	2004	11	—	—

Norway	Government Pension Fund – Global	Oil	1990	397	Global asset allocation	40% equities; 60% global fixed income
Oman	State General Reserve Fund	Oil	1980	2	—	—
Qatar	Qatar Investment Authority	Oil	2000	60	Global asset allocation	Major asset class
Russia	Oil Stabilization Fund/National Welfare Fund*	Oil	2004	163	—	96% in global fixed-income instruments
Saudi Arabia	Public Investment Fund	Oil	2008	5	Global asset allocation	—
Saudi Arabia	SAMA Foreign Holdings	Oil	NA	300	Global asset allocation	Major asset class
Singapore	Government Investment Corporation; Temasek Holdings	Non-commodity	1974	159	Global asset allocation	Major asset class
Taiwan, POC	National Stabilisation Fund	Non-commodity	2000	15	Local	Equities and corporates
United Arab Emirates	Abu Dhabi Investment Authority /Abu Dhabi Investment Council	Oil	1976	875	Global asset allocation	Major asset class
United States (Alaska)	Alaska Permanent Reserve Fund	Oil	1976	40	State-owned	All major asset class
United States (New Mexico)	New Mexico State Investment Office	Non-commodity	1958	16	—	—
United States (Wyoming)	Trust Permanent Wyoming Mineral Trust Fund	Commodity	1974	4	—	—

(continued)

Table 15.1 Continued

Country ^a	Names of SWFs	Type	Year established	US (billion)	Geographical coverage	Portfolio allocation
Venezuela	FIEM – Macroeconomic Stabilization Fund	Oil	1999	1	—	—
Vietnam	State Capital Investment Corporation	Non-commodity	2006	2	Local	—
Total ^c of which: commodity related funds				2977 2024		

^a Chad, Ecuador, Equatorial Guinea, Gabon, Mexico, Sudan, Trinidad and Tobago, and Uganda also have official funds established.

^b Figures based on news report.

^c Some funds not listed also contribute to the total sum. Figures from Sovereign Wealth Fund Institute as of 12 June 2008.

Source: Sovereign Wealth Fund Institute, IMF, *Financial Times*, Linaburg and Maduell¹² and various websites.

NA= Not available.

*= Preliminary.

it increasingly compelling to establish alternative investment vehicles that could broaden sovereign portfolio allocations to other major asset classes. In addition to investing their foreign exchange reserves in higher-return assets, SWFs have also been used to hedge against shocks in the commodity- and export-oriented sectors by holding assets whose returns are inversely correlated with the country's primary risk exposure.^{8,9} Over 60 per cent of assets under management by sovereigns are concentrated in oil- and commodity-related SWFs.¹⁰ Moreover, based on public information, most SWFs adopt diversified investment strategies covering major asset class, including equities, fixed-income instruments (and in some cases, real estates and private equity) and geographic areas (Table 15.2).

SWFs have been criticized for their perceived weak disclosure requirements. Truman¹¹ and Linaburg and Maduell¹² noted that most SWFs in emerging markets rank poorly in terms of disclosing investment strategies and governance. In contrast, Norway, Australia and Korea have ranked above 9 out of 10 in terms of disclosure in key areas like management, external managers and objectives. We classify the disclosure in terms of a few key areas, including whether the SWFs have indicated their history and objectives, published audited annual reports and disclosed their portfolio allocation

Table 15.2 Risk-sharing incentive of sovereign wealth funds

Country	Type	Year established	US (billion)	Risk-sharing/ diversification	Geographical coverage
Australia	Non-commodity	2004	59	—	Mainly local
China	Non-commodity	2007	200–311	Y	Global asset allocation
Korea	Non-commodity	2005	30	—	Global asset allocation
Kuwait	Oil	1953	250	Y	Global asset allocation
Norway	Oil	1990	397	—	Global asset allocation
Qatar	Oil	2000	60	Y	Global asset allocation
Russia	Oil	2004	163	Y	—
Saudi Arabia	Oil	NA/2008	300	Y	Global asset allocation
Singapore	Non-commodity	1974	159	Y	Global asset allocation
United Arab Emirates	Oil	1976	875	Y	Global asset allocation

Source: Sovereign Wealth Fund Institute, IMF, *Financial Times* and various websites.

NA=Not available.

(Table 15.3). We find that the largest SWFs from the emerging markets do not generally disclose adequately relative to their counterparts in the developed countries.

Although governance issues are usually considered microeconomic in nature, it can be argued that they could have significant macroeconomic effects and undermine the stability of the international financial system. Previous IMF staff research suggests that the pattern of global capital flows could be affected by an increased participation of SWFs in international financial markets.^{13–15} For instance, a shift away from the asset classes in which foreign exchange reserves have been traditionally held could impact asset prices and cross-border financial flows, while actual or rumored transactions may affect valuations in particular sectors and result in herding behavior, adding to volatility. Given the size of their portfolios, SWFs may potentially cause market disturbance and destabilize the global financial system. Such effects could be especially pronounced in shallower markets and require recipient countries to strengthen their policy framework and design appropriate policies.

Given data limitation, an extensive empirical analysis is not a viable approach. Event studies analyzing the role of SWFs are subject to sample bias, as they would include only publicly disclosed SWFs' transactions. Presented below is an analytical framework that incorporates the investment objectives and behavior of SWFs and other institutions.

The Analytical Framework

The framework builds on the literature on the transmission mechanism of financial distress, as discussed in Morris and Shin^{16,17} and Campos,¹⁸ by explicitly incorporating SWFs as a specific class of investors. It illustrates how the interplay among diverse institutions impacts the likelihood of shocks turning into crises and the transmission mechanism therein.

The existing literature on self-fulfilling financial crises provides an analytical framework in which the impact of shocks to fundamentals on financial stability takes account of market participants' rational reaction to the expectation that other participants would run for the exit when faced with those shocks. However, the framework distinguishes neither among different classes of market participants in terms of the relative impact of their actions in triggering a financial crisis nor the potentially larger joint impact of such actions.

This article explores a novel transmission mechanism by explicitly incorporating the investment objectives and behavior of SWFs. The key characteristics of the proposed framework are:

- *Uncertainty.* At $t=0$, investors are uncertain as to which state of the economy, $s \in [s, \bar{s}]$, will prevail at time $t=1$, where s has density function

Table 15.3 Disclosure indicators among the 10 largest SWFs

Country	Transparency ratings ^a	Disclosure indicators							
		History	Audited annual reports	Portfolio holdings/Asset class	Management	Policy and governance guidelines	Strategy and objective	External managers	Website/Contact
Norway	10	Y	Y	Y	Y	Y	Y	Y	Y
Australia	9	Y	Y	Y	Y	Y	Y	Y	Y
Korea	9	Y	Y	Y	Y	Y	Y	Y	Y
Ireland	7	—	Y	Y	—	—	Y	Y	Y
Singapore	6–7	Y	—	—	Y	Y	Y	—	Y
Kuwait	6	Y	—	—	Y	—	Y	—	Y
Russia	5	—	—	—	—	—	—	—	Y
Saudi Arabia	NA	—	—	—	—	—	Y	—	—
Saudi Arabia	4	Y	—	—	Y	—	—	—	—
United Arab Emirates	3	Y	Y	—	—	—	Y	—	Y
Chile	3	Y	—	—	—	—	Y	—	—
China	2	Y	—	—	Y	—	—	—	Y
Iran	2	—	—	—	—	—	—	—	—
Qatar	1	—	—	—	—	—	Y	—	Y

^aLinaburg and Maduell¹².Source: Sovereign Wealth Fund Institute, IMF, *Financial Times* and various websites.

NA=Not available.

$p(s)$. Total wealth is identified by an aggregate endowment $e(s)$, which is increasing with s . It is equal to $\sum_i e_i(s)L_{ij}$, where $e_i(s)$ identifies each investor's endowment and L_{ij} , which is equal to or greater than 1, identifies each investor's leverage.

- *Market participants.* Investors (i) have different degrees of risk aversion and leverage. SWFs are assumed to be risk averse ($\gamma_i > 0$) and not leveraged ($\gamma_{SWF} > 0$ and $L_{SWF} = 1$). Heterogeneity with regard to risk aversion and leverage drives portfolio allocation at $t=0$ and assets liquidation at $t=1$. Market participants maximize the same standard Constant Relative Risk Aversion (CRRA) utility function:

$$U(s) = E_s \left(\frac{c(s)^{1-\gamma_i}}{1-\gamma_i} \right) \quad (1)$$

- *Time dimension.* Decisions are taken at three points in time, $t=\{0,1,2\}$. At time $t=0$, market participants (including SWFs) trade contingent claims (of same type) to hedge against risk and to build their portfolios. At time $t=1$, uncertainty is realized and the state of the economy is observed by all market participants. At this point, participants decide how much of their asset holdings to liquidate subject to their expectation of a financial crisis and of other investors' liquidation strategies. At time $t=2$, market participants receive the return on their portfolios.
- *Payoffs.* Proceeds from liquidated assets at $t=1$ earn a riskless return D . Returns on holdings that are not liquidated vary. If a financial crisis occurs at $t=2$, asset prices fall sharply and earn a return R lower than D . If, instead, a financial crisis does not occur at $t=2$, investors earn a return R higher than D .
- *Financial crisis trigger.* The size of holdings that are liquidated at $t=1$ depends on (i) the state of the economy s at $t=1$, and (ii) investors' expectations, at $t=1$, of a crisis in $t=2$, which, in turn, depends on the liquidity of financial markets at $t=1$. The liquidity of financial markets at $t=2$ is the result of each investor's decision to liquidate after observing the state of economy at $t=1$, $\Omega(s, \Theta_s)$. Accordingly, liquidity (Θ_s) is assumed to be normally distributed with $\sim N(\mu(s), \sigma^2)$. A financial crisis at $t=2$ occurs only if liquidations by investors bring liquidity to below a certain level, that is $\Omega(s) \geq \Theta_s^*$, where the threshold Θ_s^* will be discussed below.
- *Information structure.* At $t=0$ investors know only the distribution of Θ_s . At $t=1$, investors receive private information y . It is assumed that $y = \Theta_s + \varepsilon$, where ε is a white noise with zero mean and variance σ_ε^2 . Individual investors then update their expectation about the probability of a crisis $(1-\delta)$; δ_i represents the individual investor's belief that a financial crisis would not occur.

The above can be formally described as:

$$\begin{aligned} & \max_{a(s, \delta)} E_s \left(\left(\frac{c(s)^{1-\gamma_i}}{1-\gamma_i} \right) \right) \\ & = \max_{a(s, \delta)} E_s \left\{ \frac{1}{1-\gamma_i} \left[[\delta_i (Ra(s, \gamma_i, L_i) - (R-D)\omega_i(s, \gamma_i, L_i, \delta_i))]^{1-\gamma} + \right. \right. \\ & \quad \left. \left. [(1-\delta_i)(R_c a(s, \gamma_i, L_i) - (D-R_c)\omega_i(s, \gamma_i, L_i, \delta_i))]^{1-\gamma} \Theta_s^* \right] \right\} \end{aligned} \quad (2)$$

subject to the resource constraint:

$$E_s[q(s)\alpha(s, \gamma)] \leq E_s[q(s)e_i(s)] \quad \omega_i(s, \delta_i, L_i) = \alpha_i(s) \quad \text{if } \alpha_i(s) < L_i$$

where $q(s)$ is the price of state-contingent claims.

The model can be solved using backward induction, starting at $t=1$. Individual investors choose the size of holdings to be liquidated ($\omega(s, \delta_i, \gamma_i, L_i)$ (or $\omega(s, \delta_i)$, as risk aversion and leverage are fixed parameters), taking the state of the economy (s) as given. The optimal size of liquidation is the solution to the following maximization problem:

$$\begin{aligned} & \max_{\omega_i(s)} \left(\frac{c_i(s)^{1-\gamma_i}}{1-\gamma_i} \mid s, \delta_i \right) \\ & = \max_{\omega(s, \delta_i)} \frac{1}{1-\gamma_i} \left\{ [\delta_i (Ra_i(s) - (R-D)\omega(s, \delta_i))]^{1-\gamma_i} + \right. \\ & \quad \left. [(1-\delta_i)(R_c a_i(s) - (D-R_c)\omega(s, \delta_i))]^{1-\gamma_i} \right\} \end{aligned} \quad (3)$$

given investors' holdings $\alpha_i(s; \gamma_i, L_i)$ (or $\alpha_i(s)$), and individual investors' expectations, δ_i , that a financial crisis will not occur; both are known before choosing the size of liquidation). Investors consume the returns from the assets, which is expressed as:

$$\omega(s, \delta_i) \cdot D + \delta_i \cdot R \cdot [\alpha_i(s) - \omega(s, \delta_i)] + (1-\delta_i) \cdot R_c \cdot [\alpha_i(s) - \omega(s, \delta_i)] \quad (4)$$

Proposition 1: The optimal size of liquidation $\omega_i^*(s, \delta_i)$ satisfies the following conditions:

- (i) $0 < \underline{\delta} < \bar{\delta} < 1$;
 - (ii) $\lambda(s, \delta_i) = 1$ iff $\delta_i \in [0, \underline{\delta}]$
 - (iii) $\lambda(s, \delta_i) = 0$ iff $\delta_i \in [\bar{\delta}, 1]$
 - (iv) $\lambda(s, \delta)$ is decreasing in δ_i and increasing in γ_i in the range $\delta \in [\bar{\delta}, \underline{\delta}]$
 - (v) $\omega(s, \delta_i)$ is increasing in L_i .
- (5)

Proof:

See Appendix A, where $\lambda(s, \delta_i) = \omega(s, \delta_i) / \alpha_i(s)$ is the fraction of assets that are liquidated. Under the first three conditions, $\lambda(s, \delta_i)$ is independent of investors' expectations of a future financial crisis. These conditions define upper and lower bounds on investors' expectations of a future financial crisis according to which investors would either liquidate their whole portfolio or not liquidate at all, regardless of risk aversion and leverage. These bounds correspond to the 'very good' and 'very bad' states of the economy, respectively.

The last condition establishes that $\lambda(s, \delta_i)$ is weakly increasing with risk aversion, the riskless rate of return and investors' expectations of a future financial crisis. For instance, investors with higher risk aversion (for example SWFs) would tend to liquidate more relative to other investors, given the same expectations and holdings.

Investors' expectations of a future financial crisis, δ_{i^*} , can be defined by a conditional cumulative density function (CDF):

$$\delta_i = 1 - G(\Theta_s^* | y_i, s)$$

where

$$G(\Theta_s^* | y_i, s) \equiv \Pr\{\Theta_s < \Theta_s^* | y_i, s\}' \quad (6)$$

with δ_i depending on the private signal (y_i) and the state of the economy (s).¹⁹ The conditional CDF $G(\cdot)$ measures the probability for the individual investor that the liquidity of the financial markets lies below a threshold Θ_s^* , that is, a crisis occurs if and only if $\Theta_s < \Theta_s^*$.

The aggregate size of liquidation—that is, the sum of all liquidations by individual investors can be expressed as:

$$\begin{aligned} \Omega(s, \Theta_s^*) &\equiv \sum_i \omega(s, \delta_i) \\ &= \sum_i \alpha_i(s) \int \lambda(s, \delta_i) dF(\delta_i | s, \Theta, \Theta_s^*) \end{aligned} \quad (7)$$

The first term $\alpha_i(s)$ indicates the share of assets held by individual investors, whereas the second represents the individual liquidation taking into account individual expectations of a crisis, given the state of the economy.

Proof:

See Appendix B.

The fact that investors' decisions about how much of their portfolio to liquidate at $t=1$ depends, *inter alia*, on their private signal (y_i) introduces endogeneity in the model. As the probability of a crisis depends on the aggregate size of liquidation, which, in turn, depends on individual investors'

decisions, the latter and the probability of a crisis need to be determined simultaneously in equilibrium.²⁰⁻²² Therefore, the distribution of beliefs is an equilibrium object to be endogenously determined. Morris and Shin,^{16,17} Campos,¹⁸ and Hellwig *et al*²¹ show that the equilibrium exists and is unique.

Proposition 2: The aggregate size of liquidation will vary according to the state of the economy, risk aversion and the liquidity of the financial markets:

$$\frac{\partial \Omega(s, \Theta_s^*)}{\partial \mu} < 0; \quad \frac{\partial \Omega(s, \Theta_s^*)}{\partial \gamma_i} > 0; \quad \frac{\partial \Omega(s, \Theta_s^*)}{\partial L_i} > 0 \quad (8)$$

Higher expected liquidity implies lower likelihood of a financial crisis and, hence, smaller size of aggregate liquidation. An increase in risk aversion triggers a larger aggregate liquidation, as investors tend to shy away from assets with riskier returns (see Appendix C).

The aggregate size of liquidation impinges upon the probability of a financial crisis at $t=2$, which, in turn, depends on the liquidity threshold Θ_s^* :

$$\begin{aligned} \phi(s) &= \Pr(\{\text{crisis occurs in a given state } s\}) \\ &= \Phi\left(\frac{\Theta_s^* - \mu(s)}{\sigma}\right) \end{aligned} \quad (9)$$

The higher the threshold, the higher the probability of a financial crisis, where the threshold is the amount of liquidity necessary to match aggregate liquidation determined at $t=1$.

Given the solution to the maximization problem at $t=1$, at $t=0$ investors choose their portfolio allocation, $a_{i,s}$, given their risk aversion and the expected probability of a financial crisis, which is the same for every investor, maximizing the problem as shown in (2). The more risk averse investors are, the larger the share of assets with relatively higher payoffs in the case of a negative shock to the economy (low s) in their portfolio. This entails that, at $t=1$, and in the case of a negative shock, the market value of the share of total wealth held by risk-averse investors increases (see Appendix D).

The model's transmission mechanism can be formulated formally as follows:

$$\phi(s) = \Pr\{\delta \leq \Theta_s^* | s\} = CDF\left(\frac{\Theta_s^* - \mu(s)}{\sigma}\right) \quad (10)$$

which indicates the probability of a financial crisis.

Proposition 3: This probability depends (inversely) on the realization of the state of the economy, through the expected liquidity of the financial market, $\mu(s)$, and the liquidity threshold Θ_s^* (see Appendix E):

$$\begin{aligned} \frac{\partial \Phi(s)}{\partial s} = & \left[\frac{\partial \Phi'(s)}{\partial \mu(s)} \cdot \frac{\partial \mu(s)}{\partial s} + \frac{\partial \Phi'(s)}{\partial \Theta_s^*} \cdot \frac{\partial \Theta_s^*}{\partial \mu(s)} \cdot \frac{\partial \mu(s)}{\partial s} \right] \\ & + \frac{\partial \Phi'(s)}{\partial \Theta_s^*} \cdot \frac{\partial \Theta_s^*}{\partial e_i(s)} \cdot \frac{\partial e_i(s)}{\partial s} < 0 \end{aligned} \quad (11)$$

This model captures the impact of the realization of a shock at $t=1$ on the expected liquidity of the financial market, $\mu(s)$, at $t=2$, and, hence on the likelihood of a financial crisis (*basic channel*). A deterioration in the state of the economy, s , *ceteris paribus*, would decrease the expected liquidity of the financial market, and, hence, increase the likelihood of a financial crisis. This model also captures two additional channels through which a shock affects the likelihood of financial crisis (Figure 15.2):

- The *information channel* captures investors' decision to liquidate on the basis, *ceteris paribus*, of their individual expectations of a financial crisis, δ_i , which depends on the realization of a shock at $t=1$, coupled with their expectations about liquidation decisions by others. A negative shock would, as in the basic channel, decrease the expected liquidity of the financial market and increase investors' expectations of a financial crisis. This would directly increase investors' liquidations. In addition, the expectation of larger liquidations by others would further contribute to increasing each investor's liquidations, and, accordingly, the aggregate size of liquidation. The latter, which corresponds to a larger Θ_s^* , would, in turn, imply a higher probability of a financial crisis.
- The *portfolio channel* captures the compounded impact of risk aversion on investors' decisions to liquidate. As shown above, when the economy is hit by a negative shock, the share of total wealth held by risk-averse investors increases. Moreover, as shown in (8), the more risk averse investors are, the larger the share of their portfolio holdings that they would liquidate for any given expectation of a financial crisis. This means that, in the case of a negative shock, the higher the risk aversion, the larger the aggregate size of liquidation, and, hence the higher the probability of a financial crisis. However, the portfolio channel would not yield a symmetric result in the case of a positive shock because the aggregate size of liquidation would decrease relatively less, as the share of total wealth held by risk-averse investors would be relatively smaller.
- The *leverage channel* captures the multiplicative impact of leveraging on the aggregate size of liquidation. Higher leverage would imply a larger impact on the aggregate size of liquidation because it increases (i) the size

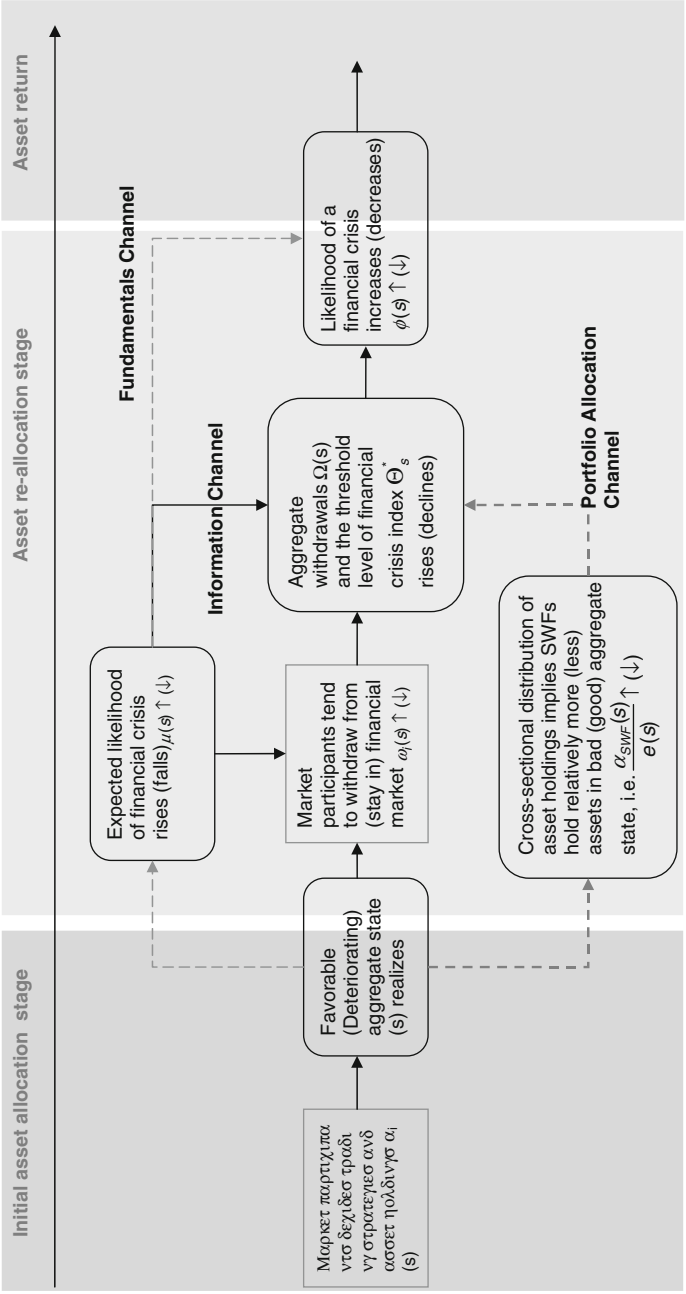


Figure 15.2 Transmission mechanism on the likelihood of a financial crisis

of investors' portfolios (when compared with the size of the portfolios in the absence of leverage), and (ii) the share of the portfolio that needs to be liquidated. The larger the aggregate size of liquidation, the higher the probability of a financial crisis.

Policy Implications

The model offers an analytical framework in which the impact of portfolio allocations by SWFs on the occurrence of a financial crisis can be analyzed. The framework explicitly accounts for several transmission channels through which such impact can be felt and the appropriate policy, including regulatory, responses discussed. Some of these issues are briefly taken up below.

Portfolio allocation and cross-sectional asset holdings

Recognizing heterogeneity when considering SWFs portfolio strategies is important. SWFs in countries at different phases of economic developments may have different investment strategies. For instance, SWFs in oil-exporting countries may invest in asset classes that are very different from countries of SWFs that demand natural resources. A uniform policy for all SWFs on investment strategies may not be applicable. Policy discussion generally underplays the portfolio allocation strategies of SWFs. Although most discussions rightly recognize their increasing size and diversification motives, the variation of cross-sectional asset holdings in the financial market and correlation of fundamentals with domestic shocks are generally not explicitly considered.²²

Asymmetric role of SWFs in financial markets

The transmission mechanism notes an asymmetric role of SWFs during financial distress and boom. Asset re-allocation by SWFs would raise concerns at times of potential financial distress, although not necessarily during periods of financial boom. Specifically, the size of SWFs becomes relatively important, and sometimes a determining role, in a potential financial distress, given a deterioration of fundamentals. Empirical works on financial stability may need to incorporate this asymmetry.

Disclosure and governance of SWFs

The appropriate disclosure of the investment strategies and governance of SWFs would mitigate information asymmetries, weaken the impact from the information channel, and, hence reduce the likelihood that financial distress escalates into a financial crisis.

Promoting financial stability

From the viewpoint of the global financial system, SWFs can facilitate a more efficient allocation of resources generated by high commodity prices,

and favorable balance of payments developments more generally, across countries and enhance market depth and liquidity, including at times of financial stress. SWFs can play a stabilizing role in global financial markets, particularly because most of them are long-term investors with mainly unleveraged positions.

Conclusions

This article discusses the recent developments of the SWFs, and illustrates, using a simple model, their role in contributing to financial stability at times of potential financial distress. The model identifies additional transmission mechanisms through which SWFs could impact financial markets, particularly in light of their increasing size, limited disclosure practices and portfolio allocation objectives. The model suggests that SWFs' investment strategies play an asymmetric role during financial distress and boom. By stressing the additional transmission mechanism, this article provides insights to policy guidelines in relation to SWFs' governance, and highlights key areas for empirical works.

References and notes

1. A special investment fund created or owned by a government to hold assets for long-term purposes; it is typically funded from reserves or other foreign currency sources, including commodity export revenues, and predominantly owns, or has significant ownership of, foreign currency claims on non-residents.
2. Jen³ and Farrell and Lund⁴ project that assets under management could exceed official reserves by 2011.
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$$\theta|y \sim N\left(\frac{\gamma\sigma^2 + \mu(s)\sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2}, \frac{\sigma^2\sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2}\right)$$

and the CDF has the properties that it is increasing in Θ , but decreasing in γ and $\mu(s)$.

20. The inference from the private signals does not depend on risk aversion or asset holdings. As a result, investors receiving same signals will behave in the same way, as shown by Morris and Shin.¹⁶⁻¹⁷
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Appendix A

The size of liquidation, $\omega(s, \delta_i, \gamma_i, L_i)$ (abbreviated as $\omega(s, \delta_i)$), will be a fraction of the asset holding, $\omega(s, \delta_i) = \lambda(s, \delta_i)a_i(s)$, where $\lambda(s, \delta_i)$ is a fraction between 0 and 1. There exist threshold levels in relation to the withdrawal rule with the following properties.

- (i) $0 < \underline{\delta} < \bar{\delta} < 1$;
- (ii) $\lambda(s, \delta_i) = 1$ iff $\delta_i \in [0, \bar{\delta}]$
- (iii) $\lambda(s, \delta_i) = 0$ iff $\delta_i \in [\bar{\delta}, 1]$
- (iv) $\lambda(s, \delta_i)$ is decreasing in δ_i and increasing in γ_i in the range $\delta_i \in [\underline{\delta}, \bar{\delta}]$
- (v) $\omega(s, \delta_i)$ is increasing in L_i .

Proof:

Taking the first derivatives with respect to the size of liquidation $\omega(s, \delta_i)$,

$$\begin{aligned} \frac{\partial U(.)}{\partial \omega} &= -\delta_i [R\alpha(s) - (R - D)\omega(s, \delta_i)]^{-\gamma_i} \\ &\quad \times (R - R_d) + (1 - \delta) [R_c \alpha(s) + (D - R_c)\omega(s, \delta_i)]^{-\gamma} (D - R_c) \end{aligned}$$

(1) If $\omega(s, \delta_i) = \alpha(s)$, it implies $\lambda(s, \delta_i) = 1$, such that the first derivate becomes

$$\begin{aligned} &-\delta_i [D\omega(s, \delta_i)]^{-\gamma_i} (R - D) + (1 - \delta_i) [D\omega(s, \delta_i)]^{-\gamma_i} (D - R_c) > 0 \\ \delta_i &< \bar{\delta} \equiv \frac{(D - R_c)}{(R - R_c)} \end{aligned}$$

(2) If $\omega(s, \delta_i) = 0$, it implies $\lambda(s, \delta_i) = 0$, such that the first derivate becomes

$$\begin{aligned} &-\delta_i'' [R\alpha_i(s)]^{-\gamma_i} (R - D) + (1 - \delta_i) [R_c \alpha_i(s)]^{-\gamma_i} (D - R_c) < 0 \\ \delta_i &> \bar{\delta} \equiv \frac{(D - R_c)}{(R - D)(R_c / R)^{\gamma_i} + (D - R_c)} \end{aligned}$$

(3) In case where $0 < \omega(s, \delta_i) < \alpha_i(s)$,

$$\begin{aligned} \frac{\partial U(.)}{\partial \omega} &= -\delta_i [R\alpha_i(s) - (R - D)\omega(s, \delta_i)]^{-\gamma_i} \\ &\quad \times (R - D) + (1 - \delta_i) [R_c \alpha_i(s) + (D - R_c)\omega(s, \delta_i)]^{-\gamma_i} (D - R_c) = 0 \\ \delta_i [R - (R - D)\lambda(s, \delta_i)]^{-\gamma_i} (R - D) - (1 - \delta_i) [R_c + (D - R_c)\lambda(s, \delta_i)]^{-\gamma_i} \times (D - R_c) &= 0 \\ \lambda(s, \delta_i) &= \frac{R - R_c A^{1/\gamma_i}}{R - D + (D - R_c) A^{1/\gamma_i}} \quad \text{if } \bar{\delta} < \delta_i < \bar{\delta} \end{aligned}$$

$$\text{where } A \equiv \left[\frac{\delta_i}{1 - \delta_i} \cdot \frac{R - D}{D - R_c} \right]$$

$\lambda(s, \delta_i)$ is decreasing with belief δ_i but increasing in γ_i .

$$\begin{aligned} \frac{\partial \lambda(s, \delta_i)}{\partial \delta_i} &= \frac{(R - R_c)A}{(1 - \delta_i)\delta_i [(R - D) + (D - R_c)A]^2 \gamma_i} < 0 \\ \frac{\partial \lambda(s, \delta_i)}{\partial \gamma_i} &= \frac{(R - R_c)A \ln(A^{\gamma_i})}{[(R - D) + (D - R_c)A]^2 \gamma_i^2} > 0 \end{aligned}$$

(4) Higher leverage L_i for a given endowment level would imply higher asset holding $\alpha_i(s)$ at $t=1$. As the size of liquidation is proportional to asset holding, $\omega_i(s, \delta_i)$ is increasing in L_i .

Appendix B

The investor's expectation that a financial crisis will *not* occur, δ_i , follows the conditional CDF $\delta_i = 1 - G(\Theta_s^* | y_i, s)$, where $G(\Theta_s^* | y_i, s) \equiv \Pr\{\Theta_s < \Theta_s^* | y_i, s\}$. The conditional CDF $G(\cdot)$ measures the probability that liquidity of the financial markets below a threshold value Θ_s^* , such that a crisis occur if and only if $\Theta_s < \Theta_s^*$.²³

Proof:

Define y_i^* to be the solution that satisfies the condition $\delta_i = 1 - G(\Theta_s^* | y_i^*, s)$. The set of signals that is equal to or lower than y_i^* is expressed as

$$F(\delta | s, \Theta, \Theta_s^*) = \int^{y_i^*(\delta_i, \Theta_s^*)} f(y) dy$$

Using Bayesian updating, the realization of δ_i given private signal y_i is normally distributed such that for a given y_i , the belief associated will follow

$$\begin{aligned} \delta_i &= 1 - G(\Theta_s^* | y_i, s) \\ &= 1 - \Phi\left(\left(\frac{\sigma^2 \sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2}\right)^{1/2} \left(\Theta_s - \frac{y_i \sigma^2 + \mu(s) \sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2}\right)\right) \end{aligned}$$

Noting that there is a one-to-one mapping between private signal y_i and the belief δ_i , the signal (y_i^*) that is consistent with the threshold Θ_s^* and belief δ_i can be found by inverting the above expression such that

$$y_i^*(\delta_i, \Theta_s^*) = \frac{(\sigma^2 + \sigma_\varepsilon^2)}{\sigma^2} \left(\Theta_s^* - \Phi^{-1}(1 - \delta_i) \times \left(\frac{\sigma^2 \sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2} \right)^{-1/2} \right) - \mu(s) \frac{\sigma_\varepsilon^2}{\sigma^2}$$

Substituting into the set of signals that is equal to or lower than y_i^* accumulated signals, and rearranging

$$\begin{aligned} F(\delta_i | s, \Theta, \Theta_s^*) &= \Phi\left(\left(\frac{\sigma^2 + \sigma_\varepsilon^2}{\sigma^2}\right) \left(\Theta_s^* - \Phi^{-1}(1 - \delta_i) \left(\frac{\sigma^2 \sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2}\right)^{-1/2}\right) - \mu(s) \frac{\sigma_\varepsilon^2}{\sigma^2} - \Theta_s \right) / \sigma_\varepsilon \\ &= \Phi\left(\frac{\Theta_s^* - \Theta_s}{\sigma_\varepsilon} + \frac{\sigma_\varepsilon^2}{\sigma^2} (\Theta_s^* - \mu(s)) - \frac{\sigma^2}{(\sigma^2 + \sigma_\varepsilon^2)^{1/2}} \Phi^{-1}(1 - \delta_i)\right) \end{aligned}$$

Taking the derivative with respect to the parameters and noting that $\Phi'(\cdot)$ is a density function, the CDF is increasing in δ and Θ_s^* but decreasing in Θ_s . The distribution of beliefs across investors is therefore an equilibrium object. The inference from the private signals on the probability of not having a crisis (δ_i) does not depend on risk aversion or asset holdings.

Appendix C

The threshold equilibrium is denoted by Θ_s^* . If realization Θ falls below the threshold, there exists a financial crisis, and vice versa. The threshold satisfies at each state s

$$\begin{aligned} \Omega(s, \Theta_s^*) &\equiv \sum_i \omega(s, \delta_i) \\ &\equiv \sum_i \alpha(s_i) \int \lambda(s, \delta_i) dF(\delta | s, \Theta, \Theta_s^*) \end{aligned} \tag{C1}$$

The withdrawal has the following properties:

$$\frac{\partial \Omega(s, \Theta_s^*)}{\partial \mu} < 0; \quad \frac{\partial \Omega(s, \Theta_s^*)}{\partial \gamma} > 0; \quad \frac{\partial \Omega(s, \Theta_s^*)}{\partial \Theta} < 1$$

Proof:

This largely relies on the proof from the previous two results, after expressing the aggregate liquidation $\Omega(s, \Theta_s^*)$ as (C1). Taking the derivative with respect to $\mu(s)$ and γ and using the chain rule will contain the terms of the derivatives of $\lambda(s, \delta_i)$ and CDF $F(\cdot)$ with respect to the parameters. Note that the results from the previous propositions, both $\lambda(s, \delta_i)$ and CDF $F(\cdot)$, are decreasing in $\mu(s)$ but increasing in γ_i . The derivative with respect to Θ could be decomposed similarly. It is noted that the withdrawal decision $\lambda_i(s, \delta)$ given Θ_s^* is independent of Θ , and the derivative of the CDF $F(\cdot)$ with respect to Θ is positive but smaller than unity.

Appendix D

An additional component of the optimization problem introduces the difference in cross-sectional asset holdings such that if the bad state of an economy were realized, the asset holding for risk-averse investors would become relatively larger in share.

Proof:

The optimization problem is expressed as

$$L(s, \delta_i, \gamma_i) = \max E \left\{ \frac{1}{1 - \gamma_i} \left[\delta_i [D a_i(s) - (R - D) \omega(s, \delta_i)^{1 - \gamma_i} + (1 - \delta_i) [R_c \cdot a_i(s) + (D - R_c) \omega(s, \delta_i)^{1 - \gamma_i}]] \right] \middle| \Theta_s^* \right\}$$

subj

$$\begin{aligned} \sum_i \alpha_i(s, \delta_{ii}) &= \sum_i e_i(s) L_i = e(s) \\ \sum_s q(s) \alpha_i(s) &= e_i(s) L_i \end{aligned}$$

From Appendix A, the liquidation decision $\omega(s, \delta_i)$ is a linear function of the asset holdings $a_i(s)$ given the feature of the CRRA utility function. Therefore, the optimization problem is rewritten as follows:

$$L(s, \delta_i, \gamma_i) = \max_{a_i(s)} \sum_z p(s) M(s, \delta_i; \gamma_i) \cdot \frac{a_i(s)^{1 - \gamma_i}}{1 - \gamma_i}$$

where

$$M(s, \delta_i, \gamma_i) = E_\delta \left\{ \frac{1}{1 - \gamma_i} \left[\delta_i [D - (R - D) \lambda(s, \delta_i)^{1 - \gamma_i} + (1 - \delta_i) [R_c + (D - R_c) \lambda(s, \delta_i)^{1 - \gamma_i}]] \right] \middle| \Theta_s^* \right\}$$

subj

$$\sum_i \alpha_i(s, \delta_{ii}) = \sum_i e_i(s) L_i = e(s)$$

$$\sum_s q(s) \alpha_i(s) = e_i(s) L_i$$

where $p(s)$ is the probability of state s . Taking the derivatives and the first-order conditions become

$$p(s) \cdot M(s, \delta_i, \gamma_i) \cdot \alpha_i(s)^{1-\gamma_i} = \eta q(s)$$

$$\sum_i \left[\frac{M(s, \delta_i; \gamma_i)}{M(s, \delta_i; \gamma_i)} \alpha_i(s; \gamma_i)^{\gamma_i/\gamma_i} \right] - e(s) = 0$$

The cross-sectional asset holdings vary across aggregate states. Investors with higher risk aversion will carry asset holdings predominantly into good states. In states with low aggregate endowment, the asset holdings are predominantly held by the risk-averse participants, that is, SWFs.

Proof:

The effect of aggregate state (s) on the relative holding in risky assets ($a_i(s)/e_i(s)L_i$) can be obtained by taking the first-order derivative and applying the implicit function theorem as follows:

$$\frac{\partial[\alpha_i(s)/e_i(s)L_i]}{\partial s} = \frac{\partial[\alpha_i(s)/e_i(s)L_i]}{\partial e_i(s)} \cdot \frac{\partial e_i(s)}{\partial s} + \frac{\partial[\alpha_i(s)/e_i(s)L_i]}{\partial \mu(s)} \cdot \frac{\partial \mu(s)}{\partial s}$$

$$+ \frac{\partial[\alpha_i(s)/e_i(s)L_i]}{\partial \Theta_s^*} \cdot \frac{\partial \Theta_s^*}{\partial s}$$

The first term shows how endowments affect the risk-sharing rule as in the general case, taking the leverage into consideration. The effect on asset allocation arising from the liquidation stage comes from the second and third terms. Using the implicit theorem,

$$\frac{\partial[a_i(s)/e_i(s)L_i]}{\partial s} = \frac{a_i(s)}{e_i(s)L_i} \left[\frac{1}{\partial e_i(s)} \left(\frac{\bar{\gamma} - \gamma}{\gamma} \right) \frac{\partial e_i(s)}{\partial s} + \frac{\bar{\gamma}}{\gamma} V(s; \gamma_i) \cdot \left(-\frac{\partial \mu(s)}{\partial s} + \frac{\partial \Theta_s^*}{\partial s} \right) \right]$$

where

$$\bar{\gamma} \equiv \left[\sum_i \frac{\alpha_i(s; \gamma_i)}{e_i(s)L_i} \cdot \frac{1}{\gamma_i} \right]^{-1}$$

$$V(s; \gamma_i) \equiv \sum_i \frac{\alpha_i(s; \gamma_i)}{e_i(s)L_i} \cdot \frac{1}{\gamma_i} \cdot \frac{\partial[\ln M(s; \delta_i; \gamma_i) - \ln V(s; \gamma_i)]}{\partial \Theta_s^*}$$

The first term in the bracket is negative for investors with a risk aversion higher than the average. Similarly, $V(s, \gamma_i)$ is also negative for investors with higher degrees

of risk aversion. Therefore, the general effect of aggregate state is reinforced by the additional component.

Appendix E

The transmission mechanism of how bad fundamentals could turn into a financial crisis consists of three effects. Taking the derivatives with respect to the state s on the probability of crisis:

Proof:

$$\begin{aligned}\frac{\partial \phi(s)}{\partial s} &= \frac{\partial \Phi'(s)}{\partial \mu s} \cdot \frac{\partial \mu(s)}{\partial s} + \frac{\partial \Phi'(s)}{\partial \Theta_s^*} \cdot \frac{\partial \Theta_s^*}{\partial s} \\ &= -\frac{1}{\sigma} \phi(\cdot) \frac{\partial \mu(s)}{\partial s} + \frac{1}{\sigma} \phi(\cdot) \frac{\partial \Theta_s^*}{\partial s}\end{aligned}$$

Following the results in Appendix C and D, the last term could be expressed as

$$\begin{aligned}\frac{d\Theta_s^*}{ds} &= \left[\sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{e_i(s)} \cdot \frac{de_i(s)}{ds} \Omega(\mu(s), \Theta_s^*) \right] / \\ &\quad \left[1 + \sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{\mu(s)} \Omega(\mu(s), \gamma_i, \Theta_s^*) \right] \\ &\quad + \left[\sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{\mu(s)} \Omega(\mu(s), \gamma_i, \Theta_s^*) + \sum_i \frac{\alpha_i(s; \gamma_i)}{e_i(s)} \cdot \frac{d\Omega(\mu(s), \gamma_i, \Theta_s^*)}{d\mu(s)} \right] \frac{d\mu(s)}{ds}\end{aligned}$$

Collecting terms to arrive at the three channels in the transmission mechanism,

$$\begin{aligned}\frac{\partial \phi(s)}{\partial s} &= \left[\frac{\partial \Phi'(s)}{\partial \mu(s)} \cdot \frac{\partial \mu(s)}{\partial s} + \frac{\partial \Phi'(s)}{\partial \Theta_s^*} \cdot \frac{\partial \Theta_s^*}{\partial \mu(s)} \cdot \frac{\partial \mu(s)}{\partial s} \right] \\ &\quad + \frac{\partial \Phi'(s)}{\partial \Theta_s^*} \cdot \frac{\partial \Theta_s^*}{\partial e(s)} \cdot \frac{\partial e(s)}{\partial s} < 0\end{aligned}$$

From above equation,

$$\begin{aligned}\frac{\partial \phi(s)}{\partial s} &= -\frac{1}{\sigma} \phi(\cdot) \frac{\partial \mu(s)}{\partial s} + \frac{1}{\sigma} \phi(\cdot) \left[\left[\sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{e_i(s)} \cdot \frac{de_i(s)}{ds} \Omega(\mu(s), \gamma_i, \Theta_s^*) \right] / \left[1 + \sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{\mu(s)} \Omega(\mu(s), \gamma_i, \Theta_s^*) \right] + \right. \\ &\quad \left. \left[\sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{\mu(s)} \Omega(\mu(s), \gamma_i, \Theta_s^*) + \sum_i \frac{\alpha_i(s; \gamma_i)}{e_i(s)} \cdot \frac{d\Omega(\mu(s), \gamma_i, \Theta_s^*)}{d\mu(s)} \right] \frac{d\mu(s)}{ds} \right] \\ &= \underbrace{-\frac{1}{\sigma} \phi(\cdot) \frac{\partial \mu(s)}{\partial s}}_{\text{fundamental channel}} + \underbrace{\frac{1}{\sigma} \phi(\cdot) \left[\sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{\mu(s)} \Omega(\mu(s), \gamma_i, \Theta_s^*) + \sum_i \frac{\alpha_i(s; \gamma_i)}{e_i(s)} \cdot \frac{d\Omega(\mu(s), \gamma_i, \Theta_s^*)}{d\mu(s)} \right] \frac{\partial \mu(s)}{\partial s}}_{\text{information channel}} \\ &\quad + \underbrace{\frac{1}{\sigma} \phi(\cdot) \left[\sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{e_i(s)} \Omega(\mu(s), \gamma_i, \Theta_s^*) \right] / \left[1 + \sum_i \frac{d\alpha_i(s; \gamma_i) / e_i(s) L_i}{\mu(s)} \Omega(\mu(s), \gamma_i, \Theta_s^*) \right] \frac{de_i(s)}{ds}}_{\text{risk-sharing channel}}\end{aligned}$$

The transmission mechanism is therefore decomposed into three channels: the fundamental channel (first term), the information channel arising from the interactions

between investors (second term) and the risk-sharing channel (third term). Following the previous Appendices A–D, the first term is negative. For investors with a risk-aversion parameter higher than the weighted average across individuals, it increases their weight in cross-sectional asset holding when bad fundamentals occur. Their withdrawal is higher and it increases the likelihood of the financial crisis. The aggregate effect across individual investors is negative given the weighted averages and the predominant effect arising from high risk-aversion investors.

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16

Modeling Autocallable Structured Products

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Since first introduced in 2003, the number of autocallable structured products in the United States has increased exponentially. The autocall feature causes the product to be redeemed if the reference asset's value rises above a pre-specified call price. Because an autocallable structured product matures immediately if it is called, the autocall feature reduces the product's duration and expected maturity. In this article, we present a flexible Partial Differential Equation framework to model autocallable structured products. Our framework allows for products with either

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discrete or continuous call dates. We value the autocallable structured products with discrete call dates using the finite difference method, and the products with continuous call dates using a closed-form solution. In addition, we estimate the probabilities of an autocallable structured product being called on each call date. We demonstrate our models by valuing a popular autocallable product and quantify the cost to the investor of adding this feature to a structured product.

Introduction

Autocallable structured products^{1,2} have become increasingly common in recent years. The first autocallable structured product on record in the United States was issued by BNP Paribas on 15 August 2003. Figures 16.1(a) and (b) plot the number and aggregate face value of autocallable structured products issued between 2003 and 2010. As the figures indicate, the number of issues increased sharply in 2007 and has continued to grow through 2010 at a 40 per cent annual growth rate. In just the first 6 months of 2010 there were more than 2500 autocallable products issued. The aggregate face value of newly issued autocallable structured products follows the same pattern, with a surge in 2007 and continued growth since then.

One reason for the rapid expansion of autocallable structured products is the ease with which the call feature can be attached to existing types of structured products.³⁻⁶ The call feature causes the structured product to be redeemed if the reference asset's price reaches or exceeds a predefined level (the call price) on a call date.

In this article we describe the call feature, explain how to value it and show an example of the valuation methodology. We use this example to discuss the cost this feature can add to a structure product. We value autocallable structured products using a general Partial Differential Equation (PDE) approach. We set up the PDE using the Black-Scholes equation and add boundary conditions representing the product's features, including the autocall feature.^{7,8}

We divide the autocallable structured products into two categories: products that have discrete call dates ('discrete autocallables') and products that have continuous call dates ('continuous autocallables'). Figures 16.2(a) and (b) demonstrate graphically the difference between discrete and continuous autocallables. Both figures plot the same underlying stock price over time. The continuous autocallable structured product is called immediately upon crossing the call price C , while the discrete autocallable must wait until t_3^c before it is called. If the underlying stock price had dropped back below C on t_3^c , the discrete autocallable structured product would not have been called. Thus, holding all else equal, a continuous autocallable structured product is more likely to be called than a discrete one. Although we only consider constant call prices in the article, the methodologies are expandable to exponentially increasing call prices. Closed-form solutions are also available. The extension is analogous to valuing a barrier option with an exponentially varying barrier.^{9,10}

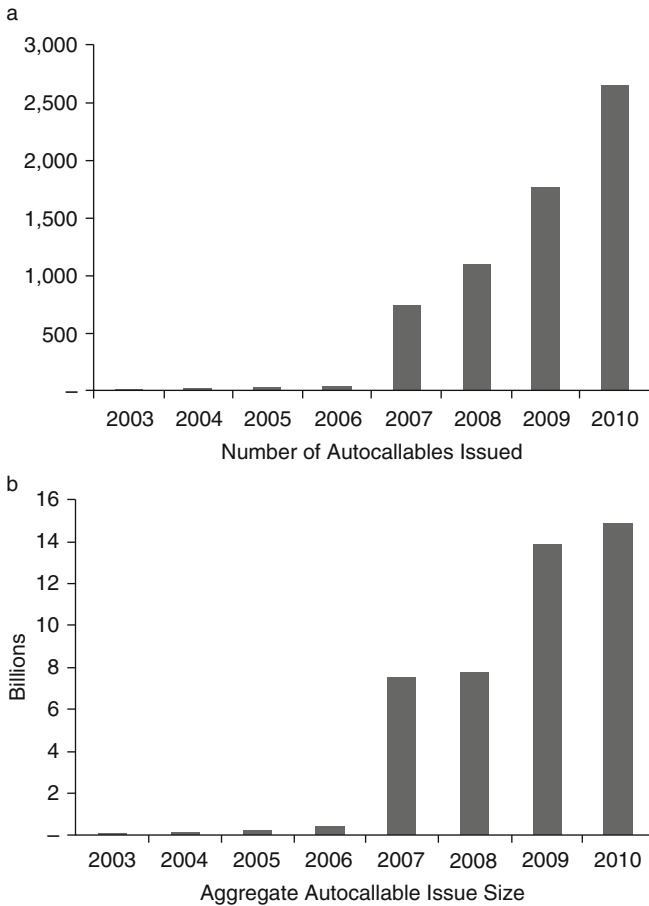


Figure 16.1 Number and total issue size of autocallable structured products, January 2003–June 2010

An autocallable structured product is fundamentally similar to a reverse-convertible¹¹ that pays a high coupon in exchange for exposing the investor to the downside risk of the reference security. Although autocallable structured products tend to be issued for longer terms than reverse-convertibles, autocallable structured products can have shorter effective durations because of the embedded call feature. See Arzac,¹² Chemmanur *et al*¹³ and Chemmanur and Simonyan¹⁴ for a discussion of why investment banks issue mandatory convertibles and why investors purchase them.

For example, a common autocallable structured product would have the following payoffs: If the reference asset's price is above the call price on one

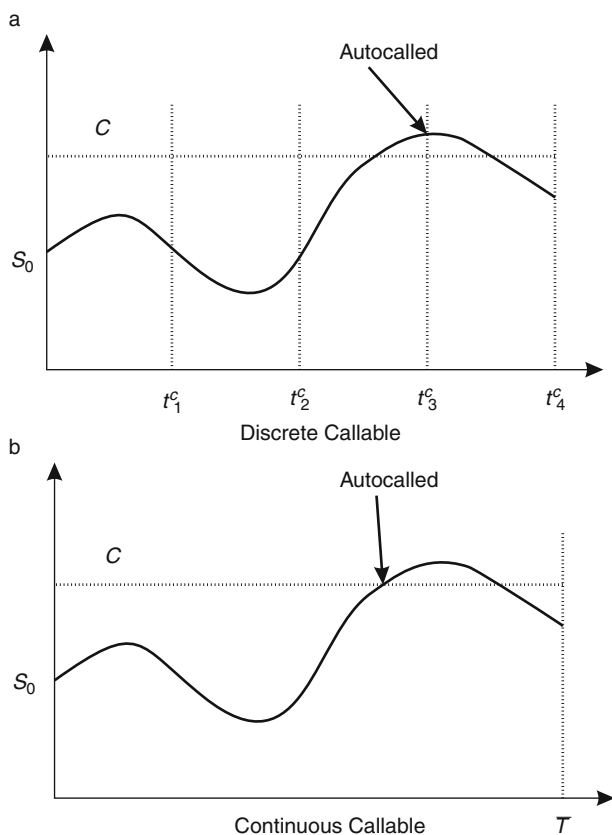


Figure 16.2 An autocall event

of the call dates, it is called, and pays a pre-specified fixed-rate return. If the reference asset's price is below the call price on every call date, the product is never called. In such a case, the investors receive the product's face value at maturity, if the final price of the reference asset is above a predetermined threshold. If the final price is below the threshold, investors receive the same negative percentage return as the reference asset.

The article proceeds as follows: In the next section, we explain our valuation framework. The first subsection discusses autocallable structured products with discrete call dates, and the second subsection presents autocallable structured products with continuous call dates. The following section implements our valuation framework for an example of structured product. We conclude in the last section. In the appendix we explain the main features of popular autocallable structured-products.

Autocallable Structured Product Valuation Models

There are three main characteristics of the call feature that will affect the value of the structured product: the timing of the call dates, the probability of being called on each call date and the determination of the payoff at maturity. In this section we set up the valuation of autocallable structured products as a PDE problem. The PDE problem is general enough to be used on both discrete and continuous autocalls.

Modeling autocallable structured products using PDE

Our valuation model follows the Black-Scholes framework with risk-neutral assumptions. The reference asset's price is a generalized Brownian motion

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t \quad (1)$$

where r is the risk-free rate, q is the dividend yield, and σ is the volatility of the price process. (Throughout this article, we assume r , q and σ are constant and continuously compounded over the product's term $[0, T]$. For simplicity, we omit the subscript t from S_t .) If we assume the price of the structured product $V(S, t)$ is a function of time $t \in [0, T]$ and the reference asset's price $S \in [0, \infty]$. The Black-Scholes formula implies that a structured product's dynamic value can be expressed as the following PDE:

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q) S \frac{\partial V}{\partial S} \\ - (r + C\bar{D}S)V = 0, \end{aligned} \quad (2)$$

where $C\bar{D}S$ is the credit default swap (CDS) spread of the issuer. (Structured products are unsecured debt securities, and hence lose value if the issuer defaults. It is therefore essential to include the issuer's credit risk $C\bar{D}S$ in the PDE to calculate the structured product's present value^{6,15}).

Many different structured product features can be modeled as variations on equation (2). For example, when the structured product is not called, the payoff at maturity $f(S_T)$ is typically a function of the value of the reference asset at maturity:

$$V(S_T, T) = f(S_T).$$

For simplicity and without loss of generality, we assume the initial principal of a structured product is equal to the reference asset's initial value S_0 . Embedded call and put options and the autocall feature can all be modeled as boundary conditions.⁶ The autocall features boundary condition is

$$V(C, t) = P_t \text{ for } t \in T_c \quad (3)$$

where C is the time-independent call price, P_t is the final payoff if the note is called, and T_c is a set of discrete or continuous call dates. Once the autocall is triggered, the structured product matures immediately and the final payout is P_t . Called structured products typically pay out a fixed rate of return. Therefore, the payoff follows

$$P_t = He^{Bt},$$

where B is the rate of return, and H is a constant.

Valuing autocallable structured products with discrete call dates

Most discrete autocallables do not have a closed-form solution. Instead, the PDE is solved and the products are valued via numerical methods such as the finite-difference method.¹⁶

For discrete autocallable structured products, the boundary conditions of the PDE are the equations

$$V(C, t) = P_t \text{ for all } t \in T_c,$$

$$V(0, t) = f(0)e^{-(r+C\bar{D}S)(T-t)}.$$

The first condition requires that the product's value never exceeds the autocall payout on a call date. The second condition guarantees that if the reference asset's price hits 0 it will remain 0. For tractability, we define it using a general function $f(0)=0$. This boundary condition is necessary as it guarantees that the structured product cannot ever be called if the reference asset becomes worthless.

The first step in solving the PDE is to simplify the complex notation and transform the equation into a standard heat equation. Using a 'dimensionless' change of variables similar to Wilmott *et al*⁸ and Hui,¹⁷ we transform the variables $\{S, t, V(S, t)\}$ into $\{x, \tau, u(x, \tau)\}$ as follows

$$S = Ce^x, \quad t = T - \frac{2\tau}{\sigma^2},$$

$$V(S, t) = Ce^{\alpha x + \beta \tau} u(x, \tau) + f(0) e^{-(r+C\bar{D}S)(T-t)},$$

where the constants are

$$k_1 = \frac{2(r-q)}{\sigma^2}, \quad \alpha = -\frac{1}{2}(k_1 - 1),$$

$$\beta = -\alpha^2 - \frac{2(r+C\bar{D}S)}{\sigma^2}.$$

After the change of variables, the Black-Scholes equation is reduced to a heat equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \quad \text{for } -\infty < x < 0, \quad \tau > 0, \quad (4)$$

the boundary conditions become

$$\begin{aligned} u(-\infty, \tau) &= 0, \\ u(0, \tau) &= C^{-1} e^{-\beta\tau} \left(P_t - f(0) e^{-(r+C\bar{D}S)\frac{2\tau}{\sigma^2}} \right) \\ \text{for } T - \frac{2\tau}{\sigma^2} &\in T_c \end{aligned}$$

and the initial condition becomes (the change of variables converts the final condition into an initial condition)

$$\begin{aligned} u(x, 0) &= C^{-1} e^{-\alpha x} (f(Ce^x) - f(0)), \\ \text{for } -\infty < x < 0. \end{aligned}$$

To further simplify notation we denote

$$h_1(\tau) = C^{-1} e^{-\beta\tau} \left(P_t - f(0) e^{-(r+C\bar{D}S)\frac{2\tau}{\sigma^2}} \right)$$

and

$$h_2(x) = C^{-1} e^{-\alpha x} (f(Ce^x) - f(0)),$$

reducing the boundary conditions and initial condition to

$$\begin{aligned} u(-\infty, \tau) &= 0, \quad u(0, \tau) = h_1(\tau) \\ \text{for } T - \frac{2\tau}{\sigma^2} &\in T_c, \quad u(x, 0) = h_2(x). \end{aligned} \quad (5)$$

The finite difference method allows us to discretize the domain of the function $u(x, \tau)$, which is a plane $(x, \tau) \in (-\infty, 0] \times [0, T\sigma^2/2]$. We discretize the plane into an $N \times M$ grid, where the size of each grid block is $\delta x \times \delta \tau$. Because x has no lower bound, we can assign x an arbitrarily large minimum value of $-N\delta x$. The bounds on τ require that $\delta \tau$ satisfy

$$M\delta\tau = \frac{T\sigma^2}{2}.$$

Generally speaking, the accuracy of the valuation increases as δx and $\delta \tau$ get smaller. $\delta \tau$ is typically set to correspond to one trading day, such that $\delta t = 2\delta \tau / \sigma^2 = 1/250$ of a year.

There are three finite difference methods: the explicit finite difference method, the implicit finite difference method and the Crank-Nicolson method. The methods differ in how they approximate the derivatives $\partial u / \tau$ and $\partial^2 u / x^2$. In this example, we use the explicit finite difference method, which approximates the derivatives as

$$\begin{aligned}\frac{\partial u}{\tau} &\sim \frac{u_n^{m+1} - u_n^m}{\delta \tau}, \\ \frac{\partial^2 u}{x^2} &\sim \frac{u_{n+1}^m - 2u_n^m + u_{n-1}^m}{(\delta x)^2}.\end{aligned}$$

Solving equation (4) with the conditions in equation (5) is equivalent to solving

$$\begin{aligned}\frac{u_n^{m+1} - u_n^m}{\delta \tau} &= \frac{u_{n+1}^m - 2u_n^m + u_{n-1}^m}{(\delta x)^2}, \\ 0 < n < N, \quad 0 < m < M\end{aligned}$$

with the conditions

$$\begin{aligned}u_n^0 &= h_2(-n\delta x), \quad 0 \leq n \leq N, \\ u_0^m &= 0, \quad 0 < m < M, \\ u_N^m &= h_1(m\delta \tau), \quad \text{for } T - 2(m\delta \tau) / \sigma^2 \in T_C.\end{aligned}$$

The formula updating $m\delta \tau$ to $(m+1)\delta \tau$ is therefore

$$\begin{aligned}u_n^{m+1} &= u_n^m + \frac{\delta \tau}{(\delta x)^2} (u_{n+1}^m - 2u_n^m + u_{n-1}^m), \\ 0 < n < N, \quad 0 < m < M.\end{aligned}$$

The solution is derived iteratively from $m=0 \rightarrow M$, which corresponds to $t=T \rightarrow 0$. For the convergence and stability of the explicit finite difference method, we require that $\delta \tau / (\delta x) \leq 1/2$. Once all of the u_n^m , for $n=1, 2, \dots, N$ are derived, we can approximate $u(x, T\sigma^2/2)$ for every x . By reversing the change of variables, we can use $u(x, T\sigma^2/2)$ to finally solve the original function $V(S, t)$ at $t=0$.

An alternative probability approach to valuing discrete autocallables

Another way to estimate the value of a discrete autocallable structured product is by calculating the probability of the autocall being exercised on each

call date, and then use the probability at each date to value the structured product. Let $p_i, i=1, \dots, n$ be the probability of the call being exercised at time t_i^c . The probability of the call never being exercised is then $1 - \sum_{i=1}^n p_i$, where each p_i is conditional on the call not being exercised at any previous call date $(t_1^c, \dots, t_{i-1}^c)$. Recall that the conditional distribution of $S_{t_i^c} | S_{t_{i-1}^c}$ follows a lognormal distribution

$$S_{t_i^c} = S_{t_{i-1}^c} e^{(r-q-\frac{1}{2}\sigma^2)\Delta t_i^c + \sigma\Delta\sqrt{t_i^c}W_i},$$

where Δt_i^c is the time between call dates $\Delta t_i^c = t_i^c - t_{i-1}^c$ and $W_i, i=1, \dots, n$ are i.i.d. standard normal variables. To simplify notation, we use $X_i = (r-q-(1/2)\sigma^2)\Delta t_i^c + \sigma\Delta\sqrt{t_i^c}W_i$ to represent the continuously compounded return from t_{i-1}^c to t_i^c . This means the ending stock price S_T can be written as

$$S_T = S_0 e^{\sum_{i=1}^n (r-q-\frac{1}{2}\sigma^2)\Delta t_i^c + \sigma\Delta\sqrt{t_i^c}W_i} = S_0 e^{\sum_{i=1}^n X_i}.$$

Because of the price's Markov property, the X_i 's are pairwise independent. Furthermore, if Δt_i^c is a constant, the X_i 's are i.i.d. normal variables. The probability of the call being exercised at time t_i^c can now be written as

$$p_i = \text{Prob}(S_{t_j^c} < C, j=1, 2, \dots, i-1, \text{ and } S_{t_i^c} \geq C)$$

$$= \text{Prob}\left(\sum_{k=1}^i X_k < \log\left(\frac{C}{S_0}\right), j=1, 2, \dots, i-1,$$

$$\text{and } \sum_{k=1}^i X_k \geq \log\left(\frac{C}{S_0}\right)\right)$$

$$= \int \dots \int g(x_1, \dots, x_n) dx_1 dx_2 \dots dx_n,$$

$$\sum_{k=1}^j x_k < \log\left(\frac{C}{S_0}\right), j=1, 2, \dots, i-1,$$

$$\sum_{k=1}^j x_k \geq \log\left(\frac{C}{S_0}\right)$$

where $g(x_1, \dots, x_n)$ is the joint probability density function (PDF) of $X_1, \dots, X_n$. Because the X_i 's are independent, the joint PDF can be expressed as the product of each X_i 's individual PDF.

We can now estimate the product's present value as the discounted expected cash flows, where the cash flow probabilities are the p_i we just calculated.

$$\begin{aligned}
 V(S_0, 0) &= \sum_{i=1}^n e^{-(r+C\bar{D}S)t_i^c} p_i P_{t_i^c} + e^{-(r+C\bar{D}S)T} \int \dots \int f(S_T) g(x_1, \dots, x_n) dx_1 \dots dx_n \\
 &\quad \sum_{k=1}^j x_k < \log\left(\frac{C}{S_0}\right), j = 1, 2, \dots, n \\
 &= \sum_{i=1}^n e^{-(r+C\bar{D}S)t_i^c} p_i P_{t_i^c} + e^{-(r+C\bar{D}S)T} \int \dots \int f\left(S_0 e^{\sum_{i=1}^n x_i}\right) g(x_1, \dots, x_n) dx_1 \dots dx_n \\
 &\quad \sum_{k=1}^j x_k < \log\left(\frac{C}{S_0}\right), j = 1, 2, \dots, n \quad (6)
 \end{aligned}$$

If the structured product's payoff at maturity is constant $f(S_T) = P_T$, the equation can be further reduced to

$$\begin{aligned}
 V(S_0, 0) &= \sum_{i=1}^n e^{-(r+C\bar{D}S)t_i^c} p_i P_{t_i^c} \\
 &\quad + e^{-(r+C\bar{D}S)T} \left(1 - \sum_{i=1}^n p_i\right) P_T. \quad (7)
 \end{aligned}$$

Valuing autocallable structured products with continuous call dates

For a continuous autocallable structured product, the boundary conditions of the PDE are continuous equations

$$V(C, t) = P_v, \quad V(0, t) = f(0) e^{-(r+C\bar{D}S)(T-t)}.$$

We apply the change of variables and simplifications from the section 'Valuing autocallable structured products with discrete call dates', yielding the heat equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \quad \text{for } -\infty < x < 0, \quad \tau > 0, \quad (8)$$

with the boundary conditions

$$u(-\infty, \tau) = 0, \quad u(0, \tau) = h_1(\tau) \quad \text{for } \tau > 0$$

and the initial condition

$$u(x, 0) = h_2(x) \quad \text{for } -\infty < x < 0.$$

The next step is to convert the two boundary conditions so that they are both zero boundaries (homogenous boundaries). To do this we introduce the transformation $v(x, \tau) = u(x, \tau) - \gamma(x, \tau)$, where $\gamma(x, \tau) = e^x h_1(\tau)$. Using this transformation, the Black-Scholes equation becomes

$$\frac{\partial v}{\partial \tau} = \frac{\partial^2 v}{\partial x^2} + e^x (h_1(\tau) - h_1'(\tau)),$$

$$\text{for } -\infty < x < 0, \quad \tau > 0, \quad (9)$$

The new, homogenous boundary conditions are

$$v(-\infty, \tau) = 0, \quad v(0, \tau) = 0, \quad \text{for } \tau > 0$$

and the new initial condition is

$$v(x, 0) = h_2(x) - e^x h_1(0), \quad \text{for } -\infty < x < 0.$$

The transformed continuous autocall PDE problem is thus a standard inhomogeneous PDE problem with homogeneous boundary conditions. By further simplifying notation, the PDE problem can be solved using methods in Evans.¹⁸ Specifically, let $h_3(x) = h_2(x) - e^x h_1(0)$ and $h_4(x, \tau) = e^x (h_1(\tau) - h_1'(\tau))$. The PDE problem is then the following general form

$$\frac{\partial v}{\partial \tau} = \frac{\partial^2 v}{\partial x^2} + h_4(x, \tau), \quad \text{for } -\infty < x < 0, \tau > 0,$$

$$v(-\infty, \tau) = 0, \quad v(0, \tau) = 0, \quad \text{and } v(x, 0) = h_3(x).$$

The solution to this PDE is

$$v(x, \tau) = \frac{2}{2\sqrt{\pi\tau}} \int_{-\infty}^0 h_3(s) \left(e^{-(x-s)^2/4\tau} - e^{-(x+s)^2/4\tau} \right) ds$$

$$+ \int_0^\tau \int_{-\infty}^0 \frac{h_4(s, r)}{2\sqrt{\pi(\tau-r)}} \left(e^{-(x-s)^2/4(\tau-r)} - e^{-(x+s)^2/4(\tau-r)} \right) ds dr. \quad (10)$$

Once $v(x, \tau)$ is solved, we can now solve our transformation $u(x, \tau) = v(x, \tau) + e^x h_1(\tau)$. Once this function is solved we can fold back and find the value of $V(S, t)$.

Example of an Autocallable Structured Product

As an example of our valuation methodology we describe a simple autocallable structured product. (This example is similar in its features to one of the more popular brands, the ‘Autocallable Optimization Securities with Contingent Protection’. More than US\$1.4 billion in face value of these products were issued in 2009. See the Appendix for more details of the different brands of autocallable structured products and their main features.) If the reference asset has a cumulative positive return on any autocall date, the structured product is called and investors will receive a positive, pre-specified yield. If the product is not called, at maturity the payoff will be:

$$V(S, T) = f(S) = \begin{cases} I = S_0, & S > L; \\ S, & \text{otherwise,} \end{cases} \quad (11)$$

where I is the structured product’s face value, S_0 is the reference asset’s initial value, S is the reference asset’s final value and L is the threshold price.

If the structured product is not called, investors will receive a 0 per cent or a negative return. Figure 16.3 illustrates the autocallable structured product’s payoff at maturity if it is not called.

To demonstrate the application of our models, we value three stylized types of autocallable structured products. The first example, our benchmark case, does not have an autocall feature, but has a constant coupon payment. The payoff structure resembles a plain vanilla reverse convertible structured

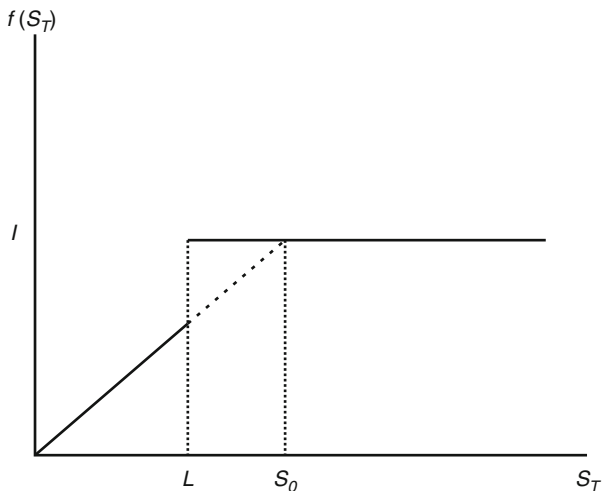


Figure 16.3 Maturity payoff if the autocallable structured product is not called

product. The second type has an autocall feature with monthly call dates, and the third type has an autocall feature with continuous call dates.

For all three examples we assume that the reference asset's initial stock price S_0 and the face value of the note I are both \$100, the call price is \$102, the risk-free rate r is 5 per cent, the volatility σ of the reference asset is 20 per cent, the dividend yield q of the reference asset is 1 per cent, the issuer's CDS spread $C\bar{D}S$ is 1 per cent, the contract length T is 1 year, and the threshold L is \$80. If the reference asset's price is over the call price on an call date (that is, $S_t \geq C=102$), the product will be called and will pay a 9.2 per cent annualized return (that is, $P_t = He^{Bt} = 100e^{0.092t}$). (This case is our benchmark case, hence we use a 9.2 per cent coupon rate that makes this example first type non-autocallable note a par value note, that is, principal = \$100.) Many autocallable products have a call price identical to the price of the stock ($C=S_0$); however, our assumption $C>S_0$ is without loss of generality. (In a continuous case, if the call price were identical to the stock price the product would likely be immediately called at issuance, defeating the point of such a call provision).

Case 1: Benchmark–Not autocallable

In this case, the valuation of autocallable structured product is relatively straightforward. Because the reference asset's final price follows a lognormal distribution

$$S_T = S_0 e^{\left(r - q - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}W},$$

the value of the structured product is the discounted expected cash flow

$$V(S_0, 0) = e^{-(r+C\bar{D}S)T} \left(\int_0^\infty f(S_T)g(S_T)dS_T + S_0TB \right).$$

where $g()$ is the PDF of S_T . We set the product's issue date value to be \$100.00 per \$100.00 face value by our choice of parameters. As many have shown (see for example Henderson and Pearson⁵) reverse convertible structured products tend to be overpriced, that is, that they are issued on average at a price that exceeds the present value of their expected future cash-flows. We use this as a benchmark example and hence set it artificially to be priced at face value.

Case 2: Autocallable at discrete call dates

Generally, autocallable structured products have discrete autocall dates. We assume that the product in this example is callable monthly.

We first implement the explicit finite difference method to calculate the product value. We set the range of x to be $[-5, 0]$ and the range of τ to be $[0, (T\sigma^2/2)]$. The resulting $n \times m$ grid has $1000 \times 500 = 500\,000$ blocks. Following equation (6), the value of the product is \$98.39 per \$100.00 face value. We also calculate the monthly probabilities of the call being exercised p_ν and show the results in Table 16.1.

Case 3: Continuously autocallable

If the call dates are continuous, we can follow the steps in the section 'Valuing autocallable structured products with continuous call dates' to get the closed-form solution.

After the first phase of change of variables, we get a homogeneous heat equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \text{ for } -\infty < x < 0, \quad \tau > 0,$$

$$u(-\infty, \tau) = 0, \quad u(0, \tau) = C^{-1}e^{-\beta\tau}P_t,$$

$$u(x, 0) = C^{-1}e^{-ax}f(Ce^x),$$

Using our notation, $h_1(\tau) = C^{-1}e^{-\beta\tau}P_t$ and $h_2(x) = C^{-1}e^{-ax}f(Ce^x)$. Applying the second phase of change of variables to make the boundary conditions equal zero. Let $\gamma(x, \tau) = e^x h_1(\tau) = C^{-1}e^{x-\beta\tau}P_t$ and a new function $v(x, \tau) = u(x, \tau) + \gamma(x, \tau)$, then the PDE changes to an inhomogeneous equation

$$\frac{\partial v}{\partial \tau} = \frac{\partial^2 v}{\partial x^2} + C^{-1}e^{x-\beta\tau}P_t \left(1 + \beta + \frac{2B}{\sigma^2}\right),$$

$$\text{for } -\infty < x < 0, \quad \tau > 0,$$

$$v(-\infty, \tau) = 0, \quad v(0, \tau) = 0,$$

$$v(x, 0) = C^{-1}e^{-ax}f(Ce^x) - \frac{H}{C}e^{x+BT}.$$

Here $h_3(x) = C^{-1}e^{-ax}f(Ce^x) - (H/C)e^{x+BT}$ and $h_4(x, \tau) = C^{-1}e^{x-\beta\tau}P_t(1 + \beta + (2B/\sigma^2))$. Applying equation (10), the solution is

$$\begin{aligned} v(x, \tau) = & \frac{S_0}{C}e^{a^2\tau - ax} \left[N\left(\frac{-x + 2a\tau}{\sqrt{2\tau}}\right) - N\left(D_1 - \frac{x}{\sqrt{2\tau}}\right) \right] \\ & - \frac{S_0}{C}e^{a^2\tau + ax} \left[N\left(\frac{x + 2a\tau}{\sqrt{2\tau}}\right) - N\left(D_1 + \frac{x}{\sqrt{2\tau}}\right) \right] \end{aligned}$$

Table 16.1 The probability of the product being called on each monthly call date, conditional on not being called at an earlier date

Month i	1	2	3	4	5	6	7	8	9	10	11	12
p_i	0.3767	0.1435	0.0781	0.0506	0.0361	0.0275	0.0218	0.0178	0.0149	0.0127	0.0110	0.0096

$$\begin{aligned}
& +e^{D_2+(1-a)x}N\left(D_3-\frac{x}{\sqrt{2\tau}}\right)-e^{D_2-(1-a)x} \\
& \times N\left(D_3+\frac{x}{\sqrt{2\tau}}\right)-\frac{H}{C}e^{BT+x\tau}N\left(\frac{-x-2\tau}{\sqrt{2\tau}}\right) \\
& +\frac{H}{C}e^{BT-x\tau}N\left(\frac{x-2\tau}{\sqrt{2\tau}}\right)+\frac{H}{C}\left(1+\beta+\frac{2B}{\sigma^2}\right) \\
& \int_0^\tau e^{D_4+x}N\left(D_5-\frac{x}{\sqrt{2(\tau-r)}}\right)-\frac{H}{C}e^{D_4-x} \\
& +N\left(D_5+\frac{x}{\sqrt{2(\tau-r)}}\right)dr,
\end{aligned}$$

where the parameters are

$$D_1 = \frac{\log(L/C) + 2a\tau}{\sqrt{2\tau}},$$

$$D_2 = \tau(\alpha - 1)^2,$$

$$D_3 = \frac{\log(L/C) + 2\tau(a - 1)}{\sqrt{2\tau}},$$

$$D_4 = BT - \left(\beta + \frac{2B}{\sigma^2} + \tau - r\right),$$

$$D_5 = -\sqrt{2(\tau - r)}.$$

The value of the of the product is $V(S, t)$ valued at $(S_0, 0)$, where the form is

$$V(S, t) = Ce^{ax+\beta\tau} (v(x, \tau) + \gamma(x, \tau)).$$

The value $V(S, 0)$ is \$99.54.

Comparing the three cases

We can now compare the values in the three different cases: \$100.00, \$98.39 and \$99.54, respectively. Investments with the autocall feature are worth less than their non-callable benchmark. The reason for it is fairly intuitive. A non-callable investment essentially guarantees a coupon payment until maturity. With an autocall feature, the coupon may be paid for a shorter period or may not be paid at all. Since both investments share the same

downside risk, adding the call feature (without adjusting the price or the coupon) lowers the value of the investment.

In this example, the continuously autocallable structured product is more valuable than the discrete autocallable structured product. Although this is not necessarily always true. Once the coupon payment of a plain vanilla reverse convertible is replaced with an autocallable feature, the investment has a higher value if it is called and the longer it takes to get called. A discrete autocallable feature is less likely to be called, but holding all else equal may be called later if it is called. Hence, it is more likely for a continuous feature to be more valuable than a discrete one but this does not have to be always the case.

Real-life example

We calculate the product value of a real 'Autocallable Optimization Securities with Contingent Protection' note issued by UBS. (The CUSIP for the product is 90267C136. See the product's pricing supplement at http://www.sec.gov/Archives/edgar/data/1114446/000139340110000136/c178916_690465-424b2.htm.) The note is linked to the stock of Bank of America. It was issued on 26 March 2010 and had a maturity of 1 year. The reference asset's price on the issue date was $S_0 = \$17.90$. The dividend yield q and implied volatility of the underlying stock σ were 0.2235 per cent and 35.21 per cent, respectively. UBS's 1-year CDS spread was 0.4531 per cent. On the issue date, the 1-year continuously compounded risk-free rate was 0.4951 per cent. The call price C equaled the initial price S_0 . If the note were called, investor would receive a return of 16.1 per cent, and if it were not called, the contingent protection level was $L = 0.7S_0$. Applying our methods, we get a product value of \$97.73 per \$100.00 invested.

Conclusion

An autocallable structured product is called by the issuers if the reference asset's price exceeds the call price on a call date. The feature has been embedded in many different types of structured products, including Absolute Return Barrier Notes and Optimization Securities with Contingent Protection.

We provide a general PDE framework to model autocallable structured products. We solve the PDE for autocallable structured products with discrete call dates, for which there is typically not a closed-form solution, by using the finite difference method. For continuous autocallables, we derive the closed-form solution. We illustrate our modeling approaches with an example. We then quantify the incremental cost of adding an autocall feature to a plain-vanilla reverse-convertible. We also show the difference between the value of an autocall feature with continuous call dates and one with discrete call dates. The Securities and Exchange Commission, as a

matter of policy, disclaims responsibility for any private publication or statement by any of its employees.

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Appendix

Descriptions of select existing autocallable structured products

Autocallable optimization securities with contingent protection

Autocallable Optimization Securities with Contingent Protection have been issued by several investment banks, including Royal Bank of Canada, UBS, JPMorgan and HSBC.

Payout if the product is called Autocallable Optimization Securities with Contingent Protection generally have monthly or quarterly call dates, with the final call date being at the product's maturity. In general, the product is called if the reference asset has a positive cumulative return on the call date. When the product is called, investors receive the product's face value plus a pre-specified annual yield.

Payout if the product is not called The payout if not called varies by issuance between yielding a 0 per cent return, or a negative return tied to the stock return of the reference asset.

Most products compare the reference asset's final price to a threshold. If the final price is at or above the threshold, investors receive the product's face value at maturity. If the final price is below the threshold, investors receive the same negative percentage return as the reference asset.

Some products compare the reference asset's *lowest* price during the product's life to a threshold. If the lowest price is at or above the threshold, investors receive the product's face value at maturity. If the lowest price is below the threshold, investors receive the same negative percentage return as the reference asset.

Autocallable absolute return barrier notes (Autocallable ARBNs)

Autocallable ARBNs are continuously callable structured products, issued by Lehman Brothers and UBS. A non-autocallable ARBN is analyzed in depth in Deng *et al.*¹⁹

Payout if the product is called An autocallable ARBN generally has continuous call dates. The product is called whenever the reference asset's price crosses either an upper or a lower barrier, advancing the return of the structured product's face value.

Payout if the product is not called The note is not called if the reference asset's price stays within the barriers. At maturity, the note pays investors the absolute return of the reference asset, which is a return bounded by the size of the barriers.

(Semi-) Annual review notes with contingent principal protection

Semi-annual and Annual Review Notes with Contingent Principal Protection have been issued by several investment banks, including JPMorgan, Credit Suisse and HSBC.

Payout if the product is called Review Notes with Contingent Principal Protection have discrete call dates. The frequency of the call dates varies, with Annual Review Notes having annual call dates and Semi-annual Review Notes having semi-annual call dates. In both cases, the final call date is generally at the product's maturity.

In general, the product is called if the reference asset has a positive cumulative return on a call date. However, some products exercise the autocall if the reference asset's cumulative return is positive *or* not too negative (for example, -10 per cent). Regardless of the autocall trigger, exercising the autocall entitles investors to receive the product's face value plus a pre-specified annual yield.

Payout if the product is not called If the product is not called, the payoff at maturity is guaranteed to be no more than the face value of the product. All of the Review Notes

we examined have one of three kinds of loss buffers, which we refer to as standard buffers, contingent buffers and fading buffers. Regardless of the buffer style, investors receive the product's face value at maturity as long as the reference asset's cumulative return is above the buffer (for example, -20 per cent). If the reference asset's return is below the buffer, investors will lose money.

Review Notes with standard buffers expose investors to any loss in the reference asset beyond the buffer. Thus, a -20 per cent standard buffer will offset a -23 per cent return on the reference asset so the investor only loses 3 per cent.

Review Notes with contingent buffers expose investors to all of the reference asset's losses if the loss is greater than the buffer. For example, if the product has a -20 per cent buffer and the reference asset has a -23 per cent return, investors will receive a -23 per cent return. However, those same investors would receive a 0 per cent return if the reference asset had a -19 per cent return.

Review Notes with fading buffers provide a standard buffer that diminishes as the reference asset's return gets worse. Figure 16.4 graphs the relationship between the protection offered by a fading buffer and the reference asset's return. The graph shows that the investor begins with a -20 per cent buffer, but the buffer becomes smaller as the reference asset's return becomes more negative. In the extreme, a 0 per cent buffer corresponding to a reference asset return of -100 per cent. Thus, an reference asset return of -23 per cent would equate to a -23.75 per cent loss for the investor.

Autocallable reverse convertible notes

Autocallable Reverse Convertible Notes (issued by Eksportfinans and HSBC) and Autocallable Reverse Exchangeable Notes (issued by JPMorgan) both make coupon payments.

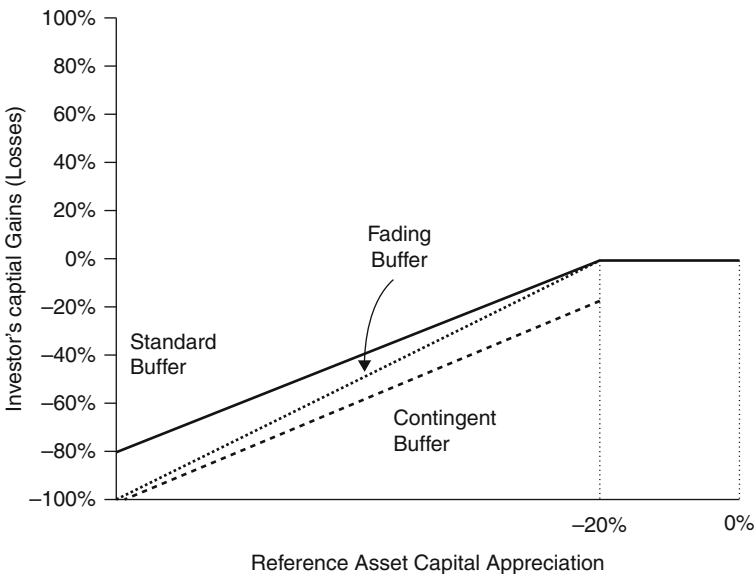


Figure 16.4 Standard buffers, contingent buffers and fading buffer

Payout if the product is called Autocallable Reverse Convertible Notes generally have a single, discrete call date early in the life of the product. If the product's reference asset has a positive cumulative return on the call date, the product is called and investors receive any accrued coupon payments and the face value of the note.

Autocallable Reverse Exchangeable Notes are similar, but tend to have multiple discrete call dates or continuous call dates after an initial non-callable period.

Payout if the product is not called If the product is not called, the payout at maturity is similar to a non-autocallable Reverse Exchangeable Note or a Reverse Convertible Note. If the reference asset's price ever crosses a barrier set below its initial price, investors receive the coupon payments plus the product's face value reduced by the lesser of a 0 per cent return or the reference asset's percentage return at maturity. If the reference asset's price never crosses the barrier, investors receive the coupon payments plus the face value of the product.

Some products compare the reference asset's final value, rather than its lowest value, to the barrier return. If the reference asset's final value is below the barrier, the investor is exposed to the reference asset's losses. Otherwise, the product returns its face value. Either way, investors receive the coupons.

Strategic accelerated redemption securities

Bank of America, Merrill Lynch and Eksportfinans have all issued Strategic Accelerated Redemption Securities.

Payout if the product is called These autocallable structured products have discrete call dates, with the final call date being at the product's maturity. The product is called if the reference asset has a non-negative return on the call date. When the product is called, investors receive a pre-specified yield on their investment.

Payout if the product is not called If the product is not called, the payout at maturity is similar to a Reverse Exchangeable Note, except that Strategic Accelerated Redemption Securities do not pay coupons. If the reference asset's ending value is not below a threshold, investors receive the face value of the product. If the ending value of the reference asset is below the threshold, investors lose a multiple of the reference asset's negative return. Although the multiple can theoretically be less than 1 or greater than 1, all of the products we saw had a multiple of 1.

Some products do not have a threshold. Instead, investors are guaranteed to lose a multiple of the reference asset's negative return if the product is not called.

Bear market strategic accelerated redemption securities

Bank of America and the Norwegian credit institution Eksportfinans have also issued Bear Market Strategic Accelerated Redemption Securities. These structured products are the same as regular Strategic Accelerated Redemption Securities, except that investors lose money if the reference asset's return is too *high* and earn a pre-specified yield if the reference asset *loses* value.

Premium mandatory callable equity-linked securities (PACERS)

PACERS, issued by Citigroup, pay coupons and have a set of discrete call periods. Each period is a set of two or three continuous call dates. If the reference asset's value on any call date is equal to or greater than its initial value, the product is called and investors receive a pre-specified yield in addition to the accrued coupons.

If the product is not called, the payout is similar to that of an Autocallable Reverse Convertible Note. Specifically, investors receive the same percentage return as the reference asset if the reference asset's ending value is below a threshold (for example, -25 per cent). Otherwise, investors receive the face value of the note. Regardless of the reference asset's value, investors receive the coupons if the product is not called.

Also similar to Autocallable Reverse Convertible Notes, some PACERS compare the reference asset's lowest value, rather than the final value, to the threshold to determine whether investors receive the face value of the PACERS or the same return as the reference asset.

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The Beta Puzzle Revisited: A Panel Study of Hedge Fund Returns

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Secretary of Campeau Commission on the improvement of the situation of financial institutions in Montreal which led to the foundation of *Institut de Finance Mathématique de Montreal*. He has published many articles and many books on financial engineering, especially in the fields of numerical methods and computational finance. Moreover, he is the founder of DESS (finance) issued by UQAM and a co-founder of the *Maîtrise en finance appliquée* at the same university. He teaches portfolio management theory and computational finance. His research focuses on modelling hedge fund returns, especially in its link with specification errors. He is an associate member of the *Chaire d'information financière et organisationnelle* located at ESG-UQAM.

Practical applications

This paper will add to the tools of the financial practitioner for estimating the conditional alphas and betas of individual hedge funds classified by strategy. Actually, the excess returns of individual funds within a strategy must be estimated in panel, our paper showing that the behaviour of individual funds sorted by strategy may differ greatly. Moreover, those returns must also be estimated by an instrumental econometric method to discard specification errors from the chosen multifactor model. We provide new instruments which are very relevant to account for the option-like dimensions of the hedge fund strategies.

Ferson and Schadt (1996) observed a beta puzzle in the mutual fund industry, which is a negative link between their beta and the market risk premium. The objective of this study is to verify if such a relation is present in the hedge fund industry. Our contribution is threefold. First, we use an instrumental variable method to estimate our conditional version of the Fama and French (F&F) model, conditional models being usually estimated by ordinary least squares. Secondly, we resort to a new set of performing instruments to estimate our conditional model: the higher moments of the risk factors that constitute the augmented version of the F&F model. Finally, we resort to a GMM panel procedure to estimate the excess returns of the HFR hedge funds pooled by strategy over the period 1997–2005. Our study reveals that there is generally no beta puzzle in the hedge fund industry. There is an apparent beta puzzle for the distressed securities and short seller strategies, but it seems rational for those strategies to increase their beta when the market risk premium is low: business opportunities are then much more important for them.

Introduction

There is yet an unsolved beta puzzle in the financial literature. While studying the returns of mutual funds, Ferson and Schadt¹ observed that these funds had a tendency to increase their conditional beta when the market risk premium decreases. According to them, the behaviour of the typical

mutual fund is perverse and that gives way to the beta puzzle. Ferson and Schadt tried to explain this puzzle by the negative correlation between changes in conditional betas and change in net new money flows. Yet, cash holdings are positively correlated with net new money flows. Thus, when cash holdings are high, beta is low, a normal relation. Here, there is an endogeneity econometric problem that must be dealt with a robust instrumental variable (IV) estimator.

In this paper, we revisit the beta puzzle for a sample of HFR hedge fund indices and a sample of HFR individual funds pooled by strategy. Our contributions are the following. First, we resort, to perform our estimations, to a GMM estimator using as instruments the higher moments of the risk factors of the augmented Fama and French (F&F) model, the basic model used in this study to do our estimation of returns. Because the choice of the conditional information is strictly an empirical matter, the estimation of conditional financial models requires resorting to IV methods. And as deplored by Watson,² the choice of instruments is too often neglected in the financial econometric literature. We will show that the higher moments of the risk factors of our conditional model are relevant instruments to do the estimation of the coefficients related to endogenous variables, which, as just said, create problems in conditional model. We will call this new form of GMM: the GMM-hm.

Secondly, we will do panel estimations of the hedge funds pooled by strategy to study the relation between the conditional beta and the risk premium. Obviously, even if we pool hedge funds by strategy, these groups remain heterogeneous due to the relative leeway of a portfolio manager. Panel estimation will allow us to take care of the fixed effects present in each group but also of the interaction between funds in each group by performing estimations using cross-section SUR.³ Really, hedge funds of a specific strategy are in interaction, be it only because they are confronted to a common environment.

This paper is organised as follows. The next section presents the innovative instruments used in this study, which are the higher moments of the risk factors, the ingredients to our GMM-hm estimations. In the subsequent section, we develop the conditional model required to study the beta puzzle observed by Ferson and Schadt.¹ In the penultimate section, we estimate these models using the HFR indices and HFR individual hedge funds grouped by strategies over the period 1997–2005 and we analyse the results. The final section concludes.

The choice of instruments^{4–7}

The basic model of this paper is the augmented version of the F&F model. We will modify this model in the next section in order to estimate a conditional form of this model that allows studying the beta puzzle.

The augmented F&F⁸⁻¹⁰ model is a purely empirical model which may be written as:

$$R_{pt} - R_{ft} = \alpha + \beta_1(R_{mt} - R_{ft}) + \beta_2SMB_t + \beta_3HML_t + \beta_4UMD_t + \varepsilon_t \quad (1)$$

with $R_{pt} - R_{ft}$ the excess return of a portfolio, R_{ft} being the risk-free return; $R_{mt} - R_{ft}$ the market risk premium; SMB a portfolio which mimics the 'small firm anomaly', which is long in the returns of selected small firms and short in the returns of selected big firms; HML a portfolio which mimics the 'income stock anomaly', which is long in returns of stocks of selected firms having a high (book value/market value) ratio (income stocks) and short in selected stocks having a low (book value/market value) ratio (growth stocks); UMD a portfolio which mimics the 'momentum anomaly', which is long in returns of selected stocks having a persistent upper trend and short in stocks having a persistent downward trend; ε_t the innovation of the equation.

To explain the return of a stock or of a portfolio of stocks, the F&F model adds to the unique factor retained by the CAPM, the market risk premium, three other factors which are assumed to represent market anomalies: the small firm anomaly, the book value to market value anomaly and the momentum anomaly.¹¹⁻¹³ Subsequently, the anomalies of the F&F model were viewed as real risk factors.

According to Capocci and Hübner,¹⁴ a positive sign for SMB in equation (1) would indicate that a portfolio manager, here a hedge fund, prefers the stocks of small firms over the stocks of larger ones, what is usually the case for hedge funds. Moreover, a positive sign for the variable HML would be symptomatic of a preference for stocks with a high book-to-market value ratio over stocks with a low book-to-market value ratio, what is also a frequent preference in the hedge fund industry. Finally, a positive sign for the variable UMD indicates that a hedge fund follows the market trend and a negative sign, that it is a 'contrarian', what is the case for some very specialised strategies like the distressed securities one. Incidentally, the factor UMD does not seem to be important in the hedge fund industry.¹⁵

We postulate that equation (1) contains specification errors. These errors might be due to multiple causes but the main plausible ones may be the omission of relevant variables, the aggregation level of the data or simply an incorrect functional form.^{16,17} In this study, we relate these errors to an aggregation bias and to the neglect of the conditioning information. Following these errors, the risk factors become endogenous and the condition of orthogonality between these factors of risk and the innovation term in equation (1) is violated: the estimators of the coefficients of equation (1) are no longer unbiased nor consistent. To purge these coefficients from these biases, we must regress in a first pass the independent variables on IVs. The estimated method used in this paper is the GMM. The problem lays in the judicious choice of these instruments.

As we said before, it is difficult to find valuable instruments for the excess returns of the mimicking portfolios. Being long in some stocks and short in others, their cash flows are actually similar to those of hedge funds. Higher moments of returns, like asymmetry and kurtosis, might have a significant influence on these returns. This suggests the use of higher moments of the variables on the RHS of equation (1) as IVs. An econometric theory is indeed in construction on this subject. Following Durbin¹⁸ and Pal,¹⁹ Dagenais and Dagenais²⁰ showed that higher moments of independent variables of a regression might be valid instruments to remove errors-in-variables, and more generally specification errors. Racicot^{21–24} transposed this method to the estimation of asset pricing models. But instead of defining higher moments as in these papers, we will adopt in this section a method more akin to asset pricing theory that defines higher moments of returns as powers of these returns.

The method of asset pricing based on higher moments is not new. Samuelson,²⁵ Rubinstein²⁶ and Kraus and Litzenberger²⁷ put the foundations of the three-moment and four-moment CAPM. The three-moment CAPM integrates asymmetry of returns in the analysis, while the four-moment CAPM adds kurtosis.

The n -moment CAPM can be written as follows^{28–30}:

$$R_i - R_f = \alpha_0 + \alpha_1(R_m - R_f) + \alpha_2(R_m - R_f)^2 + \alpha_3(R_m - R_f)^3 + \dots + \alpha_{n-1}(R_m - R_f)^{n-1} \quad (2)$$

A test on α_2 is a test on skewness preferences in asset pricing and a test on α_3 , a test on kurtosis preferences, and so on. The higher moments are consequently powers of returns in this approach. We therefore use a financial theory, the n -moment CAPM, to give an object to the method Dagenais and Dagenais²⁰ for correcting errors-in-variables. Let us return to the variable *SMB*, which we want to correct for the problem of absence of orthogonality with the innovation. In the first pass of the regression, this variable will be regressed on:

$$\widehat{SMB}_i = f(F_{it-1}, F_{it}^2, F_{it}^3, \dots, F_{it}^5, \dots) \quad (3)$$

where F_i are the variables in the RHS of the equation of F&F (equation 1) including *SMB*. They stand for the higher moments of these variables. F_{it}^2 stands for the skewness of factor F_i ; F_{it}^3 for its kurtosis, and so on. The variables appearing on the RHS of equation (3) will thus serve as IVs in our GMM estimations to revisit the beta puzzle.

A conditional version of the F&F model to study the Beta puzzle

To see if there is a beta puzzle, we would formulate a conditional version of the F&F model (equation 1). The conditional CAPM on which this version

is based is due to Ferson and Schadt,¹ Christopherson *et al.*³¹ More recently, Ferson and Qian³² revisited this model. Since hedge funds follow dynamic strategies, their alpha and beta are conditional upon public information. In this study, we are more interested in the conditional beta but we will also condition the alpha on public information.

Let us assume that a Fund has public information on a variable designated by I_t . On time t , it knows I_{t-1} . The Jensen alpha and the beta are conditional on this public information, that is:

$$\alpha_t = \alpha_0 + \varphi_0 I_{t-1} \quad (4)$$

$$\beta_t = \beta_0 + \omega_0 I_{t-1} \quad (5)$$

If we assume that I_t follows a martingale process, we may write:

$$E(I_t / \Omega) = I_{t-1} \quad (6)$$

with Ω the information set. The variable I_{t-1} is thus the best predictor of I_t .

In equations (4) and (5), I_{t-1} is written in deviation from its mean. The parameters α_0 and β_0 are the nonconditional alpha and beta. Substituting equations (4) and (5) in equation (1), we get:

$$\begin{aligned} R_{pt} - R_{ft} = & \alpha_0 + \varphi_0 I_{t-1} + \beta_0 (R_{mt} - R_{ft}) + \omega_0 (R_{mt} - R_{ft}) I_{t-1} \\ & + \beta_1 \text{SMB}_t + \beta_2 \text{HML}_t + \beta_3 \text{UMD}_t + \varepsilon_t \end{aligned} \quad (7)$$

The choice of public information I_{t-1} for estimating equation (7) is obviously an empirical matter. As suggested by Christopherson *et al.*³¹ and Ferson and Qian,³² we tried several financial and macroeconomic variables to play the role of conditioning information. Among them, we have market trend, inflation, the growth of industrial production, the dividend yield of stock indices like the S&P500, the term structure of interest rates and various indicators of credit risk like the spread between BAA and AAA corporate bond returns. We finally retained the following empirical specification for equation (4):

$$\alpha_t = \alpha_0 + \varphi_0 tb_{t-1} \quad (8)$$

with α_0 , the unconditional alpha, and tb , the American three-month Treasury bill rate.

An increase in the interest rate might be perceived as good news or bad news by hedge funds. If this increase is seen as a forthcoming deterioration of the stock market trend or as an indicator of inflation, that is bad news. But for hedge funds which follow call-like strategies, that might be good news. The sign of φ_0 is thus indeterminate in equation (8).

Let us now consider the empirical version of equation (5), that is the equation of conditional beta. Since we want to verify if the beta puzzle is present in the hedge fund industry, we will introduce the risk premium in the equation of the conditional beta. The normal relation between the conditional beta and the market risk premium should be positive. Really, if the market risk premium increases, that should induce hedge funds to bear more risk, that is to increase their beta. Nevertheless, if there is a beta puzzle, the relation between the beta and the risk premium will be negative, a perverse relation to use the terms of Ferson and Schadt.¹ We will see that there might be a beta puzzle for some hedge funds following very specialized strategies like the distressed securities one but that this puzzle may be solved easily due to the specific character of these strategies.

We thus write the empirical version of equation (5) like this:

$$\beta_t = \beta_0 + \theta_1(R_{mt-1} - R_{ft-1}) + \theta_2(R_{mt-1} - R_{ft-1})^2 \quad (9)$$

with $(R_{mt-1} - R_{ft-1})$, the market risk premium lagged on period. We also introduced in this equation of the conditional beta, the square of the market risk premium as an indicator of stock market volatility. An increase of market volatility should usually induce hedge funds to bear less risk, and therefore to decrease their beta, but it remains that an increase in market volatility might be welcomed by some hedge fund strategies whose hedging activities are particularly important. Moreover, according Treynor and Mazuy,³³ the squared risk premium might serve to detect good or bad market timing, a good market timing being associated to a positive sign for this variable. The sign of the squared market risk premium is thus theoretically indeterminate in equation (9).

The empirical counterpart of equation (7), obtained by substituting equations (8) and (9) in equation (1), is thus:

$$R_{pt} - R_{ft} = \alpha_0 + \beta_0(R_{mt} - R_{ft}) + \beta_2SMB_t + \beta_3HML_t + \beta_4UMD_t + \varphi_0tb_{t-1} + \dots + \theta_1(R_{mt} - R_{ft}) \times (R_{mt-1} - R_{ft-1}) + \theta_2(R_{mt} - R_{ft}) \times (R_{mt-1} - R_{ft-1})^2 + \varepsilon_t \quad (10)$$

where the conditioning variables tb_{t-1} , $(R_{mt-1} - R_{ft-1})$ and $(R_{mt-1} - R_{ft-1})^2$ are expressed in deviation from their mean.

Empirical results and analysis

Sample description

This study is based on two samples: the sample of the HFR indices and the sample of individual HFR funds classified by strategies. Statistical information on these samples appears in Tables 17.1 and 17.2. Our observation period of the monthly returns of these hedge funds runs from January 1997 to December 2005, for a total of 108 observations. The risk factors which

Table 17.1 Descriptive statistics of the HFR indices returns, 1997–2005^a

	Mean	Median	s.d.	MAX	MIN	Skew	Kurtosis	CAPM-beta
Distressed securities	0.95	1.00	1.82	5.34	-8.88	-1.38	10.26	0.21
Emerging	1.19	1.77	4.82	15.75	-22.07	-0.97	7.57	0.71
Equity hedge	1.10	1.18	3.07	11.54	-7.42	0.35	4.24	0.52
Equity market neutral	0.66	0.52	1.16	3.88	-3.90	0.04	5.33	0.03
Equity non hedge	1.10	1.44	4.65	12.50	-15.52	-0.27	3.95	0.85
Event driven	1.00	1.14	2.18	5.93	-9.81	-1.21	7.89	0.36
Fund of funds	0.72	0.68	1.48	5.46	-6.09	-0.24	7.41	0.20
Macro	0.76	0.80	2.04	6.24	-4.27	0.22	3.36	0.16
Market timing	1.02	0.91	2.34	6.91	-3.20	0.18	2.28	0.37
Short seller	0.33	0.23	7.13	26.43	-21.00	0.43	5.80	-1.22
Mean of indices	0.88	0.96	3.07	10.00	-10.22	-0.28	5.81	0.22
Weighted composite	0.95	0.96	2.32	7.62	-8.31	-0.27	4.91	0.41
S&P500	0.72	0.93	4.64	9.78	-14.44	-0.48	3.23	1.00

^aThe statistics appearing in this table are computed on the monthly returns of the HFR indices over the period running from January 1997 to December 2005. The weighted composite index is computed over the whole set of the HFR indices. The CAPM beta is estimated using the simple market model, that is: $R_{it} - R_{ft} = \alpha + \beta(R_{mt} - R_{ft}) + \varepsilon_{it}$, where R_i is the return of the index i , R_m is the S&P500 return, R_f is the riskless rate and ε_i is the innovation.

Table 17.2 Descriptive statistics of the returns of the individual HFR funds classified by strategies, 1997–2005^a

	Mean	Median	s.d.	MAX	MIN	Skew	Kurtosis	Number of funds
Distressed securities	0.97	0.90	1.49	37.93	-18.13	-0.02	7.27	23
Emerging	1.39	1.56	8.21	81.76	-64.59	-0.66	11.26	42
Equity hedge	1.16	0.93	5.04	60.81	-34.52	0.40	7.12	99
Equity market neutral	0.58	0.51	2.49	18.09	-24.80	-0.06	4.82	26
Equity non hedge	1.28	1.35	6.06	76.19	-61.54	0.00	5.43	47
Event driven	0.90	0.95	3.22	88.47	-54.30	-0.81	9.07	45
Fund of funds	0.82	0.74	2.19	27.45	-28.08	-0.24	8.39	101
Macro	0.85	0.65	3.81	44.27	-24.84	0.53	5.62	26
Managed futures	0.86	0.50	5.50	40.83	-27.00	0.41	4.23	70
Market timing	1.01	0.67	3.77	27.10	-14.50	0.68	4.83	8
Short seller	0.19	-0.26	8.88	47.32	-51.86	0.22	5.86	6
Mean of funds	0.91	0.77	4.61	50.02	-36.74	0.04	6.72	
Weighted composite	0.95	0.96	2.32	7.62	-8.31	-0.27	4.91	
S&P	1.29	1.14	5.28	23.30	-14.02	0.48	6.37	

^aFor each strategy, we selected, in the HFR database, all the funds whose sample begins in January 1997 or before.

appear in the F&F equation—that is the market risk premium and the three mimicking portfolios: SMB, HML and UMD—are for their part drawn from French’s website.³⁴ We used as instruments, besides the lagged endogenous variables and the higher moments of the risk factors, the Chen–Roll–Ross³⁵ factors as suggested by Watson²: the industrial production and the consumer

price index expressed in monthly and annual growth rates; the spread between long- and short-term bond returns; the spread between BBB and AAA corporate bond returns and the dividend yield of the S&P500. These factors are drawn from the databases of the *Federal Reserve Bulletin* and the Federal Reserve Bank of St-Louis.

A first glance at the data

The sample of HFR indices

Table 17.1 gives the descriptive statistics of the HFR indices over the period 1997–2005. This period was plagued by three major financial crises: (i) the Asian financial crisis (1997); (ii) the Russian/LTCM³⁶ crisis (1998); (iii) the bursting of the high-tech market bubble (2000). Our period of analysis is therefore rich in major stock market corrections. In spite of these market collapses, Table 17.1 reveals that the HFR hedge funds performed very well during this period. The mean monthly return of these indices was 0.88 per cent over this period, for an annual rate of 10.6 per cent. That rate is higher than the annual mean return of the S&P500 over the same period, which was 8.6 per cent. The low performers over this period were the short sellers and the equity market neutral indices, whereas the high performers were the emerging markets, the equity hedge and nonhedge indices.

Moreover, the standard deviation of the returns differs greatly from one index to the other. The standard deviation of the returns of the indices is generally below one of the S&P500 index except for the short seller index that incidentally has the lowest mean return over the period of analysis. As expected, the equity market neutral index has the lowest standard deviation but it still performed relatively well in spite of its low volatility.

Several researchers argue that the strategies followed by hedge funds are similar to option-based strategies. And effectively, Table 17.1 reveals that hedge fund strategies are actually similar to hedged option strategies, like the covered call and protective put ones. Similarly to the hedge fund indices, these option-based strategies have a beta which is quite low, of the order of 0.6 for at-the money options, and may yet offer all the same quite high returns which approximate those shown in Table 17.1.^{37,38} Furthermore, the nonhedged option-based strategies might give a mean return which is much higher than those ones of Table 17.1 but at the cost of a much higher beta of the order of 7. For instance, a plain vanilla call, which is at-the-money might have a beta equal to 8.

Plain vanilla puts have a negative expected return. That may explain the low mean return of the short seller index over the period of analysis. Incidentally, the CAPM beta of the short seller index, equal to -1.22 , is negative and quite high in absolute value over the period of analysis. According to the CAPM, the excess return of a portfolio having a negative beta should be low and even negative and that is the case of the short seller index. Another index that has a very low beta on Table 17.1 is the equity market

neutral index and its mean return is among the lowest in conformity with the CAPM.

Furthermore, according to Table 17.1, the composite index of hedge funds has more kurtosis than the market one, a characteristic shared by almost every strategy. A high kurtosis means that rare or extreme events are more frequent than in a normal distribution and that nonlinearities of payoffs are very present. Once more, we may relate these statistics to those associated to the cash flows of option-based strategies. They have a relatively low standard deviation but a high degree of kurtosis in comparison with the returns of the market index.

The sample of HFR individual hedge funds classified by strategies

Table 17.2 gives some descriptive statistics for individual HFR hedge funds classified by strategies over the period 1997–2005.³⁹ We notice that the mean returns of the strategies are quite comparable to those of the indices but that the standard deviations of the strategy returns are higher. That is because there is obviously less diversification for the individual funds than for the indices. Incidentally, there are five strategies over 11 that have a standard deviation of returns higher than the S&P500, which might indicate that the individual hedge funds are not so hedged than usually thought. There is more diversification on the side of the funds of funds whose standard deviation is one of the lowest among strategies reported on Table 17.2. This table also reveals that the maximum and minimum returns obtained by individual funds are much higher, in absolute value, than in the case of the indices. For instance, the minimum return over the period of analysis, that is (88.5 per cent), was observed for the event-driven strategy whose maximum return, also the highest over strategies, was 54.3 per cent, rates obviously very extreme. Similar maximum and minimum returns were observed for the emerging market strategy. The alternative investments may obviously be much riskier than the traditional ones even when viewed on the basis of a traditional measure of risk: the standard deviation. Hedging operations do not necessarily decrease this measure of risk. Really, it is often difficult to distinguish hedging operations from speculative ones.

We also note in Table 17.2 that the kurtosis of returns is generally higher for the individual funds than for the indices. Indeed, there is some evidence in the literature that diversification reduces kurtosis. But we must be cautious on that matter. Funds of funds have some degree of diversification. Nonetheless, at 8.39, the kurtosis of their returns is one of the highest of Table 17.2 over the period of analysis. Obviously, rare events seem more frequent for the individual funds than for the indices that are not tradable.

Methods of estimation and the empirical choice of instruments

In this paper, we will use two methods to estimate our models: the OLS (ordinary least squares) and the GMM (generalised method of moments).^{40,41}

The GMM method will be used to purge the estimated coefficients from their eventual specification errors. The choice of instruments was discussed in the previous section. As noted by Watson,² empirical verifications of asset pricing models which use the GMM method generally neglect the problem of the choice of instruments required by this method. Watson limits his choice to the lagged variables of the F&F model and to the Chen–Roll–Ross³⁵ factors, which are essentially financial and macroeconomic variables: credit risk spreads, term structure of interest rates, dividend yields of stock market indices, inflation rate, prices of commodities, growth of industrial production, exchange rates, and so on. As said previously, we innovate in this study by using as additional instruments the higher moments of the four risk factors of the F&F model. These moments are powers of these risk factors. We incorporate in our GMM instruments these higher moments up to the fifth order. This method is labelled the GMM-hm method.

In Table 17.3, we compare the R^2 and the adjusted R^2 of the risk factors of the F&F model, which are supposed endogenous, to two groups of instruments: on one hand, the classical instruments comprising the lagged variables of the F&F model and the Chen–Roll–Ross factors enumerated previously, and on the other hand instruments comprising, in addition to the classical instruments, the higher moments of the risk factors up to the fifth order.

As revealed by Table 17.3, the regressions of the F&F factors on the classical instruments are quite disappointing. The R^2 do not exceed 0.14 and the adjusted R^2 are very low. Moreover, regressing the risk factors on the higher moments gives quite good results. The lowest R^2 , obtained by regressing SMB on the higher moment instruments, is 0.80. Using higher moments as instruments seems to be a great improvement over using only the classical ones.

Good instruments must also have a low correlation with the innovation term of the estimated model for which they are used to remove the eventual specification errors of that model. Table 17.4 shows the regression of the innovation term of the OLS estimation of the F&F model performed

Table 17.3 R^2 of the regressions of the risk factors of the Fama and French model on classical instruments and higher moment instruments^a

	Classical instruments		Classical and higher moment instruments	
	R^2	R^2 adj.	R^2	R^2 adj.
mkt-rf	0.07	0.01	0.86	0.84
smb	0.06	0.00	0.83	0.80
hml	0.14	0.07	0.87	0.84
umd	0.05	0.00	0.86	0.82

^aThe classical instruments are the explanatory variables lagged one period and the Chen–Roll–Ross³⁵ factors. In addition to the classical instruments, the higher moment instruments include the higher moments of the regressors up to the fifth order.

Table 17.4 Adjusted R^2 obtained by performing the regressions of the innovation of the OLS estimation of the F&F model on the higher moment instruments for the period 1997–2005

HFR indices	R^2 adj.
Distressed securities	0.25
Emerging	0.18
Equity hedge	0.15
Equity market neutral	0.19
Equity nonhedge	0.25
Event-driven	0.14
Fund of funds	0.20
Macro	0.06
Market timing	0.19
Short seller	0.13
Weighted composite	0.07
Average	0.16

on the HFR indices on the higher moment instruments. This adjusted R^2 of these regressions are quite low, the average R^2 being 0.16. The maximum and minimum R^2 are respectively 0.25 for the distressed securities index and 0.06 for the macro one. Basically, these low R^2 are not a problem and the higher moments of the explanatory variables thus qualify as appropriate instruments.

The estimated conditional version of the F&F model for the HFR indices

Tables 17.5 and 17.6 give respectively the OLS and the GMM-hm estimations of equation (10) for ten HFR indices. A comparison of the two tables reveals that the coefficients of these tables are different enough to suspect specification errors in the OLS estimation of our conditional version of the F&F model. Coefficients are somewhat overstated and somewhat understated by the OLS estimation of a given strategy returns and the intensity of this problem seems in inverse relation with the R^2 of this strategy.⁴² Therefore, it seems preferable to resort to an IV method, like the GMM-hm method, to estimate the model. This procedure is unusual because the introduction of conditional variables in a financial model is generally viewed as a method *per se* to eliminate specification errors. But as a conditional model is actually an empirical model, it seems appropriate to estimate this model by an IV method in view of the differences between Tables 17.5 and 17.6.

Table 17.6 reveals that the R^2 are quite high for the majority of the indices. Only the macro strategy has a R^2 below 0.5. Two strategies, the distressed securities and the equity market neutral ones, have a R^2 comprised between 0.5 and 0.6, while the other seven strategies have a R^2 higher than 0.6. But we know that the estimation of asset pricing models is always better

Table 17.5 OLS estimation of the conditional version of the Fama and French model, HFR indices, 1997–2005^a

	c	mkt-rf	smb	hml	umd	alpha1	beta1	beta2	R ²	DW
Distressed securities	0.2850 2.06	0.2233 7.13	0.1818 5.83	0.1071 2.68	0.0015 0.07	-2.8297 3.59	-0.0049 0.90	-0.002886 3.98	0.61	1.63
Emerging	0.0729 0.21	0.7365 9.50	0.3298 4.28	0.1463 1.48	0.0611 1.17	-4.2278 2.17	0.0056 0.42	-0.0042 2.32	0.64	1.58
Equity hedge	0.5627 4.39	0.5037 17.21	0.2592 8.91	-0.0014 0.04	0.0750 3.82	2.1480 2.92	0.0083 1.65	-0.0001 0.08	0.87	1.52
Equity market neutral	0.1874 2.14	0.1157 5.77	0.0445 2.23	0.1114 4.36	0.1344 9.99	0.9350 1.86	0.0034 0.98	0.0003 0.72	0.56	1.86
Equity non hedge	0.3633 2.63	0.7768 24.63	0.4198 13.39	-0.0093 0.23	0.0184 0.87	1.0436 1.32	0.0077 1.41	-0.0004 0.48	0.94	1.88
Event driven	0.4153 3.92	0.3894 16.08	0.2281 9.47	0.1592 5.16	-0.0082 0.50	0.1408 0.23	-0.0016 0.39	-0.0018 3.20	0.83	1.80
Fund of funds	0.2399 2.62	0.2293 10.96	0.1397 6.72	0.0628 2.35	0.0564 4.01	0.3418 0.65	0.0042 1.16	-0.0014 2.91	0.72	1.59
Macro	0.0046 0.02	0.2343 5.27	0.1706 3.86	0.1688 2.98	0.0684 2.29	-1.7352 1.56	0.0211 2.76	0.0011 1.10	0.35	1.89
Market timing	0.5573 3.64	0.3855 11.01	0.0724 2.08	-0.0141 0.32	0.0636 2.71	1.4921 1.70	0.0232 3.85	0.0000 0.04	0.68	2.28
Short seller	0.6278 2.06	-1.0067 14.46	-0.4296 6.21	0.3842 4.33	-0.1422 3.04	-0.4189 0.24	-0.0170 1.42	-0.0016 0.99	0.87	2.36
Mean level of the coef.	0.3316 2.37	0.2588 12.20	0.1416 6.30	0.1115 2.39	0.0329 2.85	-0.3110 1.62	0.0050 1.49	-0.0011 1.58	0.71	1.84
Mean level of the t-stat.	0.4321 4.49	0.3952 17.98	0.1959 8.96	0.0212 0.76	0.0392 2.66	0.7677 1.39	0.0032 0.83	-0.0009 1.77	0.87	1.44

^aThe conditional version of the F&F model is given by equation (10). The variable alpha1 is the three-month T-bills rate lagged one period, expressed in deviation from its mean. The variable beta1 is the product of the market risk premium, on the one hand, and the market risk premium lagged one period and expressed in deviation from its mean, on the other hand. If the coefficient of this variable is negative, there is an apparent beta puzzle. The variable beta2 is the product of the market risk premium, on the one hand, and the squared market risk premium lagged one period and expressed in deviation from its mean, on the other hand. The last variable wants to verify the impact of market volatility on beta. The t-statistics are expressed in absolute value.

Table 17.6 GMM-hm estimation of the conditional version of the Fama and French model, HFR indices, 1997–2005^a

	c	mkt-rf	smb	hml	umd	alpha1	beta1	beta2	R ²	DW
Distressed securities	0.1110 1.10	0.2431 11.78	0.1868 8.11	0.1210 4.00	0.0157 0.95	-5.0376 6.87	-0.0018 0.62	-0.0027 6.34	0.56	1.53
Emerging	0.2838 0.99	0.6425 11.52	0.2755 6.05	0.0230 0.37	0.0477 1.26	-3.2867 1.91	0.0071 0.74	-0.0042 3.88	0.61	1.59
Equity hedge	0.6406 6.28	0.4698 28.26	0.2551 14.41	-0.0328 1.64	0.0607 6.72	3.3337 4.26	0.0083 3.36	-0.0001 0.31	0.86	1.53
Equity market neutral	0.1951 2.29	0.1184 5.71	0.0482 3.43	0.1265 5.25	0.1330 12.47	1.3365 2.23	0.0039 1.99	0.0004 2.20	0.52	1.88
Equity non hedge	0.4334 3.21	0.7655 26.57	0.4247 19.32	-0.0251 0.98	0.0087 0.43	1.8087 1.72	0.0076 2.42	-0.0004 1.42	0.93	1.89
Event driven	0.3397 5.00	0.3909 16.94	0.2250 15.33	0.1659 7.19	0.0017 0.18	-0.3633 0.78	0.0002 0.08	-0.0015 4.84	0.81	1.76
Fund of funds	0.2922 4.02	0.2122 9.87	0.1411 11.02	0.0479 2.79	0.0490 5.60	0.2558 0.40	0.0042 1.54	-0.0014 4.84	0.70	1.60
Macro	-0.1843 1.23	0.2595 8.31	0.1640 5.46	0.1860 4.01	0.0967 4.34	-4.8921 4.45	0.0211 3.63	0.0012 1.97	0.25	1.77
Market timing	0.5217 4.24	0.3638 12.35	0.0790 4.37	-0.0114 0.34	0.0583 3.19	1.0175 1.00	0.0222 5.01	-0.0002 0.56	0.66	2.27
Short seller	1.0468 5.59	-1.0059 16.66	-0.4252 8.21	0.3987 5.92	-0.1780 10.11	1.7115 0.89	-0.0173 1.66	-0.0016 1.58	0.85	2.34
Mean level of the coef.	0.3680 3.40	0.2460 14.80	0.1374 9.57	0.1000 3.25	0.0294 4.53	-0.4116 2.45	0.0056 2.11	-0.0010 2.79	0.68	1.82
Mean level of the t-stat.	0.4792 5.61	0.3925 24.27	0.1870 12.97	0.0035 0.18	0.0374 6.10	1.2839 2.16	0.0033 1.41	-0.0009 3.01	0.87	1.47

^aThe GMM-hm estimation uses as instruments the higher moments of the risk factors up to the fifth order, as explained in the second section. The conditional version of the F&F model is given by equation (10). The variable alpha1 is the three-month T-bills rate lagged one period, expressed in deviation from its mean. The variable beta1 is the product of the market risk premium, on the one hand, and the market risk premium lagged one period and expressed in deviation from its mean, on the other hand. If the coefficient of this variable is negative, there is an apparent beta puzzle. The variable beta2 is the product of the market risk premium, on the one hand, and the squared market risk premium lagged one period and expressed in deviation from its mean, on the other hand. The last variable wants to verify the impact of market volatility on beta. The *t*-statistics are expressed in absolute value.

for portfolios than for individual stocks and that may overstate the performance of these models. This is why we will repeat these estimations for the individual funds pooled by strategy in the next section.

Table 17.6 reveals that there seems to be a beta puzzle for only two strategies: the distressed securities and the short sellers one. For the short sellers strategy, the coefficient associated to the market risk premium in equation (9) is significant at the 10 per cent level, while it is not significant for the distressed one. Short sellers increase their beta when the risk premium decreases probably because the gains associated to short selling are more important during these periods because the stock market is then depressed. The hedge funds following the distressed securities strategy do the same thing because business opportunities are more important for them when the market risk premium is low. It is at this time that business failures surge. Therefore, the negative relation between the market risk premium and the beta observed for those two strategies is not really a puzzle but is associated to their style.

Always, according to Table 17.6, the relation between beta and the risk premium is positive for the other eight strategies, indicating the absence of a beta puzzle. In five cases, the beta is significant at the 5 per cent level. But it is difficult at the aggregation level of the indices to analyse dynamic strategies. Really, the portfolios constituted by the indices hide the heterogeneity of the funds composing these indices.

Table 17.6 also shows that the coefficient associated to the variable measuring market volatility in the equation of conditional beta (equation 9), that is $(R_{mt} - R_{ft})^2$, is generally negative. In the majority of hedge fund strategies, the beta is reduced when market volatility increases, a rational behaviour. Two strategies, however, increase their beta when the market volatility increases: the equity market neutral and the macro strategies. Before interpreting these results, we will wait for a confirmation of those results at the level of individual funds.

The estimated conditional version of the F&F model for the individual funds pooled by strategies

Table 17.7 gives GMM-hm estimations of the individual funds grouped by strategies. These estimations were not performed in panel but were done separately on each individual fund and the results were then averaged by strategy. Usually, the researchers resort to this method of estimation. By comparing it to Table 17.9 where estimations are done in panel, we notice that the panel results are generally much superior in terms of the R^2 and the t -statistics of the estimated coefficients. Therefore, we will only discuss the results for the panel estimations.

For the panel estimations of hedge funds pooled by strategies, we resort to cross-section SUR,⁴³ to allow for the interactions between the funds within a strategy, and to fixed effects, to account for the heterogeneity of the funds within the same strategy. This procedure generally greatly improves

Table 11.7 GMM-hm estimation of the conditional version of the Fama and French model, individual HFR funds grouped by strategies, averages of the unpooled results, 1997–2005^a

	c	mkt-rf	smb	hml	umd	alpha1	beta1	beta2	R ²	DW
Distressed securities	0.2030	0.1767	0.1181	0.1166	0.0070	-4.1868	-0.0024	-0.0021	0.24	1.70
t-stat	1.97	5.24	5.20	4.77	2.24	3.70	1.53	4.13		
s.d.	0.2281	0.0889	0.0908	0.1976	0.0362	1.9707	0.0094	0.0014		
Emerging	0.6163	0.7453	0.1939	0.0495	0.0055	-1.6185	-0.0079	-0.0070	0.28	1.73
	2.01	6.31	2.66	1.55	2.53	1.18	1.19	2.79		
Equity hedge	1.2759	0.4529	0.1887	0.2527	0.1265	6.3492	0.0304	0.0079		
	0.5589	0.5142	0.2982	0.1180	0.0774	2.5736	0.0137	-0.0002	0.45	1.84
	2.51	8.74	5.81	3.11	4.71	2.01	2.04	1.56		
Equity market neutral	0.1396	0.1819	-0.0310	0.0668	0.0890	0.7669	0.0020	-0.0001	0.26	1.83
	2.53	4.81	3.07	2.57	5.07	1.61	1.50	1.72		
Equity non hedge	0.4137	0.3567	0.1414	0.1623	0.1361	2.7306	0.0126	0.0011		
	0.3070	0.7964	0.2971	0.2369	-0.0162	-0.3823	0.0044	-0.0001	0.55	1.92
	1.64	13.24	5.24	4.55	3.94	1.76	1.85	1.53		
Event driven	0.6682	0.2838	0.2856	0.4266	0.1709	6.2911	0.0360	0.0046		
	0.2250	0.3530	0.2359	0.1631	-0.0324	-1.3193	-0.0030	-0.0017	0.39	1.75
	2.82	8.08	6.44	4.64	1.94	1.62	1.94	2.84		
Fund of funds	0.5754	0.2588	0.2490	0.1771	0.1276	3.7938	0.0113	0.0022		
	0.3726	0.2200	0.1348	0.0514	0.0523	0.6538	0.0069	-0.0013	0.38	1.74
	4.18	7.41	5.94	2.59	4.91	2.02	2.05	2.97		
Macro	0.3826	0.1553	0.0803	0.0880	0.0746	3.0009	0.0114	0.0024		
	0.2021	0.0941	0.0828	0.0840	0.0790	-0.2836	0.0288	0.0026	0.10	1.97
	2.15	2.94	1.75	1.49	2.66	1.80	2.95	3.03		
Managed futures	0.6555	0.2100	0.1236	0.1346	0.0801	6.2601	0.0209	0.0038		
	0.3818	-0.0895	0.0576	0.0691	0.1045	1.4632	0.0429	0.0029	0.06	1.97
	1.97	2.00	1.61	1.27	3.07	1.32	3.06	2.11		
Market timing	0.6165	0.2208	0.1400	0.1583	0.1165	5.3304	0.0411	0.0053		
	0.3511	0.4542	0.1829	0.0133	0.1427	1.3770	0.0371	0.0010	0.41	2.11
	3.40	9.30	3.78	2.21	4.83	2.00	4.98	1.43		
	0.6154	0.2544	0.2388	0.1298	0.0526	3.7178	0.0194	0.0008		

Short seller	0.9401	-1.2054	-0.4105	0.3995	-0.1325	4.5819	-0.0137	0.0012	0.76	1.89
	2.89	17.16	5.53	4.25	6.58	1.59	2.90	1.07		
Mean level coef.	0.9979	0.1442	0.3367	0.1772	0.2407	4.6083	0.0466	0.0018		
Mean level <i>t</i> -stat	0.3907	0.2037	0.1054	0.1244	0.0342	0.3296	0.0099	-0.0004	0.3530	1.86
	2.55	7.75	4.28	3.00	3.86	1.87	2.36	2.29		
Mean level s.d.	0.6359	0.2499	0.1921	0.2014	0.1233	4.4846	0.0238	0.0031		

^aFor each strategy, equation (10) is estimated separately for each individual fund and the results are averaged. The GMM-hm estimation uses as instruments the higher moments of the risk factors up to the fifth order, as explained in the second section. The variable α_1 is the three-month T-bills rate lagged one period, expressed in deviation from its mean. The variable β_1 is the product of the market risk premium, on the one hand, and the market risk premium lagged one period and expressed in deviation from its mean, on the other hand. If the coefficient of this variable is negative, there is an apparent beta puzzle. The variable β_2 is the product of the market risk premium, on the one hand, and the squared market risk premium lagged one period and expressed in deviation from its mean, on the other hand. The last variable wants to verify the impact of market volatility on β_1 . The *t*-statistics are expressed in absolute value.

our results over those obtained by a simple averaging of the results within a strategy, as shown by a comparison of Table 17.7, on the one hand, and Tables 17.8 and 17.9, on the other hand. Furthermore, we will only discuss here the GMM-hm estimation in panel because, as shown previously, the OLS estimation is plagued by specification errors.

Table 17.9 gives the GMM-hm estimation in panel of the individual funds returns pooled by strategy. The conditional model performs quite well for the three most important strategies of the hedge fund industry, which are the fund of funds, the equity hedge and equity nonhedge strategies.⁴⁴ The R^2 of the regressions for these strategies are respectively 0.94, 0.88 and 0.61. The estimated panel is also satisfying for the short seller strategy, whose R^2 is 0.60. Besides, the estimations are poor for four very specialised strategies whose R^2 is below 0.25: the distressed securities, equity market neutral, macro and managed futures strategies. The conditional model is particularly deceiving for the macro strategy whose R^2 is only 0.04. As said previously, the estimation of the indices, by aggregating the funds of the same strategy, may give way to a result that is not supported by the panel estimation of those same strategies. For instance, the R^2 decreases sensibly when shifting from the indices to the individual funds for five strategies: the distressed securities, emerging, equity market neutral, event-driven and market timing. The decrease of the R^2 of the event-driven strategy, from 0.81 to 0.34, is especially important. Notice that the strategies whose panel estimation is deceiving are not the most important ones but, as said before, are very specialised strategies. The F&F model does not seem to be appropriate for those strategies.

The apparent beta puzzle, which was observed for the distressed securities and the short sellers strategies when estimating the indices, is corroborated by the panel estimation. For those strategies, the coefficient of the risk premium is negative and significant at the 5 per cent level. As explained previously, there is not really a beta puzzle for these two strategies because the negative link between the beta and the risk premium is the expression of a rational behaviour. Moreover, for two other strategies, the emerging market and the event-driven ones, the coefficient of the risk premium is negative in their beta equation but not significant neither at the 5 per cent nor at the 10 per cent level. Finally, for five strategies, the coefficient associated to the risk premium in the beta equation is positive and significant at the 5 per cent level, indicating the absence of a beta puzzle. These strategies are: equity hedge, equity nonhedge, fund of funds, managed futures and market timing. For the macro strategy, the coefficient is positive and significant at the 10 per cent level.

Table 17.9 reveals that hedge funds usually reduce their beta when market volatility increases. For two strategies, the macro and managed futures ones, an increase in market volatility leads them to increase their beta. That relationship may be linked to their market timing activities but it may also reflect that volatility *per se* is a positive factor for these strategies. The volatility of macroeconomic variables is related to market volatility and macro

Table 17.8 OLS estimation of the conditional model in panel for the HFR funds pooled by strategies^a

	c	mkt-rf	smb	hml	umd	alpha1	beta1	beta2	R ²	DW
Distressed securities	0.4468 13.29	0.1178 15.46	0.1102 14.52	0.1111 11.43	-0.0214 4.20	-2.0724 10.79	-0.0030 2.29	-0.0019 10.79	0.25	1.72
Emerging	0.3398 1.97	0.7533 19.23	0.3144 8.05	0.2188 4.38	-0.0126 2.90	-2.8652 2.90	0.0044 0.65	-0.0046 5.11	0.16	2.00
Equity hedge	0.4833 39.88	0.5418 197.26	0.3013 110.07	0.1374 39.19	0.0814 44.20	1.7578 25.38	0.0143 30.25	-0.0001 1.38	0.87	2.02
Equity market neutral	0.1474 3.40	0.1487 15.12	-0.0254 2.59	0.0295 2.35	0.0613 9.31	-0.2231 0.90	0.0001 0.09	-0.0003 1.51	0.14	1.95
Equity non hedge	0.4786 11.44	0.7777 82.04	0.2902 30.72	0.2090 17.28	-0.0202 3.18	0.7408 3.10	0.0058 3.54	0.0000 0.19	0.69	2.02
Event driven	0.4214 13.87	0.2539 36.87	0.1544 22.51	0.1399 15.93	-0.0169 3.65	0.0078 0.04	-0.0035 2.97	-0.0018 11.30	0.34	1.75
Fund of funds	0.3099 25.06	0.2320 82.72	0.1358 48.63	0.0696 19.46	0.0556 29.59	0.2834 4.01	0.0062 12.90	-0.0013 19.33	0.94	2.01
Macro	0.3449 5.79	0.1190 8.74	0.0534 3.94	0.0984 5.67	0.0565 6.18	0.1889 0.55	0.0202 8.62	0.0024 7.65	0.06	1.98
Managed futures	0.2222 3.77	-0.0296 2.22	0.0858 6.45	0.1405 8.25	0.0995 11.12	-0.5586 1.66	0.0414 18.02	0.0033 10.75	0.15	2.01
Market timing	0.4073 3.90	0.3737 15.77	0.1083 4.59	0.0092 0.30	0.1164 7.33	0.9062 1.52	0.0274 6.73	0.0008 1.53	0.33	2.08
Short seller	0.5707 2.00	-1.1680 19.49	-0.4186 4.94	0.3380 5.40	-0.0578 1.74	1.4104 1.14	-0.0148 1.49	0.0010 1.06	0.65	1.93
Mean level of coef.	0.3793 11.31	0.1928 44.99	0.1009 23.36	0.1365 11.79	0.0311 11.00	-0.0385 4.73	0.0090 7.96	-0.0002 6.42	0.42	1.95
Mean level of t-stat										

^aThe conditional version of the Fama and French model is given by equation (10). For each strategy, the excess returns of the individual funds are estimated in panel. This method uses fixed effects to account for heterogeneity of the funds in a given strategy and cross-section SUR (seemingly unrelated regression) to take into account the interaction of the funds within a strategy. The variable alpha1 is the three-month T-bills rate lagged one period, expressed in deviation from its mean. The variable beta1 is the product of the market risk premium, on the one hand, and the market risk premium lagged one period and expressed in deviation from its mean, on the other hand. If the coefficient of this variable is negative, there is an apparent beta puzzle. The variable beta2 is the product of the market risk premium, on the one hand, and the squared market risk premium lagged one period and expressed in deviation from its mean, on the other hand. The last variable wants to verify the impact of market volatility on beta. The *t*-statistics are expressed in absolute value.

Table 17.9 GMM-hm estimation of the conditional model in panel for the HFR funds pooled by strategies^a

	c	mkt-rf	smb	hml	umd	alpha1	beta1	beta2	R ²	DW
Distressed securities	0.3068 6.60	0.1349 12.18	0.1259 13.44	0.1431 10.90	-0.0220 3.36	-3.7814 9.67	-0.0073 2.37	-0.0023 7.69	0.22	1.70
Emerging	0.5966 2.91	0.6874 11.47	0.2305 4.59	0.0592 0.86	-0.0396 1.12	-2.8503 2.53	-0.0144 0.87	-0.0090 5.65	0.15	2.00
Equity hedge	0.5235 31.40	0.5640 141.81	0.2939 87.38	0.1250 26.51	0.0956 40.75	2.8235 20.11	0.0325 29.30	0.0001 1.25	0.88	2.01
Equity market neutral	0.2344 3.62	0.1601 10.33	-0.0325 2.48	0.0289 1.58	0.0594 6.51	1.1529 2.11	0.0081 1.89	-0.0002 0.43	0.11	1.94
Equity non hedge	0.4307 6.66	0.8191 53.12	0.2808 21.53	0.2410 13.18	-0.0067 0.74	0.9266 1.70	0.0221 5.13	0.0000 0.01	0.61	2.02
Event driven	0.3362 8.81	0.2705 29.71	0.1678 21.80	0.1645 15.24	-0.0084 1.57	-0.7289 2.27	-0.0004 0.15	-0.0020 8.03	0.34	1.76
Fund of funds	0.3332 19.84	0.2360 58.78	0.1342 39.62	0.0528 11.12	0.0555 23.50	0.6207 4.39	0.0131 11.72	-0.0012 10.74	0.94	2.01
Macro	0.3502 4.60	0.0770 4.18	0.0526 3.39	0.0694 3.18	0.0406 3.74	-0.5854 0.90	0.0097 1.88	0.0024 4.86	0.04	1.98
Managed futures	0.2008 2.03	-0.0650 2.75	0.0403 2.01	0.0747 2.66	0.1138 8.16	-0.6383 0.76	0.0554 8.39	0.0066 10.33	0.14	2.02
Market timing	0.3442 2.36	0.4119 11.85	0.1056 3.59	-0.0060 -0.15	0.1452 7.09	1.0843 0.88	0.0572 5.90	0.0015 1.63	0.30	2.07
Short seller	0.7784 2.81	-1.2509 13.64	-0.4328 4.26	0.2778 3.42	-0.1468 3.45	0.8281 0.57	-0.0560 3.01	-0.0023 1.41	0.60	1.92
Mean level of coef.	0.4032 8.33	0.1859 31.80	0.0878 18.55	0.1119 8.05	0.0261 9.09	-0.1044 4.17	0.0109 6.42	-0.0006 4.73	0.39	1.95
Mean level of t-stat										

^aThe conditional version of the Fama and French model is given by equation (10). For each strategy, the excess returns of the individual funds are estimated in panel. This method uses fixed effects to account for heterogeneity of the funds in a given strategy and cross-section SUR (seemingly unrelated regression) to take into account the interaction of the funds within a strategy. Moreover, the GMM-hm estimation uses as instruments the higher moments of the risk factors up to the fifth order, as explained in the second section. The variable alpha1 is the three-month T-bills rate lagged one period, expressed in deviation from its mean. The variable beta1 is the product of the market risk premium, on the one hand, and the market risk premium lagged one period and expressed in deviation from its mean, on the other hand. If the coefficient of this variable is negative, there is an apparent beta puzzle. The variable beta2 is the product of the market risk premium, on the one hand, and the squared market risk premium lagged one period and expressed in deviation from its mean, on the other hand. The last variable wants to verify the impact of market volatility on beta. The t-statistics are expressed in absolute value.

funds have more business opportunities when the macroeconomic variables are more volatile. For managed futures, market volatility generally increases their profit opportunities. The coefficient of the squared market premium is also positive for the market timing strategy but it is significant only at the 10 per cent level. That might indicate in this case good market timing by the funds following this strategy.

Finally, Figures 17.1 and 17.2 show the fixed effects distribution for the two most important groups of hedge funds in our sample, which are the equity hedge and the fund of funds. At Figure 17.1, we see that the average alpha

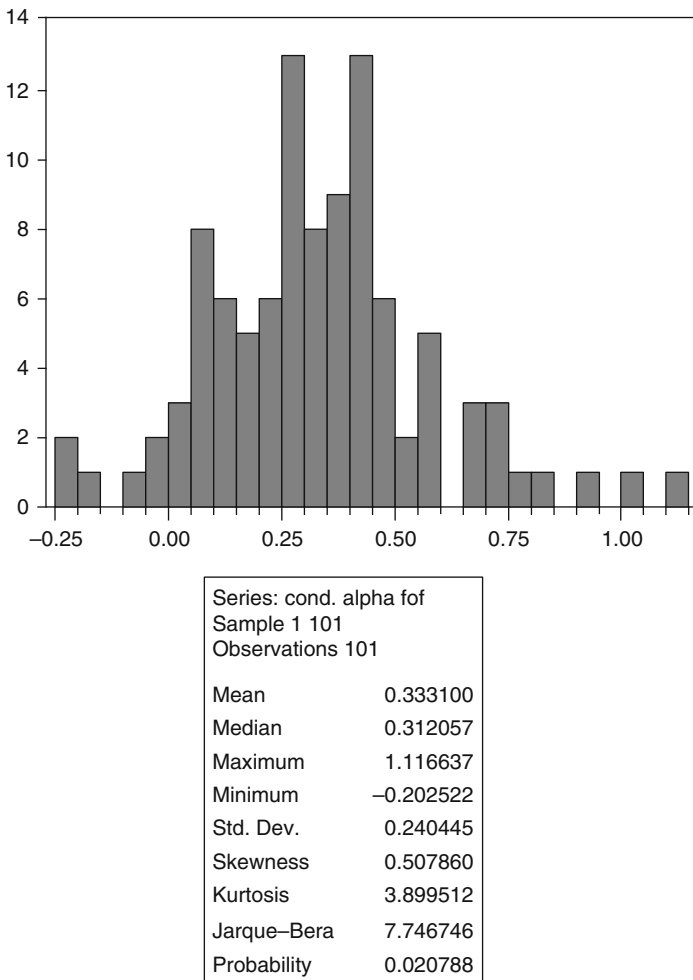


Figure 17.1 Distribution of the fixed effects in the panel GMM-hm estimation of the funds of funds strategy conditional model

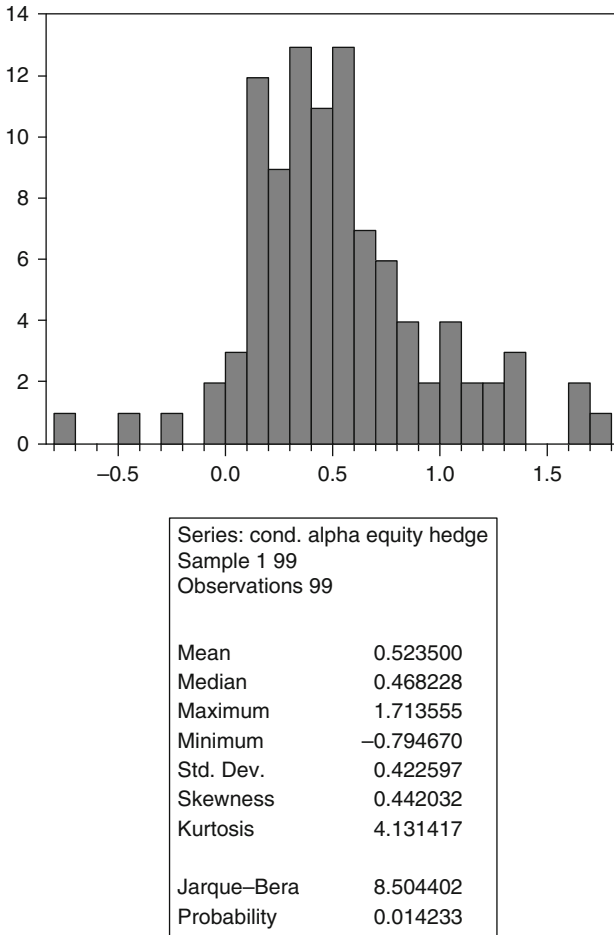


Figure 17.2 Distribution of the fixed effects in the panel GMM-hm estimation of the equity hedge strategy conditional model

for the fund of funds group is relatively moderate at 0.33 per cent but also that the standard deviation is quite high at 0.24. Therefore, the fund of funds group is very heterogeneous, the test for the fixed effects being incidentally very significant. Furthermore, a proportion of 6 per cent of these funds had an alpha under 0 over the period of analysis. We could make the same observations for the equity hedge group (Figure 17.1), except that the average alpha, at 0.52, is higher for this group. These results serve to qualify the high levels of estimated alphas in the case of the indices that might lead to suspect market inefficiencies.

Conclusion

Our contribution in this paper was threefold. First, we showed that it is preferable to estimate conditional models using an IV method because our empirical works suggest that there are many specification errors in the OLS estimation of such models, which are by nature empirical and may thus be contaminated by such errors. To do so, we proposed a new set of instruments based on the higher moments of the F&F model. These instruments were viewed as very promising to estimate asset pricing models. These estimation procedures generally greatly improve the fit of the estimations.

Secondly, we made our GMM-hm estimations in panel to estimate our conditional version of the F&F model. We showed that this procedure is much better than the simple averaging of results by strategy. It allows introducing fixed effects to account for the heterogeneity of the funds forming a strategy and cross-section SUR to take into account the interactions of the funds within a strategy.

Finally, we showed, with the help of a conditional version of the F&F model, that there does not seem to be a beta puzzle in the hedge fund industry. There is, however, an apparent beta puzzle for very specialised strategies, like the distressed securities and the short sellers one. But for those strategies, the negative link between their beta and the market risk premium is not really a puzzle because business opportunities are greater for the funds comprising these strategies when the stock market is depressed.

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43. SUR for 'seemingly unrelated regressions'.
44. For instance, at the end of the fourth quarter of 2003, the shares of the equity hedge, fund of funds and equity nonhedge strategies in the total assets of the hedge funds were respectively, 29.74, 19.57 and 5.96 per cent. The global share of these three strategies in the total assets of hedge funds was thus about 55 per cent, according to a HFR report.

18

Option Pricing Based on Mixtures of Distributions: Evidence from the Eurex Index and Interest Rate Futures Options Market

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Practical applications

This paper considers several alternatives to the standard Black and Scholes assumption of log-normally distributed security prices in an option pricing framework. Judging from empirical results for a comprehensive database of options transactions, a mixture of two or three log-normal kernels is found to perform well for price forecasting purposes, while the numerical implementation and calibration remain feasible. For non-path-dependent options, the technique could therefore serve as an alternative to complex process assumptions for the underlying, competing distributional shapes or other pricing approaches such as a direct fitting of smile curves.

The use of flexible distributional assumptions in an option pricing framework has been analysed extensively in recent years. Based on a large set of transaction data from 1999 and 2000, this paper readdresses the suitability of flexible distributions for forecasting and hedging purposes of DAX and Euro-Bund-Future options traded on the Eurex. Implied risk-neutral distributions (RND) are derived via the original Black and Scholes model, the series expansion approach of Corrado and Su and

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models based on mixtures of two and three log-normal distributions. While, especially for DAX options, the more complex RND prove superior to the log-normality assumption within the Black–Scholes model in terms of forecasting prices, delta-neutral hedge portfolios are best derived from simple Black-Scholes hedge ratios.

Introduction

In addition to the substantial body of theoretical research on the valuation of contingent claims, many empirical investigations on the pricing of derivatives have been undertaken in recent decades (see, for example, the comprehensive investigations by Bakshi *et al.*¹ and Dumas *et al.*²). In spite of the large size and steady growth of worldwide turnover in derivative contracts, both at exchanges and over the counter (OTC), analyses have focused mainly on the US market. Studies of the European Exchange (Eurex), currently the world's largest futures exchange, are rare. Apart from a few model-free analyses (see Mittnik and Rieken^{3,4}), most studies concentrate on model-specific phenomena such as volatility smile and term structure in the standard Black–Scholes⁵ model (see, for example, Eberlein *et al.*,⁶ Tompkins,⁷ Hafner and Wallmeier,⁸ Brunner and Hafner,⁹ Fengler *et al.*,¹⁰ Fengler and Wang¹¹). The literature, however, offers only a few comprehensive studies on price forecasting and hedging functionality applied to representative option databases. For the Deutsche Terminbörse, the predecessor of the Eurex, Rachev and Mittnik¹² study DAX options between February 1992 and September 1995 with up to 60 days to maturity on the basis of an array of models (eg Laplace, Student and inverse Gaussian distributions). Belledin and Schlag¹³ investigate several competing models with stochastic volatility, applied to a very limited set of DAX options traded on the Deutsche Terminbörse between July and December 1996.

The purpose of this study is to fill this conspicuous gap in the literature by undertaking a pioneering investigation of option models for price forecasting and hedging purposes of Eurex DAX and Euro-Bund-Future options, based on data from 1999 and 2000. From a methodological point of view, the main technique consists in extracting risk-neutral distributions (RND) from a given cross-sectional set of options transactions. This mapping of the market pricing mechanism abstracts from specific assumptions about the stochastic evolution of market parameters. As a 'natural' extension of the original Black–Scholes model, the paper focuses on models based on mixtures of two and three log-normal distributions for the underlying. The use of such mixtures (for an introduction, see, for example, McLachlan and Peel¹⁴) can be motivated by their outstanding suitability for approximation purposes. For example, using an increasing number of normal mixture components, any distribution can be approximated to an arbitrary degree of precision. Ritchey¹⁵ was among the first to employ a mixture of (normal) distributions in option pricing. Subsequent research, including that of Bahra,^{16–18} Melick

and Thomas,¹⁹ Guo,²⁰ Söderlind and Svensson,²¹ Gemmill and Saflekos,²² Schittenkopf and Dorffner,²³ Kim and Kim²⁴ and, very recently, Rebonato and Cardoso,²⁵ assumes that the density of the terminal distribution of the underlying is a linear combination of log-normal densities. (For a discussion of the implications of the mixture approach, for example, with respect to the calculation of moments and the structure of the volatility smile, see Brigo and Mercurio.²⁶) An economic interpretation is the assumption of multiple alternative regimes. The main advantage of the mixture approach is its high flexibility, allowing the approximation of a large array of distributions—without becoming excessively vulnerable numerically owing to the ‘semi-parametric’ character. The smooth behaviour of the tails—which rapidly decline and therefore, for example, prevent excessive kurtosis—is advantageous. Resulting valuation formulae are of closed-form, facilitating parameter estimation and covering the Black–Scholes model as a special case. For a competing benchmark, refer additionally to the Gram–Charlier series expansion approach proposed by Corrado and Su.²⁷

On average, the more complex RNDs prove superior to the log-normality assumption within the Black–Scholes model in terms of forecasting ability, especially for DAX options, whereas delta-neutral hedge portfolios are best set up using hedge ratios derived from the simple Black–Scholes formula.

The remainder of the paper is organised as follows. In the second section, valuation formulae are derived for options in a mixture-of-distributions framework. The third section briefly reviews the series expansion approach to option valuation which serves as an additional benchmark for the mixture models. The data for the empirical study are described in the fourth section. The fifth section is devoted to the presentation of the results with respect to calculated (in-sample) RNDs, the models’ (out-of-sample) forecasting ability and suitability for hedging purposes. The final section summarises the main results and provides an outlook for further research.

Option pricing based on mixtures of distributions

Overview

The value of a derivative security can be expressed as an expected discounted pay-off, whereby the expectation is taken with respect to an appropriate risk-neutral density $f_{t,T}^*$.²⁸ For the value c_t of a European call option with strike K and maturity T at time $t \leq T$, this result is formalised as

$$\begin{aligned} c_t(K) &= e^{-r(T-t)} E^* (\max\{S_T - K, 0\}) \\ &= e^{-r(T-t)} \int_K^\infty (S_T - K) f_{t,T}^*(S_T) dS_T \end{aligned} \quad (1)$$

where r denotes the appropriate domestic interest rate (continuously compounded); S_T the random spot price at the option’s maturity; and E^* the risk-neutral expectation operator. Breeden and Litzenberger²⁹ demonstrate

that, if the risk-neutral distribution is continuous, its discounted density is equal to the second derivative of the European call option price with respect to the strike price

$$f_{t,T}^*(S_T) = e^{r(T-t)} \frac{\partial^2 c_t(K)}{\partial K^2} \bigg|_{K=S_T} \quad (2)$$

Boundary conditions $\partial c_t(K)/\partial K \geq -e^{-r(T-t)}$ and $\lim_{K \rightarrow \infty} \partial c_t(K)/\partial K = 0$ as well as convexity and differentiability of $c_t(K)$ are assumed. Based on the boundary restrictions, it is easily shown that $f_{t,T}^*(S_T) \geq 0$ and $\int_0^\infty f_{t,T}^*(K=S_T) dK = 1$ holds; ie $f_{t,T}^*(S_T)$ is indeed a density function. Notably, the martingale restriction holds for each case $E^*(S_T) = \int_0^\infty S_T f_{t,T}^*(S_T) dS_T = e^{r(T-t)} S_T$.

The RND can be entirely (re)constructed from given European option prices and, in a complete market, be uniquely identified. For practical purposes, direct use of relations (1) and (2) faces the problem that option prices are observable only at a few traded strike prices. The structure of the RND between strike prices inevitably requires further assumptions. In addition, the function resulting from (2) will probably lack the desired smoothness. Methods for constructing RND from European option prices can basically be classified as follows (for a slightly different typology see, for example, Bahra¹⁷):

1. Assumption of a specific stochastic process driving the security price.
2. Assumption of a specific functional form of the terminal distribution—a more general approach, because a single terminal distribution can result from many different stochastic processes.
3. Non-parametric techniques that are data oriented and largely free of parametric assumptions; Jackwerth³⁰ distinguishes further between kernel methods, maximum entropy techniques and the group of approaches which directly fit implied volatilities and RND (methods of curve fitting).

Many models from categories (2) and (3) are reviewed by Jackwerth.³⁰ Because of its flexibility and relative ease of implementation, this paper focuses on the class of parametric approaches from category (2) as a base technique and thereby especially on the mixture-of-distributions framework. Compared with non-parametric approaches, parametric methods generally are less vulnerable to overfitting and are useful as a means of ensuring a 'reasonable' distribution.²¹ Furthermore, parametric approaches contain submodels which facilitate an analysis of the required flexibility of the RND, while for non-parametric methods, there are no 'subcases'.

Derivation of pricing formulae

Within the Black-Scholes model, the assumption of a security process following a geometric Brownian motion results in an inflexible log-normal price distribution in any interval $[t, T]$. A (finite) mixture of k log-normal

distributions can be seen as a ‘natural’ extension of the Black–Scholes framework. Modelling the (risk-neutral) security price distribution as a mixture of log-normal distributions accordingly leads to a mixture of normal distributions for the corresponding (risk-neutral) return distribution. This follows directly from the relationship between the density functions for return ($g_{t,T}^*$) and security ($f_{t,T}^*$): $g_{t,T}^*(u) = S_t e^u f_{t,T}^*(S_t e^u)$.

Formally, let $k \in \mathbb{N}$

$$\begin{aligned} f_{t,T}^{\text{Mixture } k}(S_T) &= \sum_{j=1}^k \lambda_j f_j(\alpha_j, \beta_j; S_T) \\ &= \sum_{j=1}^k \frac{\lambda_j}{S_T \beta_j} \varphi\left(\frac{\ln S_T - \alpha_j}{\beta_j}\right), S_T > 0 \end{aligned} \quad (3)$$

with

$$\sum_{j=1}^k \lambda_j = 1, \quad \lambda_j > 0 \forall j = 1, 2, \dots, k \quad (4)$$

and

$$\begin{aligned} \alpha_j &= \ln S_t + \left(\varsigma_j - \frac{\sigma_j^2}{2} \right) (T - t). \\ \beta_j &= \sigma_j \sqrt{T - t} \end{aligned} \quad (5)$$

$\varphi(x)$ stands for the density function of the standard normal distribution and λ_j denotes the ‘weight’ of the j th mixture component with parameters α_j and β_j , which themselves depend on (instantaneous) drifts ς_j and volatilities σ_j of the corresponding return distributions. It is obvious that (3) is indeed a density function with

$$f_{t,T}^{\text{Mixture } k}(S_T) \geq 0 \forall S_T > 0$$

and

$$\int_{0,t,T} f_{0,t,T}^{\text{Mixture } k}(S_T) dS_T = 1$$

In order to exclude arbitrage opportunities in a complete market, an investment in the option underlying within each interval $[t, T]$ must yield the risk-free rate, ie $E^*(S_T) = S_t e^{r(T-t)}$. With (3), this leads to the martingale constraint

$$E^*(S_T) = \sum_{j=1}^k \lambda_j \exp\left(\alpha_j + \frac{\beta_j^2}{2}\right) = S_t e^{r(T-t)} \quad (6)$$

Without losing generality, the last drift parameter of the mixture, ς_k , $k \geq 2$ can be used to guarantee (6). Hence, for ς_k it follows that

$$\sum_{j=1}^k \lambda_j e^{\varsigma_j(T-t)} = e^{r(T-t)} \Leftrightarrow$$

$$\varsigma_k = \frac{1}{T-t} \ln \left(\frac{e^{r(T-t)} - \sum_{j=1}^{k-1} \lambda_j e^{\varsigma_j(T-t)}}{1 - \sum_{j=1}^{k-1} \lambda_j} \right)$$

Ensuring the existence of $\varsigma_k \in \mathbb{R}$ requires

$$e^{r(T-t)} - \sum_{j=1}^{k-1} \lambda_j e^{\varsigma_j(T-t)} > 0 \Leftrightarrow \sum_{j=1}^{k-1} \lambda_j e^{(\varsigma_j - r)(T-t)} < 1$$

Based on the mixture of distributions, specific valuation formulae for European standard calls and puts, are then easily derived according to Cox and Ross²⁸

$$c_t^{\text{Mixture } k} = \sum_{j=1}^k \lambda_j [S_t e^{(\varsigma_j - r)(T-t)} \Phi(d_{j1}) - K e^{-r(T-t)} \Phi(d_{j2})] \quad (7)$$

$$p_t^{\text{Mixture } k} = \sum_{j=1}^k \lambda_j [K e^{-r(T-t)} \Phi(-d_{j2}) - S_t e^{(\varsigma_j - r)(T-t)} \Phi(-d_{j1})] \quad (8)$$

with

$$d_{j1} = \frac{\ln\left(\frac{S_t}{K}\right) + \left(\varsigma_j + \frac{1}{2}\sigma_j^2\right)(T-t)}{\sigma_j \sqrt{T-t}}$$

$$d_{j2} = d_{j1} - \sigma_j \sqrt{T-t} \quad (9)$$

$\Phi(x)$ as cumulative density function of the standard normal distribution and (6) to hold. Formulae (7) and (8) highlight that the mixture of densities leads to a linear combination of Black–Scholes expressions. Besides S_t , K , r and $T-t$, the value of a European call or put in a model with k mixture components depends on $k-1$ weights λ_j , $k-1$ parameters ς_j and k parameters σ_j , hence $3k-2$ parameters in total. For $k=2$, and without explicitly demanding the martingale restriction to hold, Bahra¹⁷ provides a numerically advantageous transformation of these formulae.

Competing benchmark: moment-adjusted option pricing formulae

As an alternative to the Black–Scholes and mixture models, this paper additionally refers the approach of Corrado and Su,²⁷ which uses a Gram–Charlier series expansion of the (risk-neutral) return distribution, thus incorporating non-normal skewness (γ_1) and kurtosis (γ_2) into option pricing. The value of a European call accordingly equals the Black–Scholes model value plus two correction terms that account for the higher moments of the return distribution

$$c_t^{GC} = c_t^{BS} + \gamma_1 \Theta_3 + (\gamma_2 - 3) \Theta_4 \quad (10)$$

with

$$c_t^{BS} = S_t \Phi(d_1) - Ke^{-r(T-t)} \Phi(d_2) \quad (11)$$

$$\begin{aligned} \Theta_3 = & \frac{1}{3!} S_t \sigma \sqrt{T-t} [\varphi(d_1) (2\sigma \sqrt{T-t} - d_1) \\ & + \sigma^2 (T-t) \Phi(d_1)] \end{aligned} \quad (12)$$

$$\begin{aligned} \Theta_4 = & \frac{1}{4!} S_t \sigma \sqrt{T-t} [\varphi(d_1) \\ & (d_1^2 - 1 - 3\sigma \sqrt{T-t} \\ & (d_1 - \sigma \sqrt{T-t})) \\ & + \sigma^3 (T-t)^{3/2} \Phi(d_1)] \end{aligned} \quad (13)$$

and

$$\begin{aligned} d_1 = & \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma \sqrt{T-t}} \\ d_2 = & d_1 - \sigma \sqrt{T-t} \end{aligned} \quad (14)$$

It is worth noting that in Corrado and Su²⁷ (p. 179), there is a false sign in the expression for (3) in (12). Instead of a plus, the formula shows a minus sign (see also Brown and Robinson³¹). An analogous formula p_t^{GC} for European put options can be derived by means of put-call parity. The direct derivation of a put formula, however, leads to an expression that, together with (10), does not fulfil put-call parity since, during the series expansion, terms of order five and higher are neglected.

One of the important constraints of the expansion model is that the implied (risk-neutral) return density $g_{t,T}^*(x)$ must fulfil $\int_{-\infty}^{\infty} g_{t,T}^*(x) dx = 1$ as well

as $g_{t,T}^*(x) \geq 0$ for all x from the support. The latter condition is necessary, since an approximation of a density function with the aid of a finite sum can also lead to negative function values. To guarantee that these conditions hold in (10), the parameters γ_1 and γ_2 must be restricted to certain intervals. Importantly, by means of series expansion, neither extremely skewed nor platykurtic return distributions can be captured (for more details see Jondeau and Rockinger³²).

Data

The option products of the Eurex form the basis of the study. The analysis focuses on two popular products—index options on the German blue-chip stock index DAX (Deutscher Aktienindex) and options on the Euro-Bund-Future (Eurex³³)—and their trades from January 1999 to December 2000. All option data were provided by Deutsche Börse, Frankfurt/Main, Germany.

DAX options are European-style and their underlying is the DAX performance index, based on the XETRA electronic trading system at the Frankfurt Stock Exchange. The index comprises the stocks of the 30 largest German exchange-listed companies. As a performance index, the valuation assumes a fictitious reinvestment of paid dividends into the index portfolio. Since the DAX performance index is also adjusted for any price-relevant corporate actions, it can be treated as a non-dividend paying asset. Euro-Bund-Future options are American-style and, as a special feature, are subject to a daily marking-to-market similar to their underlying (futures-style margining). The underlying future is based on fictitious notional long-term debt issued by the Federal Republic of Germany with a remaining time-to-maturity of between 8.5 and 10.5 years, and an interest coupon of six per cent per annum. Note that the underlying of Euro-Bund-Future options depends on their time to maturity: while the options can comprise maturities in any month, the Euro-Bund-Future expires on a quarterly basis. The options always refer to the next maturing Euro-Bund-Future. For example, options due in January, February and March are all based on the March Euro-Bund-Future contract.

When deriving RND from prices of American-style Euro-Bund-Future options, one does not need to control for the early exercise premium. For a futures-style option, it is never optimal to exercise it prematurely.^{34,35} Since Euro-Bund-Future contracts expire long before the bonds' specifics in the underlying (eg pull-to-par) become substantial, we do not have to account for the special interest rate-sensitive characteristics of the underlying.

For DAX options, the XETRA-DAX is used as input parameter for the underlying. The last trade price of the corresponding Euro-Bund-Future contract is assigned to the Euro-Bund-Future options. Intraday data for the XETRA-DAX were provided by Karlsruher Kapitalmarktdatenbank, Karlsruhe, Germany. Transactions data on the Euro-Bund-Future contracts were obtained from

Deutsche Börse, Frankfurt/Main, Germany. In the following, time-to-maturity is measured in calendar days and used in formulae as fractions of a year. Following standard practice, the appropriate term-dependent money market Euribor (European Interbank Offered Rate), available for intervals up to 12 months, is used as a proxy for the risk-free interest rate. Linearly interpolated rates are derived, while for times-to-maturity of more than one year, the 12-month Euribor is employed for simplicity.

In a preliminary step, the databases are corrected with respect to (1) insufficient time synchronisation between underlying and options market; and (2) options' times to maturity of less than seven calendar days or prices below ten ticks. Further cleaning steps refer to (3) the violation of distribution-free arbitrage boundaries; (4) the law-of-one-price; and (5) put-call parity. Since far in-the-money and out-of-the-money options are very rarely traded and therefore generally do not provide reliable and time-stable information, (6) moneyness— $M = (S_t - K)/K$ for calls and $M = (K - S_t)/K$ for puts—is restricted as follows: For DAX options, an interval $M \in [-0.25, 0.25]$, for Euro-Bund-Future options $M \in [-0.05, 0.05]$ is considered suitable. Note that no filtering with respect to implied volatility is applied, since this would imply a model-dependent cleaning that one explicitly wishes to avoid.

Table 18.1 provides an overview of characteristics of the cleaned databases, consisting of 1,195,820 records for DAX and 138,583 records for Euro-Bund-Future options, as well as their assigned market variables. Besides minimum and maximum, the respective 10-, 25-, 50-, 75- and 90-percentiles are displayed.

For both DAX and Euro-Bund-Future options, approximately one-third of the records in the cleaned databases belong to at-the-money (ATM) transactions; ie $M \in [-0.025, 0.025]$ for DAX options and $M \in [-0.005, 0.005]$ for Euro-Bund-Future options, while two-thirds are out-of-the money or deep out-of-the-money trades. Notably, 55 per cent (70 per cent) of the DAX (Euro-Bund-Future) options records belong to very short-term options with seven to 30 calendar days to maturity.

Results

Implied risk-neutral distributions

Given a set of N_1 (N_2) observed call (put) option prices $c_{t,T;i}^{\text{Market}}$ ($p_{t,T;i}^{\text{Market}}$), the M model-implied parameters ξ_i are obtained by minimising the sum of squared differences between these prices and the theoretical option values

$$c_{t,T;i}^{\text{Model}} \quad (p_{t,T;i}^{\text{Model}})$$

$$\begin{aligned} \arg \min_{\xi_1, \xi_2, \dots, \xi_m} & \sum_{i=1}^{N_1} (c_{t,T;i}^{\text{Market}} - c_{t,T;i}^{\text{Model}})^2 \\ & + \sum_{i=1}^{N_2} (p_{t,T;i}^{\text{Market}} - p_{t,T;i}^{\text{Model}})^2 \end{aligned} \quad (15)$$

Table 18.1 Descriptive statistics for the cleaned data bases

	Percentiles						
	Min.	10	25	50	75	90	Max.
<i>I. DAX options</i>							
DAX index ^a	4,601.07	5,041.47	5,266.49	6,438.74	7,116.26	7,420.69	8,136.16
Strike ^a	3,750	4,900	5,300	6,200	7,100	7,600	10,400
Time-to-mat. ^b	7	10	16	29	50	100	732
Interest rate ^c	2.2824	2.6068	2.9040	3.3280	4.3780	4.8384	5.2736
Option price ^a	1.00	12.00	28.70	70.50	149.00	277.10	2,345.00
Moneyiness	-0.2500	-0.1401	-0.0852	-0.0410	-0.0102	0.0112	0.2499
<i>II. Euro-Bund-Future options</i>							
Future ^a	101.88	103.49	104.86	105.80	109.36	113.85	116.33
Strike ^a	98.00	103.50	104.50	106.00	109.50	114.00	118.50
Time-to-mat. ^b	7	10	15	22	36	58	90
Option price ^a	0.10	0.13	0.20	0.35	0.60	0.95	5.67
Moneyiness	-0.0500	-0.0190	-0.0125	-0.0066	-0.0014	0.0018	0.0498

Notes: ^apoints; ^bcalendar days; ^c% per annum (continuous compounding).

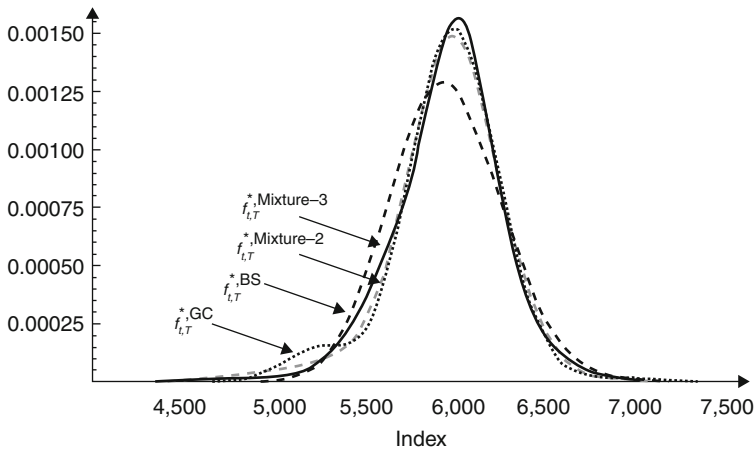
The author refers to transactions prices because settlement prices are set by the Eurex and therefore do not always reflect actual trading opportunities. Based on the least-squares approach in (15), implied parameters for each interval $[t, T]$ are calculated for each model and trading day. Accordingly, all call and put options transactions enter the minimisation procedure (15). The data input, however, is restricted to those subsamples of options transactions with at least eight different strike prices in each estimation, so as to ensure reliable support for the extracted distributions. In order to investigate whether calibration for the Black–Scholes model (BS) should constrain on information from ATM options, the Black–Scholes model is also analysed by restricting the minimisation approach in (15) to ATM option trades (BS-ATM). The results are compared with those of the mixture models with two (Mixture-2) and three (Mixture-3) kernels as well as with those of the Gram–Charlier model (GC).

The calculation of optimal parameter vectors according to (15) requires global optima of m -dimensional functions which consist of sums of highly non-linear terms. The usual optimisation and grid search algorithms tend to fail, especially for the mixture models with $m=4$ and $m=7$, respectively. Simulated annealing is therefore employed routinely for (stochastic) global optimisation (see, for example, Locatelli³⁶), together with local optimisation algorithms (cf Rebonato and Cardoso²⁵ for an alternative numerical approach to solving the optimisation problem posed in (15)). For the mixture models, the number of parameters that increase rapidly when adding new mixture components is a problem. Numerical stability is another issue which may become critical, especially for the higher moments of the distribution.^{37,38}

For each model, there are a total of 2,630 parameter vectors for the DAX distributions and 657 parameter vectors for the distributions of the Euro-Bund-Future. Figure 18.1 depicts the implied estimations for the DAX option-implied RND on a randomly selected trading day, 19th November, 1999. The higher moments are particularly conspicuous. While the log-normal security price distribution is always skewed to the right in the Black–Scholes model, skewness in all three alternative models is reversed to a right-steepness; the more pronounced the more flexible the model (−0.41 to −0.94). A similar picture is obtained for kurtosis: the log-normal distribution with a value of approximately 3.0 faces the much more leptokurtic alternative distributions with values of up to 7.3. The figure also shows that for index values of approximately 5,300 or lower, the Black–Scholes alternatives have more probability mass in the left tail. For the right tails, such a phenomenon cannot be detected.

Out-of-sample performance

Based on the premise that estimated implied distributions are constant for short-term horizons, option prices on the following trading day are forecast (sliding windows technique). Therefore, all forecasts are run term-dependent.



	BS	GC	Mixture-2	Mixture-3
Expected value	5,976.50	5,976.50	5,976.50	5,976.50
Standard dev.	309.46	319.74	320.87	352.29
Skewness	0.1555	-0.4148	-0.7057	-0.9402
Kurtosis	3.0430	4.5779	5.2152	7.2597
Median	5,968.50	5,997.80	5,997.00	6,001.78
Modus	5,952.55	6,008.71	6,007.13	6,034.23
IQD*	416.70	359.49	366.59	361.34

* Interquartile distance

Figure 18.1 Example of implicit risk-neutral security price distributions ($f_{t,T}^*$) for DAX options. Sample data from 19th November, 1999; calculation from 2,247 options transactions; options' time-to-maturity was 28 calendar days

For the DAX options, Table 18.2 provides the summary statistics for absolute and relative deviations in a one-day price forecast. Based on 2,630 implicit calibrations for each model, the table provides the overall one-day out-of-sample analysis for DAX options. In total, projected values of 1,178,789 options are compared with their realised market values, leading to the reported statistics for the model-specific mean and mean absolute errors (ME and MAE) as well as for the mean percentage and mean absolute percentage errors (MPE and MAPE) between model values and market prices.

The statistics for the ME show that the direction of deviation (given in points) is the same for all models, decreasing with an increase in distributional flexibility. The figures for the MAE serve as an indicator that the best forecast is achieved by means of the mixture models: compared with BS and BS-ATM, MAE is, on average, reduced by more than 40 per cent (accompanied by a standard deviation that is approximately 30 per cent lower).

Table 18.2 DAX options: Overall out-of-sample performance

Model	ME				MAE			
	Mean	Std. dev.	Min	Max.	Mean	Std. dev.	Min.	Max.
BS	-1.99	21.26	-1,205.65	797.64	13.10	16.87	0.00	1,205.65
BS-ATM	-1.51	20.86	-1,183.99	426.23	13.21	16.22	0.00	1,183.99
GC	-0.04	14.41	-1,162.59	784.17	6.64	12.78	0.00	1,162.59
Mixture-2	-0.75	13.13	-1,160.66	726.52	5.83	11.79	0.00	1,160.66
Mixture-3	-0.74	13.11	-1,160.64	718.47	5.80	11.78	0.00	1,160.64

Model	MPE (in %)				MAPE (in %)			
	Mean	Std. dev.	Min.	Max.	Mean	Std. dev.	Min.	Max.
BS	-3.41	48.15	-100.00	7,562.20	28.80	38.74	0.00	7,562.20
BS-ATM	-3.38	48.98	-100.00	7,562.71	29.22	39.45	0.00	7,562.71
GC	0.95	24.38	-100.00	6,074.09	12.15	21.16	0.00	6,074.09
Mixture-2	-1.34	18.54	-99.92	5,811.63	10.08	15.62	0.00	5,811.63
Mixture-3	-0.61	18.44	-100.00	5,808.42	9.81	15.62	0.00	5,808.42

Though not part of the optimisation approach, the relative deviation measures MPE and MAPE yield a similar picture, favouring the mixture models. The MAPE proves that for these models the absolute percentage error is, on average, more than 60 per cent lower than for the BS and BS-ATM case, while more than halving the standard deviation.

Additionally, option-specific characteristics are examined for their explanatory power with respect to the forecast error. Out-of-sample value differences $\hat{V}_t - V_t^{\text{Market}}$ are regressed linearly against moneyness (M), time-to-maturity (τ) and type of option (given by the indicator variable ζ with 0: call and 1: put)

$$\begin{aligned} \hat{V}_t - V_t^{\text{Market}} = & a_0 + a_1 M_i + a_{21} \tau_i^c \\ & + a_{22} \tau_i^p + a_3 \zeta_i + \varepsilon_i \end{aligned} \tag{16}$$

with $\tau^c = \tau(1 - \zeta)$ and $\tau^p = \tau\zeta$, respectively. Results for the DAX options are given in Table 18.3. In order to cope with likely heteroscedasticity in the data set, standard errors are adjusted for heteroscedasticity of unknown form according to White.³⁹

For the BS model, the option characteristics exhibit a high explanatory power for the forecast error ($R_{\text{adj}}^2 \approx 0.30$). The constant term a_0 shows a general overvaluation of approximately 2.90 points. Differentiating by calls and puts by means of ζ_i reveals a strong bias in the pricing of both option types ($a_3 \approx -7.00$ (points)). The hypothesis that moneyness does not influence the forecast error ($a_1 = 0$) cannot be rejected. The coefficients for τ^c and τ^p show the opposite ‘direction’ of the mispricing depending on time-to-maturity:

Table 18.3 DAX options: Influence of moneyness, time-to-maturity and option type on price forecasting performance

Model	a_0	a_1	a_{21}	a_{22}	a_3	R^2_{adj}
BS	2.8952 (19.3***)	-0.4476 (-0.3)	0.1047 (24.0***)	-0.1731 (-32.3***)	-6.9966 (-30.4***)	0.3116
BS-ATM	0.8506 (6.2***)	1.9298 (1.5)	0.1649 (45.8***)	-0.1245 (-23.6***)	-6.5056 (-28.8***)	0.3464
GC	-0.9485 (-7.8***)	-20.9903 (-20.6***)	0.0479 (13.8***)	-0.0814 (-17.4***)	0.9655 (5.1***)	0.0829
Mixture-2	-0.4202 (-4.0***)	-5.1908 (-5.6***)	0.0092 (3.2***)	-0.0202 (-4.6***)	-0.7224 (-4.1***)	0.0092
Mixture-3	-0.4468 (-4.3***)	-5.6732 (-6.0***)	0.0092 (3.2***)	-0.0212 (-4.9***)	-0.6548 (-3.7***)	0.0094

Statistics for two-sided coefficient tests for test value zero are given in parentheses; standard errors are adjusted for heteroscedasticity of unknown form according to White.³⁹

*** (**, *): significance at 1% (5%, 10%) level.

R^2_{adi} denotes the adjusted coefficient of determination.

among long-term options, the BS model overvalues calls and undervalues puts. No similar characteristic is found for the moneyness coefficient a_1 . As expected, similar results are obtained in the BS-ATM case.

For the GC model, far less of the variance in the regression can be explained by moneyness, time-to-maturity and option type ($R^2_{\text{adi}} \approx 0.08$). Compared with the BS reference case, moneyness is now found to have a significant influence on the forecast error. Remarkably, the imbalance in the valuation between calls and puts is much less pronounced ($a_3 \approx -0.97$ (points)).

The regressions for the mixture models prove the usefulness of flexible distributional assumptions in the case of DAX options: with ($R^2_{\text{adi}} \approx 0.01$, the forecast error cannot be properly explained by option moneyness, time-to-maturity or option type. Compared with the GC case, the moneyness coefficient a_1 , for example, is much lower in absolute terms, indicating that the bias in the GC model is far less evident within the mixture framework. Both mixture models exhibit a similar performance, calling the appropriateness of a third distribution kernel into question.

Table 18.4 provides the results of the forecasting performance of the five models for the group of Euro-Bund-Future options. Based on 657 implicit calibrations for each model, the table provides the overall one-day out-of-sample analysis for Euro-Bund-Future options. In total, projected values of 110,108 options are compared with their realised market values, leading to the reported statistics for the model-specific ME and MAE as well as for the MPE and MAPE between model values and market prices.

For all models, deviation between projected values and realised market prices are economically very small. In a similar manner to the case of DAX

Table 18.4 Euro-Bund-Future options: Overall out-of-sample performance

Model	ME				MAE			
	Mean	Std. dev.	Min.	Max.	Mean	Std. dev.	Min.	Max.
BS	-0.0058	0.0481	-1.4685	1.0310	0.0332	0.0353	0.0000	1.4685
BS-ATM	-0.0098	0.0479	-1.5302	0.9957	0.0335	0.0356	0.0000	1.5302
GC	-0.0038	0.0400	-1.4931	0.9980	0.0251	0.0313	0.0000	1.4931
Mixture-2	-0.0051	0.0402	-1.4773	0.9961	0.0255	0.0315	0.0000	1.4773
Mixture-3	-0.0059	0.0404	-1.4633	0.9790	0.0257	0.0317	0.0000	1.4633

Model	MPE (in %)				MAPE (in %)			
	Mean	Std. dev.	Min.	Max.	Mean	Std. dev.	Min.	Max.
BS	-2.06	15.60	-99.75	355.53	10.40	11.81	0.00	355.53
BS-ATM	-3.22	15.58	-99.80	343.36	10.53	11.94	0.00	343.36
GC	-0.54	10.39	-96.73	344.14	6.95	7.73	0.00	344.14
Mixture-2	-0.92	10.42	-93.70	343.48	7.04	7.74	0.00	343.48
Mixture-3	-1.13	10.44	-93.47	337.57	7.03	7.80	0.00	337.57

options, GC and mixture models lead to a reduction in mean mispricing—judging, for example, from the statistics for MAPE, which is as much as approximately 30 per cent lower than in the BS case.

The regression analysis according to equation (16), shown in Table 18.5, yields only a moderate influence of option moneyness, time-to-maturity and type on the mispricing within the BS and BS-ATM models ($R^2_{\text{adj}} \approx 0.16$). With the help of the BS alternatives, none of the three variables is found to explain the forecasting error ($R^2_{\text{adj}} \approx 0.01$). The GC, Mixture-2 and Mixture-3 models all perform very similarly.

Hedging performance

To investigate the ability of option pricing models to capture dynamic structures and the correct co-movement between option and underlying over time, hedge portfolios, built model-specifically at time t and monitored for a period of length Δt are referred to. A delta hedge is constructed as follows: sell an option at time t and simultaneously enter a position in underlying and a risk-free asset. In the following, a positive sign for a variable denotes a cash inflow due to a sale and a negative sign indicates cash flows from the portfolio due to a purchase.

In the case of a call, a sale at time t and a simultaneous purchase of δ^c_t items of the underlying at current price S_t are required. Since cash inflows from the option sale are insufficient to finance this transaction, an additional credit of $B_t = \delta^c_t S_t - c_t$ is necessary. Analogously, in the case of a put, δ^p_t units of the underlying must be sold. The cash inflow amounting to $B_t = p_t + \delta^p_t S_t$ must be put into a risk-free money market account. The number

Table 18.5 Euro-Bund-Future options: Influence of moneyness, time-to-maturity, and option type on price forecasting performance

Model	a_0	a_1	a_{21}	a_{22}	a_3	R^2_{adj}
BS	0.0125 (10.4***)	0.5594 (11.1***)	0.0001 (3.3***)	-0.0002 (-4.7***)	-0.0265 (-15.9***)	0.1587
BS-ATM	0.0082 (6.8***)	0.4999 (9.9***)	0.0001 (3.2***)	-0.0002 (-5.0***)	-0.0265 (-15.9***)	0.1573
GC	0.0000 (0.0)	-0.1064 (-2.5**)	-0.0001 (-2.9***)	-0.0000 (-1.1)	-0.0053 (-3.7***)	0.0033
Mixture-2	-0.0011 (-1.0)	-0.0969 (-2.3**)	-0.0001 (-2.6***)	-0.0000 (-0.4)	-0.0065 (-4.6***)	0.0039
Mixture-3	-0.0031 (-2.8***)	-0.1225 (-2.9***)	-0.0001 (-1.8*)	-0.0000 (-0.3)	-0.0056 (-3.9***)	0.0036

Statistics for two-sided coefficient tests for test value zero are given in parentheses, standard errors are adjusted for heteroscedasticity of unknown form according to White.³⁹

*** (**, *): significance at 1% (5%, 10%) level.

R^2_{adj} denotes the adjusted coefficient of determination.

of units of the underlying, δ_t^c and δ_t^p respectively, equals the model-specific sensitivity measure 'delta'

$$\delta_t^c = \frac{\partial c_t}{\partial S_t} \text{ and } \delta_t^p = \left| \frac{\partial p_t}{\partial S_t} \right| \quad (17)$$

The portfolio value at time $t + \Delta t$ yields the deviation from the reference value zero and measures the model-specific hedge error (HE)

$$HE_{t+\Delta t}^c = -c_{t+\Delta t} + \delta_t^c S_{t+\Delta t} - e^{r\Delta t} B_t \quad (18)$$

and

$$HE_{t+\Delta t}^p = -p_{t+\Delta t} - \delta_t^p S_{t+\Delta t} + e^{r\Delta t} B_t \quad (19)$$

respectively. The smaller the absolute value of HE, the more effectively the model provides risk immunisation. For this specific empirical analysis, average prices of the last three transactions in a certain call or put option on a given day are employed. Hedge errors are calculated over one- and five-trading-day horizons. The model performance is assessed via the mean hedging error (MHE) and mean absolute hedging error (MAHE) within an interval $[t, t + \Delta t]$.

The results of the hedging analysis for DAX options are documented in Table 18.6. In total, 29,290 (26,620) 'pairs' of calls and 29,219 (26,091) 'pairs' of puts are built for a one-day (five-day) horizon. For call hedge portfolios with a one-day horizon, the average mean hedge error amounts to 0.39

Table 18.6 DAX options: Hedging performance

Model	Calls				Puts			
	MHE		MAHE		MHE		MAHE	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
<i>1-day interval</i>								
BS	0.39	22.09	7.48	20.78	0.35	15.43	6.10	14.18
BS-ATM	0.34	19.17	7.43	17.67	0.35	15.43	6.09	14.18
GC	0.43	22.40	7.82	21.00	0.44	15.99	6.76	14.50
Mixture-2	0.49	22.78	8.19	21.26	0.50	16.22	6.92	14.68
Mixture-3	0.50	22.75	8.19	21.23	0.49	16.23	6.92	14.69
<i>5-day interval</i>								
BS	1.62	32.11	18.48	26.30	2.37	26.86	15.89	21.79
BS-ATM	1.60	30.57	18.53	24.36	2.35	26.87	15.84	21.83
GC	1.54	32.24	18.66	26.34	3.00	27.62	16.78	22.15
Mixture-2	1.72	32.93	19.27	26.76	3.32	28.07	17.08	22.52
Mixture-3	1.73	33.03	19.31	26.85	3.33	28.03	17.07	22.49

MHE, mean hedging error; MAHE, mean absolute hedging error.

index points, according to the BS model. The average deviation is slightly lower (0.34) for the BS-ATM model. Remarkably, the average mean error increases for all three more complex models (up to 0.50). A similar picture is obtained for the mean absolute hedge error, with average values around 7.00 (index points). Extending the hedge period to five trading days leads to a comparable ‘ranking’ among the models and an increase in average mean errors by approximately factor 4.0 for calls and even factor 7.0 for puts. Referring to the MAHE, the error increases by a factor of approximately 2.5. Differentiating further according to moneyness and time-to-maturity of the options (not documented here in detail), the results are very similar. With few exceptions—for example, short-term, far out-of-the-money calls—deviations are lowest for the BS and the BS-ATM models.

Table 18.7 provides the hedging results for Euro-Bund-Future options. Here, there are 1,167 (959) ‘pairs’ of calls and 1,357 (1,134) ‘pairs’ of puts for a one-day (five-day) horizon. All findings are very similar to the DAX options; this is also true for the non-documented differentiation between moneyness and time-to-maturity.

Summary and outlook

This paper has considered the question of which distributional assumption allows the most accurate forecasting of option prices and the most accurate static delta hedging strategy. Employing the Black Scholes model, the Gram Charlier density expansion model and two models with mixtures of log-normal distributions on Eurex DAX index and Euro-Bund-Future options

Table 18.7 Euro-Bund-Future options: Hedging performance

Model	Calls				Puts			
	MHE		MAHE		MHE		MAHE	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
<i>1-day interval</i>								
BS	-0.59	5.93	3.83	4.56	-0.62	5.85	3.98	4.33
BS-ATM	-0.59	5.92	3.82	4.56	-0.63	5.86	3.99	4.34
GC	-0.78	6.32	4.07	4.90	-0.77	6.24	4.28	4.59
Mixture-2	-0.80	6.37	4.07	4.97	-0.73	6.19	4.25	4.56
Mixture-3	-0.78	6.38	4.08	4.97	-0.75	6.22	4.28	4.58
<i>5-day interval</i>								
BS	-1.23	14.34	9.19	11.08	-1.72	13.90	9.06	10.68
BS-ATM	-1.23	14.33	9.17	11.07	-1.78	13.97	9.09	10.75
GC	-1.95	15.73	9.83	12.43	-2.04	15.35	9.94	11.87
Mixture-2	-2.07	15.84	9.85	12.58	-2.02	15.44	9.95	11.98
Mixture-3	-2.12	15.92	9.89	12.65	-2.04	15.40	9.94	11.94

Note: For ease of notation, all displayed values are multiplied by 100.

MHE, mean hedging error; MAHE, mean absolute hedging error.

data from 1999 and 2000, the study showed that, compared with the Black Scholes reference model, more flexible distributions led, on average, to a significant reduction in absolute and relative deviations in forecasted prices. The mixture models performed better than the Gram Charlier model, although the difference for two and three mixture kernels proved to be small. Analysing the forecasting of option prices in detail, it was found that the more complex models help to reduce systematic biases relating to option type (call vs put), moneyness and time-to-maturity. In general, the use of more advanced distributional assumptions was shown to be of practical use especially for DAX options with a well-pronounced smile structure of implied volatility. Conversely, for Euro-Bund-Future options, differences in forecasting performance for the five models were of limited economic relevance. For the latter group, one might even suspect an overfitting of the mixture model with three kernels, also leading to rather unstable distributions over time. Although the limitation in allowed skewness and kurtosis of competing RNDs as in the Gram-Charlier model, can generally be overcome with the help of mixture models, the problem of a likely overparameterisation has to be kept in mind. For example, Li⁴⁰ recommends that: 'A good SPD (state price density, author's note) is feasible with as few as three parameters: one for volatility, one for skewness, and one for kurtosis' (p. 49).

With respect to hedging performance, average absolute hedging errors compared with the Black-Scholes reference case were found to increase with the flexibility of the underlying RND for both groups of options in the majority of cases. Extending the hedging interval from one to five calendar days,

average absolute hedging errors across all models approximately doubled or even tripled. As a general assessment, one might agree with the finding of Dumas *et al.*² that: 'simpler is better' (p. 2104). Against the background of transactions costs occurring in practice, detected market anomalies in any of the models do not seem to reveal economically relevant arbitrage potential.

While the results support recent research on other international options markets (for example, by Kim and Kim²⁴), the evident limitations of the investigated mixture approach suggest the need for further investigation. Based on cross-sectional data at a given date, the static implied distributions do not consider the evolution of RND over time. In order to cope with the empirically non-flat term structure of implied volatilities, it is necessary to use separate estimations of distributions for any future time interval $[t, T]$. A straightforward extension of the mixture concept would be a 'dynamisation' by means of regime-switching models (see, for example, Bollen⁴¹). In a flexible set-up, however, time-varying transition probabilities in inevitable Monte-Carlo simulations render such an approach computationally demanding. Nevertheless, one has to bear in mind that, in practice, implied RND—and especially implied volatilities—serve as measures for pricing options relative to their underlyings. Traders are usually not particularly concerned as to whether implied volatility is really an accurate estimator of volatility over the option's lifetime.⁴² Their role as price substitutes rather suggests, for example, a dependence of implied volatilities on supply and demand.⁴³ Future studies—especially those referring to distributional rather than structural model assumptions—might consider, in particular, such empirically proven market phenomena.

Note

This paper does not represent an official statement of the firm the author is affiliated to.

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